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Akarsh Verma
Sanjay Mavinkere Rangappa
Shigenobu Ogata
Suchart Siengchin *Editors*

Forcefields for Atomistic-Scale Simulations: Materials and Applications

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Akarsh Verma · Sanjay Mavinkere Rangappa ·
Shigenobu Ogata · Suchart Siengchin
Editors

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Editors
Akarsh Verma
Department of Mechanical Engineering
University of Petroleum and Energy Studies
Dehradun, India

Shigenobu Ogata
Department of Mechanical Science
and Bioengineering
Osaka University
Osaka, Japan

Sanjay Mavinkere Rangappa
Department of Materials and Production
Engineering
KMUTNB
Bangkok, Thailand

Suchart Siengchin
Department of Materials and Production
Engineering
KMUTNB
Bangkok, Thailand

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Introduction to Molecular Dynamics Simulations



Gaurav Kumar, Radha Raman Mishra, and Akarsh Verma

Abstract The invention of novel functional materials and their investigation at the molecular level are vital in today's nanotechnology era. Atomistic modelling approaches are cost-effective and time-consuming alternatives to expensive and time-consuming experimental methods, and they are precise enough to predict the mechanical characteristics of materials. The current chapter goes through the many steps involved in a molecular dynamic's investigation. The various types of interatomic potentials and their applicability to various materials have been thoroughly examined. Following that, the integration algorithm for solving a set of Newton's equations, as well as the radius cut-off distance and temperature control, was addressed. Afterwards, many types of ensembles and boundary conditions were addressed, which helped in simulating real-world experimental settings. The approaches for minimizing energy have also been briefly explored. Finally, the limitations of molecular dynamics have been examined, as well as their applicability.

Keywords Molecular dynamics · Interatomic potential · Temperature control · Energy minimization · Ensemble

G. Kumar (✉)
Department of Mechanical Engineering, National Institute of Technology, Uttarakhand, Srinagar 246174, India
e-mail: grv.kmr@nituk.ac.in; grv.kmr2015@gmail.com

R. R. Mishra
Department of Mechanical Engineering, Birla Institute of Technology and Science, Pilani 333031, Rajasthan, India

A. Verma
Department of Mechanical Engineering, University of Petroleum and Energy Studies, Dehradun 248007, Uttarakhand, India

Department of Mechanical Science and Bioengineering, Osaka University, Osaka 560-8531, Japan

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1 Introduction

Molecular dynamics (MD) simulation integrates Newton's equations of motion over time to obtain the motion of the atoms/molecules in a system (Fig. 1) and, subsequently, quantitative and qualitative information about macroscopic behaviour of the system at the atomistic level [1]. Various properties like thermal expansion coefficient, melting point, heat capacity, thermal conductivity, radial distribution function, structural properties, etc. can be determined directly or with the help of statistical analysis of the data containing atomic positions, momenta, energies, forces, etc. over each timesteps during and after equilibration. Further, raw data can also be used to describe processes and phenomena such as crack growth and fast fracture, ion bombardment, sputtering, vapour deposition, etc. [1].

Three steps are involved in a typical molecular dynamic's simulation: (1) estimation of forces and evaluation of potential energy based on present atomic locations and velocities; (2) updating atomic coordinates and velocities as per the integrator algorithm and (3) post-processing and visualization from the raw data of the simulation [2].

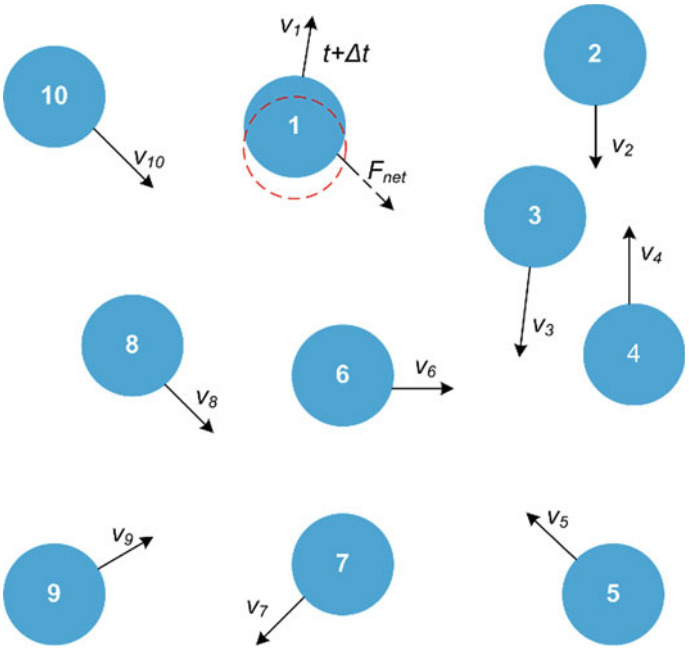


Fig. 1 Interaction of atom 1 with other nine atoms in a particular MD simulation

2 Interatomic Potential or Force Field

In classical molecular dynamics, a single point with a finite mass replaces an atom as a three-dimensional structure (Fig. 1). The balance between attraction and repulsion is established by the so-called interatomic potential or force field when atoms are close enough to feel each other [1]. The atoms/molecules in the system have potential energy due to interaction and bonding with other atoms/molecules in the system, which may be expressed mathematically by Eq. 1:

$$U = U\left(\vec{r}_1, \vec{r}_2, \vec{r}_3, \ldots .. \vec{r}_n\right) \tag{1}$$

With the provided interatomic potential, U and the initial positions, r_i of the particle, i having a mass, m_i in the system, the force experienced by each atom/molecule is calculated in Eq. 2:

$$\vec{F}_i = m_i \vec{a}_i = m_i \frac{d\vec{v_i}}{dt} = m_i \frac{d^2\vec{\boldsymbol{R}}_i}{dt^2} = \frac{d\vec{\boldsymbol{p_i}}}{dt} = -\vec{\nabla} U\left(\vec{R}_i\right) = -\frac{dU\left(\vec{R}_i\right)}{dr} \tag{2}$$

Using above equations, trajectories and motion of the atoms are determined in a small timestep of Δt over transition to lower energy states, i.e. state of equilibrium from higher energy state (Fig. 1). Forces on atoms can be calculated, and subsequently the above equation for the system's temporal evolution can be solved if a system's potential as a function of interatomic distance is known. Interatomic potentials are commonly calculated empirically, either via DFT or by fitting experimental data.

Several potentials having varying degrees of precision had been reported, each with its own set of advantages and drawbacks. The force fields used to define the bonded and non-bonded interatomic interaction are critical to any atomistic simulation. The thermo-mechanical properties of the material under investigation should be accurately reproduced by the interatomic potential. Majority of the forcefield used in molecular dynamic modelling of materials like nanocomposites, metals, and nanofillers were developed empirically and tested against experimental data or high-fidelity simulation data [3]. The choice of potential is significantly influenced by the application as well as the material. Semiempirical or empirical methods are preferred alternatives for modelling mechanical properties of materials because they can frequently simulate huge systems with tens of thousands to billions of atoms. But, dealing with diverse facets of a material's mechanical behaviour frequently involves the use of multiple simulation approaches [4].

Pair potentials are the simplest and computationally efficient way to describe the particle interactions because interatomic distance solely determines the potential energy, and the system's total energy is calculated by adding the all-atomic bond's energy over all N particles in the system. The overall energy of the system can be expressed in Eq. 3 [4]:

$$U_{total} = \frac{1}{2} \sum_{\alpha \neq \beta = 1}^{N} \sum_{\beta = 1}^{N} \varphi\left(r_{\alpha\beta}\right) \tag{3}$$

where $r_{\alpha\beta}$ is the interatomic distance amid α and β and $\varphi\left(r_{\alpha\beta}\right)$ is the potential energy function.

The popular pair potentials for the simulation of materials are Lennard–Jones (LJ) potential [5] and the Morse potential [6].

The Lennard–Jones potential can be expressed mathematically in Eq. 4:

$$\varphi\left(r_{\alpha\beta}\right) = 4\epsilon_0 \left[\left(\frac{\sigma}{r_{\alpha\beta}} \right)^{12} - \left(\frac{\sigma}{r_{\alpha\beta}} \right)^{6} \right] \tag{4}$$

where $r_{\alpha\beta}$ is the interatomic distance, ϵ_0 is the lowest energy of the potential curve and scales the energy of atomic bonds and σ is the interatomic distance at which the potential is zero and scales the length.

The Morse potential is defined in the form given in Eq. 5 [6]:

$$\varphi\left(r_{\alpha\beta}\right) = D\left[1 - \exp\left(-\beta\left(r_{\alpha\beta} - r_0\right)\right)\right]^2 \tag{5}$$

where r_0 is the nearest neighbour lattice spacing, and D and β are additional fitting parameters.

For many materials, the Morse potential is more realistic, but because of the exponential term, it is more computationally expensive than the LJ potential. In general, pair potential functions consist of two terms, each of which describes the repulsive and attractive interactions between atoms, and the sum of which gives the total energy. The pair potential is used to express the atomic interactions of noble gases (Neon, Argon, Krypton, etc.), spherical molecules and secondary bonds very well. However, the application of potential to metals, semiconductors and other solids is limited due to the absence of many-atom effect (more than two) [1]. Further, pair potential is also not used to model the elastic properties of metals [4].

When a material is made up of more than two atoms, multibody potentials are used in molecular dynamics simulations due to differences in bond strength and bond equivalent distance. Furthermore, atomic bonds may have different properties at a crystal's surface than in the bulk, and a description that uses an atom's environment to determine bond strength is required to appropriately depict the change in bond properties at a surface. As a result, the binding energy between two atoms is now a function of the locations of all other atoms in the region, rather than just the distance between them. Multibody potentials can capture this behaviour, which was not possible with pair potentials. Multibody potentials are designed to incorporate more detailed information about atom bonds than merely the distance between two neighbours. In the case of metals, embedded atom method (EAM) potentials can be used to precisely represent atom interactions [7, 8]. An EAM potential is usually expressed as given in Eq. 6:

$$U_i = \sum_{j=1}^{N_i} \varphi(r_{ij}) + f(\rho_i) \tag{6}$$

where ρ_i denotes the local electron density which depends on the atom i's local environment, and f denotes the embedding function which describes the dependence of atomic energy on the local energy density.

The EAM takes into account contributions from the pair potential as well as the embedding energy which approximates the missing N-atom effect in a pair potential. Overall, multibody potentials reproduce the elastic characteristics of metals far better than pair potentials [9]. Multibody potentials are effective for most metals, particularly for FCC metals, but not for alloys [10]. Most conventional multibody potentials can't mimic the effect of directional bonding, but modified embedded atom method (MEAM) potentials can [11].

Covalent solids are less densely packed (with a coordination number of 4) and contain strongly equal directed bonds; therefore, deviations from the pairwise additive scheme are significantly larger. As a result, the primary characteristics of these materials are bond angle and bond order, and the bond strength of two atoms depends on coordination number and bond angle. Bond order potentials are based on the bond order concept and are especially well suited for characterizing forces in covalently bonded materials such as carbon nanotubes, diamonds and zincblende structures [12]. The Tersoff potential [13, 14], Stillinger–Weber potential [15], Brenner's force fields [16, 17], Stuart reactive potential [18], reactive bond order (REBO) potential, reactive force field (ReaxFF) [19] and adaptive intermolecular reactive empirical bond order (AIREBO) potential are the examples of bond order potentials. The many-body term in the Tersoff potential is explicitly dependent on the angles of triplets of atoms, as well as the coordination effect [13, 14]. The Tersoff potential can be expressed mathematically as given in Eq. 7:

$$U_{Tersoff} = \frac{1}{2}\sum\nolimits_{i \neq j} U_R(r_{ij}) + \frac{1}{2}\sum\nolimits_{i \neq j} B_{ij} U_A(r_{ij}) \tag{7}$$

where U_R represents the potential of repulsive nature; U_A represents the potential of attractive nature and B_{ij} represents the bond order of atoms i and j.

Tersoff-type interatomic potential has been used for the modelling of h-BN nanofillers, carbon, silicon and graphene because of its parametrization simplicity [20, 21]. Tersoff potential has been used to treat heteronuclear bonds [22] to capture phonon dispersion, to allow the formation and dissociation of covalent bonds and to estimate binding energy [23]. The Tersoff potential may be used to evaluate the thermo-mechanical properties of graphene/BN nanofillers and nanocomposites. Researchers have reported the use of the Morse potential, reactive empirical bond order (REBO) potential [24] and, more recently, the adaptive intermolecular reactive empirical bond order (AIREBO) potential [25] for simulating and characterizing the thermo-mechanical properties, and fracture behaviour of carbon nanotube and graphene.

The mathematical expression for REBO potential is given in Eq. 8:

$$E_{\alpha\beta}^{REBO} = f\left(r_{\alpha\beta}\right)\left(E_{\alpha\beta}^{Repulsive} + b_{\alpha\beta}E_{\alpha\beta}^{Attractive}\right) \tag{8}$$

where $b_{\alpha\beta}$ is the bond order; $f\left(r_{\alpha\beta}\right)$ is the cut-off function that restricts the bonded interaction between the nearest adjacent atoms within the cut-off distance; $E_{\alpha\beta}^{Repulsive}$ is the energy corresponding to repulsion and $E_{\alpha\beta}^{Attractive}$ is the energy corresponding to attraction.

Bonded interaction is mainly accountable for sustaining the structural integrity of carbon-based nanofillers, although non-bonded interactions also play a crucial role in imparting structural strength to large systems like graphene. AIREBO potential is a more advanced version of REBO force field as it also accounts for non-bonded interaction as well four-body torsional interactions, which is significant for modelling curved structures like CNT. AIREBO is the most precise interatomic potential for modelling graphene's mechanical, fracture and thermal characteristics. The mathematical expression for AIREBO potentials is given in Eq. 9:

$$E_{\alpha\beta}^{AIREBO} = \frac{1}{2}\sum_{\alpha}\sum_{\alpha\neq\beta}\left[E_{\alpha\beta}^{REBO} + E_{\alpha\beta}^{LJ} + \sum_{\gamma\neq\alpha}\sum_{\delta\neq\alpha,\beta,\gamma}E_{\gamma\alpha\beta\delta}^{tors}\right] \tag{9}$$

where α, β, γ and δ refer to atoms, $E_{\alpha\beta}^{LJ}$ represents the non-bonded LJ potential and $E_{\gamma\alpha\beta\delta}^{tors}$ is the four-body torsional energy.

The total energy E_{system} in the reactive force field consists of several partial energies, as defined in Eq. 10 [19]:

$$E_{\text{system}} = E_{\text{bond}} + E_{\text{over}} + E_{\text{under}} + E_{\text{val}} + E_{\text{pen}} + E_{\text{tors}} + E_{\text{conj}} + E_{\text{vdWaals}} + E_{\text{Columb}} \tag{10}$$

where E_{bond} is the bond energy; E_{over} is the energy corresponding to over-coordination; E_{under} is the energy corresponding to undercoordination; E_{val} is the energy corresponding to valence angle; E_{pen} is the energy corresponding to penalty; E_{tors} is the torsional energy; E_{conj} is conjugation energy; E_{vdWaals} is the energy corresponding to van der Waals interactions and E_{Columb} is the energy corresponding to Coulomb interactions.

The ReaxFF estimates potential energy contributions from both bonded and non-bonded interactions. It is the sole approach that can be used to forecast the reactivity of multimolecular systems, and it is more suited in modelling bond breaking and formation mechanisms [19]. ReaxFF is used to simulate hydrocarbons, graphite, diamond, cyclohexane, buckyball, graphene, h-BN nanofillers, etc. [26]. Universal force field (UFF) is another interatomic potential besides Tersoff, REBO, AIREBO and ReaxFF which can be used to simulate any type of material [27]. The element, its hybridization state and their connectivity all influence its parameterization. Its

incorrect prediction of metal characteristics, on the other hand, is due to the fact that it ignores the response of valence electrons.

The UFF, which is provided in Eq. 11, is the superposition of several energy terms.

$$E_{UFF} = E_{stretch} + E_{bending} + E_{torsion} + E_{invn} + E_{\text{vdWaals}} + E_{\text{Columb}} \tag{11}$$

where $E_{stretch}$ is energy function corresponding to bond stretching; $E_{bending}$ is the bending energy of an angle; $E_{torsion}$ is the torsional energy and E_{invn} is the energy corresponding to inversion terms.

Commonly used force fields for atomistic modelling of polymer-based system are Condensed-phase Optimized Molecular Potentials for Atomistic Simulation Studies (COMPASS) [28], Chemistry at Harvard Macromolecular Mechanics (CHARMM) [29] and Molecular Mechanics (MM3) [30].

3 Numerical Integration: Finding Trajectories

A set of N second-order differential equations is represented by Newton's equation. It's impossible to obtain atom trajectories by solving this set of equations exactly. As a result, MD simulation employs the finite-difference approach. To calculate the trajectory of each atom, Newton's equations are numerically solved at discrete time steps. First, from the given potential, the forces acting on all atoms are computed. After that, accelerations of all atoms are computed. Subsequently, finite-difference techniques are used to determine the positions, velocities and accelerations at time, $t + \Delta t$. The leap-frog algorithm, Verlet algorithm, velocity Verlet algorithm and predictor–corrector algorithm are some of the techniques available for solving Newton's equations of motion and determination of atomic trajectories in molecular dynamics simulation. One of the most often used approaches in MD is the velocity Verlet algorithm [31]. The position, velocity and acceleration at time, $t + \Delta t$ are calculated using Eqs. 12–15 from the corresponding quantities at time t:

$$v\left(t + \frac{\Delta t}{2}\right) = v(t) + \frac{1}{2!}a(t)\Delta t \tag{12}$$

$$r(t + \Delta t) = r(t) + v(t)\Delta t + \frac{1}{2!}a(t)\Delta t^2 = r(t) + v\left(t + \frac{\Delta t}{2}\right)\Delta t \tag{13}$$

$$a(t + \Delta t) = -\left(\frac{1}{m}\right)\frac{du[r(t + \Delta t)]}{dr} = \frac{F(t + \Delta t)}{m} \tag{14}$$

$$v(t + \Delta t) = v(t) + \frac{a(t) + a(t + \Delta t)}{2}\Delta t = v\left(t + \frac{\Delta t}{2}\right) + \frac{1}{2}a(t + \Delta t)\Delta t \tag{15}$$

4 Time Step

As atom vibrates in the frequency of range of 10^{-14} s in a solid lattice, and therefore a unit of molecular dynamic simulation should be in the range of 10^{-15} s. A lower time step increases computing effort and, in most cases, improves accuracy, whereas a bigger timestep has the reverse effect. Furthermore, each atom's velocity, acceleration and force are assumed to be constant in this time step. The timestep should be enhanced as much as possible to speed up the simulation while maintaining energy conservation and run stability. The atoms should not traverse more than 1/30 of the nearest neighbour distance in this time step, which results in typical time steps ranging from 0.1 to several femtoseconds. A trade-off problem that frequently emerges in a computational process is computational load and accuracy with regard to a time step. According to Jensen et al., 2015, the optimum time step may be identified by performing a series of simulations with increasingly larger time steps in the constant volume and energy (NVE) ensemble and selecting the largest time step that preserves energy conservation. As the use of a 1.0 fs time step proved troublesome in the simulation, time steps over 0.5 fs are not reliable for predicting the mechanical behaviour of materials like a diamond [32]. Yao et al. reported that for modelling mechanical properties of carbon nanotubes, a time step of 5 fs may be utilized [33]. Velocity Verlet algorithm is used to integrate the equations of motion in order to examine the mechanical characteristics and buckling behaviour of boron nitride nanotubes under the action of torsional stress with a basic time step of 1 fs, which ensures good energy conservation [34]. For proper molecular dynamics simulation of H_2 and CO_2, the maximum time step should be 1.823 fs and 3.808 fs, respectively [35].

5 Cut-off Radius Distance

At long distances, a potential fades typically away and becomes insignificant. The idea that atomic interactions are fundamentally local is a primary assumption in developing many interatomic potentials. The idea is that atoms interact so weakly beyond a certain bond length that they make basically no contribution to the overall energy. The interatomic potential energy function must be constructed in such a way that the energy contribution attributed to a single atom is only impacted by atoms within a cut-off radius, r_{cut}, and force computation beyond r_{cut} is ignored. The r_{cut} must be less than half the size of the primary box. If this isn't the case, an atom may interact with both an atom in the primary box and its image in the image box, resulting in an unphysical double interaction. A smaller cut-off radius improves computing efficiency but reduces simulation accuracy; as a result, cut-off radius is commonly designed so that calculation accuracy is not compromised. Wu and Wei used a value of 1.92 Å as the C–C bond cut-off radius to study the influence of defects in graphene; however, Jhon et al. utilized a value of 2.0 Å to prevent unphysical findings before the

bond breaking while determining the impact of cut-off radius on the tensile strength of graphene [36, 37]. Lennard–Jones potentials have a cut-off distance of 2.5–3.2 Å, while EAM potentials have a cut-off distance of 5. Tersoff potentials have a cut-off distance ranging from 3 to 5 Å.

6 Temperature Control

Constant temperature molecular dynamics (MD) simulations are commonly used to predict the mechanical behaviour of materials atomistically [38]. MD simulations at constant temperature employ approaches that allow some atoms to absorb the extra energy pumped into a system, and so successfully manage the system temperature in a physically realistic manner [39]. The temperature of a system is the average of kinetic energies of all the atoms which can be calculated from the atomic velocities (refer to the Eq. 16) [40, 41]:

$$T = \frac{1}{k_B N_f} \sum_{i,\alpha} m^{\alpha} \left(v_i^{\alpha}\right)^2 \tag{16}$$

where k_B is Boltzmann's constant; N_f is the total translational degree of freedom; m^{α} is the mass of atom, α and v_i^{α} is the velocity of atom α in i direction.

Because temperature is connected to kinetic energy, the velocities of the particles in the simulation system must be adjusted to regulate the temperature. One technique is to employ a velocity rescaling method, also known as a velocity thermostat, which is relatively simple and efficiently adds (or subtracts) energy to (or from) the system. This approach of direct velocity rescaling, on the other hand, is quite imprecise and far from how energy is actually dissipated [39].

This approach rescales each particle's velocity directly, as observed in Eq. 17:

$$\left(\frac{v_{new}}{v_{old}}\right)^2 = \frac{T}{T_{ins}} \tag{17}$$

where v_{old} is the velocity before the rescaling and v_{new} is the rescaled velocity.

The velocity rescaling approach has been superseded by more advanced constant-temperature systems. The generalized Langevin equation (GLEQ) technique [42], the Berendsen method [43] and the Nose–Hoover thermostat [44–46] are the three most often utilized methods, and so are taken into account in this work. Anderson thermostat imposes a temperature on a system by putting it into thermal contact with a huge heat bath. For a particular temperature, the velocity of a random particle is replaced with a value picked from the Maxwell–Boltzmann distribution. Anderson thermostat has a high computational cost. The Langevin thermostat is frequently referred to as a stochastic dynamics technique because it integrates random forces into the equations of motion of the atoms. These random forces are intended to mimic

the effect of a heat bath. The end effect is a method for keeping the system's average temperature constant. The Nose–Hoover thermostat is regarded as one of the best thermostats available.

To modify the equations of motion, this thermostat uses a friction factor (μ) defined in Eq. 18:

$$\frac{d\mu(t)}{dt} = \frac{k_B N_f}{Q}(T(t) - T_0) \tag{18}$$

where Q is the effective mass of thermostat.

The modified equation of motion is defined by Eq. 19:

$$a = \frac{f(t)}{m} - \mu(t)v(t) \tag{19}$$

This is a method for creating true canonical ensembles in both momentum and coordinate space when performing constant-temperature dynamics. The Berendsen thermostat is the most widely used thermostat because of its simplicity and ease of use. The Berendsen algorithm conserves total momentum but not energy and maintains a constant temperature with good approximation. The system is connected to an external heat bath source with a predetermined temperature T_0 to maintain a consistent temperature during the simulation. At each time state, the velocities of atoms are scaled so that the rate of temperature change is proportional to the temperature difference (refer Eq. 20):

$$\frac{dT(t)}{dt} = \frac{1}{\tau}(T - T(t)) \tag{20}$$

where τ is the characteristic relaxation time, δt is the timestep size and T is the instantaneous temperature.

7 Ensembles

The use of ensembles is a prerequisite for performing the powerful MD simulations. Each atom moves and reacts differently in a simulation box of a molecular dynamics simulation, and every timestep generates a new microstate of the system. A group of alternative configurations that correspond to distinct microscopic states but have the similar macroscopic state is referred to as an ensemble. Ensembles are not all the same; differing macroscopic constraints result in various ensembles. Micro-canonical (NVE), canonical (NVT), isobaric-isothermal (NPT) and Grand Canonical (μVT) are the four most often utilized ensembles as shown in Fig. 2. Number of atoms, volume, energy, temperature, pressure and chemical potential are all represented by the letters N, V, E, T, P and μ, respectively [1, 4, 26].

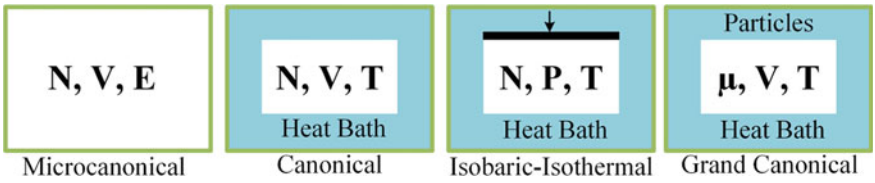


Fig. 2 Schematic of different ensemble in MD simulation

The number of atoms (N), simulation box volume (V) and total energy (E) in the microcanonical ensemble are fixed. The microcanonical ensemble is a closed system that is unable to exchange matter or energy with its environment. This ensemble is the most popular in molecular dynamics due to its ability to represent the real experimental condition very well. All other ensembles are enclosed by large external systems with set parameters, as seen in Fig. 2. The canonical ensemble, which is most commonly employed in Monte Carlo techniques, fixes the number of atoms, box volume and temperature (T). The isothermal–isobaric ensemble fixes the number of atoms, pressure and temperature, whereas the grand canonical ensemble emerges when the simulated system is immersed in an infinite heat bath and may exchange particles with an infinite particle reservoir. The chemical potential, volume and temperature are fixed in the grand canonical ensemble, but the number of atoms is permitted to fluctuate. Tsai and Tu studied a single layer of graphene sheet and graphite flakes using molecular dynamics in 2010. They used two different types of ensembles to simulate the mechanical behaviour of graphene and found that the modified NPT ensemble is more accurate than the conventional NVT ensemble for simulating the mechanical behaviour of graphene [47].

8 Boundary Conditions

When it's required to predict a material's bulk behaviour by modelling a bunch of atoms, the periodic box concept comes into play. Free surfaces are not desirable during simulation because atoms at the box's borders may experience very different resultant forces (surface effects) than those in the box's centre and may move out of the unit cell, resulting in a change in density. The periodic box concept helps to solve these problems by simulating the behaviour of a bulk material. The central box shown in Fig. 3 is actually the simulation box in which the atoms are all vibrating, in which system has been created. When the periodic boundary condition is imposed, an atom leaving the main simulation box is treated as if it were entering from the opposite side. So, when one atom goes out like this, it's the same as when another atom comes in like this. So, there's a primary simulation box, followed by periodic images. Only the atoms inside the primary box at the centre are really simulated, and all image boxes simply duplicate the primary box. The force acting between an atom and its periodic image cancels out, and if necessary, a simple coordinate

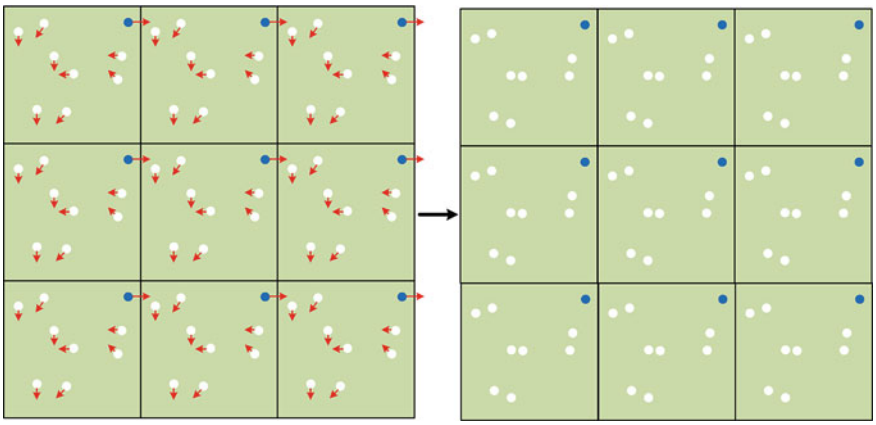


Fig. 3 Schematic of periodic boundary condition

transformation updates the locations of particles in the neighbouring image box. The primary simulation box will have length, height and breadth. A 2d simulation box will have 8 periodic images while a three-dimensional simulation box will have 26 periodic images. Other boundary conditions such as a free boundary or a fixed boundary condition, and so on may also be utilized to simulate different conditions.

9 Energy Minimization

Energy minimization is a process of reducing the system's potential energy at 0 K to the smallest attainable value. Because the sketched chemical structures are not energetically favourable, energy minimization is critical for establishing the right molecule arrangement in space. Because a molecule's potential energy includes several energy components, when an energy minimization programme is performed, it will quickly approach a minimal local energy value (Fig. 4), and it may halt if the programme used is not thorough. Subsequently, the most stable conformer is not obtained as the energy minimization has stopped before the barrier which may be overcome with the use of suitable algorithms that can increase the structure's strain energy and eventually lead to the most stable conformer, known as the global energy minimum (Fig. 4). As a result, knowing the behaviour of a stable molecule requires identifying the energy minima. The use of molecular modelling enables the most stable conformational stage to be achieved.

The steepest descent and conjugate gradient method are the most popular energy minimization algorithms. The steepest descent method is used to find local stationary spots in a multi-dimensional function. However, because of the approximation, it is less efficient, necessitating additional steps to determine the minimum. Because the direction in which the geometry is initially reduced is opposite to the direction in

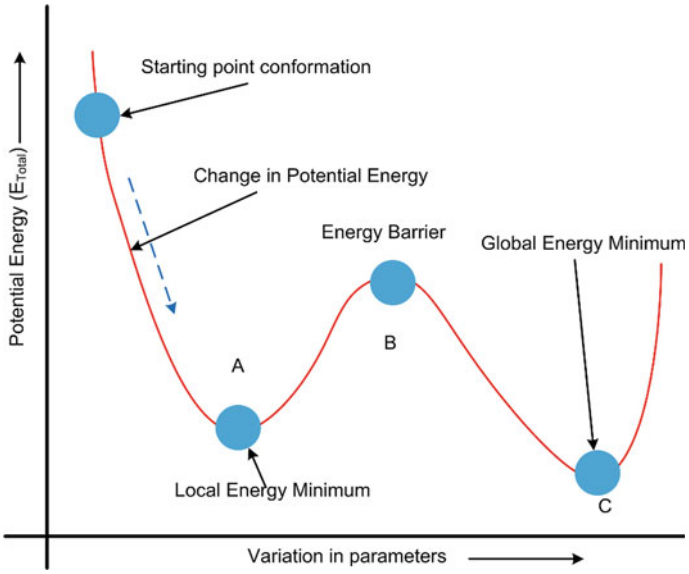


Fig. 4 Variation of potential energy of a particle with respect to parameters during energy minimization [48]

which the gradient is largest (i.e. steepest) at the beginning location, the approach is called Steepest Descent. After achieving a minimum in the first direction, a second minimization is performed, beginning at that point and progressing in the steepest remaining direction as shown in Fig. 5. This process is repeated until a minimum in all directions has been attained within a reasonable tolerance [49]. Conjugate gradient is an energy reduction approach that is more sophisticated than steepest descent.

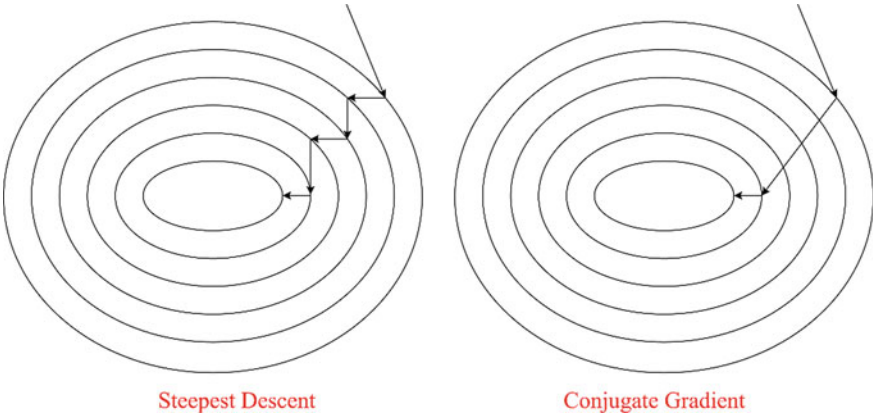


Fig. 5 Schematic of different energy minimization techniques

It accelerates convergence by using step history. In the steepest descent approach, both the directions of succeeding steps and the slopes are orthogonal. The conjugate gradient technique is more appropriately called as the conjugate direction method since the gradients are orthogonal in nature at every location and the directions of subsequent steps are conjugate [50].

10 MD Algorithm

Figure 6 shows the algorithm of a particular molecular dynamic simulation. Following are the steps involved:

1. Choose an initial set of atom velocities and initial positions for the atoms. These are typically chosen to obey a Boltzmann distribution for a given temperature, then normalized to zero net momentum for the entire system.
2. Calculate each atom's momentum using its velocity and mass.
3. From the energy expression, calculate the forces on each atom. This is generally a force field created using molecular mechanics for use in dynamical simulations.
4. Compute new positions for the atoms after a short time step Δt.
5. Compute new velocities and accelerations for the atoms.

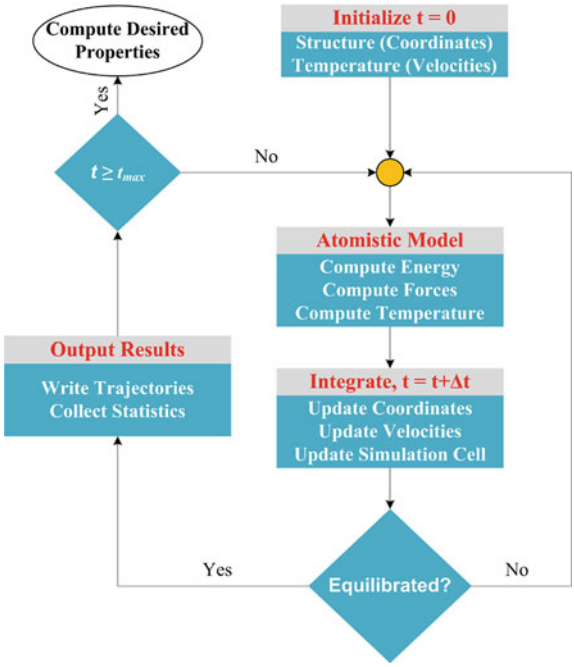


Fig. 6 Schematic of MD algorithm

6. Repeat steps 2 through 5.
7. Continue this repetition until the system reaches equilibrium. Equilibrium is not the lowest energy configuration in this situation; it is a configuration that is suitable for the system given the quantity of energy available.
8. After the system has stabilized, start saving the atomic coordinates every few rounds. A trajectory is a set of coordinates plotted over time.
9. Iterate and save data until you have enough data to get the findings you want with the precision you want.
10. Examine the trajectories in order to know more about the system. This might be found by computing radial distribution functions, diffusion coefficients, vibrational motions or any other feature that can be calculated using this data.

11 Limitations of MD

A molecular dynamics simulation is a fundamental technique because it addresses the smallest entity, an atom which is the basic building component of materials. Molecular dynamics (MD) is an important tool for modelling material deformation at extremely high strain rates that are not possible to simulate using other approaches such as the finite element method, finite volume method and so on. MD simulations, on the other hand, have a number of drawbacks. Majority of molecular dynamics simulation models can only explore materials with dimensions of a few hundred nanometres or less. Another important constraint of molecular dynamics is the time-scale limitation, which has stopped many researchers from researching interesting phenomena or making rigorous connections to laboratory operations that are generally carried out at completely different time scales. To correctly address electrons, molecular models must explicitly integrate quantum mechanical approaches, which include degrees of freedom that define electron structure [4]. Furthermore, classical molecular dynamics is ineffective for studying electricity, magnetism and optical characteristics [51–65].

12 Applications

Since its inception, the MD technique has been utilized to explore a wide range of issues in a variety of scientific domains. In chemistry and biochemistry, molecular dynamics may be used to explore molecular structures, protein folding, drug design, reactions vibrational relaxation and energy transfer. It may also be used to examine correlated many-body motion, statistical ensemble characteristics, tiny cluster structure and properties, and phase transitions in statistical mechanics and physics. In materials science, it may also be used to investigate various defects in crystals, microscopic fracture processes, melting and evaporation, and friction. Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) [66, 67]

and GROMACS are the most widely used software packages available for the MD simulations. Open Visualization Tool (OVITO) [68], AtomEye [69], Visualization for Electronic Structural Analysis (VESTA) [70], Visual Molecular Dynamics (VMD) [71] and Avogadro [72] are some of the visualization tools that may be employed.

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References

1. Lee, J.G.: Computational Materials Science: An Introduction. CRC Press (2016)
2. Rajasekaran, G., Narayanan, P., Parashar, A.: Effect of point and line defects on mechanical and thermal properties of graphene: a review. Crit. Rev. Solid State Mater. Sci. **41**, 47–71 (2016)
3. Tadmor, E.B., Miller, R.E.: Modeling Materials: Continuum, Atomistic and Multiscale Techniques. Cambridge University Press (2011)
4. Buehler, M.J.: Atomistic Modeling of Materials Failure. Springer, US (2008)
5. Wagner, N.J., Holian, B.L., Voter, A.F.: Molecular-dynamics simulations of two-dimensional materials at high strain rates. Phys. Rev. A **45**, 15 (1992)
6. Morse, P.M.: Diatomic molecules according to the wave mechanics. II. Vibrational levels. Phys. Rev. **34**, 57–64 (1929)
7. Mishin, Y., Mehl, M.J., Papaconstantopoulos, D.A., et al.: Structural stability and lattice defects in copper: ab initio, tight-binding, and embedded-atom calculations. Phys. Rev. B **63**, 224106 (2001)
8. Finnis, M.W., Sinclair, J.E.: A simple empirical N-body potential for transition metals. Philos. Mag. A **50**, 45–55 (1984)
9. Daw, M.S., Baskes, M.I.: Embedded-atom method: derivation and application to impurities, surfaces, and other defects in metals. Phys. Rev. B **29**, 6443 (1984)
10. Baskes, M.I.: Modified embedded-atom potentials for cubic materials and impurities. Phys. Rev. B **46**, 2727 (1992)
11. Swadener, J.G., Baskes, M.I., Nastasi, M.: molecular dynamics simulation of brittle fracture in silicon. Phys. Rev. Lett. **89**, 085503 (2002)
12. Johnston, H.S., Parr, C.: Activation energies from bond energies. I. Hydrogen transfer reactions. J. Am. Chem. Soc. **85**, 2544–2551 (2002)
13. Tersoff, J.: Empirical interatomic potential for carbon, with applications to amorphous carbon. Phys. Rev. Lett. **61**, 2879 (1988)
14. Tersoff, J.: New empirical model for the structural properties of silicon. Phys. Rev. Lett. **56**, 632 (1986)
15. Stillinger, F.H., Weber, T.A.: Computer simulation of local order in condensed phases of silicon. Phys. Rev. B **31**, 5262 (1985)
16. Brenner, D.W.: Empirical potential for hydrocarbons for use in simulating the chemical vapor deposition of diamond films. Phys. Rev. B **42**, 9458 (1990)
17. Brenner, D.W., Shenderova, O.A., Harrison, J.A., et al.: A second-generation reactive empirical bond order (REBO) potential energy expression for hydrocarbons. J. Phys.: Condens. Matter. **14**, 783 (2002)
18. Stuart, S.J., Tutein, A.B., Harrison, J.A.: A reactive potential for hydrocarbons with intermolecular interactions. J. Chem. Phys. **112**, 6472 (2000)
19. van Duin, A.C.T., Dasgupta, S., Lorant, F., Goddard, W.A.: ReaxFF: a reactive force field for hydrocarbons. J. Phys. Chem. A **105**, 9396–9409 (2001)

20. Kumar, R., Rajasekaran, G., Parashar, A.: Optimised cut-off function for Tersoff-like potentials for a BN nanosheet: a molecular dynamics study. Nanotechnology **27**, 085706 (2016)
21. Rajasekaran, G., Kumar, R., Parashar, A.: Tersoff potential with improved accuracy for simulating graphene in molecular dynamics environment. Mater. Res. Express **3**, 035011 (2016)
22. Tersoff, J.: Modeling solid-state chemistry: interatomic potentials for multicomponent systems. Phys. Rev. B **39**, 5566 (1989)
23. Tersoff, J.: New empirical approach for the structure and energy of covalent systems. Phys. Rev. B **37**, 6991 (1988)
24. Chandra, Y., Scarpa, F., Adhikari, S., et al.: Pullout strength of graphene and carbon nanotube/epoxy composites. Compos. B Eng. **102**, 1–8 (2016)
25. Cha, J.H., Kyoung, W., Song, K., et al.: Quantitative evaluation of the dispersion of graphene sheets with and without functional groups using molecular dynamics simulations. Nanoscale Res. Lett. **11**, 1–7 (2016)
26. Verma, A., Parashar, A., Packirisamy, M.: Atomistic modeling of graphene/hexagonal boron nitride polymer nanocomposites: a review. Wiley Interdiscip. Rev.: Comput. Mol. Sci. **8** (2018)
27. Rappé, A.K., Casewit, C.J., Colwell, K.S., et al.: UFF, a full periodic table force field for molecular mechanics and molecular dynamics simulations. J. Am. Chem. Soc. **114**, 10024–10035 (1992)
28. Sun, H.: COMPASS: an ab initio force-field optimized for condensed-phase applications overview with details on alkane and benzene compounds. J. Phys. Chem. B **102**, 7338–7364 (1998)
29. MacKerell, A.D., Wiórkiewicz-Kuczera, J., Karplus, M., MacKerell, A.D.: An all-atom empirical energy function for the simulation of nucleic acids. J. Am. Chem. Soc. **117**, 11946–11975 (1995)
30. Allinger, N.L., Yuh, Y.H., Lii, J.H.: Molecular mechanics. The MM3 force field for hydrocarbons. 1. J. Am. Chem. Soc. **111**, 8551–8566 (1989)
31. Verlet, L.: Computer "experiments" on classical fluids. II. Equilibrium correlation functions. Phys. Rev. **165**, 201 (1968)
32. Jensen, B.D., Wise, K.E., Odegard, G.M.: The effect of time step, thermostat, and strain rate on ReaxFF simulations of mechanical failure in diamond, graphene, and carbon nanotube. J. Comput. Chem. **36**, 1587–1596 (2015)
33. Yao, Z., Zhu, C.C., Cheng, M., Liu, J.: Mechanical properties of carbon nanotube by molecular dynamics simulation. Comput. Mater. Sci. **22**, 180–184 (2001)
34. Ajori, S., Ansari, R.: Torsional buckling behavior of boron-nitride nanotubes using molecular dynamics simulations. Curr. Appl. Phys. **14**, 1072–1077 (2014)
35. Choe, J.-I., Kim, B.: Determination of proper time step for molecular dynamics simulation. Bull. Korean Chem. Soc. **21**, 419–424 (2000)
36. Wu, J., Wei, Y.: Grain misorientation and grain-boundary rotation dependent mechanical properties in polycrystalline graphene. J. Mech. Phys. Solids **61**, 1421–1432 (2013)
37. Jhon, Y.I., Zhu, S.E., Ahn, J.H., Jhon, M.S.: The mechanical responses of tilted and non-tilted grain boundaries in graphene. Carbon **50**, 3708–3716 (2012)
38. He, L., Guo, S., Lei, J., et al.: The effect of Stone–Thrower–Wales defects on mechanical properties of graphene sheets—a molecular dynamics study. Carbon **75**, 124–132 (2014)
39. Frenkel, D.B.S.: Understanding Molecular Simulation: From Algorithms to Applications. Elsevier (2001)
40. Dehoff, R.T.: Thermodynamics in Materials Science. McGraw Hill Inc., New York, NY (1993)
41. Hu, Y., Sinnott, S.B.: Constant temperature molecular dynamics simulations of energetic particle—solid collisions: comparison of temperature control methods. J. Comput. Phys. **200**, 251–266 (2004)
42. Adelman, S.A., Doll, J.D.: Generalized Langevin equation approach for atom/solid-surface scattering: general formulation for classical scattering off harmonic solids. J. Chem. Phys. **64**, 2375 (2008)
43. Berendsen, H.J.C., Postma, J.P.M., van Gunsteren, W.F., et al.: Molecular dynamics with coupling to an external bath. J. Chem. Phys. **81**, 3684 (1998)

44. Hoover, W.G.: Canonical dynamics: equilibrium phase-space distributions. Phys. Rev. A **31**, 1695 (1985)
45. Nosé, S.: A unified formulation of the constant temperature molecular dynamics methods. J. Chem. Phys. **81**, 511 (1998)
46. Nosé, S.: A molecular dynamics method for simulations in the canonical ensemble. Mol. Phys. **52**(2), 255–268 (2006)
47. Tsai, J.L., Tu, J.F.: Characterizing mechanical properties of graphite using molecular dynamics simulation. Mater. Des. **31**, 194–199 (2010)
48. Roy, K., Kar, S., Das, R.N.: Computational Chemistry. Understanding the Basics of QSAR for Applications in Pharmaceutical Sciences and Risk Assessment, pp. 151–189 (2015)
49. Wiberg, K.B.: A scheme for strain energy minimization. Application to the cycloalkanes1. J. Am. Chem. Soc. **87**, 1070–1078 (2002)
50. Fletcher, R., Reeves, C.M.: Function minimization by conjugate gradients. Comput. J. **7**, 149–154 (1964)
51. Verma, A., Parashar, A.: Structural and chemical insights into thermal transport for strained functionalised graphene: a molecular dynamics study. Mater. Res. Express **5**(11), 115605 (2018)
52. Verma, A., Parashar, A., Packirisamy, M.: Effect of grain boundaries on the interfacial behaviour of graphene-polyethylene nanocomposite. Appl. Surf. Sci. **470**, 1085–1092 (2019)
53. Chaurasia, A., Verma, A., Parashar, A., Mulik, R.S.: Experimental and computational studies to analyze the effect of h-BN nanosheets on mechanical behavior of h-BN/polyethylene nanocomposites. J. Phys. Chem. C **123**(32), 20059–20070 (2019)
54. Verma, A., Kumar, R., Parashar, A.: Enhanced thermal transport across a bi-crystalline graphene–polymer interface: an atomistic approach. Phys. Chem. Chem. Phys. **21**(11), 6229–6237 (2019)
55. Verma, A., Parashar, A.: Molecular dynamics based simulations to study the fracture strength of monolayer graphene oxide. Nanotechnology **29**(11), 115706 (2018)
56. Verma, A., Parashar, A.: The effect of STW defects on the mechanical properties and fracture toughness of pristine and hydrogenated graphene. Phys. Chem. Chem. Phys. **19**(24), 16023–16037 (2017)
57. Verma, A., Parashar, A., Packirisamy, M.: Tailoring the failure morphology of 2D bicrystalline graphene oxide. J. Appl. Phys. **124**(1), 015102 (2018)
58. Singla, V., Verma, A., Parashar, A.: A molecular dynamics based study to estimate the point defects formation energies in graphene containing STW defects. Mater. Res. Express **6**(1), 015606 (2018)
59. Verma, A., Parashar, A.: Molecular dynamics based simulations to study failure morphology of hydroxyl and epoxide functionalised graphene. Comput. Mater. Sci. **143**, 15–26 (2018)
60. Verma, A., Zhang, W., van Duin, A.C.: ReaxFF reactive molecular dynamics simulations to study the interfacial dynamics between defective h-BN nanosheet and water nanodroplets. Phys. Chem. Chem. Phys. **23**, 10822–10834 (2021)
61. Verma, A., Parashar, A., Packirisamy, M.: Role of chemical adatoms in fracture mechanics of graphene nanolayer. Mater. Today: Proc. **11**, 920–924 (2019)
62. Verma, A., Parashar, A.: Characterization of 2D nanomaterials for energy storage. In: Recent Advances in Theoretical, Applied, Computational and Experimental Mechanics, pp. 221–226. Springer, Singapore (2020)
63. Verma, A., Parashar, A., Singh, S.K., Jain, N., Sanjay, M.R., Siengchin, S.: Modeling and simulation in polymer coatings. In: Polymer Coatings: Technologies and Applications, Chapter 16, pp. 309–324. Taylor & Francis Group (CRC Press), Boca Raton (2020)
64. Kataria, A., Verma, A., Sanjay, M.R., Siengchin, S.: Molecular modeling of 2D graphene grain boundaries: mechanical and fracture aspects. Mater. Today: Proc. (2021)
65. Deji, R., Verma, A., Kaur, N., Choudhary, B.C., Sharma, R.K.: Density functional theory study of carbon monoxide adsorption on transition metal doped armchair graphene nanoribbon. Mater. Today: Proc. (2021)

66. Thompson, A.P., Aktulga, H.M., Berger, R., Bolintineanu, D.S., Brown, W.M., Crozier, P.S., in't Veld, P.J., Kohlmeyer, A., Moore, S.G., Nguyen, T.D., Shan, R.: LAMMPS-A flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales. Comput. Phys. Commun. 108171 (2021)
67. Plimpton, S.: Fast parallel algorithms for short-range molecular dynamics. J. Comput. Phys. **117**(1), 1–19 (1995)
68. Stukowski, A.: Visualization and analysis of atomistic simulation data with OVITO—the Open Visualization Tool. Modell. Simul. Mater. Sci. Eng. **18**(1), 015012 (2009)
69. Li, J.: AtomEye: an efficient atomistic configuration viewer. Modell. Simul. Mater. Sci. Eng. **11**(2), 173 (2003)
70. Momma, K., Izumi, F.: VESTA: a three-dimensional visualization system for electronic and structural analysis. J. Appl. Crystallogr. **41**(3), 653–658 (2008)
71. Humphrey, W., Dalke, A., Schulten, K.: VMD: visual molecular dynamics. J. Mol. Graph. **14**(1), 33–38 (1996)
72. Hanwell, M.D., Curtis, D.E., Lonie, D.C., Vandermeersch, T., Zurek, E., Hutchison, G.R.: Avogadro: an advanced semantic chemical editor, visualization, and analysis platform. J. Cheminform. **4**(1), 1–17 (2012)

Introduction to Interatomic Potentials/Forcefields



Aditya Kataria, Akarsh Verma, Sushanta Kumar Sethi, and Shigenobu Ogata

Abstract Interatomic potentials (or forcefields) for detailed structures are produced by fitting certain functions within constraints of investigational data or with planned data from the first-principles methods. The experimental data employed for fitting comprises equilibrium lattice parameter, bulk modulus, cohesive energy, thermal expansion coefficient, dielectric constants, elastic modulus, vibration spectrum, surface energy and vacancy formation energies. All these empirical potentials are specifically for their own domain or system. It should be ensured that it is transferable and one must be cautious about applying one specific potential to other systems or situations.

1 Potentials

1.1 Introduction

The balance between attraction and repulsion occurs when atoms are close enough to one other, and it is governed by what are termed as interatomic potentials [1, 2]. After applying Newton's motion equations, atoms finally get established at the lowest potential states at balance distances. The total of all atomic forces, F, is given by Eq. 1:

A. Kataria · A. Verma (✉)
Department of Mechanical Engineering, University of Petroleum and Energy Studies, Dehradun 248007, India
e-mail: akarshverma007@gmail.com

A. Verma · S. Ogata
Department of Mechanical Science and Bioengineering, Osaka University, Osaka 560-8531, Japan

S. K. Sethi
Department of Metallurgical and Materials Science Engineering, Indian Institute of Technology Bombay, Mumbai 400076, India

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$$F = ma = m\frac{dv}{dt} = m\frac{d^2r}{dt^2} = \frac{dp}{dt}. \tag{1}$$

Here,
m is the mass of the body
a is the acceleration
v is the velocity
r is the distance
t is the time
p is the momentum.

Since r is a vector, v, a, p and F are also vectors for its first and second derivatives. If the total energy E is constant in time ($dE/dt = 0$), which is an isolated MD simulation system, F is connected with a negative potential gradient in relation to location (refer to Eq. 2).

$$F = -\Delta\phi \rightarrow F = -\Delta\phi(r_1, r_2, \ldots, r_n) \tag{2}$$

where ϕ is the potential. Therefore, forces will act on the atoms, and then solve Eq. 2 in terms of temporal evolution of the system (if we know the potential of a system built on interatomic distance). Atoms are constantly in accordance with this potential in the minimum energy line.

Potentials for certain systems are shaped by the variation of some functions with experimental data parameters or computed data from techniques of first principles. The experimental fitting data comprises lattice parameter of equilibrium, cohesive energy, bulk modules, elastic modulus, vacancy energy, a coefficient of thermal expansion, dielectric constant and a surface energy spectrum. Note that these presumed empirics are system-specific. You must be cautious to confirm that a specific potential for dissimilar systems or conditions is transferred and used. Although certain potentials may be changed in the same system and under the same conditions, as most potentials are often built through static characteristics.

Many interatomic potentials are built on the belief that atomic interactions are fundamentally local. When it comes to long-range Coulomb interactions in ionic systems, the supposition of local bonding is crushed mostly. Materials that are ionically bonded have a strong bonding, as we'll see ahead. It's safe to accept that bonding is frail outside a moderately small cut-off distance for metallic, covalent and van der Waals bonding, though. Be careful that there are always coulomb interactions between the protons and electrons in a solid. We consider that beyond some bonding distance, the interaction between molecules is so weak that it has little consequence on the global energy. As a result, the interatomic potential energy function must be built in such a way that the energy contribution attributable to a specific atom is only impacted by atoms within the cut-off radius (r_{cut}) of that atom (Fig. 1).

For example, we have a pair potential given in Eq. 3:

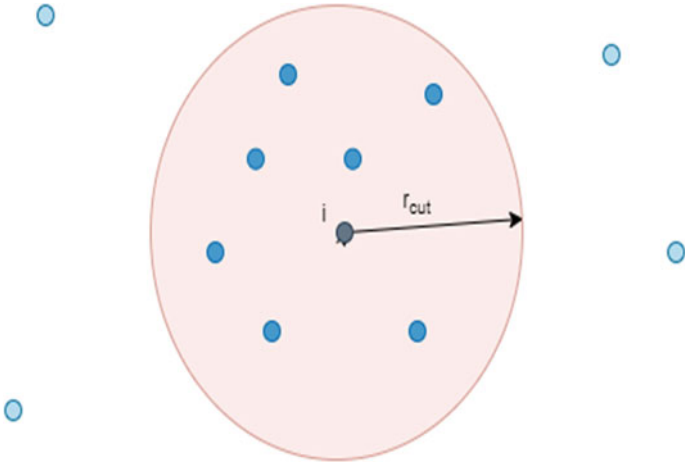


Fig. 1 The black dot 'I' represents the acting atom that exerts its interatomic potential on the atoms inside a certain cut-off radius

$$\phi_{\alpha\beta}(r) = \begin{cases} f_{\alpha\beta}(r), & r < r_{cut} \\ 0, & r > r_{cut} \end{cases} \tag{3}$$

While doing molecular dynamics (MD) simulation, atoms will probably pass close to one another with interatomic distances that are equal or tremendously close to r_{cut}. In these circumstances, the function $\phi'_{\alpha\beta}(r_{cut})$ is undefined, since $f_{\alpha\beta}(r)$ is not zeroed out at $r = r_{cut}$. Higher derivatives of the potential may be essential for particular applications. Elastic constants, for example, require the use of second derivatives of potentials. The potential can be modified in a generic way, such that stated through Eq. 4

$$\phi_{\alpha\beta} = \begin{cases} f_{\alpha\beta}(r) - t_{\alpha\beta}(r - r_{cut}), & r < r_{cut} \\ 0, & r \geq r_{cut} \end{cases} \tag{4}$$

and the smooth truncation polynomial $t_{\alpha\beta}(x)$ (refer to Eq. 5) for all $r > 0$ gives smooth truncation of the pair potential $f_{\alpha\beta}(r)$.

$$t_{\alpha\beta} = \sum_{i=0}^{m} x^i f_{\alpha\beta}^{(i)}(r_{cut}). \tag{5}$$

If we take the ith derivative of the function $f_{\alpha\beta}^{(i)}(r)$, and x^i is x increased to the ith power, we get the ith derivative $f_{\alpha\beta}^{(i)}(r)$. At $r = r_{cut}$, this form guarantees that all derivatives up to order M have the same value of zero. One way to do this is by setting all potentials to zero when any distance is larger than a certain cut-off radius. Of course, other smoothing functions are possible. Murdoch [3] described the

"mollifying functions", which when fitting a potential, is necessary to add smoothing terms. This can be tricky since it can radically vary the predictions when adding a smoothing function to an existing potential. The expected ground-state structure of crystalline materials, for example, can be changed by this process.

In metals with a face-centred cubic ground state structure, this is recurrent. Occasionally, small changes in the potentials can cause the adjacent hexagonal close-packed (hcp) structure to become the ground state. To observe how substitute cut-off techniques for the Lennard–Jones potential affect the projected ground state energy versus density curve for fcc and HCP structures, please refer to Ref. [4].

1.2 Cluster Potentials

Potential energy can be accurately calculated by multiplying positional information from 'n' atom clusters ($n = 1, 2, \ldots, N$) with another random number. Finally, by terminating the series in the correct sequence, an approximation model may then be shaped from the data. In the end, the cluster potential is what we have. Rather than using the phrase "cluster potential," the term "cluster expansion" is often used to describe the collapse of the energy into a series of cluster terms [5].

Let's say you have N atoms in your system. As a starting point, we will focus on systems with only one type of atom in them. Potential energy of such a system may always be represented in the manner given by Eq. 6 [5, 6]:

$$v_{(N)} = \phi_0 + \sum_{\alpha=1}^{N} \hat{\phi}_1(r^\alpha) + \sum_{\substack{\alpha,\beta \\ \alpha<\beta}}^{N} \hat{\phi}_2(r^\alpha, r^\beta) + \sum_{\substack{\alpha,\beta,\gamma \\ \alpha<\beta<\gamma}}^{N} \hat{\phi}_3(r^\alpha, r^\beta, r^\gamma) + \cdots . \tag{6}$$

Here, ϕ_0 is employed as a reference term, and the sequence ends with the n-body potential function $\widehat{\phi}_n$. When it comes to two-body terms and above, the summations represent multiple sums subject to the stated boundaries. In the three-body case, the summations are $\sum_\alpha \sum_\beta \sum_\gamma$ subject to the constraint, $\alpha < \beta < \gamma \leq N$. There are $\binom{N}{n} = N!/(N-n)!n!$ terms in the corresponding sums for the n-body potential $\widehat{\phi}_n$. They are permutation symmetrical with their arguments. There are a number of terms that satisfy these conditions. For example, the pair term is satisfied by the formula,$\widehat{\phi}_2(r^\alpha, r^\beta) = \widehat{\phi}_2(r^\beta, r^\alpha)$; the three-body term by $\widehat{\phi}_3\left(r^\alpha, r^\beta, r^\gamma\right) = \widehat{\phi}_3\left(r^\beta, r^\alpha, r^\gamma\right) = \widehat{\phi}_3\left(r^\alpha, r^\gamma, r^\beta\right) = \ldots$ and so on.

There are two components to Eq. (6), the "one-body" terms $\widehat{\phi}_1$ (see Eq. 7) indicate the interaction between one atom with an external potential and additional atoms outside of the system, which collectively make up the external potential energy.

$$v_{(N)}^{ext} = \sum_{\alpha=1}^{N} \widehat{\phi}_1(r^\alpha). \tag{7}$$

The rest of the equation represents system's internal potential energy, which contains both energies related with reference as well as energy connected with interatomic contacts (Eq. 8).

$$v_{(N)}^{int} = \phi_0 + \sum_{\substack{\alpha,\beta \\ \alpha<\beta}}^{N} \hat{\phi}_2\left(r^\alpha, r^\beta\right) + \sum_{\substack{\alpha,\beta,\gamma \\ \alpha<\beta<\gamma}}^{N} \hat{\phi}_3\left(r^\alpha, r^\beta, r^\gamma\right) + \cdots. \tag{8}$$

Referring to the energy of atoms in isolation from one another, the phrase refers to the atomic energy. Because of this, we can write through Eq. 9:

$$\phi_0 = N E_{free}(Z) \tag{9}$$

where Z is the atomic number of the system's atoms, and $E_{free}(Z)$ is a function that gives the energy of an isolated atom with atomic number Z. It is critical to note that the series in Eq. (8) is not unique.

If N atoms are involved, the functional forms $\widehat{\phi}_n$ $(n = 2, ..., N-1)$ can be set freely for any given system of atoms (Eq. 10). The N-body term would be defined to satisfy Eq. (9) and then enforced.

$$\hat{\phi}_N\left(r^1, \ldots, r^N\right) \equiv \hat{v}_{(N)}^{int}\left(r^1, \ \ldots, \ r^N\right) - \phi_0 - \sum_{\substack{\alpha,\beta \\ \alpha<\beta}}^{N} \hat{\phi}_2\left(r^\alpha, r^\beta\right) - \sum_{\substack{\alpha,\beta,\gamma \\ \alpha<\beta<\gamma}}^{N} \hat{\phi}_3\left(r^\alpha, r^\beta, r^\gamma\right) - \cdots. \tag{10}$$

Naturally, a cluster potential constructed this way would not be particularly useful. But there is a simple recursive process for defining the possible function that we can understand which gives them a clear physical meaning.

Look at atoms that are arranged in increasing numbers

$$\left(r^1, r^2\right), (r^1 r^2 r^3), (r^1 r^2 r^3 r^4), ...$$

in addition to their allied internal potential energy functions

$$\hat{v}_{(2)}^{int}\left(r^1, r^2\right), \hat{v}_{(3)}^{int}\left(r^1, r^2, r^3\right), \hat{v}_{(4)}^{int}(r^1, r^2, r^3, r^4), \ldots$$

that are presumed to be well-established. Equations 8 and 9 are the equations that are used for two atoms, and results in Eq. 11:

$$\widehat{\phi}_2\left(r^1, r^2; Z\right) \equiv \hat{v}_{(2)}^{int}\left(r^1, r^2\right) - 2E_{free}(Z) \tag{11}$$

it is defined as a difference between two atoms' intrinsic energies when they are separated. Equations (8) and (9) for three atoms result in Eq. 12:

$$\hat{v}_3^{int}\left(r^1, r^2, r^3\right) = 3E_{free}(Z) + \sum_{\substack{\alpha,\,\beta \\ \alpha < \beta}}^{3} \widehat{\phi}_2(r^\alpha, r^\beta) + \widehat{\phi}_3\left(r^1, r^2, r^3\right). \tag{12}$$

By inverting this relationship and putting the total out clearly, we obtain Eq. 13:

$$\begin{aligned}\hat{\phi}_3\left(r^1, r^2, r^3; Z\right) \equiv \hat{v}_3^{int}\left(r^1, r^2, r^3\right) - \hat{\phi}_2\left(r^1, r^2\right) - \hat{\phi}_2\left(r^1, r^3\right) - \hat{\phi}_2\left(r^2, r^3\right) \\ - 3E_{free}(Z).\end{aligned} \tag{13}$$

Thus, $\widehat{\phi}_3$ describes extra energy in a three-atom system that is not captured by pair interactions. As an analogy, the four-body expression is given by Eq. 14:

$$\begin{aligned}\hat{\phi}_4\left(r^1, r^2, r^3, r^4\right) \equiv \hat{v}_{(4)}^{int}\left(r^1, r^2, r^3, r^4\right) - \sum_{\substack{\alpha,\beta \\ \alpha<\beta}}^{4} \hat{\phi}_2\left(r^\alpha, r^\beta\right) \\ - \sum_{\substack{\alpha,\beta,\gamma \\ \alpha<\beta<\gamma}}^{4} \hat{\phi}_3\left(r^\alpha, r^\beta, r^\gamma\right) - 4E_{free}(Z)\end{aligned} \tag{14}$$

and so, on up to $\widehat{\phi}_n$. A Moebius inversion is the mathematical term for the recursive method described earlier [7, 8]. It is clearly defined that the Moebius inversion is not infinite for some combination of its parameters as long as $\hat{v}_{(n)}^{int}(n < N)$ [6]. Higher-order functions of the kind $\hat{\phi}_{n+1}, \hat{\phi}_{n+2,\ldots}$, would be undefined for the given set of inputs. To remove the vagueness in the definition of the cluster potential, it is necessary to build the words in the cluster potential in a recursive fashion. Therefore, such specified potential functions $\hat{\phi}_n\left(r^\alpha, r^\beta, \ldots\right)$ have the characteristic that they trend to zero anytime any one of the constituents is removed to the infinitesimal point. For example, n-body theory states that excess energy for an isolated n-body is equal to the sum of the energy of all interactions between smaller clusters $m < n$ within the system.

For theoretical derivations, Eq. (8) can alternatively be expressed in the following symmetric form, given by Eq. 15,

$$v_{(N)}^{int} = \phi_0 + \frac{1}{2!} \sum_{\substack{\alpha,\beta \\ \alpha \neq \beta}}^{N} \hat{\phi}_2\left(r^\alpha, r^\beta\right) + \frac{1}{3!} \sum_{\substack{\alpha,\beta,\gamma \\ \alpha \neq \beta \neq \gamma}}^{N} \hat{\phi}_3\left(r^\alpha, r^\beta, r^\gamma\right) + \ldots \quad (15)$$

because of the symmetry that exists amongst all possible permutations of the potential functions. Potential functions will be varied depending on which atoms are used if they are not all of the same species. The atom subscripts are added to the potential functions to make this clear (refer to Eq. 16).

$$v_{(N)}^{int} = \phi_0 + \frac{1}{2!} \sum_{\substack{\alpha,\beta \\ \alpha \neq \beta}}^{N} \hat{\phi}_{\alpha\beta}\left(r^\alpha, r^\beta\right) + \frac{1}{3!} \sum_{\substack{\alpha,\beta,\gamma \\ \alpha \neq \beta \neq \gamma}}^{N} \hat{\phi}_{\alpha\beta\gamma}\left(r^\alpha, r^\beta, r^\gamma\right) + \ldots. \quad (16)$$

As well as the atomic number of each atom, reference energy is a function of the system (Eq. 17).

$$\phi_0 = \sum_{\alpha=1}^{N} E_{free}(Z^\alpha) \quad (17)$$

where Z_α is the atomic number of atom α. This notation can be made clear by using the words,

$$\hat{\phi}_{\alpha\beta}\left(r^\alpha, r^\beta\right) = \hat{\phi}_2\left(r^\alpha, r^\beta; Z^\alpha, Z^\beta\right),$$

$$\hat{\phi}_{\alpha\beta\gamma}\left(r^\alpha, r^\beta, r^\gamma\right) = \hat{\phi}_2\left(r^\alpha, r^\beta, r^\gamma; Z^\alpha, Z^\beta, Z^\gamma\right)$$

the function $\hat{\phi}_m$ has a functional form defined by Z_α.

Like the single-species functions, the species-dependent functions have permutation symmetry. The species subscripts must likewise be permuted when permuting arguments in this situation. Hence, the pair term is satisfied by the following Eq. 18.

$$\hat{\phi}_{\alpha\beta}\left(r^\alpha, r^\beta\right) = \hat{\phi}_{\beta\alpha}\left(r^\beta, r^\alpha\right). \quad (18)$$

Similarly, the three-body term satisfies Eq. 19.

$$\hat{\phi}_{\alpha\beta\gamma}\left(r^\alpha, r^\beta, r^\gamma\right) = \hat{\phi}_{\beta\alpha\gamma}\left(r^\beta, r^\alpha, r^\gamma\right) = \hat{\phi}_{\alpha\gamma\beta}\left(r^\alpha, r^\gamma, r^\beta\right) = \ldots. \quad (19)$$

In the case of numerous species, we may, for example, define Eq. 20

$$E^{\alpha} = E_{free}(Z^{\alpha}) + \frac{1}{2!} \sum_{\substack{\beta \\ \beta \neq \alpha}}^{N} \hat{\phi}_{\alpha\beta}\left(r^{\alpha}, r^{\beta}\right) + \frac{1}{3!} \sum_{\substack{\beta,\gamma \\ \beta \neq \gamma \neq \alpha}}^{N} \hat{\phi}_{\alpha\beta\gamma}\left(r^{\alpha}, r^{\beta}, r^{\gamma}\right) + \ldots \tag{20}$$

where the total potential energy can be expressed using

$$v_{(N)}^{int} = \sum_{\alpha=1}^{N} E^{\alpha}.$$

Due to the fact that cluster potential itself is not unique, this definition of atom energy is likewise not unique, and we must exercise caution when attempting to attach a physical meaning to it, as it is not [9].

Since we have described the n-body potentials in terms of absolute atomic locations, it is time to move on to the next step. On the other hand, we know that to meet the invariance principle, we must be able to describe the potential energy in terms of interatomic distances. For each term in the series, the same condition applies to them all. We therefore have Eq. 21

$$\begin{aligned} v_{(N)}^{int} &= \phi_0 + \frac{1}{2!} \sum_{\substack{\alpha,\beta \\ \alpha \neq \beta}}^{N} \hat{\phi}_{\alpha\beta}\left(r^{\alpha\beta}\right) + \frac{1}{3!} \sum_{\substack{\alpha,\beta,\gamma \\ \alpha \neq \beta \neq \gamma}}^{N} \hat{\phi}_{\alpha\beta\gamma}\left(r^{\alpha\beta}, r^{\beta\gamma}, r^{\alpha\gamma}\right) \\ &+ \frac{1}{4!} \sum_{\substack{\alpha,\beta,\gamma,\delta \\ \alpha \neq \beta \neq \gamma\delta}}^{N} \hat{\phi}_{\alpha\beta\gamma\delta}\left(r^{\alpha\beta}, r^{\alpha\gamma}, r^{\alpha\delta}, r^{\beta\gamma}, r^{\beta\delta}, r^{\gamma\delta}\right) + \ldots \\ &= \phi_0 + \phi_{(2)} + \phi_{(3)} + \phi_{(4)} + \ldots. \end{aligned} \tag{21}$$

In this case, "$\Phi_{(n)}$" stands for the total sum of the n-body terms. When it comes to materials having covalent nature, bond angles play an essential role. Bond angles are defined by three-atom groups or clusters; therefore, it may be more convenient to describe the atomistic model in terms of an angle created between each three-atom group. Example is the angle produced by bonds $\alpha - \beta$ and $\alpha - \gamma$, with the vertex at atom α as the point of intersection between them. $r^{\alpha\beta}$, $r^{\alpha\gamma}$ and $r^{\beta\gamma}$ are equal to the three distances in trigonometry, as the rule of cosines in trigonometry gives us through Eq. 22

$$\cos\theta^{\beta\alpha\gamma} = \frac{(r^{\alpha\beta})^2 + (r^{\alpha\gamma})^2 - (r^{\beta\gamma})^2}{2r^{\alpha\beta}r^{\alpha\gamma}}. \tag{22}$$

It is obvious that a representation of the interatomic potential energy in terms of bond angles also complied with this concept. Despite the fact that bond angles are sometimes more convenient, these models can always be expressed in terms of interatomic lengths. In formal terms, the cluster potential described earlier is precise. One may question, however, what, if anything has been gained by this approach because the creation of this series needs knowledge of specific potential energies. Because this function may be utilised directly, it is not necessary to compute the n-body potentials, as long as you know the value of $\hat{v}_{(N)}^{int}\left(r^1, \ldots, r^N\right)$. For two reasons, cluster potential is a valuable method of expression. First and foremost, it might be useful for analysing results. This can be looked up in the statistical mechanics derivation, which establishes particular restrictions on the possible forms for $\hat{\phi}_n$ that ensure that the atomistic system acts appropriately in the thermodynamic limit. Second, the cluster potential may be utilised to create approximation interatomic models. Here, the series is intentionally stopped at some level $\Phi_{(n)}$, with only terms up to n remaining.

It is common for ϕ_0 to be set to zero as an arbitrarily low energy reference. Using this methodology, $n = 2$ results in two-body potential model, $n = 3$ results in three-body potential model, etc. On the basis of physical intuition or simplified theoretical models, the shapes of the appropriate potential energy functions for a function of single species $\phi_k (k = 2, \ldots, n)$ are then "guessed" and fitted to experimentally observed characteristics or calculated using first-principles methods. As a basis for this approach, it should be noted that material bonding tends to be relatively local in character. Because of this, it is fair to assume that the behaviour of the system can be captured by models that use atom clusters around the bonding sites.

We've already understood the insertion of a cut-off radius for each k-body term. Example of a pair potential is expanded to any order potential by ensuring that the k-body term ($fork \leq n$) goes to zero if any of its distance arguments exceeds r_{cut}. For a three-body system (Eq. 23)

$$\hat{\phi}_{\alpha\beta\gamma}\left(r^{\alpha\beta}, r^{\alpha\gamma}, r^{\beta\gamma}\right) = \begin{cases} f_{\alpha\beta\gamma}\left(r^{\alpha\beta}, r^{\alpha\gamma}, r^{\beta\gamma}\right), & \text{if } r^{\alpha\beta}, r^{\alpha\gamma}, r^{\beta\gamma} < r_{cut} \\ 0, & \text{other cases} \end{cases}. \tag{23}$$

The topic of potentials can be further classified into four, i.e. pair potentials, Tersoff potentials, potential for ionic solids and other forcefields.

1.3 Pair Potentials

An atom '*i*' interacts at once with all other atoms in the N-atomic system, where N is the number of atoms meaning there are two atom interactions, three atoms and so on at any time. The statement can be expressed in the form of Eq. 24:

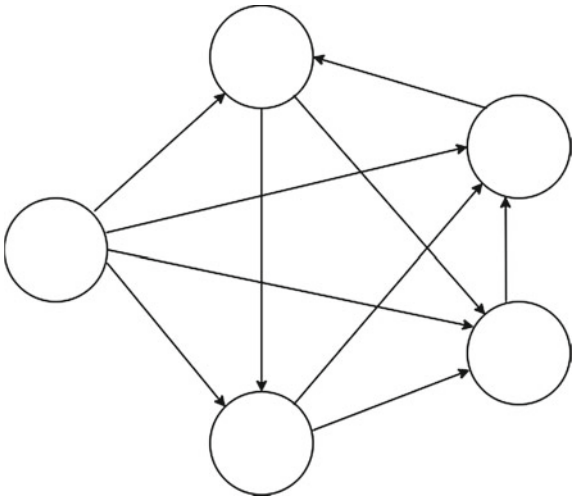


Fig. 2 An example of a 5-atom system where individual atoms exert potential energy on their neighbouring atoms

$$v = \sum_{i<j}^{N} \phi_2(r_i, r_j) + \sum_{i<j<k}^{N} \phi_3(r_i, r_j, r_k) + \ldots \tag{24}$$

where the terms represent potentials for two atoms, three atoms, etc. Here, the summation symbols show sums for all different pairs, triplets, etc. without counting any two times. Let us leave this 3N-dimensional potential and observe solely the interactions between the two atoms. Interactions are $(N-1)$ by atom, therefore the pair number (N_{pair}) is in line with N^2. The expression can be quantified through Eq. 25 (Fig. 2):

$$N_{pair} = \frac{(N-1)N}{2} \propto O(N^2). \tag{25}$$

1.3.1 Two Body Potentials

Pair potential is the simplest version of a potential that takes these two-atomic interactions into account and neglected all other potentials. There are various examples that demonstrate this relation. One of the most prominent is Lennard–Jones (LJ) potential, expressed in terms of interatomic distance, r, as defined from Eq. 26:

$$\phi_{LJ}(r) = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{7}\right]. \tag{26}$$

Here, the equation is judged on the basis of two parameters such as ε that describes the lowest energy of the potential curve and σ which is the interatomic distance at which the potential is zero.

Force can be explained in terms of repulsion and attraction, which can be formulated through Eq. 27:

$$F_{LJ} = \frac{24\varepsilon}{\sigma}\left[\left(\frac{\sigma}{r}\right)^{13} - \left(\frac{\sigma}{r}\right)^{7}\right]. \tag{27}$$

Several characteristics are highlighted when two atoms, such as the inert element Argon, from the vast distance approach each other, the boundary conditions can be expressed as at $r = \infty$, ϕ_{LJ} and F_{LJ} are zero. When they come closer the attraction of dipoles takes place and the term r^{-6} best represents this interaction of van der Waals. If they approach the situation, the Pauli repulsion occurs because of overlapping electron clouds, and this sharp growth is described by the arbitrary r^{-12} term.

At the interatomic distance of the equilibrium, $r_o = 2^{1/6}\,\sigma = 1.122\,\sigma$, two forces are equilibrated to the net force of zero ($d\phi_{\mathrm{LJ}}/dr = 0$). At 0, $r = \sigma$, the energy is dramatically increasing when the Pauli repulsion is lower down r. It is relatively easy to represent the atomic interactions of noble gases, molecules and secondary bonds using the fundamental pair potential method. It is not possible to apply this potential to metals, semiconductors and other solids since three atoms and so on are completely ignored for their many-atomic effects (about 10% of total potential in most materials).

1.3.2 Three and Four-Body Potentials

Materials having a high degree of covalent bonding require a model that takes bond angle into account. As a result, the (or 3 in the case of one atomic species) contributions must be included. For a model to be reasonably transferrable, even basic metals require more than a pair potential description. Cluster potentials in this section aren't as successful as the pair functional method for simple metals, since they don't have a large enough pair potential. Here, we examine cluster potentials involving three or four bodies in a system. Typically, these potentials are employed with covalent materials such as silicon or mixed metallic-covalent systems. There are a number of three-body potentials, but the Stillinger–Weber [10] model is perhaps the most renowned. A few important physical discoveries and ease of implementation were the major criteria for selecting functions to include in the Stillinger–Weber potential. The model represents a decent first shot at a potential with bond-angle dependency. This potential is limited to the pair and triplet contributions, which have the form (Eqs. 28–31):

$$\phi_2\left(r^{\alpha\beta}\right) = \varepsilon f_2\left(r^{\alpha\beta}/\sigma\right) \tag{28}$$

$$\phi_3\left(r^{\alpha\beta}, r^{\alpha\gamma}, r^{\beta\gamma}\right) = \varepsilon f_3\left(r^{\alpha\beta}/\sigma, r^{\alpha\gamma}/\sigma, r^{\beta\gamma}/\sigma\right) \tag{29}$$

$$f_2(\hat{r})\begin{cases} A(B\hat{r}^{-p}-\hat{r}^{-q})e^{(\hat{r}-a)^{-1}}, & \hat{r}<a, \\ 0, & \hat{r}\geq a, \end{cases} \tag{30}$$

$$f_3(\hat{r}^{\alpha\beta},\hat{r}^{\alpha\gamma},\hat{r}^{\beta\gamma}) = h(\hat{r}^{\alpha\beta},\hat{r}^{\alpha\gamma},\theta^{\beta\alpha\gamma}) + h(\hat{r}^{\beta\gamma},\hat{r}^{\alpha\beta},\theta^{\gamma\beta\alpha}) + h(\hat{r}^{\alpha\gamma},\hat{r}^{\beta\gamma},\theta^{\alpha\gamma\beta}). \tag{31}$$

The caps on 'r' mean that distance is normalised by a factor of σ. $\theta^{\beta\alpha\gamma}$ is the angle produced by the bonds α–β and α–γ with the vertex at atom α in Eq. (31), but employing the angle directly gives a handier functional form. Equation (31) also contains other angles of similar construction. For diamond-cubic materials like Silicon, this function penalises bond angles that deviate from the ideal bond angle, $\theta_t = 109.47°$ found in diamond-cubic materials. Specifically, Eq. 32 is employed:

$$h(\hat{r}_1,\hat{r}_2,\theta)\begin{cases} \lambda\exp\left[\gamma(\hat{r}_1-a)^{-1}+\gamma(\hat{r}_2-a)^{-1}\right](\cos\theta+\frac{1}{3})^2, & \hat{r}_1<a \text{ and } \hat{r}_2<a, \\ 0, & \text{other case} \end{cases} \tag{32}$$

Due to the fact that $\cos\theta_t = -1/3$, this function's lowest value ($h = 0$) occurs at the point when $\theta = \theta_t$. To investigate additional materials in diamond cubic phase, this combination of functions offers a cluster potential with seven free parameters. To examine silicon in the majority of situations, Stillinger and Weber's original parameters are utilised, which are

$$A = 7.049556277,\ \ B = 0.6022245584,\ \ p = 4,\ \ q = 0,$$

$$a = 1.80, \lambda = 21.0, \mu = 1.2.$$

The constants ε and σ are responsible for determining the energy and length scales. $a_0 = 5.429$ Å for the Si lattice constant and $E^0_{coh} = 4.63eV$ for the equilibrium cohesive energy of the diamond structure were determined by Stillinger and Weber [11]. This gives

$$\sigma = 2.0951 \text{ Å}, \quad \varepsilon = 2.1682 \text{ eV}$$

to complete the definition of the model. There is a difference between the empirically recognised value of 4.63 eV for Silicon and the value given in Ref. [11].

Moriarty (in a series of publications [12–14]) took a more basic approach to the creation of cluster potentials. Generalized pseudopotential theory (GPT) was first created to give a rigorous extension of the energy of a system of atoms into interatomic potentials up to the four-body cluster terms. GPT models were then constructed using reduced analytic forms to mimic interatomic potentials, resulting in the "model GPT" (MGPT). Despite being significantly more complicated than models such as the Stillinger–Webber model, these forms are nevertheless basic

enough to allow MD simulations involving millions of molecules over timescales measured in picoseconds. Transition metals like Mo and Ta have been the focus of MGPT work [15–21].

1.3.3 Modelling Organic Molecules: CHARMM and AMBER

Materials consisting of a large number of atoms, generally in the solid state, are the trending subject of computational materials science. The study of individual macromolecules, from organic polymers in man-made plastics to DNA, requires atomistic modelling investigations. Structure and arrangement of these molecules are essential for understanding things like the strength and reactivity of a plastic or the interactions between a medication and proteins in the body. These interatomic cluster potentials, known in the biology and chemistry sectors as "force field", were developed by several researchers through collaboration.

The two most widely utilised force field models are CHARMM (Chemistry at Harvard Macromolecular Mechanics) [22, 23] and AMBER (Assisted Model Building with Energy Refinement) [24, 25]. AMBER and CHARMM potentials are general cluster potentials at their core. Their construction is distinct since macromolecules were the focus of their development. This allows them to be more effective in their intended use. Covalent bonds between atoms in the chain and groups on the side are common in macromolecules, as are weaker interactions such as van der Waals and Coulomb effects. "Bonded" and "non-bonded" interactions in macromolecules are therefore easily distinguished. If you're familiar with CHARMM's covalent bonds, you know that they're unbreakable. These are the weaker and more fragile interactions. This impact is included in the CHARMM model's total energy, as illustrated through Eq. 33:

$$v_{CHARMM}^{int} \equiv v^{bonded} + v^{nonbonded}. \tag{33}$$

When three physical impacts on the bonds (stretching, bending and rotating) are taken into account, it is simply considered as the sum of separate harmonic terms (Eq. 34).

$$v^{bonded} = v^{stretch} + v^{bend} + v^{rotate}. \tag{34}$$

Pair potentials are used to characterise bond stretching as (Eq. 35)

$$v^{stretch} = \frac{1}{2} \sum_{\substack{\alpha,\beta \\ \alpha \neq \beta}}^{bonded} k_b^{\alpha\beta} \left(r^{\alpha\beta} - r_0^{\alpha\beta} \right)^2. \tag{35}$$

Because of the notation, we are reminded that the sum is only over atoms that were first classified as bound or "bonded" in the issue. Stiffness and equilibrium length are represented by the coefficients $k_b^{\alpha\beta}$ and $r_0^{\alpha\beta}$. It is only through their atomic species that they are reliant on the atoms α and β. "Bond-bending" is basically a three-body cluster potential of harmonic shape, summed exclusively over "bonded" atoms. It comes in the form of Eq. 36:

$$v^{bend} = \frac{1}{6} \sum_{\substack{\alpha,\beta,\gamma \\ \alpha \neq \beta \neq \gamma}}^{bonded} k_\theta^{\alpha\beta\gamma} \left(\theta^{\alpha\beta\gamma} - \theta_0^{\alpha\beta\gamma}\right)^2. \tag{36}$$

At least four bound atoms must be known in order to determine the bond rotation. Consider, for example, four atoms named α, β, γ and δ. They all lay in the same plane. Atoms grouped α, β and γ; and β, γ and δ are also in the same plane. When these planes intersect, the angle between them is what determines how the bond is rotated. Because of this, it may be expressed as a four-body cluster potential, even if the actual machinery of a CHARMM code is optimised for bond rotation energy (refer to Eq. 37).

$$v^{rotate} = \frac{1}{24} \sum_{\alpha,\beta,\gamma,\delta}^{bonded} k_\phi^{\alpha\beta\gamma\delta} \left(\phi^{\alpha\beta\gamma\delta} - \phi_0^{\alpha\beta\gamma\delta}\right)^2. \tag{37}$$

The bonded energy is now complete. We've already seen the introduction of the Lennard–Jones (LJ) potential to simulate van der Waals interactions, and we will see the Born–Mayer ionic model's Coulomb portion of interactions. A cut-off radius is generally included in the Lennard–Jones portion, although these contributions are calculated for all atom pairs, whether or not they are bound. CHARMM's fundamental model is fairly simple, relying on easy-to-implement harmonic, Lennard–Jones and Coulomb potentials.

When it comes to quantum effects, little attempt is taken to incorporate them, as the bonds between the atoms are maintained by construction. Quantum mechanics has shown that even more complicated functions may be described using harmonic approximations. Since the bond configuration is near equilibrium, the bound interactions are only valid in the vicinity of it (length, angle or rotation). It also implies that when building up a CHARMM/AMBER model, you need to know a lot about the types of bonds that are predicted. For example, stating that two carbon atoms have a "bond" is not sufficient. To estimate the harmonic portion of the bond energy, it is required to know if the C–C bond is a so-called "π" or "σ" bond. Atomic configurations are governed by geometric restrictions, which mean that even the same two orbitals can form connections of varying strength and type based on their spatial relationship. Two orbitals that could be placed side by side end up making π bonds

whereas placed end to end make σ bonds. These details are known for a large number of significant compounds. However, the lack of bond breakage and formation makes these models highly restricted when it comes to deformation difficulties in materials science. CHARMM force fields need a significant number of fitting parameters that must be correctly determined. There are several variables to consider that includes calculating the multiple parameters k_b, k_θ and k_ϕ for each conceivable combination of atomic species, as well as properly fitting the van der Waals and Coulomb components. Because of the smart utilisation of experimental and more fundamental discoveries, a large number of bonding libraries for CHARMM and AMBER have been created. AMBER and CHARMM work because of the minimal number of bonding species in macromolecular science, which keeps the number of fitting parameters to a reasonable level. Look at organic compounds that are dominated by carbon (C), oxygen (O), hydrogen (H) and nitrogen (N). Other more "exotic" components only appear on occasion. Remember that CHARMM and AMBER models aren't suitable for studying chemical processes involving the bond breakage. Their goal is to shed light on molecules that have a stable structure, configuration and mobility. The "ReaxFF" model is an extension of the CHARMM/AMBER method to fixed bonding. ReaxFF allows bonds to be broken and repaired. Due to the fact that all pairs are treated equally, the distinction between bonded and non-bonded pairs is effectively erased. Because it takes into account their closeness as well as an estimate of the bond order, the ReaxFF model determines whether two atoms are bound or not.

1.3.4 Cluster Potential Limitations and the Requirement for Interatomic Functionals

It is assumed in this section, that the energy of a system of atoms may be approximated by stopping the series in Eq. 21 at a very low order. As a result, we can now define pair, three-body and four-body potentials, as well as their properties. Some substances are bound in such a way that their n-body terms, which range in number from 2 to as many as there are atoms, converge far more slowly.

There are a number of reasons why cluster potentials might be highly inefficient. Thus, other functional forms have been developed, which we will explore in the further sections. We adopt the terminology of Carlsson [26], who coined the word "functionals" to designate any interatomic model that didn't match Eq. 21. We may further split interatomic functionals into "pair functionals" and "cluster functionals" based on Carlsson's classification. This is because the functionals are dependent on atomic locations, as their names imply.

Pair functionals require information about atom locations in pairs, whereas cluster functionals require information about atom positions in groups of three or more atoms. They come in a variety of shapes, and it's hard to come up with a general term that's informative enough. The recursive technique may be used to express any interatomic functional in the form of Eq. 21. Clustered potential and interatomic

functional vary in that the latter in its simplest standard form differs from Eq. 21 description.

1.4 Tersoff Potentials

In covalent solids, deviations from the pairwise additive system are much wider as they are less tightly packed and contain strong directed bonds of equal strength (typically just with a coordination number of 4). This set of solids includes diamonds or zinc-blende structures, defined by an angle ~109.47° and by the bonds in the sp^3 orbital hybridization. The bond angle and bond order are therefore the main features of these materials. The strength of the bond between two atoms, such as coordination number and bond angle, relies on the local environment. It can be expressed by Eq. 38:

$$\phi_{Tersoff} = \frac{1}{2}\sum_{i \neq j} \phi_R\left(r_{ij}\right) + \frac{1}{2}\sum_{i \neq j} B_{ij}\phi_A\left(r_{ij}\right) \tag{38}$$

where ϕ_r and ϕ_a are the attractive and repulsive potentials.

B_{ij} is the order of the bond between the atom i and j, which usually constitutes a decreasing coordinate number function, N_{coor}. It is signified from Eq. 39,

$$B_{ij} \propto \frac{1}{\sqrt{N_{coor}}}. \tag{39}$$

That is, B_{ij} of atom i binding with atom j is decreased by another B_{ik} bond existence. The degree of weakening relies on the location and angle of this other bond.

While both components are distance-dependent, this sort of potential generally requires over six parameters to fill. This has worked very well and has been used for several covalent binding solids and molecules such as SiC, Si, diamond, graphite, carbohydrate and hydrocarbons with an excellent parameterisation. One has to be careful with graphite since the description of its layer by layer is outside the scope of Tersoff's potentials. The bond-order function b, which alters the intensity of an attractive interaction based on the environment in which the bond resides, is the major enhancement to these pair potentials. An environment's bond coordination is quantified by the coordination function z. Pair functionals are used to establish this environment, but bond angles are also used to determine how far apart nearby atoms are.

Due to the requirement of at least three atoms to define a bond angle, the Tersoff potential uses clusters of three atoms to determine the attractive bond strengths. They can be represented as the following Eq. 40.

$$\phi_R(r) = Ae^{-\lambda_1 r}, \quad \phi_A(r) = -Be^{-\lambda_2 r}$$

$$b(z) = \left(1 + \delta^n z^n\right)^{-1/2n}$$

$$z^{\alpha\beta} = \sum_{\substack{\gamma \\ \gamma(\alpha,\beta)}} f(r^{\alpha\gamma}) g\left(\theta^{\alpha\beta\gamma}\right) e^{\lambda_3^3 \left(r^{\alpha\beta} - r^{\alpha\gamma}\right)^3}$$

$$f(r) = \begin{cases} 1 & if\ r,\ R - D \\ \frac{1}{2} - \frac{1}{2}\sin\left(\frac{\pi}{2}\frac{r-R}{D}\right) & if\ R - D < r < R + D \\ 0 & if\ r > R + D \end{cases}$$

$$g(\theta) = 1 + \frac{c^2}{d^2} - \frac{c^2}{d^2 + (h - \cos\theta)^2} \tag{40}$$

where $\theta^{\alpha\beta\gamma}$ is the angle shaped between bonds $\alpha-\beta$ and $\alpha-\gamma$, and the functions comprise 12 fitting parameters:A, B, λ_1, λ_2, λ_3, δ, n, R, D, c, d, and h. Perhaps the most frequently employed parameter values are the ones Tersoff established for Si and called "Si(B)" in Ref. [27]. These are

$$A = 3264.7\ \text{eV},\quad B = 95.373\ \text{eV},\quad \lambda_1 = 3.2394\ \text{Å}^{-1},\ \lambda 2 = \lambda 3 = 1.3258\ \text{Å}^{-1}$$

$$\delta = 0.33675,\quad n = 22.956,\quad R = 3.0\ \text{Å}\quad D = 0.2\ \text{Å},\quad c = 4.8381,$$

$$d = 2.0417 \text{ and } h = 0.0.$$

Brenner expanded Tersoff's form to better deal with hydrocarbon systems. This is because H and C have different properties, and Brenner discovered that the Tersoff form could not be fitted to this system adequately. After that, he showed that a new and improved potential (the "Tersoff-Brenner" potential) could be produced by generalising the bond order function, b. On the other hand, the Tersoff–Brenner formalism has been shown to be a second moment expansion of Tersoff–Brenner. On the contrary, Pettifor and Oleynik [28] pointed out that the Tersoff–Brenner form has two fundamental shortcomings: it is unable to appropriately pick between certain structural arrangements and it does not accurately depict hydrocarbons' 'π-bond'. However, this model has certain limitations because it was intended primarily to simulate specific structural configurations and not others. As a result of Tersoff's integration of bond order into his potential, a broad range of models that incorporate an empirical bond order function of the Tersoff–Brenner variety have been developed.

Since then, "BOP" has become a catch-all name for any model of this kind. To avoid confusion, we prefer to refer to the Tersoff and Tersoff–Brenner forms of BOP as empirical BOPs, as well as potentials such "environment-dependent interatomic potential" (EDIP) for Si [29] and "reactive bond order" (REBO) for hydrocarbons [30].

1.5 Potentials for Ionic Solids

The majority of potentials among second neighbours are short-term and exponentially approach zero. Ions can be polarised or charged in ionic materials, however, and the Coulombic component can be distributed over a very long range. Therefore, one has to consider two elements of pair interaction to explain the potential of these solids: short-range and long-range potential, which can be expressed in Eq. 41.

$$\phi_{ionic} = \phi_{short}(r_{ij}) + \frac{Z_1 Z_2}{r_{ij}} \tag{41}$$

where Z_1 and Z_2 are constituent atom charges. The first term usually consists of two contributions: one through the repulsive interactions between electrons in a near proximity, and the other because of the appealing interaction of van der Waals forces. This statement can be formulated as Eq. 42:

$$\phi_{short}(r_{ij}) = C_1 exp\left(-\frac{r_{ij}}{\rho}\right) - \frac{C_2}{r_{ij}^2} \tag{42}$$

where C_1 and C_2 are constants.

The second factor for Coulomb interaction is the long-range interaction, which decays as $1/r_{ij}$. Under periodic boundary circumstances, this term cannot be handled with the conventional energy-cut-off technique for an ionic crystal since the potential builds on a regular basis, posing a significant divergence challenge. It may, however, be dealt with effectively using a technique known as the Ewald sum. This approach produces an artificial charge distribution centred around each ion twice: once with the opposite sign as the ions and once with the same sign (the magnitude is the same for both times).

The first charge creation essentially cancels much of the original charge distribution, resulting in a short-ranged issue with no divergent problem. The second charge generation recovers the original system, which may be done well using Fast Fourier Transformations (FFT) when the charge distributions are represented in a spherically symmetric Gaussian type (Fast Fourier Transformations). Note that in the case of a large dimension, the FFT method transforms the continuous distribution into discreet charges of a limited grid in the reciprocal room. When utilising ionic potential, one should be very careful since anions like oxygen ions tend to vary their size and loading conditions, especially with metal ions transition. Potentials for mixed systems (such as the Al–Al_2O_3) likewise are known to be ineffective. Fixed-charge potentials forecast the cohesive energy of ionic materials inaccurately and therefore cannot be utilised to simulate oxidation at metal surfaces or metal/oxide interfaces, where the local ion charge might be considerably different from that of the bulk oxide.

1.5.1 Modelling Ionic Crystals: The Born–Mayer Potential

If you assume paired, environment-independent interactions, you may describe ionic crystals using a simple pair potential [31]. If you combine group VI or VII elements with alkali or alkali-earth metals, the electron transfer is so powerful that most of the ionic cohesion originates from Coulomb interactions. This is an instance when a pair potential is an adequate model. The Born–Mayer potential is a typical example of an ionic model. You may think of it as an amalgamation between long-range Coulomb interactions and short-range repulsive interactions that characterise the overlap of filled orbitals in ion cores. The model as a whole has a shape (refer Eq. 43).

$$v^{int} = \frac{1}{2}\sum_{\substack{\alpha,\beta \\ \alpha \neq \beta}}^{N} \frac{Z^{\alpha} Z^{\beta} e^2}{r^{\alpha\beta}} + \frac{1}{2}\sum_{\substack{\alpha,\beta \\ \alpha \neq \beta}}^{near} A_{\alpha\beta}\left(1 + \frac{Z^{\alpha}}{z^{\alpha}} + \frac{Z^{\beta}}{z^{\beta}}\right)\exp\left[\left(\sigma^{\alpha} + \sigma^{\beta} - r^{\alpha\beta}\right)/\rho^{\alpha\beta}\right]. \tag{43}$$

Here, electron charge is defined as e and Z as atomic number of an ion; while σ, A and ρ are fitting factors that are exclusively dependent on ionic species. The second sum, which contains the fitting parameters, is more empirical of the two. According to the notation "near" over the second sum, it is only taken over closest neighbours in the equilibrium crystal structure (or some other short cut-off distance). Repulsion owing to core overlap is reflected in this total, which is difficult to calculate precisely. For now, let's look at the well-known Coulomb energy first. Depending on the charge of the ions, it comprises both repulsive and attractive terms and is long-range. Despite its simplicity, it is exceedingly difficult to apply in a practical simulation because of its subtlety. We may explain this phenomenon by observing that, while the potential decays slowly by $1/r$, it increases by r^2, in proportion to distance r. Due to the fact that oppositely charged particles are arranged in a crystal to guarantee charge neutrality on a small scale, this sum is only convergent. In order to elucidate on this topic, look at an arrangement of atoms in the form of fcc. Atoms in the nth shell are all at some distance r_n from an atom at the origin, where for fcc, $r_n = a\sqrt{n/2}$ and a is the lattice constant. Evidently, the distance between two adjacent shells, $(r_{n+1} - r_n) \to 0$ as $\to \infty$. As a result, there is a cancellation of terms and convergence at $r \to \infty$ limit. When modelling an infinite crystal with periodic boundary conditions, this means that we cannot simply terminate the summing at any cut-off radius and get a valid result. For the same reason in DFT, equations have formulas that allow for precise assessment of these sums, but they are impractical when the periodic simulation box contains several atoms.

Although rapid multipole techniques [32] have been developed in the late 1980s to properly assess these sums with $O(NlogN)$ efficiency, the long-range interactions in ionic systems will always slow down calculations compared to short-range covalent or metallic bonds. It is also known as the "rigid-ion" model since it ignores the possibility of ion polarisation caused by adjacent ions. For example, higher-order effects of "deformable ions" may be incorporated in more complex ionic models [33].

1.6 Reactive Force Field Potentials

Reactive force fields (ReaxFF) potentials are made up of tens of parameters established by first-principles computations. These potentials can react in the local environment dynamically and represent the formation/breakdown of connections in situ. While it is approximately ten times slower to use, in hydrocarbons, nanotube systems, Si Systems, etc., their successful applications made the method highly popular and promising. Instead of CHARMM and AMBER, ReaxFF automatically determines whether two atoms are bound or not, considering all interactions equally [34]. This "reactivity" is nothing new to the materials science community. Except for CHARMM and AMBER, all the models discussed in this chapter are reactive. When it comes to chemical reactions in metals and intermetallic compounds as well as (to a certain extent) ceramics and semiconductors, they are rather straightforward. For example, ReaxFF builds a generic energy functional with several fitting parameters to solve this issue. Here, we have discussed ReaxFF for hydrocarbons in detail (i.e., for molecules comprising only the elements C and H) [34]. A broad range of systems, including metal hydrides [35], silicon and silicon dioxide, have had their ReaxFF parameters established since 2001. This approach assumes energy comparable to that used in CHARMM or AMBER, but with the inclusion of the following concepts (Eq. 44):

$$\nu^{int} = \nu^{bond} + \nu^{over} + \nu^{under} + \nu^{val} + \nu^{pen} + \nu^{tors} + \nu^{vdWalls} + \nu^{Coul} + \nu^{conj} \tag{44}$$

Each of these terms is meant to reflect a physical feature of the connection. A two-atom bond's energy is accounted for in the first term. The second term attempts to rectify this energy depending on each bond's excess or deficiency (determined by estimates of the bond order). The other terms are

1. $Vval$: a form of energy linked to body angles (a potential with three bodies),
2. $Vpen$: for certain special cases, a penalty to correct the bond-angle energy,
3. $Vtors$: bond torsion angles energy (basically a four-body potential for bond rotation),
4. $VvdWaals$: the interaction energy of the van der Waals force,
5. $VCoul$: the Coulomb interaction energy and finally,
6. $Vconj$: a term that deals with conjugated bonds.

Each term signifies a unique empirical function with its own set of fitting parameters. Thorough motivation from the theory is seen in some functional forms in some terms like van der Waals and Coulomb form, while others might copy other features of bonding which are generally convenient mathematical forms. Overall, the ReaxFF model is based on the assumption that to determine bond energy we require bond order as the key parameter.

Due to computational costs associated with quantum mechanically derived estimations of bonds' order, ReaxFF begins with an extensive curve fit to generate an

empirical formula for the bond order based only on atomic species and interatomic distances instead. We initially estimate the order of atoms α and β in hydrocarbons by using an empirical Eq. 45:

$$\widehat{BO}^{\alpha\beta} = \sum_{i=1}^{3} exp[\lambda_i \left(\frac{r^{\alpha\beta}}{r_{0,i}}\right)^{\mu_i} \tag{45}$$

This $i = 1$ is related to the σ-bond, $i = 2$ to the first π-bond, and $i = 3$ to the second π-bond, respectively. According to whether the bond is formed by C–C or C–H atoms or in the case of H–H atoms, these parameters, namely, λ_i, μ_i and$r_{0,i}$ take on various values. For C–C bond interactions, the π-bonds $i = 2, 3$ are not included in the equation at all. This is followed by a second empirical formulation of this kind that corrects the bond ordering through Eq. 46:

$$BO^{\alpha\beta} = BO^{\alpha\beta}\left(\widehat{BO}^{\alpha\beta}, \Delta^{\alpha}, \Delta^{\beta}\right) \tag{46}$$

the deviation Δ^{α} of atom α from its ideal state of valence z^{α} is calculated by adding up each of an element's bond orders as given in Eq. 47:

$$\Delta^{\alpha} = \sum_{\substack{\beta \\ \beta\alpha}} \widehat{BO}^{\alpha\beta} - z^{\alpha}. \tag{47}$$

Therefore, it is necessary to estimate each contribution to the energy using the bond orders $BO^{\alpha\beta}$ and its deviation Δ^{α}. However, there are two exceptions to this rule: the van der Waals and Coulomb energies due to the fact that these are simple pair potentials. For example, the bond energy is denoted as v^{bond} and expressed as the mathematical Eq. 48:

$$v^{bond} = \sum_{bonds} -D^{\alpha\beta} BO^{\alpha\beta} \exp\{p^{\alpha\beta}(1 - \left(BO^{\alpha\beta}\right)^{p^{\alpha\beta}}]\}. \tag{48}$$

Parameters $D^{\alpha\beta}$ and $p^{\alpha\beta}$ depend upon the two atoms species involved in a given bond.

Equation 45 appears to be at odds with our earlier considerations of bond order, as it indicates that the bond order is merely a function of bond length. We've already established that the bond order's primary purpose is to describe the surroundings of a specific bond. However, Eq. 46 and the valence variations and of the two atoms in each bond reveal the environmental dependency of the bond order. As a result of the valence variations, the energy terms V_{under} and V_{over} attempt to further include many-body environmental dependency of the bonds. In this manner, ReaxFF offers a great degree of versatility when it comes to fitting. It takes 100 times as long to compute ReaxFF models as they do with pair functionals [35].

2 Types of Materials and Their Potential Models

There are numerous applications for each sort of interatomic potential, but it is helpful to summarise them in one location.

Firstly, they are used for noble gases. van der Waals forces dominate interactions between noble gas atoms, making them the most basic of solid systems. Pair potentials in general and the 6–12 Lennard–Jones model in particular may be used to explain solid crystals of noble gases that have FCC structures. It is possible that a pair potential would be sufficient to simulate noble gases because of their electronic structures, which render them inert in their natural state. Different electronic shells competing for electrons do not cause sudden changes in bonding properties. Rather, there are only the predicted, radially symmetric interactions of induced dipoles in adjacent atoms. This estimate, despite the reasonable agreement between Lennard–Jones potential and the behaviour of noble gases, is still only an approximation. There are also many advanced models that integrate many-body effects for argon [36–39]. In ionic crystal pair potentials, Coulomb interactions are long-range, necessitating a high cut-off radius and specific handling of periodic systems [32]. In theory, this complicates the computation by necessitating huge neighbour lists and extra calculations, but it does not (in principle) restrict physical precision and transferability.

In the description of metallic bonding, empirical interatomic models have proved very successful. Simply paired functionals have proved astonishingly accurate for basic metals and transition metals with d-orbitals full or almost filled. Elements such as Al, Ni, Cu, Pd, Ag, Pt and Au are of special technical importance. As a result of their non-directional, metallic bonding, they are all FCC crystals at ambient temperature and pressure. Simple pair potential is insufficient, but the inclusion of an environmental component in the form of pairs with spherically symmetric coordinates is adequate. As a result, metals with low-temperature BCC or HCP structures perform less well with pair functionals. Fe, Mo, W, Mg, Zr and Ti are some examples. However, compared to the FCC systems described before, these pair functionals are not as accurate. A covalent bond is already present in these systems by virtue of their inclination to stabilise in the BCC or HCP phases. Due to their directionality, covalent bonds are difficult to model using the pair functional form. However, this is less of an issue when it comes to hard-sphere metals, especially those with close to the optimum c/a ratio. A pair functional can be fitted so that, for example, the bcc phase's energy is lower than that in the phase, but the amount of the energy difference tends to be smaller than anticipated by DFT calculations. Deformation tends to make these models unduly prone to changing from the ground state to a finite-state continuum (FSC).

It's a potential expansion of the pair functional method to increase its fidelity for crystals like BCC and HCP. According to the "two band" paradigm, the system's bonding may be described as the sum of two pair functional forms that reflect contributions from separate electron bands. Using a basic model, each pair of functionals is weighted to calculate the electronic occupancy in the s and d-bands correspondingly. Modelling dilute solid solutions and intermetallic alloys of simple metals with

pair functionals may be done rather well with them. But one of today's interatomic modelling problems is to build trustworthy, efficient models to explain alloy systems. This was first complicated by a paucity of data against which to fit models. The equilibrium structure and lattice constant were known, but there were relatively few experimental data points that could be utilised to successfully fit a model to alloy behaviour, therefore the model could not be robustly fitted.

Heavily weighted DFT computations are being utilised to provide fitting and validation data for empirical models, making the function of DFT in model creation more important than ever. To be sure, it's helped to enhance the fitting data, but it also showed that the basic pair functional form is not generic enough to be used throughout the full phase diagram of most alloy systems. As a rule, successful applications of the pair functional form to alloys work within a very limited section of the phase diagram for which they have been tuned [40–45]. When characterising interactions between Al and Ni, the Voter–Chen functional should not be considered a generic model for the Al–Ni alloy system [46]. On the contrary, it is ideal for use in a Ni-rich phase diagram. From dilute Al/Ni solutions to the Ni_3Al structure, it performs a good job simulating alloys. It is common for intermetallic models to be constrained in this way. The covalent nature of intermetallic alloys necessitates the use of cluster functionals or cluster potentials. It's true that they can be modelled with some success using the MEAM [47, 48] and MGPT [49] formalisms, but the number of fitting parameters and the complexity of the functional forms necessary for alloys make model creation and testing very difficult. In many metals, hydrogen and carbon are significant components that occur in low quantities. Hydride brittleness is generally caused by hydrogen, while carbon is required for the cementite phase that gives steel its strength. Some work has gone towards modelling these elements' effects on metals using pair functions [50–52], but the trustworthiness of these models is restricted. These systems need cluster functionals such as MEAM [47, 53] or ReaxFF [54], especially in the case of carbon, due to its highly covalent bonding, but also in the case of hydrogen.

Due to silicon's enormous technical relevance, many models have been created in an attempt to characterise its behaviour [55, 56]. Si bonding is notoriously difficult to simulate because of its strong covalent nature and ability to hybridise various orbitals in diverse settings. They tend to be well-suited to particular parts of configuration space but aren't very transferrable. Using diamond cubic Si melting as an example, the canonical Stillinger–Weber potential is most suited for simulating the solid–liquid interface at melting temperature since it has been calibrated to it. A wide range of applications have attempted to exploit this potential down to zero kilobytes, typically without much success. Because it can mimic the effects of orbital hybridization through its empirical description of the bond order, the Tersoff potential for Si is a relatively realistic model. Tersoff forms tend to be less dependable for Si, however, when compared with simple metals, such as gold and silver. Similarly, the Tersoff form may be used to describe Ge or C in various crystalline states [57].

Bonding is a complicated combination of metallic and covalent bonding for transition metals with partially filled d-bands. Three- and four-body cluster potentials are

ideally suited to these materials since the effects of bond angle may be easily understood. Some examples of this include Moriarty and coworkers' MGPT potentials, as well as the MEAM form [58]. They are restricted in their capacity to transmit cluster potentials, though.

It is possible to simulate diamonds and other ordered states of carbon (such as nanotubes and bucky-balls) using CHARMM or AMBER cluster potentials, but only in the limit when bond breakage is not predicted. As a result of this significant restriction, the Tersoff–Brenner [57] model for carbon is more commonly used. In spite of the fact that this model is very empirical [28], it gives a decent match to the behaviour of organised carbon. To get a more transferrable behaviour, one may use the more recently created ReaxFF model that is more computationally expensive [34].

Ceramics, geological materials, and other complex solids are covalent crystals. For ceramics, minerals, and other complicated covalent or mixed covalent–metallic structures, the TB techniques or a complete DFT analysis are the main options, although the ReaxFF framework is currently being used to study some of the simpler ones [59]. As more components are introduced to such systems, the fitting method becomes increasingly complicated. Such materials frequently include a variety of elements in high concentrations. When it comes to modelling organic compounds, CHARMM and AMBER are the most popular.

Considering that molecular conformations are frequently of greater interest than chemical processes, the absence of reactivity in these models isn't necessarily a deal-breaker. It was only in recent years that the ReaxFF model introduced the concept of reactivity in such systems. Various configurations have been subjected to ReaxFF [60–64] simulations. The authors also have a vast experience in the field of molecular dynamics and experimental characterisation of composites materials [65–111].

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References

1. Lee, J.G.: Computational materials science: an introduction. CRC Press (2016)
2. Tadmor, E.B., Miller, R.E.: Modeling materials: continuum, atomistic and multiscale techniques. Cambridge University Press (2011)
3. Murdoch, A.I.: A critique of atomistic definitions of the stress tensor. J. Elast. **88**(2), 113–140 (2007)
4. Jackson, A.N., Bruce, A.D., Ackland, G.J.: Lattice-switch Monte Carlo method: application to soft potentials. Phys. Rev. E **65**(3), 036710 (2002)
5. Martin, J.W.: Many-body forces in metals and the Brugger elastic constants. J. Phys. C Solid State Phys. **8**(18), 2837 (1975)

6. Fisher, M.E.: The free energy of a macroscopic system. Arch. Ration. Mech. Anal. **17**(5), 377–410 (1964)
7. Andrews, G.E.: Number Theory. Courier Corporation (1994)
8. Drautz, R., Fähnle, M., Sanchez, J.M.: General relations between many-body potentials and cluster expansions in multicomponent systems. J. Phys. Condens. Matter **16**(23), 3843 (2004)
9. Admal, N.C., Tadmor, E.B.: Stress and heat flux for arbitrary multibody potentials: a unified framework. J. Chem. Phys. **134**(18), 184106 (2011)
10. Stillinger, F.H., Weber, T.A.: Packing structures and transitions in liquids and solids. Science **225**(4666), 983–989 (1984)
11. Stillinger, F.H., Weber, T.A.: Computer simulation of local order in condensed phases of silicon. Phys. Rev. B **31**(8), 5262 (1985)
12. Moriarty, J.A.: First-principles interatomic potentials in transition metals: Multi-ion interactions and their analytic representation. In: Many-atom Interactions in Solids: Proceedings of the International Workshop, Pajulahti, Finland, June 5–9, 1989, vol. 48, p. 158. Springer Verlag (1990)
13. Moriarty, J.A.: Analytic representation of multi-ion interatomic potentials in transition metals. Phys. Rev. B **42**(3), 1609 (1990)
14. Moriarty, J.A.: Angular forces and melting in bcc transition metals: a case study of molybdenum. Phys. Rev. B **49**(18), 12431 (1994)
15. Xu, W., Moriarty, J.A.: Atomistic simulation of ideal shear strength, point defects, and screw dislocations in bcc transition metals: Mo as a prototype. Phys. Rev. B **54**(10), 6941 (1996)
16. Schoeck, G.: Thermal activation in plastic deformation. In: Multiscale Phenomena in Plasticity: From Experiments to Phenomenology, Modelling and Materials Engineering (pp. 33–56). Springer, Dordrecht (2000)
17. Campbell, G.H., Belak, J., Moriarty, J.A.: Atomic structure of the σ5 (310)/[001] symmetric tilt grain boundary in tantalum. Scr. Mater. **43**(7), 659–664 (2000)
18. Yang, L.H., Soderlind, P., Moriarty, J.A.: Accurate atomistic simulation of (a/2) <111> screw dislocations and other defects in bcc tantalum. Philos. Mag. A **81**, 1355–1385 (2001)
19. Pizzagalli, L., Demenet, J.L., Rabier, J.: Theoretical study of pressure effect on the dislocation core properties in semiconductors. Phys. Rev. B **79**(4), 045203 (2009)
20. Moriarty, J.A., Belak, J.F., Rudd, R.E., Söderlind, P., Streitz, F.H., Yang, L.H.: Quantum-based atomistic simulation of materials properties in transition metals. J. Phys. Condens. Matter **14**(11), 2825 (2002)
21. Moriarty, J.A., Benedict, L.X., Glosli, J.N., Hood, R.Q., Orlikowski, D.A., Patel, M.V., Söderlind, P., Streitz, F.H., Tang, M., Yang, L.H.: Robust quantum-based interatomic potentials for multiscale modeling in transition metals. J. Mater. Res. **21**(3), 563–573 (2006)
22. Brooks, B.R., Bruccoleri, R.E., Olafson, B.D., States, D.J., Swaminathan, S.A., Karplus, M.: CHARMM: a program for macromolecular energy, minimization, and dynamics calculations. J. Comput. Chem. **4**(2), 187–217 (1983)
23. Lim, M.C., Pei, Q.X., Zhong, Z.W.: Translocation of DNA oligonucleotide through carbon nanotube channels under induced pressure difference. Phys. A **387**(13), 3111–3120 (2008)
24. Lopes, P.E., Guvench, O., MacKerell, A.D.: Current status of protein force fields for molecular dynamics simulations. In: Molecular Modeling of Proteins, pp. 47–71. Humana Press, New York, NY (2015)
25. Case, D.A., Darden, T.A., Cheatham, T.E., III, et al.: AMBER 10. http://www.ambermd.org. 2008
26. Carlsson, A.E.: Beyond pair potentials in elemental transition metals and semiconductors. In: Solid state physics, vol. 43, pp. 1–91. Academic Press (1990)
27. Tersoff, J.: Empirical interatomic potential for silicon with improved elastic properties. Phys. Rev. B **38**(14), 9902–9905 (1988)
28. Pettifor, D.G., Oleynik, I.I.: Interatomic bond-order potentials and structural prediction. Prog. Mater. Sci. **49**(3–4), 285–312 (2004)
29. Bazant, M.Z., Kaxiras, E., Justo, J.F.: Environment-dependent interatomic potential for bulk silicon. Phys. Rev. B **56**, 8542–8552 (1997)

30. Brenner, D.W., Shenderova, O.A., Harrison, J.A., Stuart, S.J., Ni, B., Sinnott, S.B.: A second-generation reactive empirical bond order (REBO) potential energy expression for hydrocarbons. J. Phys. Condens. Matter **14**(4), 783 (2002)
31. Adams, J.B., Rockett, A., Kieffer, J., Xu, W., Nomura, M., Kilian, K.A., Richards, D.F., Ramprasad, R.: Atomic-level computer simulation. J. Nucl. Mater. **216**, 265–274 (1994)
32. Greengard, L., Rokhlin, V.: A fast algorithm for particle simulations. J. Comput. Phys. **135**(2), 280–292 (1997)
33. Finnis, M.: Interatomic forces in condensed matter, vol. 1. OUP Oxford (2003)
34. Kaledin, A.L., Van Duin, A.C., Hill, C.L., Musaev, D.G.: Parameterization of reactive force field: dynamics of the [Nb6O19H x](8–x)–Lindqvist polyoxoanion in bulk water. J. Phys. Chem. A **117**(32), 6967–6974 (2013)
35. Zhang, Q., Çağın, T., Van Duin, A., Goddard, W.A., III., Qi, Y., Hector, L.G., Jr.: Adhesion and nonwetting-wetting transition in the Al/$\alpha-Al_2O_3$ interface. Phys. Rev. B **69**(4), 045423 (2004)
36. Lotrich, V.F., Szalewicz, K.: Three-body contribution to binding energy of solid argon and analysis of crystal structure. Phys. Rev. Lett. **79**(7), 1301 (1997)
37. Rościszewski, K., Paulus, B., Fulde, P., Stoll, H.: Ab initio coupled-cluster calculations for the fcc and hcp structures of rare-gas solids. Phys. Rev. B **62**(9), 5482 (2000)
38. Schwerdtfeger, P., Gaston, N., Krawczyk, R.P., Tonner, R., Moyano, G.E.: Extension of the Lennard-Jones potential: theoretical investigations into rare-gas clusters and crystal lattices of He, Ne, Ar, and Kr using many-body interaction expansions. Phys. Rev. B **73**(6), 064112 (2006)
39. Aoki, M., Kurokawa, T.: A simple environment-dependent overlap potential and Cauchy violation in solid argon. J. Phys. Condens. Matter **19**(23), 236228 (2007)
40. Miracle, D.B.: Intermetallic compounds. Practice **2**, 53 (1994)
41. Mishin, Y.: Atomistic modeling of the γ and γ'-phases of the Ni–Al system. Acta Mater. **52**(6), 1451–1467 (2004)
42. Mishin, Y., Mehl, M.J., Papaconstantopoulos, D.A.: Embedded-atom potential for B2−NiAl. Phys. Rev. B **65**(22), 224114 (2002)
43. Zope, R.R., Mishin, Y.: Interatomic potentials for atomistic simulations of the Ti-Al system. Phys. Rev. B **68**(2), 024102 (2003)
44. Chung, J.S., Thorpe, M.F.: Local atomic structure of semiconductor alloys using pair distribution functions. Phys. Rev. B **55**(3), 1545 (1997)
45. Ruda, M., Farkas, D., Abriata, J.: Interatomic potentials for carbon interstitials in metals and intermetallics. Scr. Mater. **46**(5), 349–355 (2002)
46. Voter, A.F., Chen, S.P., Siegel, R.W., Weertman, J.R., Sinclair, R.: Characterization of defects in materials. In: MRS symposia proceedings, vol. 82, p. 175. Materials Research Society Pittsburgh, PA (1987)
47. Agrawal, A., Mirzaeifar, R.: Copper-graphene composites; developing the MEAM potential and investigating their mechanical properties. Comput. Mater. Sci. **188**, 110204 (2021)
48. Creemers, C., Deurinck, P.: Platinum segregation to the (111) surface of ordered Pt80Fe20: LEIS results and model simulations. Surf. Interface Anal. Int. J. Devoted Dev. Appl. Tech. Anal. Surf. Interfaces Thin Films **25**(3), 177–190 (1997)
49. Moriarty, J.A., Widom, M.: First-principles interatomic potentials for transition-metal aluminides: theory and trends across the 3d series. Phys. Rev. B **56**(13), 7905 (1997)
50. Daw, M.S.: Application of the embedded atom method to fracture, dislocation dynamics, and hydrogen embrittlement. In: Modeling Environmental Effects on Crack Growth Process, pp. 99–124. (1986)
51. Angelo, J.E., Moody, N.R., Baskes, M.I.: Trapping of hydrogen to lattice-defects in nickel. Modell. Simul. Mater. Sci. Eng. **3**(3), 289–307 (1995)
52. Ruda, M., Farkas, D., Garcia, G.: Atomistic simulations in the Fe–C system. Comput. Mater. Sci. **45**(2), 550–560 (2009)
53. Lee, B.J., Jang, J.W.: A modified embedded-atom method interatomic potential for the Fe–H system. Acta Mater. **55**(20), 6779–6788 (2007)

54. Cheung, S., Deng, W.Q., Van Duin, A.C., Goddard, W.A.: ReaxFFMgH reactive force field for magnesium hydride systems. J. Phys. Chem. A **109**(5), 851–859 (2005)
55. Balamane, H., Halicioglu, T., Tiller, W.A.: Comparative study of silicon empirical interatomic potentials. Phys. Rev. B **46**(4), 2250 (1992)
56. Cook, S.J., Clancy, P.: Comparison of semi-empirical potential functions for silicon and germanium. Phys. Rev. B **47**(13), 7686 (1993)
57. Brenner, D.W.: Empirical potential for hydrocarbons for use in simulating the chemical vapor deposition of diamond films. Phys. Rev. B **42**(15), 9458 (1990)
58. Jelinek, B., Houze, J., Kim, S., Horstemeyer, M.F., Baskes, M.I., Kim, S.-G.: Modified embedded-atom method interatomic potentials for the Mg–Al alloy system. Phys. Rev. B **75**(5), 054106 (2007)
59. van Duin, A.C.T., Merinov, B.V., Han, S.S., Dorso, C.O., Goddard, W.A., III.: ReaxFF reactive force field for the Y-doped BaZrO3 proton conductor with applications to diffusion rates for multigranular systems. J. Phys. Chem. A **112**(45), 11414–11422 (2008)
60. Endo, K., Masumoto, C., Matsumoto, D., Ida, T., Mizuno, M., Kato, N.: Fragment distribution of thermal decomposition for PS and PET with QMD calculations by considering the excited and charged model molecules. Appl. Surf. Sci. **255**(4), 856–859 (2008)
61. Zhu, R., Janetzko, F., Zhang, Y., van Duin, A.C., Goddard, W.A., Salahub, D.R.: Characterization of the active site of yeast RNA polymerase II by DFT and ReaxFF calculations. Theor. Chem. Acc. **120**(4), 479–489 (2008)
62. Salmon, E., Behar, F., Lorant, F., Hatcher, P.G., Metzger, P., Marquaire, P.M.: Thermal decomposition processes in algaenan of Botryococcus braunii race L. Part 1: experimental data and structural evolution. Org. Geochem. **40**(3), 400–415 (2009)
63. Chenoweth, K., van Duin, A.C.T., Dasgupta, S., Goddard, W.A., III.: Initiation mechanisms and kinetics of pyrolysis and combustion of JP-10 hydrocarbon jet fuel. J. Phys. Chem. A **113**(9), 1740–1746 (2009)
64. van Duin, A.C.T., Dasgupta, S., Lorant, F., Goddard, W.A.: ReaxFF: a reactive force field for hydrocarbons. J. Phys. Chem. A **105**, 9396–9409 (2001)
65. Verma, A., Parashar, A., Packirisamy, M.: Atomistic modeling of graphene/hexagonal boron nitride polymer nanocomposites: a review. Wiley Interdiscip. Rev. Comput. Mol. Sci. **8**(3), e1346 (2018)
66. Verma, A., Singh, V.K., Verma, S.K., Sharma, A.: Human hair: a biodegradable composite fiber—a review. Int. J. Waste Resour. **6**(206), 2 (2016)
67. Verma, A., Singh, V.K.: Mechanical, microstructural and thermal characterization of epoxy-based human hair–reinforced composites. J. Test. Eval. **47**(2), 1193–1215 (2019)
68. Verma, A., Parashar, A.: Structural and chemical insights into thermal transport for strained functionalised graphene: a molecular dynamics study. Mater. Res. Express **5**(11), 115605 (2018)
69. Verma, A., Negi, P., Singh, V.K.: Experimental analysis on carbon residuum transformed epoxy resin: chicken feather fiber hybrid composite. Polym. Compos. **40**(7), 2690–2699 (2019)
70. Verma, A., Gaur, A., Singh, V.K.: Mechanical properties and microstructure of starch and sisal fiber biocomposite modified with epoxy resin. Mater. Perform. Charact. **6**(1), 500–520 (2017)
71. Verma, A., Parashar, A., Packirisamy, M.: Effect of grain boundaries on the interfacial behaviour of graphene-polyethylene nanocomposite. Appl. Surf. Sci. **470**, 1085–1092 (2019)
72. Verma, A., Budiyal, L., Sanjay, M.R., Siengchin, S.: Processing and characterization analysis of pyrolyzed oil rubber (from waste tires)-epoxy polymer blend composite for lightweight structures and coatings applications. Polym. Eng. Sci. **59**(10), 2041–2051 (2019)
73. Verma, A., Negi, P., Singh, V.K.: Physical and thermal characterization of chicken feather fiber and crumb rubber reformed epoxy resin hybrid composite. Adv. Civil Eng. Mater. **7**(1), 538–557 (2018)
74. Verma, A., Negi, P., Singh, V.K.: Experimental investigation of chicken feather fiber and crumb rubber reformed epoxy resin hybrid composite: mechanical and microstructural characterization. J. Mech. Behav. Mater. **27**(3–4) (2018)

75. Chaurasia, A., Verma, A., Parashar, A., Mulik, R.S.: Experimental and computational studies to analyze the effect of h-BN nanosheets on mechanical behavior of h-BN/polyethylene nanocomposites. J. Phys. Chem. C **123**(32), 20059–20070 (2019)
76. Jain, N., Verma, A., Singh, V.K.: Dynamic mechanical analysis and creep-recovery behaviour of polyvinyl alcohol based cross-linked biocomposite reinforced with basalt fiber. Mater. Res. Express **6**(10), 105373 (2019)
77. Verma, A., Joshi, K., Gaur, A., Singh, V.K.: Starch-jute fiber hybrid biocomposite modified with an epoxy resin coating: fabrication and experimental characterization. J. Mech. Behav. Mater. **27**(5–6) (2018)
78. Verma, A., Kumar, R., Parashar, A.: Enhanced thermal transport across a bi-crystalline graphene–polymer interface: an atomistic approach. Phys. Chem. Chem. Phys. **21**(11), 6229–6237 (2019)
79. Verma, A., Singh, V.K.: Experimental investigations on thermal properties of coconut shell particles in DAP solution for use in green composite applications. J. Mater. Sci. Eng. **5**(3), 1000242 (2016)
80. Verma, A., Singh, V.K., Arif, M.: Study of flame retardant and mechanical properties of coconut shell particles filled composite. Res. Rev. J. Mater. Sci. **4**(3), 1–5 (2016)
81. Verma, A., Parashar, A., Jain, N., Singh, V.K., Rangappa, S.M., Siengchin, S.: Surface modification techniques for the preparation of different novel biofibers for composites. In: Biofibers and Biopolymers for Biocomposites, pp. 1–34. Springer, Cham (2020)
82. Rastogi, S., Verma, A., Singh, V.K.: Experimental response of nonwoven waste cellulose fabric–reinforced epoxy composites for high toughness and coating applications. Mater. Perform. Charact. **9**(1), 151–172 (2020)
83. Bharath, K.N., Madhu, P., Gowda, T.G., Verma, A., Sanjay, M.R., Siengchin, S.: A novel approach for development of printed circuit board from biofiber based composites. Polym. Compos. **41**(11), 4550–4558 (2020)
84. Verma, A., Jain, N., Parashar, A., Gaur, A., Sanjay, M.R., Siengchin, S.: Lifecycle assessment of thermoplastic and thermosetting bamboo composites. In: Bamboo Fiber Composites, pp. 235–246. Springer, Singapore (2021)
85. Singh, K., Jain, N., Verma, A., Singh, V.K., Chauhan, S.: Functionalized graphite-reinforced cross-linked poly (vinyl alcohol) nanocomposites for vibration isolator application: morphology, mechanical, and thermal assessment. Mater. Perform. Charact. **9**(1), 215–230 (2020)
86. Verma, A., Parashar, A.: Molecular dynamics based simulations to study the fracture strength of monolayer graphene oxide. Nanotechnology **29**(11), 115706 (2018)
87. Verma, A., Parashar, A.: The effect of STW defects on the mechanical properties and fracture toughness of pristine and hydrogenated graphene. Phys. Chem. Chem. Phys. **19**(24), 16023–16037 (2017)
88. Verma, A., Parashar, A., Packirisamy, M.: Tailoring the failure morphology of 2D bicrystalline graphene oxide. J. Appl. Phys. **124**(1), 015102 (2018)
89. Singla, V., Verma, A., Parashar, A.: A molecular dynamics based study to estimate the point defects formation energies in graphene containing STW defects. Mater. Res. Express **6**(1), 015606 (2018)
90. Verma, A., Parashar, A.: Molecular dynamics based simulations to study failure morphology of hydroxyl and epoxide functionalised graphene. Comput. Mater. Sci. **143**, 15–26 (2018)
91. Verma, A., Zhang, W., van Duin, A.C.: ReaxFF reactive molecular dynamics simulations to study the interfacial dynamics between defective h-BN nanosheet and water nanodroplets. Phys. Chem. Chem. Phys. **23**, 10822–10834 (2021)
92. Verma, A., Parashar, A., Packirisamy, M.: Role of chemical adatoms in fracture mechanics of graphene nanolayer. Mater. Today Proc. **11**, 920–924 (2019)
93. Verma, A., Parashar, A.: Characterization of 2D nanomaterials for energy storage. In: Recent Advances in Theoretical, Applied, Computational and Experimental Mechanics, pp. 221–226. Springer, Singapore (2020)

94. Verma, A., Jain, N., Parashar, A., Singh, V.K., Sanjay, M.R., Suchart, S.: Design and modeling of lightweight polymer composite structures. In: Lightweight Polymer Composite Structures: Design and Manufacturing Techniques, Chapter 7, pp. 193–224. Taylor & Francis Group (CRC Press), Boca Raton (2020)
95. Verma, A., Jain, N., Parashar, A., Singh, V.K., Sanjay, M.R., Siengchin, S.: Lightweight graphene composite materials. In: Lightweight Polymer Composite Structures: Design and Manufacturing Techniques, Chapter 1, pp. 1–20. Taylor & Francis Group (CRC Press), Boca Raton (2020)
96. Verma, A., Parashar, A., Singh, S.K., Jain, N., Sanjay, M.R., Siengchin, S.: Modeling and simulation in polymer coatings. In: Polymer Coatings: Technologies and Applications, Chapter 16, pp. 309–324. Taylor & Francis Group (CRC Press), Boca Raton (2020)
97. Verma, A., Jain, N., Rastogi, S., Dogra, V., Sanjay, M.R., Siengchin, S., Mansour, R. Mechanism, anti-corrosion protection and components of anti-corrosion polymer coatings. In: Polymer Coatings: Technologies and Applications, Chapter 4, pp. 53–66. Taylor & Francis Group (CRC Press), Boca Raton (2020)
98. Verma, A., Jain, N., Kalpana, Sanjay, M.R., Siengchin, S., Jawaid, M.: Natural fibers based bio-phenolic composites. In: Phenolic Polymers based Composite Materials, Chapter 10, pp. 153–168. Springer Nature, Singapore (2020)
99. Bharath, K.N., Madhu, P., Gowda, T.Y., Verma, A., Sanjay, M.R., Siengchin, S.: Mechanical and chemical properties evaluation of sheep wool fiber-reinforced vinylester and polyester composites. Mater. Perform. Charact. **10**(1), 99–109 (2021)
100. Marichelvam, M.K., Manimaran, P., Verma, A., Sanjay, M.R., Siengchin, S., Kandakodeeswaran, K., Geetha, M.: A novel palm sheath and sugarcane bagasse fiber based hybrid composites for automotive applications: an experimental approach. Polym. Compos. **42**(1), 512–521 (2021)
101. Chaudhary, A., Sharma, S., Verma, A.: Optimization of WEDM process parameters for machining of heat treated ASSAB'88 tool steel using Response surface methodology (RSM). Mater. Today Proc. (2021)
102. Chaudhary, A., Sharma, S., Verma, A.: WEDM machining of heat treated ASSAB '88 tool steel: a comprehensive experimental analysis. Mate. Today Proc. (2021)
103. Verma, A., Singh, V.K.: Experimental characterization of modified epoxy resin assorted with almond shell particles. ESSENCE Int. J. Environ. Rehabil. Conserv. 36
104. Verma, A., Samant, S.S.: Inspection of hydrodynamic lubrication in infinitely long journal bearing with oscillating journal velocity. J. Appl. Mech. Eng. **5**(3), 1–7 (2016)
105. Verma, A., Parashar, A., Jain, N., Singh, V.K., Rangappa, S.M., Siengchin, S. (2020) Surface modification techniques for the preparation of different novel biofibers for composites. In: Biofibers and Biopolymers for Biocomposites, pp.1–34
106. Bisht, N., Verma, A., Chauhan, S., Singh, V.K.: Effect of functionalized silicon carbide nanoparticles as additive in cross-linked PVA based composites for vibration damping application. J. Vinyl Add. Tech. **27**(4), 920–932 (2021)
107. Kataria, A., Verma, A., Sanjay, M.R., Siengchin, S.: Molecular modeling of 2D graphene grain boundaries: mechanical and fracture aspects. Mater. Today Proc. (2021)
108. Arpitha, G.R., Verma, A., Sanjay, M.R., Siengchin, S.: Preparation and experimental investigation on mechanical and tribological performance of hemp-glass fiber reinforced laminated composites for lightweight applications. Adv. Civil Eng. Mater. **10**(1), 427–439 (2021)
109. Deji, R., Verma, A., Kaur, N., Choudhary, B.C., Sharma, R.K.: Density functional theory study of carbon monoxide adsorption on transition metal doped armchair graphene nanoribbon. Mater. Today Proc. (2021)
110. Deji, R., Verma, A., Choudhary, B.C., Sharma, R.K.: New insights into NO adsorption on alkali metal and transition metal doped graphene nanoribbon surface: a DFT approach. J. Mol. Graph. Model. (2021). https://doi.org/10.1016/j.jmgm.2021.108109
111. Deji, R., Jyoti, R., Verma, A., Choudhary, B.C., Sharma, R.K.: A theoretical study of HCN adsorption and width effect on co-doped armchair Graphene Nanoribbon. Comput. Theor. Chem. 113592 (2022)

Current Perspective on Atomistic Force Fields of Polymers



Kiranmai Yellam, Ratna S. Katiyar, and Prateek K. Jha

Abstract The theoretical toolbox of polymer physics has expanded over the years, beginning from simple phenomenological theories to detailed field-theoretical and generic toy models of polymer chains. One major bottleneck of these methods relates to the issue of scale and resolution as both the structure and dynamics of a long polymer chain and the local physical and chemical interactions at the level of its repeat unit cannot be studied together. As a case in point, minor variations in the chemistry and/or arrangement of repeat units of a polymer may greatly influence the polymer behavior, which is difficult to capture using generic bead-spring models of the polymer chain. This is where atomistic simulation comes into picture. With gradual increase in computational capabilities, it is now possible, up to a reasonable extent, to simulate polymers with atomistic resolution. We present a state of the art review of the atomistic simulation of polymers with emphasis on the atomistic force fields used in these simulations. Various force fields used in atomistic simulations are described, along with a discussion of their strengths and weaknesses. This is followed by a discussion on some of the application areas where atomistic simulations have been successfully used in polymer science and an outlook on some other upcoming applications that may potentially benefit from atomistic simulations. This review should be of interest to computational researchers working on the use of atomistic simulations in polymer science and experimentalists seeking to find complementary computational tools for polymer design.

Keywords Polymer physics · Atomistic simulations · Force field · Molecular dynamics · Multiscale simulations

K. Yellam · P. K. Jha (✉)
Department of Chemical Engineering, Indian Institute of Technology Roorkee, Roorkee 247667, Uttarakhand, India
e-mail: prateek.jha@ch.iitr.ac.in

K. Yellam
e-mail: ykiranmai@ch.iitr.ac.in

R. S. Katiyar
Department of Chemical Engineering, Jawaharlal Nehru Engineering College, MGM University, Aurangabad 431003, India

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1 Introduction

Polymers are everywhere. In fact, if we touch anything around us, it is most likely going to contain a polymer. Despite environmental concerns that are now associated with the use of some of the polymeric materials like plastics, the global per capita use of polymers has only increased over the years [1]. Apart from the widespread industrial usage, the fundamental physics and chemistry of polymeric materials have fascinated academics quite extensively in the last century, as reflected by the fact that it has been a subject of over a dozen polymer scientists winning the Nobel Prizes. What is so special about polymers to deserve such attention? From a chemistry perspective, polymers are similar to hydrocarbons in the fact that they typically contain a chain of carbon atoms bonded together, except that the number of carbon atoms should be much higher (at least ~50 but usually > 1000) to be classified as a polymer. It turns out that a long chain of carbons in the case of a polymer imparts the possibility of a very long number of configurations arising from the rotations around the $C-C$ bond, which results in much higher configurational entropy than compared to small hydrocarbons. In addition, entangled configurations are usually attained for longer polymers [2] that results in restrictions on the chain movement. Configurational entropy and entanglements together result in the interesting viscoelastic properties of polymers [3]. Further complexities are introduced by the intra-chain and inter-chain interactions (e.g., hydrogen bonding) [4] that vary with temperature and nature of solvent [5]. These result in a rich equilibrium and non-equilibrium phase behavior of polymeric materials [6], exemplified by the coil-globule [7] and glass transition [8] behavior of polymers.

Figure 1 shows some of the commonly used models of the polymer chain. Initial models were based on the statistics of random walk [5], which may be used to interpret many possible configurations of a polymer chain. For example, in the freely jointed chain model, the polymer chain is represented by segments of the polymer chain connected together, where the angle between adjacent segments is randomly chosen. In the freely rotating chain model (Fig. 1a), adjacent segments are at a fixed angle with respect to each other. Both these models provide the same scaling of the average squared end-to-end distance $\langle R_e^2 \rangle \propto N$, where N is the number of segments. Experiments showed that the actual scaling[1] depends on the nature of solvent and temperature that inspired the Flory–Huggins lattice model (Fig. 1b). In this model, the polymer–polymer (ε_{PP}), polymer–solvent (ε_{PS}), and solvent–solvent (ε_{SS}) interactions are modeled as interactions between neighboring lattice sites. The net effect of these interactions can be quantified in terms of an effective solvent quality parameter $\chi = z(\varepsilon_{PP} + \varepsilon_{SS} - 2\varepsilon_{PS})/2k_BT$, where z is the lattice coordination

[1] The length of segment in the freely jointed chain model is called the 'Kuhn length', b_K, that is defined as twice the persistence length of the polymer chain. For an ideal chain, $\langle R_e^2 \rangle = b_K N^2$ where N is the number of segments (often referred as 'Kuhn segment'). Since the persistence length is considered independent of polymer molecular weight M, the same scaling $\langle R_e^2 \rangle \propto M$ is expected for an ideal chain. In freely rotating chain, an additional prefactor of $\frac{1+\cos\theta}{1-\cos\theta}$ appears in the equation but the same scaling behavior with the number of segments (and molecular weight) is obtained for an ideal chain.

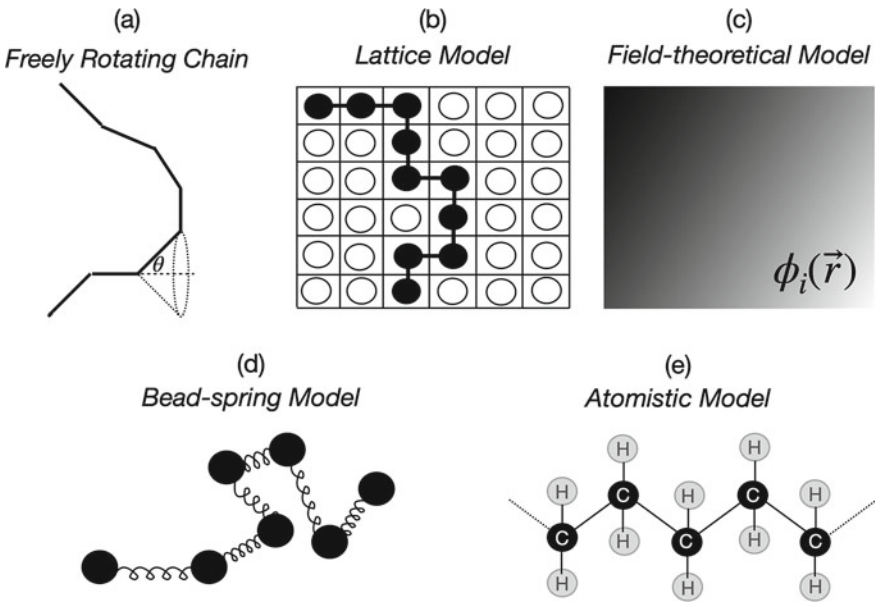


Fig. 1 Some commonly used models of polymer chain. **a** In freely rotating chain model, adjacent segments of polymer chain are at a fixed angle θ with respect to each other. **b** Lattice model is used to model polymer behavior in solution. Filled and open circles in figure represents the repeat unit and solvent molecules, respectively. **c** Field-theoretical models use density of species $\phi_i(\vec{r}) = \sum_k \delta(\vec{r} - \vec{r}_{ik})$ as variable in place of position vectors $\{\vec{r}_{ik}\}$ for units of species i. Here, $\delta(\vec{r})$ is the Dirac Delta function and the densities are appropriately normalized. **d** Beads and springs in the bead-spring model represent the units of a polymer chain and segments connecting the units, respectively. **e** Atomistic model treats each of the atom of the polymer chain as an entity

number, k_B is the Boltzmann constant, and T is the absolute temperature. In general, $\langle R_e^2 \rangle \propto N^\alpha$, where $\alpha = 6/5$ in the 'good solvent' (swollen chain or self-avoiding walk) regime found for $\chi < 1/2$, and $\alpha = 2/3$ in the 'poor solvent' (collapsed chain) found for $\chi > 1/2$, and $\alpha = 1$ ('ideal chain' or random walk) in the 'theta solvent' regime found for $\chi = 1/2$. This result is obtained using a 'mean field approximation' where the interactions are essentially averaged over all possible polymer and solvent configurations. That is, the effects of fluctuations are not considered [9]. An apparent difficulty with the lattice model is that the positions of polymer and solvent units are assumed to be fixed on sites of a lattice and they occupy the same volume. This assumption is relaxed in a general field-theoretical model [10] (Fig. 1c) where species densities are used as variables in place of their positions. However, the mean field approximation is still employed in most field-theoretical models. A more general description, which also accounts for the effects of fluctuations is the bead-spring model (Fig. 1d). Classical Rouse [11] and Zimm [12] models of the dynamics of polymer chain also employ the bead-spring model. Since the springs connecting the beads may be stretched or compressed, these models capture various relaxation modes of the polymer chain. None of the above described models capture the detailed

chemistry of polymer chain, which may only be captured by the use of an atomistic model as represented in Fig. 1e.

Before we discuss simulations of atomistic models of polymer chains (Fig. 1e), it is important to emphasize that most reported particle simulation studies of polymeric systems are based on 'coarse-grained' models that are essentially variations of the bead-spring model of Fig. 1d. Typically, such models incorporate a harmonic spring energy [13] for each spring that represents the bonded interactions,

$$U_{bonded} = \frac{1}{2}k(b - b_0)^2 \tag{1}$$

where the spring constant k is a measure of bond stiffness, b is the length of the spring at a given instant and b_0 is an arbitrary parameter referred as the 'equilibrium bond length'. Many other spring energy models are also proposed. For example, Finitely Extensible Nonlinear Elastic (FENE) spring model [14] imposes a limit on the maximum possible stretch. In addition to bonded interactions, there are also nonbonded interactions (e.g., electrostatic and van der Waals interaction) between every pair of beads in the system. Atomistic force fields, as we will discuss in this review, provide a much more detailed representation of the bonded and nonbonded interactions between atoms of various involved species.

Coarse-graining [15, 16] refers to the lumping of several atoms and associated bonds into a single unit. The widespread use of such coarse-grained models in polymer literature is both historical and based on reason. From a physics perspective, the bead-spring model may be perceived as the extension of the freely jointed chain model and the mean field lattice model, as it accounts for fluctuations and chain dynamics. At the same time, the simulations of atomistic models (Fig. 1e) of polymers were impossible in the early days of computers. Even today, it is not typically possible to simulate realistic polymer chains with a large number of repeat units (> 100) [17]. Note that a polymeric system actually involves many polymeric chains often along with solvents and other molecules, and the simulation of a single chain in isolation with all atomistic details does not quite serve the purpose. One might argue that the simulation of such long polymer chains is not necessary and even the simulation of short polymer chains with a much lesser number of repeat units will capture the same behavior. This unfortunately is not true. For instance, the scaling of $\langle R_e^2 \rangle$ with the number of repeat units N is found to be different for very small and large values of N (Fig. 2). The key quantity for the reliability of atomistic simulations is the persistence length l_p that depends on the polymer chemistry and is defined as the characteristic length along the contour for the decay of position correlations between repeat units on the chain [18]. Since atomistic simulations are limited to small N and therefore small contour length $L_c \lesssim l_p$, it is usually difficult to capture the behavior in the flexible regime [19] that is more typical of long polymers. As a consequence, chain entanglements are seldom captured [20] that greatly affect the dynamical properties of polymers such as viscosity. Probably, a more serious limitation of atomistic simulation is the fact that simulations can only be conveniently performed over time scales of the order of *ns*, whereas chain relaxation may occur

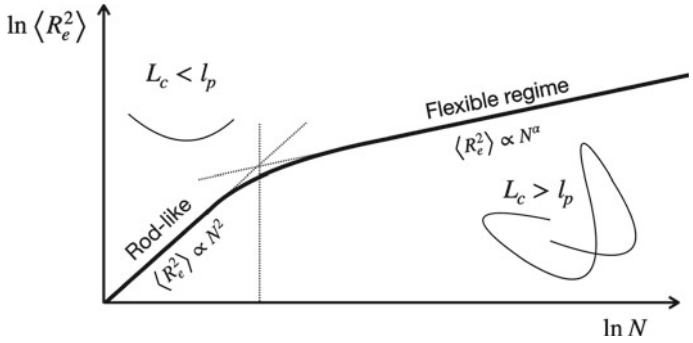


Fig. 2 Different regimes of chain flexibility, determined by the persistence length l_p and the contour length L_c. For small number of repeat units ($L_c < l_p$), the chain may be approximated as a rod with the end-to-end distance roughly proportional to the number of repeat units. For large number of repeat units ($L_c > l_p$), the exponent α appearing in the scaling $\langle R_e^2 \rangle \propto N^{\alpha}$ is determined by the nature of solvent and temperature

over time scales up to several μs or higher [21]. Even when the dynamical properties are not of interest, the approach to equilibrium in such simulations are very slow. In fact, these problems are not easily resolved in coarse-grained simulations as well, and a range of special Monte Carlo moves [22] have been designed for simulation of polymers to achieve faster equilibration.

The discussion in the above paragraph could be demotivating for someone interested in atomistic simulation of polymers. Thankfully, despite its obvious weaknesses, atomistic simulations of polymers are still quite useful as we will discuss in this review. First and foremost, atomistic simulations capture the detailed polymer chemistry and it is, therefore, apt to understand the consequences of variations in chemistry [23, 24]. Second, modern supercomputers have pushed the limits of atomistic simulations and quite long polymer chains can now be simulated, if one invests on computational resources. Third, atomistic simulations can also be used to train systematic coarse-grained models [25, 26], which may together overcome the limitations of scale and resolution. In this review, we will attempt to justify these claims. In the next section, we present a timely review of the various atomistic force fields of polymers. This is followed by a section on the applications of atomistic simulations of polymers. Methods to bridge atomistic simulations to mesoscale methods (e.g., systematic coarse-graining) is presented afterwards. We conclude this review with a perspective on the future challenges and opportunities in this area. Given the width of scope of this review, we will skip most of the mathematical details of individual force fields and methods and the readers are referred to original papers for a detailed understanding.

2 Atomistic Force Fields of Polymers

2.1 Basics of Atomistic Force Fields

Atomistic simulations may be broadly categorized into two categories: (1) stochastic minimization of thermodynamic energy functions, and (2) simulating classical equation of motion. Monte Carlo (MC) simulations come in the first category, wherein different configurations of a system are generated using various sampling algorithms. In importance sampling [27], which is the most commonly used MC algorithm, the configurations are generated according to their Boltzmann weights. It therefore requires information on the energy U of the system for a given configuration. Energy is obtained as the sum of the energies u_{ij} between all pairs of atoms $i, j \in [1, N]$ for a system containing N atoms. That is,

$$U = \frac{1}{2}\sum\nolimits_i \sum\nolimits_{j \neq i} u_{ij} \tag{2}$$

where the factor $1/2$ accounts for the double counting of interactions. Note that the above description neglects the three-body and other many body interactions. This assumption is generally not serious for most applications of atomistic simulations, but the effects of many body interactions may be substantial for dense, strongly correlated systems like ionic liquids [28]. Molecular Dynamics (MD) simulations belong to the second category, where particles (atoms) move as per the classical (Newton's) equation of motion

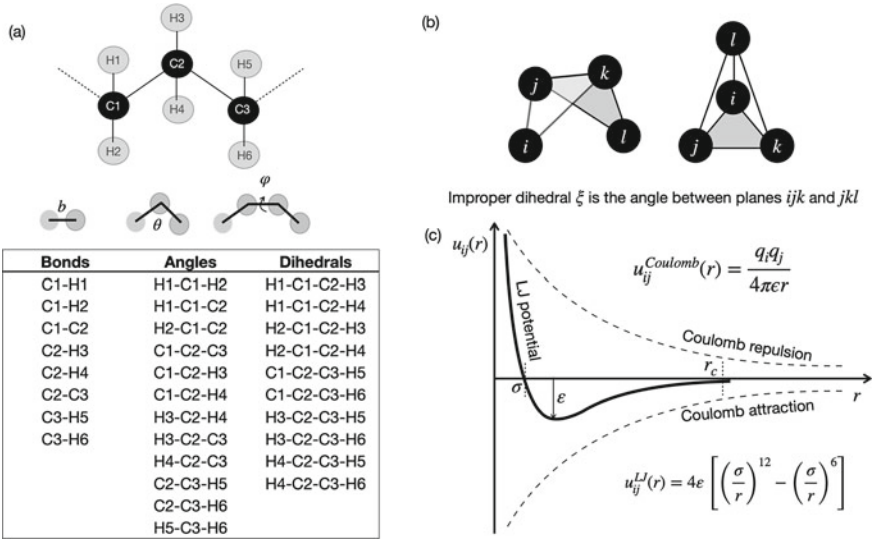
$$m\overrightarrow{a} = m\frac{d^2\overrightarrow{x}}{dt^2} = \overrightarrow{F} \tag{3}$$

Here, m is the atomic mass and $\overrightarrow{a}$ is the acceleration at a given time defined as the second derivative of position $\overrightarrow{x}$ with respect to time t. The force acting on the particle at time t is defined as $\overrightarrow{F} = -\overrightarrow{\nabla} U$. Alternatively, the force $\overrightarrow{F}_i$ acting on a particle i is computed as the sum of pairwise forces $\overrightarrow{f}_{ij}$ exerted on the particle i by other particles $j \neq i$. That is,

$$\overrightarrow{F}_i = \sum\nolimits_{j \neq i} \overrightarrow{f}_{ij} \tag{4}$$

Both MC and MD simulations may be used to analyze the thermodynamic equilibrium behavior of a system, but the real dynamics of the system is only captured in MD simulations [29].

In molecular systems, u_{ij} may be further classified into 'bonded interactions' between pairs of atoms within a molecule and 'nonbonded interactions' between all pairs of atoms in the system (Fig. 3). Strong bonded interactions generally occur because of the presence of a covalent bond between atoms connected by a bond.



Bonds	Angles	Dihedrals
C1-H1	H1-C1-H2	H1-C1-C2-H3
C1-H2	H1-C1-C2	H1-C1-C2-H4
C1-C2	H2-C1-C2	H2-C1-C2-H3
C2-H3	C1-C2-C3	H2-C1-C2-H4
C2-H4	C1-C2-H3	C1-C2-C3-H5
C2-C3	C1-C2-H4	C1-C2-C3-H6
C3-H5	H3-C2-H4	H3-C2-C3-H5
C3-H6	H3-C2-C3	H3-C2-C3-H6
	H4-C2-C3	H4-C2-C3-H5
	C2-C3-H5	H4-C2-C3-H6
	C2-C3-H6	
	H5-C3-H6	

Fig. 3 Typical bonded and nonbonded interactions in atomistic force fields. **a** Bonds, angles, and dihedrals to be considered for the molecule shown in the picture. **b** Improper dihedrals are sometimes needed for planar molecules [33], i.e., to ensure that *ijk* and *jkl* surfaces in the figure are in the same plane. **c** Typical nonbonded interactions considered in atomistic simulations. LJ potential is used to represent short-ranged interactions and is composed of a repulsive part for small distances ($r < \sigma$) to prevent overlap of atoms and an attractive part with maximum depth ε for the remaining range with a cutoff distance r_c, beyond which the potential is considered as zero. Coulomb potential is used for long-ranged electrostatic interactions between atoms of partial charges q_i and q_j in a medium of dielectric constant ϵ

Bonded interactions are further classified into bond, angle, and dihedral (torsion) interactions. Nonbonded interactions can be further classified into short-ranged and long-ranged contributions. The short-ranged contribution generally includes steric interactions and van der Waals interactions, both of which is usually represented by a Lennard Jones (LJ) potential [30]. The long-ranged contribution involves the electrostatic interactions. While the detailed charge distribution (electron cloud) is not modeled within an atomistic simulation framework, atoms of molecules are assigned 'partial charges' within atomistic force fields and the electrostatic interaction between a pair of atoms is then represented as the Coulomb interaction between partial charges of atoms. It is important to emphasize that usually all atoms of a molecule are assigned a partial charge. Since the Coulomb interactions have a slower decay and exist up to longer distances than compared to LJ interactions, their computation is typically the most expensive component of atomistic simulations. Though it is not often prescribed in atomistic force fields, several methods may be used to speed up the computation of Coulomb interactions. One of the most common strategy in systems involving periodic boundary conditions [27, 31, 32] is to divide these interactions into a small-r and large-r contributions, where the former is computed in real space and the latter is computed in reciprocal (Fourier) space.

2.2 Types of Force Fields

Force fields are sometimes categorized into three categories, referred as Class I, Class II, and Class III force fields [23]. A typical force field expression for a Class I force field is given below [34]:

$$U = \sum_{bonds} \frac{1}{2} k_b (b - b_0)^2 + \sum_{angles} \frac{1}{2} k_\theta (\theta - \theta_0)^2 + \sum_{\substack{proper \\ dihedrals}} \frac{1}{2} k_\varphi [1 + cos(n\varphi - \delta)] + \sum_{\substack{improper \\ dihedrals}} \frac{1}{2} k_\xi (\xi - \xi_0)^2 + \sum_{pairs} 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^{6} \right] + \sum_{pairs} \frac{q_i q_j}{4\pi\epsilon r} \tag{5}$$

Here, the parameters $k_b, k_\theta, k_\varphi, k_\xi, b_0, \theta_0, \xi_0, n, \varepsilon, \sigma, \{q_i\}$ are the force field parameters. In the above expression, the contribution of bond, angle, and dihedral terms are considered to be additive. Cross-terms representing the coupling between different terms, such as bond and angle terms are included in some of the force fields. Further, Eq. (5) assumes a harmonic form of the bond and angle potentials. Advanced force fields referred to as Class II force fields include an anharmonic form of these potentials in addition to all the cross-terms. Class III force field also incorporates some of the special organic chemistry features that are not captured by Class I and Class II force fields, but rarely used in polymer simulations. Most of the polymer simulations involve Class I force fields (e.g., AMBER [35], GROMOS [36], OPLS [37], CHARMM [38], DREIDING [39], etc.) but some class II force fields (e.g., COMPASS [40], PCFF [41]) are also often used. The distinction of force fields into classes is not uniformly used in recent literature and there is now a substantial overlap between the mathematical expressions and parameterization strategies between different force field classes.

The bond length b, angle θ, (proper) dihedral angle φ and phase δ, improper dihedral angle ξ, and distance r, appearing in Eq. (5) vary during the course of the simulation. Note that though the atom indices are not used as subscript (except for q_i) in the definition of parameters in Eq. (5) for the ease of representation, the magnitude of these parameters depends on the atoms involved in the interaction. It is also important to note that while the expressions for the different interaction energies are derived from physical intuition, their parameterization is usually performed by a laborious fitting procedure that involves a large number of simulations performed for different values of each of the force field parameters, to attain the final force field expression. 'Mixing rules' [42] are used to derive the parameters involving pairs of atoms of different species. For example, σ and ε value of a pair of species k and l may be defined using arithmetic and geometric mixing rules, respectively, as given below.

$$\sigma_{kl} = \frac{\sigma_k + \sigma_l}{2} \tag{6}$$

$$\varepsilon_{kl} = \sqrt{\varepsilon_k \varepsilon_l} \tag{7}$$

Different force fields vary in the choice of mathematical representation of the various bonded and nonbonded interactions and the method(s) of force field parameterization. The guiding principles of the force field development process are the (1) simplicity of mathematical expression with ease of computing derivatives and (2) applicability of the force field for a wide range of molecules.

All Atom (AA) force fields do not involve any coarse-graining and are therefore purely atomistic. United Atom (UA) force fields [43] that do not explicitly treat hydrogen atoms connected to aliphatic (nonpolar) groups signify the first level of coarse-graining. Since these hydrogens have a limited effect on the polymer behavior, UA force fields are very commonly used. However, additional improper dihedrals (Fig. 3b) terms are sometimes needed in the UA force field, since the dihedrals consisting of hydrogen atoms (Fig. 3a) are removed. In literature, both AA and UA force fields are generally referred to as atomistic force fields. Coarse-Grained (CG) force fields (Sect. 3) refer to higher levels of coarse-graining.

Most commonly used force fields in polymer simulations are GROMOS, OPLS, CHARMM, and AMBER. GROMOS is a UA force field. AMBER, CHARMM, and OPLS force fields have both AA and UA versions. All of these force fields have a mathematical form similar to Eq. (5) with mixing rules given by Eqs. (6) and (7), with some important exceptions given below [34]:

(1) OPLS-AA and GROMOS force fields use geometric mixing rule $\sigma_{ij} = \sqrt{\sigma_i \sigma_j}$ in place of Eq. (6).

(2) CHARMM force field uses a Urey-Bradley (UB) expression of the type

$$E_{UB} = \sum\nolimits_{angles} k_{UB}\left(r_{1,3} - r_{1,3;0}\right)^2 \tag{8}$$

for some of the angles. Here, $r_{1,3}$ is the distance between the terminal atoms of the angle and $r_{1,3;0}$ is a constant signifying the equilibrium distance between the terminal atoms. k_{UB} is a measure of angle stiffness.

(3) OPLS-AA uses the following expression for dihedral energies

$$E_{dihedrals} = \sum_{dihedrals} \frac{V_1}{2}(1 + cos\varphi) + \frac{V_2}{2}(1 - cos2\varphi) + \frac{V_3}{2}(1 + cos3\varphi) + \frac{V_4}{2}(1 - cos4\varphi) \tag{9}$$

Here, V_1, V_2, V_3, V_4 are fitting parameters.

Force fields and the simulation methodologies discussed until so far in this section have one major disadvantage—the chemical identity of molecule and constituent atoms is constant. This is not really surprising as we are following a classical mechanics framework where electrons are not explicitly included and their effect is only incorporated in terms of fixed average partial charges assigned to atoms.

Assumption of fixed partial charges is especially violated in polar solvents where solvent polarization effects [44] are crucial. For example, aqueous solubility of polymers may be significantly influenced by such polarization effects. Also, since the chemical bonding involves the overlap of electron clouds that are not being modeled, changes in the bonding of atoms are not possible. This becomes problematic when protonation/deprotonation of charged groups occurs in weak polyelectrolytes [45], or in studies of polymerization and polymer degradation processes that involve the formation/breakage of chemical bonds [46]. Some of the recent approaches attempt to address these shortcomings, notably constant pH molecular dynamics [47, 48] and reactive force fields [49], but only a few applications of these methods on polymeric systems have been reported. On similar lines, it is important to caution novice readers about the challenges associated with simulations involving probably the most important and one of the simplest looking molecules, water (H_2O) [50]. It might appear obvious to consider a 3-site (atom) model with an oxygen atom of fixed net negative partial charge $-2\delta q$ surrounded by two hydrogen atoms of fixed positive partial charge $+\delta q$ with fixed values of O–H bond lengths and H–O–H angle (Fig. 4a). Such models, e.g., SPC and TIP3P models [51], are indeed commonly used in literature but does not reproduce many experimental properties of water and more elaborate models (Fig. 4b–d) that consider more than three sites (e.g., TIP4P model containing 4 sites [52]) or flexible bonds and angle [53] (e.g., SPC/F) are required for certain properties/conditions. Finally, assuming fixed partial charges of atoms completely ignores the effects of polarizability, which are now being addressed by some of the polarizable water models [44]. In the context of polymer simulations, choice of water models substantially influences the simulation results of polymer solutions, as will be discussed in later sections.

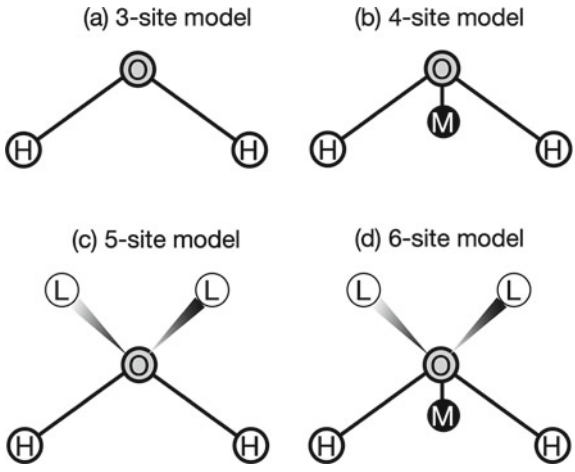


Fig. 4 Types of water models [33, 54, 55]. Dummy atom M in **b** and **d** and L in **c** and **d** are added to better capture the charge distribution

2.3 Force Field Parameterization

Force field parameterization involves an elaborate fitting protocol [56–59] that involves the use of available experimental data and quantum chemistry calculations [60] performed on a wide range of molecules. Broadly speaking, two different approaches may be used to find the force field parameters. In the first approach, the parameters of different bonded and nonbonded interactions (as represented by different terms in Eq. 5) are determined separately. For example, the parameters appearing in bond and angle terms may be obtained using bond and angle vibration (stretching) data obtained by experiments [61]. The dihedral parameters may be obtained by the analysis of torsions using quantum chemistry simulations [57, 62, 63]. Partial charges and van der Waals interaction parameters may be obtained by analysis of electrostatic potential around atoms in quantum chemistry calculations [64]. In the second approach, all force fields parameters are found by finding an optimal fit of the entire force field expression using the least square method [65, 66]. Even in the first approach, where the parameters of different terms are found separately, some fine tuning of parameters is required to find the best fit. Thus, each term in the final force field expression should not be seen in isolation and is affected by other terms in the expression.

Several parameter sets of the force fields are available that are used depending on the chemistry under consideration, which is continuously updated to incorporate even higher number of chemistries. However, none of these parameter sets are universally applicable with an exception of Universal Force Field (UFF) [67], which covers all atoms of the periodic table. That is, different parameter sets are often prescribed for different classes of molecules (e.g., carbohydrates [37, 68], lipids [69]) and for different state(s). Force field development is considered as a separate research field by itself and most reported atomistic simulation studies on polymers use existing parameter sets with minor modifications. It is therefore recommended to do an elaborate literature survey of simulations performed on similar chemistries and to perform comparisons of results obtained using two or more force fields and available experimental data before deciding on the appropriate force field of a system. At the same time, the issues related to ‘transferability’ and ‘compatibility’ of force field parameter sets must be kept in mind, while choosing force field for a simulation [33]. Transferability refers to the performance of the force field for the molecules and properties it has not been parameterized for. It is particularly important in the polymer case as the force fields used of polymers is typically parameterized using experimental and quantum chemical data of small hydrocarbons or oligomers. Compatibility comes into picture when using different parameter sets for simulations of mixtures of different classes of molecules. For instance, in simulations of polymers with drug molecules [70], a single parameter set may not be appropriate for both polymer and drug chemistries. In this case, the interaction energy between the polymer and drug molecule is derived by the use of mixing rules applied on different parameter sets.

2.4 Atom Typing and Molecular Topology

One of the important features of the force field is 'atom typing' [71], which refers to several definitions of the same atom that differentiates its hybridization state within a molecule and its local environment. For example, an sp^3 carbon atom that is forming single bonds with four atoms is typed differently from an sp^2 carbon atom forming a double bond and an *sp* carbon atom either doubly bonded to two atoms or forming a triple bond. Similarly, a carbon atom connected to an oxygen atom is often typed different from a carbon atom connected to a hydrogen atom. After selection of molecular structure and force field parameter set to be used for the simulation, one needs to first find appropriate atom types of the chosen force field of all constituent atoms and then define 'molecular topology' [72, 73], which consists of the bonded and nonbonded parameters as described in Fig. 3 and Eq. 5. Since the force field parameters of long polymers are generally not available, the force field parameters of the monomer (or parts of monomer) are generally used while developing the atom typing and molecular topology of polymer chains. This however results in some technical difficulties as exemplified in Fig. 5 for the case of cellulose that is comprised of glucose units [74]. First, an additional glycosidic linkage is present in cellulose at the expense of terminal hydrogen atoms of a glucose molecule. The parameters of the oxygen atom forming this glycosidic linkage cannot be considered the same as that present in the terminal OH group of glucose, as it now connects two carbon atoms. Also, glycosidic linkage results in two new C–O bonds, and several new angles and dihedrals consisting of the oxygen atom of the glycosidic linkage. These parameters are usually borrowed from similar atoms/bonds/angles/dihedrals present in other molecules that are parameterized using the same parameter set. The consequences of an inappropriate choice of such missing parameters must not be underestimated. In this case, the ability of cellulose to change its configuration depends mainly on the rotation around the glycosidic linkage, which therefore strongly influences chain characteristics such as the radius of gyration [74]. Second, end groups of the polymer

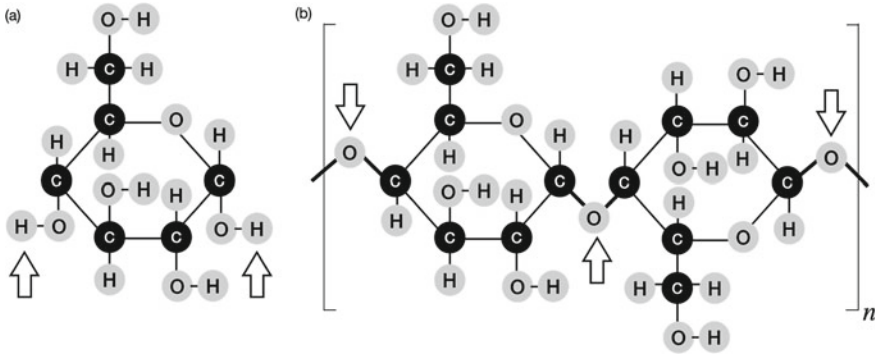


Fig. 5 **a** Glucose molecule and **b** repeat unit of cellulose. The hydrogen atoms of glucose labeled by arrows in **a** are removed to form the glycosidic linkages labeled by arrows in **b**

chain must complete valence, i.e., the end oxygen should be connected to a hydrogen or some other atom/group. Although end groups are always present in polymers, it has a limited consequence for realistic, large molecular weight polymers. For atomistic simulations, since the number of repeat units simulated is small, the end group effect is quite substantial. It is therefore suggested to perform simulations for a different number of repeat units and different possible choices of end group to decipher the end group effect on chain properties. Third, the partial charges of all atoms in cellulose cannot be taken the same as that in glucose molecule because the change in partial charge by hydrogen removal of glucose is not completely compensated by the glycosidic linkage and the effect is felt over the other atoms as well. It is therefore desired to redistribute charges over some or all of the atoms in order to maintain overall charge neutrality. Since the electrostatic forces dictate the polymer interactions with other molecules (e.g., water), improper partial charge assignment would result in spurious effects on such interactions. One promising strategy for partial charge assignment is to perform quantum chemistry calculations of longer oligomers to compute the partial charges of atoms, which would better mimic the polymer chain behavior. Note that the partial charges of the original parameter set are fitted together with various other bonded and nonbonded parameters in Eq. 5 during the actual development of the force field. Thus, it is difficult to quantify the actual benefit of using the partial charges computed from quantum chemistry calculations, along with all other parameters of the original parameter set. Though we have considered an example case of cellulose, the above mentioned three difficulties are almost always present in all atomistic simulation studies of polymers.

2.5 Time and Length Scales

The time step of atomistic MD simulations of polymers is severely limited by high frequency bond oscillations [24]. When using AA force fields, a time step of 1 fs or lesser is typically recommended. In the case of the UA force field, since C–H bonds are not considered, up to 2 fs time step may be used. At the same time, several studies have shown no significant differences between the results obtained using AA and UA force fields [43, 75]. Even higher time steps may be used during the simulation by the use of bond and angle constraints that fixes the bond length ($b = b_0$) and angle ($\theta = \theta_0$), thus disallowing high frequency bond oscillations. This is particularly useful to capture long-time dynamics of polymer chains, given that the overall relaxation of the chain is not significantly affected by these oscillations. LINCS [76], RATTLE [77], SETTLE [78], and SHAKE [79] algorithms have been used to enforce these constraints. A related issue is limitations on the chain lengths that can be simulated, as the chain relaxation slows down with the increase in molecular weight. Further limitations on chain lengths arise from the use of periodic boundary condition [80], as polymer contour length L_c should not exceed the simulation box size L (Fig. 6). It is worth pointing out that particle-image interactions may be negligible even for the $L < L_c$ case if the end-to-end distance of the chain configuration $R_e < L_c$. However,

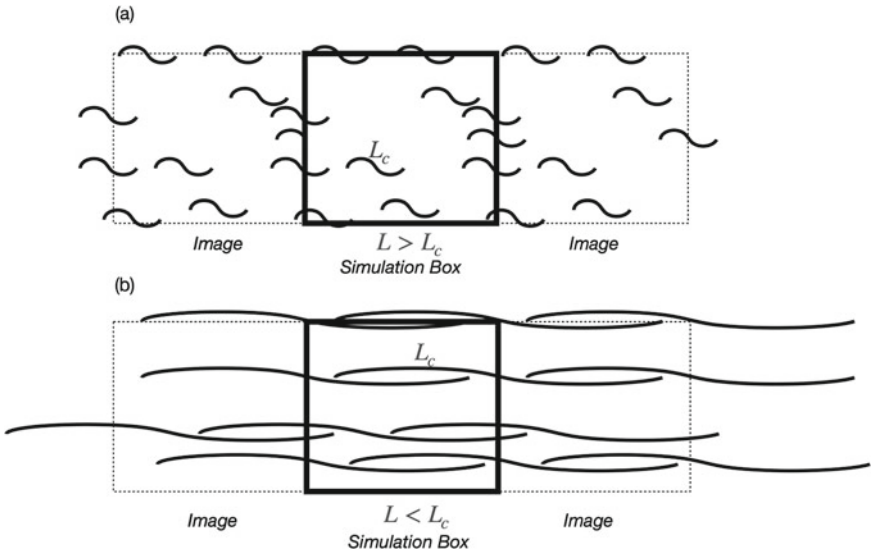


Fig. 6 Representation of simulation box and two of its images containing polymer chains. For simplicity in representation, images in other directions and different configurations of polymer chains are not shown. **a**, **b** show the cases when the simulation box size L is larger and smaller than the chain contour length L, respectively. Case **b** is not appropriate as the chains are too close and thus strongly interact with their images

since polymer chains would adopt different configurations, it is important to apply the condition on the maximum possible end-to-end length of the chain that is same as the chain contour length. Nonetheless, this criterion is not strictly followed in many reported studies as the chain end-to-end lengths commonly observed during simulations is substantially smaller than its contour length.

Polymer simulations are typically performed either in a liquid (solution) or a solid state (amorphous or crystalline). In simulations of polymer solutions, single chain simulations are sometimes performed to analyze polymer structure in dilute solution and the solvation behavior of the polymer. In this case, the major limitation on the simulation box size comes from the very large number of solvent molecules, $N_s \gg 1$, that are required to solvate the polymer chain. Most of the computational effort in these simulations is wasted in the movements of these solvent molecules. Thus, there is no significant computational advantage of the UA force field over the AA force field. CG simulations that do not treat solvent molecules explicitly are much more efficient than atomistic simulations of AA/UA force field. However, CG simulations do not capture the polymer–solvent interactions and thus have limited applicability. Simulations of multiple polymer chains along with solvent molecules are required to analyze the behavior in the semidilute limit, where again the number of solvent molecules is quite large and similar limitations apply. UA force fields are especially advantageous in simulations of amorphous and crystalline states of polymers [81], where multiple chains can be simulated without additional solvent molecules.

3 Atomistic Simulation Studies of Polymers

In the following, we sample some of the successful applications of atomistic simulations in polymeric systems, with a particular emphasis on the force fields employed in the studies. Unless otherwise mentioned, we will discuss atomistic MD studies. Also, for the sake of brevity, only the force field family is mentioned in the following and the name of parameter sets is not mentioned. Readers are referred to original articles to find further details on the force field parameter sets and the precise simulation methodology being employed.

One of the simplest applications of atomistic simulations is to find the characteristics of a single chain in a solution. Lee et al. [82] have used a revision of the CHARMM force field to simulate single chains containing 9, 18, 27 and 36 repeat units of polyethylene oxide (PEO) and polyethylene glycol (PEG) in water. Persistence length determined by simulations (3.7 Å for both PEO and PEG) was found to be in excellent agreement with the experimentally determined value (3.7 Å for PEO and 3.8 Å for PEG). Also, the simulations confirmed the ideal chain scaling obtained in experiments. Sulatha and Natarajan [83] compared the performance of two UA force fields, GROMOS and ENCAD, in the analysis of conformational and hydration behavior of polyacrylic acid (PAA) and polymethyl acrylate (PMA) containing 20 repeat units. Simulations are performed for different degrees of deprotonations of the COOH units along the chain, i.e., varying number of COO^- units. Simulations showed a stretching of chains with an increase in the degree of deprotonation for both force fields. However, the GROMOS force field was better able to distinguish between the behavior of PAA and PMA chains. Yao et al. [84] have analyzed in detail the effects of chain size, tacticity, and degree of deprotonation on the conformational and hydration behavior of PAA chains of up to 100 repeat units using the GROMOS force field. They demonstrated the collapse of long PAA chains by the addition of salt due to the electrostatic screening effect. Park et al. [85] have used Hamiltonian replica exchange molecular dynamics (HREMD) for efficient sampling of configurations of short chains of polystyrene-co-styrene sulfonate using CHARMM force field. Min et al. [86] studied the coil to globule transition of poly-2-dimethylaminoethyl methacrylate (PDMAEMA) containing 30 repeat units by simulations performed using the PCFF force field. They analyzed the hydrogen bonding pattern and the motion of water molecules in the vicinity of the polymer at different temperatures and degrees of deprotonations. Chiessi and Paradossi [87] studied the effect of dyad composition on the conformational properties of poly-N-isopropylacrylamide (PNIPAM) at different temperatures. They used an OPLS-AA force field for polymer and TIP4P/2005 model for water. Rukmani et al. [88] performed a detailed comparison of three polymer force fields (DREIDING, GAFF, and GAFF2) and five water models (SPC, SPC/E, TIP3P, TIP4P, and TIP4P/2005) in simulations of two polymers: polyacrylamide (PAM) and poly-N-isopropylacrylamide (PNIPAAm). DREIDING force field was reported to be best performing among the studied force fields. They also observed that the 4-site water models provided better agreement with experimentally observed R_g values. Zhang

et al. [89] used AMBER force field to determine the scattering behavior of a bottle-brush polymers comprised of polynorbornene (PNB) backbone and different sizes of polylactide (PLA) side chains. The simulation results showed good agreement with small angle neutron scattering (SANS) data, despite the fact that simulation time scale was much shorter than the experimentally determined relaxation time.

Single chain simulations in the solvent discussed above do not consider the inter-chain interaction between polymers and therefore represent the behavior of polymer solutions at very low concentrations. Multiple polymer chains in water (or other solvent) need to be simulated to get a detailed effect of polymer concentration. Katiyar and Jha [45] have used the CHARMM force field to study the pH-induced aggregation behavior of PAA as a function of PAA concentration and the degree of deprotonation. Similar strategies have been used to study the complexation of polymers with drug molecules [70, 90]. Jha et al. [91] used the CHARMM force field to study the ionic complexation of PAA and PDMAEMA and the salt-induced dissolution of complexes by simulations of two polyelectrolyte chains (one each) in water. Hoda and Larson [92] have performed detailed atomistic simulations using a UA force field to compare the ionic complexation phenomena in two systems: (1) PAA and polyallylamine hydrochloride (PAH) and (2) polystyrene sulfonate (PSS) and PAH. Going one step further, Sánchez et al. [93] have simulated the layer-by-layer assembly of polydiallyl dimethyl ammonium chloride (PDADMAC) and PSS using the OPLS-AA force field. Li et al. [94] simulated marine gels represented as anionic polysaccharide assemblies by considering polysaccharides of 3 repeat units using GLYCAM force field and TIP3P model for water. Effect of calcium and sodium ions on marine gels are incorporated using AMBER force field. It was demonstrated that the ionic complexation between divalent calcium ions and anionic polysaccharide chains drives the formation of big polysaccharide assemblies, which is not obtained when monovalent sodium ions are used. Morrow et al. [95] performed constant pH MD using CHARMM force field for studying pH-induced self-assembly of chitosan. Hamed et al. [96] studied the conformations of multiple PEG chains conjugated to peptide assembly using the CHARMM force field and observed that PEG forms a mushroom-like shell around the peptide helix bundle. Liu et al. [97] studied the lower critical solution temperature (LCST) transition of poly-N-isopropylacrylamide (pNIPAAm) and poly-NIPAAm-co-AAc-co-tBAAm polymers using COMPASS force field and found good agreement with experimental results. Yin et al. [98] simulated polyvinyl formal (PVF) with different water content using COMPASS force field and studied the effect of water content on cohesive energy density and water diffusivity through PVF.

Apart from solution behavior, atomistic simulations have also been used to predict miscibility and interfacial characteristics of polymer blends. Martinez de Arenaza et al. [99] computed the solubility parameter δ and Flory–Huggins interaction parameter χ for blends of poly-L-lactide (PLLA) and polyvinylphenol (PVPh) using COMPASS force field. Calculated values of solubility parameter showed good agreement with experimentally predicted values. Prathab et al. [100, 101] have used a similar approach to compute solubility parameters of many polymers and copolymers. Natarajan et al. [102] studied interfacial characteristics of the interface

between atactic polypropylene (PP) and poly-1,4-cis-butadiene (PBD) by simulations performed using the DREIDING force field. Luchinsky et al. [103] have developed an atomistic model of thermal welding by simulating interfaces formed by amorphous mixtures of polyetherimide (PEI) and polycarbonate (PC). By atomistic simulations performed using the DREIDING force field, they analyzed interfacial diffusion and strengthening process pertaining to the 'wetting' phenomena occurring during welding, which refers to the closing of the gap between interfaces.

Several atomistic simulation studies have focused on finding mechanical properties and glass transition behavior [104] of homopolymers [105–109], block copolymers [110], covalently crosslinked polymer networks (e.g., epoxy resins [111–115]), and composites [116–120]. Xiang et al. [110] studied the mechanical and glass transition behavior of PLA-PEG block copolymer using the COMPASS force field. The calculated glass transition temperature showed excellent agreement with experimental values. Sharma et al. [121] have used large scale atomistic simulations to analyze the glass transition temperature of cis and trans 1,4-polybutadiene (PB) and 1,4-polyisoprene (PI). They performed a detailed validation of existing AA and UA force fields of rubber and have done further force field refinement using quantum chemistry calculations, which resulted in an improved agreement with experiments. Masoumi and Valipour [122] have studied the effect of moisture on the glass transition temperature and Young's modulus of epoxy resins using the COMPASS force field. Bohlén and Bolton [119] used the COMPASS force field to determine the effect of a single wall carbon nanotube (SWCNT) on Young's modulus, bulk and shear modulus, pullout energy, pullout force, interfacial shear stress and interfacial bonding energy of polyvinylidene fluoride. Similar strategy has been used to understand the effect of SWCNT on binaphthyl core-based chiral phenylene dendrimers [123] and the effect of fullerene and carbon onion on poly methyl methacrylate (PMMA) [116]. Lempesis et al. [124] used nonequilibrium MD simulations of a lamellar stock model to study the mechanical response of thermoplastic polyurethanes comprising of 4,4′-diphenylmethane diisocyanate and *n*-butanediol (hard segment) and polytetramethylene oxide (soft segment). OPLS-UA force field was used for the hard segments and TraPPE-UA force field was used for the soft segments.

In contrast to studies focusing on the solution or amorphous state of polymers, the simulations of highly ordered polymer solids pose several challenges with regard to the choice of force field creation of initial geometry [125], i.e., creation of an ordered packing of chains. Moreover, crystal nucleation is a rare event as the molecules need to orient in a very specific way to form an ordered arrangement and is therefore very difficult to simulate using atomistic simulations. Rutledge and Suter [126] developed a simulation methodology and force field parameters for simulations of the solid state of the rigid-rod polymer poly-p-phenylene terephthalamide. More recently, the Rutledge group has been able to simulate the homogeneous crystal nucleation of polyethylene [81] using a UA force field. It is relatively easier to simulate the adsorption of polymers on a crystalline surface, as has been demonstrated in studies of polymer interactions with drug crystals using COMPASS force field [127–129]. In a similar vein, Daoulas et al. [130] have studied the conformational properties of a thin film of polyethylene melt adsorbed on a graphite surface using a UA force field

for polymer and graphite surface represented as a set of LJ carbon atoms centered at crystalline sites. Another relatively less common but upcoming application of atomistic simulations is in the analysis of diffusion and solubility characteristics of small species (e.g., carbon dioxide) in solid membranes comprised of polymers [131–133].

4 Multiscale Simulations of Polymers

A hierarchy of simulation methods have been proposed for polymeric systems as shown in Fig. 7. Each of these methods has a range of applicability in terms of the time and length scales they can simulate. The smallest volumes that can be considered within continuum simulations (e.g., computational fluid dynamics) must still contain a thermodynamically large number of molecules for the continuum approximation to be valid, which means they cannot resolve the properties of individual molecules. This assumption is relaxed in coarse-grained simulations of polymers that resolves a polymer molecule. However, unlike continuum simulations, coarse-grained (CG) simulations are not feasible for realistic, big systems containing very large number of molecules. Field theoretical simulations (FTS), usually based on a mean field (saddle point) approximation, is an alternative to CG simulations, which also resolve individual polymer molecules but may also be applied to a relatively longer length and time scales. Atomistic simulations share the mathematical apparatus of CG simulations but include the detailed chemistry of a molecule, thus going from molecular scale to atomic scale resolution. At the same time, they are much less computationally

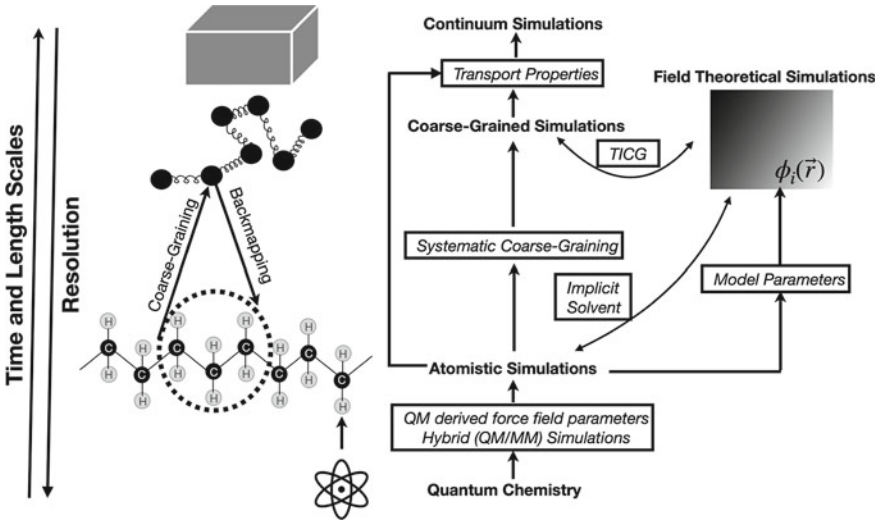


Fig. 7 Hierarchy of simulation methods and multiscale strategies

efficient than CG simulations when both of them are applicable. Finally, quantum chemistry simulations go to subatomic resolutions but is only feasible for much smaller molecules. It is worth pointing out that out of all these methods, only the quantum chemistry calculations are strictly 'first principle', i.e., they do not involve the use of model parameters that have to be separately determined from experiments or a higher resolution simulation method.

Multiscale simulation [25, 134–138] refers to the use of a combination of simulation methods applicable at two different time and length scales. Two distinct multiscale modeling paradigms have been used. The common paradigm is to use a higher resolution method to derive the model parameters of a lower resolution method and subsequently use the lower resolution method for simulations at longer length and time scales. For example, as discussed earlier, the force field parameters of atomistic simulations can be determined using quantum chemistry calculations [60]. In fact, the reliance on experimental data to develop atomistic force fields is progressively declining and newer parametrizations of atomistic force fields mostly employ quantum chemistry data. Similarly, systematic coarse-graining [19, 26, 139] may be performed using atomistic simulation data to develop CG models that preserve chemical specificity. The other paradigm is to use a hybrid simulation strategy where some part of the system (e.g., interactions at small distances) are modeled using a higher resolution scheme and the remaining system is modeled using a lower resolution scheme [140, 141]. In QM/MM simulations [142], quantum mechanical (QM) calculations are used for the inner shell of the system and molecular mechanics (MM) simulations are performed for the outer shell. Implicit solvent approaches [92, 143, 144] that treat solvent molecules in field representation may be used to speed up atomistic simulations of polymer solutions. Theoretically informed coarse-grained simulations (TICG) [145, 146] handle bonded interactions using a particle representation and nonbonded interactions using a field representation. It must be understood that none of these paradigms can preserve all features of the higher resolution method. However, one may target capturing some of the properties with a similar accuracy [147–149]. For example, in the Iterative Boltzmann Inversion (IBI) scheme of systematic coarse-graining [16, 19, 150, 151], CG interaction potentials are iteratively trained using the atomistic simulation data such that the radial distribution function obtained by CG simulations $g_{CG}(r)$ matches that obtained by atomistic simulations $g_{AA}(r)$. Alternatively, the effective forces between CG beads [152, 153] may be computed using atomistic simulations. Since fewer degrees of freedom are considered in CG simulations, a time scaling using mean square displacements is often used to match the CG time scale with atomistic time scale [154]. It may also be possible to directly compute the transport properties required in continuum simulations using atomistic simulations. For example, the penetrant diffusion coefficient obtained using atomistic simulations [155] may be directly used in continuum simulations diffusion and mass transfer through polymeric matrix.

One question that often comes to the mind of a novice researcher in this field is what level of coarse-graining is appropriate for a system? A polymer chain containing N_R repeat unit may be represented by a CG model containing $n_{CG} < N$ units with lower values of n_{CG} referring to higher levels of coarse-graining (Fig. 8). Clearly, the

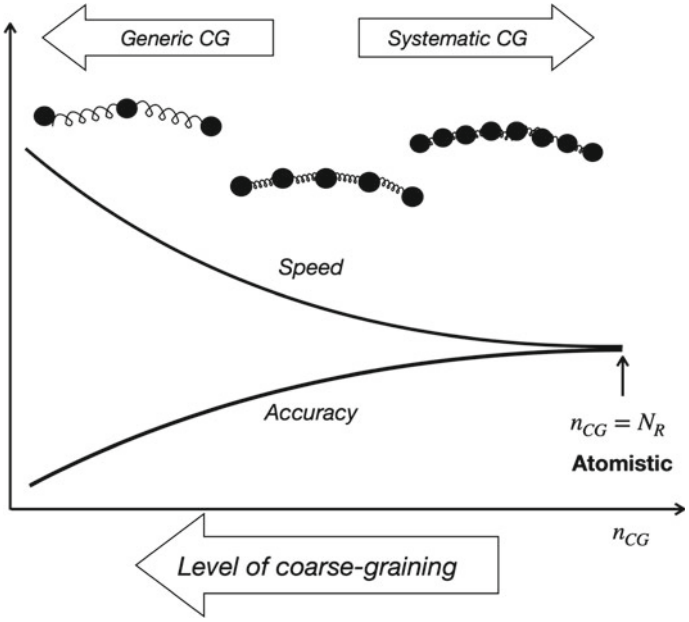


Fig. 8 Variation in speed and accuracy of CG simulations (relative to AA simulations) against the number of CG beads, n_{CG}, used to represent a polymer chain containing N_R repeat unit

relative speed (computational performance) of CG simulations with respect to atomistic simulations of the same system increases with an increase in the level of coarse-graining (decrease in n_{CG}) but the relative accuracy decreases. The optimal choice of coarse-graining level for a given application, therefore, depends on the expectations from the model. Generic CG models ('toy models') that focus on capturing the qualitative behavior of long polymer chains typically use higher levels of coarse-graining. Systematic CG models that are trained using atomistic simulations, on the other hand, typically use lower levels of coarse-graining but are limited to moderate chain lengths. Results of CG simulations may be backmapped [16, 156, 157] to atomistic resolution, i.e., the final coordinates of CG units are used to determine the atomic coordinates of the polymer. Note that the backmapping procedure must be followed by atomistic simulations using the new atomic coordinates in order to attain the local relaxation of atoms within each CG unit. One may argue that the basic premise of the CG simulations followed by backmapping is flawed as the system is evolved to equilibrium using only the CG degrees of freedoms and the local dynamics are introduced only towards the end. It is difficult to establish that the system would have a similar fate if an atomistic simulation of the same system is performed for a very long time. However, if properly validated, this strategy can be useful for overcoming the time scale limitations of atomistic simulations. Yet another promising class of CG models is the MARTINI model [158], which was originally developed for lipids and has also

been applied to polymer simulations with parameterizations performed using experimental data [159] or atomistic simulation data [160, 161]. The design philosophy of MARTINI models is to replace four heavy atoms and associated hydrogen atoms into a CG bead, which are then classified into types and subtypes based on polarity and hydrogen bonding capability, respectively. The ultimate object of these models is to bring robustness to CG models and reduce the requirements of reparameterization.

5 Summary and Future Perspectives

In this review, we have discussed the current status of polymer simulations with particular emphasis on the atomistic force fields of polymers. Multiscale simulation methods that attempt to bridge simulations at different time and length scales are also briefly discussed. As mentioned at the outset, polymers have intrigued scientists immensely in the last century for its versatile properties and rich physics. With the advent of atomistic simulations, the polymer modelers are gradually shifting their goal post from a fundamental qualitative understanding of polymer behavior to a first-principle, quantitative prediction of polymer properties and the development of actual design rules to synthesize polymers for the desired application. In this regard, one of the major bottlenecks of the existing force fields is that they are parameterized using experimentally or quantum chemistry derived parameters of small molecules. Therefore, these force fields required substantial fine tuning and validation studies in order to provide good quantitative predictions. Further, though the mathematical apparatus of atomistic simulations is quite powerful as demonstrated for a range of polymeric systems, high-resolution simulations are computationally demanding and often unable to reach the relevant length and time scales. United Atom force fields that do not explicitly treat the hydrogens attached to aliphatic carbon atoms have been particularly successful in systems devoid of solvent. The other limitation in atomistic simulations is that actual chain relaxation times may be much larger than the feasible simulation times. This is where multiscale bridging methods that connect methods at a different time and length scales and machine learning strategies become useful.

In our personal judgement, the use of CG models in polymer science is going to continue, despite the advancements in atomistic simulation methods and computational hardware. This is because of the scientific inertia and sheer breadth of the qualitative behavior that can be obtained from these simulations with limited computational power. Also, there is still a huge gap between the polymer molecular weights that can be simulated using atomistic simulations and the molecular weights of realistic polymers. However, the use of atomistic simulations to directly train a continuum simulation is also being tried. Yet another upcoming area is the use of machine learning in polymer simulations, where the machine learning models [162–165] are used to develop new force fields and directly predict polymer properties.

References

1. Geyer, R., Jambeck, J.R., Law, K.L.: Production, use, and fate of all plastics ever made. Sci. Adv. **3**, e1700782 (2021)
2. Porter, R.S., Johnson, J.F.: The entanglement concept in polymer systems. Chem. Rev. **66**, 1–27 (1966)
3. Watanabe, H.: Viscoelasticity and dynamics of entangled polymers. Prog. Polym. Sci. **24**, 1253–1403 (1999)
4. Kuo, S.-W.: Hydrogen-bonding in polymer blends. J. Polym. Res. **15**, 459–486 (2008)
5. Doi, M.: Introduction to Polymer Physics. Oxford University Press (1996)
6. Knychała, P., Timachova, K., Banaszak, M., Balsara, N.P.: 50th anniversary perspective: phase behavior of polymer solutions and blends. Macromolecules **50**, 3051–3065 (2017)
7. Nishio, I., Sun, S.-T., Swislow, G., Tanaka, T.: First observation of the coil–globule transition in a single polymer chain. Nature **281**, 208–209 (1979)
8. Angell, C.A., Sichina, W.: Thermodynamics of the glass transition: empirical aspects. Ann. N. Y. Acad. Sci. **279**, 53–67 (1976)
9. Herkt-Maetzky, C., Schelten, J.: Critical fluctuations in a binary polymer mixture. Phys. Rev. Lett. **51**, 896–899 (1983)
10. Fredrickson, G.: The equilibrium theory of inhomogeneous polymers. In: The Equilibrium Theory of Inhomogeneous Polymers. Oxford University Press (2007)
11. De Gennes, P.G.: Dynamics of entangled polymer solutions. I. The Rouse model. Macromolecules **9**, 587–593 (1976)
12. De Gennes, P.G.: Dynamics of entangled polymer solutions II. Inclusion of hydrodynamic interactions. Macromolecules **9**, 594–598 (1976)
13. Bruns, W., Bansal, R.: Molecular dynamics study of a single polymer chain in solution. II. Bead–spring model. J. Chem. Phys. **75**, 5149–5152 (1981)
14. Christainsen, R.L., Bird, R.B.: Dilute solution rheology: experimental results and finitely extensible nonlinear elastic dumbbell theory. J. Nonnewton. Fluid Mech. **3**, 161–177 (1977)
15. Ingólfsson, H.I., et al.: The power of coarse graining in biomolecular simulations. Wiley Interdiscip. Rev. Comput. Mol. Sci. **4**, 225–248 (2014)
16. Müller-Plathe, F.: Coarse-graining in polymer simulation: from the atomistic to the mesoscopic scale and back. ChemPhysChem **3**, 754–769 (2002)
17. Gartner, T.E., Jayaraman, A.: Modeling and simulations of polymers: a roadmap. Macromolecules **52**, 755–786 (2019)
18. Salerno, K.M., Bernstein, N.: Persistence length, end-to-end distance, and structure of coarse-grained polymers. J. Chem. Theory Comput. **14**, 2219–2229 (2018)
19. Huang, W., Ramesh, R., Jha, P.K., Larson, R.G.: A systematic coarse-grained model for methylcellulose polymers: spontaneous ring formation at elevated temperature. Macromolecules **49**, acs.macromol.5b02373 (2016)
20. Karatrantos, A., Clarke, N., Kröger, M.: Modeling of polymer structure and conformations in polymer nanocomposites from atomistic to mesoscale: a review. Polym. Rev. **56**, 385–428 (2016)
21. Katiyar, R.S., Jha, P.K.: Molecular simulations in drug delivery: opportunities and challenges. Wiley Interdiscip. Rev. Comput. Mol. Sci. **8** (2018)
22. Theodorou, D.N.: Progress and outlook in Monte Carlo simulations. Ind. Eng. Chem. Res. **49**, 3047–3058 (2010)
23. Allinger, N.L.: Understanding molecular structure from molecular mechanics. J. Comput. Aided. Mol. Des. **25**, 295–316 (2011)
24. Dubbeldam, D., Walton, K.S., Vlugt, T.J.H., Calero, S.: Design, parameterization, and implementation of atomic force fields for adsorption in nanoporous materials. Adv. Theory Simul. **2**, 1900135 (2019)
25. Gooneie, A., et al.: A review of multiscale computational methods in polymeric materials. Polymers (Basel) **9**, 16 (2017)

26. Brini, E., et al.: Systematic coarse-graining methods for soft matter simulations—a review. Soft Matter **9**, 2108–2119 (2013)
27. Frenkel, D., Smit, B. Understanding molecular simulation. Comput. Phys. **11**. AIP Publishing (2002)
28. Salanne, M., et al.: Including many-body effects in models for ionic liquids. Theor. Chem. Acc. **131**, 1143 (2012)
29. Allen, M.P., Tildesley, D.J.: Computer Simulation of Liquids. Oxford University Press (1989)
30. Hansen, J.-P., Verlet, L.: Phase transitions of the Lennard-Jones system. Phys. Rev. **184**, 151–161 (1969)
31. Rottler, J.: Local electrostatics algorithm for classical molecular dynamics simulations. J. Chem. Phys. **127**, 1–10 (2007)
32. Koehl, P.: Electrostatics calculations: latest methodological advances. Curr. Opin. Struct. Biol. **16**, 142–151 (2006)
33. Leach, A. R.: Molecular Modelling: Principles and Applications. Pearson Education Limited (2001)
34. Harrison, J.A., et al.: Review of force fields and intermolecular potentials used in atomistic computational materials research. Appl. Phys. Rev. **5**, 031104 (2018)
35. Barbosa, N.S.V., Zhang, Y., Lima, E.R.A., Tavares, F.W., Maginn, E.J.: Development of an AMBER-compatible transferable force field for poly(ethylene glycol) ethers (glymes). J. Mol. Model. **23**, 194 (2017)
36. Oostenbrink, C., Villa, A., Mark, A.E., Van Gunsteren, W.F.: A biomolecular force field based on the free enthalpy of hydration and solvation: the GROMOS force-field parameter sets 53A5 and 53A6. J. Comput. Chem. **25**, 1656–1676 (2004)
37. Damm, W., Frontera, A., Tirado-Rives, J., Jorgensen, W.L.: OPLS all-atom force field for carbohydrates. J. Comput. Chem. **18**, 1955–1970 (1997)
38. MacKerell, A.D., et al.: All-atom empirical potential for molecular modeling and dynamics studies of proteins. J. Phys. Chem. B **102**, 3586–3616 (1998)
39. Mayo, S.L., Olafson, B.D., Goddard, W.A.: DREIDING: a generic force field for molecular simulations. J. Phys. Chem. **94**, 8897–8909 (1990)
40. Sun, H., et al.: COMPASS II: extended coverage for polymer and drug-like molecule databases. J. Mol. Model. **22**, 47 (2016)
41. Sun, H.: Ab initio calculations and force field development for computer simulation of polysilanes. Macromolecules **28**, 701–712 (1995)
42. Al-Matar, A.K., Rockstraw, D.A.: A generating equation for mixing rules and two new mixing rules for interatomic potential energy parameters. J. Comput. Chem. **25**, 660–668 (2004)
43. da Silva, G.C.Q., Silva, G.M., Tavares, F.W., Fleming, F.P., Horta, B.A.C.: Are all-atom any better than united-atom force fields for the description of liquid properties of alkanes? J. Mol. Model. **26**, 296 (2020)
44. Jing, Z., et al.: Polarizable force fields for biomolecular simulations: recent advances and applications. Annu. Rev. Biophys. **48**, 371–394 (2019)
45. Katiyar, R.S., Jha, P.K.: Phase behavior of aqueous polyacrylic acid solutions using atomistic molecular dynamics simulations of model oligomers. Polymer (Guildf). **114**, 266–276 (2017)
46. Doherty, D. C., Holmes, B. N., Leung, P. & Ross, R. B. Polymerization molecular dynamics simulations. I. Cross-linked atomistic models for poly(methacrylate) networks. Comput. Theor. Polym. Sci. **8**, 169–178 (1998)
47. Mongan, J., Case, D.A.: Biomolecular simulations at constant pH. Curr. Opin. Struct. Biol. **15**, 157–163 (2005)
48. Donnini, S., Tegeler, F., Groenhof, G., Grubmüller, H.: Constant pH molecular dynamics in explicit solvent with λ-dynamics. J. Chem. Theory Comput. **7**, 1962–1978 (2011)
49. Kowalik, M., et al.: Atomistic scale analysis of the carbonization process for C/H/O/N-based polymers with the reaxFF reactive force field. J. Phys. Chem. B **123**, 5357–5367 (2019)
50. Onufriev, A.V., Izadi, S.: Water models for biomolecular simulations. WIREs Comput. Mol. Sci. **8**, e1347 (2018)

51. Mark, P., Nilsson, L.: Structure and dynamics of the TIP3P, SPC, and SPC/E water models at 298 K. J. Phys. Chem. A **105**, 9954–9960 (2001)
52. Abascal, J.L.F., Vega, C.: A general purpose model for the condensed phases of water: TIP4P/2005. J. Chem. Phys. **123**, 234505 (2005)
53. Ferguson, D.M.: Parameterization and evaluation of a flexible water model. J. Comput. Chem. **16**, 501–511 (1995)
54. Nada, H., van der Eerden, J.P.J.M.: An intermolecular potential model for the simulation of ice and water near the melting point: a six-site model of H_2O. J. Chem. Phys. **118**, 7401–7413 (2003)
55. Mahoney, M.W., Jorgensen, W.L.: A five-site model for liquid water and the reproduction of the density anomaly by rigid, nonpolarizable potential functions. J. Chem. Phys. **112**, 8910–8922 (2000)
56. Nerenberg, P.S., Head-Gordon, T.: New developments in force fields for biomolecular simulations. Curr. Opin. Struct. Biol. **49**, 129–138 (2018)
57. McAliley, J.H., Bruce, D.A.: Development of force field parameters for molecular simulation of polylactide. J. Chem. Theory Comput. **7**, 3756–3767 (2011)
58. Fröhlking, T., Bernetti, M., Calonaci, N., Bussi, G.: Toward empirical force fields that match experimental observables. J. Chem. Phys. **152**, 230902 (2020)
59. Monticelli, L., Tieleman, D.P.: Force fields for classical molecular dynamics. Biomol. Simul. 197–213 (2013)
60. Reith, D., Kirschner, K.N.: A modern workflow for force-field development—bridging quantum mechanics and atomistic computational models. Comput. Phys. Commun. **182**, 2184–2191 (2011)
61. Rosi-Schwartz, B., Mitchell, G.R.: Extracting force fields for disordered polymeric materials from neutron scattering data. Polymer (Guildf). **37**, 1857–1870 (1996)
62. Canongia Lopes, J.N., Pádua, A.A.H.: CL&P: a generic and systematic force field for ionic liquids modeling. Theor. Chem. Acc. **131**, 1129 (2012)
63. Satyanarayana, K.C., Bohlén, M., Lund, A., Rychwalski, R.W., Bolton, K.: Analysis of the torsion angle distribution of poly(vinylidene fluoride) in the melt. Polymer (Guildf) **53**, 1109–1114 (2012)
64. Fang, C.-E., Tsai, Y.-C., Scheurer, C., Chiu, C.-C.: Revised atomic charges for OPLS force field model of poly(ethylene oxide): benchmarks and applications in polymer electrolyte. Polymers **13** (2021)
65. Vanommeslaeghe, K., Yang, M., MacKerell, A.D.: Robustness in the fitting of molecular mechanics parameters. J. Comput. Chem. **36**, 1083–1101 (2015)
66. Hopfinger, A.J., Pearlstein, R.A.: Molecular mechanics force-field parameterization procedures. J. Comput. Chem. **5**, 486–499 (1984)
67. Rappe, A.K., Casewit, C.J., Colwell, K.S., Goddard, W.A., Skiff, W.M.: UFF, a full periodic table force field for molecular mechanics and molecular dynamics simulations. J. Am. Chem. Soc. **114**, 10024–10035 (1992)
68. Danne, R., et al.: doGlycans–tools for preparing carbohydrate structures for atomistic simulations of glycoproteins, glycolipids, and carbohydrate polymers for GROMACS. J. Chem. Inf. Model. **57**, 2401–2406 (2017)
69. Jämbeck, J.P.M., Lyubartsev, A.P.: Derivation and systematic validation of a refined all-atom force field for phosphatidylcholine lipids. J. Phys. Chem. B **116**, 3164–3179 (2012)
70. Jha, P.K., Larson, R.G.: Assessing the efficiency of polymeric excipients by atomistic molecular dynamics simulations. Mol. Pharm. **11**, 1676–1686 (2014)
71. Vanommeslaeghe, K., MacKerell, A.D.: Automation of the CHARMM general force field (CGenFF) I: bond perception and atom typing. J. Chem. Inf. Model. **52**, 3144–3154 (2012)
72. Malde, A.K., et al.: An automated force field topology builder (ATB) and repository: version 1.0. J. Chem. Theory Comput. **7**, 4026–4037 (2011)
73. Zoete, V., Cuendet, M.A., Grosdidier, A., Michielin, O.: SwissParam: a fast force field generation tool for small organic molecules. J. Comput. Chem. **32**, 2359–2368 (2011)

74. Huang, W., Dalal, I.S., Larson, R.G.: Analysis of solvation and gelation behavior of methylcellulose using atomistic molecular dynamics simulations. J. Phys. Chem. B **118**, 13992–14008 (2014)
75. Chen, C., et al.: A comparison of united atom, explicit atom, and coarse-grained simulation models for poly(ethylene oxide). J. Chem. Phys. **124**, 234901 (2006)
76. Hess, B., Bekker, H., Berendsen, H.J.C., Fraaije, J.G.E.M.: LINCS: a linear constraint solver for molecular simulations. J. Comput. Chem. **18**, 1463–1472 (1997)
77. Andersen, H.C.: Rattle: A "velocity" version of the shake algorithm for molecular dynamics calculations. J. Comput. Phys. **52**, 24–34 (1983)
78. Miyamoto, S., Kollman, P.A.: Settle: an analytical version of the SHAKE and RATTLE algorithm for rigid water models. J. Comput. Chem. **13**, 952–962 (1992)
79. Ryckaert, J.-P., Ciccotti, G., Berendsen, H.J.C.: Numerical integration of the cartesian equations of motion of a system with constraints: molecular dynamics of n-alkanes. J. Comput. Phys. **23**, 327–341 (1977)
80. Yeh, I.C., Hummer, G.: System-size dependence of diffusion coefficients and viscosities from molecular dynamics simulations with periodic boundary conditions. J. Phys. Chem. B **108**, 15873–15879 (2004)
81. Yi, P., Locker, C.R., Rutledge, G.C.: Molecular dynamics simulation of homogeneous crystal nucleation in polyethylene. Macromolecules **46**, 4723–4733 (2013)
82. Lee, H., Venable, R.M., MacKerell, A.D., Pastor, R.W.: Molecular dynamics studies of polyethylene oxide and polyethylene glycol: hydrodynamic radius and shape anisotropy. Biophys. J. **95**, 1590–1599 (2008)
83. Sulatha, M.S., Natarajan, U.: Origin of the difference in structural behavior of poly(acrylic acid) and poly(methacrylic acid) in aqueous solution discerned by explicit-solvent explicit-ion MD simulations. Ind. Eng. Chem. Res. **50**, 11785–11796 (2011)
84. Yao, G., Zhao, J., Ramisetti, S.B., Wen, D.: Atomistic molecular dynamic simulation of dilute poly(acrylic acid) solution: effects of simulation size sensitivity and ionic strength. Ind. Eng. Chem. Res. **57**, 17129–17141 (2018)
85. Park, S., Zhu, X., Yethiraj, A.: Atomistic simulations of dilute polyelectrolyte solutions. J. Phys. Chem. B **116**, 4319–4327 (2012)
86. Min, S.H., Kwak, S.K., Kim, B.-S.: Atomistic simulation for coil-to-globule transition of poly(2-dimethylaminoethyl methacrylate). Soft Matter **11**, 2423–2433 (2015)
87. Chiessi, E., Paradossi, G.: Influence of tacticity on hydrophobicity of poly(N-isopropylacrylamide): a single chain molecular dynamics simulation study. J. Phys. Chem. B **120**, 3765–3776 (2016)
88. Rukmani, S.J., Kupgan, G., Anstine, D.M., Colina, C.M.: A molecular dynamics study of water-soluble polymers: analysis of force fields from atomistic simulations. Mol. Simul. **45**, 310–321 (2019)
89. Zhang, Z., et al.: Atomistic structure of bottlebrush polymers: simulations and neutron scattering studies. Macromolecules **47**, 5808–5814 (2014)
90. Katiyar, R.S., Jha, P.K.: Molecular insights into the effects of media-drug and carrier-drug interactions on pH-responsive drug carriers. Mol. Pharm. **15**, 2479–2483 (2018)
91. Jha, P.K., Desai, P.S., Li, J., Larson, R.G.: pH and salt effects on the associative phase separation of oppositely charged polyelectrolytes. Polymers (Basel). **6**, 1414–1436 (2014)
92. Hoda, N., Larson, R.G.: Explicit- and implicit-solvent molecular dynamics simulations of complex formation between polycations and polyanions. Macromolecules **42**, 8851–8863 (2009)
93. Sánchez, P.A., Smiatek, J., Qiao, B., Sega, M., Holm, C.: Atomistic simulation of oligoelectrolyte multilayers growth. In: Nagel, W.E., Kröner, D.H., Resch, M.M. (Eds.) High Performance Computing in Science and Engineering, vol. 15, pp. 215–228. Springer International Publishing (2016)
94. Li, X., et al.: Cross-linked polysaccharide assemblies in marine gels: an atomistic simulation. J. Phys. Chem. Lett. **4**, 2637–2642 (2013)

95. Morrow, B.H., Payne, G.F., Shen, J.: pH-responsive self-assembly of polysaccharide through a rugged energy landscape. J. Am. Chem. Soc. **137**, 13024–13030 (2015)
96. Hamed, E., Ma, D., Keten, S.: Multiple PEG chains attached onto the surface of a helix bundle: conformations and implications. ACS Biomater. Sci. Eng. **1**, 79–84 (2015)
97. Liu, M.S., et al.: Conformational transitions and dynamics of thermal responsive poly(N-isopropylacrylamide) polymers as revealed by molecular simulation. Eur. Polym. J. **55**, 153–159 (2014)
98. Yin, Q., Zhang, L., Jiang, B., Yin, Q., Du, K.: Effect of water in amorphous polyvinyl formal: insights from molecular dynamics simulation. J. Mol. Model. **21**, 2 (2015)
99. Martinez de Arenaza, I., Meaurio, E., Coto, B., Sarasua, J.-R.: Molecular dynamics modelling for the analysis and prediction of miscibility in polylactide/polyvinilphenol blends. Polymer (Guildf) **51**, 4431–4438 (2010)
100. Prathab, B., Subramanian, V., Aminabhavi, T.M.: Molecular dynamics simulations to investigate polymer-polymer and polymer-metal oxide interactions. Polymer (Guildf). **48**, 409–416 (2007)
101. Prathab, B., Aminabhavi, T.M., Parthasarathi, R., Manikandan, P., Subramanian, V.: Molecular modeling and atomistic simulation strategies to determine surface properties of perfluorinated homopolymers and their random copolymers. Polymer (Guildf). **47**, 6914–6924 (2006)
102. Natarajan, U., Misra, S., Mattice, W.L.: Atomistic simulation of a polymer-polymer interface: interfacial energy and work of adhesion. Comput. Theor. Polym. Sci. **8**, 323–329 (1998)
103. Luchinsky, D.G., et al.: Welding dynamics in an atomistic model of an amorphous polymer blend with polymer–polymer interface. J. Polym. Sci. **58**, 2051–2061 (2020)
104. Barrat, J.-L., Baschnagel, J., Lyulin, A.: Molecular dynamics simulations of glassy polymers. Soft Matter **6**, 3430–3446 (2010)
105. Mishra, S., Keten, S.: Atomistic simulation based prediction of the solvent effect on the molecular mobility and glass transition of poly (methyl methacrylate). Appl. Phys. Lett. **102**, 41903 (2013)
106. Wang, X., Tang, C., Wang, Q., Li, X., Hao, J.: Selection of optimal polymerization degree and force field in the molecular dynamics simulation of insulating paper cellulose. Energies **10** (2017)
107. Saha, S., Bhowmick, A.K.: An Insight into molecular structure and properties of flexible amorphous polymers: a molecular dynamics simulation approach. J. Appl. Polym. Sci. **136**, 47457 (2019)
108. Mansfield, K.F., Theodorou, D.N.: Atomistic simulation of a glassy polymer surface. Macromolecules **23**, 4430–4445 (1990)
109. Ewen, J.P., et al. A Comparison of classical force-fields for molecular dynamics simulations of lubricants. Materials **9**, (2016)
110. Xiang, H., et al.: Study of the mechanical and thermal properties of poly(Lactic Acid) and poly(Ethylene Glycol) block copolymer with molecular dynamics. Int. J. Polym. Anal. Char. **15**, 235–244 (2010)
111. Wu, C., Xu, W.: Atomistic molecular modelling of crosslinked epoxy resin. Polymer (Guildf). **47**, 6004–6009 (2006)
112. Wu, C., Xu, W.: Atomistic molecular simulations of structure and dynamics of crosslinked epoxy resin. Polymer (Guildf). **48**, 5802–5812 (2007)
113. Masoumi, S., Arab, B., Valipour, H.: A study of thermo-mechanical properties of the cross-linked epoxy: an atomistic simulation. Polymer (Guildf) **70**, 351–360 (2015)
114. Shokuhfar, A., Arab, B.: The effect of cross linking density on the mechanical properties and structure of the epoxy polymers: molecular dynamics simulation. J. Mol. Model. **19**, 3719–3731 (2013)
115. Koo, B., Subramanian, N., Chattopadhyay, A.: Molecular dynamics study of brittle fracture in epoxy-based thermoset polymer. Compos. Part B Eng. **95**, 433–439 (2016)
116. Izadi, R., Ghavanloo, E., Nayebi, A.: Elastic properties of polymer composites reinforced with C60 fullerene and carbon onion: molecular dynamics simulation. Phys. B Condens. Matter **574**, 311636 (2019)

117. Sina, A., Nima, R.: Effects of cement-polymer interface properties on mechanical response of fiber-reinforced cement composites. J. Nanomech. Micromech. **7**, 4017002 (2017)
118. Zaminpayma, E.: Molecular dynamics simulation of mechanical properties and interaction energy of polythiophene/polyethylene/poly(p-phenylenevinylene) and CNTs composites. Polym. Compos. **35**, 2261–2268 (2014)
119. Bohlén, M., Bolton, K.: Molecular dynamics studies of the influence of single wall carbon nanotubes on the mechanical properties of Poly(vinylidene fluoride). Comput. Mater. Sci. **68**, 73–80 (2013)
120. Verma, A., Parashar, A., Packirisamy, M.: Atomistic modeling of graphene/hexagonal boron nitride polymer nanocomposites: a review. WIREs Comput. Mol. Sci. **8**, e1346 (2018)
121. Sharma, P., Roy, S., Karimi-Varzaneh, H.A.: Validation of force fields of Rubber through glass-transition temperature calculation by microsecond atomic-scale molecular dynamics simulation. J. Phys. Chem. B **120**, 1367–1379 (2016)
122. Masoumi, S., Valipour, H.: Effects of moisture exposure on the crosslinked epoxy system: an atomistic study. Model. Simul. Mater. Sci. Eng. **24**, 35011 (2016)
123. Mo, Z., Zhu, X., Wang, G., Han, W., Guo, R.: Molecular dynamics simulation on the interaction between single-walled carbon nanotubes and binaphthyl core-based chiral phenylene dendrimers. J. Mater. Res. **29**, 2156–2161 (2014)
124. Lempesis, N., in 't Veld, P.J., Rutledge, G.C.: Atomistic simulation of a thermoplastic polyurethane and micromechanical modeling. Macromolecules **50**, 7399–7409 (2017)
125. Luo, C.: A challenging topic of computer simulations: polymorphism in polymers. Polym. Cryst. **3**, e10109 (2020)
126. Rutledge, G.C., Suter, U.W.: Detailed atomistic simulation of oriented pseudocrystalline polymers and application to a stiff-chain aramid. Macromolecules **24**, 1921–1933 (1991)
127. Zhu, W., Romanski, F.S., Dalvi, S.V, Dave, R.N., Silvina Tomassone, M.: Atomistic simulations of aqueous griseofulvin crystals in the presence of individual and multiple additives. Chem. Eng. Sci. **73**, 218–230 (2012)
128. Zhu, W., Romanski, F.S., Meng, X., Mitra, S., Tomassone, M.S.: Atomistic simulation study of surfactant and polymer interactions on the surface of a fenofibrate crystal. Eur. J. Pharm. Sci. **42**, 452–461 (2011)
129. Gupta, K.M., Yani, Y., Poornachary, S.K., Chow, P.S.: Atomistic simulation to understand anisotropic growth behavior of Naproxen crystal in the presence of polymeric additives. Cryst. Growth Des. **19**, 3768–3776 (2019)
130. Daoulas, K.C., Harmandaris, V.A., Mavrantzas, V.G.: Detailed atomistic simulation of a polymer melt/solid interface: structure, density, and conformation of a thin film of polyethylene melt adsorbed on graphite. Macromolecules **38**, 5780–5795 (2005)
131. Zhang, L., Hu, Z., Jiang, J.: Metal-organic framework/polymer mixed-matrix membranes for H_2/CO_2 separation: a fully atomistic simulation study. J. Phys. Chem. C **116**, 19268–19277 (2012)
132. Hong, B., Panagiotopoulos, A.Z.: Atomistic simulation of CO_2 solubility in poly(ethylene oxide) oligomers. Mol. Phys. **112**, 1540–1547 (2014)
133. Kim, D.J., Park, C.H., Tocci, E., Nam, S.Y.: Experimental and modeling study of blended membranes for direct methanol fuel cells. J. Memb. Sci. **564**, 308–316 (2018)
134. Glotzer, S.C., Paul, W.: Molecular and mesoscale simulation methods for polymer materials. Annu. Rev. Mater. Res. **32**, 401–436 (2002)
135. de Pablo, J.J.: Coarse-grained simulations of macromolecules: from DNA to nanocomposites. Annu. Rev. Phys. Chem. **62**, 555–574 (2011)
136. Peter, C., Kremer, K.: Multiscale simulation of soft matter systems—from the atomistic to the coarse-grained level and back. Soft Matter **5**, 4357 (2009)
137. Mushrif, S.H., Vasudevan, V., Krishnamurthy, C.B., Venkatesh, B.: Multiscale molecular modeling can be an effective tool to aid the development of biomass conversion technology: a perspective. Chem. Eng. Sci. **121**, 217–235 (2015)
138. Praprotnik, M., Site, L.D., Kremer, K.: Multiscale simulation of soft matter: from scale bridging to adaptive resolution. Annu. Rev. Phys. Chem. **59**, 545–571 (2008)

139. Öttinger, H.: Systematic coarse graining: “four lessons and a caveat” from nonequilibrium statistical mechanics. MRS Bull. **32**, 936–940 (2007)
140. Wang, Y., Li, Z., Xu, J., Yang, C., Karniadakis, G.E.: Concurrent coupling of atomistic simulation and mesoscopic hydrodynamics for flows over soft multi-functional surfaces. Soft Matter **15**, 1747–1757 (2019)
141. Rzepiela, A.J., Louhivuori, M., Peter, C., Marrink, S.J.: Hybrid simulations: combining atomistic and coarse-grained force fields using virtual sites. Phys. Chem. Chem. Phys. **13**, 10437–10448 (2011)
142. Gołębiowski, J.R., Kermode, J.R., Haynes, P.D., Mostofi, A.A.: Atomistic QM/MM simulations of the strength of covalent interfaces in carbon nanotube–polymer composites. Phys. Chem. Chem. Phys. **22**, 12007–12014 (2020)
143. Carrillo, J.M.Y., Dobrynin, A.V.: Detailed molecular dynamics simulations of a model NaPPS in water. J. Phys. Chem. B **114**, 9391–9399 (2010)
144. Arnarez, C., et al.: Dry martini, a coarse-grained force field for lipid membrane simulations with implicit solvent. J. Chem. Theory Comput. **11**, 260–275 (2015)
145. Pike, D.Q., Detcheverry, F.A., Müller, M., De Pablo, J.J.: Theoretically informed coarse grain simulations of polymeric systems. J. Chem. Phys. **131**, 1–10 (2009)
146. Jha, P.K., Zwanikken, J.W., Detcheverry, F.A., de Pablo, J.J., Olvera de la Cruz, M.: Study of volume phase transitions in polymeric nanogels by theoretically informed coarse-grained simulations. Soft Matter **7**, 5965–5975 (2011)
147. Ayton, G., Voth, G.A.: Bridging microscopic and mesoscopic simulations of lipid bilayers. Biophys. J. **83**, 3357–3370 (2002)
148. McCarty, J., Clark, A.J., Copperman, J., Guenza, M.G.: An analytical coarse-graining method which preserves the free energy, structural correlations, and thermodynamic state of polymer melts from the atomistic to the mesoscale. J. Chem. Phys. **140**, 204913 (2014)
149. Ayton, G., Smondyrev, A.M., Bardenhagen, S.G., McMurtry, P., Voth, G.A.: Interfacing molecular dynamics and macro-scale simulations for lipid bilayer vesicles. Biophys. J. **83**, 1026–1038 (2002)
150. Li, X., Kou, D., Rao, S., Liang, H.: Developing a coarse-grained force field for the diblock copolymer poly(styrene-b-butadiene) from atomistic simulation. J. Chem. Phys. **124**, 204909 (2006)
151. Fukuda, M., Zhang, H., Ishiguro, T., Fukuzawa, K., Itoh, S.: Structure-based coarse-graining for inhomogeneous liquid polymer systems. J. Chem. Phys. **139** (2013)
152. Fu, Y., et al.: Molecular dynamics and dissipative particle dynamics simulations for prediction of miscibility in polyethylene terephthalate/polylactide blends. Mol. Simul. **39**, 415–422 (2013)
153. Scocchi, G., Posocco, P., Fermeglia, M., Pricl, S.: Polymer−clay nanocomposites: a multiscale molecular modeling approach. J. Phys. Chem. B **111**, 2143–2151 (2007)
154. Depa, P., Chen, C., Maranas, J.K.: Why are coarse-grained force fields too fast? a look at dynamics of four coarse-grained polymers. J. Chem. Phys. **134**, 14903 (2011)
155. Katiyar, R.S., Jha, P.K.: Mimicking the dissolution mechanisms of pH-responsive drug release formulations in atomistic MD simulations. Adv. Theory Simul. **2**, 1900053 (2019)
156. Mackenzie, R., Booth, J., Alexander, C., Garnett, M.C., Laughton, C.A.: Multiscale modeling of drug-polymer nanoparticle assembly identifies parameters influencing drug encapsulation efficiency. J. Chem. Theory Comput. **11**, 2705–2713 (2015)
157. Glagolev, M.K., Lazutin, A.A., Vasilevskaya, V.V.: Macroscopic properties of hypercrosslinked polystyrene networks: an atomistic and coarse-grained molecular dynamics simulation. Macromol. Symp. **348**, 14–24 (2015)
158. Marrink, S.J., Risselada, H.J., Yefimov, S., Tieleman, D.P., Vries, A.H.D.: Coarse grained model for biomolecular simulations: the MARTINI force field. J. Phys. Chem. B **111**, 7812–7824 (2007)
159. Rossi, G., Monticelli, L., Puisto, S.R., Vattulainen, I., Ala-Nissila, T.: Coarse-graining polymers with the MARTINI force-field: polystyrene as a benchmark case. Soft Matter **7**, 698–708 (2011)

160. Grunewald, F., Rossi, G., de Vries, A.H., Marrink, S.J., Monticelli, L.: Transferable MARTINI model of poly(ethylene Oxide). J. Phys. Chem. B **122**, 7436–7449 (2018)
161. Panizon, E., Bochicchio, D., Monticelli, L., Rossi, G.: MARTINI coarse-grained models of polyethylene and polypropylene. J. Phys. Chem. B **119**, 8209–8216 (2015)
162. Mannodi-Kanakkithodi, A., Pilania, G., Huan, T.D., Lookman, T., Ramprasad, R.: Machine learning strategy for accelerated design of polymer dielectrics. Sci. Rep. **6**, 20952 (2016)
163. Wei, Q., Melko, R.G., Chen, J.Z.Y.: Identifying polymer states by machine learning. Phys. Rev. E **95**, 32504 (2017)
164. Kim, C., Batra, R., Chen, L., Tran, H., Ramprasad, R.: Polymer design using genetic algorithm and machine learning. Comput. Mater. Sci. **186**, 110067 (2021)
165. Sha, W., et al.: Machine learning in polymer informatics. InfoMat **3**, 353–361 (2021)

Forcefields and Modeling of Polymer Coatings and Nanocomposites



Uday Shankar, Sushanta K. Sethi, and Akarsh Verma

Abstract Since the discovery of molecular dynamics (MD) simulations, their usage benefited a broad range of researchers and scientists in applied physics, biomaterials, and material science, owing to its greater accuracy and savings in materials wastage. In MD simulations, the interactions among different atoms or molecules present in a system are defined by a so-called forcefield (FF). The past decades have proven that the MD simulations that have carefully employed such FFs can provide useful insights into many biological or chemical questions. This study focuses on a few recently developed and mostly used FFs that researchers employed to gain an atomic-level description. Besides the FF definition, the article discusses the terms composing a classical FF, the optimized potential parameters, the most widely used potentials to predict interactions, and the lines of research to improve them. This article is focused on readers working especially in polymeric coatings and nanocomposites.

1 Introduction

The first MD and Monte Carlo (MC) simulations of model liquids, undertaken by Alder and Wainwright [1] and Metropolis et al. [2], respectively, originate in the 1950s. Since then, fast technological advancements have led to this field's massive growth, and MD simulations are now being used in many academic areas. Moreover,

U. Shankar
Department of Polymer and Process Engineering, IIT Roorkee, Roorkee, India

S. K. Sethi (✉)
Department of Metallurgical Engineering and Materials Science, IIT Bombay, Mumbai, India
e-mail: sksethi@iitb.ac.in; ssethi@pe.iitr.ac.in

A. Verma
Department of Mechanical Engineering, University of Petroleum and Energy Studies, Dehradun 248007, India

Department of Mechanical Science and Bioengineering, Osaka University, Osaka 560-8531, Japan

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their application is no longer limited to professionals; many researchers currently employ computer simulations to evaluate and validate their data when it is too complicated to be represented by basic analytical models. That's also relevant when it comes to neutron scattering. The practical measurements in this example correspond directly with the characteristics acquired from computations. The spatial and time scales that can be predicted closely align with those which can be computed. Because of these characteristics, including the accessibility of trustworthy and straightforward software programs, this form of computation, display, and analysis of findings has become commonplace among neutron users.

MD and MC simulations relying on well-designed force fields (FFs) can help to validate and anticipate the characteristics of many coating-related uses. However, utmost attempts to create FFs for this group of materials have primarily centered on calculating molecular structures and intramolecular interactions rather than densities or lattice energies. As a result, we have already begun research on an all-atom FF for MD and MC simulations of polymeric materials, which will be employed in the coatings industry.

It is essential to keep in mind that the FF was produced and established using practical estimations of the enthalpies of vaporization (ΔvapHm°) or sublimation (ΔsubHm°) of the examined chemicals, with considerable consideration of phase purity of solid samples. Undoubtedly, while the structural characteristics of the FF can be analyzed using a sizeable structural database, the identical cannot be valid for the energetic part since dependable ΔsubHm° and ΔvapHm° data, which can be used as measurements of the condensed phase cohesion energy, are scant. Some available ΔsubHm° values can be reliably designated to a specific crystal structure. However, these ab initio approaches are frequently excessively costly due to the required spatial and/or time scales. In these instances, we must resort to empirical force field (FF)-based techniques and apply a greater level of approximation. They make it possible to simulate systems with large numbers of atoms in fractions of a second. Instead, FF value should be found through experimentation. Neutron scattering is used to validate the technique once again, and various experimental data can be considered in order to validate the FF parameters.

2 Forcefields (FFs) and MD Simulations

Lipids, proteins, nucleic acids, carbohydrates, and small organic molecules can all be studied using conventional MD simulations. Because of the size limitation of the selected molecules and the time scales of the considered processes to simulate the system at an atomic level, each atom present in a molecule is represented by a point (with a pre-defined coordinate) in space with mass and partial charge. As a result, the electrons are not taken into account expressly, and the molecule is believed to be in its electronic ground state. Here, Newton's equations of motion can describe the system's dynamics. A potential-energy function of the atomic coordinates, i.e., FF, is used to define the interactions between the atoms in the system, as stated by Levitt and Lifson in 1969:

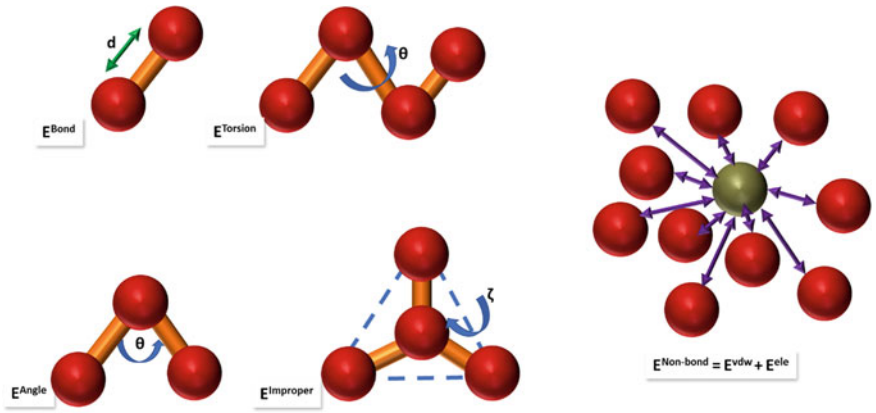


Fig. 1 Schematic representation of different energy terms in a classical fixed-charge forcefield. E^{Bond}, $E^{Torsion}$, E^{Angle}, $E^{Improper}$, E^{vdW}, and E^{ele} represent the bond stretching, bond-angle bending, dihedral angle torsion, improper dihedral angle bending, vdW, and interactions energies, respectively

$$E^{\mathrm{Pot}}\left(\overrightarrow{r}\right) = E^{\mathrm{Bond}}\left(\overrightarrow{r}\right) + E^{\mathrm{Angle}}\left(\overrightarrow{r}\right) + E^{\mathrm{Torsion}}\left(\overrightarrow{r}\right) + E^{\mathrm{Improper}}\left(\overrightarrow{r}\right) + E^{\mathrm{ele}}\left(\overrightarrow{r}\right) + E^{vdW}\left(\overrightarrow{r}\right) \quad (1)$$

where E_{pot} and $\overrightarrow{r}$ represent the potential-energy function and the coordinates of all atoms in the system, respectively. E_{bond}, E_{angle}, $E_{torsion}$, and $E_{improper}$ refer to the bond stretching, bond-angle bending, dihedral-angle torsion, and improper dihedral-angle bending that occurs in a molecule during the simulation process. E_{ele} and E_{vdW} are defined as the nonbonded energy terms of electrostatic and vdW interactions among the atoms. The FF terms are schematically represented in Fig. 1. The previous editions of the FFs included added H-bond interactions that were quickly eliminated. Despite the significance of polarization effects, partial charges of the atoms have generally been fixed throughout computations because of the high computational cost and additional complexities that are associated with polarizable FFs. Due to the remarkable success of fixed-charge FFs, we will concentrate on this class of FFs in the following sections and provide an insight into the main forcefield families, covering parametrization methodologies and historical evolution. Detailed studies of different FFs can be found in refs [3–7].

2.1 Intramolecular Terms

The bond stretching is described using a harmonic function, which regulates the covalent bond's length. The r0 values can be predicted from X-ray diffraction analysis, whereas infrared or Raman spectra can help to evaluate the spring constant.

The harmonic potential is inadequate for more than 10% bond shifts from the equilibrium value. Furthermore, the employment of the harmonic equation indicates that the bond cannot be fragmented, limiting the study of chemical reactions. It is the most significant disadvantage of FF-based MD simulations over ab initio MD. Sometimes, distinct functional forms (especially the Morse potential) can increase accuracy. Nonetheless, as those models take longer to compute and because the harmonic approximation is appropriate in most cases, most present possibilities choose a simpler harmonic equation. A harmonic potential is also commonly used to illustrate angle bending, while in a few instances, a trigonometric potential is recommended:

$$U_{\text{Bending}} = \frac{1}{2} K_a (\cos\theta - \cos\theta_0)^2 \tag{2}$$

Various parameters are typically applied to improve the match to vibrational spectra. Very often employed addition is the Urey-Bradley potential.

$$U_{UB} = \sum_{\text{Angles}} \frac{1}{2} K_{UB} (\text{s} - s_0)^2 \tag{3}$$

Two external atoms separated by a distance, denoted by s, make up the angle.

A torsional or dihedral keyword must be inserted if there is a presence of more than four atoms in a molecule. Although angle bending and bond stretching are high-frequency movements, which can be substituted with a stiff approximation because they are typically unimportant, torsional movements are less rigid than bond stretching movements. Yet, they must maintain the molecule's stiffness and repeat the substantial conformational changes caused by bond rotations. As a result, they're vital in evaluating a macromolecule's local structure or the relative stability of various molecular conformations. Figures 1 and 2 depict the description of a dihedral angle and the accompanying torsional potential in the 1,2-dichloroethane molecule. The torsional energy is commonly expressed by a cosine function, where the phase is the torsional angle, and n is the number of minima or maxima between 0 and $2Vn$ of the potential barrier. It is possible to generate a dihedral potential with distinct depth minima by blending two or more terms with different n. However, there are other ways to describe the dihedral potential in the research. For instance, the OPLS potential adopts the below equation:

$$U_{\text{Tors}} = \sum_{\text{Torsion}} K_0 + \frac{K_1}{2}(1 + \cos\varnothing) + \frac{K_2}{2}(1 - \cos 2\varnothing) + \frac{K_3}{2}(1 + \cos 3\varnothing) \tag{4}$$

The torsional parameters are often calculated ab initio and subsequently modified via experiment results like molecule geometries or vibrational spectra. Lastly, certain groups, for example, sp^2 hybridized carbons in carbonyl groups or aromatic rings, require an additional term to achieve planarity. Since the standard torsion

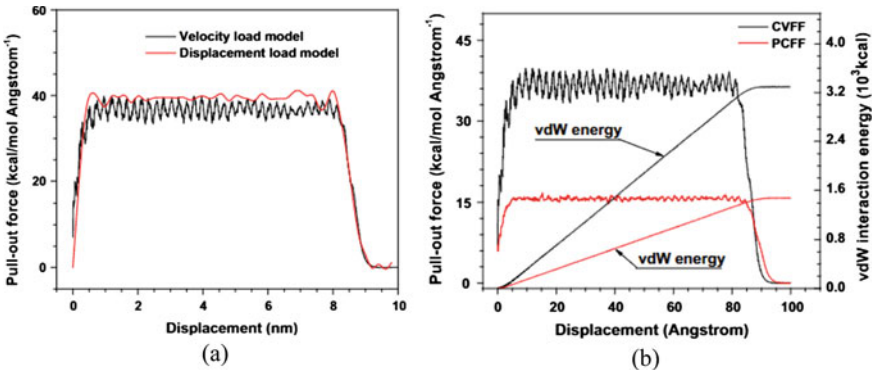


Fig. 2 Illustration of **a** comparative studies of required pull-out forces for a CNT nanocomposite system using velocity and displacement load model and **b** variation of the estimated pull-out forces and vdW energy during the pull-out simulation. (Refurbished with permission from [8])

terms discussed above are insufficient to ensure planarity, this additional term indicates the out-of-plane motions' positive contribution to energy. The improper torsion expression is commonly expressed as

$$U_{\text{imp}} = \sum_{\text{Impropers}} \frac{k_{\text{imp}}}{2}[1 + \cos(2\omega - \pi)] \tag{5}$$

where ω is the improper angle representing the deviation from planarity.

2.2 Intermolecular Terms

The balance of repulsive and attractive forces produces vdW interactions between two atoms. The atoms' electron clouds overlap, causing repulsion, though the interactions among the induced dipoles exhibit an attractive factor, which changes with r^{-6}. In order to depict such interactions, LJ 12 potential is very often being utilized, which is not very common to obtain a few "softer" terms to define the repulsive portion, especially the exponential function that is majorly applicable in the Buckingham potential and is more likely to be used over LJ function for the simulations of organic molecular crystals. Besides, other potential functions are being rarely used, such as LJ 9–6 potential, i.e., used in COMPASS FF or even a buffered LJ 14–7 term used in MMFF. The vdW interactions occur among any two atoms associated with distinct molecules; however, they also interfere among atoms related to the same molecule, which are significantly apart. Although this is essential to pre-define some variables (i.e., σ_{ij} and ϵ_{ij}) for every set of atoms, nonetheless for suitability, most FFs need separate atomic variables, as well as specific methods to integrate them. For

instance, in LJ potential, the corresponding well depth describing the interactions among two atoms obtained by the geometric mean gives $\sigma_{ij} = (\sigma_i \sigma_j)^{1/2}$ and $\epsilon_{ij} = (\epsilon_i \epsilon_j)^{1/2}$, whereas the point where the potential becomes zero can be provided by the geometric, $\sigma_{ij} = (\sigma_i \sigma_j)^{1/2}$, or the arithmetic mean, $\sigma_{ij} = 1/2\ (\sigma_i + \sigma_j)$, depending on the FF taken. The final term in Eq. 1 describes the electrostatic interactions. Although high-level quantum mechanical computations may be used to obtain the molecular electronic density with higher accuracy, the difficulty of minimizing this density to a reasonable explanation that can be employed in an MD simulation is not simple. The most common method is to attribute each nucleus a partial atomic charge and apply Coulomb's equation to calculate its input to the overall energy. Experimental thermodynamic data can be used to calculate partial charges; however, this method is only practicable for tiny molecules. There are also other quick ways for calculating the charge distribution using the electronegativities of the atoms. However, the most frequent method for obtaining trustworthy partial charges is to make an ab initio calculation and then deduce them from the quantum mechanical potential. However, as atomic charges are not experimentally observable, several various approaches for determining them have been established, and they do not necessarily give the same distribution of partial charges. It is worth noting that fixed partial charges cannot wholly explain polarization effects in condensed phases. Despite polarizable FFs, these non-pairwise additive effects are usually considered on an average basis by employing effective charges. Consequently, the molecular model's dipole moment is typically more prominent than the observed dipole moment of the isolated molecule. For instance, most water models use charge distributions of = 2.3–2.6 D as a substitute for the practical value of 1.85 D for vapor. One more thing to remember is that electrostatic interactions are long-ranged, necessitating special handling when truncating them in our force computations. Last but not the least, partial charges are commonly allocated to atomic sites for computational and practical reasons. However, nothing prevents them from being placed outside of the atom locations. Water is another nice illustration, as we may obtain water models with only three sites (with partial charges assigned to the O and H atoms) and models with four, five, and even six sites.

Intramolecular electrostatic interactions among atoms separated by >three bonds are addressed much as if they were intermolecular in several cases, but interactions between 1–2 and 1–3 pairs are ignored. Due to the small distances concerned, the potential becomes strongly repulsive or attractive, avoiding numerical issues. Intramolecular concepts are thought to describe the interactions between those atoms. Treatment of 1–4 interactions is less common; however, they are frequently somewhat eliminated by scaling factors. In contrast, the scale can vary from one FF to the next. If variables from different FFs are combined, this should be factored in. Moreover, because the rotational barriers are determined by the specific mixing of the dihedral potential with the 1–4 vdW and electrostatic interactions, it is usually not recommended to change or mix variables from multiple origins.

3 Popular Forcefields Used in Modeling and Simulation of Different Coating Composites

With the advent of the molecular mechanics (MM) approach, the first FFs were developed in the 1960s. Their main objective was to forecast isolated molecules' molecular structures, vibrational spectra, and enthalpies. Therefore, they were primarily designed to treat tiny organic compounds, but several have lasted to this day and are still being explored and used. The MM potentials (MM2, MM3, and MM4) established by Allinger's group are significant examples. These FFs were designed to examine hydrocarbons, but they were eventually expanded to include a broad range of organic and functionalized compounds. Following this, the scientific focus has shifted to considerably more complicated systems, resulting in more broadly applied FFs. The Dreiding and Universal (UFF) FFs, which include parameters for all atoms present in the periodic table, are prime instances. CHARMM, AMBER, GROMOS, OPLS, and COMPASS are some more prominent FFs. Despite their simplicity, the first three are commonly applied in biomolecule simulations, whereas OPLS and COMPASS were designed to mimic condensed matter. It is indeed worth noting that many of these FFs are constantly developing, with multiple versions accessible, e.g., CHARMM19, CHARMM22, CHARMM27; GROMOS96, GROMOS45A3, GROMOS53A5, GROMOS53A6; AMBER91, AMBER94, AMBER96, AMBER99, and AMBER02. First-generation FFs are those that use an energy formula analogous to Eq. 1. The cross-terms are included in second-generation FFs. COMPASS, UFF, MM2, MM3, and MM4 are coming under second generation, and CFF (consistent FF) and MMFF (Merck molecular FF) are widespread in these fields. The first broad polarizable FFs emerged in the 1990s. Such advancements can be seen in the polarizable intermolecular potential function (PIPF), DRF90, and AMOEBA FFs. Furthermore, polarizable forms of a few of the above-mentioned general FFs have been constructed. As a result, polarization has been added to CHARMM, which uses either fluctuating charges or the shell model and AMBER, OPLS, and GROMOS. The reference contains a more thorough list of polarizable FFs. Several customized potentials are generated, and those general FFs explain a specific chemical system. Water deserves special note in that regard; because of its relevance, various water models have been presented following Barker and Watts' original MC simulation. Most of these models are given in the bibliography, along with a comprehensive assessment of their merits and flaws. Vega et al. [9] likened the most common stiff non-polarizable water potentials (TIP3P, TIP4P, TIP5P, SPC, and SPC/E), concluding that a new edition of the second one, termed TIP4P/2005, best describes 9 out of the 10 experimental features studied. Comparing the available broad FFs is significantly more difficult; as a result, it will be highly reliant on the system and characteristics simulated. There are a few contrasts in the publications, especially on the precision of the CHARMM, AMBER, and OPLS FFs in biomolecular simulations; however, the findings are not conclusive. As a result, Price and Brooks discovered that the three produced strikingly identical protein structure and dynamics outcomes. Yeh and Hummer observed major changes

in the end-to-end diffusion coefficient and the architectures of two peptides investigated with CHARMM and AMBER. Aliev and Courtier-Murias discovered that the structure of short open-chain peptides is strongly influenced by the FF employed. Different structural tendencies are favored based on the FF used in simulations of insulin using CHARMM, AMBER, OPLS, and GROMOS, yielding a comparable conclusion. Paton and Goodman conducted the most thorough assessments of current opportunities. They used seven distinct FFs to calculate the interaction energies of 22 small- and medium-sized molecule complexes and 143 nuclear acid bases and amino acids (MM2, MM3, AMBER, OPLS, OPLSAA, MMFF94, and MMFF94s). They compared them to energies calculated using high-level ab initio calculations. Their findings demonstrate that all potentials examined, particularly OPLSAA and MMFF94s, precisely characterize electrostatic and vdW interactions but underrate the amount of H-bond interactions. The significant insight is that the data and process used in the parametrization of each FF have advantages and drawbacks. As a result, the ultimate selection is based on the situation. Though, as Jorgensen and Tirado-Rives point out, "all leading FFs now provide reasonable results for a wide range of properties of isolated molecules, pure liquids, and aqueous solutions".

$$U = \sum_{\text{bond}} \frac{1}{2} k_b (r - r_0)^2 + \sum_{\text{angle}} \frac{1}{2} a (\theta - \theta_0)^2 + \sum_{\text{torsions}} V_n [1 + \cos(n\varnothing - \delta)] + \sum_{\text{improper}} V_{\text{imp}} \sum_{LJ} 4\varepsilon_{ij} \left(\frac{\sigma_{ij}^{12}}{r_{ij}^{12}} - \frac{\sigma_{ij}^{6}}{r_{ij}^{6}} \right) + \sum_{\text{elec}} \frac{q_i q_j}{r_{ij}} \tag{6}$$

The first four components of the equation denote intramolecular inputs to the total energy, i.e., bond stretching, angle bending, dihedral, and improper torsions. On the other hand, the last two components denote repulsive and vdW interactions (12–6 LJ potential) and Coulombic interactions.

3.1 Non-Reactive Forcefields

Although the LJ potential is the most basic and extensively utilized interatomic potential, it has been used in various simulations for several years. It can be employed on its own or in combination with other possibilities. This clearly describes the basic processes for multiple applications, liquids and polymers computations, and interactions among solids. This LJ potential is mainly beneficial to predicting adhesion and cohesion among inert atoms, molecules, and solids. Mathematically, a typical LJ potential form is as follows:

$$E_{LJ} = \sum_{i \neq j} \left(4\varepsilon_{ij} \left[\left(\frac{\sigma_{ij}}{\varepsilon_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{\varepsilon_{ij}} \right)^{6} \right] \right) \tag{7}$$

wherein r_{ij} refers to the interatomic distance among ith and jth atoms. The interatomic separation at zero energy and the minimal interaction energy are denoted by σ_{ij} and $\mathfrak{E}_{ij}$, respectively. Hence, this LJ potential is a two-body potential that primarily depends upon the distance between two atoms. This potential is composed of one attractive vdW interaction and another repulsive term that arises when the two atoms come closer. When the two atoms come closer, the $\frac{1}{r_{ij}^6}$ term dominates, and the obtained LJ function becomes −ve (i.e., attractive), denoted by $\mathfrak{E}_{ij}$. Upon decreasing the distance further, the $\frac{1}{r_{ij}^{12}}$ term governs and thereby the potential shifts from −ve to +ve at $r_{ij} = \sigma_{ij}$, with a gradually growing repulsive wall at distances less than σ_{ij}.

It is also called LJ 12–6 potential, which refers to the factors' repulsive and attractive terms. By studying the polarization effect of electron density, the attractive 6 components may be directly determined from fundamental principles. The estimation of repulsive 12 terms, on the other hand, cannot be done independently, although 12 is a popular option for the factor and is utilized in various potentials to describe intermolecular interactions. Several LJ potential variants, including 9–3, are mostly employed to explain particle interactions with an even wall. One of the first interatomic potentials utilized in computations was the LJ potential. In 1964, Rahman et al. [10] reported one of the first computational studies of liquid Ar. Subsequently, Verlet et al. [11] reported another computational investigation wherein the authors compared the LJ potential to neutron scattering data. This study validated the concept of this emerging scientific method. Though the quantum mechanical principles did not fully utilize this LJ potential, it proved to be capable of reproducing the experimental data in MD simulations. Properties such as radial distribution function (RDF), diffusion coefficient, and other thermodynamic properties as a function of density and temperature were obtained accurately. This simplest LJ potential has been shown to be incredibly helpful for gaining fundamental insights into different materials. For instance, this potential was utilized to model different interacting surfaces and further answered many basic queries in tribology [12–15]. LJ potentials have also been employed in conjunction with MD simulations to investigate the relevance of continuum mechanics to nanoscale contacts [13, 16].

3.2 Reactive Forcefields

Baskes et al. [17] have first detailed the embedded-atom method (EAM), which is a many-body interaction potential and is especially best adapted for modeling metals. The term "embedded" refers to an individual atom being embedded in the atmosphere created by the other atoms in the unit. The atmosphere is made up of two parts: (1) the embedded atom interacts pairwise with the host material's atom, and (2) the metal's valence electrons interact with the embedded atom in a many-bodied manner. Mathematically, the total energy of a system via EAM potential that contains N atoms is given by

$$E_{\text{Tot}} = \sum_{i}^{N} F_i\left(\rho_{h,i}\right) + \frac{1}{2}\sum_{j,j\neq 1} \varnothing_{ij}\left(r_{ij}\right) \tag{8}$$

wherein F_i refers to the required energy to embed atom i at its location, $\rho_{h,i}$ refers to the approximate local electron density obtained by adding contributions from all atoms in the local region except the embedded atom, and ϕ_{ij} refers to the core-core repulsion among ith and jth atoms.

4 Case Studies

4.1 Computational Studies in Coating Application

Computational techniques play a significant role in the characterization of different coating-related applications by providing simulated results. A few case studies on optimizing different coating materials employing MD simulations are described below.

4.1.1 Graphene-Based Coatings

The interaction of graphene with water has gathered significant attention owing to its astounding mechanical, electronic and optical properties [18, 19], and wide potential applications in electricity generation, water filtration, desalination, design of fuel cells, biochemical sensing, etc. In this line, a considerable amount of work has been done to investigate the wetting phenomena of graphene. Andrews et al. [20] have examined the dynamic spreading behavior of water droplets on graphene-coated substrates or free-standing graphene layers. The authors have observed that the dynamics of wetting, like standard solids, can be split into the formation of patches, and then spread over the surface. This spreading obeys the universal inertial spreading rule where $r \sim t^{1/2}$, irrespective of the nature of the supporting substrate and the number of graphene layers, wherein r is the spreading radius and t is the spreading time of the water droplet.

Liu et al. [21] described a linear correlation between microscopic solid–liquid interaction potential and surface energy using super hydrophilic surfaces, which is also reasonable for hydrophobic and hydrophilic surfaces. Mathematically,

$$\sigma_{sg-sl} - \sigma_{lg} = \alpha\left(\varepsilon - \varepsilon^0\right) \tag{9}$$

wherein σ terms are denoted for the macroscopic energies among solid, liquid, and gas phases. ε and ε^0 are denoted for the LJ interaction parameter and critical point

when the water droplet begins to spread thoroughly throughout the surface. For the graphene substrate, this $\varepsilon^0 = 0.14$ kcal mol^{-1}. This linear correlation is valid for several substrates exhibiting distinct lattice lengths, and the slope of this direct relationship depends upon the surface atom densities (i.e., the greater the densities of the surface atom, the larger will be the slopes, α).

Huang et al. [19] have studied the dependence of wetting phenomena of a graphene substrate on surface morphology, tension, and vibration. The authors have adopted the density profile method for the prediction of WCA and found it to be 88.27°. The authors also found that the wrinkled morphology of the graphene layer may potentially help to enhance the hydrophobic nature of the graphene substrate. Further, on increasing the number of water molecules and thereby the water droplet size on certain wrinkled graphene, the wrinkles originate an energy barrier which prevents the water droplets to spread over it, and thereby makes the contact lines of the droplet demonstrate a "pinning effect". Upon increasing the water molecules' concentration and thereby the droplet size further to overcome this energy barrier, the droplet contact line spreads promptly to the nearby available crests. The WCA first increased, and subsequently, reduced and then increased again. It was also found that the phobic nature of graphene can be significantly enhanced by tension and vibration. The WCA enhances linearly on increasing the strain till the tensile strain is <10%, and remains stable at about 110° when the strain becomes >10%. Further, on increasing the amplitude and decreasing the period of the vibration, the CA was found to increase.

4.1.2 Poly(Dimethylsiloxane) (PDMS)-Based Coatings

PDMS is a hydrophobic material that is widely used for developing self-clean coating materials [22]. An adequate amount of research articles has been published investigating the phobic nature of PDMS experimentally [23, 24]. Sethi et al. [22] investigated the wetting nature of PDMS by using MD simulations and COMPASS forcefield and further observed a good match with experimental outcomes. The authors have reported the dipole moment orientations of individual water molecules present at the solid–liquid interface and found that the orientations are completely random. These random orientations suggested a negligible penetration tendency of water molecules inside the PDMS due to its strong repellence nature. Further, the WCA of PDMS was found to decrease up to 78 ± 2° and 60 ± 2° by addition of 50 wt.% of PVAc and PVOH, respectively. In another investigation, Sethi et al. [25] copolymerized PVAc with PDMS at different wt. fractions (20, 40, 60, and 80 wt.%) and observed a decrease in CA from 99 ± 1° to 71 ± 2° on increasing the grafting wt.% from 20 to 80. In both the studies, the authors have utilized COMPASS forcefield, which further has been found to be in very close proximity to experimental outcomes.

4.1.3 Poly(Vinylacetate) (PVA)-Based Coatings

Researchers have used PVA alongside other hydrophobic materials to design durable hydrophobic coating materials. For example, Kumar et al. [26] investigated the wetting nature of PVA and found it to be hydrophilic with an estimated CA of 58.27° against water, which was close to the estimates (62 ± 2°) of Sethi et al. [22]. Further, Kumar et al. [26] added PFO (perfluoro octane) at different wt. fractions to enhance its phobic nature. The WCA increased to 121.18° by the addition of 74.7 wt.% of PFO in PVA, thereby suggesting an impactful change of the hydrophilic to hydrophobic nature of the PVAc-PFO system with the addition of PFO. Both Kumar and Sethi et al. have employed COMPASS forcefield and reported a very good match of estimated results with experimentally reported values.

4.1.4 Poly (Vinylidene Fluoride) (PVDF)-Based Coatings

PVDF is an extensively used material for the fabrication of water-replant coatings and oil–water separators since it exhibits astounding repellence nature against water. Kitabata et al. [27] investigated the impact of the microscopic structure of PVDF-based coatings on the wetting behavior through molecular simulations using GROMACS and VMD. The authors have employed two types of PVDF surfaces such as (i) crystalline and (ii) amorphous, and further used the obtained trajectories to estimate the phobic nature of both the surfaces. The authors have used water nanodroplets with a varying diameter of $d = 4$, 5.5, 7, 10, and 13 nm. They obtained different WCA values for different surfaces wherein the crystalline structured surface demonstrated higher water repellence nature for all varying diameter water nanodroplets. Thusly concluded that compared to crystalline, the amorphous surface demonstrates a relatively lower hydrophobic nature. The authors suggested that the water molecules interact more favorably with amorphous surfaces compared to the crystalline surface, thereby reducing the excess free energy, i.e., interfacial tension, which in turn makes the amorphous surface less hydrophobic. Darvishi et al. [28] have investigated the phobic characteristics of PVDF against water and oil for the fabrication of a water–oil separator. They acquired fluid CAs of 132° and 4° for the systems of PVDF-water and PVDF-oil systems.

4.2 Computational Studies in Nanocomposite-Based Materials

Computational methods are beneficial for fundamental understandings before experimental studies. Such techniques require a high computational configuration to use any theoretical tools.

This section briefly reviewed the simulated studies estimating the mechanical characteristics of graphene, carbon nanotubes (CNTs), and coiled CNT-reinforced nanocomposites. Among the simulation studies on nanocomposites' physical performance offered are representative volume element approaches, multiscale simulations, and MD simulations. A few researchers have employed numerous techniques to compare simulated outcomes of the effective elastic characteristics of the reinforced composite. Since this is crucial for the researcher to connect MD simulations and continuum mechanics, researchers have attempted to do so in a few articles to obtain more accurate outcomes in the field of nanocomposites.

4.3 Poly (Ethylene) (PE)-Based Nanocomposites

Poly(ethylene) (PE) is a thermoplastic polymer with low cost and decent mechanical behavior owing to which it is widely used as a base matrix to design different nanocomposites. Verma et al. [29] have studied the impact of grain boundaries on the interfacial properties of bi-crystalline PE/graphene-based nanocomposites. The authors have employed MD simulations with reactive FF and revealed that higher energy states in bi-crystalline graphene help to improve the interatomic interactions at the PF/graphene interphase. Similarly, Chaurasia et al. [30] have investigated the reinforcing capabilities of hexagonal boron nitride (h-BN) nanosheets in PE-based nanocomposites. The authors have used the same reactive FF to investigate the impact of aspect ratio, wt. fraction, morphology, dispersion of h-BN, and strain rate loading on mechanical behavior. The authors have also performed an experimental analysis and made a good comparison of simulated results with experimental outcomes. Both the case studies suggest the excellent accuracy of reactive FF in nanocomposite applications.

4.4 Coiled Carbon Nanotube-Reinforced Nanocomposites

Scientists have combined the unique behavior of CNTs with morphological features to enhance the performance of CNTs, i.e., called coiled CNTs (CCNTs) [31, 32]. Such CCNTs exhibit higher load transfer and bonding strengths with polymer matrix compared to CNTs when used for developing nanocomposites [33, 34]. These potentials of CCNTs over CNTs have recently attracted overwhelming attention for CCNT-reinforced nanocomposites. MD and MM are empirical approaches to investigate and model such carbon-based nanomaterials. Mortazavi et al. examined the impact of filler shape, concentration, and property contrast and, in particular, interphase thickness and property contrast on effective thermal conductivity and elastic stiffness of nanocomposites by using 3D finite element (FE) simulation.

Mortazavi et al. utilized 3D FE simulation to investigate the effect of filler shape, concentration, and features difference, particularly interphase thickness and characteristics comparison, on effective thermal conductivity and elastic stiffness of nanocomposites.

The authors had considered the cylinder, sphere, and thin disk-shaped filler geometries and observed that although the effect of interphase is significant for spherical-shaped fillers when it deviates from sphere shape, it seemed to be less effective. Coil and tube diameters, helix angle, and the number of coils for CCNTs are crucial factors influencing nanocomposites' overall features. Khani et al. [35] used a 3D FE model for CCNTs to investigate the impact of interphase, volume fraction, orientation, and geometric parameters on the elastic properties of CCNT-reinforced nanocomposite. Further, the authors have proposed an algorithm to study the importance of spring fillers as an alternative to single-wall CNTs for nanocomposite applications.

4.5 Computational Studies on Graphene-Based Nanocomposites

A 2D carbonic honeycomb-like structural material called graphene is regarded as a fascinating material owing to its numerous characteristics like attractive physicochemical and mechanical properties. Such crystalline material can be synthesized by several techniques such as physical, mechanical, or chemical oxidation.

Estimation of melting temperature (T_m) of liquid carbon and graphite is a novel research area of the invention which can be performed both experimentally and computationally in computations; the T_m estimation could be employed by using empirical parameters such as Tersoff, Brenner, and adaptive intermolecular reactive empirical bond order (AIREBO) potentials. Two explanatory observations are possibly discussed, considering the melting phenomena of graphite and graphene through the sublimation process and demonstrating the mutual difference. Graphene forms linear chains during the melting process via simulated sublimation in an argon environment. However, further provide distinct values among T_m for graphene and graphite. Owing to graphite's intact structure, the adjacent bonds get converted into bridges when the atoms among adjacent planes are heated, which further causes its decomposition. For graphene, the exfoliation causes an adequate separation of the individual sheets which causes no formation of bridges indicating its higher T_m than graphite.

Such melting of graphene was studied through the AIREBO model and further examined via the sublimation process at a higher external pressure reducing the uncertainties and simultaneously increasing the temperature during the computations. This study has been performed on 6,400 and 4,000 graphene graphite particles, respectively. At certain instances during the experimentation, ~3,020 particles were involved in cluster formation, but the linear chains were formed by the remaining. Such studies clearly suggest that the computational-based experimental models may

not be able to predict the graphene/graphite's melting points accurately but may surely help to find a range of melting points with some fundamental insights.

5 Comparison of Estimated Results of Nanocomposites Using Different Forcefields

Most researchers perform a comparative study of situation predictions with experimental outcomes. But sometimes, it is important to validate the MD simulation estimates with varying forcefields. Xiong et al. [8] presented a comparative study of MD simulation predictions with varying forcefields in this direction. Figure 2a demonstrates a comparative study of the pull-out force required for CNT using PCFF and COMPASS forcefield. Due to the pull-out force (F_{pull}), the total work done (w) is equal to the vdW pairwise energy change ΔE_{int} () for the nanocomposite system for a certain pull-out distance (Δ_x), mathematically:

$$\Delta E_{\text{int}} = \int_0^{\Delta x} F_{\text{pull}}(x)dx \tag{10}$$

The authors demonstrated a good agreement of the obtained outcomes with the results obtained earlier by Li et al. [36], where they have employed COMPASS forcefield. When the same experiment was undertaken using CVFF FF, a significant difference was observed. This noteworthy difference was attributed to the accuracy of interatomic interaction representations involved in different potentials. Figure 2b demonstrates the estimated pull-out forces and vdW interaction energy obtained during the pull-out simulation. Moreover, a close match of the slope of vdW energy was also observed for both simulation estimates using CVFF and PCFF forcefields. The authors have a vast experience in the field of molecular dynamics and Density Functional Theory-based simulations [37–49].

6 Conclusion

The advancements in computational approaches for several polymers, fillers, and nanocomposites are rapidly evolving day by day. Major developments are undergoing, especially in designing distinct FF for different materials. All the studies discussed in this article demonstrate a significant role of FF in obtaining interfacial properties on mechanical performance and some fundamental understandings for coatings and nanostructured reinforced nanocomposites. This study suggests how the MD simulations are able to estimate the interactions and structural changes during contact with other materials. For example, an insignificant change in graphene's structure as discussed could not be configured in wet-lab conditions, whereas the

same interaction was clearly observed in computational techniques. However, such functionalities of MD simulations could potentially help to provide atomic-level descriptions such as structural changes or defects. Scientists and researchers work on improvising the FFs substantially on the state-of-the-art accuracy for MD simulations. These improvements may surely be useful in developing novel methodologically treatable algorithms which probably predict the interactions with greater accuracy and insights.

References

1. Alder, B.J., Wainwright, T.E.: Phase transition for a hard sphere system. J. Chem. Phys **27**, 1208 (1957)
2. Metropolis, N., Rosenbluth, A.W., Rosenbluth, M.N., Teller, A.H., Teller, E.: Equation of state calculations by fast computing machines. J. Chem. Phys. **21**, 1087 (2004)
3. Hünenberger, P.H., van Gunsteren, W.F.: Empirical classical interaction functions for molecular simulation. Comput. Simul. Biomol. Syst. 3–82 (1997)
4. Anon: Computational biochemistry and biophysics. Comput. Biochem. Biophys. (2001)
5. Ponder, J.W., Case, D.A.: Force fields for protein simulations. Adv Protein Chem. **66**, 27–85 (2003)
6. Mackerell, A.D.: Empirical force fields for biological macromolecules: overview and issues. J. Comput. Chem. **25**, 1584–1604 (2004)
7. Jorgensen, W.L., Tirado-Rives, J.: Potential energy functions for atomic-level simulations of water and organic and biomolecular systems. Proc. Natl. Acad. Sci. **102**, 6665–6670 (2005)
8. Xiong, Q.L., Meguid, S.A.: Atomistic investigation of the interfacial mechanical characteristics of carbon nanotube reinforced epoxy composite. Eur. olym. J. **69**, 1–15 (2015)
9. Jungwirth, P., Beattie, J.K., Djerdjev, A.M., Warr, G.G., Hänni-Ciunel, K., Schelero, N., Von Klitzing, R., Bratko, D., Daub, C.D., Luzar, A., Link, O., Vöhringer-Martinez, E., Lugovoj, E., Liu, Y., Siefermann, K., Faubel, M., Grubmüller, H., Gerber, R.B., Miller, Y., Abel, B., Tobias, D.J., Sengupta, N., Tarek, M., Frölich, A., Gabel, F., Jasnin, M., Lehnert, U., Oesterhelt, D., Stadler, A.M., Tehei, M., Weik, M., Wood, K., Zaccai, G., Qvist, J., Persson, E., Mattea, C., Halle, B., Ghosh, A., Kramer Campen, R., Sovago, M., Bonn, M., Born, B., Kim, S.J., Ebbinghaus, S., Gruebele, M., Havenith, M., Willard, A.P., Chandler, D., Salmeron, M., Bluhm, H., Tatarkhanov, M., Ketteler, G., Shimizu, T.K., Mugarza, A., Deng, X., Herranz, T., Yamamoto, S., Nilsson, A., Gallagher, M., Omer, A., Darling, G.R., Hodgson, A., Bishop, C.L., Pan, D., Liu, L.M., Tribello, G.A., Michaelides, A., Wang, E.G., Slater, B., Bertin, M., Meyer, M., Stähler, J., Gahl, C., Wolf, M., Bovensiepen, U., Ricci, M.A., Bruni, F., Giuliani, A., Lombardo, T.G., Giovambattista, N., Debenedetti, P.G., Coudert, F.-X., Cailliez, F., Vuilleumier, R., Fuchs, A.H., Boutin, A., Perkin, S., Goldberg, R., Chai, L., Kampf, N., Klein, J., Lee, M., Sung, B., Hashemi, N., Jhe, W., Reed, S.K., Madden, P.A., Vega, C., et al.: What ice can teach us about water interactions: a critical comparison of the performance of different water models. Faraday Discuss. **141**, 251–276 (2008)
10. Rahman, A.: Correlations in the motion of atoms in liquid argon. Phys. Rev. **136**, A405 (1964)
11. Verlet, L.: Computer "experiments" on classical fluids. I. Thermodynamical properties of Lennard-Jones molecules. Phys. Rev. **159**, 98–103 (1967)
12. Müser, M.H., Wenning, L., Robbins, M.O.: Simple microscopic theory of Amontons's laws for static friction. Phys. Rev. Lett. **86**, 1295 (2001)
13. Luan, B., Robbins, M.O.: The breakdown of continuum models for mechanical contacts. Nature **435**, 929–932 (2005)
14. Cheng, S., Luan, B., Robbins, M.O.: Contact and friction of nanoasperities: effects of adsorbed monolayers. Phys. Rev. E Stat. Nonlinear, Soft Matter Phys. **81**, 016102 (2010)

15. Cheng, S., Robbins, M.O.: Defining contact at the atomic scale. Tribol. Lett. **39**, 329–348 (2010)
16. Luan, B., Robbins, M.O.: Contact of single asperities with varying adhesion: comparing continuum mechanics to atomistic simulations. Phys. Rev. E - Stat. Nonlinear, Soft Matter Phys. **74**, 026111 (2006)
17. Daw, M.S., Baskes, M.I.: Embedded-atom method: Derivation and application to impurities, surfaces, and other defects in metals. Phys. Rev. B **29**, 6443 (1984)
18. Raj, R., Maroo, S.C., Wang, E.N.: Wettability of graphene. Nano Lett. **13**, 1509–1515 (2013)
19. Huang, C., Xu, F., Sun, Y.: Effects of morphology, tension and vibration on wettability of graphene: a molecular dynamics study. Comput. Mater. Sci. **139**, 216–224 (2017)
20. Andrews, J.E., Sinha, S., Chung, P.W., Das, S.: Wetting dynamics of a water nanodrop on graphene. Phys. Chem. Chem. Phys. **18**, 23482–23493 (2016)
21. Liu, J., Wang, C., Guo, P., Shi, G., Fang, H.: Linear relationship between water wetting behavior and microscopic interactions of super-hydrophilic surfaces. J. Chem. Phys. **139** (2013)
22. Sethi, S.K., Soni, L., Shankar, U., Chauhan, R.P., Manik, G.: A molecular dynamics simulation study to investigate poly(vinyl acetate)-poly(dimethyl siloxane) based easy-clean coating: an insight into the surface behavior and substrate interaction. J. Mol. Struct. **1202**, 127342 (2020)
23. Sethi, S.K., Manik, G.: Recent progress in super hydrophobic/hydrophilic self-cleaning surfaces for various industrial applications: a review. Polym. Plast. Technol. Eng. **57**, 1932–1952 (2018)
24. Sethi, S.K., Manik, G., Sahoo, S.K.: Fundamentals of Superhydrophobic Surfaces. Superhydrophobic Polymer Coatings. Elsevier, pp 3–29 (2019)
25. Sethi, S.K., Shankar, U., Manik, G.: Fabrication and characterization of non-fluoro based transparent easy-clean coating formulations optimized from molecular dynamics simulation. Prog. Org. Coatings **136**, 105306 (2019)
26. Kumar, N., Manik, G.: Molecular dynamics simulations of polyvinyl acetate-perfluorooctane based anti-stain coatings. Polymer (Guildf). **100**, 194–205 (2016)
27. Kitabata, M., Taddese, T., Okazaki, S.: Molecular dynamics study on wettability of poly(vinylidene fluoride) crystalline and amorphous surfaces. Langmuir **34**, 12214–12223 (2018)
28. Darvishi, M., Foroutan, M.: Molecular investigation of oil-water separation using PVDF polymer by molecular dynamic simulation. RSC Adv. **6**, 74124–74134 (2016)
29. Verma, A., Parashar, A., Packirisamy, M.: Effect of grain boundaries on the interfacial behaviour of graphene-polyethylene nanocomposite. Appl. Surf. Sci. **470**, 1085–1092 (2019)
30. Chaurasia, A., Verma, A., Parashar, A., Mulik, R.S.: Experimental and computational studies to analyze the effect of h-BN nanosheets on mechanical behavior of h-BN/polyethylene nanocomposites. J. Phys. Chem. C **123**, 20059–20070 (2019)
31. Wu, J., He, J., Odegard, G.M., Nagao, S., Zheng, Q., Zhang, Z.: Giant stretchability and reversibility of tightly wound helical carbon nanotubes. J. Am. Chem. Soc. **135**, 13775–13785 (2013)
32. Liu, L., Liu, F., Zhao, J.: Curved carbon nanotubes: from unique geometries to novel properties and peculiar applications. Nano Res. **75(7),** 626–657 (2014)
33. Park, S.H., Yun, D.J., Theilmann, P., Bandaru, P.R.: Superior electrical and mechanical characteristics observed through the incorporation of coiled carbon nanotubes, in comparison to non-coiled forms, in polymers. Polymer (Guildf). **54**, 1318–1322 (2013)
34. Lau, K.T., Lu, M., Hui, D.: Coiled carbon nanotubes: synthesis and their potential applications in advanced composite structures. Compos. Part B Eng. **37**, 437–448 (2006)
35. Khani N, Yildiz M, Koc B (2016) Elastic properties of coiled carbon nanotube reinforced nanocomposite: a finite element study. Mater Des 109:123–132
36. Li, Y., Liu, Y., Peng, X., Yan, C., Liu, S., Hu, N.: Pull-out simulations on interfacial properties of carbon nanotube-reinforced polymer nanocomposites. Comput. Mater. Sci. **50**, 1854–1860 (2011)
37. Verma, A., Parashar, A., Packirisamy, M.: Atomistic modeling of graphene/hexagonal boron nitride polymer nanocomposites: a review. Wiley Interdiscip. Rev. Comput. Mol. Sci. **8**(3), e1346 (2018)

38. Verma, A., Parashar, A.: Structural and chemical insights into thermal transport for strained functionalised graphene: a molecular dynamics study. Mater. Res. Express **5**(11), 115605 (2018)
39. Verma, A., Kumar, R., Parashar, A.: Enhanced thermal transport across a bi-crystalline graphene–polymer interface: an atomistic approach. Phys. Chem. Chem. Phys. **21**(11), 6229–6237 (2019)
40. Verma, A., Parashar, A.: Molecular dynamics based simulations to study the fracture strength of monolayer graphene oxide. Nanotechnology **29**(11), 115706 (2018)
41. Verma, A., Parashar, A.: The effect of STW defects on the mechanical properties and fracture toughness of pristine and hydrogenated graphene. Phys. Chem. Chem. Phys. **19**(24), 16023–16037 (2017)
42. Verma, A., Parashar, A., Packirisamy, M.: Tailoring the failure morphology of 2D bicrystalline graphene oxide. J. Appl. Phys. **124**(1), 015102 (2018)
43. Singla, V., Verma, A., Parashar, A.: A molecular dynamics based study to estimate the point defects formation energies in graphene containing STW defects. Mater. Res. Express **6**(1), 015606 (2018)
44. Verma, A., Parashar, A.: Molecular dynamics based simulations to study failure morphology of hydroxyl and epoxide functionalised graphene. Comput. Mater. Sci. **143**, 15–26 (2018)
45. Verma, A., Zhang, W., van Duin, A.C.: ReaxFF reactive molecular dynamics simulations to study the interfacial dynamics between defective h-BN nanosheet and water nanodroplets. Phys. Chem. Chem. Phys. **23**, 10822–10834 (2021)
46. Verma, A., Parashar, A., Packirisamy, M.: Role of chemical adatoms in fracture mechanics of graphene nanolayer. Mater. Today: Proc. **11**, 920–924 (2019)
47. Kataria, A., Verma, A., Sanjay, M.R., Siengchin, S.: Molecular modeling of 2D graphene grain boundaries: mechanical and fracture aspects. Mater. Today: Proc. (2021)
48. Deji, R., Verma, A., Kaur, N., Choudhary, B.C., Sharma, R.K.: Density functional theory study of carbon monoxide adsorption on transition metal doped armchair graphene nanoribbon. Mater. Today: Proc. (2021)
49. Deji, R., Verma, A., Choudhary, B.C., Sharma, R.K.: New insights into NO adsorption on alkali metal and transition metal doped graphene nanoribbon surface: a DFT approach. J. Mol. Graph. Model. (2021). https://doi.org/10.1016/j.jmgm.2021.108109

Development, Availability, and Applications of EAM Potentials for Characterization of Complex HCP Materials



Divya Singh

Abstract Due to complex crystal structure in conjunction with anisotropic material properties, limited research has been reported on hcp materials as compared to materials with body (bcc) and face-centred (fcc) crystals. Despite these challenges, hcp materials such as zirconium (Zr), titanium (Ti), zinc (Zn), and magnesium (Mg) are emerging with potential applications in the field of nuclear, aerospace, and biomedical implants. Properties of hcp materials such as strength and fracture toughness are affected by the orientation of grain boundaries, orientation of crack in the anisotropic crystal structure, and position of point defects in such complex materials. The deformation mechanism of hcp metals is radically different, which produces challenges as well as limitations in experimentally characterizing hcp metals containing complex slip systems and anisotropy as compared to simple slip systems of bcc and fcc. These hcp materials are extensively used in nuclear reactors and biomedical implants, which also introduces some phenomena at a very short time scale in the range of picoseconds that also limit experimental techniques. However, these complex material behaviour shown by hcp materials can be easily captured by computational-based atomistic techniques. This chapter focusses on the development, availability, importance, and application of EAM potentials which are employed for computational-based Molecular Dynamics-based simulations to capture interesting behaviour of hcp materials.

Keywords EAM potential · HCP materials · Anisotropy · Material characterization · Molecular dynamics · Yield strength · Dislocations · Grain boundaries · Cracks

D. Singh (✉)
Department of Materials Science and Engineering, University of California, Irvine, USA
e-mail: dsingh4@uci.edu

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1 Introduction

HCP materials have a complex crystal structure which leads to a unique combination of deformation modes. The ideal c/a ratio for an hcp material is 1.633. However, it has been observed for most of the hcp materials that the c/a ratio deviates from the ideal value. The variation in the c/a ratio is a very important factor and has to be taken care of while carrying out computational simulations on hcp materials. Literature indicates that the interatomic potentials developed for hcp materials have paid special emphasis on producing the exact non-ideal c/a ratio. It has been observed that the c/a ratio affects the structural units and coincident lattice sites at the grain boundaries as well as the formation and binding energies of different types of point defects. The probability of active deformation modes in different hcp materials can vary with the variation in c/a ratios. However, it has been found that single crystal hcp materials show varied deformation media depending upon the loading directions, temperature, presence of dislocations, and point defects. This can be theoretically related to the Schmid factor of all the posibble slip and twinning systems in hcp materials. The Schmid factor for various modes of deformation many times indicates that multiple deformation modes can be active. This is sometimes hard to predict that experimentally. However, it is convenient to get such insights through MD simulations by employing accurate interatomic potentials. The first principle-based DFT simulations and classical mechanics-based MD simulations have proven to give accurate results regarding the point defect formation and migration energies and phenomena like tension–compression asymmetry in hcp crystals. Computational-based simulation techniques have been used to generate bicrystalline and polycrystalline materials as well. These grain-boundary structure and energies have been found to be in agreement with the experimentally observed boundaries. Ti, an hcp material, is widely utilized as a biomaterial and in aviation industries. Similarly, Mg is another important material gaining popularity as a biocompatible material for bone implants. Zr is the most widely accepted nuclear material for manufacturing fuel rods and claddings. In order to improve the performance and reduce the failures in such critical applications, we need to explore the underlying phenomenon behind the failures. The classical mechanics-based MD simulations and first principle-based DFT simulations are capable of identifying the phenomenon at the atomic level giving new insights for improving material properties. However, DFT simulations have computational and simulation size constraints. Therefore, MD-based simulation techniques have become popular to study a range of defects—including point, line defects, and grain boundaries as well as complex deformation mechanisms under various loading conditions, temperature, etc. It has been discovered through computational studies that alloying significantly affects the generalized stacking fault energies and plastic behaviour in hcp materials. Grain boundaries have been found to have sinking capabilities for point defects. The tensile and fracture behaviour of hcp materials is dependent upon the misorientation angles, boundary plane, and the misorientation axis. Not only this but also alloying was found to affect the point defects migration

energies and crack tip stress shielding in bicrystalline Zr. The experimental procedure to characterize a complex crystal structure is much more challenging. Certain phenomena like irradiation, corrosion, etc. which are of utmost importance in nuclear and biomaterials have limitations concerning time and spatial scales. MD simulations are capable of working at time scales of picoseconds and femtoseconds.

development, availability, importance, and application of EAM potentials which are employed for computational-based Molecular Dynamics-based simulations to capture interesting behaviour of hcp materials.

2 Embedded Atom Method (EAM) Potential

The energy of an atomic system as estimated by a many-body EAM potential can be expressed mathematically as follows:

$$E = F_\alpha\left(\sum\nolimits_{j\neq i} \rho_\beta r_{ij}\right) + \frac{1}{2}\sum\nolimits_{j\neq i} \emptyset_{\alpha\beta}\left(r_{ij}\right) \tag{1}$$

where F_α is the embedding energy which is a function of the atomic electron density ρ, $\emptyset$ is a pair potential interaction, and α and β are the element types of atoms I and J.

The total energy in a many-body EAM potential is a summation of two energies. The first term indicates the energy on an atom due to the surrounding electron cloud it is embedded in. The second term represents the pair-wise interaction between two atoms separated by some distance.

2.1 Development of EAM Potentials for HCP Materials

As early as 1991, Igarashi et al. [1] developed Finnis–Sinclair-type EAM potential for eight hcp materials, namely—Co, Zr, Ti, Ru, Hf, Zn, Mg, and Be. The potential developed is fitted semi empirically to cohesive energy, unrelaxed vacancy formation energy, five independent second-order elastic constants, and two equilibrium conditions. Each potential, thus developed, is true to the specific non-ideal c/a ratio corresponding to the particular stable hcp configuration of each material. Ackland et al. [2] developed a many-body potentials to study the defects, surface, and displacement threshold properties for α-Zr. This potential utilizes the non-ideal c/a ratio specific to Zr and third neighbour interactions. In this potential, in order to account for the close-range repulsive forces during collision simulations, the pair part of the potential is modified for the small interatomic spacing. For this, the pair part of the potential was matched to the 'universal' potential function of Ziegler, Biersack, and Littmark (Biersack and Ziegler 1982) [3] at small interatomic spacing. The potential was

verified for defect and surface energies by calculating point defect energies, surface and stacking fault energies, and displacement threshold energies using MD-based simulations. In their work, Igarashi et al. [1] neglected the inner strain contribution to the elastic constants in an attempt to match the experimental values of five elastic constants and c/a ratio. In order to bridge that gap, Goldstein and Jonsson [4] reported an EAM that accounts for inner strains. The fitting scheme employed by them was able to match all the elastic constants within 5%, with an exception for C44 which was greater than 25%, although the c/a ratio was well replicated. Further, Pasianot and Monti [5] developed an EAM potential that could reproduce the room temperature elastic behaviour of Zr and be capable of representing a self-interstitial relaxation volume consistent with the relatively low measured value. The potential predicted the stability of 2D small vacancy clusters. Mendeleev and Ackland [6] proposed 3 many-body interatomic potentials (#1, #2, and #3), keeping in mind that the previously defined EAM potentials were incapable of capturing certain distinctive properties of Zr like high stacking fault energies and phase transformation at high enough temperatures. Starting from the potential developed by Ackland et al. [2], the pair-wise potential and the embedding energy were refitted to a pair correlation function (PCF) from an X-ray diffraction data, using the Born–Green–Bogoliubov (BGB) equation. This potential #1 predicts the melting and thus phase transformation around 400 K below the experimental value. The data corresponding to the phase transformation needed to be explicitly fitted in the existing potential. Thus, starting from the potential #1, hcp to bcc transformation data were properly fit to obtain potential #2. However, potentials #1 and #2 do not give good estimates of defect formation energy in hcp Zr when compared to first-principles data. In order to create potential #3, starting from potential #2, the phase transformation information was deleted and the first principle date on interstitial and defect formation energy was incorporated. This potential accurately represents the defect energies but gives poor phase transformation properties. Varvenne et al. [7] in their work showed that although potential #3 was more suited for more reliably predicting the defect formation energies, it could not predict stable vacancy clusters and dislocation loops in zirconium. Potential #2 was, however, able to predict these properties accurately. A comparison of properties from the works of AWB, IKB, PM, and Mendeleev and Ackland (potentials #1, #2, and #3) are tabulated in Table 1. The EAM potential by Mendeleev and Ackland is, by far, the most widely used potential for studying Zr.

Ackland [8] developed an equivalent many-body EAM potential for Titanium by Finnis–Sinclair-type formulation of the second-moment approximation to the tight-binding theory. This potential takes into account the anisotropy of shear constants and the non-ideal c/a ratio. The parametrization approach used to construct this short-range many-body potential is that the parameters are fitted assuming an fcc material with a negative stacking fault energy. This is possible since the potential being developed is short ranged and includes only twelve nearest and six second neighbours in the fcc structure, with two more distant neighbours with comparatively weak contribution and it is already established that the radial distribution of the first eighteen neighbours in fcc and hcp structure is similar. Once the potential is generated, it is

Table 1 A comparison of properties from the works of AWB, IKB, PM, and Mendeleev and Ackland (potentials #1, #2, and #3) [6]

Property	Target Value	AWB	PM	IKV	#1	#2	#3
a (hcp) (Å)	3.232	3.249	3.232	3.232	3.231	3.220	3.234
c (hcp) (Å)	5.182	5.183	5.149	5.149	5.186	5.215	5.168
E_{coh}	−6.32	−6.250	−6.250	−6.250	−6.017	−6.469	−6.635
C_{11}	155	160	146	155	196	165	147
C_{12}	67	76	70	67	88	65	69
C_{44}	36	36	32	36	47	48	44
C_{13}	65	70	65	65	81	63	74
C_{33}	172	174.7	164.8	173	212	180	168
E_f^v	2.077	1.817	1.780	1.830	1.550	2.310	1.762
Octahedral interstitial	2.84	4.13	2.81	8.01	3.23	3.51	2.88
Basal octahedral interstitial	2.88	3.98	2.63	9.10	3.02	2.87	2.90
Basal Crowdion interstitial	2.95	3.82	2.56	8.50	3.25	3.35	2.91
I_2 basal SFE	12.5	3.3	4.4	1.7	6.3	6.8	12.4
Prism SFE	9.1	Unstable	1.01	Unstable	Unstable	23.3	10.9
$T_{\alpha\text{--->}\beta}$ (K)	1136	2054>T_m	1211>T_m	1251	588	1233	1369
$\Delta H_{\alpha\text{--->}\beta}$ (eV/atom)	0.040	0.054	0.041	0.037	0.019	0.039	0.060
$\Delta V_{\alpha\text{--->}\beta}/V_\alpha$ (%)	−0.4	+1.3	+1.9	+4.9	−1.5	−0.8	+0.9
T_m (hcp,K)		1778	1045	1765	1555	1913	1363

validated by testing for point defect energies, stacking fault energies, surface energies, and stresses**.** Three EAM potentials were developed by Mendeleev et al. [9] for studying phase transformation, melting, and defect formation accurately in Ti—Ti1, Ti2, and Ti3. Potential Ti1 is capable of accurately reproducing the melting and phase transformation of Ti hcp to bcc phases and is suited for bcc Ti. Potentials Ti1 and Ti2 are capable of capturing the defect properties and are employed to study plasticity or radiation damage in hcp Ti. Potential Ti2 was obtained by excluding the bcc melting temperature data from Ti1. Potential Ti2, thus, underestimates the bcc melting temperature while still correctly predicting the hcp-bcc transformation temperature. The drawback with potential Ti2 is that it does not capture the monotonic increase in both c and a lattice parameter with an increase in temperature. Potential Ti3 was developed to take care of this deficiency. Table 2 have the four sets of properties generated using all the three potentials along with the ideal target values for comparison. Table 2 enlists basic T = 0 K properties of Ti, Table 3 has point defect formation and migration energy value, Table 4 entails the planar defect formation energies, and Table 5 has information regarding phase transformation.

Table 2 Properties of crystal phases at T = 0 for Ti [8]

	Target value	Ti1	Ti2	Ti3
a (hcp) (Å)	2.951	2.947	2.949	2.
c/a	1.588	1.597	1.593	1.589
E_{coh} (eV/atom)	4.85	5.346	5.247	5.402
C_{11} (hcp, GPa)	176.1	161	160	165
C_{12} (hcp, GPa)	86.9	80	70	88
C_{44} (hcp, GPa)	50.8	53	54	58
C_{13} (hcp, GPa)	68.3	86	70	83
C_{33} (hcp, GPa)	172.5	169	165	166
C_{66} (hcp, GPa)	44.6	40	45	39
$\Delta E_{hcp\text{--->}\omega}$ (eV/atom)	0.005	0.005	0.007	0.016
a (fcc) (Å)	4.115	4.182	4.213	4.131
$\Delta E_{hcp\text{--->}fcc}$ (eV/atom)	0.059	0.059	0.052	0.053
a (bcc) (Å)	3.26	3.251	3.256	3.242
$\Delta E_{hcp\text{--->}bcc}$ (eV/atom)	0.099	0.029	0.074	0.089

Table 3 Formation energies (eV) of point defects in hcp Ti at T = 0 [8]

	Target value	Ti1	Ti2	Ti3
E_f^v	1.97	2.74	1.78	1.73
E_f^O	2.13	2.30	2.17	2.23
E_f^{BO}	2.25	2.30	2.28	2.29
E_f^{BS}	2.45	2.36	2.37	2.35
E_f^S	2.48	3.15	2.79	3.01

Table 4 Formation energies (mJ/m2) of planar defects in hcp Ti at T = 0 [8]

Defect	Target Value	Ti1	Ti2	Ti3
I_I basal stacking fault defect energy	148	130	119	118
I_2 basal stacking fault defect energy	259	257	236	236
E basal stacking fault defect energy	353	383	351	352
Prism stacking fault defect energy	250	208	257	255
[0001] free surface energy	124	86	120	141

Zope and Mishin [10] developed an EAM potential for the Ti-Al system applicable to α-Ti (hcp), Al, and Ti-Al. The EAM potential was generated by fitting a large number of data derived from experiments as well as ab initio calculations. The ab initio calculations were carried out in the framework of density functional theory using the linearized augmented plane wave (LAPW) method. Most of the

Table 5 Phase transformation data. for the hcp-bcc transformation temperatures; the top value is obtained from MD simulation and the bottom values are obtained from the LSMC [8]

	Target value	Ti1	Ti2	Ti3
$T_{\alpha--->\beta}$ (K)	1155	1150±5	1148±2	1148±2
		1152±3	1150±2	1149±1
			1151±2	
$\Delta H_{\alpha--->\beta}$ (eV/atom)	0.0435	0.022	0.032	0.041
$\Delta V_{\alpha--->\beta}/V_{\alpha}$ (%)		−0.70	−0.03	0.041
T_m (hcp, K)		1765±1	1277±1	1189±1
ΔH_m (hcp, eV/atom)		0.143	0.124	0.122
T_m (bcc,K)	1941	1918±1	1322±1	1210±1
ΔH_m (bcc, eV/atom)	0.157	0.130	0.097	0.083

empirical potentials are created by fitting methodology where the potential is fitted to various experimentally derived values of crystal properties. The most basic properties to which a potential is fitted are equilibrium lattice parameter a_0, the cohesive energy E_0, three elastic constants, and the vacancy formation energy Ev. In the case of non-cubic and anisotropic materials, like hcp crystals, additional values of elastic constants and equilibrium ratios are included in the dataset. In addition, planar fault energies and phonon frequencies or low-index surface energies are also used as complementary information in the fitting parameters. The calculation of such sensitive properties that have a direct connection to atomic behaviour from experiments is time and effort intensive. Therefore, a large set of accurate data on such metals' properties is lacking. Not only this but experiments can capture the behaviour of material only in a confined region. Ab into methods have proved handy in expanding this database as they can be tuned easily to calculate these properties for different atomic configurations.

Equilibrium lattice properties, vacancy characteristics, as well as stacking fault and surface energies in Ti computed using this EAM potential for Ti are compared to experimental data in Table 6.

Liu et al. [11] developed an EAM potential for Mg using the force matching method by fitting experimental and ab initio data. The developed potential fits the crystal properties and reproduces a reliable vacancy formation energy for Mg. The thermal properties as imitated by the potential are in good agreement with the experimental data for solid state, but not for the liquid state. The potential by Liu et al underestimates the melting point of hcp Mg by 180 K. It also overestimated the higher melting point of the bcc structure, which leads to a stable bcc structure up to the melting point of the hcp structure. Sun et al. [12] reflected on this drawback and developed an EAM potential for Mg which could predict the melting points and phases of Mg more accurately. They also employed the force matching method in conjunction with fitting the equilibrium melting temperatures as proposed by Sturgeon and Laird [13].

Table 6 Properties of Ti predicted by the present EAM potential in comparison to experimental data [10]

	Experiment	EAM
a_0 (Å)	2.951	2.951
c/a	1.588	1.585
E_0 (eV/atom)	4.850	4.850
C_{11} (GPa)	176	178
C_{12} (GPa)	87	74
C_{13} (GPa)	68	77
C_{33} (GPa)	190	191
C_{44} (GPa)	51	51
E^f_v (eV)	1.55	1.83
E^m_v (basal) (eV)		0.80
E^m_v (non-basal) (eV)		0.83
Q (eV)	3.14	2.62
γ_{I1} (mJ/m^2)		31
γ_{I2} (mJ/m^2)	290, 300	56
γ_E (mJ/m^2)		82
γ_S (0001) (mJ/m^2)	2100, 1920	1725

As mentioned earlier, due to its unique crystal structure and anisotropy, the hcp materials exhibit distinctive tensile and fracture behaviour. The lack of active slip systems and the presence of a large number of twin planes and directions make the study of hcp materials interesting. Not only this, but the presence of different types of grain boundaries—low and high angle, tilt and twin boundaries, and in bicrystalline or polycrystalline hcp material—affects the mechanical behaviour to a large extent. Thus, the nature of deformation in a single crystal can vary substantially from that of a bicrystal or polycrystal. The failure mechanism is also dependent on the crystal orientation or direction in which tension or compression is applied due to the presence of crystal anisotropy. The presence of point defects, in conjunction with cracks and grain boundaries, also greatly dominates the failure mechanism in any hcp material. In the subsequent sections, it is attempted to study the atomistic and continuum methods used to study single and bicrystal hcp materials for tensile, fracture, and irradiation properties, till date. Ouyang et al. [14] developed a long-range interaction analytical embedded atom method (namely la-EAM). This potential has been developed for the simulation of defects and phase changes in Zr. The potential is created by fitting the lattice constants, cohesive energy, mono-vacancy formation energy, and elastic constants of alpha-Zirconium. This EAM potential is validated for bulk, surface, and defect properties of Zr against experimental and theoretical results. These properties again include lattice constants, cohesive energy, elastic constants, and phonon dispersion curves of different phases of Zr. The potential is also capable

of reproducing point, line, and surface defects. In addition, it also gives good agreement of phase transformation energy barrier with experimental values. This potential also shows good accountability for long-range interactions of Zr atoms.

3 Application of EAM Potentials—MD Simulations to Predict Properties of HCP Materials

3.1 Yield Behaviour of Single Crystals

The molecular dynamics method has been widely used by researchers to catch the anisotropic behaviour of deformation in single crystal hcp materials. The asymmetric effect of tension and compression has been explored in these materials. The directional dependency of deformation behaviour has also been studied in many single crystal hcp materials. The results from computer simulations using EAM potentials have been found to be consistent with the experimental predictions. Twinning has been identified to be an important deformation feature in any hcp material. The activation of slip or twinning as a factor of loading directions is also explored in these works and methods have been developed to examine the slip–twin interactions.

Since hcp materials have a unique crystal structure and a varied combination of twinning and dislocations as deformation modes, it is of interest to have knowledge about the different slip and twinning systems active in them. Molecular Dynamics simulations provide a convenient way to explore these complicated features in hcp materials, which can be hard to achieve easily in experiments. Such simulation results give a close look into various atomistic events and factors that can be controlled in experiments to produce materials of required strength, ductility, etc.

Although a number of possible deformation modes present themselves in hcp Mg, its ductility and formability at room temperatures are low. An in-depth knowledge about the actual mechanism that occurs during failure in an hcp material is required in order to improve features like ductility and formability. Fan and El-Awady [15] have used atomistic simulations to look closely at the effect of orientation on the deformation of magnesium nanocrystals during tension, compression, and bending. In order to do so, the crystal c-axis orientation angle (θ) is incremented by 15° from 0° to 90° relative to the loading direction. These simulations were carried out to understand conditions for the activation of the various deformation modes which would be challenging to do in an experimental setup. With the rotation of the c-axis from the loading direction, a significant change in the deformation modes is observed. For tensile loading, at $\theta \leq 45°$, the deformation mode is TT, which changes to compression twinning at $\theta > 45°$. In contrast to this, during compression loading, dislocation slip is the dominant mode of failure. At $\theta < 15°$, dislocation slip is active on the pyramidal plane, and for $15° < \theta < 60°$, it is active on the basal plane. The prismatic slip is active during compression when θ exceeds 60°. As rotation increases, the

nucleation stress decreases for both tensile and compressive loading. The tension–compression asymmetry, which is a prevalent feature in any hcp crystal, was well captured in this study. During pure bending, there are both tensile and compressive loads acting on the crystal. The pure bending simulations revealed that yielding was a factor of the degree of weakness of the deformation mode present on the compression and tension side of the simulation cell. In addition, they also studied the relation between the Schmid factor, crystal orientation, and the active mode of failure. The agreement of CRSS from the Schmid factor analysis with the nucleation stress from MD prediction for each orientation validates the deformation features captured by the MD simulations. Thus, the EAM potential employed in these MD simulations gives visual insights into the deformation which would be hard to produce in experiments as such. Chang et al. [16] studied the effect of strain rate on tensile properties, structure evolution, and activation of twin variants during tensile loading of Ti single crystal nanowires along [0001] direction. In another study, Chang et al. [17] also studied the effect of loading direction/crystal orientation on the deformation mechanism of Ti single crystal nanowires. It was observed in their work that the crystal with [0001] and [11–20] have the highest and lowest strain rate sensitivity. At a strain rate above 10^{10} s^{-1}, failure is caused by the rearrangement of disordered atoms causing the superplastic behaviour in the nanocrystal of Ti. However, below 10^{10} s^{-1}, the nanocrystals behave differently. The active modes of deformation in crystal oriented along [0001] is {10–12} twinning and that in [−1100] orientation is prismatic slip <a> for all strain rates. These modes of failure have the highest Schmid factor for the corresponding orientations. Thus, the results using this EAM potential correlate well with the theoretical predictions. Although the Schmid factor for all the {10–12} twin variants are the same, only some of these can be actually activated, which was not known from theoretical estimation. The number of activated twin variants is directly proportional to the applied strain rate. The crystal with [11–20] loading orientation has a complex failure mechanism. Partial dislocations and deformation faults along with {10–11} twinning were observed. The 1/3 [−1100] Shockley partials drive transformations between compression twins and deformation faults. This leads to plasticity and increases the fracture strain for this orientation much higher than in the other two orientations. All this information can be useful when trying to understand the various complex failure features that occur in anisotropic hcp crystals. Rawat and Mitra [18] investigated the c-axis compression of a single crystal Ti under uniaxial stress conditions. It has been indicated earlier that the accuracy of the results of any molecular dynamics simulations is dependent on the potential used. In their work, Rawat and Mitra [18] compared the deformation behaviour of two interatomic potentials for Ti—namely Kim [19] and Henning [20]. They characterized the failure mechanism in terms of stress–strain response, structural growth, activation of compression twins along with variants, dislocation slip and density, and the twin volume fraction. They observed from the stress–strain response that the yield stress from both the potentials was drastically different. Not only this, but the stress–strain curve had a completely different trend having one and two peaks with Kim [19] and Henning [20] potential, respectively. The presence of the different number of peaks is indicative of the fact that both the potentials capture

microstructural growth differently. They also studied the strain rate sensitivity of both potentials by carrying out simulations with two different strain rates—10^9 s^{-1} and 10^8 $s^{-1.}$ They observed that both the potentials gave quite different results in terms of the factors studied. However, the Henning [20] potential was found to be consistent with the Schmid factor for activation of the twin system but not the Kim [19] potential. The stress–strain curve for both potentials followed the same trend at both the strain rates. In another work, Rawat and Mitra [21] have attempted to study the compression twinning under uniaxial strain conditions in directions other than the [0001] axis. In this work also, they compared the results of both potentials. The basal–prismatic and prismatic–basal interfaces were captured by both the potentials during loading along [1–10] direction. The activation of {10–12} twin variants in this study was consistent with the calculated Schmid factors. It was observed that processes like nucleation, coalescence, and detwinning were all involved in the evolution of final twins. The random nature of activation of the twin variants is suggested by the fact that even the twins with the same Schmid factor do not activate simultaneously. Also, the size of the variants is different. Another important observation was the presence of other modes of deformation like phase transformation along with twin activation. Rawat and Mitra [22] also studied the c-axis compression of single crystal Ti under uniaxial strain conditions. They compared four most commonly used potentials for Ti—Ackland [8], Mishin [10], Kim [19], and Henning [20]—to study the deformation mechanism. Based on their findings, they recommended the Kim potential for capturing the c-axis compression of a single crystal Ti. The stacking fault energy calculated with Kim potential was the closest to the experimental findings. Both twins and dislocations were observed during the microstructural growth in Ti. Both {10–11} and {11–22} or either of the twins activate depending upon the type of potential employed. As before, the activation of twins even with the same Schmid factor was not simultaneous. Honggang et al. [23] studied the c-axis deformation of magnesium single under tension by using molecular dynamics-based simulations at different temperatures. The EAM potential used in this work was developed by Liu et al. [11] by fitting to both ab initio forces and experimental data. The potential is known to give good results for various bulk structural properties. The temperature has a significant effect on the activation of twins during deformation. It was found that at low temperatures, both {10–12} and {10–11} twins get activated. However, as the temperature increased, the number of {10–12} twins decreased. The primary deformation mechanism in c-axis tension in single crystal Mg was found to be the {10–12} twin. In addition, shear bands also presented themselves during the failure mechanism. At temperatures above 500 K, the thermally activated non-basal slip was also observed. Luque et al. [24] performed tension and compression simulations on Mg single crystal nanopillars of two sizes along two orientations using MD simulations, using potentials of Sun [12] and Liu [11]. On loading the sample along [0001] direction, the crystal deformed by twinning. In contrast, for the second orientation, with loading direction being [1–11], the mode of deformation observed was a basal slip. The effect of sample size on twin nucleation stress could be noticed. The failure mechanism, under tension, for both the potentials was identical for loading along [1–11] direction, where both leading and trailing partials were present. However,

the mechanism of activation of basal slip varied depending on the type of potential utilized during compression. Sun potential captures the basal slip to be activated by both leading and trailing partials whereas for the Liu potential, only the leading partial can be observed. The tension/compression asymmetry of the Liu potential can be correlated with the difference in the values of stacking fault energies and Peierls stress which are calculated using Sun on the basal plane.

3.2 Dislocations and Generalized Stacking Fault Energy in HCP Materials—MD-Based Studies

Another outstanding deformation feature in an hcp material is the activation of <c+a> dislocation, which controls the plastic behaviour of an hcp material. A <c+a> dislocation can exist both on Py-I and Py-II planes. Wu and Curtin [25] carried out simulations on Mg that revealed the cross slip of <c+a> dislocations from pyramidal I to pyramidal II plane that takes place through cross slip of individual partial dislocations. They used nudged elastic band (NEB) methodology to observe the cross slip of <c+a> screw dislocations in Mg, which is generic to all hcp materials. They identified that the core energy of dislocation is smaller on pyramidal II with respect to the pyramidal I plane. The cross slip process exhibits three unique stages: first is nucleation which is followed by dislocation propagation and ends with the annihilation of the dislocation. These stages are found to be independent of the dislocation length. However, the energies along the MEP for each stage are directly dependent on the dislocation line length for II-I cross slip. It is due to this increased energy barrier that a relatively long dislocation is unlikely to cross slip from pyramidal II to I plane. However, due to the low energy barrier for a slip from pyramidal I to II plane, dislocation of any length can easily cross slip. Tang and El-Awady [26] have concentrated their study on the <c+a> pyramidal dislocation during the <c>-axis uniaxial loading of a Mg single crystal. It was elucidated in their work that although there are four possible planes for <c+a> dislocations, the pyramidal I plane is favoured, being a super plane with the highest areal density along with the lowest general stacking fault energy. The <c+a> dislocations form on the pyramidal I plane by sequential leading and trailing partials. In addition, the pyramidal I plane also experienced local shuffling which was reasoned for the nucleation and slip of <c+a> pyramidal I dislocation. The appearance of slip on pyramidal II planes, for moderate strains, was found to be consistent with experiments. The slip on pyramidal II plane was a result of the transition of pyramidal I slip to pyramidal II planes. This transition occurred as a result of cross slip from one plane to another, double cross slip, or cooperative slip of edge segments on two equal and alternative pyramidal I plane. At very low temperatures, <c+a> dislocations exhibited climb on both pyramidal I and II planes. As a result of this phenomenon of the climb, interstitials and vacancies were formed in the crystal. During this study, the formation of <a> pyramidal I dislocation was observed to be formed by the reaction of two pyramidal <c+a> dislocations. The

<c+a> dislocations are not well understood in the hcp crystals. A lot of researchers have shown interest in understanding the nature of both pyramidal I and pyramidal II. MD and molecular statics (MS) simulations and correlating the work with experimentally observed phenomenon. The dissociation of <c+a> dislocations into partials have been studied by Kumar et al. [27] in Mg where the trend of general stacking fault energy (GSFE) calculated with DFT, EAM, and MEAM potentials indicated the possibility of dissociation of <c+a> into partials, as observed experimentally. The correct GSFE trend corresponding to the splitting into partials was obtained after an atomic shuffling was allowed during the calculation of GSFE. MS simulations were performed using MEAM potential by introducing a full partial on the pyramidal II plane by using an elastic Volterra displacement field, which dissociated into two partials.

A similar kind of work was done by Ghazisaeidi et al. [28], where they studied the core structures of edge and screw <c+a> dislocations using Kim [29], Liu [11], and Sun [12] potential in Mg. These core structures can be seen in Fig. 1. They also calculated the GSFE curve for pyramidal II plane using DFT, MEAM, and common Liu [11] and Sun [12] potentials and predicted the dissociation of full <c+a> dislocation into partials on pyramidal II plane.

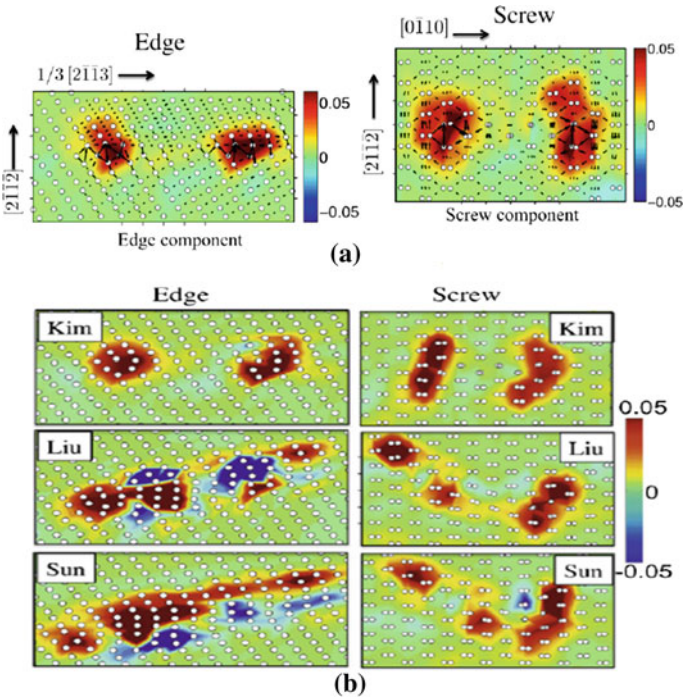


Fig. 1 Core structures of edge and screw dislocations using (**a**) DFT and (**b**) other potentials (edge/screw component of Nye tensors are plotted for edge/screw dislocations) [28]

Buey and Ghazisaeidi [30] used MD simulations to study the glide behaviour of <c+a> screw dislocations in Mg at different temperatures. It was found that pyramidal II was a preferred slip plane 300 K. On application of stress at lower temperatures, the cross slip was observed between the two pyramidal planes. The screw core energies were estimated and pyramidal II planes were found to have lower core energies in comparison to pyramidal I plane, similar to the findings of Wu and Curtin [25]. This made pyramidal II plane the preferred configuration. The dissociation of full dislocations into partials was also observed, which was asymmetric in nature. In addition to this, they discovered a compact core metastable structure during the cross slip from pyramidal II to pyramidal I plane at 0 K and 70 K. Tang and El-Awady [31] explored the glissile <c+a> dislocations with near-edge, near-screw, and screw characters on pyramidal I planes. The Peierls stress estimated in this work indicates a clear anisotropy between each dislocation character. The screw-type <c+a> had the lowest Peierls stress, whereas the near-edge had the highest one. It was seen in their study that local shuffling significantly lowered the Peierls stress for near-edge-type dislocation. However, no such significant effect was seen for near-screw and screw dislocations. Thus, these atomistic simulations using the EAM potential proved that shuffling can mediate slip for a near-edge <c+a> dislocation. The already established tension–compression asymmetry was also reproduced in this work, from the fact that <c+a> dislocation movement is initiated at a high stress when under compression than when under tension. A lot of other researchers have also shown interest in properties of <c+a> edge and screw dislocations using molecular dynamics simulations using Sun [12] and Liu [11] potentials for Mg. Nogaret et al. [32] calculated the gamma surfaces for pyramidal I and pyramidal II planes using MD simulations. The stable and unstable stacking faults energies were found to be in agreement with experiments. In contrast to experiments, where the <c+a> dislocations were found to glide on the {11–22} pyramidal II plane under large stress, in this work the dislocations were found to be active on the {1–101} pyramidal I plane. The lower stacking fault energy justifies the pyramidal I glide over pyramidal II glide. The unstable stacking faults in the <a> direction on the pyramidal I are higher than those previously predicted on prism planes. This justifies the movement of <a> dislocation on the prism plane rather than the pyramidal I plane. Different types of <c+a> screw and edge dislocation cores were studied for their energies and positive and negative critically resolved shear stress (CRSS) under tension and compression at 0 K and 300 K. Zhang et al. [33] discovered a new outlook on plasticity in hcp Mg in his MD-based work where they observed slip of <c+a> dislocations on new slip planes, {12–11}, which is found to have low Peierls stress and comparable GSFE to pyramidal I and II planes. The maximum GSFE is observed for the (12−11) plane and is close to GSFE values for the Py-II plane. In contrast, Py-I and prism planes have comparatively lower maximum GSFE values. In addition, they also observed recurrent cross slip between the dislocation cores which indicated a low cross slip energy barrier in comparison to the Peierls stress. The cross slip of <c+a> dislocation from pyramidal II plane to pyramidal I suggested pyramidal I plane to be more stable. Fan et al. [34] used MD-based simulations to study the effect of temperature on the mobility of <c+a> dislocations on pyramidal planes

in Mg. They concluded from their work that the pyramidal I planes are active at lower temperatures, whereas at higher temperatures both orders of pyramidal planes are active. This conclusion was drawn from the trend of CRSS calculated for both the planes with respect to the change in temperature. The activation of both the pyramidal slip planes explains the higher plasticity of Mg at elevated temperatures. The unlikely stability of <c+a> dislocation on the pyramidal I plane was studied by Wu and Curtin [35]. Through molecular dynamics simulation, they studied the intrinsic transformation of <c+a> dislocations in Mg into either a sessile, basal-dissociated <c+a> or a sessile basal-dissociated ⟨c⟩ dislocation plus an ⟨a⟩ dislocation, by thermal activation. The dislocation core energy computed in their work indicated the stability of the climb dissociated <c+a> basal configuration, which is sessile in nature and thus points towards limited plasticity of hcp Mg. In a similar work, Rasool et al. [36] explored the dissociation of <c+a> dislocations from pyramidal II to sessile basal dislocation in Mg causing poor ductility and the possibility of improvement of plastic behaviour by alloying Mg with Y, which stabilizes the pyramidal <c+a> dislocations. For this, they carried out MD simulations to calculate the effect of solute on Mg-3%Y on the transition mechanism, energy barrier, and time, and no significant effects were observed. They concluded that the improvement in ductility was not related to the stability of <c+a> pyramidal dislocations, but to the one described in the work of Wu et al. [25] which stated that favourable solutes have the ability to increase <c+a> cross slip and multiplication at a faster rate when compared to the PB transition rate. This helps to circumvent the adverse consequences of PB transition. Fan and El-Awady [37] have attempted to reduce the anonymity of pyramidal slip in Magnesium by calculating the CRSS of pyramidal I and pyramidal II dislocations during c-axis compression, using MD-based simulations. The critical stress for <c+a> dislocations for c-axis compression on pyramidal II plane was found to be seven times higher than that for the pyramidal I plane, making the activation of pyramidal I plane the major contributor to the plasticity of Mg single crystal during c-axis compression.

3.3 MD Simulations to Generate and Investigate Bicrystalline and Polycrystalline HCP Metals

The presence of line defects like grain boundaries (GB) in any real material affects the mechanical properties of polycrystalline materials. Grain boundaries, being at higher stress, can act as a source of dislocations or twinning. Grain boundaries can be either low or high angle boundaries,

symmetric or asymmetric, and tilt or twist boundaries. The misorientation angle between them is a major factor in defining the effect of the boundaries on the overall deformation mechanism of a crystal. Molecular dynamics have been employed to define grain boundaries in bicrystals and polycrystals of hcp materials in order to consider a more realistic material for elucidating the failure mechanism using

computer-based simulations. Due to anisotropy in the hcp crystal, the grain boundaries about different grain-boundary planes or axes make an interesting field of study. However, the extent of study on a bicrystalline or polycrystalline hcp is not as extensive as the single crystal of hcp materials using computational techniques. One of the most recent works on grain boundaries on an hcp material has been done by Divya and Avinash [38]. They have elucidated the generation of GBs along two axes [0001] and [0–110] using MD-based simulations and have calculated the GB energies and correlated them with the GB structures [38]. The disordered structure at the ATGBs increases their energy with respect to the corresponding STGB with the same misorientation angle. The GBs with low misorientation angles have distinct and separate dislocation cores which merge as the misorientation angle is increased. The grain-boundary energies along the [0001] axis are somewhat similar in trend to those obtained for Ti by Wang et al. [39]. The GB energies along [0–110] for Zr were compared to the ones calculated by Solanki and Bhatia [40]. The EAM potential used in their work gives a good estimate of the GB structures and energies. In a separate work to understand the tensile deformation mechanism in α-Zr, GBs were again generated along [0001] and [0–110] tilt axes by Divya et al. [41]. A more detailed structure of lower and higher angle of the STGBs and ATGBs was shown in this work. Wang and Ye [39] studied the grain-boundary energies in hcp Ti with tilt GBs along [0001] using the embedded atom method interatomic potential. It was found that the core structures of high angle tilt GBs were similar to low angle tilt GBs. The dislocation cores have either T or A units or a combination of both. The GB energies monotonically increase with an increase in the misorientation angle. In general, if a dislocation core unit has the possibility of either T and A types core, the GB energy corresponding to the A-type core is comparatively higher. Hammerschmidt et al. [42] exclusively studied the GB structure of a Ti bicrystal with a misorientation angle of 27.8 and a sigma 7 CSL. Zheng et al. [43] studied the grain boundaries in Ti along the [0001] tilt axis for both symmetric and asymmetric tilt grain boundaries. Wang and Beyerlin [44] studied the STGBs along [0–110] tilt axis in Mg and Ti. Two types of STGBs are obtained—coherent STGBs and STGBs with grain-boundary dislocations (GBDs). Two STGBs are perfect defect-free crystals with tilt angles of 0 and 90, respectively, and imaginary GB plane as basal and (11–20) planes. Between them are four STGBs with tilt angles of 28.4°, 39.06°, 58.36°, and 72.88° that correspond to the cusps in GB energies vs θ plots and are called as coherent STGBs. Out of these, twin boundaries correspond to GB planes (−2114), (−2112), (−2111) with 39.06°, 58.36°, and 72.88°, as misorientation angles along the [0–110] axis, respectively. The two defect-free GBs along with the coherent GBs are called as special GBs. The other GBs which lie intermittent to these angles are STGBs with GBDs. These other STGBs have GB energies higher than the nearest special STGBs. The same phenomenon can be seen for both Ti and Mg. In a similar study, Wang et al. [45] modelled also GBs along the [1–210] axis where again six minimum energy coherent boundaries were observed. Two were the defect-free GBs for tilt angles 0 and 90, respectively. Out of the other four coherent boundaries, three were twin boundaries. All lower energy coherent boundaries in both the works of Wang are free of dislocations. The twin boundaries along [1–210] axis correspond to GB planes (−1013)

(−1012) (−1011) with rotation angles 31.99, 43.11, and 61.91, respectively. The last coherent boundary corresponds to the (−2021) plane with a tilt angle of 75.06. The bicrytsals with tilt angles in between these coherent GBs have higher energies than them that are made of grain-boundary dislocations (GBDs). Saitoh et al. [46] studied the grain boundaries in Mg along the [1–100] axis through MD-based simulations, based on coincidence site lattice (CSL). The CSL plays a major role in determining the rotation angle for each crystal-forming STGBs. A range of misorientations starting from 11.7° to 146° has been generated by them. Farkas [50] employed EAM potential to study the coincident and near coincident symmetrical tilt grain boundaries in Ti and Co along [1–100]. The near coincidence boundary is an important feature, especially in the case of hcp materials. Farkas [47] developed a structure unit-type model to study the effect of c/a ratio on the grain-boundary structure. The structures obtained for both Ti and Co were found to be very similar, but the energy of the GB in Co was estimated to be higher than Ti by a factor of two. Since the coincidence ratio in the grain-boundary plane depends on the *c/a* ratio in an hcp material, a structural unit-type model can accurately predict the grain-boundary structure variation with a change in c/a ratio. The model can predict a series of symmetrical tilt boundaries with the grain-boundary plane oriented perpendicular to a fixed crystallographic direction and varying *c/a* ratios. It was concluded in this particular work that near coincident boundaries have a similar trend to that of the exact coincidence boundaries provided a coincident periodic structure in the grain-boundary plane exists. They also found that the structural units were independent of the c/a ratio. Working on similar lines, Wu et al. [48] have predicted the misorientation dependence of GB energies by utilizing a disclination structural unit model in Ti and Co for grain boundaries along the [1–100] axis. They also estimated the dipole field stress and established that it depends upon the misorientation angle similar to the GB energies. In addition to the above work, the twin GBs boundaries in hcp bicrystals have been also simulated using MD simulations. Tohei et al. [49] generated a {1011} twin boundary in alumina using atomistic simulations. The {1011} twin boundary was constructed as a supercell by bringing together two single crystals with a GB plane of (1011) while rotating one crystal at 180° with respect to the other along the normal to this plane. Further, rigid body translations were employed to obtain a minimum energy stable structure along all the three principle axis directions. Yang et al. [50] constructed {10–12} twin boundary in a Mg bicrystal nanopillar using MD simulations. MD simulations have been used to model polycrystalline hcp materials as well. The two most widely used methodologies employed to generate polycrystalline structures are either Voronoi tessellation [51] or the one given by Kim et al. [52]. Different textures of polycrystals have been reported in the literature in which grains are oriented about a common axis. [0001] basal, [1–100] prismatic, and [11–20] textured polycrystalline hcp have been generated in different works employing either of the methods. Thus, intricate details of different types of GB structure and energies even for an hcp material with a highly complex crystal structure can be obtained by using a good fitting EAM potential in the MD and MS simulations.

3.4 *Fracture Properties Single Crystals, Bicrystalline, and Polycrystalline HCP Metals Using MD-Based Simulations*

In the most recent work on a single crystal of α-Zr, by Divya et al. [53], the change in deformation mechanism with change in the crack plane has been studied through MD-based simulations. Three crack planes, viz. basal, prismatic, and pyramidal as depicted in Fig. 2, were generated and further subjected to a tensile load in a direction perpendicular to the crack surface.

Simulations were also done with rotation of the single crystal along the c-axis, such that the angle between loading direction and the crack surfaces varied. This rotation was found to have a negligible impact on the deformation mode or the yield strength. It was found in the MD simulations that the deformation of a single crystal with cracks in the basal and prismatic planes is dominated by dislocations evolution. However, the mode of deformation for a crystal having a crack in the pyramidal plane is governed by twinning. These simulation results were validated with the Schmid factor and dislocation densities. Twinning was exhibited by crystal for which the Schmid factor for dislocations gliding corresponds to zero. For basal and prismatic planes, the crystal with higher critical stress (the UTS) had a lower Schmid factor and vice versa. The crack tip behaviour was also observed to be affected by the deformation modes. Dislocations and twin dislocations can be observed to be emitted from the crack tips. The change in crack length is maximum for twinning and minimum for crystals having perfect dislocations. The dislocations help in blunting the crack tip whereas twinning causes the breaking of bonds at the crack tip. The EAM potential was able to capture important information on the crack properties of a Zr (or in general an hcp crystal) in terms of change in the crack plane. Such crucial information would otherwise be difficult to gain directly through current experimental methodologies. Tang et al. [54] also studied the effect of central crack oriented in different planes. As expected, the rate and nature of fatigue crack growth were closely dependent on the crack orientation. A difference in crack growth rate and

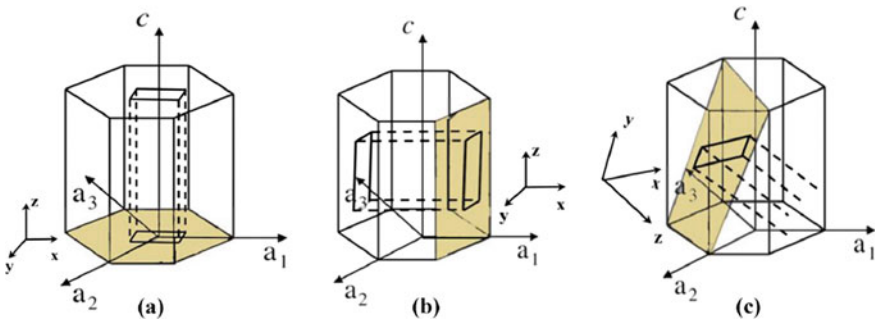


Fig. 2 Schematic of different orientations of the crack plane in a single crystal of Zr (**a**) basal, (**b**) prismatic, and (**c**) pyramidal planes [53]

resistance to fatigue crack growth was observed based on the crack plane orientation. Both twins and dislocations were observed to have emitted from the crack tip due to stress concentration in all the cases. Temperature effects on fatigue crack growth were also a function of the crack plane orientation. For the second part of the study, they used Edge Crack (EC) specimens. In a continued effort to understand fatigue crack growth in Mg, Tang et al. [55] studied the effect of the crack plane on the rate of mode I fatigue crack growth in a single crystal of magnesium. Unlike the centrally embedded crack in the study by Divya et al. [53] and their own previous study, Tang et al. [55] al used an edge crack model for fatigue studies. Orientation of the crack was again found to dominate the mode of failure and fracture behaviour including the fatigue growth rate. The value of fatigue growth constant as well as deformation mode varied for each orientation. In a separate work, Tang et al. [56] studied the fracture in a single crystal Mg with an initial central crack as a factor of crack orientation, temperature, uniaxial loading strain rate, and the sample length scale using MD-based simulations. The ultimate stress goes on decreasing as the size of the specimen was increased. The size effect is more prominent at small sizes and becomes smaller with the increased specimen size. An increase in strain rate, crack size, and temperature all reduce the specimen strength. An increase in temperature weakens the deformation of twinning. Thus, sharpening of the crack tip is reduced and crack growth diminishes with increases in temperature. The crack growth in single crystal Ti was explored as a function of strain rate and temperature change for uniaxial loading conditions by Zhang et al. [57]. An edge crack is considered in this work and once again the crack tip was the primary source of deformation initiation.

All the results obtained in this study through MD simulations were found to be in agreement with the experimental results. Voids and stress concentration in pure Ti is found to be responsible for materials failure, as also observed in experiments. Phase transformation was also present along with stacking faults during the deformation. As the temperature is increased from 200 K to 400 K, the emission of stacking faults increases. On the other hand, an increase in the strain rate leads to more symmetric stacking faults. For a single crystal Mg, the Linear Elastic Fracture Mechanics (LEFM) predictions along with atomistic simulations are used to study an underlying mechanism behind the competition between brittle cleavage and ductile fracture, with the change in the crack plane by Wu and Curtin [58]. The predicted critical stress intensity factor for both Griffith cleavage (K_{Ic}) and dislocation emission (K_{Ie})by LEFM were in agreement with the atomistic simulations. Rawat and Joshi [59] studied the effect of multiaxial loading, with variation in both loading and constrains direction, in a Mg single crystal containing a pre-existing crack.

3.5 MD-Based Simulations to Study Point Defects Formation and Migration in HCP Metals

Molecular dynamics simulations have been extensively used to understand the point defect formation and migration/diffusion energies in hcp materials. Given the unique crystal structure, the point defects formation is an intriguing area, which has been explored by various researchers with respect to the positioning of the defects in the hcp lattice structure. In addition, some limited work has been done to study irradiation and survival of vacancy and interstitial defects post-irradiation in single crystals and bicrystals. The interaction of point defects with grain boundaries and their effect on the formation and diffusion energies have also been studied. In 1990, Kapinos et al. [60] studied the displacement cascades by applying a thermal spike mode using MD in hcp Zr. The diffusional rearrangement in the depletion zone of the cascade is investigated. Vacancy clusters were the prominent result of this work. Burger's vector of the vacancy loops up to 5% spatial distribution of vacancy clusters in the depleted zone after thermal spike was in the {10–10} planes with Burgers vector <10–10> or 1/3 <11–20>. As the vacancy cluster concentration increased, the loops are primarily bound to the basal plane with a Burgers vector of 1/6 <20–23>. Wen et al. [61] studied the formation and diffusion of self-interstitials in Ti. They calculated the self-interstitial formation energies at all possible points in the hcp lattice. These are the tetrahedral (T), octahedral (O), basal tetrahedral(BT), basal octahedral (BO), basal crowdion (CB), non-basal crowdion (CN), c-dumbbell (DC), and the basal dumbbell (DB). They concluded the basal dumbbell position to be the most stable. In order to study the effect of anisotropy on the defect formation and migration energies, they calculated the values under compressive stress in three directions [0001], [1–100], and [11–20]. Under compressive stress, the basal dumbbell position of the interstitial atom was again found to be the most stable. The change in the migration energy, for both in-plane and out-of-plane diffusion, was significant when the load was applied along [11–20] direction when compared to the load along [0001] or [1–100] directions. Osetsky et al. [62] also attempted to study the anisotropy of vacancy and interstitial diffusion in hcp Zr through molecular statics (MS) and MD-based simulations. The potential employed for the work established the basal split and basal crowdion as the most stable interstitial sites. Here, MS was used to calculate migration barriers for vacancy and the <11–20> crowdion. MD simulations were employed to study the diffusive jumps and calculate diffusion coefficients from these results, at various temperatures. The diffusion coefficients are calculated by the. They defined the anisotropy ratio AR1 as the ratio of Arrhenius plots fitted to a and c components of the interstitial diffusion coefficient. AR2 estimates AR for the a and c components of the self-diffusion coefficient individually in each particular simulation run. AR was indicative that the vacancy diffusion was nearly isotropic in the temperature range of 1050 K to 1650 K. In the initial range of the temperature, the vacancy diffusion did show some inclination towards basal plane diffusion. The interstitial diffusion shows anisotropy by varying its trend from 1D, 2D, and 3D, depending upon the temperature. The diffusion anisotropy difference (DAD) between the two types of

point defects—vacancy and interstitial—has been studied by Woo and Liu [63] using molecular dynamics simulations in Zr. The six defect positions were categorized broadly into three kinds of interstitial positions—stable, metastable, and unstable. The formation and in-plane and out-of-plane migration energies for vacancies have been listed and were also calculated in this study. Mendelev and Bokstein [64] have studied the self-diffusivity in Zr using EAM potential #Zr. The defect formation energies, the migration energies, and the diffusivity have all been estimated as a function of temperature.

References

1. Igarashi, M., Khantha, M., Vitek, V.: N-body interatomic potentials for hexagonal close-packed metals. Philosophical Magazine B **63**(3), 603–627 (1991). https://doi.org/10.1080/13642819108225975
2. G. J. Ackland, S. J. Wooding & D. J. Bacon (1995) Defect, surface and displacement-threshold properties of α-zirconium simulated with a many-body potential. Philos. Mag. A 71(3), 553–565. https://doi.org/10.1080/01418619508244468
3. Biersack, J.P., Ziegler, J.F. (1982) The stopping and range of ions in solids. In: Ryssel, H., Glawischnig, H. (eds) Ion Implantation Techniques. Springer Series in Electrophysics, vol 10. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-68779-2_5
4. Goldstein A.S., Jónsson H.: An embedded atom method potential for the h.c.p. metal Zr. Philos Mag B 71(6), 1041–1056 (1995). https://doi.org/10.1080/01418639508241894
5. Pasianot, C., Monti A.M.: A many body potential for α-Zr. Application to defect properties. J. Nuclear Mater. 264(1–2), 198–205 (1999). https://doi.org/10.1016/S0022-3115(98)00477-2
6. Mendelev, M.I., Ackland, G.J.: Development of an interatomic potential for the simulation of phase transformations in zirconium. Philos. Mag. Lett. **87**(5), 349–359 (2007). https://doi.org/10.1080/09500830701191393
7. Varvenne, Céline., Mackain, Olivier, Clouet, Emmanuel: Vacancy clustering in zirconium: an atomic-scale study. Acta Mater. **78**, 65–77 (2014). https://doi.org/10.1016/j.actamat.2014.06.012
8. Ackland, G.J.: Theoretical study of titanium surfaces and defects with a new many-body potential. Philos. Mag. A **66**(6), 917–932 (1992). https://doi.org/10.1080/01418619208247999
9. Mendelev, M.I., Underwood, T.L., Ackland, G.J.: Development of an interatomic potential for the simulation of defects, plasticity, and phase transformations in titanium. J. Chem. Phys. **145**(15), 154102 (2016). https://doi.org/10.1063/1.4964654
10. Zope, R.R., Mishin, Y.: Interatomic potentials for atomistic simulations of the Ti-Al system. Phys. Rev. B **68**(2), 024102 (2003). https://doi.org/10.1103/physrevb.68.024102
11. Liu X.-Y., et al.: EAM potential for magnesium from quantum mechanical forces. Modell Simul. Mater. Sci. Eng. 4, 293 (1996)
12. Sun, D.Y., Mendelev, M.I., Becker, C.A., Kudin, K., Haxhimali, T., Asta, M., Hoyt, J.J., Karma, A., Srolovitz, D.J.: Crystal-melt interfacial free energies in hcp metals: a molecular dynamics study of Mg. Phys. Rev. B **73**(2), 024116 (2006). https://doi.org/10.1103/physrevb.73.024116
13. Sturgeon, J.B., Laird, B.B.: Adjusting the melting point of a model system via Gibbs-Duhem integration: application to a model of aluminum. Phys. Rev. B **62**(22), 14720–14727 (2000). https://doi.org/10.1103/physrevb.62.14720
14. Ouyang, Yifang, Jizheng, Wu., Zheng, Minghui, Chen, Hongmei, Tao, Xiaoma, Yong, Du., Peng, Qing: An interatomic potential for simulation of defects and phase change of zirconium. Comput. Mater. Sci. **147**, 7–17 (2018). https://doi.org/10.1016/j.commatsci.2018.01.049

15. Fan, H., El-Awady, J.A.: Molecular dynamics simulations of orientation effects during tension, compression, and bending deformations of magnesium nanocrystals. ASME. J. Appl. Mech. 82(10), 101006 (2015). https://doi.org/10.1115/1.4030930.
16. Chang, Le., Zhou, Chang-Yu., Wen, Lei-Lei., Li, Jian, He, Xiao-Hua.: Molecular dynamics study of strain rate effects on tensile behavior of single crystal titanium nanowire. Comput. Mater. Sci. **128**, 348–358 (2017). https://doi.org/10.1016/j.commatsci.2016.11.034
17. Chang, Le., Zhou, Chang-Yu., Liu, Hong-Xi., Li, Jian, He, Xiao-Hua.: Orientation and strain rate dependent tensile behavior of single crystal titanium nanowires by molecular dynamics simulations. J. Mater. Sci. Technol. **34**(5), 864–877 (2018). https://doi.org/10.1016/j.jmst.2017.03.011
18. Rawat, Sunil, Mitra, Nilanjan: Molecular dynamics investigation of c-axis deformation of single crystal Ti under uniaxial stress conditions: evolution of compression twinning and dislocations. Comput. Mater. Sci. **141**, 19–29 (2018). https://doi.org/10.1016/j.commatsci.2017.09.015
19. Kim, Y.-M., Lee, B.-J., Baskes, M.I.: Modified embedded-atom method interatomic potentials for Ti and Zr. Phys. Rev. B **74**(1), 014101 (2006)
20. Hennig, R.G., Lenosky, T.J., Trinkle, D.R., Rudin, S.P., Wilkins, J.W.: Classical potential describes martensitic phase transformations between the, and titanium phases. Phys. Rev. B **78**(5), 054121 (2008)
21. Rawat S., Mitra N.: Twinning, phase transformation and dislocation evolution in single crystal titanium under uniaxial strain conditions: a molecular dynamics study. Comput. Mater. Sci. 172, 109325 (2020). https://doi.org/10.1016/j.commatsci.2019.109325
22. Rawat, Sunil, Mitra, Nilanjan: Compression twinning and structural phase transformation of single crystal titanium under uniaxial compressive strain conditions: comparison of interatomic potentials. Comput. Mater. Sci. **126**, 228–237 (2017). https://doi.org/10.1016/j.commatsci.2016.09.034
23. Honggang, Q.I., Yafang, G.U.O., Xiaozhi, T.A.N.G., Shuang, X.U.: Atomistic simulation of the structural evolution in magnesium single crystal under c-axis tension[J]. Acta Metallurgica Sinica (English Letters) **24**(6), 487–494 (2011)
24. Luque, A., et al.: Deformation modes in magnesium (0 0 0 1) and (01$\bar{1}$1) single crystals: simulations versus experiments. Modell. Simul. Mater. Sci. Eng. **21**, 045010 (2013)
25. Wu, Z., Curtin, W.A.: Mechanism and energetics of ⟨c + a⟩ dislocation cross-slip in hcp metals. PNAS 113(40), 11137–11142 (2016). https://doi.org/10.1073/pnas.1603966113
26. Tang, Y., El-Awady, J.A.: Formation and slip of pyramidal dislocations in hexagonal close-packed magnesium single crystals. Acta Mater. 71, 319–332 (2014). https://doi.org/10.1016/j.actamat.2014.03.022
27. Kumar, A., Morrow, B., Mccabe, R.J., Beyerlein, I.J.: First-principles study of <c+a> dislocations in Mg. Mater. Sci. (2016)
28. Ghazisaeidi, M., Hector, L.G., Curtin, W.A.: First-principles core structures of ⟨c+a edge and screw dislocations in Mg. Scripta Mater. **75**, 42–45 (2014). https://doi.org/10.1016/j.scriptamat.2013.11.013
29. Kim, Y.-M., Kim, N.J., Lee, B.-J.: Atomistic modeling of pure Mg and Mg–Al systems. Calphad **33**(4), 650–657 (2009). https://doi.org/10.1016/j.calphad.2009.07.004
30. Buey, D., Ghazisaeidi, M.: Atomistic simulation of <c+a>screw dislocation cross-slip in Mg. Scripta Mater. **117**, 51–54 (2016). https://doi.org/10.1016/j.scriptamat.2016.02.001
31. Tang, Y., El-Awady, J.A.: Highly anisotropic slip-behavior of pyramidal I c+a dislocations in hexagonal close-packed magnesium. Mater. Sci. Eng.: A 618:424–432 (2014). https://doi.org/10.1016/j.msea.2014.09.032
32. Nogaret, T., Curtin, W.A., Yasi, J.A., Hector, L.G., Trinkle, D.R.: Atomistic study of edge and screw ⟨c+a⟩ dislocations in magnesium. Acta Mater **58**(13), 4332–4343 (2010). https://doi.org/10.1016/j.actamat.2010.04.022
33. Zhang, J., Zhang, Y., El-Awady, J.A., Tang, Y.: The plausibility of <c + a> dislocation slip on -12-11 planes in Mg. Scripta Mater. **156**, 19–22 (2018). https://doi.org/10.1016/j.scriptamat.2018.07.007

34. Fan, H., Wang, Q., Tian, X., El-Awady, J.A.: Temperature effects on the mobility of pyramidal <c+a>dislocations in magnesium. Scripta Mater. **127**, 68–71 (2017). https://doi.org/10.1016/j.scriptamat.2016.09.002
35. Wu, Z., Curtin, W.: The origins of high hardening and low ductility in magnesium. Nature 526, 62–67 (2015). https://doi.org/10.1038/nature15364
36. Ahmad, R., Wu, Z., Groh, S., Curtin, W.A.: Pyramidal II to basal transformation of ⟨c + a⟩ edge dislocations in Mg-Y alloys. Scripta Mater. 155:114–118 (2018). https://doi.org/10.1016/j.scriptamat.2018.06.026
37. Fan, H., El-Awady, J.A.: Towards resolving the anonymity of pyramidal slip in magnesium. Mater Sci Eng: A **644**, 318–324 (2015). https://doi.org/10.1016/j.msea.2015.07.080
38. Singh, D., Parashar, A.: Effect of symmetrical and asymmetrical tilt grain boundaries on radiation-induced defects in zirconium 2018. J. Phys. D: Appl. Phys. 51:265301
39. Wang, Y.C., Ye, H.Q.: On the tilt grain boundaries in hcp Ti with [0001] orientation. Philos. Mag. A **75**(1), 261–272 (1997). https://doi.org/10.1080/01418619708210294
40. Bhatia, M.A., Solanki, K.N.: Energetics of vacancy segregation to symmetric tilt grain boundaries in hexagonal closed pack materials. J. Appl. Phys. 114:244309 (2013). https://doi.org/10.1063/1.4858401
41. Singh, D., Parashar, A., Kedharnath, A., et al.: Effect of symmetrical and asymmetrical tilt grain boundaries on the tensile deformation of zirconium bicrystals: a MD-based study. J Mater Sci **54**, 3082–3095 (2019). https://doi.org/10.1007/s10853-018-3032-7
42. Hammerschmidt, T., Kersch, A., Vogl, P.: Embedded atom simulations of titanium systems with grain boundaries. Phys. Rev. B 71, 205409 (2005)
43. Zheng, Z., Balint, D.S., Fionn, P.E.: Dunne, investigation of slip transfer across HCP grain boundaries with application to cold dwell facet fatigue. Acta Mater. 127, 43–53. https://doi.org/10.1016/j.actamat.2017.01.021
44. Wang, J., Beyerlein, I.J.: Atomic structures of [01¯10] symmetric tilt grain boundaries in hexagonal close-packed (HCP) crystals. Metall. Mater. Trans. A **43**, 3556–3569 (2012). https://doi.org/10.1007/s11661-012-1177-6
45. Wang, J., Beyerlein, I.J.: Atomic structures of symmetric tilt grain boundaries in hexagonal close packed (hcp) crystals. Modell. Simul. Mater. Sci. Eng. 20, 024002 (2012)
46. Saitoh, K.-I., Kuramitsu, K., Sato, T., Takuma, M., Takahashi, Y.: Molecular dynamics study on deformation mechanism of grain boundaries in magnesium crystal: based on coincidence site lattice theory. J. Mater. 4153464, 10 (2018). https://doi.org/10.1155/2018/4153464
47. Farkas, D.: Grain-boundary structures in hexagonal materials: coincident and near coincident grain boundaries. Metall. Mater. Trans. A **25**, 1337–1346 (1994). https://doi.org/10.1007/BF02665467
48. Wu, M.S., Nazarov, A.A., Zhou, K.: Misorientation dependence of the energy of [1$\underline{1}$00] symmetrical tilt boundaries in hcp metals: prediction by the disclination-structural unit model. Philos. Mag. **84**(8), 785–806 (2004). https://doi.org/10.1080/14786430310001646817
49. Tohei, Tetsuya, Watanabe, Yuito, Takahashi, Nobuaki, Nakagawa, Tsubasa, Shibata, Naoya, Ikuhara, Yuichi: First principles study of oxygen diffusion in a α-alumina {10$\underline{1}$1} twin grain boundary. Philos. Mag. **95**(35), 3985–3999 (2015). https://doi.org/10.1080/14786435.2015.1110257
50. Yang, X., Xu, S., Chi, Q.: Plastic deformation behavior of Bi-crystal magnesium nanopillars with a {1012} twin boundary under compression: molecular dynamics simulations. Materials 12(5), 750 (2019). https://doi.org/10.3390/ma12050750
51. Derlet, P.M., Swygenhoven, H.V.: Atomic positional disorder in fcc metal nanocrystalline grain boundaries. Phys. Rev. B. **67**, 014202 (2003)
52. Kim, D.-H., Manuel, M.V., Ebrahimi, F., Tulenko, J.S., Phillpot, S.R.: Deformation processes in [112¯0]-textured nanocrystalline Mg by molecular dynamics simulation. Acta Mater. **58**(19), 6217–6229 (2010). https://doi.org/10.1016/j.actamat.2010.07.036
53. Singh, Divya, Sharma, Pankaj, Jindal, Sahil, Kumar, Prince, Kumar, Piyush, Parashar, Avinash: Atomistic simulations to study crack tip behaviour in single crystal of bcc niobium and hcp zirconium. Curr. Appl. Phys. **19**(1), 37–43 (2019). https://doi.org/10.1016/j.cap.2018.11.002

54. Tang, T., Kim, S., Horstemeyer, M.F.: Fatigue crack growth in magnesium single crystals under cyclic loading: molecular dynamics simulation. Comput. Mater. Sci. 48(2), 426–439 (2010). https://doi.org/10.1016/j.commatsci.2010.02.003
55. Tang, T., Kim, S., Jordon, J.B., Horstemeyer, M.F., Wang, P.T.: Atomistic simulations of fatigue crack growth and the associated fatigue crack tip stress evolution in magnesium single crystals. Comput. Mater. Sci. 50(10), 2977–2986 (2011). https://doi.org/10.1016/j.commatsci.2011.05.016
56. Tang, T., Kim, S., Horstemeyer, M.F., Wang, P. (2011) A molecular dynamics study of fracture behavior in magnesium single crystal. In: Sillekens, W.H., Agnew, S.R., Neelameggham, N.R., Mathaudhu, S.N. (eds.) Magnes. Technol. Springer, Cham (2011). https://doi.org/10.1007/978-3-319-48223-1_65
57. Zhang, Bowen, Liucheng Zhou, Yu., Sun, Weifeng He, Chen, Yazhou: Molecular dynamics simulation of crack growth in pure titanium under uniaxial tension. Mol. Simul. **44**(15), 1252–1260 (2018). https://doi.org/10.1080/08927022.2018.1485150
58. Wu, Z., Curtin, W.A.: Brittle and ductile crack-tip behavior in magnesium. Acta Mater. **88**, 1–12 (2015). https://doi.org/10.1016/j.actamat.2015.01.023
59. Rawat, S., Joshi, S.P.: Effect of multiaxial loading on evolution of 101¯2 twinning in magnesium single crystals. Mater. Sci. Eng.: A **659**, 256–269 (2016). https://doi.org/10.1016/j.msea.2016.02.049
60. Kapinos, V.G., Osetskii, Yu.N., Platonov, P.A.: Simulation of defect cascade collapse in hcp zirconium. J. Nuclear Mater. **184**(2), 127–143 (1991). https://doi.org/10.1016/0022-3115(91)90503-Y
61. Wen, M., Woo, C., Huang, H.: Atomistic studies of stress effects on self-interstitial diffusion in α-titanium*. J. Comput.-Aided Mater. Des. **7**, 97–110 (2000). https://doi.org/10.1023/A:1026557026817
62. Osetsky, Y.N., Bacon, D.J., de Diego, N.: Anisotropy of point defect diffusion in alpha-zirconium. Metall. Mater. Trans. A 33, 777–782 (2002). https://doi.org/10.1007/s11661-002-0144-z
63. Woo, C.H., Liu, X.: Atomistic calculation of point-defect diffusion anisotropy and irradiation growth in α -zirconium. Philos. Mag. **87**(16), 2355–2369 (2007). https://doi.org/10.1080/14786430601185085
64. Mendelev, Mikhail I., Bokstein, Boris S.: Molecular dynamics study of self-diffusion in Zr. Philos. Mag. **90**(5), 637–654 (2010). https://doi.org/10.1080/14786430903219020

EAM Potentials for Characterisation of HCP Nuclear Materials



Divya Singh

Abstract Molecular dynamics (MD)-based simulations to study material behaviour are based on the classical mechanic approach. In recent times, molecular dynamics has emerged as a vigorous technique to study metals for their structure and behaviour. MD is a semi empirically computer modelling technique based on Newton's second law of motion for an N-body system. The atoms are treated as classical particles and the interaction and time evolution of a system of particles is mapped by solving for their potential energy using semi empirically derived interatomic potential. The position, velocity and acceleration of each atom in space are further calculated by Newton's second law of motion at every integration time step. The predictive power of the simulation depends on the reliability of the interatomic potential. The parameters of the potential function are derived by fitting to known experimental properties such as lattice constants and elastic constants. MD simulations can be employed to study atomistic phenomena occurring at small timescales of picoseconds and femtoseconds. When compared to first principle simulations which can only handle a few thousand atoms and small simulation cells, MD-based simulations can handle millions of atoms. This makes MD-based simulations a great option to study phenomena like radiation damage and defect generation in nuclear materials. HCP materials like Zr and Ti are widely accepted as nuclear materials. In order to study irradiation defects in such materials, it is imperative to take caution to use interatomic potential which can correctly reproduce point defect formation and migration energies. The anisotropy and complex crystal structure of HCP materials make irradiation studies even more challenging. This chapter presents examples of the application of EAM potentials to understand the point defect formation and migration energies and diffusion anisotropy, and presents an assessment of available potentials for studying the radiation response of HCP nuclear materials.

Keywords EAM potential · HCP materials · Anisotropy · Point defects · Molecular Dynamics · Migration energy · Irradiation · Dislocation loop

D. Singh (✉)
Department of Materials Science and Engineering, University of California, Irvine, California, United States
e-mail: dsingh4@uci.edu

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1 Introduction

Irradiation-induced defects arise in the materials at a timescale of femtoseconds or picoseconds and a length scale of angstroms. It is impossible with the current experimental techniques to gain an in situ picture of radiation-induced displacement cascades and the atomistic level changes and interactions that occur therein. In addition, handling post irradiated samples has several health hazards. Not only this, but experiments on post irradiated samples might not give accurate results given the complex handling processes. In the past, catastrophic failures in nuclear reactors have occurred due to a lack of knowledge on the effect of irradiation on material properties and strength. In such a scenario, molecular dynamics simulations play an important role in the qualitative understanding of the effects of irradiation on the formation of point defects, their interaction with existing defects in crystals and characterising material strength and life. The accuracy of EAM potentials to reproduce accurate point defect formation and migration energies is of utmost importance when it comes to understanding the effect of irradiation on material properties. Molecular dynamics simulations have been extensively used to understand the point defect formation and migration/diffusion energies in hcp materials. Given the unique crystal structure, point defect formation is an intriguing area, which has been explored by various researchers with respect to the positioning of the defects in the hcp lattice structure. In addition, some limited work has been carried out to study the effect of irradiation and survival of point defects post irradiation in hcp single crystals and bi-crystals. The interaction of point defects with grain boundaries and their effect on the formation and diffusion energies have also been studied. Several important conclusions can be drawn from such studies which add to the knowledge base of the complexity created by irradiation-induced defects on tensile and fracture properties of nuclear materials.

2 Application of EAM Potentials for Irradiation Studies in HCP Materials

In 1990, Kapinos et al. [1] studied the displacement cascades by applying a thermal spike mode using MD in hcp Zr. The diffusional rearrangement in the depletion zone of the cascade is investigated. Vacancy clusters were the prominent result of this work. The Burger's vector of the vacancy loops up to 5% spatial distribution of vacancy clusters in the depleted zone after thermal spike was in the {10–10} planes with Burgers vector $< 10\text{–}10 >$ or $1/3 < 11\text{–}20 >$. As the vacancy cluster concentration increased, the loops are primarily bound to the basal plane with a Burgers vector of $1/6 < 20\text{–}23 >$. Wen et al. [2] studied the formation and diffusion of self-interstitials in Ti using Ackland's interatomic potential [3] for alpha-Ti. They calculated the self-interstitial formation energies at all possible points in the hcp lattice**.** These are: the tetrahedral (T), octahedral (O), basal tetrahedral (BT), basal octahedral (BO), basal

crowdion (CB), non-basal crowdion (CN), c-dumbbell (DC) and the basal dumbbell (DB).

Further, they concluded the basal dumbbell position to be the most stable. The diffusion anisotropy of intrinsic point defects is an important factor governing the behaviour of the HCP metals under irradiation. In order to study the effect of anisotropy on the defect formation and migration energies, they calculated the values under compressive stress in three directions [0001], [1–100] and [11–20]. Under compressive stress, the basal dumbbell position of the interstitial atom was again found to be the most stable. The change in the migration energy, for both in-plane and out of plane diffusion, was significant when the load was applied along [11–20] direction when compared to the load along [0001] or [1–100] directions. In addition, they also demonstrate that applied hydrostatic stress can cause a significant alteration in the diffusion anisotropy of HCP metals and alloys. The study brings out an important discovery that creep-like deformation can be produced by hydrostatic stress through a stress-dependent change of growth rate caused due to irradiation.

Osetsky et al. [4] also attempted to study the anisotropy of vacancy and interstitial diffusion in another hcp material, Zr through molecular statics (MS) and MD-based simulations. The many-body interatomic potential developed by Ackland et al. [5] was employed for the work. This potential established the basal split and basal crowdion as the most stable interstitial sites. MS was used to calculate migration barriers for vacancy and the <11–20> crowdion. MD simulations were employed to study the diffusive jumps and calculate diffusion coefficients from these results at various temperatures. Diffusion study results were indicative of the fact that the vacancy diffusion was nearly isotropic in the temperature range of 1050 K to 1650 K. In the initial range of the temperature, the vacancy diffusion did show some inclination towards basal plane diffusion. The interstitial diffusion shows anisotropy by varying its trend from 1-D, 2-D and 3-D, depending upon the temperature. The diffusion anisotropy difference (DAD) between the two types of point defects—vacancy and interstitial—has been studied by Woo and Liu [6] using molecular dynamics simulations in Zr. The information from this study can be used to directly estimate the growth rate of interstitial loops of different burger's vector in a Zirconium foil irradiated in a high voltage electron microscope and stress-free in reactor deformation rates. Ackland's potential [5] was used in this work and all the calculations using this potential corroborate well with experimental observations. Mendelev and Bokstein [7] have studied the self-diffusivity in Zr. The defect formation energies, the migration energies and the diffusivity have all been estimated as a function of temperature. An EAM potential developed by Mendelev and Ackland [8] has been employed in this study. This potential has been chosen for obvious reasons of its capability to reproduce reasonable values for the vacancy and interstitials formation energies for hcp Zr. Zhou et al. [9] studied very high energy displacement cascades in Zr with PKA energy up to 80 keV by using EAM potential#3 by Mendelev and Ackland [8]. In this work, the probability of defect clustering and sub cascade formation is discussed in detail. This potential is good for radiation studies in Zr as it is in a screened Coulomb potential form at a short distance and can reliably model high energy collisions. Another advantage of using this potential is the reproduction of

accurate stacking fault energies of Zr. The number of Frenkel pairs again follows the power-law stated earlier. The fraction of point defects in the cluster increases with the increase in PKA energy and for both vacancies and interstitials. Cascade and sub cascade formation were observed in this study which is a topic of recent research interest in radiation damage. One of the major works on irradiation damage in hcp materials was studied by Wooding et al. [10, 11]. Wooding and his team, in separate works, studied the displacement cascades in Ti [10] and Zr [11] ranging from energies of 0.2 keV to 5 keV. The direction of the primary knock-on atom (PKA) does not have any observable effects on the final damaged state. Even at 5 keV of PKA, no substantial vacancy or interstitial clustering was observed. Later, they studied high energy displacement cascades in Zr [12] at up to a PKA energy of 20 keV and compared the results with that obtained for Ti for PKA energy up to 10 keV. In both cases, the number of Frenkel pairs and Nf, increased with the increase in PKA energy. The number of Frenkel pairs obtained in displacement cascades produced by using the appropriate EAM potential for both Ti and Zr follows the theoretical power law given by equation 1. According to the power law, the number of Frenkel pairs linearly increases with the increase in the PKA energy. The power law is given by the equation

$$N_F = A\left[E_p\right]^m \tag{1}$$

where N_F = Number of Frenkel pairs.

E_P = PKA energy.

A = 4.5–6.

m− = 0.7–0.8

In addition, for the defects generated during the displacement cascade in both Zr and Ti. The NRT or Kinchin–Pease formula is given by equation 2 and is used to theoretically predict the number of Frenkel pairs generated.

$$N_{NRT} = \frac{0.8E_{dam}}{2\overline{E_d}} \tag{2}$$

E_{dam} = energy available for displacement approximately equal to the PKA energy.

$\bar{E}_d$ = mean value of displacement threshold energy in the crystal.

These trends indicate that the EAM potentials employed in the work are capable of reproducing close to ideal irradiation-induced environments in MD simulations.

Defect clustering is remarkably increased with high energy cascades in this study when compared to simulations on Ti with 5 keV energy. Vacancy clusters in Zr are more prominent than Ti, for similar PKA energies. The clusters in Zr contain up to 30 vacancies and 25 interstitials. The clusters have a perfect Burgers vector of ½ [1–210]. The 30-vacancy cluster collapse is also observed on the prism plane. The clustering is in agreement with the dislocation loops experimentally observed in irradiated Zr. Nandi and Eapen [13] performed molecular dynamics simulations to examine the defect accumulation and microstructure evolution in hcp zirconium. They used

EAM potential with interactions modified for distances shorter than 0.1 Å to generate Cascades with a 3 keV primary knock-on atom (PKA). Their studies revealed that sequential cascading can produce not only point defects, but also extended defects such as stacking faults and dislocation loops. Their sequential cascade simulations show the emergence of stacking faults both in the basal and prism planes, and a Shockley partial dislocation on the basal plane. Surface point defect energies of vacancies and adatoms in Zr and Ti were calculated in another work by Pascuet et al. [14]. In this comprehensive study, the important takeaway points were the dependency of the formation energies of vacancies and adatoms on surface density. A lower surface density corresponds to lower adatoms and vacancies formation energies, but higher adatom migration energies. The surface actively acts as a sink for both types of defects and traps them as the energy required for their reemission is quite high. The surface diffusion energies are found to be lower than the bulk diffusion. Sali Di et al. [15] observed void formation in Zr crystal due to the combined effect of irradiation cascade and applied strain. The same potential#3 of Mendelev and Ackland [8] was used in this study for reasons stated earlier. This study is significant because irradiation-induced vacancy clusters under the strain field could act as initiation sites for dislocation nucleation. It was observed that the size of the voids keeps on increasing due to the movement of an atom out of the void core through dislocation movement. The strain conditions in this work represent the condition observed at the hydride-zirconium matrix interface and address a very critical issue of delayed hydride cracking in Zr under irradiation through MD simulations. The voids that form could potentially influence crack propagation by delayed hydride cracking process by acting crack nucleation sites.

The mobility of the point defects and defect clusters also has an influence on the overall properties of the crystal. The mobility of single, 2, 3 and 4 SIA clusters have been studied by Diego et al. [16] at various temperatures of 100 K, 300 K, 500 K and 700 K by using MD-based simulation technique. The mobility of the SIA and vacancy clusters was dependent on both the cluster size and temperature. The interaction of these kinds of vacancy clusters with existing screw dislocation in Zr was simulated through molecular dynamics by Woo and Liu [17] using the conjugate gradient method and modified flexible Green's function. Intuitively enough, the vacancy clusters were found to be stable in the absence of any dislocation in the (0001) planes. The vacancy clusters are observed to collapse into faulted loops on interaction with c-type screw dislocation when their size is large enough. In order to elucidate the atomic mechanism of the <c> type dislocation loop generation under irradiation damage in Zr, Dai et al. [18] conducted the simulations, wherein they studied the interaction of the <a> type vacancy and interstitial loops with cascades run in their close vicinity. They concluded in their work that, at a lower PKA energy value of 10 keV, no <c> axis loops were formed from either <a> type vacancy or interstitial loops. When a series of PKA at 50 keV is simulated, the existing interstitial loop transforms into <c> type loop. However, no such observation was made for a vacancy loop under the same conditions. These <c> type loops either have Burger's vector 1/3 < 11–23 > or 1/6 < 04–43 > . In another work, Zhou et al. [19] studied the formation of a dislocation loop in Zr in a single collision cascade in the presence of

a pre-existing edge dislocation. Cascade simulations are run in pristine and crystal with edge dislocation to understand the role of pre-existing edge defects on cluster formation. It was observed in their results that the size of the cluster in the pristine crystal is smaller than that of the crystal with edge defects. In any real material, line defects like grain boundaries are intrinsically present. The presence of grain boundary affects the behaviour of point defects, in terms of defect formation and migration energies, which, in turn, can affect the overall atomic-scale behaviour of materials under irradiation. The role of different types of grain boundaries on the irradiation behaviour of Zr has been studied by a number of researchers, owing to its widest use as a structural material in nuclear power plants Other hcp materials like Mg and Ti were also explored in terms of point defect and GB interaction. In general, from the trend of vacancy formation, vacancy migration and binding energy in close vicinity of the GBs, it was discovered in each work that the grain boundaries, both symmetric and asymmetric, tilt or twist act as sinks for the point defects. The sinking effect depends on the nature of the GB and its energy. From the trend of the formation and migration energies, it was deduced that each preferential absorption of interstitials over vacancies exists. Solanki and Bhatia [20] studied the effect of c/a ratio on the vacancy binding energies by considering symmetric tilt grain boundaries in Mg, Ti and Zr. In addition to the role of GBs as a sink for the point defects, they concluded that with an increase in the c/a ratio, the binding energies tend to increase. The boundaries which had symmetrical atoms along the GB plane, show higher vacancy formation energies indicating a lack of vacancy diffusion properties. In a recent work, by Divya and Avinash [21], a detailed analysis of point defect formation energies and displacement cascades with a wide range of symmetrical and asymmetrical GBs has been done for GBs along two axes, namely: [0001] and [0–110]. They elucidated through their work that as the GB energy increased, the formation energies goes on decreasing. Another work by Arjhangmehr and Feghhi [22] studied the trend of formation energies in both pristine and damaged GBs in Zr. By employing the EAM potential developed by Mendelev and Ackland (#3) [8], the GBs considered in their work were symmetrical and twist boundaries. The trend of point defect formation energy was similar to the general trend. However, the damaged GBs facilitated the formation and migration of excess vacancies in the bulk near the boundaries. The negative vacancy formation energy near the interstitial loaded (damaged) GB indicated that the vacancies are unstable that spontaneously combine with the pre-loaded interstitials in the GB centre. In addition, they also studied the diffusion barrier energies. The lower angle tilt boundaries were preferential absorbers of interstitials in contrast to the higher angle boundaries that favoured the absorption of vacancies. In both works [21, 22], the number of surviving defects in the bulk was reduced in the presence of GBs due to absorption by the GBs when energy cascades were simulated. Similar work was carried out with identical results for Zr, but a different set of GBs, by Hatami et al. [23]. Kapustin et al. [24] explored through their simulations that the GBs act as sinks as well as barriers to the movement of vacancies and interstitials and lead to clustering. In another work, they [25] tried to study the effect of alloying elements on the defect formation in Zr-1% Nb alloys with GBs. The Nb atom was found to have a higher tendency to move out as interstitial atoms. The effect of Nb

% (0–5%) on the formation and migration energies of point defects in presence of different types of GBs along [0001] axis was carried out in comprehensive work by Divya et al. [26]. It was concluded in their work that alloying with Nb has positive effects on the sinking property of the GB. The migration energy was calculated by Nudged Elastic Band (NEB) method. The vacancy migration energy for 2.5 wt% Nb was found to be minimum. Studies have been carried out to study the effect of point defects on the tensile and fracture properties of Zr single and bi-crystals [27, 28]. Li et al. [27] generated point defects including vacancies, interstitials and Frenkel pairs and subjected the single-crystal Zr to a uniaxial load along [0001] axis. The effect of increasing the concentration of each type of defect was the reduction in yield stress, as well as yield strain. of the single crystal. However, the interstitials and Frenkel pairs decreased the strength more drastically when compared to the vacancies. Dislocation loops of interstitial types were observed in this work. No effect of the point defects on the young's modulus was observed. Divya et al. [28] have elucidated the effect of point defects on the tensile strength of Zr bi-crystal with tilt axes about both [0001] and [0–110] axes. They simulated single and multiple PKAs at varying distances from the GBs to capture the effect of both spatial and number of defects on the yield strength of Zr bi-crystals. It was concluded from their work that. the increase in yield strength was a factor of the intensity by which the point defects interacted with the moving twins and dislocations in the crystal. When the dislocations were straight and concentrated on the boundary, the probability of interaction between the point defects and dislocations was minimal and no apparent increase in the yield strength is observed. Basal and pyramidal vacancy type loops were observed in this work. Further, the same authors attempted to study the effect of point defects on the fracture strength of Zr bi-crystals. In this work also, it was concluded that both the number of defects and their spatial distribution affect the ultimate tensile strength in a Zr bi-crystal with a central crack. Two very important observations in this work were—the irradiated crystal had higher UTS than the pristine one. However, this increase is limited to a particular number of defects beyond which the UTS begins to deplete**.** Vacancy loops again were a feature of this work. Vacancy clustering and their binding energies along with dislocation loop formation energies in Zr have been studied by Varvenne et al. [29] using ab Initio and the two EAM potentials #1 and #2 given by Mendelev and Ackland [8]. The corresponding results are compared and it is concluded in their work that potential #2 gives the most reliable results corresponding to the stability of large vacancy clusters. The most stable vacancy clusters generated were the dislocation loops in the prismatic plane, which is in agreement with the experimental results.

3 Conclusion

In conclusion, computational simulation techniques have identified some important features governing the effect of irradiation on mechanical properties of hcp materials. Phenomenons taking place at the atomic level affect the properties at the

macroscale, but are not visible through experiments due to time and spatial concerns like irradiation-induced defects that can be easily and accurately captured through simulation techniques. There is still scope for improving upon the current interatomic potentials or developing better interatomic potentials for MD simulations. The reasons behind the existing disparity in the activation of deformation mechanisms of hcp materials are still not fully explored. In addition, the studies on the effect of irradiation in hcp materials through computational techniques are limited and need further examination.

References

1. Kapinos, V.G., Osetskii, Yu.N., Platonov, P.A.: Simulation of defect cascade collapse in hcp zirconium. J. Nucl. Mater. **184**(2), 127–143 (1991)
2. Wen, M., Woo, C.H., Huang, H.: Atomistic studies of stress effects on self-interstitial diffusion in α-titanium. J. Comput.-Aided Mater. Design **7**, 97–110 (2000)
3. Ackland, G.J.: Theoretical study of titanium surfaces and defects with a new many-body potential. Philos. Mag. A **66**(6), 917–932 (1992)
4. Osetsky, Y.N., Bacon, D.J., de Diego, N.: Anisotropy of point defect diffusion in alpha-zirconium. Metall. Mater. Trans. A **33**, 777–782 (2002)
5. Ackland, G.J., Wooding, S.J., Bacon, D.J.: Defect, surface and displacement-threshold properties of α-zirconium simulated with a many-body potential. Philos. Mag. A **71**(3), 553–565 (1995)
6. Woo, C.H., Liu, X.: Atomistic calculation of point-defect diffusion anisotropy and irradiation growth in α -zirconium. Phil. Mag. **87**(16), 2355–2369 (2007)
7. Mendelev, M.I., Bokstein, B.S.: Molecular dynamics study of self-diffusion in Zr. Phil. Mag. **90**(5), 637–654 (2010)
8. Mendelev, M.I., Ackland, G.J.: Development of an interatomic potential for the simulation of phase transformations in zirconium. Philos. Mag. Lett. **87**(5), 349–359 (2007). https://doi.org/10.1080/09500830701191393
9. Zhou, W., Tian, J., Feng, Q., Zheng, J., Liu, X., Xue, J., Qian, D., Peng, S.: Molecular dynamics simulations of high-energy displacement cascades in hcp-Zr. J. Nucl. Mater. **508**, 540–545 (2018)
10. Wooding, S.J., Bacon, D.J.: Phil. Mag. A **76**, 1033 (1997)
11. Wooding, S.J., Bacon, D.J., Phythian, W.J.: A computer simulation study of displacement cascades in α-titanium. Philos. Mag. A **72**(5), 1261–1279 (1995)
12. Wooding, S.J., Howe, L.M., Gao, F., Calder, A.F., Bacon, D.J.: J. Nucl. Mater. **254**, 191 (1998)
13. Nandi, P.K., Eapen, J.: Cascade overlap in hcp zirconium: defect accumulation and microstructure evolution with radiation using molecular dynamics simulations. MRS Online Proc. Libr. **1514**, 37 (2013)
14. Pascuet, M.I., Pasianot, R.C., Monti, A.M.: Computer simulation of surface-point defects interaction in hcp metals. J. Mol. Catal. A: Chem. **167**(1–2), 165–170 (2001)
15. Di, S., Yao, Z., Daymond, M.R., Gao, F.: Molecular dynamics simulations of irradiation cascades in alpha-zirconium under macroscopic strain. Nucl. Instrum. Methods Phys. Res. Sect. B **303**, 95–99 (2013)
16. de Diego, N., Osetsky, Y.N., Bacon, D.J.: Mobility of interstitial clusters in alpha-zirconium. Metall Mater Trans A **33**, 783–789 (2002)
17. Woo, C.H., Liu, X.: Generation of c-component edge dislocations in α-zirconium during neutron irradiation—an atomistic study. J. Nuclear Mater. **393**(3), 513–517 (2009)

18. Dai, C., Long, F., Saidi, P., Béland, L.K., Yao, Z., Daymond, M.R.: Primary damage production in the presence of extended defects and growth of vacancy-type dislocation loops in hcp zirconium. Phys. Rev. Mater. **3**, 043602
19. Zhou, W., Tian, J., Zheng, J., et al.: Dislocation-enhanced experimental-scale vacancy loop formation in hcp Zirconium in one single collision cascade. Sci Rep **6**, 21034 (2016)
20. Bhatia, M.A., Solanki, K.N.: Energetics of vacancy segregation to symmetric tilt grain boundaries in hexagonal closed pack materials. J. Appl. Phys. **114**, 244309 (2013)
21. Singh, D., Parashar, A.: Effect of symmetrical and asymmetrical tilt grain boundaries on radiation-induced defects in zirconium. J. Phys. D: Appl. Phys. **51**, 265301 (2018)
22. Arjhangmehr, A., Feghhi, S.: Irradiation deformation near different atomic grain boundaries in α-Zr: An investigation of thermodynamics and kinetics of point defects. Sci Rep **6**, 23333 (2016)
23. Hatami, F., Feghhi, S.A.H., Arjhangmehr, A., Esfandiarpour, A.: Interaction of primary cascades with different atomic grain boundaries in α-Zr: An atomic scale study. J. Nucl. Mater. **480**, 362–373 (2016)
24. Kapustin, P., Svetukhin, V., Tikhonchev, M.: The interaction between atomic displacement cascades and tilt symmetrical grain boundaries in α-zirconium. Radiat. Eff. Defects Solids **172**(5–6), 411–419 (2017)
25. Kapustin, P., Svetukhin, V., Tikhonchev, M.: Simulation of atomic displacement cascades in the binary alloy Zr-1%Nb near symmetrical tilt grain boundaries by molecular dynamics method. Radiat. Eff. Defects Solids **173**(7–8), 608–616 (2018)
26. Singh, D., Parashar, A., Kapoor, R.: Effect of Nb precipitate on defect formation and migration energies in bi-crystalline Zr. Mater. Chem. Phys. **235**, 121729 (2019)
27. Li, Y., Chen, H., Chen, Y., Wang, Y., Shao, L., Xiao, W.: Point defect effects on tensile strength of α−zirconium studied by molecular dynamics simulations. Nuclear Mater. Energy **20**, 100683 (2019)
28. Singh, D., Parashar, A., Kedharnath, A., et al.: Effect of symmetrical and asymmetrical tilt grain boundaries on the tensile deformation of zirconium bicrystals: a MD-based study. J Mater Sci **54**, 3082–3095 (2019)
29. Varvenne, C., Mackain, O., Clouet, E.: Vacancy clustering in zirconium: an atomic scale study. Acta Mater. **78**, 65–77 (2014)

EAM Inter-Atomic Potential—Its Implication on Nickel, Copper, and Aluminum (and Their Alloys)



Swati Chaturvedi, Akarsh Verma, Sandeep Kumar Singh, and Shigenobu Ogata

Abstract In recent years, our atomic structure knowledge of solids has advanced. There are many problems, which are related to material properties on which Embedded Atom Model (EAM) can be used. Knowledge of structure, phase transitions, and diffusion is required for the EAM calculations. Dynamics, vibrations, and segregation are other topics whose information is required for EAM. Four different types of calculations are used to implement embedded atom model. Basic properties of point defects can be computed with the help of EAM. These were calculated for FCC metals like Cu and Ni that will be discussed in this chapter. Migration energy, formation energy, self-interstitial geometry, and divacancy binding energy are the properties that are computed. Other properties were migration energy and vacancy formation which has been calculated through EAM very efficiently. One of the advantages of EAM is its capacity to replicate energy of vacancy formation. Studies were conducted to study the structure and elastic behavior of grain boundaries. These studies included the computation of shear modulus, which acted in the planes of the boundary and Young's modulus for both the axis. For certain fundamental properties, correct and reliable measurements are present which are close to that obtained from experimental methods. Apart from this, certain phenomena also exist in EAM which include phase transition and reconstructions. Thus, embedded atom model has provided a great method for observing materials' physics. In alloys, variations in composition take place near defects, which has been successfully studied by the EAM. EAM calculations were extensively done to study the mechanical properties of metals.

S. Chaturvedi · A. Verma (✉)
Department of Mechanical Engineering, University of Petroleum and Energy Studies, Dehradun 248007, India
e-mail: akarshverma007@gmail.com

A. Verma · S. Ogata
Department of Mechanical Science and Bioengineering, Osaka University, Osaka 560-8531, Japan

S. K. Singh
Department of Mechanical and Industrial Engineering, Indian Institute of Technology Roorkee, Roorkee 247667, India

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Keywords Embedded atom model · Bulk properties · Alloy · Mechanical properties · Atomistic scale simulations

1 Introduction

Considering materials' physical properties is an interesting idea when we look on the atomic scale structure. In the recent years, our atomic structure knowledge of solids has advanced. Complex systems that include atomic scale structures (surfaces, grain boundaries, and interfaces) can now be studies. A solid motive was provided by these advancements in experimental study to explore the concepts of these structures. Approximate approaches were required by most cases. However, pair potential is a simpler method. Pair potential does not take into account the concept of metallic bonding. As a result, it is not a valid approach. Embedded atom method (EAM) included the concept of interactions of many atoms, which was neglected by the pair potential approach [1–5]. Simulations of various problems were possible through the development of EAM. The concept of total energy can be used to answer queries regarding the structures and thermodynamics. Schrodinger equation related to many electrons can be used to calculate the total energy for any arrangement of atoms in a solid. Complications related to problems could be reduced by the local density estimation, which also considers the crucial concepts of physics. These calculations were not suitable for the low symmetry systems, ever after developments in areas of mathematics and computer science. Pair-potential estimation was not correct, as the bonds that exited between atoms were dependent on each other.

2 Embedded Atom Model

Daw and Baskes suggested the EAM concept. In this method, we consider the energy that is given by remaining atoms of the structure as we embed an atom into the local density of electrons. We also consider the electrostatic interaction present. The initial estimation that was used for the mathematical problem is given by the Eq. 1:

$$E_{coh} = \sum_{i} G_i \left(\sum_{j(\neq i)} \rho_j^a \left(R_{ij} \right) \right) + \frac{1}{2} \sum_{i.j(j \neq i)} U_{ij} \left(R_{ij} \right) \tag{1}$$

where G stands for the embedding energy needed to embed an atom, spherically averaged atomic electron density is designated by ρ^a, and U stands for electrostatic interaction that takes place between the two atoms. The energy due to interaction between the background electron cloud and atom is termed as embedding energy. The overlapping of electron density on the nucleus due to the presence of other atoms could be used for assessing the background density of atoms. The concept of

metallic bonding is an important characteristic of EAM. Neighboring atoms create host electron clouds in which the atom is enclosed. This interaction is expressed in a complex way. Embedding function could be used to express the interaction that takes place between the atoms. Coordination affects bonding, but can be expressed in an easier manner. In closed packed metals, for doing semi-empirical calculations, EAM is used. This approach was successful in solving problems related to bulk and interface. These problems include interdiffusion that takes place on alloys, grain boundary structure and segregation and defects [1]. The effective medium theory helped in solving problems related to bulk and surfaces. The groundwork of EAM is the understanding that embedding energy can be considered to express the cohesive energy. In this approach, the atom is enclosed in the electron cloud produced by the remaining atoms. Atoms (that are present near a defect) are enclosed into electrons clouds, which have distinct density profile as compared to atoms that are present in bulk. The functions were obtained and tested by making them suitable to properties and using them on the surfaces, respectively. EAM is helpful for analyzing the interactions that takes place in N atoms. This approach of N-atom was more direct [2].

2.1 Basics

In metallic system, atom's position impacts cohesive energy. Our aim is to obtain an expression for the same. We will consider solid's electron density ($p(r)$). Thus, we have the Eq. 2:

$$G[\rho] = \int dr\gamma\big(p(r), \nabla p(r), \nabla^2 p(r), \ldots\big), \tag{2}$$

where $G\,[p]$ is an energy function and was divided into small volumes. Then solid's cohesive energy could be defined through the Eq. 3:

$$E_{coh} = G[\rho] + \frac{1}{2}\sum_{ij}{}' \frac{Z_i Z_j}{R_{ij}} - \sum_i \int dr \frac{Z_{i\rho}}{|r - R_i|} + \frac{1}{2}\int dr \int dr_2 \frac{\rho(r_1)\rho(r_2)}{r_{12}} - E_{atoms} \tag{3}$$

where summation is done over the solid's nuclei. Prime denotes that the equality of indices is removed. The integration is done over r. The ith nucleus's position is indicated by R_i. Similarly, ith charge's position is indicated by Z_i. Atoms which are isolated, their combined energy is indicated by E_{atoms}. The r_{12} indicated $|r_1 - r_2|$, which denotes the distance between the atoms and is given by the Eq. 4.

$$r_{12} = |r_1 - r_2| = \sqrt{\left(x_i - x_j\right)^2 + (y_i - y)^2 + \left(z_i - z_j\right)^2} \tag{4}$$

Individual atom's density was linearly superpower to describe the solid's electron density, and is given by the Eq. 5.

$$\left[\rho_s(r) \equiv \sum i\rho_i^a(r - R_i)\right] \tag{5}$$

Defining embedding energy through the Eq. 6:

$$G_i\left(\overline{\rho}_i\right) \equiv G\left[\rho_i^a + \overline{\rho}_i\right] - G\left[\rho_i^a\right] - G\left[\overline{\rho}_i\right] \tag{6}$$

Using the above equations, we could derive the Eq. 7:

$$E_{coh} = \sum_i G_i\left(\sum_{j(\neq i)} \rho_j^a\left(R_{ij}\right)\right) + \frac{1}{2}\sum_{i.j(j\neq i)} U_{ij}\left(R_{ij}\right) + E_{err} \tag{7}$$

3 Applications

There are many problems, which are related to material's properties for which EAM can be used. Knowledge of structure, phase transitions and diffusion is required for EAM. Dynamics, vibrations and segregation are other topics whose information is required for embedded atom model. Four different types of calculations were used to implement embedded atom model. These include (a) Molecular dynamics, (b) Vibrational normal mode analysis, (c) energy minimization, and (d) Monte Carlo. Earlier, embedded atom model calculations had limitations as it allowed simulations of molecular dynamics of up to 3500 atoms only. Time required for simulations of molecular dynamics usually was 10 picoseconds. The cost required for calculations varied linearly with numbers of atoms. 10^5 iterations per atom were possible with the Monte Carlo simulations. These were based on EAM and were helpful in finding the transition temperature. These simulations could also be used for characterizing the annealed structures. The vibrational normal mode and dynamical matrix could be obtained by the matrix diagonalization and analytically, respectively. For cells up to two hundred and sixty, normal mode calculations were possible [1].

4 Bulk Properties

In this section, the authors will consider the bulk properties to explain that Embedded Atom model provides sensible explanation of metallic bonding. Let us first consider the phonons.

4.1 Phonons

With the help of force constant tensor K_{ij}, we can get the phonon spectrum. We get the Eq. 8:

$$K_{ij} = \frac{\partial^2 E_{tot}}{\partial R_i \partial R_j} = -A_{ij} + G_i^n(\overline{\rho}_l)\rho_j^a(R_{ij})\widehat{r}_{Jl}g_i + G_j^n(\overline{\rho}_J)\rho_i^a(R_{ij})\widehat{r}_{Jl}g_j + \sum_{k(\neq i,j)} G_k^n(\overline{\rho}_k)\rho_j^a(R_{jk})\rho_i^a(R_{ik})\widehat{r_{Jk}}\widehat{r_{lk}} \quad (8)$$

where

$A_{ij} = \Psi_i^n \widehat{r_{ij}}\widehat{r_{ij}} + \Psi_{ij}^{'}/R_{ij}(1 - \widehat{r_{ij}}\widehat{r_{ij}})$

$\psi_{ij} = G_i^i(\overline{\rho_i})\rho_j^{a,}\left(R_{ij}\right) + G_j^i\left(\overline{\rho_j}\right)\rho_i^{a,}\left(R_{ij}\right) + U_{ij}\left(R_{ij}\right)$

$g_i = \sum_j \rho_j^{a,}\left(R_{ij}\right)\widehat{r_{ij}}.$

The differentiation is denoted by prime. A is a tensor that shows what is contributed to K from the potential of effective pair. With many lines of symmetry, bulk phonons of FCC Ni and Cu can be calculated [6, 7].

4.2 Thermodynamic Properties

For metals like Ni and Cu [8, 9], certain thermodynamic properties were calculated by the EAM. Thermodynamic properties were the melting point and thermal expansion. Other properties that were computed were free energies for both the solid and liquid. These were computed as a function of temperature. Computation of function related to thermodynamics over the entire range of temperature for both phases required many calculations. To understand the thermal behavior of metals, melting point results were required. To determine the density of both phases, Monte Carlo simulations could be performed. These simulations were performed at constant pressure and temperature.

Monte Carlo simulations could only be performed at values of temperature equal to Debye temperature or above it. Apart from this, free energy cannot be computed from the Monte Carlo simulations. If the value of free energy at any one point is known, values at other temperature of free energy could also be determined by the

enthalpy values using the thermodynamic expression written in the Eq. 9:

$$\frac{d}{dT}\left(\frac{G}{T}\right) = -\frac{H}{T^2} \tag{9}$$

Quasi-harmonic calculations were useful for calculating the free energy and thermodynamic functions for solids at low temperature. Quantum–mechanical effects were also considered in these calculations, which were important at lower values of temperature. Harmonic approximations were not accurate at higher values of temperature because the values of amplitude were high for thermal vibrations. As a result, these calculations couldn't be used at higher temperatures. Monte Carlo simulations could determine the enthalpies at higher temperature, and free energy could be determined by using thermodynamic relation. For liquid state, value of free energy at any value of temperature was required to calculate value at other temperatures using enthalpies from the isobaric simulations. This was ensured by keeping the chemical potential, temperature and volume constant during the Monte Carlo simulations. Adjusting the chemical potential produces zero pressure. To calculate the linear coefficient of thermal expansion, quasi-harmonic calculations were used. This was done at room temperature. This was initiated by the motivation that the values of thermal expansion coefficient was very poor and negative for body centered cubic (BCC) metal, as observed in Finnis Sinclair interactions [8].

When computer simulations are used for calculating the thermodynamic properties, problems such as they do not give information regarding entropy as a result free energy cannot be calculated. The curves of Gibbs free energy that were obtained for both solid and liquid, were compared. It was found that both experimental and theoretical results aligned with each other. The function of embedded atom model, which was obtained for Cu, was helpful for calculating the Gibbs free energy. However, the function obtained from Ni was not much useful. The same thing was observed for calculating free energy for liquid phase. The melting point was obtained by looking at the intersection obtained from Gibbs free energy curve for both the liquid and solid phases. Experimental values were higher than theoretical values for all metals except Ni. The temperature at which the melting of the sample takes place was to be determined. For that, there were two blockages. Firstly, the simulations performed by computers were restricted by time, as it is possible that the solid would not melt during that time. Secondly, there would be barrier to nucleation of melting which would be caused by energy at the interface of liquid and solid. When freezing takes place, the interface area between the solid and liquid should remain constant. Consider taking a solid cube, which is in touch with the liquid cube. In this geometry, there is no nucleation, which was caused by the energy of interface. If the temperature is not close to melting point, then molecular dynamics run is enough for movement of interface [9].

4.3 Liquids

The structure factors above melting point of several liquid transition metals were obtained. These were computed by doing simulations of molecular dynamics. They were done at higher temperatures. Densities at equilibrium and zero pressure were found to be within one or two percent of the values got from the experimental analysis. From theory of liquids, structure factors were calculated. These were based on pair interactions [10]. For density to be zero, the embedding energy should have value zero at the starting [10].

4.4 Defects

Basic properties of point defects were computed with the help of EAM. These were calculated for FCC metals like Cu, Ni, Pt, Ag, and Au [11]. Migration energy, vacancy formation energy, self-interstitial geometry, and divacancy binding energies were the properties that were computed. For determining the functions, vacancy formation energies were used. One of the advantages of EAM is its capacity to replicate energy of vacancy formation.

5 Grain Boundaries

Energy minimization techniques were used for calculating the structure of grain boundaries. For that some statements need to be taken into account. When atoms are added or removed from boundaries, lower energy state could be achieved. As we have already discussed in depth about the differences that exist between the calculated and those obtained from the experiments for grain's atomic structure. These experimental studies were obtained with the help of diffraction through X-rays or through electron microscopy at high resolution. We will also study about the boundaries of twist and tilt in Au and Al [12].

5.1 Structure

A significant focus is there on the structure of twist boundaries. The atomic positions were determined through experiments at the boundary of symmetric twist. This was done by inverting intensities of X-ray diffraction. The positions were then calculated with the help of embedded atom model. The maximum difference was found to be 0.11 Å, while minimum was 0.05 Å. These results were compared with those obtained from the experimental method. Monte Carlo simulations were used for

obtaining amplitude from the vibrations that took place in atoms. The amplitude of vibrations was less in bulk and more on the boundary. This depends on the direction in which the vibrations took place and on the environment of distinct atoms. It was found that if we squared the amplitude it was two times more in the case of the boundary as compared to that of bulk. A study integrating both experimental and theoretical methods was done for twist boundaries present in Au. Embedded atom model was used to calculate the structure of atom. This structure was then used to calculate the X-ray diffraction that was expected from boundary. These results obtained from this study were then compared from results obtained experimentally from X-ray diffraction. It was found that experimental and calculated values were close enough which showed that embedded atom model explains successfully the structure of grain boundary [13, 14].

Electron microscopy having high-resolution transmission when used for determining the structure was indirect in nature. In this method, a structure (which is the candidate) is taken and the simulation of image is done which is visible. The image obtained from simulations is then compared with image obtained from experimental methods. In conditions related to bulk and microscopic, atomic columns are found that are indicated by the bright spots. However, a channel is corresponded at boundary, which is signified by maximum intensity. If analysis is not done in depth, this can be misjudged as atom's column and it could have concluded a wrong structure. To conclude the microscopy at high resolution, it is required to perform the atomistic simulations along simulation of image. For example, let us take the work done in Al $\sum 9$ tilt boundary. It was observed in high-resolution transmission microscopy that these boundaries consisted of micro-facets, which were flat in geometry. It was also found that there are two different structures, which were symmetrical in nature along with the presence of boundary plane. This plane had their own micro-facets. One had symmetry of glide plane while other had symmetry of mirror plane. The energy structures, which were computed with the help of embedded atom model, were close enough to the observed structures [15, 16].

5.2 Thermal Effects

Embedded atom model was used to see the probability of a transition related to thermal roughening, which took places at grain boundary having high angle. For $\sum 5[001]$ boundary of twist, equilibrium properties related to finite temperature were calculated. The transition that took place in these boundaries between the two crystals showed the rotation. This took place in atoms of unit cell. Calculations related to energy minimization were done, and it was found out that very less amount of energy was required for the displacement of boundary plane. A plane did this, which was in a one-unit cell of boundary. This resulted in the formation of non-planar interface. Moreover, the interaction that took place between the unit cells was weak till the location of the boundary didn't changed by more than single plane. To observe surface roughening, solid-on-solid models were studied, which made it easy to figure

out distribution of energetics that took place in boundary locations. The energetics of various structures was then fitted for calculating the parameters related to the model. The structures used contained distinct boundary location distribution. The solution of solid-on-solid model was studied to find the temperature required for roughening transition. However, the embedded atom functions, which were used for gold, determined the values in term of quantity that was found to occur at 100 or 500 K. This depended on the amount of interactions, which were taking place. In any case, it was found that the temperature which was experimentally accessible, at them roughening transition took place. After this, X-ray diffraction was studied and how they were affected by the roughening transition was studied. The results showed that the some grain boundaries reflections disappeared. This was a way for finding the transition [17, 18].

5.3 Many-Body Interaction

In several cases regarding the calculations, which were made from grain boundary structure with the help of embedded atom model, it was found that same outputs were obtained when we use the pair-potential treatment. Thus, there was no domination by interactions due to many bodies to the properties. The results obtained form embedded atom model were close to results obtained from pair potentials; hence, it was found that at the boundary the overall density was nearly equal to that of bulk. Apart from this, studies were made to study the role, which was played by interactions that took place in many bodies to the structure that was computed of the $\sum 13[001]$ twist boundary that was symmetric in Au. Making the pair potential, which closely resembles to that of embedded atom model interactions and again calculating the locations did this. The two methods that were used for calculating locations and the results obtained from these two methods had similar results, which showed that no significant role was played in this structure boundary by the many body. The main point to be noted was that for the case of pair potential the expansion, which was normal to the boundary, was larger. However, for embedded atom model it was smaller which showed that with embedded atom model it was possible to have more density at the boundary. The structures, which were obtained from the first principle of pair potential, were then compared with the structures obtained from embedded atom model of $\sum 99$, $\sum 9$, and $\sum 11$ tilt boundaries present in Al. The results obtained from experimental and theoretical methods were reasonable for both embedded atom model and for those obtained from pair potential. Although, embedded atom model was more reasonable with experimental method. The differences, which were there in the results, obtained for the structure of grain boundary from embedded atom model calculations and pair potential showed that for the structure of grain boundary, there was less importance of many body interactions [19].

5.4 Elastic Properties

Studies were conducted to study the elastic behavior of grain boundaries in particular. These studies included the computation of shear modulus, which acted in the planes of the boundary and young modulus for both the axis. From the results of the studies conducted, it was found that the shear moduli obtained for the bulk material was larger than the shear moduli of grain boundaries. On the other hand, the value of Young's modulus, was smaller for the bulk than for the grain boundaries. These results that were obtained were then compared with the changes that took place in elastic constants of the bulk crystal that were subjected to stress. This study was purely on the basis of the fact that the structural characteristics that were common to the grain boundaries were inclusive of the matter that was common in the boundary. Thus, the areas which were subjected to tensile strain, could be related with grain boundary but should not be subjected to Poisson contraction. As results obtained from experiments included Poisson contraction, thus it was not possible to deduce elastic constants from experiments. Thus, embedded atom model was used for calculating the changes that took place in elastic constants that had strain for both the conditions, i.e., experimental and with no Poisson contraction. The studies, which were conducted showed, that embedded atom model could easily predict the changes that took place in elastic constants having strain and those that were easily accessible from experimental methods. For the other case which is without Poisson contraction, it was reported that there was increase in Young modulus due to tensile strain. On the other hand, there was decrease in the shear moduli due to tensile strain. The same observations were reported for changes that took place in elastic constants for the grain boundaries. Although this similarity forecasted the changes that took place in the signs of the elastic constant at the grain boundary. It was not much useful for forecasting the changes that took place in the magnitude. The studies showed that these results had no use of the expressions obtained from the embedded atom model, and similar results could also be obtained from pair interactions [20–22].

6 Surfaces

Embedded atom model have got metals surfaces as base for proving themselves. For certain fundamental properties, correct, and reliable measurements are present which are close to that obtained from experimental methods. Apart from this, certain phenomena, which are very unusual, also exist which include phase transition and reconstructions. Thus, embedded atom model has provided a great method for observing physics.

Embedded atom model has the ability to describe metal surfaces, which is possible due to the interactions that took place in many body. The environment influences these interactions that took place at the inter-atomic level. Keeping these things in mind, we will now discuss surface phonons and structural phase transitions that take place

in Au and Pt. With the help of these, it would become possible for us to forecast the trends in terms of interactions that took place at many atom, or may be in the form of functions of embedded atom model. The studies were made to obtain the energy and structure of the surfaces having low-index for Cu and Ni. This was done by reducing the overall energy that was present in slabs of thick size. This was done with the help of (100), (110), (111) surfaces [5]. The experimental surface energy for Cu came around to be 1790 erg/cm^2, and for Ni it came around 2380 erg/cm^2. For (100) face, the surface of Cu and Ni was found to be 1280 erg/cm^2 and 1580 erg/cm^2, respectively. For (110) face, the surface energy was found to be 1400 erg/cm^2 for Cu, and 1580 erg/cm^2 for Ni. For (111) face, the surface energy was found to be 1170 erg/cm^2 for Cu, and for Ni it was found to be 1450 erg/cm^2. Normally, the values of surface energies reported were low. The error could be reduced by ignoring the slope of the background density that was witnessed by the atoms at the surface. The changes that took place in the spacing's of the interlayer were computed. For face (100), the value of changes in the top layer spacing for Cu was found to be -0.03 and for Ni it came around -0.00. Similarly, the value of changes in the second layer spacing for Cu was found to be -0.01, and for Ni it was found to be -0.00. For face (110), the value of changes in the top layer spacing for Cu was found to be -0.06 and for Ni it came around -0.03. Similarly, the value of changes in the second layer spacing for Cu was found to be -0.00 and for Ni it was found to be -0.00. For face (111), the value of changes in the top layer spacing for Cu was found to be -0.03 and for Ni it came around -0.01. Similarly, the value of changes in the second layer spacing for Cu was found to be -0.00 and for Ni it was found to be -0.00. These values were computed only for geometries that were unreconstructed. The distances were expressed in angstrom (Å). It was found that changes in the top layer were less. Moreover, high values were reported for the surfaces (110) which were tougher than the (100) and (111) smoother surfaces. These values calculated were close enough to that found from the experimental methods. The values obtained for relaxation had same sign as that obtained from the experiments, but had less magnitude. Other parameters, which were calculated with the help of embedded atom model gave better results quantitatively and resembled close enough to that obtained from the experimental results [3]. Equivalent crystal is a method, which was also used to calculate the relaxations of surfaces [23, 24]. This method was close to embedded atom method mathematically. These interactions which take place between atoms of the surface affect the surface phonons. The bulk phonon spectrum is produced well by embedded atom model. The tensor Aij shows the addition, which is made to Kij, which is a force constant tensor. This contribution is made by effective pair potential Ψij, which is environment dependent. The other parts present in Kij also consist of contributions made by many atoms, which are environment dependent. Short bond lengths are formed due to the changes, which takes place in effective interactions. Formation of deeper potential wells also takes place at the surface. The frequencies of vibrations, which take place at the surface, are modified at the charge density. The force constant tensor, which is obtained by the model of central potential, is different from that obtained by the embedded atom model. The structure is enhanced when we include interactions due to many atoms [25].

Reasonable results were found for both experimental and theoretical results, when we discuss the bulk phonons for FCC metals. With the help of similar methods, normal modes were calculated. This was done for Cu (100) that had slabs of 28 layers. For this both the surfaces were considered, i.e., ideal as well as relaxed [26, 27]. Splitting of the surface modes was calculated. This was done from the edge of the bulk modes. It was reported that the calculated results were close to experimental results. Two-step process is required to understand the changes that take place in the force constants on the surface. The surface which is unrelaxed is created by truncating the bulk. Secondly, relaxation of the surface, which is terminated from the bulk, takes place. Due to these two processes, changes are caused in the constants. These changes include softening and stiffening in the force constants, which are present in the interlayer by around fifteen percent [25]. Over many years, people have started gaining interest in the changes that takes place in the force constant on surfaces of nobel metals. Helium scattering and other experiments were performed to measure the dispersions that take place in the surface phonons. The important study is to find how much softening is required for noticing the positions of resonance manner longitudinally of the force that takes place in interlayer. It was found that around fifteen to seventy percent of softening is needed [25]. Interactions, which take place in angular and central manner, were studied with the help of model of force constant for the studying the surface of Ag [28–31]. The cross sections of inelastic helium-scattering were fitted to calculate force constants at the surface. With the help of this method, around forty eight percent softening took place for Cu. On the other hand, if we would have used a simpler model, i.e., central potential model we would have got only fifteen percent reduction. There were changes of about ten-fifteen percent in the force constants that were at the surface [32, 33].

A study which consisted of both experimental and theoretical methods, was conducted which helped in finding two phases that were ordered for hydrogen to get adsorbed on Pt. This was around 100 K that is the minimum critical temperature, which is known for hydrogen on the surface of a transition metal [34–37]. It was found that adsorption of atoms of hydrogen took place on the surface. The adsorption also took at places that were between the planes of two metals. Sites of surface adsorption had similar energy as compared to sites of octahedral subsurface. At very low temperature, the ordered structures were in the form of arrays of hexagonal form that consisted of hydrogen which occupied the sites at octahedral subsurface. For understanding the effect at different temperature range, Monte Carlo simulations were conducted. The results obtained were then compared with results obtained theoretically. The results obtained for phase diagram for both experimental and theoretical method were quite reasonable. It was considered that surface sites are the only sites, which are occupied by atoms of hydrogen. When we take a surface, which is clean, each atom possessed embedding energy. Atoms of hydrogen, which had been adsorbed, would then contribute density to other neighbors. Studies were conducted to study the interaction that took place between atoms of hydrogen and nickel. The difference that was present in the binding energy was very small for H/Ni [38].

There were many queries when the phenomena of reconstruction were studied which included why only some metals experiences reconstruction. Taking the help of embedded atom model could solve this. The energy difference that was calculated for the surface (that had missing row), and for surfaces experiencing reconstruction were found to be similar for the geometry of atom that were fully relaxed and when the atoms were present on ideal lattice sites. The concepts such as pair interactions, which took place at the surface, could be easily applied if we do not take into account the relaxation. As a result, we got a simpler method to compare the missing row surface and surface that were unreconstructed. When we considered the geometry of both the surfaces, it was found that the numbers of pairs were almost same in both the cases. Both the structure had same energy could be explained from the fact that most of the energy comes form the interactions that took place in pair. It was found from experimental analysis that phase transition occurred in missing row reconstruction at very high temperatures [39, 40]. To observe reconstruction's thermal properties, Monte Carlo simulations were done on the (110) slab. These simulations consisted both jumps, which were larger for sites that were not occupied as well smaller jumps for vibrations and relaxation [41]. For correct computation of critical temperature, including relaxation was necessary. Apart from this including vibration was also important for correct results. When calculations of lattice gas were performed, it was found that effects that were caused by vibrations were not taken into account. Apart from this the interactions, which took place in atoms, were also affected by relaxations. The atoms which were responsible for the formation of reconstructed layer witnessed attraction and repulsion along the row. This could also be understood with the help of interactions that takes place pair and trio wise. The stability, which was possessed by the geometries of many clusters, could also be understood by the making a function of quantity of atoms present in the cluster. The results obtained from experiments make sure that the results obtained from calculations are correct. After annealing, a linear geometry is formed by group of three and five atoms. Although a close-packed geometry is formed by a group of four-six atoms. Apart from this, formation of group that consisted of seven atoms also took place. The atom, which acquired the seventh position, never acquired the place next to central atom but always acquired next to the corner atom. From the calculations that were done on theoretical basis for cluster energies, it was found that there was great importance of relaxations. If we don't take into account the relaxations, linear geometry is not preferred. However, geometry of close-packed-island is always preferred. Moreover, the difference, which was present in real, was less then the difference due to made by relaxations. The atoms, which were forming the reconstructed layer, were also studied by model of pair interaction. The embedded atom model was used to obtain energetics. The energetics that was obtained was then used to find the interactions that took place in 2, 3, or 4 bodies. For this particular case, it was found that the interaction was almost negligible for 3 or 4 bodies. The clusters of Ni atoms, which were used for forming the reconstructed layer, were also studied. For three adatoms in Pt, it was found that linear geometry was more preferable as compared to closed packed island arrangement. The results, which were obtained, were close enough to that obtained form FIM. However, for nickel it was found that linear arrangement

is the most stable arrangement for all atoms that were used for forming the layer of reconstructions. This difference occurred due to the difference, which was present in the lengths of the bond and lattice constant. The stability comparison between linear arrangement and closed packed arrangement was greatly affected by the substrate relaxations [34].

7 Alloys

In alloys, variations in composition take place near defects that are studied by EAM. EAM gives two-fold advantages while calculating the properties as compared to other methods. Origin of the electron cloud does not affect the interaction that takes place between atom and electron clouds. The interactions are assumed to depend only on the density of local electron. We assume the same embedding function for alloys as we assumed for metals. As a result, it is easier to use this method in case of alloys. Atomistic simulations could be performed with EAM, which are one of its advantages. Chemical effect is one of the factors that determine the alloys energetic. There are other factors also play role near the defect. Distinction in strength of the bond of distinct chemical species is referred as chemical effect. At the interface, interaction of chemical species takes place. This interaction occurs with local stress fields. These improvements can easily be integrated with atomistic model. However with lattice model, it is difficult to do the integration. Atomistic simulations could be performed which provides simplicity in doing computations, which is one of the advantages of embedded atom model. Near the defect, many factors are taken into consideration for energetics. When alloys are treated with embedded atom model, there is a possibility of occurring of errors. Many complexities occur when addition of alloying happens which should be handled with care. When transfer of large charge takes place, it could be solved using embedded atom model [42].

7.1 Surface Segregation in Dilute Limit

The first is study of surface segregation in alloys in dilute limit. Many studies have studied segregation in FCC metals forming alloys. These FCC metals included were Cu, Au, Ni, Pt and Ag. An impurity atom near a (100) surface was taken into account and its energetics at zero temperature was calculated which showed the capacity of functions of EAM to predict segregation in a correct manner. When the values of substitution al impurity atom's energy came negative, it indicated that in the atomic layer the impurity would be enhanced. These energies can change if the concentration is significant near the surface as they were calculated in dilute limit. Two interesting things were observed. When compared, both the planes (first and second) had different sign of segregation energy. This behavior was observed in the

system of Ni-Cu. Second observation was that; the segregation energy was small for first layer as compared to second for Cu or Ni in Pd or Pt hosts [43, 44].

7.2 Ni-Cu Alloy

Entire range of compositions was taken into consideration to study the Ni-Cu alloy. For this study, the functions used were specifically designed for system of Ni-Cu. Monte Carlo simulations were used to study bulk properties of Ni-Cu alloy. Excess enthalpy was in accordance, when compared with thermodynamic data as showed by the simulations. The values of lattice constants deviations were small and positive as obtained from the EAM. On the other hand, the values obtained from experiment were small and negative. Over the full range of composition, the values of lattice constant noted were correct and within 0.03 Å. In alloys, short-range order agreed well with diffuse neutron scattering data. In dilute limit, segregation heat was observed to get an overview of the segregation quantitatively. It was observed that when copper first layers had Ni, the segregation energy was positive. On the other hand it was negative when Cu was in the first layers of Ni surfaces. The energies showed that Cu is repelled to the second and third atomic planes. On the other hand, Ni was attracted to these planes [45–50].

7.3 Compositional Ordering

In the binary systems, behavior of segregation was studied. Finite concentrations and temperature were used for studying the profiles of composition of (111) surfaces. It was found that there was increase in bulk properties for the atomic plane that was at first position. On the other hand, for second and other planes the value was near to the value for bulk. Simulations were also performed to find the short-range order for the surface of alloy. The short-range order helps us in finding the abundance of the different places of adsorption, which were present on the surface. A site which was three-fold and hollow was also found at the middle of the triangle.

The simulations performed by computer provided knowledge regarding the composition of each plane. These simulations also provided additional information such as short-range order of alloys. The counting of different triangle was done to find their abundance. The values that were obtained were then compared with the values that would have been obtained if the upper layer would have atoms of noble metals [51]. Thus, if we consider that Pi denotes the probability of searching a cluster made of N-atoms that had i number of atoms of noble metal. Then we can deduce the Eq. 10:

$$P_N^i = \left\{[N!/(N-1)!i!]x^i(1-x)^{N-i}\right\}g_N^i \tag{10}$$

where x denotes the fraction of atoms of noble metals, which are present on the surface layer. Apart from short-range order, there was long-range order also present on the surface. This could be seen with the help of deposition that takes place on the surface of Cu (100) by Au. On the other hand, it was found that only Cu was present on the second layer of plane. There was a distance of 0.08 Å between the first and second layers of copper. This proved that ordered surface alloys were present. When the deposition of Au on Copper (100) surface starts, Au atoms would prefer to deposit on the layer of atoms of copper surface which forms the layer of reconstructed surface. To find if ordering were present in the layers of surface, Monte Carlo simulations were done. The surfaces (100), (110), and (111) were exposed to a temperature of 300 K. Only single content of Au was present in the slab bulk. In the beginning of the simulations, there were only Cu and Au atoms in the slab. It was found that there were two structures, i.e., ordered structure and primitive structure. A rippled surface of Au was found that was above the atoms of copper [52–54].

7.3.1 Long-Range Order Loss

In this sub-section we consider Ni_3Al, which is an ordered alloy as the bulk phase. When we take distinct stoichiometry, formation of anti-site defects takes place, which causes the segregation of boundary. This in turn results in the loss of long-range order at the boundary. Three geometries of boundaries were studied for this case. Temperatures at which the simulations were performed included 500 and 1000 K. For the twist boundaries, the compositional structure that was obtained from simulations was very simple. The entire composition of region close to the boundary was taken into consideration and studied as bulk composition function. For samples which were rich in Ni composition, it was found that region of grain boundary was more enhanced with concentrations of nickel. Apart from this, it was found from simulations that most of the aluminum goes to the nickel plane. For the sample, which was rich with nickel content, it was expected that Ni atoms would replace Al atoms in the plane of mixed composition. The plastic deformation was affected by the compositional disordering, which took place at the boundary [26, 55].

7.4 Segregation at Strain Field

Strain plays an important role in the facet of segregation. We consider a Ni-Cu alloy system and calculate the segregation, which took place at an edge dislocation. The location that is being studied has (a/2) [108] as its burger vector and the line of dislocation has the direction [1 $\bar{1}$2]. For the remaining two directions, use of free surfaces was done. The peaks had a composition of around 40% copper. The slip plane had an expanded side. On this side, enhancement of copper took place. It also had a compressed side, on which the depletion took place. Simulations were performed taking 10%, 50%, and 90% concentration of copper, and

$\sum 5(36.9°)$, $\sum 13(22.6°)$, and $\sum 61(10.4°)$ twist boundaries were simulated. The temperature at which the simulations were performed was 800 K. It was found that there was enrichment in copper composition for all the three planes. The net expansion was also determined. There was expansion in the region of boundary for all the three planes. It was found that for boundaries having higher angle, the segregation was the strongest. It was also observed that for alloys that were rich in copper content, the segregation was the weakest; and on the other hand, for nickel rich alloys, the segregation was the strongest [56].

8 Mechanical Properties

Embedded atom model calculations have been done to study the mechanical properties of metals [57].

8.1 Dislocation

In the FCC materials, stress, impurities, and temperature affects the dislocation motion. Let us take the example of Nickel. It is difficult to do accurate simulation of dislocation because of the boundary conditions. By placing two half planes, an edge dislocation can be created. In the x and z orientations, boundaries are periodic and lengths are kept fixed while doing the calculations. In certain calculations, in y-direction the surfaces were kept unrestrained. Another kind of boundary condition was also taken in the experiment to remove unphysical motion. The reason for this motion was free surfaces, which were present in the y-direction. In another boundary condition, additional force was given to each atom on the free surface. The energies were minimized. As a result, the atoms were relaxed. At 0 K, no stress was applied. This helped in setting up the calculations. There was separation of about fifteen angstrom between the two half planes. This was close to the separation, which was observed from continuum theory. Standard molecular dynamics calculations were performed, to understand the impacts of stress and temperature. An initial arbitrary velocity was chosen from a Boltzmann distribution, which was then given to each atom, which helped them in attaining the desired temperature [58].

8.2 Fracture

Continuum anisotropic theory was used for finding the displacement that took place in a prefect lattice, which was then further used to find the initial positions. The continuum crack had the origin at the middle of the two planes of atom. This was taken in both x- and y-directions. When calculations were made, it was found that

the choice of origin did not have much impact. The elastic constants were taken from atomic model, which helped in finding the continuum displacement. In the first region, atoms were made to relax and calculations were performed. On the other hand in second region, atoms were fixed at their position and continuum theory was used. Around seventy thousand atoms were present in first region. It was found that there was no significant impact of increasing the size on the mode of fracture [59, 60].

9 Conclusion and Future Perspective

It is important to explore the behavior of materials at atomistic scale level. Nowadays, efforts are made both experimentally as well as theoretically to understand the relation between materials at atomic level and macroscopic level. But, the field of atomistic scale can rarely be explored experimentally. Atomistic simulations are now becoming a helpful resource for understanding and observing the properties of the material at atomistic scale level. To evaluate properties through the atomistic scale simulations, we employ EAM potentials. In this embedded atom model, we consider the energy that is given by remaining atoms of the structure as we embed an atom into local density of electrons. It computes defects, which occurs in metal. It uses the concept of metallic bonding. However, this method works for systems which do not have directional bonding, which is its limitation. Other limitations include: it cannot handle when transfer of charge takes place and cannot treat effects of Fermi-surface. Methods are being developed for solving the case of directional bonding [61]. Basic properties of point defects were computed with the help of EAM. These were calculated for FCC metals like Cu, Ni, Pt, Ag, and Au. Migration energies and formation energy, self-interstitial geometry and divacancy binding energies were the properties that were computed. Other properties were migration energy and vacancy formation energy. For determining the functions, vacancy formation energies were used. One of the advantages of EAM is its capacity to replicate energy of vacancy formation. In alloys, variations in composition take place near defects, which are being studied by EAM.

To ensure that atomistic simulations could be applied to maximum areas for studying the properties of material, it is necessary to overcome the limitations caused in length and time. Recently, a scheme related to multi-scale simulations is being attempted. This scheme combines the database of grain boundary properties and simulation technique at the meso-scale. Another scheme, which uses molecular dynamics, Monte Carlo and simulations of molecular static are also being incorporated. This is a hybrid type of simulation. Apart from these advancements that are made, they are still some areas which need more improvement for applying the atomistic scale methods to a large scale. The 2NN MEAN is a technique, which can be applied to a larger scale for studying the properties of the material. We hope with these advancements in materials at research field, the atomistic level studies could further be enhanced [62]. The authors have a vast experience in the field of molecular dynamics and experimental characterization of composites materials [63–109].

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References

1. Daw, M.S., Baskes, M.I.: Semi empirical, quantum mechanical calculation of hydrogen embrittlement in metals. Phys. Rev. Lett. **50**(17), 1285 (1983)
2. Daw, M.S., Baskes, M.I.: Embedded-atom method: Derivation and application to impurities, surfaces, and other defects in metals. Phys. Rev. B **29**(12), 6443 (1984)
3. Foiles, S.M., Baskes, M.I., Daw, M.S.: Embedded-atom-method functions for the fcc metals Cu, Ag, Au, Ni, Pd, Pt, and their alloys. Phys. Rev. B **33**(12), 7983 (1986)
4. Ning, T., Yu, Q., Ye, Y.: Multilayer relaxation at the surface of fcc metals: Cu, Ag, Au, Ni, Pd, Pt, Al. Surf. Sci. **206**(1–2), L857–L863 (1988)
5. Daw, M.S.: Model of metallic cohesion: the embedded-atom method. Phys. Rev. B **39**(11), 7441 (1989)
6. Ningsheng, L., Wenlan, X., Shen, S.C.: Embedded atom method for phonon frequencies in transition metals. Solid State Commun. **75**(11), 877–880 (1990)
7. Li, M., Liu, S.J., Chen, N.X.: Modified Möbius inversion transform and phonon dispersion relations for copper. Phys. Lett. A **177**(2), 134–138 (1993)
8. Foiles, S.M., Daw, M.S.: Calculation of the thermal expansion of metals using the embedded-atom method. Phys. Rev. B **38**(17), 12643 (1988)
9. Foiles, S.M., Adams, J.B.: Thermodynamic properties of fcc transition metals as calculated with the embedded-atom method. Phys. Rev. B **40**(9), 5909 (1989)
10. Foiles, S.M.: Application of the embedded-atom method to liquid transition metals. Phys. Rev. B **32**(6), 3409 (1985)
11. Adams, J.B., Foiles, S.M., Wolfer, W.G.: Self-diffusion and impurity diffusion of fee metals using the five-frequency model and the embedded atom method. J. Mater. Res. **4**(1), 102–112 (1989)
12. Vitek, V.: Atomistic studies of the structure of grain boundaries and dislocations. In: Stability of Materials, pp. 53–97. Springer, Boston, MA (1996)
13. Yoo, M.H., King, A.H.: Slip, twinning, and fracture at a grain boundary in the L1 2 ordered structure—A$\Sigma = 9$ tilt boundary. J. Mater. Res. **3**(5), 848–855 (1988)
14. Foiles, S.M.: Calculation of the atomic structure of the$\sum = 13$ ($\theta = 22.6$)[001] twist boundary in gold. Acta Metallurgica **37**(10), 2815–2821 (1989)
15. Mills, M.J., Thomas, G.J., Daw, M.S., Cosandey, F.: Structure of a near-coincidence $\Sigma 9$ tilt grain boundary in aluminum. In: MRS Online Proceedings Library (OPL), p. 159 (1989)
16. Mills, M.J.: High resolution transmission electron microscopy and atomistic calculations of grain boundaries in metals and intermetallics. Mater. Sci. Eng., A **166**(1–2), 35–50 (1993)
17. Biegelsen, D.K., Smith, D.J., Tong, S.Y. (eds.): Atomic-scale Imaging of Surfaces and Interfaces: Symposium Held November 30–December 2, 1992, Boston, Massachusetts, USA (vol. 295). Materials Research Society (1993)
18. Binder, K., Block, B., Das, S.K., Virnau, P., Winter, D.: Monte carlo methods for estimating interfacial free energies and line tensions. J. Stat. Phys. **144**(3), 690–729 (2011)
19. Dahmen, U., Hetheringtont, C.J.D., O'keefe, M.A., Westmacottt, K.H., Mills, M.J., Daw, M.S., Vitek, V.: Atomic structure of a $\Sigma 99$ grain boundary in aluminium: a comparison between atomic-resolution observation and pair-potential and embedded-atom simulations. Philos. Mag. Lett. **62**(5), 327–335 (1990)

20. Wolf, D., Lutsko, J.F.: Structurally induced supermodulus effect in superlattices. Phys. Rev. Lett. **60**(12), 1170 (1988)
21. Vitek, V.: Pair potentials in atomistic computer simulations. MRS Bull. **21**(2), 20–23 (1996)
22. Adams, J.B., Wolfer, W.G., Foiles, S.M.: Elastic properties of grain boundaries in copper and their relationship to bulk elastic constants. Phys. Rev. B **40**(14), 9479 (1989)
23. Smith, J.R., Banerjea, A.: Equivalent-crystal theory of oscillatory surface relaxation. Phys. Rev. B **37**(17), 10411 (1988)
24. Smith, J.R., Banerjea, A.: New approach to calculation of total energies of solids with defects: surface-energy anisotropies. Phys. Rev. Lett. **59**(21), 2451 (1987)
25. Nelson, J.S., Daw, M.S., Sowa, E.C.: Cu (111) and Ag (111) surface-phonon spectrum: the importance of avoided crossings. Phys. Rev. B **40**(3), 1465 (1989)
26. Wuttig, M., Franchy, R., Ibach, H.: The Rayleigh phonon dispersion curve on Cu (100) in the ΓX direction. Solid State Commun. **57**(6), 445–447 (1986)
27. Wuttig, M., Franchy, R., Ibach, H.: The Rayleigh phonon dispersion on Cu (100): a stress induced frequency shift? Zeitschrift für Physik B Condensed Matter **65**(1), 71–74 (1986)
28. Ravelo, R., El-Batanouny, M.: Molecular dynamics study of the surface vibration of the rconstructed Au (111) surface. J. Electron Spectrosc. Relat. Phenom. **54**, 255–264 (1990)
29. Bortolani, V., Franchini, A., Nizzoli, F., Santoro, G.: Explanation of the anomalous peak observed in He-atom scattering from Ag (111). Phys. Rev. Lett. **52**(6), 429 (1984)
30. Bortolani, V., Santoro, G., Harten, U., Toennies, J.P.: Surface phonon calculations for noble metals: comparison with he-surface scattering experiments. Surf. Sci. **148**(1), 82–89 (1984)
31. Tong, S.Y., Chen, Y., Bohnen, K.P., Rodach, T., Ho, K.M.: The structure and lattice dynamics of unreconstructed Fcc (001) and (111) metal surfaces. Surf. Rev. Lett. **1**(01), 97–107 (1994)
32. Mohamed, M.H., Kesmodel, L.L., Hall, B.M., Mills, D.L.: Surface phonon dispersion on Cu (111). Phys. Rev. B **37**(5), 2763 (1988)
33. Hall, B.M., Mills, D.L., Mohamed, M.H., Kesmodel, L.L.: Surface lattice dynamics of Cu (111). Phys. Rev. B **38**(9), 5856 (1988)
34. Wright, A.F., Daw, M.S., Fong, C.Y.: Structures and energetics of Pt, Pd, and Ni adatom clusters on the Pt (001) surface: Embedded-atom-method calculations. Phys. Rev. B **42**(15), 9409 (1990)
35. Eguiluz, A.G., Campbell, D.A., Maradudin, A.A., Wallis, R.F.: Static response of a jellium surface: the image potential and indirect interaction between two charges. Phys. Rev. B **30**(10), 5449 (1984)
36. Muscat, J.P.: Ordered hydrogen overlayers on metal surfaces. Phys. Rev. B **33**(12), 8136 (1986)
37. Einstein, T.L., Schrieffer, J.R.: Indirect interaction between adatoms on a tight-binding solid. Phys. Rev. B **7**(8), 3629 (1973)
38. Behm, R.J., Christmann, K., Ertl, G.: Adsorption of hydrogen on Pd (100). Surf. Sci. **99**(2), 320–340 (1980)
39. Moritz, W., Wolf, D.: Structure determination of the reconstructed Au (110) surface. Surf. Sci. **88**(2–3), L29–L34 (1979)
40. Binnig, G., Rohrer, H., Gerber, C., Weibel, E.: (111) facets as the origin of reconstructed Au (110) surfaces. Surf. Sci. **131**(1), L379–L384 (1983)
41. Daw, M.S., Foiles, S.M.: Order-disorder transition of Au and Pt (110) surfaces: the significance of relaxations and vibrations. Phys. Rev. Lett. **59**(24), 2756 (1987)
42. Hamilton, J.C.: Prediction of surface segregation in binary alloys using bulk alloy variables. Phys. Rev. Lett. **42**(15), 989 (1979)
43. Varschavsky, A., Donoso, E.: Order strengthening in α-Cu Al alloys as influenced by grain size and solute content. Mater. Sci. Eng., A **101**, 231–240 (1988)
44. Van Langeveld, A.D., Hendrickx, H.A.C.M., Nieuwenhuys, B.E.: The surface composition of Pd-Cu alloys: a comparative investigation of photoelectric work function measurements, Auger electron spectroscopy and calculations based on a broken bond approximation. Thin Solid Films **109**(2), 179–192 (1983)

45. Egelhoff, W.F., Jr.: X-ray photoelectron and Auger-electron forward scattering: a new tool for studying epitaxial growth and core-level binding-energy shifts. Phys. Rev. B **30**(2), 1052 (1984)
46. Ng, Y.S., Tsong, T.T., McLane, S.B., Jr.: Absolute composition depth profile of a NiCu alloy in a surface segregation study. Phys. Rev. Lett. **42**(9), 588 (1979)
47. Nilsson, A., Morris, M.A., Chadwick, D.: Growth and properties of Cu on Ni (100). Surf. Sci. **152**, 247–253 (1985)
48. Brongersma, H.H., Sparnaay, M.J., Buck, T.M.: Surface segregation in Cu-Ni and Cu-Pt alloys; a comparison of low-energy ion-scattering results with theory. Surf. Sci. **71**(3), 657–678 (1978)
49. Brongersma, H.H., Ackermans, P.A.J., Van Langeveld, A.D.: Composition of Cu-Ni alloy surfaces. Phys. Rev. B **34**(8), 5974 (1986)
50. Sakurai, T., Hashizume, T., Jimbo, A., Sakai, A., Hyodo, S.: New result in surface segregation of Ni-Cu binary alloys. Phys. Rev. Lett. **55**(5), 514 (1985)
51. Zehner, D.M., Goodman, D.W.: Physical and Chemical Properties of Thin Metal Overlayers and Allow Surfaces (No. CONF-8612128-). Materials Research Society, Pittsburgh, PA (1987)
52. Palmberg, P.W., Rhodin, T.N.: Atomic arrangement of Au (100) and related metal overlayer surface structures. I. J. Chem. Phys. **49**(1), 134–146 (1968)
53. Graham, G.W.: An investigation of "Cu (100)-c (2× 2) Au": LEED, AES, ISS, ARUPS and XPS results. Surf. Sci. **184**(1–2), 137–162 (1987)
54. Wang, Z.Q., Li, Y.S., Lok, C.C., Quinn, J., Jona, F., Marcus, P.M.: Atomic and electronic structure of a surface alloy-comparison with the bulk alloy. Solid State Commun. **62**(3), 181–185 (1987)
55. Lee, T.C., Robertson, I.M., Birnbaum, H.K.: Interaction of dislocations with grain boundaries in Ni_3Al. Acta Metall. Mater. **40**(10), 2569–2579 (1992)
56. Foiles, S.M.: Calculation of grain-boundary segregation in Ni-Cu alloys. Phys. Rev. B **40**(17), 11502 (1989)
57. Wang, G.:. First Principles Based Multiscale Modeling of Single Crystal Plasticity: Application to BCC Tantalum. California Institute of Technology (2002)
58. Jones, R.H., Gerberich, W.W.: Modeling Environmental Effects on Crack Growth Processes (No. CONF-8510160-). The Metallurgical Society Inc., Warrendale, PA (1985)
59. Campbell, G.H., Wien, W.L., King, W.E., Foiles, S.M., Rühle, M.: High-resolution electron microscopy investigation of the (710) twin in Nb. Ultramicroscopy **51**(1–4), 247–263 (1993)
60. Hoagland, R.G., Daw, M.S., Hirth, J.P.: Some aspects of forces and fields in atomic models of crack tips. J. Mater. Res. **6**(12), 2565–2577 (1991)
61. Carlsson, A.E.: Beyond pair potentials in elemental transition metals and semiconductors. In: Solid State Physics, vol. 43, pp. 1–91. Academic Press (1990)
62. Lee, B.J., Ko, W.S., Kim, H.K., Kim, E.H.: The modified embedded-atom method interatomic potentials and recent progress in atomistic simulations. Calphad **34**(4), 510–522 (2010)
63. Verma, A., Parashar, A., Packirisamy, M.: Atomistic modeling of graphene/hexagonal boron nitride polymer nanocomposites: a review. Wiley Interdiscip. Rev. Comput. Mol. Sci. **8**(3), e1346 (2018)
64. Verma, A., Singh, V.K., Verma, S.K., Sharma, A.: Human hair: a biodegradable composite fiber–a review. Int. J. Waste Resour. **6**(206), 2 (2016)
65. Verma, A., Singh, V.K.: Mechanical, microstructural and thermal characterization of epoxy-based human hair–reinforced composites. J. Test. Eval. **47**(2), 1193–1215 (2019)
66. Verma, A., Parashar, A.: Structural and chemical insights into thermal transport for strained functionalised graphene: a molecular dynamics study. Mater. Res. Express **5**(11), 115605 (2018)
67. Verma, A., Negi, P., Singh, V.K.: Experimental analysis on carbon residuum transformed epoxy resin: chicken feather fiber hybrid composite. Polym. Compos. **40**(7), 2690–2699 (2019)
68. Verma, A., Gaur, A., Singh, V.K.: Mechanical properties and microstructure of starch and sisal fiber biocomposite modified with epoxy resin. Mater. Perform. Char. **6**(1), 500–520 (2017)

69. Verma, A., Parashar, A., Packirisamy, M.: Effect of grain boundaries on the interfacial behaviour of graphene-polyethylene nanocomposite. Appl. Surf. Sci. **470**, 1085–1092 (2019)
70. Verma, A., Budiyal, L., Sanjay, M.R., Siengchin, S.: Processing and characterization analysis of pyrolyzed oil rubber (from waste tires)-epoxy polymer blend composite for lightweight structures and coatings applications. Polym. Eng. Sci. **59**(10), 2041–2051 (2019)
71. Verma, A., Negi, P., Singh, V.K.: Physical and thermal characterization of chicken feather fiber and crumb rubber reformed epoxy resin hybrid composite. Adv. Civil Eng. Mater. **7**(1), 538–557 (2018)
72. Verma, A., Negi, P., Singh, V.K.: Experimental investigation of chicken feather fiber and crumb rubber reformed epoxy resin hybrid composite: mechanical and microstructural characterization. J. Mech. Behav. Mater. **27**(3–4), (2018)
73. Chaurasia, A., Verma, A., Parashar, A., Mulik, R.S.: Experimental and computational studies to analyze the effect of h-BN nanosheets on mechanical behavior of h-BN/polyethylene nanocomposites. J. Phys. Chem. C **123**(32), 20059–20070 (2019)
74. Jain, N., Verma, A., Singh, V.K.: Dynamic mechanical analysis and creep-recovery behaviour of polyvinyl alcohol based cross-linked biocomposite reinforced with basalt fiber. Mater. Res. Express **6**(10), 105373 (2019)
75. Verma, A., Joshi, K., Gaur, A., Singh, V.K.: Starch-jute fiber hybrid biocomposite modified with an epoxy resin coating: fabrication and experimental characterization. J. Mech. Behav. Mater. **27**(5–6) (2018)
76. Verma, A., Kumar, R., Parashar, A.: Enhanced thermal transport across a bi-crystalline graphene–polymer interface: an atomistic approach. Phys. Chem. Chem. Phys. **21**(11), 6229–6237 (2019)
77. Verma, A., Singh, V.K.: Experimental investigations on thermal properties of coconut shell particles in DAP solution for use in green composite applications. J. Mater. Sci. Eng. **5**(3), 1000242 (2016)
78. Verma, A., Singh, V.K., Arif, M.: Study of flame retardant and mechanical properties of coconut shell particles filled composite. Res. Rev. J. Mater. Sci. **4**(3), 1–5 (2016)
79. Verma, A., Parashar, A., Jain, N., Singh, V.K., Rangappa, S.M., Siengchin, S.: Surface modification techniques for the preparation of different novel biofibers for composites. In: Biofibers and Biopolymers for Biocomposites, pp. 1–34. Springer, Cham (2020)
80. Rastogi, S., Verma, A., Singh, V.K.: Experimental response of nonwoven waste cellulose fabric–reinforced epoxy composites for high toughness and coating applications. Mater. Perform. Char. **9**(1), 151–172 (2020)
81. Bharath, K.N., Madhu, P., Gowda, T.G., Verma, A., Sanjay, M.R., Siengchin, S.: A novel approach for development of printed circuit board from biofiber based composites. Polym. Compos. **41**(11), 4550–4558 (2020)
82. Verma, A., Jain, N., Parashar, A., Gaur, A., Sanjay, M.R., Siengchin, S.: Lifecycle assessment of thermoplastic and thermosetting bamboo composites. In: Bamboo Fiber Composites, pp. 235–246. Springer, Singapore (2021)
83. Singh, K., Jain, N., Verma, A., Singh, V.K., Chauhan, S.: Functionalized graphite-reinforced cross-linked poly (vinyl alcohol) nanocomposites for vibration isolator application: morphology, mechanical, and thermal assessment. Mater. Perform. Char. **9**(1), 215–230 (2020)
84. Verma, A., Parashar, A.: Molecular dynamics based simulations to study the fracture strength of monolayer graphene oxide. Nanotechnology **29**(11), 115706 (2018)
85. Verma, A., Parashar, A.: The effect of STW defects on the mechanical properties and fracture toughness of pristine and hydrogenated graphene. Phys. Chem. Chem. Phys. **19**(24), 16023–16037 (2017)
86. Verma, A., Parashar, A., Packirisamy, M.: Tailoring the failure morphology of 2D bicrystalline graphene oxide. J. Appl. Phys. **124**(1), 015102 (2018)
87. Singla, V., Verma, A., Parashar, A.: A molecular dynamics based study to estimate the point defects formation energies in graphene containing STW defects. Mater. Res. Express **6**(1), 015606 (2018)

88. Verma, A., Parashar, A.: Molecular dynamics based simulations to study failure morphology of hydroxyl and epoxide functionalised graphene. Comput. Mater. Sci. **143**, 15–26 (2018)
89. Verma, A., Zhang, W., van Duin, A.C.: ReaxFF reactive molecular dynamics simulations to study the interfacial dynamics between defective h-BN nanosheet and water nanodroplets. Phys. Chem. Chem. Phys. **23**, 10822–10834 (2021)
90. Verma, A., Parashar, A., Packirisamy, M.: Role of chemical adatoms in fracture mechanics of graphene nanolayer. Mater. Today Proc. **11**, 920–924 (2019)
91. Verma, A., Parashar, A.: Characterization of 2D nanomaterials for energy storage. In: Recent Advances in Theoretical, Applied, Computational and Experimental Mechanics, pp. 221–226. Springer, Singapore (2020)
92. Verma, A., Jain, N., Parashar, A., Singh, V.K., Sanjay, M.R., Siengchin, S.: Design and modeling of lightweight polymer composite structures. In: Lightweight Polymer Composite Structures: Design and Manufacturing Techniques, Chapter 7, pp. 193–224. Taylor & Francis Group (CRC Press), Boca Raton (2020)
93. Verma, A., Jain, N., Parashar, A., Singh, V.K., Sanjay, M.R., Siengchin, S.: Lightweight graphene composite materials. In: Lightweight Polymer Composite Structures: Design and Manufacturing Techniques, Chapter 1, pp. 1–20. Taylor & Francis Group (CRC Press), Boca Raton (2020)
94. Verma, A., Parashar, A., Singh, S.K., Jain, N., Sanjay, M.R., Siengchin, S.: Modeling and simulation in polymer coatings. In: Polymer Coatings: Technologies and Applications, Chapter 16, pp. 309–324. Taylor & Francis Group (CRC Press), Boca Raton (2020)
95. Verma, A., Jain, N., Rastogi, S., Dogra, V., Sanjay, M.R., Siengchin, S., Mansour, R.: Mechanism, anti-corrosion protection and components of anti-corrosion polymer coatings. In: Polymer Coatings: Technologies and Applications, Chapter 4, pp. 53–66. Taylor & Francis Group (CRC Press), Boca Raton (2020)
96. Verma, A., Jain, N., Kalpana, Sanjay, M.R., Siengchin, S., Jawaid, M.: Natural fibers based bio-phenolic composites. In: Phenolic Polymers based Composite Materials, Chapter 10, pp. 153–168. Springer Nature, Singapore (2020)
97. Bharath, K.N., Madhu, P., Gowda, T.Y., Verma, A., Sanjay, M.R., Siengchin, S.: Mechanical and chemical properties evaluation of sheep wool fiber-reinforced vinylester and polyester composites. Mater. Perform. Char. **10**(1), 99–109 (2021)
98. Marichelvam, M.K., Manimaran, P., Verma, A., Sanjay, M.R., Siengchin, S., Kandakodeeswaran, K., Geetha, M.: A novel palm sheath and sugarcane bagasse fiber based hybrid composites for automotive applications: an experimental approach. Polym. Compos. **42**(1), 512–521 (2021)
99. Chaudhary, A., Sharma, S., Verma, A.: Optimization of WEDM process parameters for machining of heat treated ASSAB'88 tool steel using Response surface methodology (RSM). Materials Today: Proceedings (2021)
100. Chaudhary, A., Sharma, S., Verma, A.: WEDM machining of heat treated ASSAB '88 tool steel: a comprehensive experimental analysis. Materials Today: Proceedings (2021)
101. Verma, A., Singh, V.K.: Experimental characterization of modified epoxy resin assorted with almond shell particles. ESSENCE-Int. J. Environ. Rehab. Conser. 36
102. Verma, A., Samant, S.S.: Inspection of hydrodynamic lubrication in infinitely long journal bearing with oscillating journal velocity. J. Appl. Mech. Eng. **5**(3), 1–7 (2016)
103. Verma, A., Parashar, A., Jain, N., Singh, V.K., Rangappa, S.M., Siengchin, S.: Surface modification techniques for the preparation of different novel biofibers for composites. In: Biofibers and Biopolymers for Biocomposites, pp. 1–34 (2020)
104. Bisht, N., Verma, A., Chauhan, S., Singh, V.K.: Effect of functionalized silicon carbide nanoparticles as additive in cross-linked PVA based composites for vibration damping application. J. Vinyl Add. Tech. **27**(4), 920–932 (2021)
105. Kataria, A., Verma, A., Sanjay, M.R., Siengchin, S.: Molecular modeling of 2D graphene grain boundaries: mechanical and fracture aspects. In: Materials Today: Proceedings (2021)
106. Arpitha, G.R., Verma, A., Sanjay, M.R., Siengchin, S.: Preparation and experimental investigation on mechanical and tribological performance of hemp-glass fiber reinforced laminated composites for lightweight applications. Adv. Civil Eng. Mater. **10**(1), 427–439 (2021)

107. Deji, R., Verma, A., Kaur, N., Choudhary, B.C., Sharma, R.K.: Density functional theory study of carbon monoxide adsorption on transition metal doped armchair graphene nanoribbon. In: Materials Today: Proceedings (2021)
108. Deji, R., Verma, A., Choudhary, B.C., Sharma, R.K.: New insights into NO adsorption on alkali metal and transition metal doped graphene nanoribbon surface: a DFT approach. J. Mol. Graph. Model. (2021). https://doi.org/10.1016/j.jmgm.2021.108109
109. Deji, R., Jyoti, R., Verma, A., Choudhary, B.C., Sharma, R.K.: A Theoretical study of HCN adsorption and width effect on co-doped armchair Graphene Nanoribbon. Comput. Theor. Chem. 113592 (2022)

Defect Energy Calculations of Nickel, Copper and Aluminium (and Their Alloys): Molecular Dynamics Approach



Swati Chaturvedi, Akarsh Verma, Sushanta Kumar Sethi, and Shigenobu Ogata

Abstract Stacking faults are defined as regions present in the crystal where the normal sequence of stacking is disturbed. This takes place around a localized area. These defects come under the planar defect. An important role is played by these defects in deciding the characteristic properties of face-centered cubic (FCC) metals and alloys, which are close-packed. The vacancy formation energy can be defined as the change in energy, which takes place when the breaking of the bond takes place between one atom and its ligands and the formation of new bonds of atoms with ligands. Interstitial defects are the point type defect. In this type of defect, a site that is normally not occupied is occupied by an atom of the same or different type. Nickel is a pure FCC metal, and its stacking fault energy (γ) ranges from 79 to 415 mJ m^{-2}. When we remove atoms from crystalline sites, vacancy is formed. Various loading conditions were used for studying the stacking fault energy of FCC metals. Density functional theory (DFT) based on first principle calculations and the embedded atom model (EAM) were used for calculating the generalized stacking fault energy of nickel, copper and aluminium. The super cell approximation based on orbital-free DFT was used for calculating the formation energies of vacancies. Ni-based super alloys are gaining wide popularity because they have excellent strength and creep resistance. This chapter reports on the work done by researchers on evaluating various defect energies of nickel, copper and aluminium through computational techniques.

Keywords Stacking fault energy · Vacancy formation energy · Interstitial formation energy · Nickel · Aluminium · Copper

S. Chaturvedi · A. Verma (✉)
Department of Mechanical Engineering, University of Petroleum and Energy Studies, Dehradun 248007, India
e-mail: akarshverma007@gmail.com

A. Verma · S. Ogata
Department of Mechanical Science and Bioengineering, Osaka University, Osaka 560-8531, Japan

S. K. Sethi
Department of Metallurgical and Materials Science Engineering, Indian Institute of Technology Bombay, Mumbai 400076, India

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1 Introduction

1.1 Stacking Fault Energy

Stacking faults are defined as the regions present in the crystal where the sequence of normal stacking is disturbed. This generally takes place around a localized area. These types of defects come under the planar defect category. An important role is played by these defects in deciding the characteristic properties of face-centered cubic (FCC) metals and alloys that are closely packed in nature. A small difference in energy is noticed for ABAB… and ABCABC…. sequences [1]. When the growth of the crystal takes place, it is possible that one layer of atoms may fall into the incorrect place. We can take the example of copper (Cu) where the normal sequence of stacking is ABCABC … and a wrong sequence of stacking can be ABCBCABC… which is bound to occur when removal of one plane takes place. Stacking faults takes place when some work is done on the material; hence, they are mostly found in materials that are deformed rather than materials that are annealed [2]. The occurrence of slip happens usually along the vectors whose direction is towards the interatomic spacing, which is the shortest on an atomic plane of the closed packed crystal. In FCC metals, this is {1 1 1} type plane. The direction of the slip is $[10\bar{1}]$, which is denoted by b_1. On the other hand, the atom, which acquires position B that is on the A plane's top, has less availability of lattice translation. After this, it can acquire position C. This results in the slip movement, which is macroscopic along [101]. This is produced due to the zigzag path of B → C → B having vectors $b_2 = a/6[2\ \bar{1}\ \bar{1}]$ and $b_3 = a/6[1\ 1\ \bar{2}]$, where a denotes the lattice of the crystal. These dislocations, which are partial and have Burgers vector of magnitude b_2 and b_3, have a force of repulsion between them, which is denoted by Eq. (1)

$$F = \mu \frac{b_2 b_3 \cos 60^\circ}{2\pi d} \tag{1}$$

where μ denotes the shear modulus, d denotes the equilibrium spacing, and b_2 and b_3 denotes the burger vector with an angle of 60°. In the region between these partials, the formation of a sheet of stacking faults takes place in the plane of slip. The separation of partials is restricted because this region has higher energy as compared to a normal lattice. Taking the energy of the fault per unit area as γ J/m^2. Then we can denote the force, which is separating the partials as γ N/m. When we equate F with γ, we get the value of equilibrium width d of the stacking fault [1].

In FCC metals and alloys, the trade-off that takes place between short-range ordering and energy of stacking fault decides whether the slip is wavy or planar in nature. When the stacking fault energy is low or the short-range ordering is low, then the distribution of dislocations is planar [3, 4]. When the value of stacking fault energy is low, and the stacking fault is broader in nature. Only when the partials combine, then only the dislocations can cross slip or climb. Therefore, in metals and alloys having lower stacking fault energy, we can witness planar slip as there

is more homogenous deformation [5]. For alloys of copper, it has been observed that when it is alloyed with zinc and aluminium, the nature of slip is planar, but the nature of slip of pure copper is wavy. Thus, this change in slip is attributed to a decrease which is caused by stacking fault energy when allowed with aluminium and zinc. It is been shown experimentally that when alloyed with Nickel, the stacking fault energy increases which results in wavy slip [3]. However, it has been observed that until the content is 10%, the nature of the slip is planar. This is because of strong competition between short-range ordering and stacking fault energy to rule the behaviour of deformation. The cross-slipping is prevented due to the restriction in the glid planes because of high short-range ordering. When the content of nickel in copper increase, there is the change from planar nature to a wavy slip [3].

Over the years, various methods have been developed to determine stacking fault energy experimentally. The loop annealing approach states that the force which drives shrinking is due to stacking fault energy. This shrinking is caused due to the vacancy formation. Thus, the stacking fault energy is directly proportional to the rate at which the loops climb [6, 7]. The texture approach tells that the normal slip together with cross-slip could describe the difference, which is developed in the deformation texture of FCC metals. As the rate at which the cross-slip incidences are dependent on the stacking fault energy of the FCC metals, this approach requires knowledge of the shear modulus of crystals, which is not accessible, easily [8, 9].

1.2 Vacancy Formation Energy

The vacancy formation energy can be defined as the change in energy, which takes place when the breaking of the bond takes place between one atom and its ligands and the formation of new bonds of atoms with ligands. The breaking takes place in the parent material and forming takes place in the reference system. The vacancy formation energy is useful as it allows us to find the probability of formation of vacancy and with how much concentration. It is also useful for studying the environment of bonding that takes place in the materials [10–12]. A vital role is played by the vacancy in the phase transition that takes place from crystalline to amorphous state [13, 14]. The term "void" is used to denote the space, which is left by the atom. The vacancy formation energy (E_f) can be calculated through Eq. (2)

$$E_f = [E_v - E_0] + \mu_i \tag{2}$$

where E_v denotes the total energy of the vacancy containing and E_0 denotes the total energy during the initial configuration. μ_i denotes the chemical potential of the atom i which is removed from the initial configuration leading to the formation of the vacancy. The chemical environment from where the atom has been removed decides μ_i [15].

1.3 Interstitial Formation Energy

Interstitial defects are the point type defects. In this type of defect, a site that is normally not occupied is occupied by an atom of the same or different type. When the atoms are of the same type as that of the lattice site, it is known as a self-interstitial defect. The self-interstitial formation energy can be defined as the summation of electrostatic energy and band structure energy. These self-interstitial metals have a split structure. In these split structures, two atoms are present that share the same site of the lattice. At the lattice site, the center of mass of two atoms is present and displacement takes place symmetrically along the directions of the principle lattice. For example, in a FCC metal (which includes copper and nickel) the structure of the ground state is a split [100] interstitial structure. In this structure, displacement of two atoms is done along the positive and negative directions. They are also referred to as dumbbell interstitial [16, 17]. Figure 1 reveals some of the methods used for determining the stacking fault, vacancy formation and interstitial formation energy in this chapter.

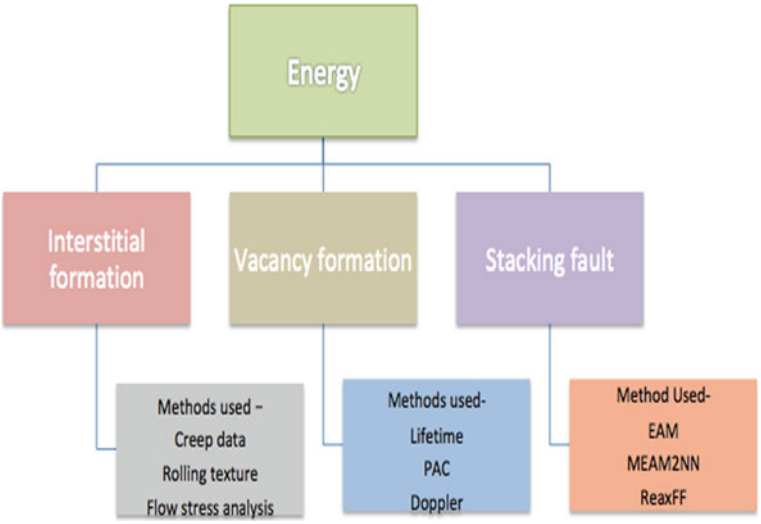


Fig. 1 Methods used for determining the stacking fault, vacancy formation and interstitial formation energies

2 Nickel

2.1 Stacking Fault Energy

Nickel is a pure FCC metal, and its stacking fault energy (SFE) γ ranges from 79 to 415 mJ m^{-2} [18]. Various methods were used in the past for finding these values, and the results are given in Table 1.

To study the two types of defects which are present in nickel that is dissociated glide dislocations and faulted dipoles, the weak beam technique of electron microscopy has been used. In the literature, strong-beam bright-field images were used to study the faulted dipoles present in nickel. However, it was found that large errors occur when we interpret such images. For these studies, the moulds were made of alumina and the specimens were made from crystals of pure nickel. The straining of the single crystals was done at room temperature. All six 220 reflections were studied to image the faulted dipoles and glide dislocation. The geometry used was weak beam diffraction, and the direction of the electron beam was close to $[\overline{1}11]$. The faulted dipoles had four partial dislocations. Apart from this, it had two Shockley and two stair-rods, which were connected using the stacking fault three ribbons. It was found that theoretically two configurations were possible Z and S. Dipoles were responsible for the formation of faulted dipoles, which had two dissociated 60 dislocations. These dislocations consisted of opposite Burgers vectors. The gliding of the Shockley created the central stacking fault partial on the inclined {111} plane. Formation of either Z or S dipoles could take place due to each reaction. However, the central fault's nature would differ in each pair of Z and S. It was found that all dislocations had a pure edge as when weak beam conditions were used to image the

Table 1 Determination of the SFE of nickel

γ (mJ m^{-2})	Method	Reference number
95	Creep data	[19]
150	Extrapolated (Ni–Co) node data	[20]
300	Extrapolated (Ni–Co) node data	[21]
450	Rolling texture	[9]
$\leq 180 \pm 55$	Faulted dipole	[22]
≥ 375	Absence of tetrahedra	[23]
$\leq 160 \pm 16$	Extrapolated (Ni–Co) tetrahedra data	[24]
≤ 160	Extrapolated (Ni–Co) tetrahedra data	[25]
240	Rolling texture	[26]
79	Dissociated edge dipole	[27]
234	Flow stress analysis	[28]

faulted dipoles, they were found out of contrast. The dimensions of dipole, thickness and depth of the foil were responsible for determining image peak separation. The results showed that for larger values of image peak separation, there were smaller values of dipole width. Furthermore, the true value of stacking fault energy lied at the upper range, when the dipole width was equal to the image peak separation [18].

2.2 Vacancy Formation Energy

When we remove atoms from the crystalline sites, a vacancy is formed. For pure nickel, the vacancy formation enthalpy was found to be $\left(H_v^f = 1.73 \pm 0.07\right)$ eV. To study the vacancy-like defects, positron lifetime measurements were used. This tool allows us to study the binding conditions. This binding takes place between vacancies and impurities of the atom. Sample tubes were taken that had a diameter of 5 mm and a length of around 45 mm. Spot welding was used to close one end of the sample tube. After this, a hydraulic operated press was used to flatten the sample tube and the end (that was open) was welded. The nickel reflector shield provided the thermal heat flux. A positron lifetime spectrometer covered with Teflon tapes was used. The prevacancy state was used to determine the source term. It was found that there was an increase in the source lifetime and a decrease in the intensity of the source as the temperature was increased. The temperature used was in the range of 490–1600 K for carrying out the experiments of positron lifetime. The results showed that at the room temperature, the value of bulk lifetime τ_f was 101 ps. On the other hand, at temperature 1243 K, the value of bulk lifetime τ_f was 113 ps. Thus, there was an increase in the value of a bulk lifetime. The reason behind the increase in bulk lifetime was attributed to the thermal expansion and fluctuations in the matrix atoms because of the increase in temperature [29]. The bulk lifetimes in nickel were low. Table 2 shows the enthalpies of vacancy formation, which were used by various authors using the technique of positron annihilation [29].

Table 2 Vacancy formation enthalpy

Method	Formation enthalpy (eV)	Reference number
Lifetime	1.54	[30]
PAC	1.72	[31]
PAC	1.74	[32]
PAC	1.55	[33]
Doppler	1.73	[34]
Doppler	1.72	[35]
Doppler	1.80	[36]
Theorite	1.79	[37]
Theorite	1.76	[38]

In the lattice, two positions are possible for interstitial impurities. The first position is tetrahedral, and the second is octahedral. In FCC lattice, generally the octahedral site is preferred. This is due to size and potential [29].

2.3 Interstitial Formation Energy

When we insert atoms in the central octahedral sites present in FCC cells, it leads to the formation of an interstitial. EAM potential was used to test convergence on this defect type. The formation energy of n interstitials (i), E_{ni}^{f} is given by Eq. (3)

$$E_{ni}^{f} = E_{N-n} - \left(E_N \times \frac{N-n}{N} \right) \tag{3}$$

where E_N denotes the total energy of the crystalline system, N denotes the number of nickel atoms and $E_{N\text{-}n}$ denotes the total energy of the system having n interstitials.

A k-ART version was used along with a Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) to understand the effect of various force fields on the defect kinetics of Nickel. The four potentials used were the embedded atom method (EAM), the modified embedded atom method (MEAM), the modified embedded atom model (which was based on second neighbor formalism (2NN)), and the reactive force field (ReaxFF). The same ground state was provided by these four potentials for the self-interstitial. Apart from this, an additional atom shared the site by forming a dumbbell. This dumbbell was at the center of the lattice site along with two atoms, which were at a distance of 2.12 Å in the {100} directions. Table 3 shows the formation of energy through various potentials [39].

Two self-interstitial systems were simulated where the defects had a separation of 4.98 Å with respect to the distance between four nearest neighbors (NN) in crystalline FCC nickel. The formation of two parallel dumbbells takes place in the nearest neighbor positions as the defect comes close to each other. The barrier is crossed with 0.19 eV energy that helps in achieving the rotation of two dumbbells. The barrier crossing is characterized by the turning of two dumbbells at 22.5°. The two self-interstitial atom system moves by two successive self-interstitial shift rotations with a 0.27 eV barrier and with the help of a second low-energy diffusion mechanism. A

Table 3 Formation energies in Nickel using different potentials

Potential	Formation energy (in eV) [Reference number]	
EAM	4.54 [39]	5.05 [40]
MEAM1NN	4.37 [39]	4.24 [41]
MEAM2NN	4.21 [39]	4.88 [42]
ReaxFF	4.78 [39]	–
DFT	–	4.07 [43]

barrier of 0.29 eV was also observed in the mechanism of third low energy diffusion. It was found that in compact lattices, the interstitial diffusion is a very complex process. The number of self-defects increases as the number of mechanisms and barriers also increases. Simulations showed that the diffusion is rapid along the direction (110) for FCC metals' small self-interstitial clusters [39].

3 Copper

3.1 Stacking Fault Energy

For determining the mechanical properties and mechanisms of deformation, stacking faults play an important role in the FCC metals. The stacking fault energy of copper has been found to be 42 J/m^2. Various loading conditions were used for studying the stacking fault energy of copper. The projector augmented wave method was used. For this method, a super cell was constructed. This super cell had 24 closed-packed (111) atomic layers along the direction [111]. The top and bottom of the super cell had a separation of about 15 Å. This separation was necessary to avoid the interaction that took place between the periodic images. To find the value of stacking fault energy, large values of strain $\in$ (up to 0.1) were used. Three approaches to loading were used, which were a simple strain, volume conserving strain and strain with uniaxial tension [44]. Table 4 shows the stacking fault energy determined using various methods.

The behaviour of stacking fault energy was studied under the volumetric strain. This was done to understand the behavior of deformation, under the conditions of

Table 4 Stacking fault energy using various methods

Method	SFE (mJ/m^2)	Reference number
DFT-GGA-PBE	41	[44]
Morse potential	30	[45]
DFT-LDA	50	[46]
Tight binding linear muffin tin orbital Green's function	56	[47]
EMT/EAM potential	78	[48]
DFT-GGA-PW91	39	[49]
DFT-GGA-PW91	33	[50]
Measurements of intersections of grain boundaries and coherent twins boundaries	42	[51]
Measurements of radius of curvature of extended dislocations by TEM	40	[20]
Measurements of dependence of critical stress for cross slip with strain rate	50	[52]
Measurement of partial dislocations spacing	41	[53]

high pressure and shock loading. It was found that the stacking fault energy value decreased as the value of volumetric strain was increased. The value of stacking fault energy was positive when the range of strain applied was large. The variation of stacking fault energy was studied as a function of strain because the copper is anisotropic in nature. The longitudinal strain was applied along the [111] direction. In a simple strain procedure, the dimensions of the cell were rescaled by $(1 + \in)$ along the [111] direction. For the case of uniaxial stress, the length of the unit cell was changed by $(1 + \in)$ and the dimensions of the cell were rescaled in x and u directions so as to ensure zero stress along these directions. For the case when the volume is conserved, the strain is applied along the [111] direction. The value of stacking fault energy was calculated for all three cases. The stacking fault energy was a linear function of [111] longitudinal strain. For the case where the volume was conserved, the values of stacking fault energy could be well fitted in a cubic polynomial. When the value of compressive strain was -0.015, it was found that the maximum value of stacking fault energy occurred [44].

3.2 Vacancy Formation Energy

The thermodynamic and kinetic properties of a metal are greatly affected by the presence of balances in metals. To understand the process of mechanical deformations or treatment of heat, it is necessary to understand the energy of the formation of a vacancy. Various methods such as positron annihilation, ab initio full potential, vacancy-vacancy and vacancy solute interaction in copper using the Korringa-Kohn-Rostoker (KKR) function method have been done. At equilibrium conditions, the numbers of vacancies in elemental copper are very less. Thus, for understanding the metal-vacancy system, it is required to take a single vacancy in bulk or take a large super cell where the separation of vacancies is well enough. However, large supercells are necessary to understand the interaction that takes place between vacancy-vacancy and vacancy-solute in alloys. For large supercells, the conventional band-structure method becomes inefficient as the computation efforts increase to N^3 (where N denotes the number of atoms present in the unit cells). Thus, the method, which is efficient for such cases is the LSFG method, which can be used to deal with ordered and disordered systems. LSFG methods had their origin from atomic sphere approximation where vacancy formation energy is overestimated by a factor of 2. We will consider a bulk crystal, which contains N_a atoms and N_v vacancies. We then define N as the total number of atoms present in the site of lattice and is denoted by Eq. (4)

$$N = N_a + N_v \tag{4}$$

Then the vacancy formation energy, E_{1V}^F, can be given by Eq. (5), where V denotes constant volume V.

Table 5 The monovacancy formation energy (H_{1V}^{F}) and the binding energies of two vacancies which are first $E_{2V}^{B}(1)$ and second $E_{2V}^{B}(2)$ neighbors in FCC copper

Method	H_{1V}^{F}	$E_{2V}^{B}(1)$	$E_{2V}^{B}(2)$
LSGF ASA + M	1.33	−0.096	0.080
FP-KKR-GF	1.41	−0.076	–

$$E_{1V}^{F} = \left(\frac{\partial E(N_v)}{\partial N_v} | N_v = 0 \right) V, N_a \tag{5}$$

The convergence of LSFG in the ASS + M was evaluated by calculating the vacancy formation energy in copper. Varying the size of the super cell did this. The monovacancy formation energy and the binding energies, which are the first and second of two vacancies, are given in Table 5. For containing the coordination shells for the FCC supercells, the zone of local interaction was chosen [54].

3.3 Interstitial Formation Energy

Various interstitial configurations were used for calculating the formation energies of copper. These configurations include (100) dumbbell, where two atoms were placed symmetrically around a normal lattice site which was vacant, (110) dumbbell or crowdion, (110) dumbbell, tetrahedral interstitial where one atom is placed at the interstitial site and last octahedral site where one atom is placed at the octahedral site [55]. Table 6 shows the formation energies of the various configuration of self-interstitial in copper.

It was found that (100) dumbbell interstitial was the most stable configuration with the lowest formation energy. Dynamic simulations were used for confirming the stability of (100) dumbbell. In this system, which had defects was made to evolve towards equilibrium. The values were calculated for the cut-off radius $r_c = 2.872$ a. It was found that the self-interstitial formation energy of copper was around 2.61 eV. Using the empirical inter-atomic potentials, various simulations were performed in

Table 6 Formation energies of various configurations of self-interstitial in copper

Configuration	Formation energy (eV) Cu[a]
(100) dumbbell	2.652
(110) dumbbell	2.856
(111) dumbbell	– [d]
Tetrahedral interstitial	2.914
Octahedral interstitial	2.902

[a] Calculated for a cutoff radius $r_c = 2.872$ a
[d] Unstable configuration in molecular statics

Table 7 Formation energy of a (100) dumbbell self-interstitial in Cu for various cut-off radii

Cutoff radius, r_c (a)	Formation energy, $E_i^f\,(eV)$
2.217	2.650
2.548	2.875
2.872	2.652
3.196	2.612

Table 8 Interstitial formation energy using different potentials

Method	Formation energy (eV)	Reference number
Born–Mayer repulsive potential	4.14	[56]
Modified Morse potential	3.4	[57]

copper [55]. Table 7 shows the formation energy of a (100) dumbbell self-interstitial in copper for various cut-off radii.

With Born–Mayer repulsive potential, it was found that (100) dumbbell configuration was the most stable configuration with a formation energy of 4.14 eV. However, when the non-central potential was used, it was found that both the (111) dumbbell and tetrahedral interstitial have the configuration of the lowest energy. With Modified Morse potential, it was found that the most stable configuration was (100) dumbbell with a formation energy of about 3.4 eV [55]. Table 8 shows the interstitial formation energy using different potentials.

4 Aluminium

4.1 Stacking Fault Energy

Density functional theory (DFT) and the embedded atom method (EAM) were used for calculating the generalized stacking fault energy of aluminium. For the plane wave basis, a kinetic energy cutoff of 12 Ry was used. When the forces of Hellmann–Feynman on an individual atom were less than 0.001 Ry/au, a fully relaxed atomic structure was taken into consideration. The calculated value of the equilibrium lattice constant (a_0) was found to be 3.94 Å. On the other hand, the experimental value of the equilibrium constant was found to be 4.05 Å, and the calculated value of bulk modulus was found to be 82.52 GPa. The experimental value of bulk modulus was found to be 76.93 GPa. Thus, there was good agreement between the results obtained from calculations and experiments. A super cell which had six and nine atomic layers in the (111) direction was employed to simulate the process of block shearing. A

Table 9 Fault vectors and energies for four stacking faults obtained from DFT and EAM calculations

	Vector	DFT (J/m^2)	EAM (J/m^2)
Intrinsic stacking	1/6 $[12\bar{1}]$	0.164	0.120
Unstable stacking	1/10 $[12\bar{1}]$	0.224	0.141
Unstable stacking	1/4 [101]	0.250	0.663
Run-on stacking	1/3 $[12\bar{1}]$	0.400	1.354

slip of $a_0/\sqrt{6}$ in the (112) direction was indicated by intrinsic stacking fault configuration. This leads to stacking order ABC|BCAB. The lowest energy barrier was indicated by the unstable stacking fault energy. The in-plane lattice constant was used for performing the theoretical calculations. In order to get accurate generalized stacking fault energy, both the atomic and volume relaxation were carried out. The value of intrinsic stacking fault energy came out to be 0.164 J/m^2. For embedded atom model calculations, the Ercolessi-Adams potential was used for the calculation of generalized stacking fault energy. A grid of 40 points was used for calculating generalized stacking fault energy γ (f) for the (111) plane. To ensure the easy computation of dislocation properties, calculated values of DFT generalized stacking fault energy were made to fit on symmetrized polynomial. The projections of the DFT and EAM generalized stacking fault energy surfaces were observed in the $[12\bar{1}]$ and [101] directions. For both calculations related to DFT and embedded atom model, it was found that the value for unstable stacking fault energy was larger for [101] as compared to $[12\bar{1}]$. The emission of dislocation from the crack tip was affected by the anisotropy of the unstable stacking fault energy. Table 9 shows the result obtained from DFT and EAM calculations for stacking fault energies [58].

4.2 Vacancy Formation Energy

The super cell approximation based on orbital-free density functional theory was used for calculating the formation energies of vacancies in aluminium. The Kohn–Sham density functional theory (KS-DFT) was used for the first-principles modeling. For simulating the radiating effect, orbital-free density functional theory (OF-DFT) was used. OF-DFT uses electron density; on the other hand, KS-DFT uses single-electron orbital. Kinetic energy density functional (KEDF) and exchange-correlations density functional (XCDF) were used for calculating the formation energy of vacancies. Local density approximations (LDA) and generalized gradient approximations (GGA) were used for describing the exchange–correlation energy functional. The Wang-Govind-Carter (WGC) was adopted as it was the most accurate functional. In the fully relaxed cells, defects were introduced. To calculate the formation energies, the structural relaxation was performed without any doubt symmetry. For defect of single vacancy, it was done by the elimination of one atom, which was at the center of the super cell; and the formation energy was given by Eq. (6)

Table 10 Vacancy formation energies of FCC-Al obtained by the KS-DFT and OF-DFT methods with different XCDFs and KEDFs

Method	KEDF	XCDF	E_v^f (eV)
KS-DFT [60–69]	–	LDA	0.66–0.73
KS-DFT [64, 66, 68]	–	GGA	0.54–0.55
OF-DFT [60, 61, 70, 71]	WGC	LDA	0.48–0.72
OF-DFT [61]	WGC	GGA	0.387
Experiment [72]	–	–	0.67 ± 0.03

$$E_v^f = E_{(n-1)Al} - \frac{n-1}{n} E_{nAl} \tag{6}$$

where $E_{(n-1)Al}$ denotes the super cell's total energy that contains (n-1) Al atoms and one vacancy, and the total energy of the perfect aluminium super cell was denoted by E_{nAl} that had n Al atoms. Firstly, the size dependence was studied for the calculation of the formation energy of vacancy. Super cell N_s size was the x-axis. The distance $d = N_s a_0$ denoted the distance between a specified defect and its image defect that was located at the neighboring cell. When there was an increment in N_s, it was found that the value of d increased. This leads to the reduction of defect-defect interaction. The exponential function was used so as to fit the formation energy of vacancy. It was found that the formation energy converged by a factor of 3 meV for $4 \times 4 \times 4$ super cell for WGC-KEDF. But in WT-KEDF, the convergence was not quick [59]. Table 10 shows the results of the formation energy of vacancy for FCC-Al that was obtained using the method of OF-DFT.

4.3 Interstitial Formation Energy

The super cell approximation based on orbital-free density functional theory was used for calculating the formation energy of interstitial in aluminium. Kohn–Sham density functional theory (KS-DFT) was used for the first-principles modeling. For simulating the radiating effect, orbital-free density functional theory (OF-DFT) was used. OF-DFT uses electron density; on the other hand, KS-DFT uses a single-electron orbital. Kinetic energy density functional (KEDF) and exchange-correlations density functional (XCDF) were used for calculating the formation energy of vacancies. Local density approximations (LDA) and generalized gradient approximations (GGA) were used for describing the exchange–correlation energy functional. The Wang-Govind-Carter (WGC) and Wang-Teter were adopted as they were the most accurate functional. In aluminium, for an interstitial defect a single atom of Al was added into the super cell. This was done at various interstitial positions. Then the formation energy of interstitial E_i^f was given by Eq. (7)

$$E_i^f = E_{(n+1)Al} - \frac{n+1}{n} E_{nAl} \tag{7}$$

Table 11 Interstitial formation energies (in units of eV) obtained from the KS-DFT and OF-DFT method with various XCDFs and KEDFs

Method	KEDF	XCDF	$E^f_{(100)}$	$E^f_{(110)}$	$E^f_{(111)}$	E^f_{oct}	E^f_{tet}
KS-DFT [59]	–	LDA	2.6073	2.9809	3.1821	2.9485	3.2941
KS-DFT [59]	–	GGA	2.4597	2.7508	3.0183	2.7945	3.1072
0F-DFT [59]	WGC	LDA	2.5862	–	–	–	–
OF-DFT [59]	WGC	LDA	2.6031	2.8854	3.0986		3.1466
OF-DFT [59]	WGC	GGA	2.3925	–	–	–	–
OF-DFT [59]	WGC	GGA	2.4200	2.6856	2.8784	2.6724	2.9217
OF-DFT [59]	WT	LDA	2.4257	2.7000	2.8507	2.6672	2.8749
OF-DFT [59]	WT	GGA	2.2943	2.5422	2.6789	2.5302	2.6998
KS-DFT [69]	–	LDA	2.9	–	–	3.1	3.5
Experiment [72]	–	–	–	–	3.0	–	–

where $E_{(n+1)Al}$ denotes the total energy of the super cell that had n atoms of Al, which were present at the lattice sites, and one interstitial atom of aluminium. The total energy of the perfect aluminium super cell was denoted by E_{nAl}, which had n Al atoms. (100) dumbbell, (110) dumbbell, (111) dumbbell, octahedral and tetrahedral were the five types of self-interstitials that were considered, and the energies related to these types were denoted by $E^f_{(100)}$, $E^f_{(110)}$, $E^f_{(111)}$, E^f_{oct} and E^f_{tet}, respectively. Convergent behaviour was shown by the interstitial energy of formation when compared with the size of a super cell. The energy of formation for (100) dumbbell interstitial was studied as a function of N_s. A super cell having a size of $7 \times 7 \times 7$ was taken into consideration for the criterion of rough convergence of 5 meV. Table 11 shows the interstitial formation energy of FCC-Al, which was obtained from OF-DFT and KS-DFT. It was found that the error was less than 0.15 eV when OF-DFT was used with WGC-KEDF [59].

5 Nickel–Copper Alloys

5.1 Stacking Fault Energy

When alloying of copper is done with nickel, it is found that the stacking fault energy increases. As the content of nickel in copper is increased, it is observed that there is a transition from planar slip to wavy. For creating stacking fault energy in molecular dynamic simulations, many options are available. Options include removing the entire plane or just displacing a single plane with respect to the remaining crystal. Suitable dimensions were taken to create a simulation box and were populated with FCC array of pure nickel atoms. In the x and y directions, periodic boundary conditions were imposed. On the other hand, in the z-direction free boundary conditions

Table 12 Stacking fault probability (α) for Cu–Ni [74]

Average content of solute atoms (at. %)	$\alpha\left(\times 10^{-3}\right)$
9.6	5.7
23.3	6.5
48.1	9.5
75.0	3.8

were imposed. Some portion of the simulation box was then alloyed with copper. Nickel and copper form a substitutional solid solution; therefore, the Cu atoms from their lattice sites were displaced by the Ni atoms. The system's stacking fault energy, γ_{SF}, was given by Eq. (8)

$$\gamma_{SF} = \frac{E_f - E_i}{A} \tag{8}$$

where E_f denotes the systems' energies after stacking fault was created. On the other hand, E_i denotes the systems' energies before stacking fault formation, and the area of the stacking fault was denoted by A. The results showed that the stacking fault energy increased as the content of nickel in copper increased. This effect was because of the change, which was caused by the electron-to-atom ratio. When the concentration of the alloying content was increased, it leads to a decrease in the e/a ratio [73]. Table 12 shows the stacking fault probability (α) for Cu–Ni.

5.2 *Vacancy Formation Energy*

In the substitutional alloys, vacancy properties are gaining popularity. In the disordered alloys, vacancy properties were studied by the ab initio approach and by studying vacancies that were embedded in a medium. First, a state is chosen with vacancies as the reference state. This not only simplifies the problem but also is applicable for the substitutional alloys above the transition temperature. From the first law of thermodynamics we can define energy E by Eq. (9)

$$dE = \sum YdX = Tds - PdV + \sum_i \mu_i dN_i + \in_{SRO} d\eta_{SRO} + \Delta E_d dN_d + \cdots \tag{9}$$

where intensive and extensive variables are denoted by the symbols Y and X, respectively. The subscript i denotes the species of atom, T denotes the temperature, P denotes the pressure, E_d denotes the defect energy, N_d denotes the number of defects and SRO denotes the short-range order. $\in_{SRO}$ denotes the effective interaction of energy and η_{SRO} is the parameter of extensive short-range order. In binary A-B random alloys, many atoms A and B surround vacancies. In FCC alloy, the neighbor

Table 13 Vacancy formation enthalpy of FCC Cu and Ni

Method	H_{vac}^{Cu}	H_{vac}^{Ni}	Reference number
LDA	1.26	–	[76]
PW91	0.99	–	[76]
PBE [77, 78]	1.06	1.41	[75]
PBE [77, 78] and 16-atom cell	1.03	1.46	[79]
PBE [77, 78] and 32-atom cell	1.02	1.46	[79]
PBE [77, 78] and 32-atom cell	1.04	1.44	[80]
PBE [77, 78] and 108-atom cell	1.06	–	[76]
AM05 [81] and 32-atom cell	1.28	1.75	[75]
AM05 [81] and 32-atom cell	1.26	1.69	[80]
AM05 [81] 108-atom cell	1.29	–	[76]
Experiment	0.92–1.27	1.45–1.8	[82]
Experiment	1.04–1.49	–	[83]
Experiment	–	1.2–1.68	[84]
Experiment (LGT DD + PAS)	1.06	–	[76]

of a vacancy can vary from twelve A and zero B atoms. The vacancy formation enthalpy is affected by the composition of shells. The alloy A-B is treated as a ternary with the vacancy in the approach of cluster expansion. Thus, in an alloy the vacancy formation enthalpy is a weight sum over surrounding and atoms, which are removed. Table 13 shows the vacancy formation enthalpy of FCC Cu and Ni. The energy that is required to put the atom that is vacated back into the alloy is represented by the chemical potential. In Cu–Ni alloy, it was found that the chemical potential of Cu (Ni) started decreasing when the composition started becoming richer in Cu (Ni). This indicates that it was difficult to put back a Cu–Ni atom in a Ni-rich alloy as compared to the Cu-rich alloy [75].

Table 14 Stacking fault energies for the Cu–Al alloys

Alloy (wt.%)	Alloy (at.%)	Stacking fault energy (mJ m^{-2})	Reference number
100 Cu	100 Cu	41 ± 9	[53, 87]
Cu–2Al	4.6Al	35 (estimate)	[86]
Cu–4.1Al	9.2Al 10Al	14 (estimate) 10.4 ± 1.8	[88]
Cu–7.85Al	16.7 Al 16Al	5 (estimate) 5.8	[89]

6 Aluminium–Copper Alloys

6.1 Stacking Fault Energy

In FCC crystals, the stacking fault energy γ plays an important role in determining the mechanical and physical properties. To determine how the composition affects γ, many studies have been carried out in the past. It has been found in many alloys, the y decreases on increasing the electron-to-atom ratio of a solid solution. In FCC crystals, γ is also affected by the temperature. Bridgman technique was used to grow single crystals of Cu-6.44 at.% Al and Cu-13.43 at.% Al alloys in a graphite mould. The homogeneity of the crystal was maintained. It was found that in Cu-6.44 at.% Al and Cu-13.43 at.% Al, there was a reversible decrease in dislocation width as the temperature increased [85]. To study the effect of stacking fault energy in fatigue crack propagation of Cu–Al alloys, three alloys were taken into consideration having the composition: Cu–2Al, Cu–4Al, and Cu–8Al. From the analysis done, it was found that the content of Aluminium was around 1.98, 4.10 and 7.85, respectively. Cold rolling and annealing was used to find the grain sizes. Table 14 gives the value of stacking fault energy for Cu–Al alloys. It was found that as the aluminium content was increased there was an increase in the planarity of slip, which was responsible for the decrease in stacking fault energy [86].

6.2 Vacancy Formation Energy

In all the diffusion-controlled and thermal diffusion processes, an important role is played by the interaction that takes place between the vacancy-solute. For the production and agglomeration of defects, which are induced by radiation, an important role is played by the interaction that happens between the solute atoms and vacancies. There are enormous difficulties when calculations are done through the first principles. Knowledge of solidus temperature is used for calculating the concentration of vacancies in alloys. The vacancy concentration in a pure solvent, C_v^0, at temperature T can be given by Eq. (10)

Table 15 Theoretically calculated and experimental values of the effective vacancy formation energy (E_v^e) for binary Cu–Al alloys

Admixture and its content in atomic %	E_v^e(eV)		Reference number
	Calculated	Experimental	
2.0	1.13	1.17	[91]
4.6	1.12	1.10	[92]
8.3	1.12	1.02	[93]
8.5	1.10	1.06	[91]
13.7	1.04	1.00	[91]
14.0	1.04	1.03	[91]
15.0	1.02	1.00	[92]
16.6	1.01	0.93	[91]
18.0	1.01	1.00	[92]

$$C_v^0 = A \exp\left(-\frac{E_v}{kT}\right) \tag{10}$$

where A denotes the constant entropy factor, E_v denotes the formation energy of vacancy, T denotes the temperature and k denotes the constant. Table 15 shows the effective vacancy formation energies for Cu–Al alloy. The values of vacancy formation energy, which were theoretically calculated, were in good agreement with those obtained from the experiments. The experimental analysis of the interaction that took place between solute-vacancy in the Cu–Al alloy was not intensive [90].

7 Nickel–Aluminium Alloys

7.1 Stacking Fault Energy

For Ni–Al based superalloys, generalized stacking fault energy surfaces (Γ-surfaces) were calculated using the first-principles calculations. Applications that need strength and resistance use Ni-based superalloys. To exhibit strength at high temperature, an important requirement is the mixing of γ'-precipitates in a γ-matrix. For greater efficiency, the components are exposed to high temperature. To understand the mechanical and physical properties of the alloys, efforts are currently been made. Modeling at the atomistic and the continuum scale play an important role. The deformation mechanism of a material can be understood by the generalized stacking fault energy as it is two-dimensional surface energy [94]. A large number of faults are formed in the alloys that have low stacking fault energies. This leads to a greater accumulation of partial dislocations. This in turn leads to high ductility and fracture toughness [95]. In the phase-field model for dislocations, the Γ-surfaces were used as an input.

It is a continuum approach and describes the dislocation's motion in a system, which is under applied stress [96]. First-principles based quantum mechanics simulations are used for calculating the Γ-surfaces. To understand atomistic mechanisms, which manifest themselves at scales of micro-structural length for the Ni-based alloy, this concept has been applied. To calculate the Γ-surfaces, the titled cell method was used. In Ni–Al alloy, the Γ-surfaces were calculated for both phases $-\gamma$ and γ'. A crystal was cut along one of its planes and the upper half was displaced by a displacement vector u with respect to the lower half. The plane chosen was closed packed {111} plane. This plane was of the primary slip system. The fault vector taken had the composition of vectors of orthogonal coplanar $[\bar{1}10]$ and $[\bar{1}\bar{1}2]$. Central force interaction was used in a model of hard sphere for calculating the stacking fault energy. Density functional theory (DFT) and force field methods [97, 98] were used for calculating the Γ-surfaces. DFT was used for calculating the total energies of super cells, which were then implemented in CASTER (a code of plane wave). It uses the ultra soft pseudo potentials in which core electrons are replaced that are present in the system. The spacing of k-point taken for calculations was 0.08°. The cell axes were taken parallel to the crystallographic axes $[\bar{1}10]$, $[\bar{1}01]$ and [111]. The planar faults were elastically isolated because of the periodic boundary conditions. FCC Ni was used to carry out trial simulations, which helped in determining the minimum number of layers required. For cells with 6, 9, 12 and 15 (111) layers, intrinsic and unstable stacking fault (ISF and USF2) energies were calculated. This helped in reaching the conclusion that a minimum of six layers are required to ignore image effects. Table 16 shows the intrinsic stacking fault and unstable stacking fault energy of FCC Ni [99].

There was less equivalency in k-point sampling, as the energy of a tilted cell was compared with energy of non-tilted cell. Multiple systems were taken to perform a test where the comparison between energy of a perfect crystal and a3 axis was done. Table 17 shows the error that resulted due to in equivalent k-point sampling. Before

Table 16 Intrinsic stacking fault (ISF) and unstable stacking fault (USF2) energies (mJ/m^2) of FCC Nickel

	Number of layers			
	6	9	12	15
Γ_{ISF}	149	153	152	154
Γ_{USF_2}	1087	1091	1095	1096

Table 17 Energy difference due to equivalent k-point sampling is shown between perfect crystals with and without the a3 cell vector being tilted in the direction a$[\bar{2}11]$

System	Energy difference	
	meV/atom	mJ m^{-2}
Ni_3Al	0.11	0.29

Table 18 Complex stacking fault (CSF), anti-phase boundary (APB) and super lattice intrinsic stacking fault (SISF)

Reference number		[99]	[100]	[101]	[102]	[103]
Ni_3Al	Γ_{SISF}	74	39	80	79	37
	Γ_{APB}	162	152	210	220	172
	Γ_{CSF}	191	170	225	267	187

extracting a final energy, it was necessary for the system to be in a relaxed state when fault was introduced into a super cell [99].

Two options are available for calculating the Γ-surfaces in DFT codes. The stacking sequence changes from …ABCABC… to …ABC|BCA, where | denotes the fault's position, when an intrinsic stacking fault is introduced. When the fault is introduced, there is no change in the short range bonding that takes place near the fault [99]. Table 18 shows the values of complex stacking fault (CSF), anti-phase boundary (APB) and super lattice intrinsic stacking fault (SISF).

7.2 Vacancy Formation Energy

Ni-based super alloys are gaining wide popularity because they have excellent strength and creep resistance. Thus, they are structural materials of high performance [104, 105]. Dislocation motions are affected by the defects such as vacancies and impurities. Embedded atom model (EAM) and quantum mechanics based DFT [106] calculations are performed in defected Ni–Al alloy system. Embedded atom model was successful for modeling the various properties of materials [107, 108]. Various non-uniform precision techniques also appear to be promising. These non-uniform precision techniques include LOFT (Learn on the fly). The scheme used in this technique is quantum mechanical/molecular mechanical (QM/MM) approach [109, 110]. Vacancies are important defects, which allow us to enable the plastic behaviour through dislocation climb. To fit the experimental values that were obtained for vacancy formation energy were reproduced by fitting the EAM potential. A $3 \times 3 \times 3$ super cell was used having 128 atoms for the γ phase. In this system, one atom was removed to create an isolated vacancy. The formation energy of vacancy was then calculated that was done by the optimization of three intermediate images by QM/MM calculations [111]. Table 19 shows the vacancy formation energy in γ phase of Nickel.

Ni–Al has excellent chemical, physical and mechanical properties [112, 113]. To know more about interatomic bonding [114–117], electronic structure [118–123],

Table 19 Vacancy formation energy in γ phase of Nickel

	DFT	EAM
E_v^f (eV)	1.45	1.57

properties of cohesion [124–128], energy-related to defects [129–144] and properties related to optics [135–139] various ab-initio studies have been done. The differences that are observed in the lattice and density of Ni–Al could only be justified by assuming that nickel atoms (which are present at the anti-site in sub-lattice aluminium) exist in off-stoichiometric alloys that are Ni–Al based [140, 141]. This can also be applied to alloys and metals having chemical bonding between them [142–144]. The local strengths of bond could be explained by ordering of vacancies [145–148]. The authors have a vast experience in the field of molecular dynamics and experimental characterization of composites materials [149–195].

8 Conclusion

Stacking faults are defined as regions present in the crystal where the sequence of stacking that is normal is disturbed. This takes place around a localized area. These defects come under planar defects. The vacancy formation energy can be defined as the change in energy, which takes place when the breaking of the bond takes place between one atom and its ligands, and the formation of new bonds of atoms with ligands. Interstitial defects are point type defects. In this type of defect, a site that is normally not occupied is occupied by an atom of the same or different type. An important role is played by these defects in deciding the characteristic properties of FCC metals and alloys, which are close-packed.

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References

1. Smallman, R.E.: Modern Physical Metallurgy. Elsevier (2016)
2. Chetty, N., Weinert, M.: Stacking faults in magnesium. Phys. Rev. B **56**(17), 10844 (1997)
3. Wang, Z.Y., Han, D., Li, X.W.: Competitive effect of stacking fault energy and short-range clustering on the plastic deformation behavior of Cu-Ni alloys. Mater. Sci. Eng. A **679**, 484–492 (2017)
4. Weidner, A., Man, J., Skrotzki, W., Polák, J.: Slip localization and dislocation structure at early stages of fatigue damage in austenitic stainless steel (316L). ICF12, Ottawa, 2013 (2009)
5. Dou, Y., Luo, H., Jiang, Y., Tang, X.: Effects of alloying elements on the stacking fault energies of Ni58Cr32Fe10 alloys: a first-principle study. Metals **9**(11), 1163 (2019)
6. Kannan, V.C., Thomas, G.: Dislocation climb and determination of stacking-fault energies in Al and Al-1% Mg. J. Appl. Phys. **37**(6), 2363–2370 (1966)
7. Dobson, P.S., Goodhew, P.J., Smallman, R.E.: Climb kinetics of dislocation loops in aluminium. Phil. Mag. **16**(139), 9–22 (1967)

8. Thornton, P.R., Mitchell, T.E., Hirsch, P.B.: The dependence of cross-slip on stacking-fault energy in face-centred cubic metals and alloys. Phil. Mag. **7**(80), 1349–1369 (1962)
9. Dillamore, I.L., Smallman, R.E.: The stacking-fault energy of FCC metals. Phil. Mag. **12**(115), 191–193 (1965)
10. Kim, H.S., Cook, J.B., Lin, H., Ko, J.S., Tolbert, S.H., Ozolins, V., Dunn, B.: Oxygen vacancies enhance pseudocapacitive charge storage properties of MoO 3–x. Nat. Mater. **16**(4), 454–460 (2017)
11. Long, S.O., Powell, A.V., Vaqueiro, P., Hull, S.: High thermoelectric performance of bornite through control of the Cu (II) content and vacancy concentration. Chem. Mater. **30**(2), 456–464 (2018)
12. Schuler, T., Barouh, C., Nastar, M., Fu, C.C.: Equilibrium vacancy concentration driven by undetectable impurities. Phys. Rev. Lett. **115**(1), 015501 (2015)
13. Sun, Z., Zhou, J., Pan, Y., Song, Z., Mao, H.K., Ahuja, R.: Pressure-induced reversible amorphization and an amorphous–amorphous transition in Ge2Sb2Te5 phase-change memory material. Proc. Natl. Acad. Sci. **108**(26), 10410–10414 (2011)
14. He, S., Zhu, L., Zhou, J., Sun, Z.: Metastable stacking-polymorphism in Ge2Sb2Te5. Inorg. Chem. **56**(19), 11990–11997 (2017)
15. Cheng, Y., Zhu, L., Wang, G., Zhou, J., Elliott, S.R., Sun, Z.: Vacancy formation energy and its connection with bonding environment in solid: a high-throughput calculation and machine learning study. Comput. Mater. Sci. **183**, 109803 (2020)
16. Ehrhart, P.: Atomic defects in metals. Landolt-Bornstein, New Series (1991)
17. Bandyopadhyay, A.K., Sen, S.K.: Calculation of self-interstitial formation energy (both split and non-split) in noble metals. Physica Status Solidi (b) **157**(2), 519–530 (1990)
18. Carter, C.B., Holmes, S.M.: The stacking-fault energy of nickel. Philos. Mag. J. Theor. Exp. Appl. Phys. **35**(5), 1161–1172 (1977)
19. Thornton, P.R., Hirsch, P.B.: The effect of stacking fault energy on low temperature creep in pure metals. Phil. Mag. **3**(31), 738–761 (1958)
20. Howie, A., Swann, P.R.: Direct measurements of stacking-fault energies from observations of dislocation nodes. Phil. Mag. **6**(70), 1215–1226 (1961)
21. Thomas, G.: Electron microscopy and strength of crystals (1963)
22. Häussermann, F., Wilkens, M.: Bestimmung der Stapelfehlerenergie kubisch-flächenzentrierter Metalle aus der Analyse des elektronenmikroskopischen Beugungskontrastes von Stapelfehlerdipolen. Physica Status Solidi (b), **18**(2), 609–624 (1966)
23. J∅ssang, T., Hirth, J.P.: The energies of stacking-fault teirahedra in fcc metals. Philos. Mag. **13**(124), 657–670 (1966)
24. Clarebrough, L.M., Humble, P., Loretto, M.H.: Faulted defects and stacking-fault energy. Can. J. Phys. **45**(2), 1135–1146 (1967)
25. Humble, P., Loretto, M.H., Clarebrough, L.M.: The nature of defects in quenched nickel. Phil. Mag. **15**(134), 297–303 (1967)
26. Beeston, B.E.P., Dillamore, I.L., Smallman, R.E.: The stacking-fault energy of some nickel-cobalt alloys. Metal Sci. J. **2**(1), 12–14 (1968)
27. Forwood, C.T., Humble, P.: Dislocation dipoles in elastically anisotropic crystals. Aust. J. Phys. **23**(5), 697–718 (1970)
28. Sastry, D.H., Tangri, K.: Dislocation mechanisms for plastic flow of nickel in the temperature range 4.2–1200 K. Philos. Mag. J. Theoret. Exp. Appl. Phys. **32**(3), 513–525 (1975)
29. Wolff, J., Franz, M., Kluin, J.E., Schmid, D.: Vacancy formation in nickel and α-nickel-carbon alloy. Acta Mater. **45**(11), 4759–4764 (1997)
30. Lynn, K.G., Snead, Jr., C.L., Hurst, J.J.: Positron lifetime studies of pure Ni from 4.2 to 1700K. J. Phys. F Metal Phys. **10**(8), 1753 (1980)
31. Dlubek, G., Brümmer, O., Meyendorf, N.: A study of mono and divacancies in Cu and Au by positron annihilation. Appl. Phys. **13**(1), 67–70 (1977)
32. Nanao, S., Kuribayashi, K., Tanigawa, S., Doyama, M.: Studies of defects at thermal equilibrium and melting in Cu and Ni by positron annihilation. J. Phys. F Met. Phys. **7**(8), 1403 (1977)

33. Matter, H., Winter, J., Triftshäuser, W.: Phase transformations and vacancy formation energies of transition metals by positron annihilation. Appl. Phys. **20**(2), 135–140 (1979)
34. Campbell, J.L., Schulte, C.S., Jackman, J.A.: Temperature dependence of positron trapping in silver and nickel. J. Phys. F Met. Phys. **7**(10), 1985 (1977)
35. Maier, K., Peo, M., Saile, B., Schaefer, H.E., Seeger, A.: High–temperature positron annihilation and vacancy formation in refractory metals. Philos. Mag. A **40**(5), 701–728 (1979)
36. Siegel, R.W.: Positron Annihilation, ed. by P.G. Coleman, S.C. Sharma, L.M. Diana (1982)
37. Klemradt, U., Drittler, B., Hoshino, T., Zeller, R., Dederichs, P.H., Stefanou, N.: Vacancy-solute interactions in Cu, Ni, Ag, and Pd. Phys. Rev. B **43**(12), 9487 (1991)
38. Dederichs, P.H., Hoshino, T., Drittler, B., Abraham, K., Zeller, R.: Total-energy calculations for point defects in metals. Physica B **172**(1–2), 203–209 (1991)
39. Mahmoud, S., Trochet, M., Restrepo, O.A., Mousseau, N.: Study of point defects diffusion in nickel using kinetic activation-relaxation technique. Acta Mater. **144**, 679–690 (2018)
40. Foiles, S.M., Baskes, M.I., Daw, M.S.: Embedded-atom-method functions for the fcc metals Cu, Ag, Au, Ni, Pd, Pt, and their alloys. Phys. Rev. B **33**(12), 7983 (1986)
41. Baskes, M.I.: Determination of modified embedded atom method parameters for nickel. Mater. Chem. Phys. **50**(2), 152–158 (1997)
42. Lee, B.J., Shim, J.H., Baskes, M.I.: Semiempirical atomic potentials for the fcc metals Cu, Ag, Au, Ni, Pd, Pt, Al, and Pb based on first and second nearest-neighbour modified embedded atom method. Phys. Rev. B **68**(14), 144112 (2003)
43. Tucker, J.D., Allen, T.R., Morgan, D.: Ab initio defect properties for modeling radiation-induced segregation in Fe-Ni-Cr alloys. In: Proceedings of International Symposium on Environmental Degradation of Materials in Nuclear Power Systems, 13th, Whistler, BC, Canada, vol. 23, pp 1004–1014 (2007)
44. Branicio, P.S., Zhang, J.Y., Srolovitz, D.J.: Effect of strain on the stacking fault energy of copper: a first-principles study. Phys. Rev. B **88**(6), 064104 (2013)
45. Cotterill, R.M.J., Doyama, M.: Energy and atomic configuration of complete and dissociated dislocations. I. Edge dislocation in an fcc metal. Phys. Rev. **145**(2), 465 (1966)
46. Schweizer, S., Elsässer, C., Hummler, K., Fähnle, M.: Ab initio calculation of stacking-fault energies in noble metals. Phys. Rev. B **46**(21), 14270 (1992)
47. Rosengaard, N.M., Skriver, H.L.: Calculated stacking-fault energies of elemental metals. Phys. Rev. B **47**(19), 12865 (1993)
48. Heino, P., Perondi, L., Kaski, K., Ristolainen, E.: Stacking-fault energy of copper from molecular-dynamics simulations. Phys. Rev. B **60**(21), 14625 (1999)
49. Ogata, S., Li, J., Yip, S.: Ideal pure shear strength of aluminum and copper. Science **298**(5594), 807–811 (2002)
50. Qi, Y., Mishra, R.K.: Ab initio study of the effect of solute atoms on the stacking fault energy in aluminum. Phys. Rev. B **75**(22), 224105 (2007)
51. Fullman, R.L.: Interfacial free energy of coherent twin boundaries in copper. J. Appl. Phys. **22**(4), 448–455 (1951)
52. Peissker, E.: Critical stress for cross slip and stacking fault energies of copper base mixed crystals. Fiz Tverd Tela **13**(4), 419–431 (1965)
53. Stobbs, W.M., Sworn, C.H.: The weak beam technique as applied to the determination of the stacking-fault energy of copper. Phil. Mag. **24**(192), 1365–1381 (1971)
54. Korzhavyi, P.A., Abrikosov, I.A., Johansson, B., Ruban, A.V., Skriver, H.L.: First-principles calculations of the vacancy formation energy in transition and noble metals. Phys. Rev. B **59**(18), 11693 (1999)
55. Lam, N.Q., Dagens, L., Doan, N.V.: Calculations of the properties of self-interstitials and vacancies in the face-centred cubic metals Cu, Ag and Au. J. Phys. F Met. Phys. **13**(12), 2503 (1983)
56. Huntington, H.B.: Mobility of interstitial atoms in a face-centered metal. Phys. Rev. **91**(5), 1092 (1953)

57. Schober, H.R., Zeller, R.: Structure and dynamics of multiple interstitials in FCC metals. J. Nucl. Mater. **69**, 341–349 (1978)
58. Lu, G., Kioussis, N., Bulatov, V.V., Kaxiras, E.: Generalized-stacking-fault energy surface and dislocation properties of aluminum. Phys. Rev. B **62**(5), 3099 (2000)
59. Qiu, R., Lu, H., Ao, B., Huang, L., Tang, T., Chen, P.: Energetics of point defects in aluminum via orbital-free density functional theory (2016). arXiv:1611.07631
60. Wang, Y.A., Govind, N., Carter, E.A.: Orbital-free kinetic-energy density functionals with a density-dependent kernel. Phys. Rev. B **60**(24), 16350 (1999)
61. Ho, G., Ong, M.T., Caspersen, K.J., Carter, E.A.: Energetics and kinetics of vacancy diffusion and aggregation in shocked aluminium via orbital-free density functional theory. Phys. Chem. Chem. Phys. **9**(36), 4951–4966 (2007)
62. Chetty, N., Weinert, M., Rahman, T.S., Davenport, J.W.: Vacancies and impurities in aluminum and magnesium. Phys. Rev. B **52**(9), 6313 (1995)
63. Turner, D.E., Zhu, Z.Z., Chan, C.T., Ho, K.M.: Energetics of vacancy and substitutional impurities in aluminum bulk and clusters. Phys. Rev. B **55**(20), 13842 (1997)
64. Carling, K., Wahnström, G., Mattsson, T.R., Mattsson, A.E., Sandberg, N., Grimvall, G.: Vacancies in metals: from first-principles calculations to experimental data. Phys. Rev. Lett. **85**(18), 3862 (2000)
65. Baskes, M.I., Asta, M., Srinivasan, S.G.: Determining the range of forces in empirical many-body potentials using first-principles calculations. Philos. Mag. A **81**(4), 991–1008 (2001)
66. Carling, K.M., Wahnström, G., Mattsson, T.R., Sandberg, N., Grimvall, G.: Vacancy concentration in Al from combined first-principles and model potential calculations. Phys. Rev. B **67**(5), 054101 (2003)
67. Uesugi, T., Kohyama, M., Higashi, K.: Ab initio study on divacancy binding energies in aluminum and magnesium. Phys. Rev. B **68**(18), 184103 (2003)
68. Lu, G., Kaxiras, E.: Hydrogen embrittlement of aluminum: the crucial role of vacancies. Phys. Rev. Lett. **94**(15), 155501 (2005)
69. Iyer, M., Gavini, V., Pollock, T.M.: Energetics and nucleation of point defects in aluminum under extreme tensile hydrostatic stresses. Phys. Rev. B **89**(1), 014108 (2014)
70. Gavini, V., Bhattacharya, K., Ortiz, M.: Quasi-continuum orbital-free density-functional theory: A route to multi-million atom non-periodic DFT calculation. J. Mech. Phys. Solids **55**(4), 697–718 (2007)
71. Radhakrishnan, B., Gavini, V.: Effect of cell size on the energetics of vacancies in aluminum studied via orbital-free density functional theory. Phys. Rev. B **82**(9), 094117 (2010)
72. Erhart, P., Jung, P., Schult, H., Ullmaier, H.: Atomic Defects in Metals. Landolt–Börnstein Berlin, Heidelberg, New York, London, Paris (1991)
73. Janani, R.D., Salman, S.A., Priyadharshini, K.P., Karthik, V.: Effect of composition on the stacking fault energy of copper-nickel alloys using molecular dynamics simulations. Mater. Today Proc. **39**, 1796–1800 (2021)
74. Nakajima, K., Numakura, K.: Effect of solute atoms on stacking faults Cu-Ni and Cu-Mn systems. Phil. Mag. **12**(116), 361–368 (1965)
75. Zhang, X., Sluiter, M.H.: Ab initio prediction of vacancy properties in concentrated alloys: the case of fcc Cu-Ni. Phys. Rev. B **91**(17), 174107 (2015)
76. Glensk, A., Grabowski, B., Hickel, T., Neugebauer, J.: Breakdown of the Arrhenius law in describing vacancy formation energies: the importance of local anharmonicity revealed by ab initio thermodynamics. Phys. Rev. X **4**(1), 011018 (2014)
77. Lomer, W.M.: Vacancies and Other Point Defects in Metals and Alloys. Institute of Metals, London (1958)
78. Perdew, J.P., Burke, K., Ernzerhof, M.: Generalized gradient approximation made simple. Phys. Rev. Lett. **77**(18), 3865 (1996)
79. Delczeg, L., Delczeg-Czirjak, E.K., Johansson, B., Vitos, L.: Assessing common density functional approximations for the ab initio description of monovacancies in metals. Phys. Rev. B **80**(20), 205121 (2009)

80. Nazarov, R., Hickel, T., Neugebauer, J.: Vacancy formation energies in fcc metals: influence of exchange-correlation functionals and correction schemes. Phys. Rev. B **85**(14), 144118 (2012)
81. Armiento, R., Mattsson, A.E.: Functional designed to include surface effects in self-consistent density functional theory. Phys. Rev. B **72**(8), 085108 (2005)
82. Ehrhart, P., Robrock, K.H., Schober, H.R.: Basic defects in metals. In: Modern Problems in Condensed Matter Sciences, vol. 13, pp. 3–115. Elsevier (1986)
83. Schaefer, H.E.: Investigation of thermal equilibrium vacancies in metals by positron annihilation. Physica Status Solidi A Appl. Res. **102**(1), 47–65 (1987)
84. Megchiche, E.H., Pérusin, S., Barthelat, J.C., Mijoule, C.: Density functional calculations of the formation and migration enthalpies of monovacancies in Ni: comparison of local and nonlocal approaches. Phys. Rev. B **74**(6), 064111 (2006)
85. Saka, H., Sueki, Y., Imura, T.: On the intrinsic temperature dependence of the stacking-fault energy in copper-aluminium alloys. Philos. Mag. A **37**(2), 273–289 (1978)
86. Higo, Y., Pickard, A.C., Knott, J.F.: Effects of grain size and stacking fault energy on fatigue-crack-propagation thresholds in Cu-Al aluminium alloys. Metal Sci. **15**(6), 233–240 (1981)
87. Cockayne, D.J.H., Jenkins, M.L., Ray, I.L.F.: The measurement of stacking-fault energies of pure face-centred cubic metals. Phil. Mag. **24**(192), 1383–1392 (1971)
88. Cockayne, D.J.H., Ray, I.L.F., Whelan, M.J.: Investigations of dislocation strain fields using weak beams. Phil. Mag. **20**(168), 1265–1270 (1969)
89. Gallagher, P.C.J.: The influence of alloying, temperature, and related effects on the stacking fault energy. Metall. Trans. **1**(9), 2429–2461 (1970)
90. Górecki, T., Górecki, C., Książek, K., Wacke, S.: Effective vacancy formation energy and the solute–vacancy binding energy in binary Cu–based alloys. Visnyk of the Lviv University, Series Khimichna **48**(1), 88–94 (2007)
91. Sueoka, O.: Study on the vacancy formation in α-phase Cu-Al alloys by positron annihilation. J. Phys. Soc. Jpn. **39**(4), 969–975 (1975)
92. Fukushima, H., Doyama, M.: Positrons as the probe of phase transformations in Cu-Mn alloy. J. Phys. F Met. Phys. **8**(2), 205 (1978)
93. Fukushima, H., Doyama, M.: Positron trapping in the liquid and solid copper alloys. Solid State Commun. **25**(11), 945–948 (1978)
94. Vitek, V.: On the stability of stacking faults in BCC crystals. Phil. Mag. **21**(174), 1275–1278 (1970)
95. Achmad, T.L., Fu, W., Chen, H., Zhang, C., Yang, Z.G.: Effects of alloying elements concentrations and temperatures on the stacking fault energies of Co-based alloys by computational thermodynamic approach and first-principles calculations. J. Alloy. Compd. **694**, 1265–1279 (2017)
96. Wang, Y.U., Jin, Y.M., Cuitino, A.M., Khachaturyan, A.G.: Nanoscale phase field microelasticity theory of dislocations: model and 3D simulations. Acta Mater. **49**(10), 1847–1857 (2001)
97. Brandl, C., Derlet, P.M., Van Swygenhoven, H.: General-stacking-fault energies in highly strained metallic environments: Ab initio calculations. Phys. Rev. B **76**(5), 054124 (2007)
98. Hunter, A., Zhang, R.F., Beyerlein, I.J.: The core structure of dislocations and their relationship to the material γ-surface. J. Appl. Phys. **115**(13), 134314 (2014)
99. Hasan, H., Mlkvik, P., Haynes, P.D., Vorontsov, V.A.: Generalised stacking fault energy of Ni-Al and Co-Al-W superalloys: density-functional theory calculations. Materialia **9**, 100555 (2020)
100. Vamsi, K.V., Karthikeyan, S.: Effect of off-stoichiometry and ternary additions on planar fault energies in Ni3Al. Superalloys **2012**(1), 521–530 (2012)
101. Mryasov, O.N., Gornostyrev, Y.N., Van Schilfgaarde, M., Freeman, A.J.: Superdislocation core structure in L12 Ni3Al, Ni3Ge and Fe3Ge: Peierls-Nabarro analysis starting from ab-initio GSF energetics calculations. Acta Mater. **50**(18), 4545–4554 (2002)
102. Schoeck, G., Kohlhammer, S., Fahnle, M.: Planar dissociations and recombination energy of [110] superdislocations in Ni3Al: generalized Peierls model in combination with ab initio electron theory. Philos. Mag. Lett. **79**(11), 849–857 (1999)

103. Rao, Y., Smith, T.M., Mills, M.J., Ghazisaeidi, M.: Segregation of alloying elements to planar faults in γ'-Ni3Al. Acta Mater. **148**, 173–184 (2018)
104. Durand-Charre, M.: The microstructure of superalloys. Routledge (2017)
105. Reed, R.C.: The Superalloys: Fundamentals and Applications. Cambridge, Cambridge University Press (2008)
106. Burke, K.: Perspective on density functional theory. J. Chem. Phys. **136**(15), 150901 (2012)
107. Finnis, M.: Interatomic forces in condensed matter, vol. 1. OUP Oxford (2003)
108. Prakash, A., Guénolé, J., Wang, J., Müller, J., Spiecker, E., Mills, M.J., Povstugar, I., Choi, P., Raabe, D., Bitzek, E.: Atom probe informed simulations of dislocation–precipitate interactions reveal the importance of local interface curvature. Acta Mater. **92**, 33–45 (2015)
109. Csányi, G., Albaret, T., Payne, M.C., De Vita, A.: "Learn on the fly": a hybrid classical and quantum-mechanical molecular dynamics simulation. Phys. Rev. Lett. **93**(17), 175503 (2004)
110. Bernstein, N., Kermode, J.R., Csanyi, G.: Hybrid atomistic simulation methods for materials systems. Rep. Prog. Phys. **72**(2), 026501 (2009)
111. Bianchini, F., Kermode, J.R., De Vita, A.: Modelling defects in Ni–Al with EAM and DFT calculations. Modell. Simul. Mater. Sci. Eng. **24**(4), 045012 (2016)
112. Miracle, D.B.: The physical and mechanical properties of NiAl, overview no. 104. Acta Metall. Mater. **41**, 649–684 (1993)
113. Noebe, R.D., Bowman, R.R., Nathal, M.V.: Physical and mechanical properties of the B2 compound NiAl. Int. Mater. Rev. **38**(4), 193–232 (1993)
114. Carlsson, A.E.: Cluster interactions and physical properties of Al–transition-metal alloys. Phys. Rev. B **40**(2), 912 (1989)
115. Hong, T., Freeman, A.J.: Effect of antiphase boundaries on the electronic structure and bonding character of intermetallic systems: NiAl. Phys. Rev. B **43**(8), 6446 (1991)
116. Lu, Z.W., Wei, S.H., Zunger, A.: Theory of bonding charge density in β′ NiAl. Acta Metall. Mater. **40**(9), 2155–2165 (1992)
117. Moriarty, J.A., Widom, M.: First-principles interatomic potentials for transition-metal aluminides: theory and trends across the 3 d series. Phys. Rev. B **56**(13), 7905 (1997)
118. Hackenbracht, D., Kubler, J.: Electronic magnetic and cohesive properties of some nickel-aluminium compounds. J. Phys. F Met. Phys. **10**(3), 427 (1980)
119. Koch, J.M., Koenig, C.: First-order effect of the extension of the perturbation around vacancies in metallic compounds. Philos. Mag. B **57**(4), 557–568 (1988)
120. Schultz, P.A., Davenport, J.W.: Calculations of systematics in B2 structure 3d transition metal aluminides. J. Alloy. Compd. **197**(2), 229–242 (1993)
121. Zou, J., Fu, C.L.: Structural, electronic, and magnetic properties of 3d transition-metal aluminides with equiatomic composition. Phys. Rev. B **51**(4), 2115 (1995)
122. Botton, G.A., Guo, G.Y., Temmerman, W.M., Humphreys, C.J.: Experimental and theoretical study of the electronic structure of Fe Co, and Ni aluminides with the B2 structure. Phys. Rev. B **54**(3), 1682 (1996)
123. Muller, D.A., Singh, D.J., Silcox, J.: Connections between the electron-energy-loss spectra, the local electronic structure, and the physical properties of a material: a study of nickel aluminum alloys. Phys. Rev. B **57**(14), 8181 (1998)
124. Lu, Z.W., Wei, S.H., Zunger, A., Frota-Pessoa, S., Ferreira, L.G.: First-principles statistical mechanics of structural stability of intermetallic compounds. Phys. Rev. B **44**(2), 512 (1991)
125. Burton, B.P., Osburn, J.E., Pasturel, A.: Theoretical calculations of the NiAl-NiTi phase diagram based on first-principles linear-muffin-tin-orbital and full-potential linearly-augmented plane-wave cohesive-energy calculations. Phys. Rev. B **45**(14), 7677 (1992)
126. Pasturel, A., Colinet, C., Paxton, A.T., Van Schilfgaarde, M.: First-principles determination of the Ni-Al phase diagram. J. Phys. Condens. Matter **4**(4), 945 (1992)
127. Mehl, M.J., Klein, B.M., Papaconstantopoulos, D.A.: Intermetallic compounds, vol. 1, Principles, ed. by J.H. Westbrook, R.L. Fleischer
128. Rzyman, K., Moser, Z., Watson, R.E., Weinert, M.: Enthalpies of formation of AlNi: experiment versus theory. J. Phase Equilibria **19**(2), 106–111 (1998)

129. Fu, C.L., Ye, Y.Y., Yoo, M.H., Ho, K.M.: Equilibrium point defects in intermetallics with the B2 structure: NiAl and FeAl. Phys. Rev. B **48**(9), 6712 (1993)
130. Fu, C.L.: Origin of ordering in B2-type transition-metal aluminides: comparative study of the defect properties of PdAl, NiAl, and FeAl. Phys. Rev. B **52**(5), 3151 (1995)
131. Farnsworth, H.E., Tuul, J.: Influence of lattice-defect density on the chemisorption and oxide formation on a clean (100) crystal face of Nickel as determined by low-energy electron diffraction. J. Phys. Chem. Solids **9**(1), 48–56 (1959)
132. Medvedeva, N.I., Mryasov, O.N., Gornostyrev, Y.N., Novikov, D.L., Freeman, A.J.: First-principles total-energy calculations for planar shear and cleavage decohesion processes in B2-ordered NiAl and FeAl. Phys. Rev. B **54**(19), 13506 (1996)
133. Medvedeva, N.I., Gornostyrev, Y.N., Novikov, D.L., Mryasov, O.N., Freeman, A.J.: Ternary site preference energies, size misfits and solid solution hardening in NiAl and FeAl. Acta Mater. **46**(10), 3433–3442 (1998)
134. Bester, G., Meyer, B., Fähnle, M.: Atomic defects in the ordered compound B2-CoAl: a combination of ab initio electron theory and statistical mechanics. Phys. Rev. B **60**(21), 14492 (1999)
135. Peterman, D.J., Rosei, R., Lynch, D.W., Moruzzi, V.L.: Optical properties and electronic structure of $\beta' -$ NiAl. Phys. Rev. B **21**(12), 5505 (1980)
136. Egorushkin, V.E., Kul'mentyev, A.I., Rubin, P.E.: Electron structure and optical properties of Ni-Al. Solid State Commun. **57**(10), 821–823 (1986)
137. Knab, D., Koenig, C.: Optical properties of TAl compounds (T= Fe Co, Ni) and CoGa: role of the 3d semi-core states of Ga. J. Phys. Condens. Matter **2**(2), 465 (1990)
138. Kim, K.J., Harmon, B.N., Lynch, D.W.: Calculation of the optical spectra of β'-NiAl and CoAl. Phys. Rev. B **43**(3), 1948 (1991)
139. Farberovich, O.V., Vlasov, S.V., Portnoi, K.I., Lozovoi, A.Y.: Electronic structure of NiAl by the FLAPW-method. Physica B **182**(3), 267–277 (1992)
140. Demkowicz, M.J., Bhattacharyya, D., Usov, I., Wang, Y.Q., Nastasi, M., Misra, A.: The effect of excess atomic volume on He bubble formation at fcc–bcc interfaces. Appl. Phys. Lett. **97**(16), 161903 (2010)
141. Taylor, A., Doyle, N.J.: Further studies on the nickel–aluminium system. I. β-NiAl and δ-Ni2Al3 phase fields. J. Appl. Crystallogr. **5**(3), 201–209 (1972)
142. Van Sande, M., Van Landuyt, J., Avalos-Borja, M., Villaseñor, G.T., Amelinckx, S.: A reinvestigation of the γ phase in Cu-Al alloys: a new long-period superstructure. Mater. Sci. Eng. **46**(2), 167–173 (1980)
143. Betterton, J., Reynolds, J.: The factors affecting the formation of 21/13 electron compounds in alloys of copper and of silver. J. Inst. Metals **80**(11), 609–616 (1952)
144. Jones, H., Chako, N.: The theory of Brillouin zones and electronic states in crystals. Phys. Today **15**(1), 66 (1962)
145. Korzhavyi, P.A., Ruban, A.V., Lozovoi, A.Y., Vekilov, Y.K., Abrikosov, I.A., Johansson, B.: Constitutional and thermal point defects in B 2 NiAl. Phys. Rev. B **61**(9), 6003 (2000)
146. Smirnov, A.A.: Metallofizika Kiev 14, 3 1992 Phys. Metals **12**, 377 (1993)
147. Cottrell, A.H.: Constitutional vacancies in NiAl. Intermetallics **3**(5), 341–345 (1995)
148. Cottrell, A.H.: Vacancies in FeAl and NiAl. Intermetallics **5**(6), 467–469 (1997)
149. Verma, A., Parashar, A., Packirisamy, M.: Atomistic modeling of graphene/hexagonal boron nitride polymer nanocomposites: a review. Wiley Interdisc. Rev. Comput. Mol. Sci. **8**(3), e1346 (2018)
150. Verma, A., Singh, V.K., Verma, S.K., Sharma, A.: Human hair: a biodegradable composite fiber–a review. Int. J. Waste Res. **6**(206), 2 (2016)
151. Verma, A., Singh, V.K.: Mechanical, microstructural and thermal characterization of epoxy-based human hair–reinforced composites. J. Test. Eval. **47**(2), 1193–1215 (2019)
152. Verma, A., Parashar, A.: Structural and chemical insights into thermal transport for strained functionalised graphene: a molecular dynamics study. Mater. Res. Exp. **5**(11), 115605 (2018)
153. Verma, A., Negi, P., Singh, V.K.: Experimental analysis on carbon residuum transformed epoxy resin: chicken feather fiber hybrid composite. Polym. Compos. **40**(7), 2690–2699 (2019)

154. Verma, A., Gaur, A., Singh, V.K.: Mechanical properties and microstructure of starch and sisal fiber biocomposite modified with epoxy resin. Mater. Perform. Charact. **6**(1), 500–520 (2017)
155. Verma, A., Parashar, A., Packirisamy, M.: Effect of grain boundaries on the interfacial behaviour of graphene-polyethylene nanocomposite. Appl. Surf. Sci. **470**, 1085–1092 (2019)
156. Verma, A., Budiyal, L., Sanjay, M.R., Siengchin, S.: Processing and characterization analysis of pyrolyzed oil rubber (from waste tires)-epoxy polymer blend composite for lightweight structures and coatings applications. Polym. Eng. Sci. **59**(10), 2041–2051 (2019)
157. Verma, A., Negi, P., Singh, V.K.: Physical and thermal characterization of chicken feather fiber and crumb rubber reformed epoxy resin hybrid composite. Adv. Civil Eng. Mater. **7**(1), 538–557 (2018)
158. Verma, A., Negi, P., Singh, V.K.: Experimental investigation of chicken feather fiber and crumb rubber reformed epoxy resin hybrid composite: mechanical and microstructural characterization. J. Mech. Behav. Mater. **27**(3–4) (2018)
159. Chaurasia, A., Verma, A., Parashar, A., Mulik, R.S.: Experimental and computational studies to analyze the effect of h-BN nanosheets on mechanical behavior of h-BN/polyethylene nanocomposites. J. Phys. Chem. C **123**(32), 20059–20070 (2019)
160. Jain, N., Verma, A., Singh, V.K.: Dynamic mechanical analysis and creep-recovery behaviour of polyvinyl alcohol based cross-linked biocomposite reinforced with basalt fiber. Mater. Res. Exp. **6**(10), 105373 (2019)
161. Verma, A., Joshi, K., Gaur, A., Singh, V.K.: Starch-jute fiber hybrid biocomposite modified with an epoxy resin coating: fabrication and experimental characterization. J. Mech. Behav. Mater. **27**(5–6) (2018)
162. Verma, A., Kumar, R., Parashar, A.: Enhanced thermal transport across a bi-crystalline graphene–polymer interface: an atomistic approach. Phys. Chem. Chem. Phys. **21**(11), 6229–6237 (2019)
163. Verma, A., Singh, V.K.: Experimental investigations on thermal properties of coconut shell particles in DAP solution for use in green composite applications. J. Mater. Sci. Eng. **5**(3), 1000242 (2016)
164. Verma, A., Singh, V.K., Arif, M.: Study of flame retardant and mechanical properties of coconut shell particles filled composite. Res. Rev. J. Mater. Sci. **4**(3), 1–5 (2016)
165. Verma, A., Parashar, A., Jain, N., Singh, V.K., Rangappa, S.M., Siengchin, S.: Surface modification techniques for the preparation of different novel biofibers for composites. In: Biofibers and Biopolymers for Biocomposites, pp. 1–34. Springer, Cham (2020)
166. Rastogi, S., Verma, A., Singh, V.K.: Experimental response of nonwoven waste cellulose fabric–reinforced epoxy composites for high toughness and coating applications. Mater. Perform. Charact. **9**(1), 151–172 (2020)
167. Bharath, K.N., Madhu, P., Gowda, T.G., Verma, A., Sanjay, M.R., Siengchin, S.: A novel approach for development of printed circuit board from biofiber based composites. Polym. Compos. **41**(11), 4550–4558 (2020)
168. Verma, A., Jain, N., Parashar, A., Gaur, A., Sanjay, M.R., Siengchin, S.: Lifecycle assessment of thermoplastic and thermosetting bamboo composites. In: Bamboo Fiber Composites, pp. 235–246. Springer, Singapore (2021)
169. Singh, K., Jain, N., Verma, A., Singh, V.K., Chauhan, S.: Functionalized graphite-reinforced cross-linked poly (vinyl alcohol) nanocomposites for vibration isolator application: morphology, mechanical, and thermal assessment. Mater. Perform. Charact. **9**(1), 215–230 (2020)
170. Verma, A., Parashar, A.: Molecular dynamics based simulations to study the fracture strength of monolayer graphene oxide. Nanotechnology **29**(11), 115706 (2018)
171. Verma, A., Parashar, A.: The effect of STW defects on the mechanical properties and fracture toughness of pristine and hydrogenated graphene. Phys. Chem. Chem. Phys. **19**(24), 16023–16037 (2017)
172. Verma, A., Parashar, A., Packirisamy, M.: Tailoring the failure morphology of 2D bicrystalline graphene oxide. J. Appl. Phys. **124**(1), 015102 (2018)

173. Singla, V., Verma, A., Parashar, A.: A molecular dynamics based study to estimate the point defects formation energies in graphene containing STW defects. Mater. Res. Exp. **6**(1), 015606 (2018)
174. Verma, A., Parashar, A.: Molecular dynamics based simulations to study failure morphology of hydroxyl and epoxide functionalised graphene. Comput. Mater. Sci. **143**, 15–26 (2018)
175. Verma, A., Zhang, W., van Duin, A.C.: ReaxFF reactive molecular dynamics simulations to study the interfacial dynamics between defective h-BN nanosheet and water nanodroplets. Phys. Chem. Chem. Phys. **23**, 10822–10834 (2021)
176. Verma, A., Parashar, A., Packirisamy, M.: Role of chemical adatoms in fracture mechanics of graphene nanolayer. Mater. Today Proc. **11**, 920–924 (2019)
177. Verma, A., Parashar, A.: Characterization of 2D nanomaterials for energy storage. In: Recent Advances in Theoretical, Applied, Computational and Experimental Mechanics, pp. 221–226. Springer, Singapore (2020)
178. Verma, A., Jain, N., Parashar, A., Singh, V.K., Sanjay, M.R., Siengchin, S.: Design and modeling of lightweight polymer composite structures. In: Lightweight Polymer Composite Structures: Design and Manufacturing Techniques, pp. 193–224. Taylor & Francis Group (CRC Press), Boca Raton, Chapter 7 (2020)
179. Verma, A., Jain, N., Parashar, A., Singh, V.K., Sanjay, M.R., Siengchin, S.: Lightweight graphene composite materials. In: Lightweight Polymer Composite Structures: Design and Manufacturing Techniques, pp. 1–20. Taylor & Francis Group (CRC Press), Boca Raton, Chapter 1 (2020)
180. Verma, A., Parashar, A., Singh, S.K., Jain, N., Sanjay, M.R., Siengchin, S.: Modeling and simulation in polymer coatings. In: Polymer Coatings: Technologies and Applications, pp. 309–324. Taylor & Francis Group (CRC Press), Boca Raton, Chapter 16 (2020)
181. Verma, A., Jain, N., Rastogi, S., Dogra, V., Sanjay. M.R., Siengchin, S., Mansour, R.: Mechanism, anti-corrosion protection and components of anti-corrosion polymer coatings. In: Polymer Coatings: Technologies and Applications, pp. 53–66. Taylor & Francis Group (CRC Press), Boca Raton, Chapter 4 (2020)
182. Verma, A., Jain, N., Kalpana, Sanjay M.R., Siengchin, S., Jawaid, M.: Natural fibers based bio-phenolic composites. In: Phenolic Polymers Based Composite Materials, pp. 153–168. Springer Nature, Singapore, Chapter 10 (2020)
183. Bharath, K.N., Madhu, P., Gowda, T.Y., Verma, A., Sanjay, M.R., Siengchin, S.: Mechanical and chemical properties evaluation of sheep wool fiber-reinforced vinylester and polyester composites. Mater. Perform. Charact. **10**(1), 99–109 (2021)
184. Marichelvam, M.K., Manimaran, P., Verma, A., Sanjay, M.R., Siengchin, S., Kandakodeeswaran, K., Geetha, M.: A novel palm sheath and sugarcane bagasse fiber based hybrid composites for automotive applications: an experimental approach. Polym. Compos. **42**(1), 512–521 (2021)
185. Chaudhary, A., Sharma, S., Verma, A.: Optimization of WEDM process parameters for machining of heat treated ASSAB'88 tool steel using Response surface methodology (RSM). Mater. Today Proc. (2021)
186. Chaudhary, A., Sharma, S., Verma, A.: WEDM machining of heat treated ASSAB '88 tool steel: a comprehensive experimental analysis. Mater. Today Proc. (2021)
187. Verma, A., Singh, V.K.: Experimental characterization of modified epoxy resin assorted with almond shell particles. ESSENCE-Int. J. Environ. Rehabil. Conserv. 36
188. Verma, A., Samant, S.S.: Inspection of hydrodynamic lubrication in infinitely long journal bearing with oscillating journal velocity. J. Appl. Mech. Eng. **5**(3), 1–7 (2016)
189. Verma, A., Parashar, A., Jain, N., Singh, V.K., Rangappa, S.M., Siengchin, S.: Surface modification techniques for the preparation of different novel biofibers for composites. In: Biofibers and Biopolymers for Biocomposites, pp. 1–34 (2020)
190. Bisht, N., Verma, A., Chauhan, S., Singh, V.K.: Effect of functionalized silicon carbide nanoparticles as additive in cross-linked PVA based composites for vibration damping application. J. Vinyl Add. Tech. **27**(4), 920–932 (2021)

191. Kataria, A., Verma, A., Sanjay, M.R., Siengchin, S.: Molecular modeling of 2D graphene grain boundaries: mechanical and fracture aspects. Mater. Today Proc.
192. Arpitha, G.R., Verma, A., Sanjay, M.R., Siengchin, S.: Preparation and experimental investigation on mechanical and tribological performance of hemp-glass fiber reinforced laminated composites for lightweight applications. Adv. Civil Eng. Mater. **10**(1), 427–439 (2021)
193. Deji, R., Verma, A., Kaur, N., Choudhary, B.C., Sharma, R.K.: Density functional theory study of carbon monoxide adsorption on transition metal doped armchair graphene nanoribbon. Mater. Today Proc. (2021)
194. Deji, R., Verma, A., Choudhary, B.C., Sharma, R.K.: New insights into NO adsorption on alkali metal and transition metal doped graphene nanoribbon surface: a DFT approach. J. Mol. Graph. Model. (2021). https://doi.org/10.1016/j.jmgm.2021.108109
195. Deji, R., Jyoti, R., Verma, A., Choudhary, B.C., Sharma, R.K.: A theoretical study of HCN adsorption and width effect on co-doped armchair graphene nanoribbon. Comput. Theor. Chem. 113592 (2022)

Tersoff and REBO Potentials



Rajesh Kumar and Jashveer Singh

Abstract Interatomic potentials are used to describe the motion of the individual atoms in atomistic simulations. A perfect approach of the interatomic forces in a system of atoms requires massive quantum mechanical calculations, which are not computationally viable in large-scale simulations. This chapter discusses the developmental history of Tersoff and REBO potentials along with the challenges and limitations associated with these potentials. Also, a glimpse of their advantages, usage, and applications have been presented.

Keywords Interatomic potentials · Molecular dynamics · Mechanical properties · Atomistic scale simulations

1 Introduction

To predict atoms' movements in an environment, one must understand how they interact with each other. To address a large system with quantum mechanical precision is impractical, even though calculating interactions between individual atoms is possible. Density functional theory, for example, can only be applied to systems of a few hundred to a few thousand atoms. For large-scale systems, the interaction models of atoms must be simpler and more efficient. A computationally efficient interatomic potential function governs the interactions among atoms in large-scale atomistic modelling such as molecular dynamics, making it possible to simulate many more complex systems for longer periods of time than in quantum mechanical simulations.

Analytical functions that may be used to compute the P.E. of atoms in systems are called interatomic potentials, which are straightforward to implement. It wasn't until the mid-twentieth century that atomistic simulations were first employed for predicting trajectory using simple pair potentials, in which mechanical energy is

R. Kumar (✉) · J. Singh
Department of Mechanical Engineering, National Institute of Technology Hamirpur, Hamirpur 177005, Himachal Pradesh, India
e-mail: rajeshpawar4u@gmail.com; rpawar@nith.ac.in

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calculated by adding the bond energies of atomic pairs [1]. When it comes to forecasting the behaviour and characteristics of crystalline materials, however, pairwise computed P.E. functions have significant limits [2]. Dislocation and deformation concepts of solid materials might've been improved by atomic simulations at the time if the pair potentials and inadequate computer capacity had not restricted their use to very small-scale systems. Cost-effective pair potentials have led to substantial improvements in physical and chemical sciences, as well as the development of dependable tools for the study of materials science and chemistry that use solid atomistic modelling [1]. Larger and more complex systems could be simulated attributable to ever-increasing computer power, and more delicate and computationally costly potential functions could be developed as a result. New theories about interatomic bonding led to the development of many-body potentials in the 1980s, including embedded atom method potentials and bond order potentials [3–5]. However, it was quick and accurate enough with large-scale simulation because of its many-body dependency to correct the problems with pair potentials. Because of the improved precision and predictive capacities of the potentials, simulations of a broad variety of materials processes were enabled. Atomistic simulations of melting, dislocation movement, corrosion, and effects of irradiation may be performed using many-body potentials since they manage bond breakage and formation in a realistic manner.

For a suitable interatomic potential function, computational efficiency and the capacity to accurately forecast material characteristics and phases need the use of a number of parameters with no simple set values. Thus, the parameters of an interatomic potential should match a well-known set of physical features. The ground state and several key structures' lattice parameters, cohesive or formation energies, bulk moduli, and elastic constants are often included in the fitting method's most critical characteristics. The melting point, defect formation energies, and surface energies may all be taken into account when evaluating the potential. Experiments or density functional theory simulations may be utilised to get the values of the different characteristics needed in the fitting procedure. It is not an easy undertaking to fit an interatomic potential that precisely predicts all of the previous qualities. Many features that are less important must be sacrificed in order to provide better descriptions of the properties that are most important to the applications for which the potential is being developed. It is possible for a good potential to function pretty well in a broad range of structures and systems, and this potential may be transferred to systems that were not included in the fitting process [6].

In this chapter, we review two important potentials developed for covalently bonded interactions in the nineties namely Tersoff interatomic potential developed by J. Tersoff and REBO (reactive bond order potential) potential by D. W. Brenner. Some of these potentials are examined, along with why they exist and various parametrizations, as well as their advantages and disadvantages.

2 Tersoff Potential

It is necessary to incorporate many-body components in the potential energy calculation to account for the atomic geometry of an atom's surroundings. The total potential energy of a system of N atoms may, thus, be represented in the most generic form.

$$U_{tol} = \sum_{i}^{N} U_1(r_i) + \sum_{i,j}^{N} U_2(r_i,\ r_j) + \sum_{i,j,k}^{N} U_3(r_i,\ r_j,\ r_k) \tag{1}$$

where r_i is the atom i's position vector and U_1, U_2, and U_3 are the one, two, and three-body potentials, respectively. Only when the system is exposed to an external field, such as an electric field, can the one-body term appear. When this is not the case, the potential energy is independent of the atomic locations and may be expressed in a more generic form using interatomic lengths and angles.

$$U_{tol} = \sum_{i,j}^{N} U_2(r_{ij}) + \sum_{i,j,k}^{N} U_3(r_{ij},\ r_{jk}, \theta_{ijk}) \tag{2}$$

where r_{ij} is the distance among i and j and θ_{ijk} is angle atoms i from j and k.

In this section, we will review the Tersoff potential.

Tersoff is a three-body potential functional that explicitly incorporates an angular contribution from force. Today, the potential is extensively employed in a variety of silicon, carbon, germanium, and other materials applications. It is written in the following form:

$$E = \sum_{i} E_i = \frac{1}{2} \sum_{i \neq j} V_{ij} \tag{3}$$

$$V_{ij} = f_C(r_{ij})\left[f_R(r_{ij}) + b_{ij} f_A(r_{ij})\right] \tag{4}$$

where

E_i Site energy.
V_{ij} Bonding energy.
f_A Attractive pair potential.
f_R Repulsive pair potential.
f_C Smooth cut-off function.

$$f_R(r) = A \exp(-\lambda_1 r)$$
$$f_A(r) = -B \exp(-\lambda_2 r) \tag{5}$$

$$f_C(r_{ij}) = \begin{cases} 1, r_{ij} < R_{ij} \\ \frac{1}{2} + \frac{1}{2}\cos\left(\frac{\pi(r_{ij}-R_{ij})}{S_{ij}-R_{ij}}\right), R_{ij} < r_{ij} < S_{ij} \\ 0, r_{ij} > S_{ij} \end{cases} \quad (6)$$

$$b_{ij} = \left(1 + \beta^n \xi_{ij}^n\right)^{\frac{1}{2}}$$

$$\xi_{ij} = \sum f_C(r_{ik}) g(\theta_{ijk}) \exp\left[\lambda_3^3 (r_{ij} - r_{ik})^3\right]$$

$$g(\theta) = 1 + \frac{c^2}{d^2} - \frac{c^2}{\left[d^2 + (h - \cos\theta)^2\right]} \quad (7)$$

To ensure that only the first-neighbour shell is included, the parameters R and D are selected (for carbon $R = 1.8$ Å). In the range $R - D < r < R + D$, the function goes from 1 to 0.

Where A (eV), B (eV), R_{ij} (Å), S_{ij} (Å), λ_1 (Å^{-1}), λ_2 (Å^{-1}), λ_3 (Å^{-1}), β c, d, h, and n are the Tersoff potential parameters.

The existence of the b_{ij} word is the most distinguishing characteristic of this potential's description. Bonding strength decreases when there are a large number of neighbours nearby. Parameters of the Tersoff potential are developed by various researchers. Here is a list containing the details in Table 1.

3 Applications

The Tersoff-type potentials have attracted more attention among researchers for simulating the covalently bonded interactions. The Tersoff potential model calculations were done to study the mechanical properties of different organic and inorganic nanofillers (e.g. BN and SiC) [12].

4 Mechanical Properties

4.1 Defects

Mechanical characteristics of h-BCN with atomic ratios of B/C/N = 1:1:1 and the modified Tersoff potential were studied in relation to strain rate, vacancy defect, and temperature. The findings indicated that h-BCN had a fracture strength of 81–94 GPa. A pristine graphene sheet has a strength of about 35% lower than this. The effect of temperature was also studied and found that fracture stress and strain of h-BCN drop when the temperature rises [13], and Researchers also looked at the effects of single-walled carbon nanotube topological defects and heteroatomic inclusions on mechanical characteristics. It was determined that the mechanical characteristics

Table 1 Parameters available in the literature for atomic interaction

Parameter	Albe and Möller [7]	Sekkal et al. [8]	Matsunaga and Iwamoto [9] ($\chi_{i\text{-}j} = 1.1593$)		Sevik et al. [10]	Lindsay and Borido [11]
	B–N	B–N	B–B	N–N	B–N	C–C
m	3.0	3.0	3	3	3	3
γ	1.0	1.0	1	1	1	1
λ3	1.9925	0.0	0	0	0	0
c	1092.9287	38,049	0.52629	79,934	25,000	38,049
d	12.38	4.384	0.001587	134.32	4.3484	4.3484
h	−0.5413	−0.57058	0.5	−0.9973	−0.89	−0.57058
n	0.36453367	0.72751	3.9929	12.4498	0.72751	0.72751
β	0.000011134	0.00000015724	0.0000016	0.10562	0.0000001257	0.00000015724
λ2	2.784247207	2.22	1.5856	2.5115	2.199	2.2119
B	3613.431337	346.7	183.49	219.45	340	430
R	2.0	1.95	1.95	2.15	1.95	1.95
D	0.1	0.2	0.15	0.15	0.05	0.15
λ1	2.998355817	3.468	1.9922	5.7708	3.568	3.468
A	4460.833973	1393.6	277.02	11,000	1380	1393.6

were affected by inclusions such as nitrogen and boron atoms and single vacancy defects. MD findings indicated that flaws had a negative impact on the ductility and failure strain of the material [14]. In terms of defects, sp3 bonds might be considered unique. Effect of defect field sp^3 bonds, nanopores, strain, and temperature on the mechanical properties of BN–Gr–BN was investigated. Compared to the bonds formed by square nanopores, those in the boron nitride area of nanostructures were more likely to break under the influence of temperature change, according to the Tersoff potential modelling results from MD simulations [15]. An explanation of the thermo-mechanical property of silicon nanowires was also done by Hasheminia et al. [16]. Mechanical properties of silicon nanowires by MD using the Tersoff potential with different temperatures from 200 to 500 K were investigated. According to their findings, when the system's temperature rose, Young's modulus dropped from 80 to 69 GPa [16]. The effects of intrinsic defects (like stone–wales and nanopore) and doping on mechanical properties of hybrid graphene monolayers using MD with the Tersoff potential were investigated. The study showed that the effects of individual defects and doping on mechanical properties were beyond the elastic regime [17], and we looked at how defects in the boron nitrogen sites (defects of carbon and oxygen), as well as single boron and triangular boron and nitrogen holes, affected the mechanical characteristics of (h-BN). For the 2D h-BN, results indicated that the ZZ direction was stiffer than the AC direction, and the strength of the material decreased as defect concentration increased [18]. The effect of point defect and carbon doping on the mechanical properties of defective single-walled BN nanotube was investigated with ZZ and AC chirality. With the Tersoff potential using MD simulations, results showed that on removing different numbers of atoms from the nanotube's surface, higher Young's modulus was observed in AC structures than ZZ structures in every type of vacancy defect [19]. The atomic crack's formation in graphene nanosheets was studied using the Tersoff potential molecular dynamics approach. Crack development and failure mechanisms depend heavily on the edge of the nanosheet, according to the findings. And on increasing the temperature from 300 to 350 K, the movement of carbon atoms was increased. So, on increasing the temperature, the atomic stability of graphene nanosheets decreased [20].

Mechanical properties of various nanomaterials evaluated using the Tersoff potential are shown in Table 2.

The effect of BN nanotubes wall thickness on mechanical properties of composite was observed with the help of molecular dynamics using the Tersoff potential. It was observed that the thickness of the nanotube wall varies with the nanotube radius. It was found that elastic modulus is affected by nanotube radius, and beyond the threshold nanotube radius, elastic modulus remains fairly constant. In this research paper, this was calculated for CNT and BNNT reinforced metal matrix composite which is shown in the figure with the help of graphs representing the effect of nanotube radius on the elastic modulus of composite [21]. The effect of vacancy defects and functionalization defects of boron nitride nanotube on Young's modulus was examined using molecular dynamics simulations. The system was modelled with reactive many-body bond order Tersoff potential. For this purpose, tensile loading was applied and various stress–strain curves were obtained. It was found that with increasing

Table 2 Mechanical properties of various nanomaterials evaluated using the Tersoff potential

Author(s) and year	Young's Modulus (GPa)	Fracture stress (GPa)	Fracture strain
Zhang et al. (2019) [13]	734–580	81.4–93.5	0.13–0.17
Roy et al. (2021) [14]	1088	223	0.37
Yao and Fan (2020) [15]	666	68	0.16
Hasheminia et al. (2021) [16]	179.8	45.92	0.318
Sagar and chinthapenta (2020) [18]	635^{AC} 650^{ZZ}	116^{AC} 119^{ZZ}	0.37^{AC} 0.31^{ZZ}
Albooyeh et al. (2020) [19]	918.3^{AC} 917.9^{ZZ}	129.52^{AC} 116.81^{ZZ}	0.32^{AC} 0.3^{ZZ}

defect concentration, a decrease in Young's modulus was reported [22]. Mortazavi et al. [23] also used MD simulations to predict the deterioration in mechanical properties of graphene nanosheets with increasing concentration of defects. Along with defects, studies have also been carried out to study the effect of nitrogen and boron doping on the mechanical behaviour of graphene nanosheets using Tersoff potentials [24, 25]. These studies report that Young's modulus is unaffected by doping, although failure stress and strain are reduced as doping concentrations rise.

5 Thermal Properties

Graphene and boron nitride nanosheets, as well as hybrid nanoparticles, were added to octadecane paraffin in an MD research to assess its thermal characteristics. The study showed that on increasing temperature, the diffusion coefficient also increased. And on the addition of nano-additives, the thermal conductivity of paraffin was increased [26]. Thermal properties of carbon and silicon nanomaterials using coarse-graining MD simulations with the Tersoff potential were investigated [27]. Researchers used the NEMD simulation approach for the study of alpha quartz's thermal conductivity and thermal expansion coefficient. The temperature range during the simulation was maintained between 200 and 1000 K with the Tersoff potential. Results showed that in terms of TC, the Tersoff potential did not show better results but in terms of the anisotropic behaviour of the crystalline quartz, TEC Tersoff achieved superior results [28]. In this study, the thermal transport properties of group-III Nitrides (single-layer BN, Al–N, and Ga–N crystals) were investigated using optimised transferrable Tersoff-type potential. This study's results indicated a high degree of agreement between experimental and theoretical investigations on the lattice thermal transport characteristics [29]. The effect of cell defects including stone–wales and single vacancy defects on the thermal properties of carbon honeycombs was investigated using MD simulation with the Tersoff potential. Results showed reduced thermal conductivity of carbon honeycomb containing stone–wales and

single vacancy defects [30]. To determine the thermal conductivity of h-BN, homogeneous NEMD simulations with the Tersoff potential were used as a basis. Defects and concentrations in the material were investigated for their impact on thermal conductivity. Researchers discovered that the thermal conductivity of h-BN was not temperature dependent while testing it at various amorphous concentrations between 300 and 500 K [31]. The effect of geometry, size, strain, structural defects, and temperature on the thermal resistance of hybrid graphene–hexagonal boron nitride nanosheets was investigated using MD simulation with the Tersoff potential. As it turned out, nitrogen atoms proved to be a superior choice for the graphene–boron nitride interface due to their increased thermal conductivity. Increased temperature and structural size reduced thermal conductivity, which in turn impacts thermal resistance as a result of defects [32]. By using NEMD, we looked at how strain affected the thermal conductivity between a graphene–boron nitride hexagonal heterostructure. Consideration was given to tensile, compressive, and shear strain. According to these findings, the graphene/h-boron nitride heterostructure's interfacial thermal conductance (ITC) may be manipulated by varying its strain loading direction [33]. Researchers used the Tersoff potential calculations to study the thermal conductivity of CNCs and h-BN. There were lower values found for h-BN than that for Gr due to the nanocomposite's dependence on ITC for much of its heat transmission [34]. Graphene/hexagonal boron nitride thermal conductivity is affected by the angle of inter-layer rotation. In the absence of rotation, the graphene/h-BN interface's interfacial thermal conductance reached 509 MW/m^2K at 500 K and declined monotonically with increasing rotation angle, indicating a 50% reduction in interfacial thermal conductance at angle 26.3 [35] and the thermal conductivity of amorphous monolayer carbon and boron nitride [36]. It was discovered that heat conductivity of the graphene/hexagonal boron nitride interface was influenced by defects and strain by use of the Tersoff potential in Molecular dynamics simulations [37].

Molecular dynamics simulations were used to investigate the thermal conductivity of the unique one-dimensional nanocomposite, a coaxial (10,0) CNT within a (19,0) BNNT. It was utilised to run MD simulations utilising the Tersoff potential for B–N interactions in LAMMPS open-source software. For high temperatures, nanocomposite offers good thermal conductivity and greater thermal stability because of its BNNT sheath [38]. MD simulations with an optimised Tersoff potential were used by Mahdizadeh et al. to examine the thermo-mechanical characteristics of BN nanoribbons. As a function of temperature and nanoribbon length, the thermal conductivity and heat transport parameters of BN nanoribbons were computed. Thermal conductivity of BN nanoribbons reduced when the temperature rose to 1000 K, according to the study's conclusion. For nanoribbon length more than 30 nm, the thermal conductivity decreased with nanoribbon size increased and converged to ~500 $Wm^{-1}\ K^{-1}$ [39]. The effect of graphene coating on the thermal conductivity of aluminium using molecular dynamics was examined by Topral K. and Yilmaz A. They applied non-equilibrium molecular dynamics (NEMD) simulation using C++ message passing interface (MPI) for the calculation of thermal conductivity of graphene-coated aluminium. The effect of various parameters like length, graphene,

and the number of graphene layers was analysed. Enhanced thermal conductivity was reported on increasing the number of graphene layers [40].

6 REBO Potential

The utilisation of ideal interatomic forces and energies is critical to the credibility of MD and Monte Carlo simulations, which have had an influence on everything from crystal formation to medication discovery. Analytical potential energy expressions and semi-empirical electronic structure approaches are used to describe the interaction of atoms or to achieve a total energy first-principle calculation, respectively. Bonds might be made and broken by altering the atomic hybridization of carbon and hydrocarbon molecules using a conventional potential energy expression [41].

7 Abell–Tersoff Bond Order Potentials

Abell utilised chemical pseudopotential theory to create the general analytic form used for intramolecular potential energy measurement in his work [42].

Using a local basis of unperturbed orbitals, Abell proved that E_b may be easily expressed as a sum over the closest neighbours.

$$E_b = \sum_i \sum_{j(>i)} \left[V^R(r_{ij}) - b_{ij} V^A(r_{ij}) \right] \tag{8}$$

There are pair-additive interactions between the core and the valence electrons, and these are the functions $V^R(r)$ and $V^A(r)$. Since bij denotes how far apart atoms I and j are from one another, it may be calculated using Huckel or an analogous level of electronic structure theory.

For regular structures, Abell used a Bethe lattice to develop the straightforward formula for the crucial quantity governing the value of the bond order.

$$b\alpha(N)^{-\frac{1}{2}} \tag{9}$$

A universal binding energy curve may be generated by combining the energy of the repulsive pair term with that of the attractive pair term, and this gives rise to systems that favour open or close-packed structures depending on their slopes. In addition, the form predicts a rise in bond length and a decrease in the individual bond energies as coordination occurs. Long-standing chemical inclinations support this. Using the second-moment approximation relating local coordination to the widening of orbital energy, the simplified bond order expression equation may be constructed (9) [43].

This method by Abell, which Tersoff adapted, allowed him to model bonding in ambient and high-pressure phases of group IV elements, as well as a range of surface and solid-state defect energies, in an analytical parametrized form [44–46]. Both the local coordination and the bond angles were predicted to have an effect on the value of the bond order. Both of these are necessary to accurately replicate elastic properties and defect energies and to maintain open lattices in the face of shearing distortion.

8 Analytic Bond Order Form

The total potential energy is calculated using the general Abell–Tersoff form Eq. (8). The empirical bond order function utilised here is represented as a sum of terms, as in the earlier hydrocarbon bonding formulation [41].

$$\overline{b_{ij}} = \frac{1}{2}\left[b_{ij}^{\sigma-\pi} + b_{ji}^{\sigma-\pi}\right] + b_{ij}^{\pi} \tag{10}$$

Functions $b_{ij}^{\sigma-\pi}$ and $b_{ji}^{\sigma-\pi}$ are determined by atoms i and j's local coordination and bond angles. The sum of two terms is also a form of the function.

$$b_{ij}^{\pi} = \prod_{ij}^{RC} + b_{ij}^{DH} \tag{11}$$

To establish the value of the first term Π_{ij}^{RC}, we need to know whether or not the connection between the two atoms of a system has a radical character. In the second equation, b_{ij}^{DH}, the dihedral angle for carbon–carbon double bonds dictates the value. Using this formula and Eq. (8), the binding energy of a collection of hydrogen and carbon atoms owing to covalent bonding may be calculated. As in traditional valence force fields, no fixed atomic hybridizations are assumed; instead, atomic bonding is governed only by local bonding neighbours and non-local conjugation. Rehybridization may be modelled as covalent bonds are broken and reformed in a classical potential. This is because the local bonding environment drives effective interatomic interactions, rather than vice versa.

After DW Brenner presented a second generation of the potential, which allowed for the breaking and formation of covalent bonds with accompanying changes in atomic hybridization, it became a strong approach for modelling complicated chemistry in huge, many-atom systems. Compared to the previous version, this new version includes better analytic features and a larger database [47].

9 Applications

When the atomic hybridization is changed appropriately, the REBO potential may be utilised to form and break bonds in carbon and hydrocarbon molecules [34].

10 Mechanical Properties

Molecular dynamics simulation has been used with REBO potential by many researchers to evaluate the mechanical properties of different materials. To better understand the nanomechanical behaviour of C_{60} fullerene, a technique known as molecular dynamics (MD) was used. There were force–displacement curves and virial stresses calculated throughout the simulation. Additionally, the atomic virial stress tensor was determined. Results indicated that the fullerene's maximum normal virial stress in the compression direction was 260 GPa for single C_{60} and crystalline C_{60} before plastic deformation. [48]. The mechanical behaviour of Si/CNT nanocomposites was also investigated using molecular dynamics simulation with REBO potential. Effective Young's modulus was calculated using stress and strain responses, which were then compared to an asymptotic solution of continuum mechanics for silicon/carbon nanotube nanocomposites. According to the results, encapsulating SiNWs with CNTs improves their mechanical reliability significantly. The effective Young's modulus and failure strengths of the Si/CNT nanocomposites were much greater than those of pure SiNWs because of the strong load-bearing capabilities of the CNTs [49]. The growth and characteristics of polymers made from pure acetylene plasma were studied using molecular dynamics simulations. For this purpose, REBO potential was used [50]. Classical molecular dynamics simulations and first-principles computations were used to investigate the mechanical properties of MoS_2 NWs under tensile loading. Because of the presence of phase change, MoS_2 NWs exhibited a complex mechanical response throughout the loading process, which include the linear, plateau, stress hardening, and fracture stages. The impact of several parameters like strain rate and temperature on mechanical properties of MoS_2 NWs such as phase transformation, Young's modulus, fracture strength, and fracture strain was also investigated. Mechanical properties of MoS_2 NWs were shown to be greatly influenced by temperature while being insignificantly affected by strain rate [51]. A double-layer MoS_2 film collision friction model was given to account for collisions in space caused by microgravity. A modified REBO (Reactive Empirical Bond Order) potential was used to characterise the atom-to-atom interactions in the MoS_2 film. The effects of vibration frequency and amplitude on the frictional behaviour of the MoS_2 film were studied. Increased vibration amplitude at various frequencies caused a rise in average friction force, resulting in a large indentation depth [52]. The interfacial strength between C–Al bonds and the mechanical characteristics of nanocomposites were investigated using the molecular dynamics (MD) modelling approach and pullout tests and uniaxial tensile testing. Because it has the

greatest pullout force, Ni turns out to be the best coating metal for boosting interfacial strength out of the three contenders. Young's modulus is up 26.39% when compared to the uncoated version of the ring, as well. The interfacial bonding can be enhanced up to 87.65%, 92.67%, 111.27%, and 145.34% when the proportion of Ni coating is 25%, 50%, 75%, and 100%, respectively. There is no influence on Young's modulus, but the tensile strength lowers from 58.94 to 51.88 GPa with the varying amounts of Ni [53]. Molecular Dynamics (MD) simulation has been used to investigate the mechanical properties of iron reinforced with single, double, and triple wall CNTs. The study found that adding CNTs to pure iron increased its strength and stiffness by up to 80.4% and 57.4%, respectively. The mechanical characteristics of a composite including single, double, and triple-walled carbon nanotubes were also studied in relation to fibre volume percentage and temperature. We also looked at the influence of production defects like stone–wales and bi-vacancies on mechanical characteristics [54]. The molecular dynamics approach is used to investigate the mechanical characteristics of multilayer MoS_2 during compression and stretching in order to determine the optimal intra- and inter-layer potentials. Sulphur atoms in the neighbouring layer generated the whole increase in inter-layer repulsive force during compression, according to the results. The two intra-layer potentials were plastic fracture with structural defects or lattice distortions and brittle fracture with rapid destruction [55].

11 Thermal Properties

By inserting graphene nanoribbons into the armchair graphene nanoribbons using non-equilibrium molecular dynamics, we were able to measure the thermal conductivity of these nanoribbons. H-termination on the edge, breadth, and H coverage on heterostructure nanoribbon thermal conductivity were all studied in more detail. The thermal conductivity of heterostructure nanoribbons was studied in relation to hydrogen atoms in intermediate graphene nanoribbons. The thermal conductivity of heterostructure nanoribbons is at least one order of magnitude lower than that of graphene nanoribbons. Hydrogen-functionalization may be utilised to lower graphene's thermal conductivity and hence regulate its thermal transport qualities [56]. The interfacial thermal conductivity of 16 layers of MoS_2/Si was studied using the molecular dynamics approach. The thermodynamic characteristics of MoS_2/Si were computed using the phonon properties. MoS_2/Si was compared to Si bulk and MoS_2 layers in terms of heat capacity, total energy, the Helmholtz free energy, and entropy (S_T). MoS_2/thermodynamic Si's faction curves follow a similar temperature trend to the Si and MoS_2 layers, as seen by these results. The thermal conductance between 16 layers of MoS_2 and the Si bulk is 17.05 MWm^2K^{-1} at 300 K, based on molecular dynamics simulations [57]. Non-equilibrium molecular dynamics was used to calculate the thermal conductivity of bilayer graphene (BLG) using different in-layer and inter-layer potentials. When different potentials were used in simulations, the values and temperature-dependence of thermal conductivity

show significant huge disparity [58]. The two-dimensional IV's thermal conductivity and Molecular dynamics simulations were used to study a graphene-based material. Materials Studio was used to create the material model, which was then imported into LAMMPS for simulations and non-equilibrium molecular dynamics computations of thermal conductivity (NEMD). The temperature conductivity of two-dimensional IV molecules may be simulated using molecular dynamics. Materials were examined in depth [59]. A multiscale technique based on non-equilibrium molecular dynamics and the finite element method was used to study the influence of various nanosized fillers (carbon-nitride and graphene) on the effective thermal conductivity of polyethylene-based nanocomposites. The results revealed that the thermal conductivity of amorphous polyethylene at ambient temperature was around 0.36 ± 0.05 W/mK using the reactive bond order interatomic potential. There were around 0.47, 0.56, and 0.74 W/mK values for C_2N–polyethylene, C_3N–polyethylene, and graphene–polyethylene nanocomposite at constant volume fractions of 1% [60]. It was shown that the bending of graphene, h-BN, and molybdenum disulfide (MoS_2) nanostructures caused by torsional bending and subsequent wrinkling lowered their heat conductivity. Torsion-induced creases have been shown to impair nanostructures' heat conductivity. MoS_2 thermal conductivity was the most affected by these creases, whereas h-BN thermal conductivity was affected the least [61].

12 Conclusion and Future Perspective

In summary, a comprehensive description of Tersoff and REBO potentials was presented in the preceding sections. These potentials were developed to describe the covalently bonded atomic interactions in organic and inorganic systems from simple single elemental interactions to multicomponent interactions for a combination of solids and molecules. Their invention was motivated by the need to create a computationally efficient model that can accurately predict chemical reactivity, as well as an accurate prediction of bond breaking and formation processes while keeping sound quantum mechanics. The explanations of long-range interactions, torsion, charge transfer, and descriptions of alloys were added to the original forms afterwards. A broad range of phenomena in nanomaterials systems have been described using this family of potentials. These potentials show good system transferability. However, these potentials have been parameterized for only a few material systems which have limited their use. As discussed earlier, parameterization of a potential entails a balancing between accuracy and computing efficiency. Additionally, semi-automated approaches like force matching have made parametrization easier. Because of the intricacy of the spline functions used to interpolate the bond order correction terms, adding more than three components remains a substantial difficulty. Either the precision of the potential may be compromised or new multidimensional interpolation functions can be developed to solve this.

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Conflicts of Interest "There are no conflicts of interest to declare by the authors."

References

1. Sinnott, S.B., Brenner, D.W.: Three decades of many-body potentials in materials research. MRS Bull. **37**(5), 469–473 (2012). https://doi.org/10.1557/mrs.2012.88
2. Valverde, J.R.: Molecular modelling: principles and applications. Brief. Bioinform. **2**(2), 199–200 (2001). https://doi.org/10.1093/bib/2.2.199
3. Daw, M.S., Baskes, M.I.: Embedded-atom method: derivation and application to impurities, surfaces, and other defects in metals. Phys. Rev. B **29**(12), 6443–6453 (1984). https://doi.org/10.1103/PhysRevB.29.6443
4. Finnis, M.W., Sinclair, J.E.: A simple empirical N-body potential for transition metals. Philos. Mag. A **50**(1), 45–55 (1984). https://doi.org/10.1080/01418618408244210
5. Tersoff, J.: New empirical approach for the structure and energy of covalent systems. Phys. Rev. B **37**(12), 6991–7000 (1988). https://doi.org/10.1103/PhysRevB.37.6991
6. Brenner, D.W.: The art and science of an analytic potential. Phys. Status Solidi **217**(1), 23–40 (2000). https://doi.org/10.1002/(SICI)1521-3951(200001)217:1%3c23::AID-PSSB23%3e3.0.CO;2-N
7. Kumar, R., Parashar, A.: Dislocation assisted crack healing in h-BN nanosheets, Phys. Chem. Chem. Phys. **19**, 21739–21747 (2017)
8. Sekkal, W., Bouhafs, B., Aourag, H., Certier, M.: Molecular-dynamics simulation of structural and thermodynamic properties of boron nitride. J. Phys. Condens. Matter **10**(23), 4975–4984 (1998). https://doi.org/10.1088/0953-8984/10/23/006
9. Matsunaga, K., Iwamoto, Y.: Molecular dynamics study of atomic structure and diffusion behavior in amorphous silicon nitride containing boron. J. Am. Ceram. Soc. **84**(10), 2213–2219 (2001). https://doi.org/10.1111/j.1151-2916.2001.tb00990.x
10. Kumar, R., Rajasekaran, G., Parasha, A.: Optimised cut-off function for Tersoff-like potentials for a BN nanosheet: a molecular dynamics study, Nanotechnology **27**, 085706 (2016)
11. Lindsay, L., Broido, D.A.: Optimized Tersoff and Brenner empirical potential parameters for lattice dynamics and phonon thermal transport in carbon nanotubes and graphene. Phys. Rev. B—Condens. Matter Mater. Phys. **81**(20), 1–6 (2010). https://doi.org/10.1103/PhysRevB.81.205441
12. Rajasekaran, G., Kumar, R., Parashar, A.: Tersoff potential with improved accuracy for simulating graphene in molecular dynamics environment. Mater. Res. Express **3**(3), 35011 (2016). https://doi.org/10.1088/2053-1591/3/3/035011
13. Zhang, Y.-Y., Pei, Q.-X., Sha, Z.-D., Zhang, Y.-W.: A molecular dynamics study of the mechanical properties of h-BCN monolayer using a modified Tersoff interatomic potential. Phys. Lett. A **383**(23), 2821–2827 (2019). https://doi.org/10.1016/j.physleta.2019.05.055
14. Roy, A., Gupta, K.K., Naskar, S., Mukhopadhyay, T., Dey, S.: Compound influence of topological defects and heteroatomic inclusions on the mechanical properties of SWCNTs. Mater. Today Commun. **26**, 102021 (2021). https://doi.org/10.1016/j.mtcomm.2021.102021
15. Yao, W., Fan, L.: Research on the correlation of mechanical properties of BN–graphene–BN/BN vertically-stacked nanostructures in the presence of interlayer sp3 bonds and nanopores with temperature. Phys. Chem. Chem. Phys. **22**(10), 5920–5928 (2020). https://doi.org/10.1039/D0CP00179A
16. Hasheminia, F., Bahari, Y., Rajabpour, A., Arabha, S.: Elucidation of thermo-mechanical properties of silicon nanowires from a molecular dynamics perspective. Comput. Mater. Sci. **200**, 110821 (2021). https://doi.org/10.1016/j.commatsci.2021.110821
17. Gupta, K.K., Mukhopadhyay, T., Roy, A., Dey, S.: Probing the compound effect of spatially varying intrinsic defects and doping on mechanical properties of hybrid graphene monolayers. J. Mater. Sci. Technol. **50**, 44–58 (2020). https://doi.org/10.1016/j.jmst.2020.03.004

18. Sagar, T.C., Chinthapenta, V.: Effect of substitutional and vacancy defects on the electrical and mechanical properties of 2D-hexagonal boron nitride. J. Mol. Model. **26**(8), 192 (2020). https://doi.org/10.1007/s00894-020-04452-y
19. Albooyeh, A.R., Dadrasi, A., Mashhadzadeh, A.H.: Effect of point defects and low-density carbon-doped on mechanical properties of BNNTs: a molecular dynamics study. Mater. Chem. Phys. **239**, 122107 (2020). https://doi.org/10.1016/j.matchemphys.2019.122107
20. Eftekhari, S.A., Toghraie, D., Hekmatifar, M., Sabetvand, R.: Mechanical and thermal stability of armchair and zig-zag carbon sheets using classical MD simulation with Tersoff potential. Phys. E Low-Dimension. Syst. Nanostruct. **133**, 114789 (2021). https://doi.org/10.1016/j.physe.2021.114789
21. Vijayaraghavan, V., Zhang, L.: Consistent computational modeling of mechanical properties of carbon and boron nitride nanotubes. Jom **72**(11), 3968–3976 (2020). https://doi.org/10.1007/s11837-020-04287-1
22. Griebel, M., Hamaekers, J., Heber, F.: A molecular dynamics study on the impact of defects and functionalization on the Young modulus of boron—nitride nanotubes. Comput. Mater. Sci. **45**(4), 1097–1103 (2009). https://doi.org/10.1016/j.commatsci.2009.03.029
23. Mortazavi, B., Ahzi, S.: Thermal conductivity and tensile response of defective graphene: a molecular dynamics study. Carbon N. Y. **63**, 460–470 (2013). https://doi.org/10.1016/j.carbon.2013.07.017
24. Kumar, R.: Effect of grain boundaries' doping on the mechanical properties of nitrogen-doped bicrystalline graphene. Diam. Relat. Mater. **121**, 108771 (2022). https://doi.org/10.1016/j.diamond.2021.108771
25. Izadifar, M., Abadi, R., Jam, A.N., Rabczuk, T.: Investigation into the effect of doping of boron and nitrogen atoms in the mechanical properties of single-layer polycrystalline graphene. Comput. Mater. Sci. **138**, 435–447 (2017). https://doi.org/10.1016/j.commatsci.2017.06.038
26. Tafrishi, H., Sadeghzadeh, S., Molaei, F., Siavoshi, H.: Nanoparticles, graphene and boron nitride, 14785–14793 (2020). https://doi.org/10.1039/d0ra01847c
27. Mohammadi, K., Madadi, A.A., Bajalan, Z., Pishkenari, H.N.: Analysis of mechanical and thermal properties of carbon and silicon nanomaterials using a coarse-grained molecular dynamics method. Int. J. Mech. Sci. **187**, 106112 (2020). https://doi.org/10.1016/j.ijmecsci.2020.106112
28. Molaei, F., Moghadam, M.S., Nouri, S.: Investigation of thermal properties of crystalline alpha quartz by employing different interatomic potentials: a molecular dynamic study. Phys. Earth Planet. Inter. **316**, 106724 (2021). https://doi.org/10.1016/j.pepi.2021.106724
29. Karaaslan, Y., Yapicioglu, H., Sevik, C.: Assessment of thermal transport properties of group-III nitrides: a classical molecular dynamics study with transferable Tersoff-type interatomic potentials. Phys. Rev. Appl. **13**(3), 34027 (2020). https://doi.org/10.1103/PhysRevApplied.13.034027
30. Du, Y., Zhou, J., Ying, P., Zhang, J.: Effects of cell defects on the mechanical and thermal properties of carbon honeycombs. Comput. Mater. Sci. **187**, 110125 (2021). https://doi.org/10.1016/j.commatsci.2020.110125
31. Wu, X., Han, Q.: Thermal conductivity of monolayer hexagonal boron nitride: from defective to amorphous. Comput. Mater. Sci. **184**, 109938 (2020). https://doi.org/10.1016/j.commatsci.2020.109938
32. Eshkalak, K.E., Sadeghzadeh, S., Jalaly, M.: Thermal resistance analysis of hybrid graphene-boron nitride nanosheets: the effect of geometry, temperature, size, strain and structural defects. Comput. Mater. Sci. **174**, 109484 (2020). https://doi.org/10.1016/j.commatsci.2019.109484
33. Liu, F., et al.: Strain effects on the interfacial thermal conductance of graphene/h-BN heterostructure. Nano Mater. Sci. (2021). https://doi.org/10.1016/j.nanoms.2021.05.009
34. Li, M., Liu, J., Yu, W., Zhang, Y., Zhou, H.: Atomistic molecular dynamic simulations of the thermal transport across h-BN/cellulose nanocrystal interface. Int. J. Heat Mass Transf. **171**, 121043 (2021). https://doi.org/10.1016/j.ijheatmasstransfer.2021.121043
35. Ren, W., Ouyang, Y., Jiang, P., Yu, C., He, J., Chen, J.: The impact of interlayer rotation on thermal transport across graphene/hexagonal boron nitride van der Waals heterostructure. Nano Lett. **21**(6), 2634–2641 (2021). https://doi.org/10.1021/acs.nanolett.1c00294

36. Microsoft Word—2021.10.12 CPB v2.docx_Enhanced Reader.pdf
37. Song, J., et al.: Effect of strain and defects on the thermal conductance of the graphene/hexagonal boron nitride interface. Phys. Chem. Chem. Phys. **22**(20), 11537–11545 (2020). https://doi.org/10.1039/D0CP01727B
38. He, T., Li, T., Huang, Z., Tang, Z., Guan, X.: Mechanical and thermal properties of the coaxial carbon nanotube@boron nitride nanotube composite. Phys. E Low-Dimension. Syst. Nanostruct. **107**(November 2018), 182–186 (2019). https://doi.org/10.1016/j.physe.2018.11.037
39. Mahdizadeh, S.J., Goharshadi, E.K., Akhlamadi, G.: Thermo-mechanical properties of boron nitride nanoribbons: a molecular dynamics simulation study. J. Mol. Graph. Model. **68**, 1–13 (2016). https://doi.org/10.1016/j.jmgm.2016.05.008
40. Toprak, K., Yilmaz, A.: Non-equilibrium molecular dynamics for calculating the thermal conductivity of graphene-coated aluminum. Kocaeli J. Sci. Eng. **3**(1), 27–32 (2020). https://doi.org/10.34088/kojose.663888
41. Brenner, D.W.: Empirical potential for hydrocarbons for use in simulating the chemical vapor deposition of diamond films. Phys. Rev. B **42**(15), 9458–9471 (1990). https://doi.org/10.1103/PhysRevB.42.9458
42. Abell, G.C.: Empirical chemical pseudopotential theory of molecular and metallic bonding. Phys. Rev. B **31**(10), 6184–6196 (1985). https://doi.org/10.1103/PhysRevB.31.6184
43. Brenner, D.W., Shenderova, O.A., Areshkin, D.A.: Quantum-based analytic interatomic forces and materials simulation, Rev. Comput. Chem. 207–239 (1998). https://doi.org/10.1002/9780470125892.ch4
44. Tersoff, J.: New empirical model for the structural properties of silicon. Phys. Rev. Lett. **56**(6), 632–635 (1986). https://doi.org/10.1103/PhysRevLett.56.632
45. Tersoff, J.: Empirical interatomic potential for silicon with improved elastic properties. Phys. Rev. B **38**(14), 9902–9905 (1988). https://doi.org/10.1103/PhysRevB.38.9902
46. Tersoff, J.: Empirical interatomic potential for carbon, with applications to amorphous carbon. Phys. Rev. Lett. **61**(25), 2879–2882 (1988). https://doi.org/10.1103/PhysRevLett.61.2879
47. Brenner, D.W., Shenderova, O.A., Harrison, J.A., Stuart, S.J., Ni, B., Sinnott, S.B.: A second-generation reactive empirical bond order (REBO) potential energy expression for hydrocarbons. J. Phys. Condens. Matter **14**(4), 783–802 (2002). https://doi.org/10.1088/0953-8984/14/4/312
48. d0nr00584c 9849..9858 ++_Enhanced Reader.pdf
49. Shim, J.S., Lee, G.H., Cui, C.Y., Beom, H.G.: Mechanical behaviors of Si/CNT core/shell nanocomposites under tension: a molecular dynamics analysis. Nanomaterials **11**(8) (2021). https://doi.org/10.3390/nano11081989.
50. Brault, P., Ji, M., Sciacqua, D., Poncin-Epaillard, F., Berndt, J., Kovacevic, E.: Insight into acetylene plasma deposition using molecular dynamics simulations. Plasma Process. Polym. n/a(n/a), e2100103 (2021). https://doi.org/10.1002/ppap.202100103
51. Ying, P., Zhang, J., Zhou, J., Liang, Q., Zhong, Z.: Mechanical behaviors of MoS nanowires under tension from molecular dynamics simulations. Comput. Mater. Sci. **179**, 109691 (2020). https://doi.org/10.1016/j.commatsci.2020.109691
52. Tong, R., Han, B., Zhang, X., Zhang, T., Zeng, Q., Liu, G.: Molecular dynamics simulation on collision frictional properties of a molybdenum disulfide (MoS2) film in microgravity environment. Microgravity Sci. Technol. **33**(4), 47 (2021). https://doi.org/10.1007/s12217-021-09896-2
53. Wang, X., et al.: Enhanced interfacial strength of graphene reinforced aluminum composites via X (Cu, Ni, Ti)-coating: molecular-dynamics insights. Adv. Powder Technol. **32**(7), 2585–2590 (2021). https://doi.org/10.1016/j.apt.2021.05.041
54. Ishraaq, R., Chhetri, S., Gautam, O., Nahid, S.M., Afsar, A.M.: Molecular dynamics simulation for the analysis of mechanical properties and effect of Stone-Wales and bi-vacancy defect on carbon nanotube reinforced iron composites. AIP Conf. Proc. **2324**(1), 30011 (2021). https://doi.org/10.1063/5.0037829
55. Yi, C., Hu, C., Bai, M., Lv, J., Tang, D.: Molecular dynamics study on the mechanical properties of multilayer MoS2 under different potentials. Nanotechnology **31**(21) (2020). https://doi.org/10.1088/1361-6528/ab73b3

56. Kim, J.-C., Wi, J.-H., Ri, N.-C., Ri, S.-I.: Thermal conductivity of graphene/graphane/graphene heterostructure nanoribbons: non-equilibrium molecular dynamics simulations. Solid State Commun. **328**, 114249 (2021). https://doi.org/10.1016/j.ssc.2021.114249
57. Wang, J., Wu, S., Xie, H., Xiong, L.: Theoretical study on thermal properties of molybdenum disulfide/silicon heterostructures. Comput. Mater. Sci. **200**, 110835 (2021). https://doi.org/10.1016/j.commatsci.2021.110835
58. Zhang, X., Chen, Z., Chen, H., Xu, L.: Comparative studies of thermal conductivity for bilayer graphene with different potential functions in molecular dynamic simulations. Results Phys. **22**, 103894 (2021). https://doi.org/10.1016/j.rinp.2021.103894
59. acm_3478301
60. Razzaghi, L., Khalkhali, M., Rajabpour, A., Khoeini, F.: Effect of graphene and carbon-nitride nanofillers on the thermal transport properties of polymer nanocomposites: a combined molecular dynamics and finite element study. Phys. Rev. E **103**(1), 13310 (2021). https://doi.org/10.1103/PhysRevE.103.013310
61. Arabha, S., Rajabpour, A.: Effect of planar torsional deformation on the thermal conductivity of 2D nanomaterials: a molecular dynamics study. Mater. Today Commun. **22**, 100706 (2020). https://doi.org/10.1016/j.mtcomm.2019.100706

Reactive Forcefield (ReaxFF): Application to Predict 2D Nanomaterials Synthesis



Rajesh Kumar

Abstract The use of atomistic simulations for mechanical characterization and energetic stability has been in use for the last few decades. These simulations have been equally employed in conventional as well as advanced materials. Traditionally, the prevalent forcefields have not been much successful in predicting the synthesis and growth reactions for materials. However, recently developed ReaxFF, a quantum chemistry-based forcefield, has been successfully employed for predicting growth and synthesis reactions along with defect dynamics, with excellent success. These simulations have been particularly employed for nanomaterials such as graphene, h-BN, MoS_2, and WSe_2. This chapter discusses the origin, development and application of ReaxFF for the synthesis reactions of nanomaterials.

Keywords ReaxFF · Bulk properties · Defect · Nanomaterials · Atomistic scale simulations

1 Introduction

Material characterization has been one of the fundamental requirements of the design of any device or structure for ages. Starting from earlier estimates of bulk failure, the advancement in technologies has enabled the designers to observe nanoscale phenomena with great detail. Technologically enabled designers can now accurately predict every minute detail of material behaviour ranging from synthesis to failure. Initially, experimental techniques have been at the forefront of material synthesis and characterization. However, advanced material classes such as nanomaterials coupled with the availability of high-performance computing have empowered the researchers to devise computational techniques for predicting material behaviour in advance even before synthesis. These days, atomistic and macro modeling computational techniques such as density functional theory (DFT) [1], molecular dynamics (MD)

R. Kumar (✉)
Department of Mechanical Engineering, National Institute of Technology, Hamirpur 177005, Himachal Pradesh, India
e-mail: rajeshpawar4u@gmail.com; rpawar@nith.ac.in

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[2], Monte Carlo (MC) [3] simulations and finite element methods (FEM) [4] are proving very useful in complementing experimental techniques. These techniques in conjunction with artificial intelligence are proving very useful in new material discovery and predicting the best practical conditions for their synthesis. Beginning with the prediction of synthesis reaction mechanisms, these techniques go on to forecast the failure behaviour of any desired material. For example, DFT-based calculations are very accurate in predicting structural and energetic stability of materials under the desired conditions but are limited by length and time scales. These shortcomings may be over by the use of MD, MC and FEM-based methods. Though, MD, MC and FEM-based methods have to be used with great precautions due to the issue of accuracy and transferability. Usually, they can be used only for the systems to which they have been fitted to. Among numerical techniques, DFT calculations are excellently accurate and viable for smaller systems, MD is preferred for larger systems with dynamic evolution and FEM is preferred for macroscale simulations for global properties. MD simulations are also limited by length and time scales, but these are well suited for predicting nanoscale atomistic phenomena governing the global behaviour of materials. For MD simulations, potential or force field functions are used to calculate the potential energy of the system which in turn can be further utilized to calculate force and other dynamic parameters of the system. These potential functions are fitted using DFT-based structural and energetic calculations for a particular system type and are suitable to use for those kinds of systems only.

A potential function may be very simple like a pair potential of LJ [5, 6] or a very complex but accurate potential function like reactive force field (ReaxFF) potential [7]. Other simpler potential functions include Tersoff potential [8–13], REBO potential [13], AIREBO potential [14], EAM potential [15], etc. The accuracy of potential comes with its complexity and higher computational resource consumption. For example, though ReaxFF is a complex potential with high computational resource consumption, its accuracy is one of the closest to DFT-based calculations. Owing to its accuracy, ReaxFF is finding applications in synthesis to fracture simulations. Nevertheless, we have restricted ourselves to the use of ReaxFF for nanomaterial synthesis in this chapter. The next subsection discusses the development of ReaxFF with a mathematical treatment.

2 Reactive Force Field (ReaxFF)

ReaxFF is a bond order dependent force field that was originally developed by van Duin et al. in 2001 [7]. ReaxFF has no discontinuity in energy or forces, even during reactions. Therefore, ReaxFF can predict the bond formation and dissociation of chemical bonds efficiently [7]. Also, ReaxFF includes the van der Waals and Coulomb interactions which are very important in predicting the structures and properties of materials. ReaxFF can also be used for a variety of large molecules. It bridges the gap between quantum chemistry and non-reactive empirical force field-based molecular

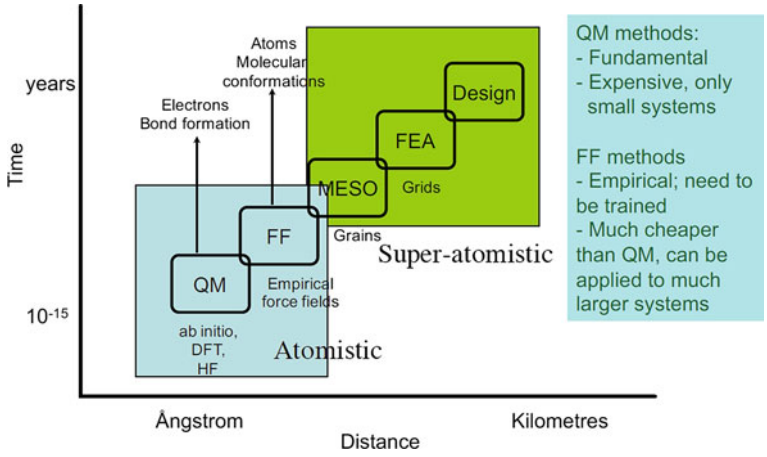


Fig. 1 Comparison of ReaxFF with different types of simulations for different time and length scales. Reproduced from [16]

simulation methods and provides a reasonable model for simulating material reactions at a part cost of quantum mechanical approaches. A comparison of ReaxFF for different length and time scales has been shown in Fig. 1.

The overall system energy in ReaxFF. ReaxFF utilizes bond distance and bond order as one of the key components along with a relationship of bond order and bond energy. In ReaxFF, the potential energy of the system comprises the following terms:

$$E_{system} = E_{bond} + E_{over} + E_{under} + E_{lp} + E_{val} + E_{pen} + E_{tors} + E_{conj} + E_{vdWaals} + E_{Columb} \quad (1)$$

where E_{bond} is the instantaneous interatomic distance-dependent bond order energy, E_{over} and E_{under} are the energy contribution due to the over and under-coordination of atoms, E_{lp} is the energy attributed to lone pairs on atoms, E_{val} is the valence angle distortion energy, E_{pen} is the penalty energy two double bonds sharing an atom in the valence angle like in allene, E_{tors} is the energy associated with the torsion of an angle, E_{conj} is for the conjugation energy, $E_{vdWaals}$ is the energy due to van der Waals interactions and $E_{Coulomb}$ is the Coulomb interaction energy. More details on energy terms and bond order can be found in reference [7].

For bond order calculations, ReaxFF uses the sum of three exponential terms, one representing σ bond and the other two representing π bonds as given below

$$BO_{ij} = BO_{ij}^{\sigma} + BO_{ij}^{\pi} + BO_{ij}^{\pi\pi} = \exp\left[p_{bo,1}.\left(\frac{r_{ij}}{r_{\sigma}}\right)^{p_{bo,2}}\right] + \exp\left[p_{bo,3}.\left(\frac{r_{ij}^{\pi}}{r_{\sigma}}\right)^{p_{bo,4}}\right] + \exp\left[p_{bo,5}.\left(\frac{r_{ij}^{\pi\pi}}{r_{\sigma}}\right)^{p_{bo,6}}\right]$$

where $p_{bo,n}$ are the bond parameters which are different for the different atom types. The BO_{ij}^{σ} is close to 1 at a distance of 1.5 Å between two carbon atoms and approaches 0 at a distance of 2.5 Å. The first π bond contribution is equal to 1 at 1.2 Å and renders insignificant when the distance is 1.75 Å. The third contribution is equal to 1 when the distance is less than 1.0 Å and decreases sharply to 0 at 1.4 Å. An overview of ReaxFF energy components along with the current progress of ReaxFF is shown in Fig. 2.

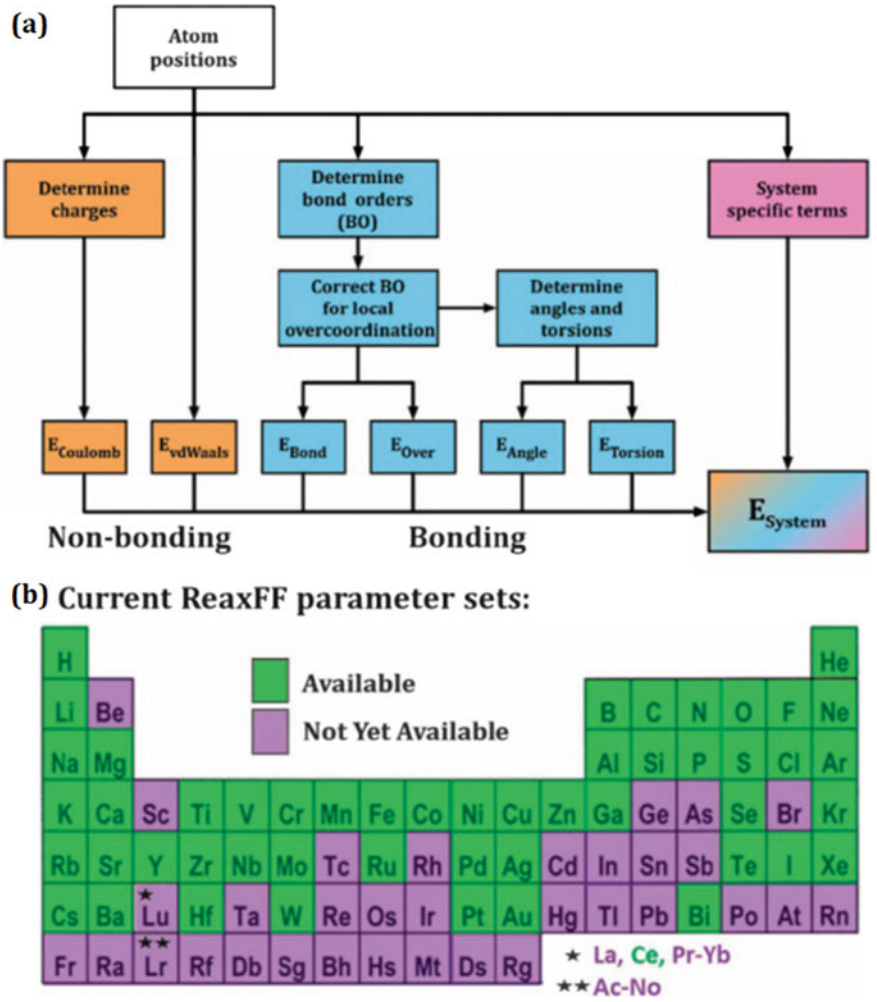


Fig. 2 **a** Overview of the ReaxFF total energy components and **b** elements currently described in available parameter sets. Reproduced from reference [17] under creative commons license

3 Applications of ReaxFF for the Synthesis of 2D Nanomaterials

Initially, ReaxFF was developed for chemical reactions in hydrocarbons [7]. However, it was gradually extended to metallic and nonmetallic materials [18, 19] and metallic elements [20–22], and became more and more widely used for different systems. Recently, it has been employed for determining the mechanical and fracture properties of 2D nanomaterials such as graphene, h-BN and molybdenum disulfide (MoS_2) [23–27]. It has also been employed to estimate the displacement threshold energy and knock-on cross section of h-BN nanosheets [28]. Hence, ReaxFF has a wide range of applications from reaction mechanism prediction to mechanical and fracture properties of materials. Nowadays, prediction of synthesis behaviour of 2D materials such as graphene, h-BN and MoS2 nanosheets, using ReaxFF is gaining momentum. Various researchers have applied ReaxFF force field for predicting the synthesis of 2D nanomaterials such as graphene, h-BN, WSe_2, WS_2, $MoSe_2$ and MoS_2 [29–48] via chemical vapour deposition (CVD) and other processes. Not only the nanosheets, researchers have employed ReaxFF simulation for predicting the synthesis of nanotubes as well [49, 50]. These recent trends in the increasing use of ReaxFF simulations indicate that prediction of optimized conditions for nanomaterials' synthesis before actual synthesis may save time and crucial resources. So, in the following subsection, a concise summary of all the materials whose synthesis and growth have been predicted, are provided.

3.1 Bulk Growth

As mentioned above, the use of ReaxFF for simulating the synthesis reactions for nanomaterials has been gaining traction over recent years. An account for most of the studies carried out with ReaxFF so far has been provided below.

Hong et al. [29, 46] provided a detailed analysis of the chemical synthesis of large-scale and defect-free MoS2 layers via CVD with direct sulfidation of MoO3 surfaces using S2 gas precursors. In another study on MoS_2, Chen et al. [33] predicted the formation of MoS_2 direct from elemental Mo and S using reactive molecular dynamics simulations using ReaxFF and REBO potential. Hong et al. [47] synthesized MoS_2 by CVD using MoO3 reactants and gas-phase sulphur precursors. Their results provided a better understanding of the sulfidation process for the scalable synthesis of defect-free MoS_2 and other Transition metal dichalcogenide (TMDC) materials. Zhang and van Duin [30] used ReaxFF-based atomistic-scale simulations to simulate the graphene growth on a silicon carbide substrate using thermal decomposition and CVD. Meng et al. [31] explored the fundamentals of graphene growth on nickel crystal via CVD. Their observations may be useful for the control of the CVD growth of graphene for producing defect-free graphene nanosheets. Lu and

Yang [40] used MD simulations of graphene growth using chemical vapour deposition on nickel through polycyclic aromatic hydrocarbons. Zhang et al. [41] studied the role of Cu substrate for graphene growth using ReaxFF MD simulations. They also discussed the effects of temperature and the carbon deposition rate on the quality of graphene films. Dong et al. [42] predicted the formation of layered graphene clusters from polyimides under extreme conditions and predicted different reaction paths under NVT and NPT conditions. Neyts et al. [44] simulated single-layer graphene growth on nickel crystal under non-equilibrium high flux conditions using hybrid ReaxFF and MC-based simulations. They concluded that the graphene film growth may proceed via a combination of deposition–segregation mechanisms on a nickel substrate. Rasuli et al. [45] utilized faced Ni(100) surface for graphene growth using CVD. They suggested that high-quality graphene deposition is theoretically possible on faceted Ni(100) surface high flux energy. In another recent study, Liu et al. [32] predicted the preferred morphology of hexagonal boron nitride domain structure on Ni using ReaxFF-based MD simulations. They concluded that careful control of synthesis conditions is required to produce desired nanostructures of h-BN. McLean et al. [34] simulated the h-BN nucleation mechanism in CVD via ReaxFF-based simulations and concluded that the chemical reactions between gas-phase B_2O_2 and NH_3 precursors form the BN nanostructures. Liu et al. [35] used first principles-based reactive MD simulations to predict the nucleation and formation of hexagonal boron nitride on nickel. They discussed the crystal quality and the role of temperature and substrates in h-BN grain growth. In addition to the nanosheets, McLean et al. [49] used ReaxFF-based simulations of catalytic CVD for predicting the growth of boron nitride nanotubes via network fusion. They concluded that BNNTs predicted by network fusion mechanism are different than the CNTs produced CVD and have sharper tips and wider diameters as compared to CVD synthesized CNTs. Another study by McLean [50] used boron oxide for nucleation growth of BNNTs in a Ni-catalyzed CVD process. Nayir et al. [36] explored the synthesis of 2D WSe_2 using ReaxFF-based molecular dynamics simulations. These simulations provided valuable insights into the edge-controlled growth of single crystals of 2D WSe_2 under reactive environments close to experimental conditions. In another study on WSe_2 Nayir et al. [37] discussed the synthesis of 2D-WS2 and its interaction with sapphire as a substrate for crystal growth. Yet another study of Nayir et al. [39] using large-scale molecular dynamics simulations investigated the synthesis, processing and characterization of 2D $MoSe_2$. Berdiyorov et al. [48] produced stabile silicene within bilayer graphene: using MD and density-functional tight-binding calculations. They concluded from their results that silicene can be favourably realized by intercalating bilayer graphene with Si.

3.2 Defect and Growth of Nanomaterials Using ReaxFF

As discussed above, ReaxFF simulations are proving very useful in predicting the synthesis and growth reactions for various nanomaterials. Moreover, reactive MD

simulations are also useful in predicting defect growth and its mechanisms. Various researchers have used these simulations for this purpose as well. A summary of such works has been given below.

Gao et al. [51] predicted the defect evolution behaviour of monovacancy and line vacancies in monolayer MoS_2. It was concluded from their study that high temperature and large vacancy concentrations cause the evolution of point vacancies into line vacancies. Ostadhossein et al. [25] predicted the formation energy of MoS2 nanosheets using the ReaxFF potential proposed by them. According to their results, this potential provided valuable insights into how defect and loading conditions can lead to different morphological changes to modify the electronic properties of 2D crystals of MoS_2. Weiwei Zhang and Adri C. T. van Duin [30] also investigated the defect and grain boundary structures of graphene nanosheets. Meng et al. [43] used the ReaxFF atomistic simulations to predict the defect healing in graphene grown by CVD on a Ni substrate. It was concluded from their simulations that the stepped surface is beneficial to synthesize better graphene for CVD growth on Ni substrate. Vashisth et al. [52] employed ReaxFF simulations for simulating the formation of laser-induced graphene (LIG) multifunctional polymer nanocomposites. Their simulations explored the mechanisms of LIG formation that can be leveraged for bulk LIG applications such as sensors, electrocatalysts and targeted heating for welding polymers. Chaban et al. [53] used ReaxFF simulations for simulating the explosion of nitrated CNTs and provided a detailed chemical mechanism of the decomposition process, explicitly in the time domain. Ryu et al. [54] employed ReaxFF simulations in their experimental and DFT-based work to predict the mechanisms of triangular hole formations under electron irradiation. It was reported from their results that the hole growth is initiated by a vacancy of boron and grows in such a way that retains an overall triangular shape. These observations are compared to density functional theory calculations and molecular dynamics simulations. Hickey et al. [38, 55] explored the grain boundaries and vacancy formations in WS_2 films using reactive molecular dynamics simulations. It was concluded from their simulations that vacancy arrays can result from the steric hindrance of bulky gas-phase precursors at narrowly separated growth. ReaxFF has also been extensively utilized to characterize 2D nanomaterials and their polymer composites as well [56–60].

4 Conclusion and Future Perspective

Atomistic-based molecular dynamics simulations are becoming an indispensable tool for exploring the structural-property relationships. Non-reactive dynamics-based simulations are useful in predicting the mechanical and thermal behaviour of materials. These simulations are very important in saving time and resources being non-destructive in nature. However, predicting the synthesis path reactions is a new area and has been brought into focus by the development of ReaxFF. Hence, ReaxFF has become a very important tool with excellent accuracy in exploring the synthesis behaviour of nanomaterials. For future aspects, this field may be extended to more

existing materials including conventional materials such as metals and nanocomposites. ReaxFF is computationally intensive as compared to traditional non-reactive forcefields such as EAM (embedded atom method) and requires properly optimized algorithms so that it may be used for bigger systems for higher time scales so that mass production reactions may be included.

Conflicts of Interest "There are no conflicts of interest to declare by the author."

References

1. Künne, L.: Recent developments and applications of modern density functional theory. Z. Phys. Chem. **204**, 263–264 (1998)
2. Rapaport, D.C., Rapaport, D.C.R.: The Art of Molecular Dynamics Simulation. Cambridge University Press (2004)
3. Mooney, C.Z.: Monte Carlo Simulation. Sage (1997)
4. Zienkiewicz, O.C., Taylor, R.L., Nithiarasu, P., Zhu, J.: The Finite Element Method. McGraw-Hill London (1977)
5. Jones, J.E., Chapman, S.: On the determination of molecular fields.—I. From the variation of the viscosity of a gas with temperature. Proc. R. Soc. London. Ser. A Contain. Pap. Math. Phys. Character **106**, 441–462 (1924)
6. Jones, J.E., Chapman, S.: On the determination of molecular fields.—II. From the equation of state of a gas. Proc. R. Soc. London. Ser. A, Contain. Pap. Math. Phys. Character **106**, 463–477 (1924)
7. van Duin, A.C.T., Dasgupta, S., Lorant, F., Goddard, W.A.: ReaxFF: a reactive force field for hydrocarbons. J. Phys. Chem. A **105**, 9396–9409 (2001)
8. Kumar, R.: Effect of grain boundaries' doping on the mechanical properties of nitrogen-doped bicrystalline graphene. Diamond Relat. Mater. **121**, 108771 (2022)
9. Kumar, R., Rajasekaran, G., Parashar, A.: Optimised cut-off function for Tersoff-like potentials for a BN nanosheet: a molecular dynamics study. Nanotechnology **27**, 085706 (2016)
10. Lindsay, L., Broido, D.A.: Erratum: optimized Tersoff and Brenner empirical potential parameters for lattice dynamics and phonon thermal transport in carbon nanotubes and graphene. Phys. Rev. B **81**, 205441, **82**, 209903 (2010)
11. Kumar, R., Parashar, A.: Dislocation assisted crack healing in h-BN nanosheets. Phys. Chem. Chem. Phys. **19**, 21739–21747 (2017)
12. Rajasekaran, G., Kumar, R., Parashar, A.: Tersoff potential with improved accuracy for simulating graphene in molecular dynamics environment. Mater. Res. Express **3**, 035011 (2016)
13. Tersoff, J.: New empirical approach for the structure and energy of covalent systems. Phys. Rev. B **37**, 6991–7000 (1988)
14. Stuart, S.J., Tutein, A.B., Harrison, J.A.: A reactive potential for hydrocarbons with intermolecular interactions. J. Chem. Phys. **112**, 6472–6486 (2000)
15. Daw, M.S., Baskes, M.I.: Embedded-atom method: derivation and application to impurities, surfaces, and other defects in metals. Phys. Rev. B **29**, 6443–6453 (1984)
16. van Duin, A.C.T.: In: ReaxFF Workshop, NSCCS, London, 27–28 Sept 2016
17. Senftle, T.P., Hong, S., Islam, M.M., Kylasa, S.B., Zheng, Y., Shin, Y.K., Junkermeier, C., Engel-Herbert, R., Janik, M.J., Aktulga, H.M., Verstraelen, T., Grama, A., van Duin, A.C.T.: The ReaxFF reactive force-field: development, applications and future directions. npj Comput. Mater. **2**, 15011 (2016)
18. Van Duin, A.C., Strachan, A., Stewman, S., Zhang, Q., Xu, X., Goddard, W.A.: ReaxFFSiO reactive force field for silicon and silicon oxide systems. J. Phys. Chem. A **107**, 3803–3811 (2003)

19. Han, S.S., Kang, J.K., Lee, H.M., Van Duin, A.C., Goddard III, W.A.: The theoretical study on interaction of hydrogen with single-walled boron nitride nanotubes. I. The reactive force field ReaxFF HBN development. J. Chem. Phys. **123**, 114703 (2005)
20. Zhang, Q., Çağın, T., Van Duin, A., Goddard, W.A., III., Qi, Y., Hector, L.G., Jr.: Adhesion and nonwetting-wetting transition in the Al/α− Al 2 O 3 interface. Phys. Rev. B **69**, 045423 (2004)
21. Cheung, S., Deng, W.-Q., Van Duin, A.C., Goddard, W.A.: ReaxFFMgH reactive force field for magnesium hydride systems. J. Phys. Chem. A **109**, 851–859 (2005)
22. Nielson, K.D., Van Duin, A.C., Oxgaard, J., Deng, W.-Q., Goddard, W.A.: Development of the ReaxFF reactive force field for describing transition metal catalyzed reactions, with application to the initial stages of the catalytic formation of carbon nanotubes. J. Phys. Chem. A **109**, 493–499 (2005)
23. Kumar, R., Mertiny, P., Parashar, A.: Effects of different hydrogenation regimes on mechanical properties of h-BN: a reactive force field study. J. Phys. Chem. C **120**, 21932–21938 (2016)
24. Kumar, R., Parashar, A.: Effect of geometrical defects and functionalization on the interfacial strength of h-BN/polyethylene based nanocomposite. Polymer **146**, 82–90 (2018)
25. Kumar, R., Parashar, A.: Fracture toughness enhancement of h-BN monolayers via hydrogen passivation of a crack edge. Nanotechnology **28**, 165702 (2017)
26. Verma, A., Kumar, R., Parashar, A.: Enhanced thermal transport across a bi-crystalline graphene–polymer interface: an atomistic approach. Phys. Chem. Chem. Phys. **21**, 6229–6237 (2019)
27. Ostadhossein, A., Rahnamoun, A., Wang, Y., Zhao, P., Zhang, S., Crespi, V.H., van Duin, A.C.T.: ReaxFF reactive force-field study of molybdenum disulfide (MoS2). J. Phys. Chem. Lett. **8**, 631–640 (2017)
28. Kumar, R., Parashar, A., Mertiny, P.: Displacement thresholds and knock-on cross sections for hydrogenated h-BN monolayers. Comput. Mater. Sci. **142**, 82–88 (2018)
29. Hong, S., Sheng, C., Krishnamoorthy, A., Rajak, P., Tiwari, S., Nomura, K.-I., Misawa, M., Shimojo, F., Kalia, R.K., Nakano, A., Vashishta, P.: Chemical vapor deposition synthesis of MoS2 layers from the direct sulfidation of MoO3 surfaces using reactive molecular dynamics simulations. J. Phys. Chem. C **122**, 7494–7503 (2018)
30. Zhang, W., van Duin, A.C.T.: Atomistic-scale simulations of the graphene growth on a silicon carbide substrate using thermal decomposition and chemical vapor deposition. Chem. Mater. **32**, 8306–8317 (2020)
31. Meng, L., Sun, Q., Wang, J., Ding, F.: Molecular dynamics simulation of chemical vapor deposition graphene growth on Ni (111) surface. J. Phys. Chem. C **116**, 6097–6102 (2012)
32. Liu, S., Comer, J., van Duin, A.C.T., van Duin, D.M., Liu, B., Edgar, J.H.: Predicting the preferred morphology of hexagonal boron nitride domain structure on nickel from ReaxFF-based molecular dynamics simulations. Nanoscale **11**, 5607–5616 (2019)
33. Chen, R., Jusufi, A., Schilowitz, A., Martini, A.: Formation of MoS2 from elemental Mo and S using reactive molecular dynamics simulations. J. Vac. Sci. Technol. A **38**, 022201 (2020)
34. McLean, B., Webber, G.B., Page, A.J.: Boron nitride nucleation mechanism during chemical vapor deposition. J. Phys. Chem. C **122**, 24341–24349 (2018)
35. Liu, S., van Duin, A.C.T., van Duin, D.M., Liu, B., Edgar, J.H.: Atomistic insights into nucleation and formation of hexagonal boron nitride on nickel from first-principles-based reactive molecular dynamics simulations. ACS Nano **11**, 3585–3596 (2017)
36. Nayir, N., Wang, Y., Shabnam, S., Hickey, D.R., Miao, L., Zhang, X., Bachu, S., Alem, N., Redwing, J., Crespi, V.H., van Duin, A.C.T.: Modeling for structural engineering and synthesis of two-dimensional WSe2 using a newly developed ReaxFF reactive force field. J. Phys. Chem. C **124**, 28285–28297 (2020)
37. Nayir, N., Shin, Y.K., Wang, Y., Sengul, M.Y., Hickey, D.R., Chubarov, M., Choudhury, T.H., Alem, N., Redwing, J., Crespi, V.H., van Duin, A.C.T.: A ReaxFF force field for 2D-WS2 and its interaction with sapphire. J. Phys. Chem. C **125**, 17950–17961 (2021)
38. Reifsnyder Hickey, D., Nayir, N., Chubarov, M., Choudhury, T.H., Bachu, S., Miao, L., Wang, Y., Qian, C., Crespi, V.H., Redwing, J.M., van Duin, A.C.T., Alem, N.: Illuminating invisible

grain boundaries in coalesced single-orientation WS2 monolayer films. Nano Lett. **21**, 6487–6495 (2021)
39. Nayir, N., Wang, Y., Ji, Y., Choudhury, T.H., Redwing, J.M., Chen, L.-Q., Crespi, V.H., van Duin, A.C.T.: Theoretical modeling of edge-controlled growth kinetics and structural engineering of 2D-MoSe2. Mater. Sci. Eng. B **271**, 115263 (2021)
40. Lu, Y., Yang, X.: Molecular simulation of graphene growth by chemical deposition on nickel using polycyclic aromatic hydrocarbons. Carbon **81**, 564–573 (2015)
41. Zhang, L., Zhu, Y., Teng, W., Xia, T., Rong, Y., Li, N., Ma, H.: A molecular dynamics simulation of the graphene growth on Cu(111) surface. Comput. Mater. Sci. **130**, 10–15 (2017)
42. Dong, Y., Rismiller, S.C., Lin, J.: Molecular dynamic simulation of layered graphene clusters formation from polyimides under extreme conditions. Carbon **104**, 47–55 (2016)
43. Meng, L., Wang, Z., Jiang, J., Yang, Y., Wang, J.: Defect healing of chemical vapor deposition graphene growth by metal substrate step. J. Phys. Chem. C **117**, 15260–15265 (2013)
44. Neyts, E.C., van Duin, A.C.T., Bogaerts, A.: Formation of single layer graphene on nickel under far-from-equilibrium high flux conditions. Nanoscale **5**, 7250–7255 (2013)
45. Rasuli, R., Mostafavi, K., Davoodi, J.: Molecular dynamics simulation of graphene growth on Ni(100) facet by chemical vapor deposition. J. Appl. Phys. **115**, 024311 (2014)
46. Hong, S., Krishnamoorthy, A., Rajak, P., Tiwari, S., Misawa, M., Shimojo, F., Kalia, R.K., Nakano, A., Vashishta, P.: Computational synthesis of MoS2 layers by reactive molecular dynamics simulations: initial sulfidation of MoO3 surfaces. Nano Lett. **17**, 4866–4872 (2017)
47. Hong, S., Krishnamoorthy, A., Sheng, C., Kalia, R.K., Nakano, A., Vashishta, P.: A reactive molecular dynamics study of atomistic mechanisms during synthesis of MoS2 layers by chemical vapor deposition. MRS Adv. **3**, 307–311 (2018)
48. Berdiyorov, G.R., Neek-Amal, M., Peeters, F.M., van Duin, A.C.T.: Stabilized silicene within bilayer graphene: a proposal based on molecular dynamics and density-functional tight-binding calculations. Phys. Rev. B **89**, 024107 (2014)
49. McLean, B., Webber, G.B., Page, A.J.: Boron nitride nanotube nucleation via network fusion during catalytic chemical vapor deposition. J. Am. Chem. Soc. **141**, 13385–13393 (2019)
50. McLean, B., Webber, G.B., Page, A.J.: Boron nitride nanotube nucleation during Ni-catalyzed boron oxide chemical vapor deposition. J. Phys. Chem. C **123**, 27875–27883 (2019)
51. Gao, C., Yang, X., Jiang, M., Chen, L., Chen, Z., Singh, C.V.: Defect evolution behaviours from single sulfur point vacancies to line vacancies in monolayer molybdenum disulfide. Phys. Chem. Chem. Phys. **23**, 19525–19536 (2021)
52. Vashisth, A., Kowalik, M., Gerringer, J.C., Ashraf, C., van Duin, A.C.T., Green, M.J.: ReaxFF simulations of laser-induced graphene (LIG) formation for multifunctional polymer nanocomposites. ACS Appl. Nano Mater. **3**, 1881–1890 (2020)
53. Chaban, V.V., Pal, S., Prezhdo, O.V.: Laser-induced explosion of nitrated carbon nanotubes: nonadiabatic and reactive molecular dynamics simulations. J. Am. Chem. Soc. **138**, 15927–15934 (2016)
54. Ryu, G.H., Park, H.J., Ryou, J., Park, J., Lee, J., Kim, G., Shin, H.S., Bielawski, C.W., Ruoff, R.S., Hong, S., Lee, Z.: Atomic-scale dynamics of triangular hole growth in monolayer hexagonal boron nitride under electron irradiation. Nanoscale **7**, 10600–10605 (2015)
55. Reifsnyder Hickey, D., Yilmaz, D.E., Chubarov, M., Bachu, S., Choudhury, T.H., Miao, L., Qian, C., Redwing, J.M., van Duin, A.C.T., Alem, N.: Formation of metal vacancy arrays in coalesced WS2 monolayer films. 2D Mater. **8**, 011003 (2020)
56. Verma, A., Parashar, A.: Molecular dynamics based simulations to study failure morphology of hydroxyl and epoxide functionalised graphene. Comput. Mater. Sci. **143**, 15–26 (2018)
57. Kumar, R., Parashar, A.: Atomistic modeling of BN nanofillers for mechanical and thermal properties: a review, Nanoscale **8**, 22–49 (2016)
58. Verma, A., Parashar, A.: Reactive force field based atomistic simulations to study fracture toughness of bicrystalline graphene functionalised with oxide groups. Diamond Relat. Mater. **88**, 193–203 (2018)
59. Verma, A., Parashar, A., Packirisamy, M.: Tailoring the failure morphology of 2D bicrystalline graphene oxide. J. Appl. Phys. **124**(1), 015102 (2018)

60. Verma, A., Parashar, A.: Structural and chemical insights into thermal transport for strained functionalised graphene: a molecular dynamics study. Mater. Res. Express **5**(11), 115605 (2018)

Reinforcing Potential of 2D Nanofiller in Polyethylene: A Molecular Dynamics Approach



Ankur Chaurasia, Sandeep Kumar Singh, and Avinash Parashar

Abstract In this chapter, the authors have emphasized the reinforcing potential of the 2D nanofiller such as graphene and hexagonal boron nitride nanosheet in the polymer matrix nanocomposites using a molecular dynamics approach. The effect of interfacial and non-interfacial factors including state of dispersion, interface functionalization, concentration, and morphology of nanofiller on mechanical and interfacial properties of the nanocomposites are discussed in this chapter. Structural defects (such as single vacancy, double vacancy, stone–wales, and grain boundaries) and chemical modification with different functional groups in the 2D nanofiller significantly affect the load transfer capability of the interface, tensile strength, young's modulus, and failure strain of the nanocomposites. This chapter will help in understanding the underlying load transfer mechanism of the 2D nanofiller reinforced polyethylene nanocomposites.

1 Introduction

Two-dimensional (2D) nanofillers are emerging as potential nanofillers for reinforcing polymers and metallic systems. Nowadays these nanofillers are replacing the conventional nanofiller due to their peculiar properties such as superior specific strength, aspect ratio, high surface area, strong interfacial interaction, and compatibility [1]. 2D nanofillers are single-layer atomic structures with the atoms arranged in a honeycomb structural configuration [2, 3]. Graphene and h-BN are the two hottest 2D nanofillers in the current time for the development of a new class of polymeric nanocomposite [4]. Graphene is an allotrope of carbon with sp^2 hybridized carbon (C) atoms arranged in a honeycomb structure [5, 6] as shown in Fig. 1. In addition, h-BN is structurally analogous to graphene with boron and nitrogen

A. Chaurasia (✉) · S. K. Singh · A. Parashar
Department of Mechanical and Industrial Engineering, Indian Institute of Technology Roorkee, Roorkee 247667, India
e-mail: achaurasia@me.iitr.ac.in

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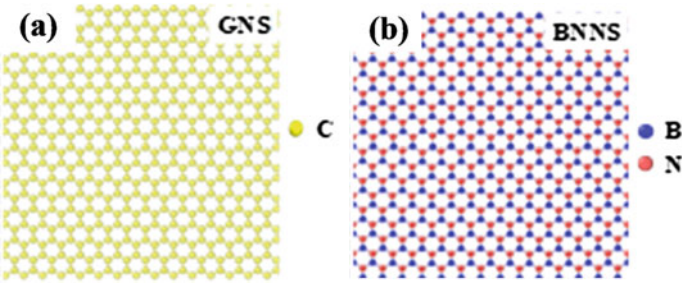


Fig. 1 Atomistic configuration of **a** graphene nanosheet **b** boron nitride nanosheet

atoms arranged in a honeycomb structure [7, 8]. Graphene and h-BN exhibit superior mechanical strength [9, 10], thermal conductivity [11, 12], tribology [13], flame retardancy [14, 15], optical (ultraviolet [UV] absorption and luminescence) properties [16–18], superior piezoelectricity [19], no cytotoxic [20], and electrochemical [21] properties.

Despite being the similar atomistic configuration of h-BN with graphene, h-BN exhibits many distinct properties such as wide bandgap (electrically insulative) [22], chemical resistant, resistance to oxidation at higher temperatures (>800° in air) [23], biological compatibility, and radiation shielding [18]. The disparity in the properties of the h-BN can be attributed to the partial ionic character of the covalent bond between boron and nitrogen in h-BN [1]. Quantitative comparison among graphene and h-BN nanofillers are summarized in Table 1.

Owing to these superior properties, graphene and h-BN are used in transparent electrodes [26], supercapacitor [27], hydrogen storage [28], thermal packaging [29], battery, lasers protection [30], gas barrier [31], nanosensors, optoelectronic application [32], solid-state devices, solar cells [33], nano actuators, clean energy devices, and biosensors [34]. In addition to this, graphene and h-BN are widely used as

Table 1 Quantitative comparison of properties of graphene and h-BN [24, 25]

Properties	Graphene	h-BN
Fracture stress (GPa)	130	165
Thermal conductivity (W/m K)	5000	1800
Color	Black	White
Bond length (Å)	1.42	1.44
Band Gap (eV)	0	5–6
Mode I Fracture Toughness (MPa $\sqrt{m}$)	4	5.5
Young's Modulus (GPa)	1	0.8
Specific surface area (m^2/g)	2630	1900
Thermal stability (°C)	500	1000
Shear Modulus (GPa)	280	112

potential filler materials to enhance the mechanical and thermal performance of the polymer and metal nanocomposites [35]. Apart from these applications, hexagonal boron nitride nanosheets are used in electronic packaging and the development of neutron shielding and corrosion-resistant nanocomposites [18].

The synthesis of these 2D nanofillers/nanosheets are mainly classified by chemical and physical methods. The graphene is synthesized by chemical vapor deposition (CVD), epitaxial growth of 2D nanofiller on substrate material, arc discharge, ion exchange, mechanical peeling, and molecular beam epitaxy [36]. On the other hand, h-BN nanosheet is synthesized in the range of temperature, i.e., high-, medium-, and low-temperature range. Laser ablation, arc discharge, plasma jet techniques, and ion beam irradiation methods are referred to as high-temperature synthesis techniques, whereas CVD, micromechanical exfoliation, ball milling, sonication method, chemical route of boric acid and urea, autoclave, chemical blowing method (CBM), carbothermal method, metal-catalyst-free approach, and mechano-thermal method are referred to as medium and low-temperature synthesis techniques [37]. However, these synthesis techniques inevitably introduce the various kind of defects in the 2D nanosheets that includes the of mono and di vacancies, Stone-Thrower-Wales (STW), dislocations, grain boundary (GB) [38]. The snapshot of these defects are shown in the Fig. 2. STW defects form due to rotation of central carbon–carbon (C–C) bond in graphene and boron-nitrogen (B-N) bond in h-BN nanosheet. The rotation of the central bond yield two pentagon-heptagon pair (5–7–7–5). The grain boundary defects form where two crystal growth front propagate and meet each other. The grain boundary consists of a regular array of pentagon-heptagon pair (5|7) or square-octagon (4|8) dislocation pair as illustrated in the Fig. 2.

Despite the defects in graphene and h-BN nanosheet, it is considered as potential nanofiller for reinforcing the polymer nanocomposites. Superior intrinsic strength and monoatomic structure and smooth surface of graphene and h-BN nanosheet make it a suitable candidate as a potential nanofiller in various types of polymers [1, 39]. Polymer exhibits excellent properties such as easy processability, excellent interfacial compatibility, resistance to chemical, ductility, and economically cheaper [5]. The extraordinary properties of polymer make it a suitable candidate to be used as a matrix in nanocomposites. However, the lower density of polymer and poor mechanical properties, as well as thermally nonconductive behavior of polymer, limit its application in the nanocomposites [40]. In order to improve mechanical and thermal performance of the polymer, 2D nanofiller such as graphene and h-BN

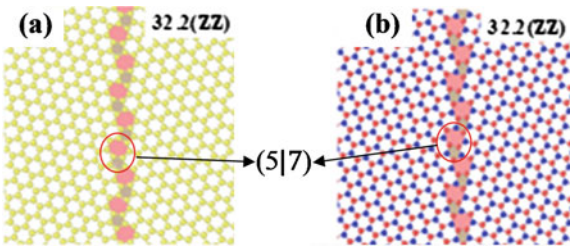


Fig. 2 Atomistic configuration of **a** bi-crystalline graphene and **b** BNNS nanosheet

play an important role. A few techniques to synthesize polymer nanocomposites are thermoforming, in-situ polymerization, solution blending method, three roll shear mixing, and co-extrusion [8]. The key issue of nanocomposites is the agglomeration of nanofiller in the polymer matrix, this can be attributed to the planar structure of the graphene and h-BN. The higher surface energy and π-π interlayer interaction force them to agglomerate. In order to overcome agglomeration of 2D nanofiller, surface functionalization of graphene and h-BN with different functional group was attempted by many researchers [41–43].

Numerous works have been performed to access the mechanical and thermal properties of 2D reinforced polymer nanocomposites using molecular dynamics approach. Although, the number of articles are available on characterizing the mechanical and thermal properties in conjunction with the molecular dynamics approach, as such, no compilation of the conclusive study was performed on effect of 2D nanofiller (graphene and h-BN) on mechanical and thermal properties of polymer nanocomposites using molecular dynamics approach. In this chapter, most of the molecular dynamics-based studies are compiled to illustrate the controlling parameter of mechanical and thermal properties of 2D reinforced polymer nanocomposites. All the studies compiled in the present chapter were performed using computationally less intensive classical mechanics-based MD simulations in conjunction with different interatomic force field potentials such as reactive force field (ReaxFF), adaptive interatomic reactive bond order (AIREBO), and Tersoff potential.

2 Modeling of Nanocomposites Using MD Approach

In the last couple of years, a significant work has been performed on characterizing the graphene and h-BN reinforced polymer nanocomposites using classical mechanics-based MD approach. Molecular dynamics is semi empirical futuristic approach for modeling 2D nanofiller and various thermosetting and thermoplastic polymers at atomistic configuration that includes few thousands of atoms. Classical mechanics-based MD simulation is many body systems that includes the movement of atom/molecules in the confined space. In MD-based approach, atoms are considered as a hard sphere and interatomic forces are triggered with the help of interatomic potentials [44, 45]. In order to update the position and velocity of the atoms in the MD simulation, Newton's second law of motion is utilized. The governing equation of Newton's law of motion in the MD simulation is as follows:

$$m_\alpha a_\alpha = -\left(\frac{\partial E}{\partial r_\alpha}\right) = F_\alpha \tag{1}$$

where, $\alpha = 1, 2, 3 \ldots N$.

Here, m_α is the mass of corresponding atom α; F_α is the force on atom α; r_α is the position of atom α, E is the total potential energy of the atomic system.

Time and position of the atoms are updated after each timestep, different integration scheme such as Velocity Verlet algorithm, Leapfrog algorithm, and Beeman's algorithm are used to update the time and position of the atoms. Velocity verlet algorithm is the most efficient and widely used integration scheme nowadays [46]. To run the MD simulation, ensembles are the prior requirement and it defines the different microstates of the system. Few of the most important widely used ensembles are NVE (constant number of atoms, volume, and energy), isobaric–isothermal NPT (constant number of atoms, pressure, and temperature), NVT (constant number of atoms, volume, and temperature), isenthalpic NPH (constant number of atoms, pressure, and enthalpy). In addition to this, NσT (stress loading at constant temperature) and NσE (finite strain) are also used for some applications. NσT is used for applying stress at constant temperature and N σ E is used to define the fill stress tensor in solids.

In order to capture the interatomic interaction between atoms, force field potentials are used in MD simulation. Interatomic potential/force field potential is the backbone of the MD simulation and the accuracy and fidelity of the MD simulation significantly depends on the interatomic potentials. Interatomic potential generally describes the total potential energy of the atomic system. In order to calculate of the total energy (U_{TOTAL}) of the system, the sum of all individual energy contribution over the all atoms in the atomic system and is expressed in Eq. 2

$$U_{TOTAL} = 1/2 \sum_{\alpha=\beta=1}^{N} \sum_{\beta=1}^{N} \varnothing(r_{\alpha\beta}) \tag{2}$$

where U_{TOTAL} is the total energy of the system, $r_{\alpha\beta}$ is the distance between the α *and* β atom, $\varnothing$ is the potential energy between the atom α *and* β. Interatomic potential is developed with the help of experiments and computationally intensive density functional theory (DFT).

2.1 Force Field Potentials for Bonded and Non-bonded Interaction in Polymer Nanocomposites

Interatomic potentials are derived from the experimental data fitting or DFT simulations. In order to model nanocomposites, bonded interaction in nanofiller and polymer matrix is triggered with different interatomic potentials. Many force field potentials are available to model the 2D nanofiller such as graphene and h-BN. Graphene-based nanofiller are modeled with the help of AMBER, Tersoff, CHARMM (Chemistry at Harvard Macromolecular Mechanics), reactive bond order (REBO) [47], adaptive interatomic reactive bond order (AIREBO) [48], and interatomic force field potentials. Among these potentials, AMBER [49] and CHARMM [50] interatomic potentials are not much more accurate and does not encounter the bond breaking and formation during the simulation [37].

However, reactive bond order-based potential such as reactive force field potential (ReaxFF) [51, 52] and AIREBO potentials are much more accurate and easily capture the bond breaking and formation. These potentials are considered as most accurate potential to capture the mechanical and fracture behavior of the carbon based nanofiller. The details of the REBO and AIREBO potentials are expressed in the Eq. 3 and Eq. 4, respectively.

$$E_{AIREBO} = \frac{1}{2}\sum_{i}\sum_{j\neq i}\left[E_{ij}^{REBO} + E_{ij}^{LJ} + \sum_{k\neq i}\sum_{l\neq i,j,k} E_{kijl}^{torsion}\right] \tag{3}$$

$$E_{ij}^{REBO} = f(r_{ij})(E_{ij}^{Repulsion} + b_{ij}E_{ij}^{Attraction}) \tag{4}$$

Here, E is the potential energy, E_{ij}^{LJ} and $E_{kijl}^{torsion}$ are the non-bonded interaction energy and four body torsional term, respectively; $E_{ij}^{Repulsion}$ and $E_{ij}^{Attraction}$ are the repulsion and attraction energy contribution, respectively; $f(r_{ij})$ is the cut off function that defines the bonded interaction between neighboring atoms within the cut off radius.

Tersoff interatomic potential offers easy parameterization and reliable results close to the experimental and density functional values [53]. Tersoff potential is widely used for characterizing mechanical and thermal behavior of graphene and h-BN [54]. The mathematical expression of Tersoff potential energy contribution are expressed in Eqs. 5–8.

$$E = \sum_{i} E_i = \left(\sum_{i\neq j} V_{ij}\right)/2 \tag{5}$$

$$V_{ij} = f_c\left(r_{ij}\right)\left[a_{ij}f_R\left(r_{ij}\right) + b_{ij}f_A\left(r_{ij}\right)\right] \tag{6}$$

$$f_c\left(r_{ij}\right) = \begin{cases} 1, r_{ij} < R_{ij} \\ \left(\frac{1}{2} + \frac{1}{2}\cos\left(\frac{\pi\left(r_{ij}-R\right)}{S_{ij}-R_{ij}}\right)\right), R_{ij} < r_{ij} < S_{ij} \\ 0, r_{ij} > R_{ij} \end{cases} \tag{7}$$

where E is the total energy of the system and it comprises of E_i and V_{ij} which represents the site energy and bond energy, respectively; f_A and f_R is the attraction and repulsion forces between atom, i, and j is the atoms in the system; r_{ij} is the radial distance between atom i, and j; a_{ij} is the range limiting range; b_{ij} is the bond order functions, and f_c is the cut off function. Tersoff potential recently employed by the Kumar and Parashar [55] to investigate the crack healing phenomenon in preexisted defected BNNS. Later on, original Tersoff potential was modified by Brenner [56].

Apart from non-reactive potentials, reactive force field potential (ReaxFF) is also used for modeling nanofiller and matrix developed by research group of Adri van

Duin [51]. Among all potentials discussed, ReaxFF potentials is assumed as most accurate and stiffer. At beginning stages, ReaxFF was used for modeling hydrocarbons, thereafter research fraternity have extended its horizon graphene, h-BN based nanofillers [57–60]. In ReaxFF potential, total energy of the system can be calculated from several energy contribution term and are expressed in Eq. 8

$$E_{sys} = E_{BONDING} + E_{OVER} + E_{UNDER} + E_{VAL} + E_{TORS} + E_{CONJ} + E_{VDW} + E_{CO} + E_{PEN} \tag{8}$$

where E_{sys} is the system energy; E_{CO} is the coulomb energy; $E_{BONDING}$ is the energy triggered to the bonded interaction, E_{OVER} and E_{UNDER} is the over and under coordination number; E_{VAL} is the valance angle, E_{PEN} is the energy related to penalty; E_{CONJ} is the conjugate energy; E_{VDW} is the Van der Walls interaction. The reactive force field potential is most accurate and high fidelity potential. Unlike other potentials, ReaxFF potential is well capable to encounter the bond breaking and bond formation in the nanofillers as well as polymer matrix. Bond order play an important role in the association and dissociation of the bond during the atomistic simulation.

In addition to the available force field for the modeling of nanofiller, various force fields have been used previously to model polymer matrix. Most commonly used force field potentials for polymer are COMPASS (condensed phase optimized molecular potential for atomistic simulation studies), PCFF (Polymer consistent force field), CHARMM, AIREBO, etc. PCFF and COMPASS are widely used potentials for the polymer matrix. COMPASS force field potential is developed with the help of ab initio and empirical parameterization. The total energy calculation (E_{sys}) in COMPASS potential are derived from the following Eq. 9

$$E_{sys} = E_{Valance} + E_{Nonbonded} \tag{9}$$

where $E_{Valance}$ includes the interaction of bond angle, torsion angle, etc., whereas, $E_{Nonbonded}$ comprises of van der Wall interaction and electrostatic non-bonded interaction.

In molecular dynamics simulation, the mechanical properties such as young's modulus, yield strength, and fracture stresses are calculated with the help of strain equation (Eq. 10) and virial stress equation (Eq. 11) as follows:

$$\epsilon = \frac{L_{Final} - L_{Initial}}{L_{Initial}} \tag{10}$$

$$\sigma_{ij} = (1/\varphi_{\alpha})\left(\frac{1}{2}m^{\alpha}v_i^{\alpha}v_j^{\alpha} + \sum_{\beta=1,n} r_{\alpha\beta}^{i} f_{\alpha\beta}^{i}\right) \tag{11}$$

where ϵ is the strain and $L_{Initial}$ and L_{Final} is the initial and final length of the simulation box in the MD simulation. The virial stress equation is defined in Eq. 11,

which constitutes two components, one referred to the masa and velocity of the atom and second one referred to the position and interatomic forces of the atoms. σ_{ij} is the virial stress; i and j is the indices in Cartesian coordinates and m^{α} is the mass of atom α and $r^{i}_{\alpha\beta}$ is the interatomic distance between the atom α *and* β atoms.

3 Mechanical Response of 2D Reinforced Polymer Nanocomposites

In this section, recent works on exploring mechanical properties of the graphene and h-BN reinforced polymer nanocomposites using molecular dynamics approach is discussed in detail.

3.1 Graphene-Based Polyethylene Nanocomposites

In this subsection, the effect of the graphene nanosheet on mechanical properties of polymer nanocomposite is comprehensively discussed. The works included here are solely based on continuum-based molecular dynamics approach. The effect of various interfacial factors such as covalent and non-covalent interaction between nanofiller and matrix (via functionalization), interphase, interfacial surface area, state of dispersion, and non-interfacial factors such as mechanical interlocking, interface spacing, and process parameters on mechanical properties of the polymer nanocomposites is exclusively discussed.

Several researchers have performed computational studies on the mechanical behavior of the graphene reinforced polymer nanocomposites using molecular dynamics approach (Cite). Li et al. [61] studied the effect of most common defects such as single vacancy, double vacancy, and Stone–Wales on the mechanical behavior of the graphene reinforced polyethylene nanocomposites. Single and double vacancy adversely affect the interfacial strength of the nanocomposite, whereas Stone–Wales defects in the graphene nanosheet increases the interfacial interaction between polymer and nanosheet, and hence enhance the interfacial strength of the nanocomposites. Verma and Parashar [62] employed the molecular dynamics approach to study the effect of bi-crystalline grain boundary (GB) on interfacial strength of the graphene reinforced polyethylene (PE) nanocomposites. Total six GBs misorientation angles were considered, three are in armchair direction (17.9°, 21.8°, and 27.8°) and the other three are in the zigzag direction (9.43°, 13.2°, and 21.8°). The stress–strain obtained from the all configuration of nanocomposites simulation are provided in Fig. 3.

In addition to the defective graphene nanosheet, functionalized graphene nanosheets have also been embedded in the PE matrix and their mechanical response was accessed using MD approach. Bačová et al. [63] functionalize the graphene

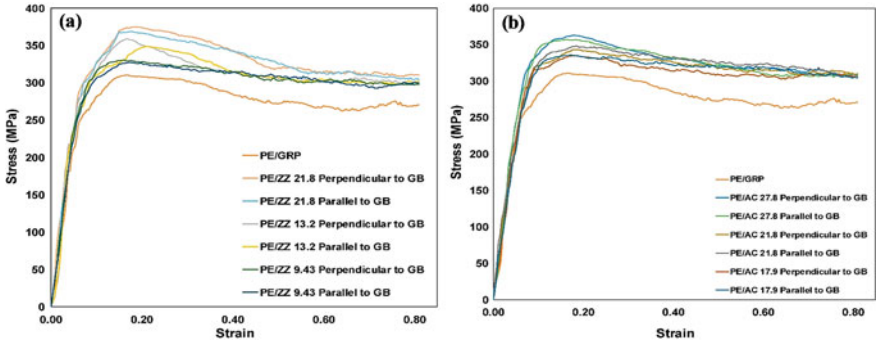


Fig. 3 Stress–strain response in bi-crystalline graphene nanosheet (GBs oriented in armchair and zigzag direction with different GB misorientation angle) (reproduced with permission from Ref. [62])

nanosheet with carboxyl (−COOH) and hydrogen (−H) groups at the edges of the nanosheet and embedded in PE (non-polar) and polyethylene oxide (polar) matrices. Static and dynamic properties of polymer chains were studied in bulk and interface region, and revealed the behavior of the polymer matrix dynamically affected by the edge functionalized graphene. In order to enhance the interfacial adhesion PE and graphene, Nikkhah et al. [64] introduces the various functional group such as ethylamino, isocynato, carboxyl, amino, and hydroxyl at the PE surface with different surface densities. The concept of work separation was used to quantify the interfacial adhesion between graphene and functionalized PE. They revealed that, the increasing grafting density of the functional group at the PE surface enhanced the interfacial adhesion that may be attributed due to the low surface roughness, and enhanced the atomic density at the interface surface as shown in Fig. 4.

Later on, this work was extended to functionalize the graphene nanosheet with different functional groups such as amino, carboxyl, methylene, and hydroxyl, and their interaction with the PE matrix was captured [65]. It can be concluded from Fig. 5 that the amine group has the highest interaction at the interface, thus results in the retardation of chain movement at the interface, making them difficult to move.

Awasthi et al. [66] elaborated the load transfer mechanism between the graphene sheet and polyethylene matrix using Consistent valance force field (CVFF) potentials under various constraints as shown in Fig. 6a. Separation studies were carried out for both opening and sliding modes, and peak traction and energy of separation were estimated for each separation mode condition. It can be observed from the force–displacement curve in Fig. 6b that the maximum interaction can occur when the PE chains were fixed near the interface. During the graphene sheet pulling process, PE chains get adhered with the sheet due to vdW interaction and separation was occurred in bulk PE itself, not at the interface.

Rahman and Foster [67] modeled the dispersed graphene nanosheet in the amorphous PE matrix using AIREBO and PCFF potentials. They captured the effect of

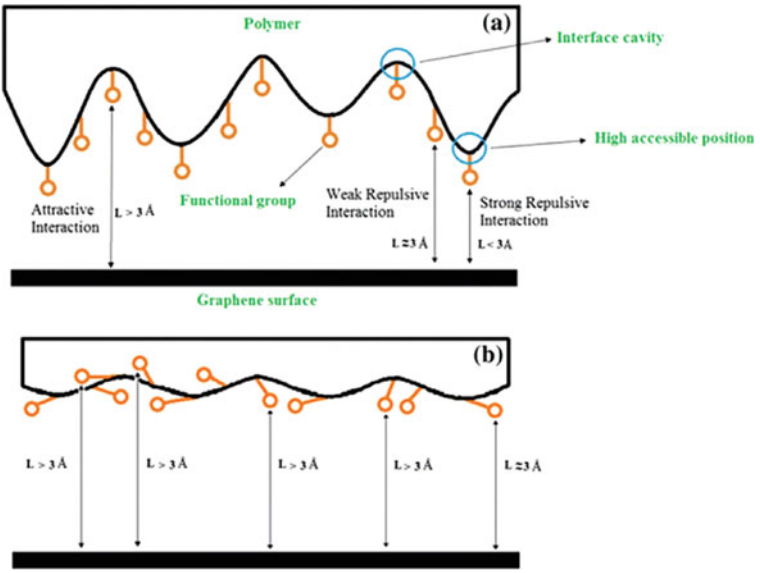


Fig. 4 Schematic representation of the functionalized PE at the surface with functional groups and their interaction with graphene nanosheet **a** unrelaxed state and, **b** relaxed state (attractive force of vdW > 3 Å and repulsive force of vdW < 3 Å (reproduced with permission from Ref. [64])

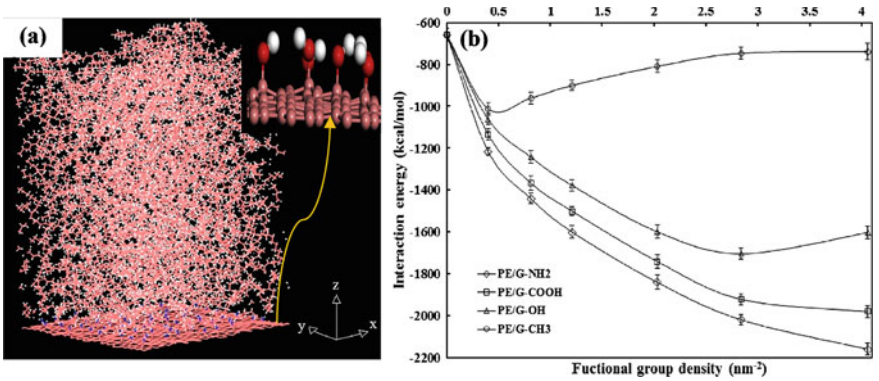


Fig. 5 **a** Schematic representation of the functionalized graphene with different functional groups and their interaction with PE matrix **b** interaction energy as a function of grafting density functional group (reproduced with permission from Ref. [65])

dispersion of graphene on the young's modulus and yield strength of the nanocomposites. Randomly oriented graphene nanosheet with superior local interaction exhibits high stiffness and tensile strength of the nanocomposites. On the way to the plastic deformation, randomly oriented graphene nanosheets were firstly pulled out from

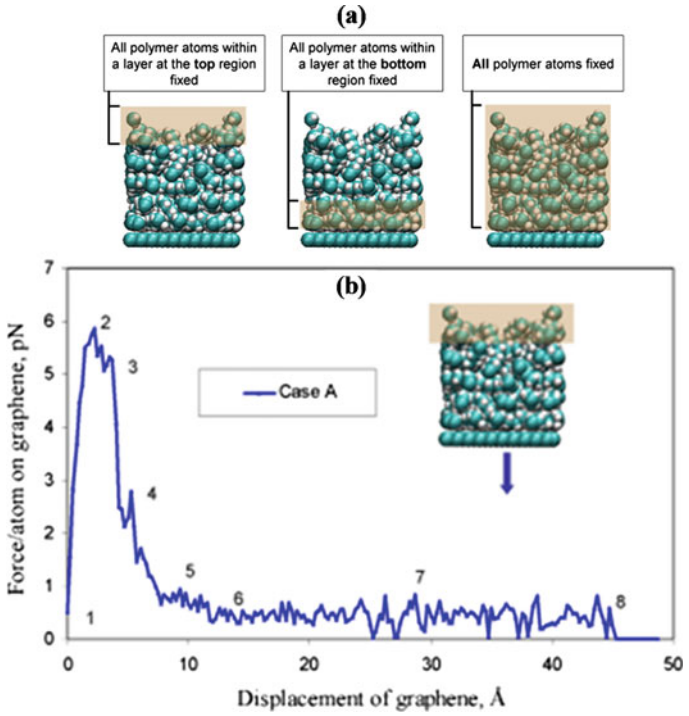


Fig. 6 **a** Schematic representation of the separation process of graphene–PE interface with different constraints **b** force–displacement curve at the graphene–PE interface (reproduced with permission from Ref. [66])

the PE matrix. In the earlier work performed by Liu et al. [68], the effect of interfacial and non-interfacial parameters such as polymer chain length, pulling velocity, and wrinkled graphene on the interfacial strength of polyethylene and poly-methyl methacrylate nanocomposites reinforced with graphene nanofiller. They applied the MD approach in conjunction with PCFF interatomic potential. It can be observed in Fig. 7 that the wrinkling of graphene and pulling velocity notably affect the interfacial shear strength and interaction energy of the graphene reinforced PE nanocomposites.

3.2 Hexagonal Boron Nitride-Based Polyethylene Nanocomposites

In the earlier section, the authors have discussed primarily the effect of interfacial and non-interfacial factors on the mechanical and interfacial properties of the graphene reinforced polymer nanocomposites using the MD approach. Owing to the analogous

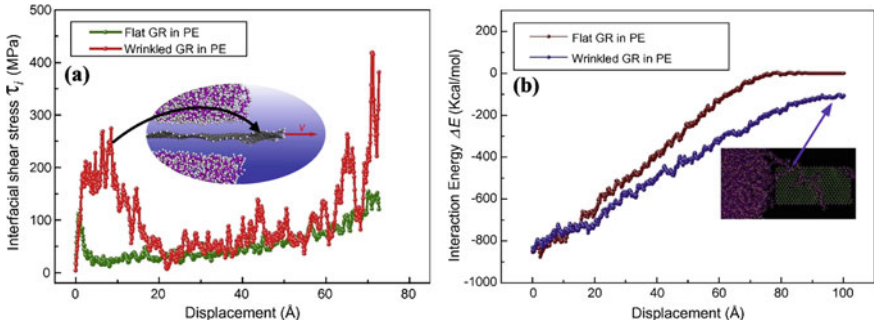


Fig. 7 Effect of wrinkled graphene on **a** interfacial shear stress and **b** interfacial energy in graphene reinforced PE nanocomposites (reproduced with permission from Ref. [68])

structure of graphene, h-BN nanosheet is also a promising 2D nanofiller for the development of polymer nanocomposites. Apart from the numerous experimental works on h-BN-based nanocomposites performed by the researchers, very few attempts have been made on the h-BN-based nanocomposites using the molecular dynamics approach. In order to make it potentially viable for the future requirement, h-BN can be functionalized through strong π-π interaction, cation-π interaction, and highly reactive sites of boron (nucleophilic) and nitrogen (electrophilic) atoms at the edges. The partial ionic nature of B-N bonds and high bonding potential of boron and nitrogen atoms in h-BN strongly participate toward the strong compatibility with the polymer.

Kumar et al. [69] performed MD simulation in conjunction with the Reaxff potentials to study the effect of different geometrical defects and functionalization of h-BN on the mechanical properties of the h-BN reinforced polyethylene nanocomposites. It was predicted from the simulation that, specific defects such as single vacancy, double vacancy, and STW significantly enhance the tensile strength of the nanocomposites, whereas hydrogenated h-BN has a negligible effect on the mechanical response of the polymer nanocomposites as shown in Fig. 8. Defects sites of STW have the homo-elemental bond such as B–B and N–N, which are at higher energy states and thus causing the enhanced interfacial interaction between defective h-BN and PE matrix.

They also reported the underlying mechanism behind the enhancement in tensile strength by estimating the interfacial shear strength of each atomistic model. The interfacial shear stress of the h-BN/PE nanocomposites is shown in Fig. 9. It can be observed from Fig. 9b that interfacial shear stress is higher for the defected nanosheet as compared to the pristine h-BN/PE nanocomposites.

Meng et al. [70] employed the combined experimental and computational approach to study the various exfoliation mechanism of the h-BN nanosheet in the PE matrix as displayed in Fig. 10. MD approach was employed to predict the essential work and force required for exfoliation of h-BN nanosheet from the PE matrix. They predicted that a large interfacial area between PE and h-BN is required to transfer the

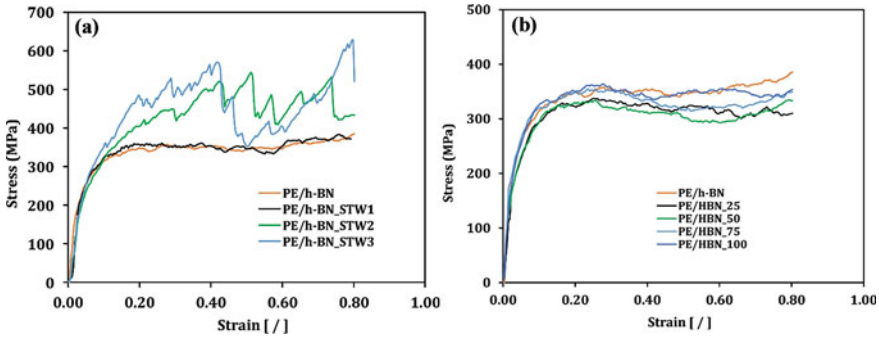


Fig. 8 Stress–strain response of **a** STW defected h-BN/PE nanocomposites **b** hydrogenated h-BN/PE nanocomposites at various degrees of grafting (reproduced with permission from Ref. [69])

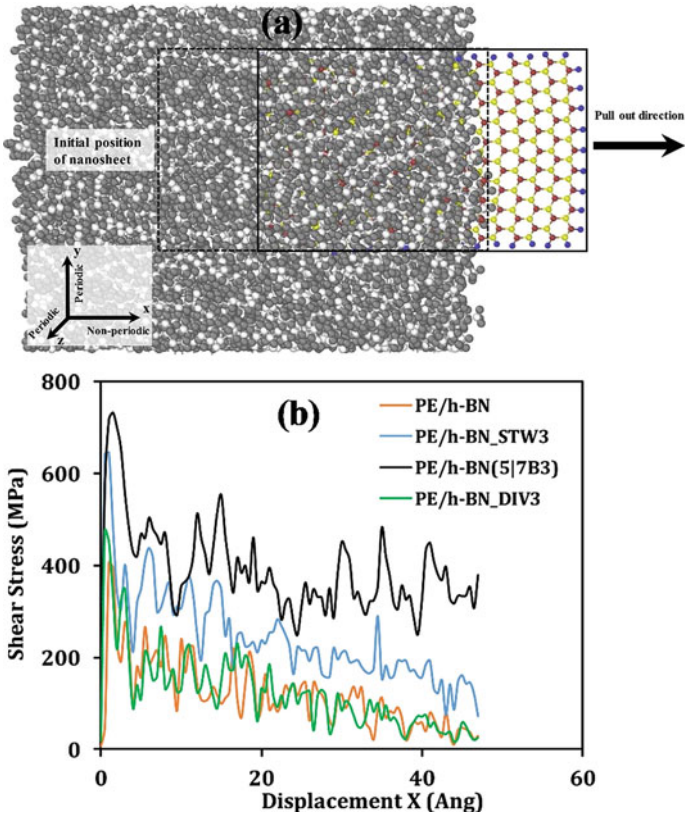


Fig. 9 **a** Schematic of pulling off the defected h-BN nanosheet from the PE matrix **b** Interfacial shear stress of the defected h-BN reinforced PE nanocomposites (reproduced with permission from Ref. [69])

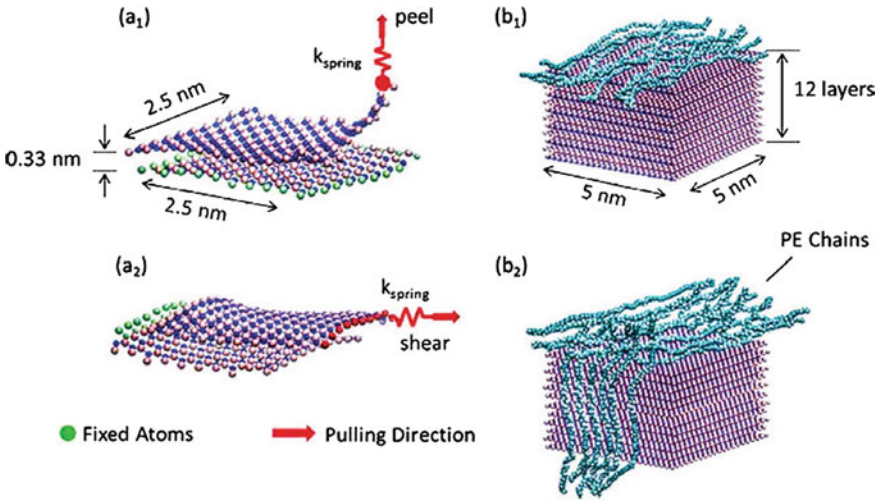


Fig. 10 Initial configurations of h-BN adhesion study for **a1** peeling and **a2** shearing conditions using SMD method; as well as the initial PE chains arrangement for studying polymer crystallization behavior **b1** on top or **b2** on both top and side of the h-BN stacking (reproduced with permission from Ref. [70])

stress from matrix to h-BN nanofiller and to exceed the threshold limit of exfoliation energy/force. Once the force required exceeds the threshold barrier, the successful exfoliation of h-BN platelets to a monolayer nanosheet is achieved.

Chaurasia et al. [36] studied the effect of different morphology, aspect ratio, concentration, and strain rate loading on the mechanical properties of the nanocomposites using Reaxff potentials. The bonded interaction of the PE matrix and h-BN nanofiller was modeled with the help of the Reaxff potentials, whereas the non-bonded interaction between the PE matrix and h-BN nanosheet was captured with the help of the Lennard-Joes potential. The effect of aspect ratio, weight concentration, and strain rate loading on the mechanical properties of the nanocomposites are shown in Fig. 11. It was concluded from Fig. 11 that the tensile strength of the nanocomposites increases with the aspect ratio, weight concentration, and strain rate loading.

It was predicted also from the simulation that the morphology of the h-BN nanosheet significantly affects the mechanical properties of the h-BN/PE nanocomposites. The maximum tensile strength of the nanocomposites was reported for rectangular h-BN nanosheets as shown in Fig. 12. The superior tensile strength of the rectangular h-BN nanosheet is attributed to the higher interactive surface area causing enhanced load transfer capability.

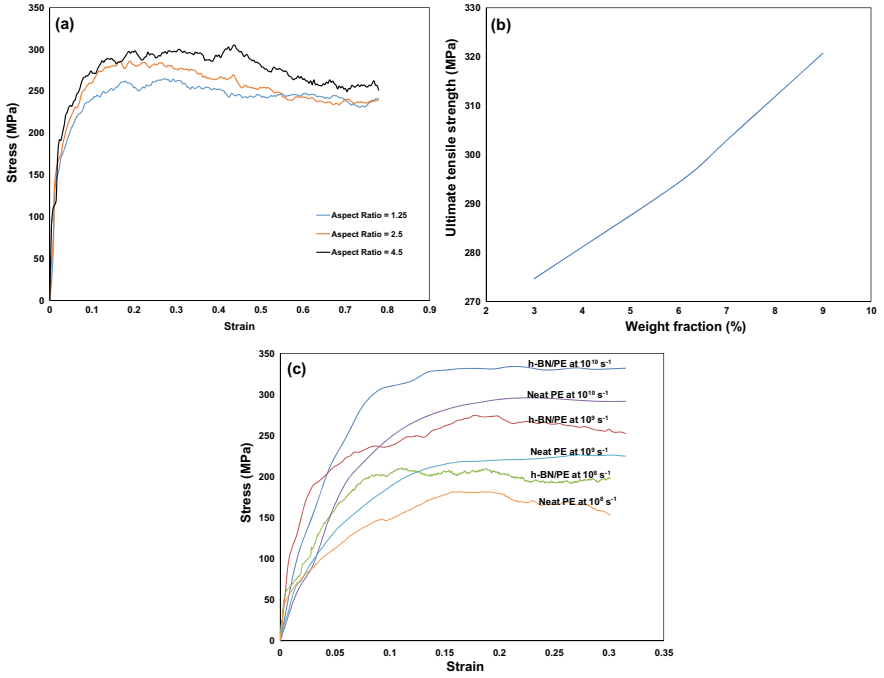


Fig. 11 Stress–strain response of the h-BN/PE nanocomposites with respect to **a** aspect ratio of h-BN nanosheet, **b** weight concentration of h-BN nanosheet, and **c** strain rate loading

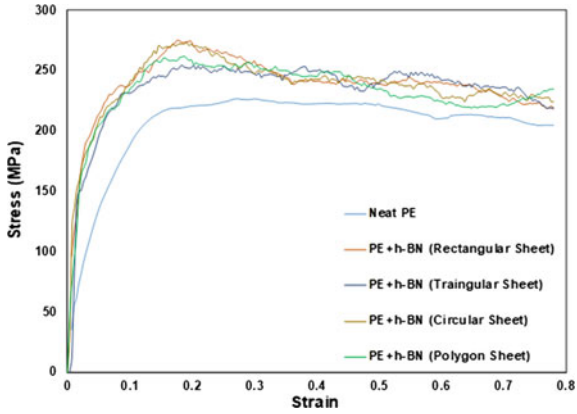


Fig. 12 Stress–strain response of the h-BN/PE nanocomposites with respect to different morphology nanosheets

4 Conclusions

In this chapter, the authors have compiled and comprehensively discussed the reinforcing potential of the 2D nanofiller (e.g., graphene and h-BN) in the polyethylene

nanocomposites. Geometrical defects and chemical modification with different functional groups were employed by many researchers to tailor the mechanical properties of the nanocomposites including tensile strength, young's modulus, failure strain, and interfacial shear strength. All the discussion reported in this chapter is solely based on classical mechanics-based molecular dynamics approach [71–73]. It can be concluded from the chapter that structural defects such as single vacancy, double vacancy, STW, and grain boundary in 2D nanofiller significantly affect the mechanical properties of the polymer nanocomposites. Defected nanosheets have high energy state atoms at the defected sites which helps in improving the interfacial interaction between the nanofiller and polymer matrix. It can also be concluded that various interfacial (e.g., state of dispersion, interface functionalization) and non-interfacial factors notably affect the interfacial/mechanical properties of the nanocomposites. In addition to this, the effect of different morphology of the nanosheet on the mechanical properties of the nanocomposites is also discussed in this chapter. This chapter will help in conceptualizing the design guidelines for the development of the 2D reinforced polyethylene nanocomposites.

References

1. Rasul, M.G., Kiziltas, A., Arfaei, B., Shahbazian-Yassar, R.: 2D boron nitride nanosheets for polymer composite materials. npj 2D Mater. Appl. **5** (2021)
2. Geim, A.K., Novoselov, K.S.: The rise of graphene. Nat. Mater. **6**, 183–191 (2007)
3. Verma, A., Parashar, A.: The effect of STW defects on the mechanical properties and fracture toughness of pristine and hydrogenated graphene. Phys. Chem. Chem. Phys. **19**, 16023–16037 (2017)
4. Rajasekaran, G., Narayanan, P., Parashar, A.: Effect of point and line defects on mechanical and thermal properties of graphene: a review. Crit. Rev. Solid State Mater. Sci. **41**, 47–71 (2016)
5. Verma, A., Kumar, R., Parashar, A.: Enhanced thermal transport across a bi-crystalline graphene–polymer interface: an atomistic approach. Phys. Chem. Chem. Phys. **21**, 6229–6237 (2019)
6. Verma, P.K., Sharma, B.B., Chaurasia, A., Parashar, A.: Inter-granular fracture toughness of bi-crystalline graphene nanosheets. Diam. Relat. Mater. **102**, 107667 (2020)
7. Coleman, J.N., Lotya, M., Neill, A.O., Bergin, S.D., King, P.J., Khan, U., Young, K., Gaucher, A., De, S., Smith, R.J., Shvets, I.V., Arora, S.K., Stanton, G., Kim, H., Lee, K., Kim, G.T., Duesberg, G.S., Hallam, T., Boland, J.J., Wang, J.J., Donegan, J.F., Grunlan, J.C., Moriarty, G., Shmeliov, A., Nicholls, R.J., Perkins, J.M., Grieveson, E.M., Theuwissen, K., Mccomb, D.W., Nellist, P.D., Nicolosi, V.: Produced by liquid exfoliation of layered materials. Science (80-) **331**, 568–571 (2011)
8. Chaurasia, A., Parashar, A., Mulik, R.S.: Effect of hexagonal boron nitride nanoplatelet on crystal nucleation, mechanical behavior, and thermal stability of high-density polyethylene-based nanocomposites. Macromol. Mater. Eng. **305**, 2000248 (2020)
9. Chang, I.-L., Chen, J.-A.: The molecular mechanics study on mechanical properties of graphene and graphite. Appl. Phys. A **119**, 265–274 (2015)
10. Chaurasia, A., Parashar, A., Mulik, R.S.: Evaluation of interfacial shear strength of h-BN/PE nanocomposites using molecular dynamics. In: Advances in Engineering Design, pp. 105–13. Springer, Singapore (2021)

11. Chaurasia, A., Parashar, A.: Experimental and atomistic insight on the thermal transport properties of h-BN/high density polyethylene nanocomposite. Int. J. Heat Mass Transf. **170**, 121039 (2021)
12. Verma, A., Parashar, A.: Structural and chemical insights into thermal transport for strained functionalised graphene: a molecular dynamics study. Mater. Res. Express **5**, 115605 (2018)
13. Min, C., He, Z., Liang, H., Liu, D., Dong, C., Song, H., Huang, Y.: High mechanical and tribological performance of polyimide nanocomposite reinforced by fluorinated graphene oxide. Polym. Compos. **41**, 1624–1635 (2020)
14. Huang, G., Chen, S., Tang, S., Gao, J.: A novel intumescent flame retardant-functionalized graphene: nanocomposite synthesis, characterization, and flammability properties. Mater. Chem. Phys. **135**, 938–947 (2012)
15. Verma, A., Parashar, A.: Molecular dynamics based simulations to study failure morphology of hydroxyl and epoxide functionalised graphene. Comput. Mater. Sci. **143**, 15–26 (2018)
16. Kubota, Y., Watanabe, K., Tsuda, O., Taniguchi, T.: Deep ultraviolet light-emitting hexagonal boron nitride synthesized at atmospheric pressure. Science (80-) **317**, 932–934 (2007)
17. Verma, A., Parashar, A., Packirisamy, M.: Tailoring the failure morphology of 2D bicrystalline graphene oxide. J. Appl. Phys. **124**, 015102 (2018)
18. Kumar, R., Parashar, A., Mertiny, P.: Displacement thresholds and knock-on cross sections for hydrogenated h-BN monolayers. Comput. Mater. Sci. **142**, 82–88 (2018)
19. Chaurasia, A., Parashar, A.: Molecular dynamics study of anisotropic shock response in mono- and bicrystalline boron nitride nanosheets: implications for shock-resistant solid-state devices. ACS Appl. Nano Mater. **5**, 2787–2800 (2022)
20. Chen, Y., Qi, Y., Tai, Z., Yan, X., Zhu, F., Xue, Q.: Preparation, mechanical properties and biocompatibility of graphene oxide/ultrahigh molecular weight polyethylene composites. Eur. Polym. J. **48**, 1026–1033 (2012)
21. Parveen, N., Mahato, N., Ansari, M.O., Cho, M.H.: Enhanced electrochemical behavior and hydrophobicity of crystalline polyaniline@graphene nanocomposite synthesized at elevated temperature. Compos. Part B Eng. **87**, 281–290 (2016)
22. Gibb, A.L., Alem, N., Chen, J.H., Erickson, K.J., Ciston, J., Gautam, A., Linck, M., Zettl, A.: Atomic resolution imaging of grain boundary defects in monolayer chemical vapor deposition-grown hexagonal boron nitride. J. Am. Chem. Soc. **135**, 6758–6761 (2013)
23. Li, L.H., Cervenka, J., Watanabe, K., Taniguchi, T., Chen, Y.: Strong oxidation resistance of atomically thin boron nitride nanosheets. ACS Nano **8**, 1457–1462 (2014)
24. Sharma, B.B., Parashar, A.: A review on thermo-mechanical properties of bi-crystalline and polycrystalline 2D nanomaterials. Crit. Rev. Solid State Mater. Sci. **45**, 134–170 (2020)
25. Verma, A., Parashar, A.: Molecular dynamics based simulations to study the fracture strength of monolayer graphene oxide. Nanotechnology **29**, 115706 (2018)
26. Singh, S.K., Parashar, A.: Effect of lattice distortion and nanovoids on the shock compression behavior of (Co-Cr-Cu-Fe-Ni) high entropy alloy. Comput. Mater. Sci. **209**, 111402 (2022)
27. Sun, Y., Wu, Q., Shi, G.: Graphene based new energy materials. Energy Environ. Sci. **4**, 1113 (2011)
28. Kumar, R., Mertiny, P., Parashar, A.: Effects of different hydrogenation regimes on mechanical properties of h-BN: a reactive force field study. J. Phys. Chem. C **120**, 21932–21938 (2016)
29. Park, O.K., Owuor, P.S., Jaques, Y.M., Galvao, D.S., Kim, N.H., Lee, J.H., Tiwary, C.S., Ajayan, P.M.: Hexagonal boron nitride-carbon nanotube hybrid network structure for enhanced thermal, mechanical and electrical properties of polyimide nanocomposites. Compos. Sci. Technol. **188**, 107977 (2020)
30. Wang, J., Chen, Y., Li, R., Dong, H., Ju, Y., He, J., Fan, J., Wang, K., Liao, K.-S., Zhang, L., Curran, S.A., Blau, W.J.: Graphene and carbon nanotube polymer composites for laser protection. J. Inorg. Organomet. Polym. Mater. **21**, 736–746 (2011)
31. Chaurasia, A., Mulik, R.S., Parashar, A.: Deformation dynamics of h-BN reinforced polyethylene nanocomposite under shock/impact loading. Int. J. Mech. Sci. **225**, 107379 (2022)
32. Wang, X.B., Weng, Q., Wang, X., Li, X., Zhang, J., Liu, F., Jiang, X.F., Guo, H., Xu, N., Golberg, D., Bando, Y.: Biomass-directed synthesis of 20 g high-quality boron nitride nanosheets for thermoconductive polymeric composites. ACS Nano **8**, 9081–9088 (2014)

33. Ma, R., Wan, X., Zhang, T., Yang, N., Luo, T.: Role of molecular polarity in thermal transport of boron nitride-organic molecule composites. ACS Omega **3**, 12530–12534 (2018)
34. Pumera, M.: Graphene in biosensing. Mater. Today **14**, 308–315 (2011)
35. Chaurasia, A., Mulik, R.S., Parashar, A.: Polymer-based nanocomposites for impact loading: a review. Mech. Adv. Mater. Struct. **0**, 1–26 (2021)
36. Chaurasia, A., Verma, A., Parashar, A., Mulik, R.S.: Experimental and computational studies to analyze the effect of h-BN nanosheets on mechanical behavior of h-BN/polyethylene nanocomposites. J. Phys. Chem. C **123**, 20059–20070 (2019)
37. Verma, A., Parashar, A., Packirisamy, M.: Atomistic modeling of graphene/hexagonal boron nitride polymer nanocomposites: a review. WIREs Comput. Mol. Sci. **8** (2018)
38. Verma, A., Parashar, A.: Reactive force field based atomistic simulations to study fracture toughness of bicrystalline graphene functionalised with oxide groups. Diam. Relat. Mater. **88**, 193–203 (2018)
39. Chaurasia, A., Jalan, S.K., Parashar, A.: An atomistic approach to study the dynamic and structural response in 2D nanofiller reinforced polyethylene nanocomposites under ultra-short shock pulse loading. Mech. Mater. **169**, 104305 (2022)
40. Zhang, X., Shen, L., Wu, H., Guo, S.: Enhanced thermally conductivity and mechanical properties of polyethylene (PE)/boron nitride (BN) composites through multistage stretching extrusion. Compos. Sci. Technol. **89**, 24–28 (2013)
41. Sandeep, S.K., Chaurasia, A., Parashar, A.: Atomistic simulations to study shock and ultrashort pulse response of high entropy alloy. Mater. Today: Proc. (2022). https://doi.org/10.1016/j.matpr.2022.03.607
42. Kim, H., Kobayashi, S., Abdurrahim, M.A., Zhang, M.J., Khusainova, A., Hillmyer, M.A., Abdala, A.A., MacOsko, C.W.: Graphene/polyethylene nanocomposites: effect of polyethylene functionalization and blending methods. Polymer (Guildf) **52**, 1837–1846 (2011)
43. Dewapriya, M.A.N., Rajapakse, R.K.N.D., Nigam, N.: Influence of hydrogen functionalization on the fracture strength of graphene and the interfacial properties of graphene-polymer nanocomposite. Carbon N. Y. **93**, 830–842 (2015)
44. Singh, S.K., Parashar, A.: Defect dynamics and uniaxial tensile deformation of equi and non-equi-atomic configuration of multi-elemental alloys. Mater. Chem. Phys. **266**, 124549 (2021)
45. Sharma, A.K., Sharma, S.S., Singh, S.K., Parashar, A.: Atomistic simulations to study the effect of helium nanobubble on the shear deformation of nickel crystal. J. Nucl. Mater. **557**, 153245 (2021)
46. Kumar Singh, S., Parashar, A.: Atomistic simulations to study crack tip behaviour in multi-elemental alloys. Eng. Fract. Mech. **243**, 107536 (2021)
47. Jeong, B.W., Lim, J.K., Sinnott, S.B.: Tensile mechanical behavior of hollow and filled carbon nanotubes under tension or combined tension-torsion. Appl. Phys. Lett. **90**, 1–4 (2007)
48. Stuart, S.J., Tutein, A.B., Harrison, J.A.: A reactive potential for hydrocarbons with intermolecular interactions. J. Chem. Phys. **112**, 6472–6486 (2000)
49. Wang, J.M., Wolf, R.M., Caldwell, J.W., Kollman, P.A., Case, D.A.: Development and testing of a general amber force field. J. Comput. Chem. **25**, 1157–1174 (2004)
50. Senftle, T.P., Hong, S., Islam, M.M., Kylasa, S.B., Zheng, Y., Shin, Y.K., Junkermeier, C., Engel-Herbert, R., Janik, M.J., Aktulga, H.M., Verstraelen, T., Grama, A., van Duin, A.C.T.: The ReaxFF reactive force-field: development, applications and future directions. npj Comput. Mater. **2**, 15011 (2016)
51. Van Duin, A.C.T., Dasgupta, S., Lorant, F., Goddard, W.A.: ReaxFF: a reactive force field for hydrocarbons. J. Phys. Chem. A **105**, 9396–9409 (2001)
52. Weismiller, M.R., van Duin, A.C.T., Lee, J., Yetter, R.A.: ReaxFF reactive force field development and applications for molecular dynamics simulations of ammonia borane dehydrogenation and combustion. J. Phys. Chem. A **114**, 5485–5492 (2010)
53. Lindsay, L., Broido, D.A.: Optimized Tersoff and Brenner empirical potential parameters for lattice dynamics and phonon thermal transport in carbon nanotubes and graphene. Phys. Rev. B Condens. Matter Mater. Phys. **81**, 1–6 (2010)

54. Kumar, R., Rajasekaran, G., Parashar, A.: Optimised cut-off function for Tersoff-like potentials for a BN nanosheet: a molecular dynamics study. Nanotechnology **27**, 085706 (2016)
55. Kumar, R., Parashar, A.: Dislocation assisted crack healing in h-BN nanosheets. Phys. Chem. Chem. Phys. **19**, 21739–21747 (2017)
56. Brenner, D.W.: Empirical potential for hydrocarbons for use in simulating the chemical vapor deposition of diamond films. Phys. Rev. B **42**, 9458–9471 (1990)
57. Chaurasia, A., Parashar, A.: Effect of BNNR on mechanical properties of polyethylene nanocomposites. Mater. Today Proc. **47**, 2875–2877 (2021)
58. Chaurasia, A., Singh, S.K., Parashar, A.: Atomistic scale insight to investigate the strain rate effect on mechanical response of boron nitride nanosheet reinforced nanocomposites. Mater. Today: Proc. (2022). https://doi.org/10.1016/j.matpr.2022.04.337
59. Chaurasia, A., Parashar, A., Mulik, R.S.: Enhancement in mechanical properties of polyethylene using h-BN nanofiller. In: Saran, V.H. (ed.) Advances in Systems Engineering, pp. 127–33. Springer, Singapore (2021)
60. Medhekar, N.V., Ramasubramaniam, A., Ruoff, R.S., Shenoy, V.B.: Hydrogen bond networks in graphene oxide composite paper: structure and mechanical properties. ACS Nano **4**, 2300–2306 (2010)
61. Li, M., Zhou, H., Zhang, Y., Liao, Y., Zhou, H.: The effect of defects on the interfacial mechanical properties of graphene/epoxy composites. RSC Adv. **7**, 46101–46108 (2017)
62. Verma, A., Parashar, A., Packirisamy, M.: Effect of grain boundaries on the interfacial behaviour of graphene-polyethylene nanocomposite. Appl. Surf. Sci. **470**, 1085–1092 (2019)
63. Bačová, P., Rissanou, A.N., Harmandaris, V.: Edge-functionalized graphene as a nanofiller: molecular dynamics simulation study. Macromolecules **48**, 9024–9038 (2015)
64. Nikkhah, S.J., Moghbeli, M.R., Hashemianzadeh, S.M.: Interfacial adhesion between functionalized polyethylene surface and graphene via molecular dynamic simulation. J. Mol. Model. **21**, 121 (2015)
65. Javan Nikkhah, S., Moghbeli, M.R., Hashemianzadeh, S.M.: Investigation of the interface between polyethylene and functionalized graphene: a computer simulation study. Curr. Appl. Phys. **15**, 1188–1199 (2015)
66. Awasthi, A.P., Lagoudas, D.C., Hammerand, D.C.: Modeling of graphene–polymer interfacial mechanical behavior using molecular dynamics. Model. Simul. Mater. Sci. Eng. **17**, 015002 (2009)
67. Rahman, R., Foster, J.T.: Deformation mechanism of graphene in amorphous polyethylene: a molecular dynamics based study. Comput. Mater. Sci. **87**, 232–240 (2014)
68. Liu, F., Hu, N., Ning, H., Liu, Y., Li, Y., Wu, L.: Molecular dynamics simulation on interfacial mechanical properties of polymer nanocomposites with wrinkled graphene. Comput. Mater. Sci. **108**, 160–167 (2015)
69. Kumar, R., Parashar, A.: Effect of geometrical defects and functionalization on the interfacial strength of h-BN/polyethylene based nanocomposite. Polymer (Guildf) **146**, 82–90 (2018)
70. Meng, J., Tajaddod, N., Cranford, S.W., Minus, M.L.: Polyethylene-assisted exfoliation of hexagonal boron nitride in composite fibers: a combined experimental and computational study. Macromol. Chem. Phys. **216**, 847–855 (2015)
71. Verma, A., Zhang, W., Van Duin, A.C.: ReaxFF reactive molecular dynamics simulations to study the interfacial dynamics between defective h-BN nanosheets and water nanodroplets. Phys. Chem. Chem. Phys. **23**(18), 10822–10834 (2021)
72. Verma, A., Parashar, A., Packirisamy, M.: Role of chemical adatoms in fracture mechanics of graphene nanolayer. Mater. Today: Proc. **11**, 920–924 (2019)
73. Singla, V., Verma, A., Parashar, A.: A molecular dynamics based study to estimate the point defects formation energies in graphene containing STW defects. Mater. Res. Express **6**(1), 015606 (2018)

Atomistic Simulations to Study Thermal Effects and Strain Rate on Mechanical and Fracture Properties of Graphene like BC3



Akarsh Verma and Sachin Sharma

Abstract In recent years, experimentally fabricated two-dimensional materials evoked great attention in nanodevice manufacturing industries. In this chapter, the author analyzed the mechanical and fracture behavior of boron-carbide with BC3 configuration. It is a graphene-like two-dimensional (2D) structure. Due to its exceptional mechanical, thermal, and electrical behavior, it can be used in a wide range of applications in future by substituting graphene. However, during the experimental production process, there could be many types of defects in the material, which leads to remarkably deterioration in the mechanical and fracture properties of the material. So, it's essential to predict the mechanical and fracture behavior of graphene-like BC3 in defected conditions. That is why the author investigated the behavior of pristine and defected BC3 nanosheets with the help of classic molecular dynamics (MD) simulations. Tersoff potential was used for finding the interactions among C–C and B–B atoms. The author concluded that the temperature and strain rate significantly affect the mechanical and fracture properties of graphene-like BC3. The author also concluded that the crack length also affects the fracture toughness of the material.

Keywords Tersoff potential · Molecular dynamics · Fracture toughness · Two-dimensional materials · Mechanical properties

1 Introduction

2D materials possess some exceptional properties such as plasticity, ductility and chemical inertness. Polymers' simple and also affordable processability make them

A. Verma (✉) · S. Sharma
Department of Mechanical Engineering, University of Petroleum and Energy Studies, Dehradun 248007, India
e-mail: akarshverma007@gmail.com

A. Verma
Department of Mechanical Science and Bioengineering, Osaka University, Osaka 560-8531, Japan

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a suitable prospect to offer the objective of the matrix in various types of nanocomposite [1, 2] in conjunction with 2D materials like graphene, etc. In order to enhance the toughness of polymers, the properties of nanofiller/reinforcement play a decisive function. Polymers-based nanocomposites show high warm distortion temperature levels, reduced slow scratch and defect resistance, noise damping, and tailored mechanical toughness [3–5].

A few techniques to contrive these nanofillers with polymers remain in situ intercalative polymerization, co-extrusion, thermoforming, three roll milling, vacuum-assisted routines, layer-by-layer assembly, solution, and thaw processing/intercalation [6–9]. As the atomistic structure of graphene/BC3/C3N includes uniformly spread carbon atoms, their molecular level interactions with polymers include weak van der Waals (vdW) force, hydrophobic-hydrophobic interaction as well as stacking. In this training course of activity, functionalized sheets are dispersed into monomer systems, and then polymer chains are straight established from sheet surface area, ensuring a better dispersion and excellent interaction between them. Regardless of having extraordinary mechanical behavior, graphene displays better transferability when piled in between the polymers. Polymer properties such as chemical inertness, plasticity, ductility, cost-effectiveness, and easy processability make them an ideal candidate to serve the function of the matrix in various kinds of nanocomposite.

Graphene [10] is a two-dimensional (2D) material with hexagonal units of sp^2 bound atoms. It is among the most useful carbon-based products with varied applications in thermal user interface products [11], biomedicine, electronic devices [12], and so forth. Additionally, graphene depicts really remarkable mechanical behavior [9, 13–16]. Graphene layers with nanopores can be made use of in medicine as well as drug distribution inside the body [17] Porous graphene, moreover, is an excellent prospect for dimension selectivity in nanofillers and also membranes [18, 19]. Nonetheless, in mechanical terms, it is proved utilizing molecular dynamics simulation that pores minimize the failing strain as well as failing stress and strain in the material [9]. The cracks of graphene are of critical significance in numerous applications. For instance, carbon nanotubes (CNTs) are bent graphene sheets in which the sides identify graphene growth behavior [20]. In addition to edges, the density of nanosheets can play a crucial function in establishing mechanical properties. For instance, researching the effective density boron nitride (BN) revealed that the mechanical strength of these nanosheets depends on the density [21].

An additional carbon-based graphene-like material is boron-carbide (BC3) whose framework is comparable to C3N [22–24]. Recently, research has actually been carried out on the specific attributes of BC3 in which mechanical actions, digital, optical, and thermal transportation residential or commercial properties making use of DFT are measured. BC3 monolayers are among the most encouraging prospects for energy storage space, nanoelectronics as well as catalysis. Because of the time-consuming, complicated, and expensive procedure of checking out material properties at the nanoscale by speculative techniques, computer simulations are feasible options. Modeling at the nanoscale has the possibility to acquire an essential understanding of the mechanical behaviors of 2-dimensional nanomaterials. Mechanical

properties are among the crucial elements for the application of nanomaterials as well as play a vital role since they are considerably related to the safety and security and also security of the structure under the enforced mechanical pressures occurring during their lifetime.

In recent times, some researchers have checked the features of butt-joint interfaces of graphene and graphene-like products due to the fact that they have different habits in comparison to their basic products [25]. In one research using MD, the mechanical behavior of defected hybrid graphene-boron nitride nanosheets has actually been computed, and also the searching's for have actually disclosed that defective crossbreed graphene-boron nitrite has reduced mechanical properties [24]. These mechanical features are dependent on the size, form, and setting of the problems. A lately conducted research study suggests that crossbreed nanostructures of graphene-boron nitrite with splits along the elbow chair instructions have even more satisfying mechanical behavior than those with fractures along the zigzag direction. In an additional research study, the geometry and also user interface of graphene-boron nitrite have been explored and also based on the outcomes, C–B bonds reveal a better result for electric goals, C–N bonds are better for mechanical applications.

Recently, the emphasis of scientists has actually been changed to one more appealing crossbreed nanosheet made from C3N as well as BC3 owing to its electric behavior and electronic properties, particularly its semiconductivity [20]. Additionally, creating deliberate issues to boost the band gap may have several repercussions that can impact the performance of this crossbreed. Likewise, throughout the synthesis of crossbreed sheets, the appearance of flaws is exceptionally potent because of their nano-sized particles. Distinctions in atomic ranges in the left and the right components of crossbreed sheet can offer rise to developing problems, especially in the interface. Researching the mechanical results of these imperfections can preclude any kind of functional disadvantages.

In the present chapter, the mechanical properties of zigzag and armchair configurations of BC3 have been explored. In addition, the effect of STW defects [14], thermal and strain rate on the mechanical properties of this material has been contrasted. In addition to this, the fracture toughness of BC3 was also calculated. To the factor we evaluated, there was no comparable research study on the mechanical residential or commercial properties of crossbreed C3N-BC3.

2 Simulation Details and Modeling

Classical molecular dynamics simulations are employed as an alternative to experimental studies to investigate the mechanical properties of pristine graphene-like BC3 nanosheets (Fig. 1) and also with different cracks and notches. All molecular dynamics modeling in this study were carried out using open-source software LAMMPS (Large-Scale Atomic/Molecular Massively Parallel Simulator) [26] and post-processing was done by using Open visualization tool (OVITO) [27, 28]. In order to determine the bonding interactions between carbon–carbon atoms, the optimized

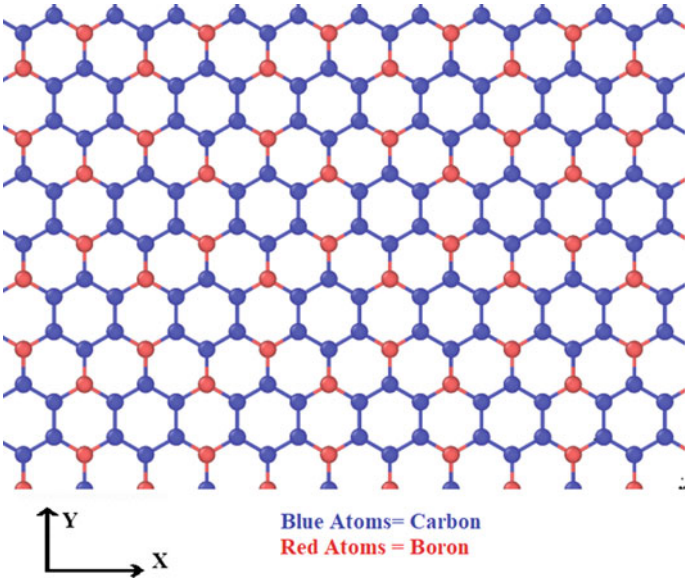


Fig. 1 Schematic diagram of graphene-like BC3

Tersoff potential proposed by Lindsay and Broido [29] is employed. Furthermore, the Tersoff potential parameters used to introduce the binding interactions between the boron–carbon atoms were adopted from Kinaci et al. Tersoff potentials have already been used to predict mechanical and electronic behavior of many 2-d nanomaterials like graphene, silicene, phosphine, etc. [30, 31]. So, it can be said that Tersoff potentials can be used for predicting the mechanical properties of graphene-like BC3 nanomaterial.

The computer models were created for a nanosheet simulation box consisting of 35,041 individual atoms. The boundary conditions are periodic along with the in-plane directions, i.e., X and Y, in Z-direction, it is fixed. Thus, these simulations are representative of a graphene-like BC3nanosheet with infinite planar dimensions. Initially, the computational box was relaxed by means of the Nose–Hoover barostat and thermostat (NPT) approach to get a stress-free condition. Then, an engineering strain rate of 1×10^8 s^{-1} was applied, and small-time step size (time increment) of 0.5 fs was adopted for all calculations in the MD simulations. The structure is also tested for lower and higher strain rates from 5×10^6 to 5×10^9 s^{-1} and compared the results. The atomic positions are rescaled according to the simulation box size dimensions to prevent void formation throughout the simulation. The stress values were determined using the Virial theorem at every time step [32, 33]. Uniaxial loading conditions were applied such that the periodic simulation box through the structure width was arranged by employing the NPT method to guarantee infinitesimal stresses in this direction.

3 Results and Discussion

At a temperature of 1 K, the 2D graphene-like BC3 nanosheet shows Young's modulus (E) of 799.43 GPa in armchair instructions as well as 815.02 GPa in zigzag positioning. A defect-free BC3 nanosheet with dimensions of 300 × 300 Å was utilized to explore the impact of temperature (200, 300, 400, 500, 600, 700 and 900 K). The models were packed in tension in both armchair as well as zigzag positionings and the extracted uniaxial stress–strain results for pristine as well as defected graphene-like BC3 nanosheets are illustrated in numerous figures. At a temperature of 1 K, the 2D graphene-like BC3 nanosheet reveals Young's modulus (E) of 799.43 GPa in armchair direction and 815.02 GPa in zigzag orientation. A defect-free BC3 nanosheet with dimensions of 300 × 300 Å was utilized to explore the effect of temperature levels (200, 300, 400, 500, 600, 700, and 900 K). The designs were packed in tension in both armchair and also zigzag positioning, and also the drawn-out uniaxial stress–strain outcomes for faulty and pristine graphene-like BC3 nanosheets are depicted on several graphs.

3.1 Mechanical Properties of BC3

Temperature, strain rate and various defects (point defects, STW defects, cracks, etc.) significantly affect the properties of any material. To examine the effect of various thermal and mechanical effects on the mechanical properties of BC3, various simulations were performed. The results are discussed in the next subsections.

3.1.1 Thermal Effects

Influence of several temperatures on the mechanical behavior of pristine graphene-like BC3 nanosheets has been studied. The extracted uniaxial stress–strain responses predicted by MD calculations for pristine graphene-like BC3 nanosheets loaded in armchair orientation at 300, 500, 700 and 900 K are depicted in Fig. 2a. Furthermore, the acquired results for aforementioned defect-free material loaded in zigzag orientation at 200, 300, 500, 700 and 900 K have been plotted in Fig. 2b.

The MD results illustrated in Fig. 2a, b show that increasing the temperature will decrease the ultimate tensile strength and strain-to-failure of the nanosheet. Distances between the neighbors atoms increase by raising temperature, decreasing the atom's interaction energy. Therefore, the tensile strength has a tendency to decrease as the temperature increases. A minor decline in the elasticity modulus is observed at higher temperatures. The elements that influence the interatomic spacing and also force in between atomic bands (like temperature level) will also impact the modulus of flexibility of product. It can be seen that young's modulus of BC3 sheets lowers with enhancing temperature since the binding energy between atoms reduces as the

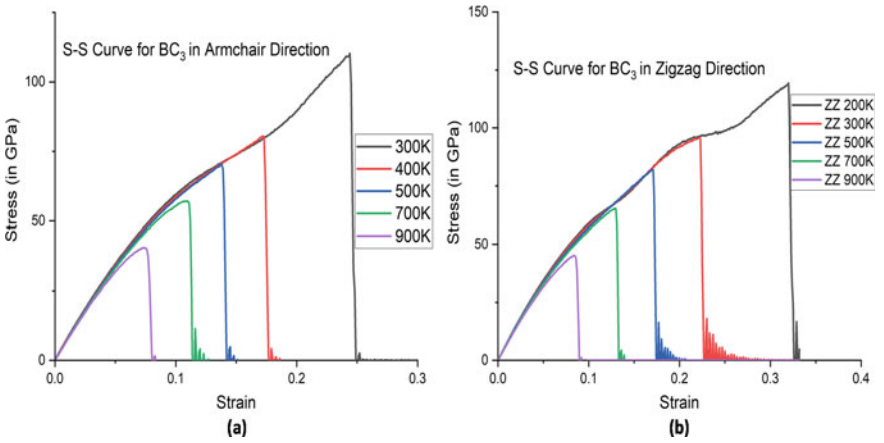


Fig. 2 Stress–strain curve for graphene-like BC3 at various temperatures **a** in armchair direction **b** in zigzag direction

temperature level boosts. The kinetic energy is transferred right into strain energy with the deformation throughout tensile loading. Increasing the temperature lowers strain energy since of the increasing kinetic energy.

For a BC3 monolayer loaded in armchair direction, at room temperature (300 K), the ultimate stress is 116.5 GPa at a strain-to-failure of 0.215. The tensile stress is 73.5 GPa when the temperature is increased to 900 K which is 37% lower than the ultimate stress at 300 K. The maximum tensile stress occurs at 200 K at a strain of 0.23 which is 120.2 GPa while the strain-to-failure at highest simulated temperature is 0.121, which is about 46.7 percent less than the maximum strain at 200 K. For a BC3 monolayer loaded along the zigzag direction, at 300 K, the ultimate stress is 122.1 GPa at a strain-to-failure of 0.31, see Fig. 2b. When the temperature is increased to 900 K, the tensile strength is 48.592 GPa which is 55 percent lower than the ultimate stress at 200 K. In armchair orientation, the maximum tensile stress occurs at 200 K which is 110.3 GPa at a strain-to-failure of 0.245. The elevation of temperature will cause the nanosheet to elongate less compared to sheets at room temperature since the molecules have less molecular motion.

According to the results shown in Fig. 2, for all studied cases, the initial portion of each stress–strain graph is a straight line indicating the validity of linear elasticity in this region. The modulus of elasticity of the material can be estimated through the slope at this zone. As the values of Young's modulus (E) reveal the particular importance in material properties and especially for micro/nano-electromechanical systems, therefore, in this section, Young's modulus (E) at aforementioned temperatures for BC3 nanosheets in armchair and zigzag orientations are estimated and presented in Tables 1, and 2 respectively. For estimating Young's modulus, lines fitted to the stress–strain curves for the strain levels up to 0.04. According to the results of Tables 1 and 2, the lower modulus of elasticity (E) occurs at higher temperatures.

Table 1 Young's modulus (E) of the pristine BC3 nanosheet along the armchair direction

T (K)	200	300	500	700	900
E (GPa)	747.12	732	736.02	704.05	687.42

Table 2 Young's modulus (E) of the pristine BC3 nanosheet along the zigzag direction

T (K)	200	300	400	500	600
E (GPa)	742.13	771.37	734.49	749.21	732.12

3.1.2 Effect of Strain Rate

After predicting the thermal effects, simulations were performed to study the strain rate behavior on the mechanical properties of graphene-like BC3. Figure 3 illustrates the ultimate tensile strength of crystalline boron carbide sheets as a function of strain rate at different temperatures in armchair orientation. As the strain rate increases, the ultimate tensile strength indicates an increasing trend. However, the strain rate effect on the tensile strength of graphene-like BC3 nanosheet in zigzag orientation is more significant than the armchair orientation. Also, this effect is more pronounced in higher temperatures than in lower temperatures.

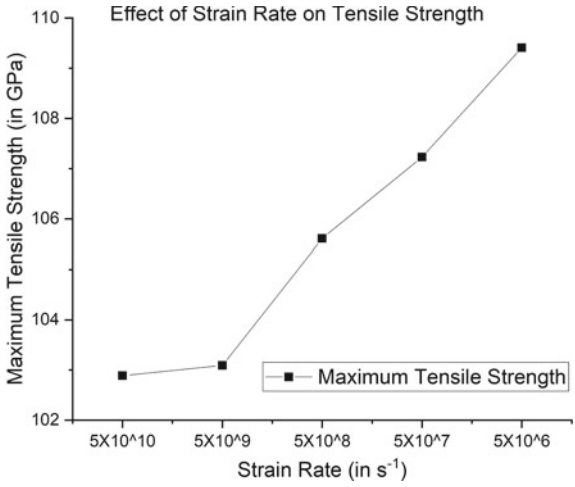


Fig. 3 Variation of tensile strength as a function of strain rate in graphene-like BC3

3.2 *Fracture Properties of BC3*

After performing simulations for predicting mechanical behavior of BC3, simulations were performed to get the insights of fracture behavior of graphene-like BC3 nanomaterial. These simulations were performed at 1 K temperature to remove thermal effects on the edges of cracks [34].

3.2.1 Fracture Toughness of BC3

The propagation of cracks is influenced by the crack orientation. Subsequently, cracks were investigated wherein the imposed loading is perpendicular to the crack orientation. As described in the introduction, to construct the models subjected to cracks, we use disconnected interatomic bonds for the atoms located at both sides of the crack. Several crack lengths including L/10, L/12, and L/15 are examined for armchair and zigzag direction, in which L is the length of one side of the simulation box in graphene-like BC3 monolayer. Figure 5 shows the maximum fracture stress values after applying tensile loads at different temperatures. The MD solutions for the cracked models are compared with the data for crack-free graphene-like BC3 nanosheets (0L). Obviously, the cracks significantly decrease the tensile strength of two-dimensional BC3. As expected, the tensile strength decreases with increasing crack length. Furthermore, the temperature increase has a weakening effect on the ultimate tensile strength of the two-dimensional graphene-like BC3 material. Table 3 shows the fracture toughness values for various crack lengths in both orientations. The average value of fracture toughness for BC3 is 7.57 MPa$\sqrt{\text{m}}$. In order to evaluate the fracture toughness of BC3, a critical stress intensity factor was estimated with the help of Eq. 1:

$$K_{\text{lc}} = Y\sigma\sqrt{\pi a} \tag{1}$$

where Y is the dimensionless parameter ($Y = 1$ in this atomistic model), σ is the far field normal stress value at the instance of first bond rupture and a is half of the central crack length. In order to fix the crack length, simulations were performed with different crack lengths (Figs. 4 and 5).

Table 3 Fracture toughness

Crack length	Fracture toughness (MPa$\sqrt{\text{m}}$)	
	In armchair direction	In zigzag direction
L/10	6.982	7.692
L/12	7.071	7.778
L/15	6.900	7.605

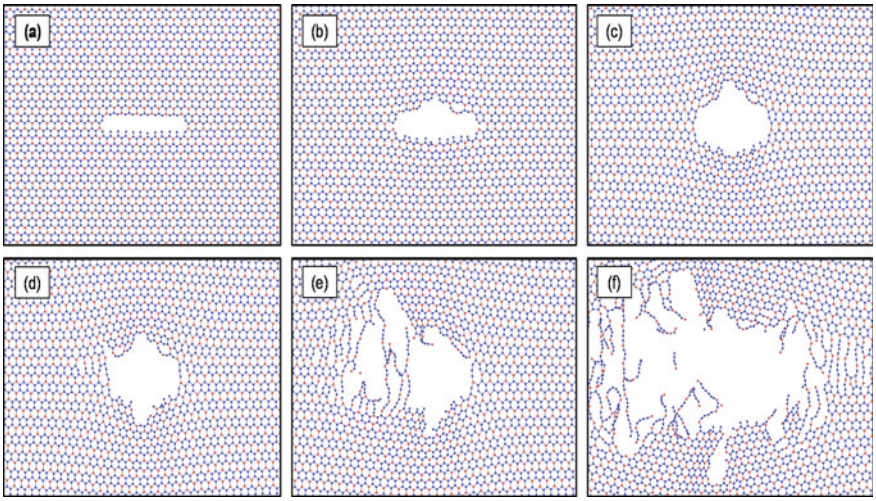


Fig. 4 Crack morphology in graphene-like BC3 at various strain values

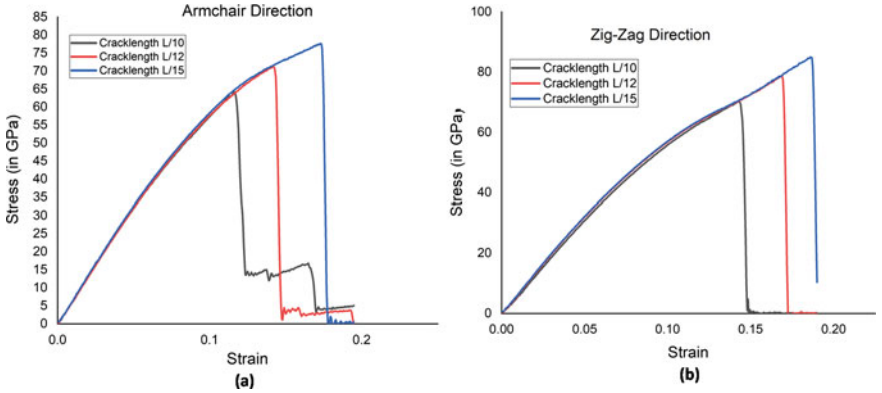


Fig. 5 Effect of crack length on the fracture stresses in graphene-like BC3 **a** in armchair direction **b** in a zigzag direction

3.3 Thermal Effects on Fracture Toughness

As shown in Fig. 6, the stress intensity factor decreases with increasing temperature. Simulations were performed for five samples with uncorrelated initial conditions in order to determine the error bars in the acquired results. Crack propagation started in a single-layer graphene-like BC3 film for a crack length of L/12 at 200, 300, 500, 700 and 900 K, respectively, crack length and other parameters held constant during the simulations. Cracks are visualized with OVITO and depicted in Fig. 6. Stress

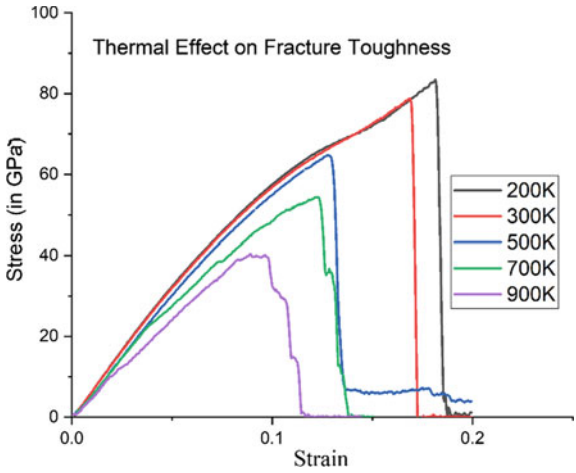


Fig. 6 Effect of temperature on the fracture stresses in graphene-like BC3

concentrations can be observed near the crack tip. Strains at each stage are depicted in each picture as well.

3.4 STW Defects in BC3

There may be several topological defects in BC3such as vacancies, dislocations (Edge or Screw) and Stone–Wales (SW) defects, also known as Stone-Thrower-Wales (STW) defects. Among them, STW defects are generated without deleting any atom from the lattice position but are introduced by rotating single C–C bonds or B–C bonds 90 degrees in its perpendicular direction from its midpoint in counterclockwise (CCW) direction. Some STW defects of SLBC are shown in Fig. 7 for the illustration.

The stress–strain responses obtained from pristine BC3 and BC3 with STW defects are plotted in Fig. 8. It can be inferred from Fig. 8 that a significant reduction in the strength was observed for BC3 containing various STW defects as compared to the pristine sheet of BC3 along with both the ZZ and AC directions. STW defects increase the C–C or C–B bond lengths in BC3, which ultimately weakens the bond strength in BC3. This can be attributed as the reason behind degradation of the mechanical properties of BC3. The authors also have a vast experience in the field of molecular dynamics and experimental characterization of composites materials [35–77].

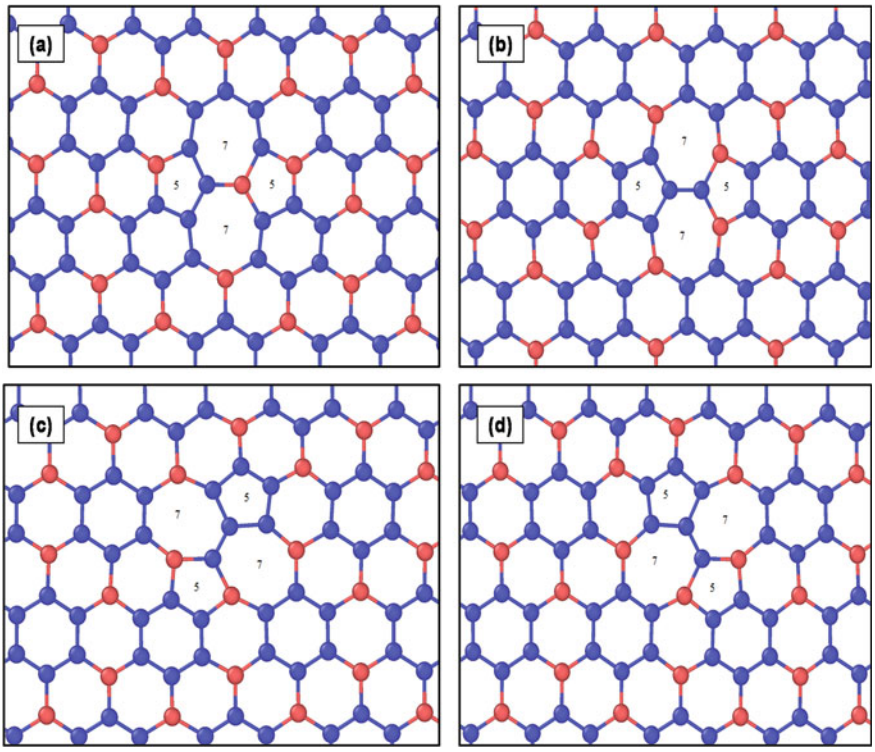


Fig. 7 Various STW defects in graphene-like BC3

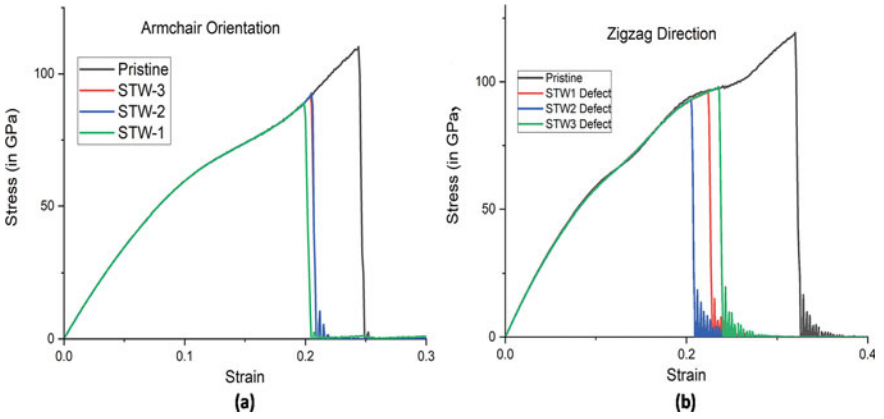


Fig. 8 Effect of various STW defects on the mechanical properties of graphene-like BC3 **a** in armchair direction **b** in zigzag direction

4 Conclusions

In this article, Tersoff potential was used for simulating the mechanical and fracture properties of graphene-like BC3 nanomaterial.

- Tersoff potential can be used for capturing the interactions of C–C and B–C atoms.
- Temperature significantly affects the mechanical properties of graphene-like BC3.
- Change in strain rate leads to the alteration in mechanical behavior of graphene-like BC3.
- Fracture toughness depends on crack length and temperature both.
- STW defects significantly affect the mechanical strength of graphene-like BC3.

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Conflicts of Interest "There are no conflicts of interest to declare by the authors."

References

1. Chaurasia, A., Mulik, R.S., Parashar, A.: Polymer-based nanocomposites for impact loading: a review. Mech. Adv. Mater. Struct. **0**, 1–26 (2021)
2. Xie, F., Lu, Z., Yang, Z., Hu, W., Yuan, Z.: Mechanical behaviors and molecular deformation mechanisms of polymers under high speed shock compression: a molecular dynamics simulation study. Polymer (Guildf) **98**, 294–304 (2016)
3. Wu, Z., Curtin, W.A.: Brittle and ductile crack-tip behavior in magnesium. Acta Mater. **88**, 1–12 (2015)
4. Zhang, Y., Jiang, S., Zhu, X., Zhao, Y.: A molecular dynamics study of intercrystalline crack propagation in nano-nickel bicrystal films with (0 1 0) twist boundary. Eng. Fract. Mech. **168**, 147–159 (2016)
5. Zhang, J.: Molecular dynamics study of crack propagation behavior and mechanisms in Nickel (2011)
6. Verma, P.K., Parashar, A.: Sequential multiscale model to study crack tip behavior in bi-crystalline graphene. J. Appl. Phys. (2020)
7. Verma, A., Parashar, A., Packirisamy, M.: Effect of grain boundaries on the interfacial behaviour of graphene-polyethylene nanocomposite. Appl. Surf. Sci. **470**, 1085–1092 (2019)
8. Chaurasia, A., Verma, A., Parashar, A., Mulik, R.S.: Experimental and computational studies to analyze the effect of h-BN nanosheets on mechanical behavior of h-BN/polyethylene nanocomposites. J. Phys. Chem. C **123**(32), 20059–20070 (2019)
9. Zarghami Dehaghani, M., Hamed Mashhadzadeh, A., Salmankhani, A., Karami, Z., Habibzadeh, S., Ganjali, M.R., Saeb, M.R.: Fracture toughness and crack propagation behavior of nanoscale beryllium oxide graphene-like structures: a molecular dynamics simulation analysis. Eng Fract. Mech. **235**, 107194 (2020)
10. Novoselov, K.S., Jiang, D., Schedin, F., Booth, T.J., Khotkevich, V.V., Morozov, S.V., Geim, A.K.: Two-dimensional atomic crystals. Proc. Natl. Acad. Sci. U. S. A. **102**, 10451–10453 (2005)
11. Mortazavi, B., Ahzi, S.: Thermal conductivity and tensile response of defective graphene: a molecular dynamics study. Carbon N. Y. **63**, 460–470 (2013)

12. Wu, Y., Farmer, D.B., Xia, F., Avouris, P.: Graphene electronics: materials, devices, and circuits. Proc. IEEE **101**, 1620–1637 (2013)
13. Verma, A., Parashar, A., Packirisamy, M.: Atomistic modeling of graphene/hexagonal boron nitride polymer nanocomposites: a review *WIREs*. Comput. Mol. Sci. **8**, e1346 (2018)
14. Verma, A., Parashar, A.: The effect of STW defects on the mechanical properties and fracture toughness of pristine and hydrogenated graphene. Phys. Chem. Chem. Phys. **19**, 16023–16037 (2017)
15. Min, K., Aluru, N.R.: Mechanical properties of graphene under shear deformation. Appl. Phys. Lett. **98**, 013113 (2011)
16. Ni, Z., Bu, H., Zou, M., Yi, H., Bi, K., Chen, Y.: Anisotropic mechanical properties of graphene sheets from molecular dynamics. Phys. B Condens. Matter **405**, 1301–1306 (2010)
17. Stankovich, S., Dikin, D.A., Dommett, G.H.B., Kohlhaas, K.M., Zimney, E.J., Stach, E.A., Piner, R.D., Nguyen, S.B.T., Ruoff, R.S.: Graphene-based composite materials. Nature **442**, 282–286 (2006)
18. Ramanathan, T., Abdala, A.A., Stankovich, S., Dikin, D.A., Herrera-Alonso, M., Piner, R.D., Adamson, D.H., Schniepp, H.C., Chen, X., Ruoff. R.S., Nguyen, S.T., Aksay, I.A., Prud'Homme, R.K., Brinson, L.C.: Functionalized graphene sheets for polymer nanocomposites. Nat. Nanotechnol. **3**, 327–331 (2008)
19. Chaurasia, A., Parashar, A., Mulik, R.S.: Enhancement in mechanical properties of polyethylene using h-BN nanofiller. In: Saran, V.H., Misra, R.K. (eds.) BT—Advances in Systems Engineering, pp 127–133. Springer Singapore, Singapore (2021)
20. Kaxiras, E.: Atomic and Electronic Structure of Solids. Cambridge University Press (2003)
21. Standley, B., Bao, W., Hang, Z., Bruck, J., Chun, N.L., Bockrath, M.: Graphene-based atomic-scale switches. Nano Lett. **8**, 3345–3349 (2008)
22. Zahedi, R.K., Shirazi, A.H.N., Alimouri, P., Alajlan, N., Rabczuk, T.: Mechanical properties of graphene-like BC3; a molecular dynamics study. Comput. Mater. Sci. **168**, 1–10 (2019)
23. Yang, J., Sun, H., He, J., Tian, Y., Chen, C.: Diamond-like BC3 as a superhard conductor identified by ideal strength calculations. J. Phys. Condens. Matter **19** (2007)
24. Dadrasi, A., Fooladpanjeh, S., Eshkalak, K.E., Sadeghzadeh, S., Saeb, M.R.: Crack pathway analysis in graphene-like BC3 nanosheets: towards a deeper understanding. J. Mol. Graph. Model. **107**, 107980 (2021)
25. Verma, A., Kumar, R., Parashar, A.: Enhanced thermal transport across a bi-crystalline graphene–polymer interface: an atomistic approach. Phys. Chem. Chem. Phys. **21**, 6229–6237 (2019)
26. Crozier, P.S.: LAMMPS molecular dynamics simulator. Tutorial (2011)
27. Stukowski, A.: Visualization and analysis of atomistic simulation data with OVITO–the open visualization tool. Model. Simul. Mater. Sci. Eng. **18**, 015012 (2010)
28. Singh, S.K., Parashar, A.: Defect dynamics and uniaxial tensile deformation of equi and non-equi-atomic configuration of multi-elemental alloys. Mater. Chem. Phys. **266**, 124549 (2021)
29. Lindsay, L., Broido, D.A.: Optimized Tersoff and Brenner empirical potential parameters for lattice dynamics and phonon thermal transport in carbon nanotubes and graphene. Phys. Rev. B **81**, 205441 (2010)
30. Pei, Q.-X., Zhang, Y.-W., Sha, Z.-D., Shenoy, V.B.: Tuning the thermal conductivity of silicene with tensile strain and isotopic doping: a molecular dynamics study. J. Appl. Phys. **114**, 033526 (2013)
31. Bakó, I., Stirling, A., Bálint, S., Pápai, I.: Association of frustrated phosphine–borane pairs in toluene: molecular dynamics simulations. Dalt. Trans. **41**, 9023 (2012)
32. Subramaniyan, A.K., Sun, C.T.: Continuum interpretation of virial stress in molecular simulations. Int. J. Solids Struct. **45**, 4340–4346 (2008)
33. Sharma, A.K., Sharma, S.S., Singh, S.K., Parashar, A.: Atomistic simulations to study the effect of helium nanobubble on the shear deformation of nickel crystal. J. Nucl. Mater. **557**, 153245 (2021)
34. Kumar Singh, S., Parashar, A.: Atomistic simulations to study crack tip behaviour in multi-elemental alloys. Eng. Fract. Mech. **243**, 107536 (2021)

35. Deji, R., Jyoti, R., Verma, A., Choudhary, B.C., Sharma, R.K.: A theoretical study of HCN adsorption and width effect on co-doped armchair graphene nanoribbon. Comput. Theor. Chem. 113592 (2022)
36. Verma, A., Singh, V.K., Verma, S.K., Sharma, A.: Human hair: a biodegradable composite fiber–a review. Int. J. Waste Res. **6**(206), 2 (2016)
37. Verma, A., Singh, V.K.: Mechanical, microstructural and thermal characterization of epoxy-based human hair–reinforced composites. J. Test. Eval. **47**(2), 1193–1215 (2019)
38. Verma, A., Parashar, A.: Structural and chemical insights into thermal transport for strained functionalised graphene: a molecular dynamics study. Mater. Res. Exp. **5**(11), 115605 (2018)
39. Verma, A., Negi, P., Singh, V.K.: Experimental analysis on carbon residuum transformed epoxy resin: chicken feather fiber hybrid composite. Polym. Compos. **40**(7), 2690–2699 (2019)
40. Verma, A., Gaur, A., Singh, V.K.: Mechanical properties and microstructure of starch and sisal fiber biocomposite modified with epoxy resin. Mater. Perform. Charact. **6**(1), 500–520 (2017)
41. Deji, R., Verma, A., Choudhary, B.C., Sharma, R.K.: New insights into NO adsorption on alkali metal and transition metal doped graphene nanoribbon surface: a DFT approach. J. Mol. Graph. Model. (2021). https://doi.org/10.1016/j.jmgm.2021.108109
42. Verma, A., Budiyal, L., Sanjay, M.R., Siengchin, S.: Processing and characterization analysis of pyrolyzed oil rubber (from waste tires)-epoxy polymer blend composite for lightweight structures and coatings applications. Polym. Eng. Sci. **59**(10), 2041–2051 (2019)
43. Verma, A., Negi, P., Singh, V.K.: Physical and thermal characterization of chicken feather fiber and crumb rubber reformed epoxy resin hybrid composite. Adv. Civil Eng. Mater. **7**(1), 538–557 (2018)
44. Verma, A., Negi, P., Singh, V.K.: Experimental investigation of chicken feather fiber and crumb rubber reformed epoxy resin hybrid composite: mechanical and microstructural characterization. J. Mech. Behav. Mater. **27**(3–4) (2018)
45. Arpitha, G.R., Verma, A., Sanjay, M.R., Siengchin, S.: Preparation and experimental investigation on mechanical and tribological performance of Hemp-glass fiber reinforced laminated composites for lightweight applications. Adv. Civil Eng. Mater. **10**(1), 427–439 (2021)
46. Jain, N., Verma, A., Singh, V.K.: Dynamic mechanical analysis and creep-recovery behaviour of polyvinyl alcohol based cross-linked biocomposite reinforced with basalt fiber. Mater. Res. Exp. **6**(10), 105373 (2019)
47. Verma, A., Joshi, K., Gaur, A., Singh, V.K.: Starch-jute fiber hybrid biocomposite modified with an epoxy resin coating: fabrication and experimental characterization. J. Mech. Behav. Mater. **27**(5–6) (2018)
48. Deji, R., Verma, A., Kaur, N., Choudhary, B.C., Sharma, R.K.: Density functional theory study of carbon monoxide adsorption on transition metal doped armchair graphene nanoribbon. Mater. Today Proc. (2021)
49. Verma, A., Singh, V.K.: Experimental investigations on thermal properties of coconut shell particles in DAP solution for use in green composite applications. J. Mater. Sci. Eng. **5**(3), 1000242 (2016)
50. Verma, A., Singh, V.K., Arif, M.: Study of flame retardant and mechanical properties of coconut shell particles filled composite. Res. Rev. J. Mater. Sci. **4**(3), 1–5 (2016)
51. Verma, A., Parashar, A., Jain, N., Singh, V.K., Rangappa, S.M., Siengchin, S.: Surface modification techniques for the preparation of different novel biofibers for composites. In: Biofibers and Biopolymers for Biocomposites, pp. 1–34. Springer, Cham (2020)
52. Rastogi, S., Verma, A., Singh, V.K.: Experimental response of nonwoven waste cellulose fabric–reinforced epoxy composites for high toughness and coating applications. Mater. Perform. Charact. **9**(1), 151–172 (2020)
53. Bharath, K.N., Madhu, P., Gowda, T.G., Verma, A., Sanjay, M.R., Siengchin, S.: A novel approach for development of printed circuit board from biofiber based composites. Polym. Compos. **41**(11), 4550–4558 (2020)
54. Verma, A., Jain, N., Parashar, A., Gaur, A., Sanjay, M.R., Siengchin, S.: Lifecycle assessment of thermoplastic and thermosetting bamboo composites. In: Bamboo Fiber Composites, pp. 235–246. Springer, Singapore (2021)

55. Singh, K., Jain, N., Verma, A., Singh, V.K., Chauhan, S.: Functionalized graphite-reinforced cross-linked poly (vinyl alcohol) nanocomposites for vibration isolator application: morphology, mechanical, and thermal assessment. Mater. Perform. Charact. **9**(1), 215–230 (2020)
56. Verma, A., Parashar, A.: Molecular dynamics based simulations to study the fracture strength of monolayer graphene oxide. Nanotechnology **29**(11), 115706 (2018)
57. Verma, A., Jain, N., Singh, K., Singh, V.K., Rangappa, S.M., Siengchin, S.: PVA-based blends and composites. In: Biodegradable Polymers, Blends and Composites, pp. 309–326. Woodhead Publishing (2022)
58. Verma, A., Parashar, A., Packirisamy, M.: Tailoring the failure morphology of 2D bicrystalline graphene oxide. J. Appl. Phys. **124**(1), 015102 (2018)
59. Singla, V., Verma, A., Parashar, A.: A molecular dynamics based study to estimate the point defects formation energies in graphene containing STW defects. Mater. Res. Exp. **6**(1), 015606 (2018)
60. Verma, A., Parashar, A.: Molecular dynamics based simulations to study failure morphology of hydroxyl and epoxide functionalised graphene. Comput. Mater. Sci. **143**, 15–26 (2018)
61. Verma, A., Zhang, W., van Duin, A.C.: ReaxFF reactive molecular dynamics simulations to study the interfacial dynamics between defective h-BN nanosheet and water nanodroplets. Phys. Chem. Chem. Phys. **23**, 10822–10834 (2021)
62. Verma, A., Parashar, A., Packirisamy, M.: Role of chemical adatoms in fracture mechanics of graphene nanolayer. Mater. Today Proc. **11**, 920–924 (2019)
63. Verma, A., Parashar, A.: Characterization of 2D nanomaterials for energy storage. In: Recent Advances in Theoretical, Applied, Computational and Experimental Mechanics, pp. 221–226. Springer, Singapore (2020)
64. Verma, A., Jain, N., Parashar, A., Singh, V.K., Sanjay. M.R., Siengchin, S.: Design and modeling of lightweight polymer composite structures. In: Lightweight Polymer Composite Structures: Design and Manufacturing Techniques, pp. 193–224. Taylor & Francis Group (CRC Press), Boca Raton, Chapter 7 (2020)
65. Verma, A., Jain, N., Parashar, A., Singh, V.K., Sanjay, M.R., Siengchin, S.: Lightweight graphene composite materials. In: Lightweight Polymer Composite Structures: Design and Manufacturing Techniques, pp. 1–20. Taylor & Francis Group (CRC Press), Boca Raton, Chapter 1 (2020)
66. Verma, A., Parashar, A., Singh, S.K., Jain, N., Sanjay, M.R., Siengchin, S.: Modeling and simulation in polymer coatings. In: Polymer Coatings: Technologies and Applications, pp. 309–324. Taylor & Francis Group (CRC Press), Boca Raton, Chapter 16 (2020)
67. Verma, A., Jain, N., Rastogi, S., Dogra, V., Sanjay, M.R., Siengchin, S., Mansour, R.: Mechanism, anti-corrosion protection and components of anti-corrosion polymer coatings. In: Polymer Coatings: Technologies and Applications, pp. 53–66. Taylor & Francis Group (CRC Press), Boca Raton, Chapter 4 (2020)
68. Verma, A., Jain, N., Kalpana, Sanjay M.R., Siengchin, S., Jawaid, M.: Natural fibers based bio-phenolic composites. In: Phenolic Polymers Based Composite Materials, pp. 153–168. Springer Nature, Singapore, Chapter 10 (2020)
69. Bharath, K.N., Madhu, P., Gowda, T.Y., Verma, A., Sanjay, M.R., Siengchin, S.: Mechanical and chemical properties evaluation of sheep wool fiber-reinforced vinylester and polyester composites. Mater. Perform. Charact. **10**(1), 99–109 (2021)
70. Marichelvam, M.K., Manimaran, P., Verma, A., Sanjay, M.R., Siengchin, S., Kandakodeeswaran, K., Geetha, M.: A novel palm sheath and sugarcane bagasse fiber based hybrid composites for automotive applications: an experimental approach. Polym. Compos. **42**(1), 512–521 (2021)
71. Chaudhary, A., Sharma, S., Verma, A.: Optimization of WEDM process parameters for machining of heat treated ASSAB'88 tool steel using response surface methodology (RSM). Mater. Today Proc. (2021)
72. Chaudhary, A., Sharma, S., Verma, A.: WEDM machining of heat treated ASSAB '88 tool steel: a comprehensive experimental analysis. Mater. Today Proc. (2021)

73. Verma, A., Singh, V.K.: Experimental characterization of modified epoxy resin assorted with almond shell particles. ESSENCE-Int. J. Environ. Rehabil. Conserv., 36
74. Verma, A., Samant, S.S.: Inspection of hydrodynamic lubrication in infinitely long journal bearing with oscillating journal velocity. J. Appl. Mech. Eng. **5**(3), 1–7 (2016)
75. Verma, A., Parashar, A., Jain, N., Singh, V.K., Rangappa, S.M., Siengchin, S.: Surface modification techniques for the preparation of different novel biofibers for composites. In: Biofibers and Biopolymers for Biocomposites, pp.1–34 (2020)
76. Bisht, N., Verma, A., Chauhan, S., Singh, V.K.: Effect of functionalized silicon carbide nanoparticles as additive in cross-linked PVA based composites for vibration damping application. J. Vinyl Add. Tech. **27**(4), 920–932 (2021)
77. Kataria, A., Verma, A., Sanjay, M.R., Siengchin, S.: Molecular modeling of 2D graphene grain boundaries: mechanical and fracture aspects. Mater. Today Proc. (2021)

Computational Modelling of Deformation and Failure of Bone at Molecular Scale



Akarsh Verma and Shigenobu Ogata

Abstract Bone is a hierarchical composite material made up of ductile collagen phase, brittle hydroxyapatite mineral crystals and water. These phases organise at multiple length scales and form an extremely tough yet lightweight structure. Furthermore, bone tissue has an intrinsic ability to adapt its morphology by adding or removing the bone according to external mechanical loading. However, the decline in the bone quality significantly affects the skeleton's load-bearing capacity which leads to bone fractures. Bone fractures are common in elderly population suffering from osteoporosis and associated disease such as diabetes. To prevent fractures associated with poor bone quality, an understanding of the deformation and failure mechanism of bone at each length scale is required. In this article, we review the deformation and failure mechanism of bone tissue at the molecular scale. The insights gained from this study could be helpful in the development of treatment plans and therapeutic interventions for specific bone disorders.

Keywords Bone · Mineralised collagen fibril · Bone diseases · Molecular mechanics · Collagen mechanics

1 Introduction

Bone is a load-bearing hard tissue with multiple mechanical and biological functionalities like supporting the whole body, protecting vital organs, storage of mineral, etc., and it also enables leverage and movement in the human body [1]. Bone tissue has an intrinsic ability to adapt its morphology by adding or removing the bone according to external mechanical load and biochemical signal, and this functionality helps repair

A. Verma (✉)
Department of Mechanical Engineering, University of Petroleum and Energy Studies, Dehradun 248007, India
e-mail: akarshverma007@gmail.com

A. Verma · S. Ogata
Department of Mechanical Science and Bioengineering, Osaka University, Osaka 560-8531, Japan

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microcracks and fractures generated during daily activity. It has a complex arrangement at different length scales, which provides high mechanical performance, the different parameters contributing to higher mechanical performance except bone mineral density (BMD) are termed bone quality. Degradation in bone quality significantly affects the skeleton's load-bearing capacity, which causes serious problems, i.e., injury, loss of flexibility, and reducing the quality of life, which collectively leads to health, economic and social issues. Bone fractures are common in older age groups suffering from osteoporosis and associated disease such as diabetes. The broader aim of bone research is to predict the fracture assessment, alteration in bone quality due to disease, assess the performance of drugs on bone quality, and design artificial bone grafts and implants. Therefore, it is crucial to understand the mechanism of bone failure for the development of techniques to treat bone disorders and designing artificial bone graft and implants [2, 3]. Bone tissue is organised in a complex hierarchical fashion from molecular to macro scale which provides it remarkable toughness. Understanding the deformation and failure mechanics of bone at the molecular level could be helpful in the development of treatment plans for specific bone disorders. Therefore, in this review, we summarised the previous works performed on bone tissue focusing on the molecular mechanics of deformation under different states.

2 Hierarchical Structure of Bone

Bone is a hierarchical composite material predominantly made up of mineral crystals, collagen matrix and water which form its molecular to macroscopic length scale structures [4]. These constituent phases are arranged at seven hierarchies and form a complex highly augmented structure that provides high mechanical performance despite being lightweight [5–7].

At the molecular scale, bone tissue is formed of ductile organic matrix, brittle mineral phase and water. These three constituents are 32–44%, 33–43% and 15–20%, respectively of bone volume [8]. The organic matrix primarily consists of type 1 collagen (~ 90%), and the remaining contain other types of collagen and more than a hundred non-collagenous proteins such as osteocalcin, osteopontin, etc. [7]. The mineral consists of Hydroxyapatite [HA] crystal, which has a poor crystalline structure. Water is present in the form of free water and bound water [9]. Bone mineral is primarily composed of Hydroxyapatite (HA), with a Ca/P ratio less than 1.67 and it is responsible for the stiffness of bone. The HA crystal has a weak crystalline structure and is present in the form of a plate. The average length, width and thickness of HA crystals are 50 nm, 25 nm and 3 nm, respectively [10, 11]. Type 1 collagen is the primary constituent of the matrix phase, which contains about 90% of the matrix. The remaining 10% are non-collagenous protein and other types of collagen. Type 1 collagen is fibrous protein and it is made up of three polypeptide chains which are arranged in triple helix form. Triple helix molecule contains two $\alpha 1$ and one $\alpha 2$ chain. The diameter of the triple helix is 1.5 nm, and the length is

300 nm. The triple helix is assembled with crosslink and hydrogen bonds in the form of cylindrical fibril with a diameter of about 30–60 nm. In the type 1 collagen fibril, two types of crosslinks are found: (1) Enzymatic crosslink and (2) Non-Enzymatic crosslink. In the initial phase of crosslinking, the immature divalent bond (pyrroles) form the enzymatic crosslink which further post-translation converted into a mature trivalent crosslink, such as pyridinoline (PYD, or hydroxylysyl pyridinoline) and deoxypyridinoline (DPD, or lysyl pyridinoline). In addition to the type 1 collagen, other types of collagen present in bone are: type II, III, V, VI. IX, X, XII, XIV and XXIV [12].

At the nanoscale, the bone tissue is formed by the mineralisation of collagen fibrils. Further by self-assembly, it forms mineralized collagen fibre and is known as a fundamental building block of bone. The collagen fibrils have approximately 100 nm diameter and up to numerous micrometres in length. In the mineralized collagen fibril, the c-axis of HA crystal is oriented along the collagen fibril's longitudinal axis.

At the submicron scale, mineralised collagen fibres (of 1 μm diameter) are assembled in the form of a planer arrangement called lamellae (of 3–7 μm plate thickness). The lamellae contain the bone cell (osteocytes), and it is connected with the channel (~100 nm diameters) known as canaliculi. The different assemblies of lamella formed two types of bone: cortical bone and trabecular bone at the microscale. In cortical bone, lamellae wrap into the concentric layer (3–8 lamellae) around a central vascular canal (porosity) to form osteon. These osteons appear cylindrically of varying diameter 200–300 μm and are almost aligned along the bone's long axis. The gap amongst the osteons is filled by interstitial bone in the form of the matrix, whereas the osteons and the interstitial bone are separated by a thin line (of 2–5 μm thickness), called cementline while the old osteon does not have cementline.

At the mesoscale, the bone either represents the cortical bone or trabecular bone. Cortical bone consists of an osteon, interstitial matrix, cementline and haversian canal. At this scale, the structural feature of cortical bone is characterized by porosity, cortical thickness, the moment of inertia, and cross-section area. The trabecular bone formed by a porous network of rod and plate-like trabeculae is sandwiched with the cortical shell. Trabecular bone is mostly present in the epiphysis and metaphysis regions of long bones and the central regions of irregular short, and flat bones.

3 Molecular Mechanics of Bone Under Different States

3.1 Mineralisation

Bone is a natural composite made up of collagen, mineral and water. These phases organize at different hierarchies and provide a unique combination of strength and toughness to the bone. The deformation behaviour and mechanical properties of bone at various length scales have been reported extensively in the literature.

However, there are limited studies on the deformation behaviour of bone at molecular level. At nanoscale, collagen fibril is the building block of bone tissue. The interaction of collagen fibril, mineral and the amount of these phases' influences strength and toughness of bone. In a previous study [13], deformation behaviour of collagen microfibril with varying mineral (0–40%) density has been investigated using molecular dynamics simulation. To model collagen and hydroxyapatite structure, CHARMM force field [14] was modified. From simulation, it was observed that the minerals predominantly deposit in gap region whereas a small amount in overlap region. Furthermore, uniaxial tensile test revealed the deformation behaviour of the collagen fibril with different mineral contents. With the increase in mineralisation, modulus of collagen fibril increases. Moreover, a significant different deformation mechanism is observed for non-mineralised and mineralised collagen fibrils. For, non-mineralised fibrils, at small deformations, deformation is localised within gap region. On the other hand, with the increase in mineralisation, gap region becomes stiffer and deformation predominantly takes place within overlap region. Furthermore, another study has shed light on the large deformation mechanism associated with tensile loading of single collagen fibril [15]. At low strain (~5%), deformation is associated with uncrimping of the collagen fibril. In this stage, fibril enters in the elastic regime. Most of the deformation occurs in the overlap region for mineralised fibrils due to presence of minerals. Further, due to sliding between minerals collagen, load is transferred to collagen molecules. At ~20% strain, molecular slippage occurs between minerals and collagen and further minerals dislocate to transfer stresses to gap to overlap region. The backbone stretching of collagen is followed by the catastrophic failure of fibril. It was also observed that adding minerals to collagen fibril can increase the strength by 10 times and toughness by 35 times. From these studies, it is quite evident that mineralisation significantly contributes to strength and toughness of collagen fibril. Also, the amount of mineral content present in bone influences its deformation behaviour.

3.2 Effect of Hydration

Water consists of 10–20% volume of bone and it is present in two compartments-within pores and bound to organic matrix. Water is known to have effect on mechanical properties of bone at different length scales. However, its role on molecular mechanic of bone is not well explored. A previous study has investigated the effect of hydration state on deformation behaviour of mineralised and non-mineralised collagen fibril [16]. Tensile deformation behaviour of collagen fibril was investigated for 0, 2 and 4 wt% water content. It was observed that with the increase in the hydration, Young's modulus decreases for both mineralised and non-mineralised collagen fibrils. For higher hydration level (~4 wt%), water distributes in gap region and gap region undergoes higher deformation. Whereas at lower hydration levels, overlap region deforms more. With the increase in hydration, gap/overlap ratio increases

40% for non-mineralised and 16% for mineralised collagen fibrils. These observations could be helpful in understanding bone disorders which are associated with the alteration in water compartment.

3.3 Cross-Linking

The fibrillar structure of collagen is stabilized by post-translational modifications that forms intermolecular and interfibrillar crosslinks. Primarily there are two types of intermolecular crosslinks. The first is the enzymatic cross links which are formed by the enzyme lysyl-oxidase acting on specific lysine amino acids of the non-helical ends of the collagen. The second cross-links are produced by advanced glycation end products (AGEs). Enzymatic crosslinks are confined in the terminal regions whereas AGEs can form in various locations along the length of the collagen. In a previous study [17], effect of mature and immature crosslinks on deformation behaviour of collagen molecule was investigated through steered molecular dynamics in LAMMPS [18]. The mechanical response of crosslinks was evaluated using ReaxFF reactive force field. The collagen fibril undergoes three deformation stages. Initial elastic deformation is associated with uncoiling of collagen molecules. Second stage corresponds to linear region which is dominated by molecular sliding. Further, in third stage, stretching of tropocollagen molecule backbone until complete rupture of fibril. The density of crosslink influences the stiffness of fibrils by altering the number of interconnected molecules whereas mechanical properties of crosslinks influence the failure strength and strain of fibril. It was also observed that mechanical properties improve with maturation of crosslinks. Strength was two-fold and toughness was three-fold higher in fibril with mature crosslinks when compare with immature crosslinks. Also, fibrils with trivalent crosslinks exhibit brittle-like behaviour. The deformation mechanics of mineralised collagen fibril with crosslinks could provide insights on the mechanism of bone disorders associated with alteration in crosslinks profiles such as Type 2 Diabetes (T2D) [19].

3.4 Viscoelasticity and Deformation Rate

Bone exhibits viscoelastic properties; which are crucial to understand the fracture behaviour and failure of bone. The viscoelastic nature of bone helps it to dissipate energy and subsequently reduces the chances of fracture. Various investigation has been performed at micro and macro length scales to understand viscoelastic behaviour of bone. Previous studies have shown water as well as organic matrix influences the viscoelastic response of bone. However, very few studies are available at fibrillar level. A previous molecular dynamics study has investigated the viscoelastic properties of mineralised collagen fibril under dry and wet condition for shear and longitudinal loads [20]. It was reported that dynamic modulus increases

with an increase in the mineralisation. In the presence of intrafibrillar water, elastic component decreases but a considerable increase in viscosity was observed. It was found that water plays an important role in the mechanics of mineralised collagen fibril and improves its ability to dissipate the energy.

Furthermore, bone exhibits creep behaviour due to its viscoelastic nature. Creep is the deformation that occurs under prolonged, constant loads and can lead to permanent deformation of bone. The understanding of creep behaviour of bone becomes important in diseases such as spinal deformities and loosening of implants. Though there are various studies available on creep behaviour of bone at macro scale; but the insights at molecular level are still limited. The creep response of collagen molecules was also investigated in a previous study [21] using steered molecular dynamics simulation with CHARMM force field [22]. Two collagen molecules were connected through lysine-lysine crosslinks. Further, axial pulling force from 10 to 2000 pN was applied. The steady state strain was found higher for higher forces and it was less for larger molecular sizes. Further, the force wave speed was found in correlation with one-dimensional elastic wave speed in solid bars. A considerable creep behaviour was observed at molecular level in collagen fibrils indicating the viscoelastic nature of bone also arises from molecular length scale.

Since bone is a viscoelastic material, its deformation behaviour is also influenced by the rate of applied loading. Molecular mechanism of deformation behaviour of tropocollagen molecules at different loading rates was investigated in a previous study [23]. The collagen peptide sequence $[(\text{Gly-Pro-Hyp})_{10}]_3$ was stretched with pulling rate from 0.01 to 100 m/s and mechanical properties and deformation behaviour was analysed. Computational results show Young's modulus of tropocollagen depends on loading rate. For loading rate lower than 0.5 m/s, mechanical properties of tropocollagen are independent of loading rate. However, beyond this loading rate, a steep increase in Young's modulus was observed. Different molecular deformation mechanism was observed for different loading rates. For slow loading rates, triple helix uncoils completely, then undergoes some recoiling in the opposite direction followed by finally straightening. For intermediate loading rates, molecule uncoils linearly with an increase in strain. For high loading rate, triple helix does not uncoil during the stretching. These observations can explain the loading rate-dependent mechanical response observed at macro length scale [24].

3.5 Interfaces Within Bone

The hierarchical structure of bone consists of multiple interfaces between mineral and protein phases at different length scales. These interfaces provide toughness to bone tissue by dissipating energy. A previous study [25] has investigated the mechanical behaviour of interface between osteopontin (OPN) and hydroxyapatite (HA) using molecular dynamics simulation. Four different types of 18-residue osteopontin peptides with different acidic amino acids were investigated. GROMOS force field was used for OPN peptide. Further pulling force was applied on each peptide in

thickness and interfacial direction. Results show the interfacial mechanical properties are primarily governed by attractive electrostatic interaction between acidic amino acids in OPN peptide and calcium in hydroxyapatite. Higher number of acidic amino acids causes higher energy dissipation. The formation of new bonds between acidic amino acids and hydroxyapatite causes stick–slip type motion between OPN peptide and hydroxyapatite surface which results in high energy dissipation.

Another study investigated the adsorption of water at hydroxyapatite and collagen interface [26]. Two different collagen configurations were investigated; a well folded triple helix representing healthy bone and a partially unfolded triple helix representing diseased bone. It was observed that the presence of collagen strengthens the adsorption of water by hydroxyapatite. Further, the water adsorption density was significantly different between folded and partially unfolded collagen. The presence of collagen disrupts the hydrogen bond network that holds water in the solution, thus allowing more adsorption with hydroxyapatite. It was observed that triple helix structure of collagen allows a reduction in area per molecule by 30%, which means healthy bone has higher amount of bound water compared to diseased bone. Aging promotes the weaker water adsorption on hydroxyapatite surface and decrease in bound water with aging.

4 Bone Diseases and Disorders

The mechanical properties of bone are known to have deterioration with aging and diseases such as osteoporosis and diabetes [19, 27, 28] and can adversely affect individuals' quality of life. To develop treatment plans for these bone disorders, the knowledge of alterations that occurred in bone structure at different length scale is required. The changes that occurred in bone structure at molecular level under different diseased states are summarised in the following section.

4.1 Osteogenesis Imperfecta

Osteogenesis imperfecta or brittle bone disease is a genetic disorder that causes increased bone fragility, low bone mass and complications related to other connective tissues. Based on clinical features, four main types of OI have been identified. 80–90% of OI cases are caused by mutation of *COL1A1* or *COL1A2* genes. These gene codes are associated with type 1 collagen which is the most abundant protein in human body. Previous studies have shown cortical bone obtained from OI patients has poor collagen organisation, high mineral content and loss of mineral heterogeneity and high vascular canal density [29–32]. Missense mutation which alters a glycine codon in gene is the most common cause of OI. Previous computational study at molecular level has reported the effect of mutation on collagen fibril and further on mechanical properties [33]. This causes poor fibril packaging, reduction in

cross-linking density and poor mineral crystal growth. Due to the presence of glycine replacements, the normal fibril packaging is hindered and loss of lateral crystallinity is observed. Further, poorer lateral fibril packaging of collagen may affect crystal growth. It was also observed that molecular softening and weakening of intermolecular adhesion occurs as disease severity increases. All these factors lead to reduction in mechanical strength at collagen fibril level.

4.2 Osteoporosis

Osteoporosis is a fragile state of the skeleton in which a bone becomes weak which causes skeletal fractures either spontaneous or with minimal trauma during daily activity. The disease has no noticeable symptoms, so individuals do not know they have osteoporosis before the bone fracture. Fractures can be causing pain, life-altering, disability and loss of independence [34]. To understand the specific abnormalities and specific pathophysiological mechanisms in osteoporosis, a traditional disease model is employed [35]. Primary osteoporosis encompasses two separate things, one associated with menopausal estrogen loss and the other with ageing [36]. Riggs et al. [37] suggested Type I osteoporosis (occurs after menopause) is associated with loss of trabecular bone and whereas in Type II osteoporosis, the loss of bone occurs in both cortical and trabecular compartment in men and women. The increase in porosity of bone has a detrimental effect on its mechanical performance [27, 38, 39].

4.3 Type 2 Diabetes (T2D)

Type 2 diabetes (T2D) is a common chronic metabolic disease that has a massive burden on the public health system. Presently, T2D affects more than 387 million adults (which was about 10% population) worldwide and is projected to reach 700 million by 2045 [40]. Until recently, the skeleton is not included in the list of target organs affected by T2D. It is now well reputable that T2D is a self-governing risk factor for bone fractures. Which is not decrease the body mass index (BMI) and other factors related to osteoporosis [41]. New data from the epidemiological, pathophysiologic report and the studies from advanced investigation techniques have increased to understand how T2D alters metabolism and fracture risk of bone. Epidemiological data direct that T2D contracted in older population have a higher chance of fractures. It includes a 50–80% increased extremity fracture risk, and about 20% of all other clinical fractures in both man and women [42]. The T2D increased fracture risk despite the slightly higher value in bone mineral density in patients with T2D than in nondiabetic [19, 43].

The mechanical response of T2D bone is studied by various researchers at micro to macro length scales and it was found that accumulation of advanced glycation

end products in organic matrix decreases the toughness of bone [27, 44–46]. The investigations on molecular mechanics of T2D bone are limited. In a previous study, the effect of T2D on energy dissipation behaviour of bone was investigated using steered molecular dynamics simulation [47]. The molecular mechanism of energy dissipation associated with osteocalcin (OC) at hydroxyapatite mineral interface was provided. It was found that the α-helix of protein plays important role in energy dissipation mechanism. In the case of T2D, glycation of osteocalcin protein through covalent crosslinking between arginine and N-terminus region disrupts the alpha-helix structure, which significantly reduces the energy dissipation capacity of bone. The glycation of protein reduces the affinity of osteocalcin towards hydroxyapatite crystals. The absence of alpha helix structure due to glycation causes reduction in stick–slip mechanism or sacrificial bonding which are the main energy dissipation mechanism under healthy conditions.

4.4 Aging

Fracture resistance of bone is known to decline with aging. A previous study has investigated the effect of aging on the post translational modifications (PTM) of collagen molecules in mice model [48]. With aging, regions of triple helix of collagen undergo deamidation process where rearrangement of hydrogen bonding with water causes an increase in water molecules at the periphery of triple helix. The deamidation occurs when nitrogen on collagen backbone attacks the carbonyl sidechain to form an intermediate ring which further alters the functional groups of sidechains. To get more insight into the effect of deamidation on Type I collagen, molecular dynamics simulations were performed. Effect of deamidation, on hydrogen bonding patterns and stability of collagen molecules was investigated. The increased binding of water by the sidechain causes localize disruption of hydrogen bonding along the backbone of collagen. This hydrogen bonding is an important factor in the stability of helix. The alteration in optimal hydrogen bonding affects the stability of triple helix which causes unravelling of helix at deamidated sites. This indicates the redistribution of stability of triple helix with deamidated when compared with native molecule. The increase in water molecules interacting with deamidated sidechain may also cause swelling (increase in the diameter) of mineralised collagen fibrils (MCFs). This could affect the interstitial water between mineral crystals and collagen fibrils. The movement of water from interstitial spaces to extrafibrillar matrix can increase the possibility of formation of microcracks and which would result in reduction in fracture resistance of bone at a macroscopic length scale.

4.5 Collagen Denaturation

Collagen in bone is present in well-organised triple helix form. Many diseases and external stimulation such as loading and temperature can denature the native state of collagen. The disturbance in native collagen structure reduces mechanical competence of bone tissue at macroscale. Therefore, the understanding of changes that occurs in secondary structure of collagen caused by denaturation can be useful in the treatment of bone disorders.

Fourier Transform Infrared (FTIR) is extensively used technique to study the secondary structure of proteins. In a previous study [24], FTIR was used to quantify the thermal treatment-induced changes in the secondary structure of collagen. The amide I band (1590–1710 cm^{-1}) was used to investigate the secondary structure collagen. The subcomponents of Amide I band were centred at 1610, 1630, 1645, 1660, 1675 and 1690 cm^{-1}. In native state of collagen, a high-intensity peak is observed at 1660 cm^{-1}; with thermal treatment the intensity of this peak decreases whereas intensity of peaks centred at 1630, 1645, 1675 and 1690 cm^{-1} increases. The subcomponent at 1690 cm^{-1} arises from helices of aggerated collagen-like peptides and vanished with hydration of collagen [49]. Thermal treatment of bone increases this peak which indicates dehydration of collagen. The sub band component centred at 1645 cm^{-1} arises from the random coil structures. The considerable increase in this component with heat treatment indicates helix to coil transformation of collagen. FTIR results indicate that with the heat-treatment of bone, unordered structure in collagen increases which reduces the mechanical properties of bone at macro-scale. The alteration in collagen phase of bone occurs in the case of many diseases such as osteogenesis imperfecta [32, 50], Type 2 diabetes [19] which subsequently reduces the ability of bone to resist fracture.

5 Collagen Inspired Bio-composites

Collagen is the basic building block of biological tissues such as bone, skin, tendon and muscle. Its hierarchical structure provides it with remarkable strength and toughness. The structure of collagen has been utilised to develop bioinspired composites. Also, various computational studies at molecular level has been performed. In a previous study, effect of carbon nano tube (CNT) content on deformation behaviour of collagen-based biocomposite has been explored [51, 52]. The deformation mechanics of collagen fibril alters with alteration in mineral and water content which causes expansion and contraction of gap and overlap regions. The presence of CNT in non-mineralised collagen reduces the deformation of the gap region which increases the elastic modulus of collagen-CNT-based biocomposite. Also, the presence of CNT increases the thermal stability of composite when compared with mineralised collagen. Besides this, the authors also have a vast experience in the field of molecular dynamics and experimental characterisation of composites materials [53–99].

6 Conclusion and Future Perspectives

Bone fracture is common in elderly population and causes loss of mobility, increase mortality and reduces the quality of life, which collectively leads to health, economic and social issues. The aim of bone research is to prevent these fractures and improve overall bone health. The inability of bones to take intended load leads to fragility fractures which are primarily caused by deterioration in bone quality. The bone quality decreases with the advancement in aging and diseases. From previous studies, it is evident that the mechanical response of bone at molecular level significantly alters with alteration in mineralisation, hydration, crosslinking, and conformation of collagen molecules. These changes are associated with aging and various bone disorders. Therefore, molecular scale investigations can be helpful in the development of therapeutic intervention to treat bone disorders. Also, bone exhibits a remarkable combination of strength and toughness which arises from its complex hierarchical structure. Each hierarchy have its distinct deformation as well as damage mechanism. A thorough understanding of mechanical behaviour of each length scale is required for prevention of bone fracture and development of treatment plans. Though there are a significant amount of literature available on mechanical response of healthy as well as diseased bones at macro scale, but there are still limited studies on the molecular mechanics of bone under different conditions. Therefore, molecular level investigations along with experimental validation are required on bone tissue under different diseased conditions. Also, multi-scale modelling approaches can connect the mechanics of deformation between different length scales. These modelling strategies along with experimental observations could be helpful in the development of treatment plans to improve bone health and reduce bone fractures associated with poor mechanical competence of bone.

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References

1. Katsamenis, O.L., Jenkins, T., Thurner, P.J.: Toughness and damage susceptibility in human cortical bone is proportional to mechanical inhomogeneity at the osteonal-level. Bone **76**, 158–168 (2015). https://doi.org/10.1016/j.bone.2015.03.020
2. Li, S., Emrah, A.A., Silberschmidt, V.V.: Fracture process in cortical bone: X-FEM analysis of microstructured models (2013). https://doi.org/10.1007/s10704-013-9814-7
3. An, B., Liu, Y., Arola, D., Zhang, D.: Fracture toughening mechanism of cortical bone: an experimental and numerical approach. J. Mech. Behav. Biomed. Mater. **4**, 983–992 (2011)

4. Weiner, S., Wagner, H.D.: THE MATERIAL BONE: Structure-Mechanical Function Relations (1998)
5. Koester, K.J., Ager, J.W., Ritchie, R.O., Ager, J.W., III., Ritchie, R.O.: The true toughness of human cortical bone measured with realistically short cracks. Nat. Mater. **7**, 672–677 (2008). https://doi.org/10.1038/nmat2221
6. Ural, A., Zioupos, P., Buchanan, D., Vashishth, D.: The effect of strain rate on fracture toughness of human cortical bone: a finite element study. J. Mech. Behav. Biomed. Mater. **4**, 1021–1032 (2011). https://doi.org/10.1016/j.jmbbm.2011.03.011
7. Ural, A., Vashishth, D.: Hierarchical perspective of bone toughness–from molecules to fracture. Int. Mater. Rev. **59**, 245–263 (2014)
8. Sabet, F.A., Najafi, A.R., Hamed, E., Jasiuk, I.: Modelling of bone fracture and strength at different length scales. A Rev. 20–30 (2016). https://doi.org/10.1098/rsfs.2015.0055.
9. Burr, D.B.: Changes in bone matrix properties with aging. Bone **120**, 85–93 (2019). https://doi.org/10.1016/j.bone.2018.10.010
10. Schwarcz, H.P., Abueidda, D., Jasiuk, I.: The Ultrastructure of Bone and Its Relevance to Mechanical Properties, vol. 5 (2017). https://doi.org/10.3389/fphy.2017.00039.
11. Stock, S.R.: The mineral—collagen interface in bone. Calcif. Tissue Int. 262–280 (2015). https://doi.org/10.1007/s00223-015-9984-6
12. Tzaphlidou, M.: The role of collagen in bone structure: an image processing approach. Micron **36**, 593–601 (2005)
13. Nair, A.K., Gautieri, A., Chang, S.W., Buehler, M.J.: Molecular mechanics of mineralized collagen fibrils in bone. Nat. Commun. **4**, 1724–1729 (2013). https://doi.org/10.1038/ncomms2720
14. Brooks, B.R., Bruccoleri, R.E., Olafson, B.D., States, D.J., Swaminathan, S.A., Karplus, M.: CHARMM: a program for macromolecular energy, minimization, and dynamics calculations. J. Comput. Chem. **4**, 187–217 (1983)
15. Depalle, B., Qin, Z., Shefelbine, S.J., Buehler, M.J.: Large deformation mechanisms, plasticity, and failure of an individual collagen fibril with different mineral content. J. Bone Miner. Res. **31**, 380–390 (2016). https://doi.org/10.1002/jbmr.2705
16. Fielder, M., Nair, A.K.: Effects of hydration and mineralization on the deformation mechanisms of collagen fibrils in bone at the nanoscale. Biomech. Model. Mechanobiol. **18**, 57–68 (2019). https://doi.org/10.1007/s10237-018-1067-y
17. Depalle, B., Qin, Z., Shefelbine, S.J., Buehler, M.J.: Influence of cross-link structure , density and mechanical properties in the mesoscale deformation mechanisms of collagen fibrils. J. Mech. Behav. Biomed. Mater. 1–13 (2014). https://doi.org/10.1016/j.jmbbm.2014.07.008
18. Plimpton, S.: Fast parallel algorithms for short-range molecular dynamics. J. Comput. Phys. **117**, 1–19 (1995)
19. Sihota, P., Yadav, R.N., Dhaliwal, R., Bose, J.C., Dhiman, V., Neradi, D., et al.: Investigation of mechanical, material, and compositional determinants of human trabecular bone quality in type 2 diabetes. J. Clin. Endocrinol. Metab. (2021)
20. Milazzo, M., David, A., Buehler, M.J.: Molecular origin of viscoelasticity in mineralized collagen fibrils, 3390–400 (2021). https://doi.org/10.1039/d0bm02003f
21. Ghodsi, H., Darvish, K.: Characterization of the viscoelastic behavior of a simplified collagen micro-fibril based on molecular dynamics simulations. J. Mech. Behav. Biomed. Mater. **63**, 26–34 (2016). https://doi.org/10.1016/j.jmbbm.2016.06.006
22. MacKerell, A.D., Jr., Bashford, D., Bellott, M., Dunbrack, R.L., Jr., Evanseck, J.D., Field, M.J., et al.: All-atom empirical potential for molecular modeling and dynamics studies of proteins. J. Phys. Chem. B **102**, 3586–3616 (1998)
23. Gautieri, A., Buehler, M.J., Redaelli, A.: Deformation rate controls elasticity and unfolding pathway of single tropocollagen molecules. J. Mech. Behav. Biomed. Mater. **2**, 130–137 (2009). https://doi.org/10.1016/j.jmbbm.2008.03.001
24. Uniyal, P., Sihota, P., Kumar, N.: Effect of organic matrix alteration on strain rate dependent mechanical behaviour of cortical bone. J. Mech. Behav. Biomed. Mater. 104910 (2021)

25. Lai, Z.B., Wang, M., Yan, C., Oloyede, A.: Molecular dynamics simulation of mechanical behavior of osteopontin-hydroxyapatite interfaces. J. Mech. Behav. Biomed. Mater. **36**, 12–20 (2014). https://doi.org/10.1016/j.jmbbm.2014.04.002
26. Welborn, V.V.: Environment-controlled water adsorption at hydroxyapatite-collagen interfaces. RSC (2021). https://doi.org/10.1039/d1cp01028j
27. Yadav, R.N., Sihota, P., Uniyal, P., Neradi, D., Bose, J.C., Dhiman, V., et al.: Prediction of mechanical properties of trabecular bone in patients with type 2 diabetes using damage based finite element method. J. Biomech. **123**, 110495 (2021)
28. Naresh, R., Uniyal, P., Sihota, P., Kumar, S., Dhiman, V., Goni, V.G., et al.: Effect of ageing on microstructure and fracture behavior of cortical bone as determined by experiment and Extended Finite Element Method (XFEM). Med. Eng. Phys. **93**, 100–112 (2021). https://doi.org/10.1016/j.medengphy.2021.05.021
29. Carriero, A., Zimmermann, E.A., Paluszny, A., Tang, S.Y., Bale, H., Busse, B., et al.: How tough is brittle bone? investigating osteogenesis imperfecta in mouse bone. J. Bone Miner. Res. **29**, 1392–1401 (2014). https://doi.org/10.1002/jbmr.2172
30. Carriero, A., Doube, M., Vogt, M., Busse, B., Zustin, J., Levchuk, A., et al.: Altered lacunar and vascular porosity in osteogenesis imperfecta mouse bone as revealed by synchrotron tomography contributes to bone fragility. Bone **61**, 116–124 (2014). https://doi.org/10.1016/j.bone.2013.12.020
31. Dong, X.N., Zoghi, M., Ran, Q., Wang, X.: Collagen mutation causes changes of the microdamage morphology in bone of an OI mouse model. Bone **47**, 1071–1075 (2010). https://doi.org/10.1016/j.bone.2010.08.013
32. Andriotis, O.G., Chang, S.W., Vanleene, M., Howarth, P.H., Davies, D.E., Shefelbine, S.J., et al.: Structure—mechanics relationships of collagen fibrils in the osteogenesis imperfecta mouse model (2015)
33. Gautieri, A., Uzel, S., Vesentini, S., Redaelli, A., Buehler, M.J.: Molecular and mesoscale mechanisms of osteogenesis imperfecta disease in collagen fibrils. Biophysj **97**, 857–865 (2009). https://doi.org/10.1016/j.bpj.2009.04.059
34. Looker, A.C., Fan, B., Shepherd, J.A.: FRAX-based estimates of 10-year probability of hip and major osteoporotic fracture among adults aged 40 and over: United States 2013 and 2014. Natl. Health. Stat. Rep. 1–16 (2017)
35. Ettinger, M.P.: Aging bone and osteoporosis: strategies for preventing fractures in the elderly. Arch. Intern. Med. **163**, 2237–2246 (2003)
36. Marcus R, Majumder S.: The nature of osteoporosis. Osteoporosis 3–17 (2001)
37. Riggs, B.L., Wahner, H.W., Seeman, E., Offord, K.P., Dunn, W.L., Mazess, R.B., et al.: Changes in bone mineral density of the proximal femur and spine with aging: differences between the postmenopausal and senile osteoporosis syndromes. J. Clin. Invest. **70**, 716–723 (1982)
38. Kumar, N., Kumar, A., Uniyal, P., Ramalingaiah, B., Sharma, S., Goni, V.G., et al.: Mimicking high strength lightweight novel structures inspired from the trabecular bone microarchitecture. Philos. Trans. R. Soc. A. **378**, 20190448 (2020)
39. Uniyal, P., Sihota, P., Tikoo, K., Kumar, N.: Anatomical variation in intracortical canal network microarchitecture and its influence on bone fracture risk. J. Mech. Behav. Biomed. Mater. **123**, 104770 (2021)
40. Idf Diabetes Atlas (2017)
41. Schwartz, A.V.: Diabetes mellitus: does it affect bone? Calcif. Tissue Int. **73**, 515–519 (2003)
42. Rubin, M.R., Patsch, J.M.: Assessment of bone turnover and bone quality in type 2 diabetic bone disease: current concepts and future directions. Bone Res. **4**, 1–9 (2016). https://doi.org/10.1038/boneres.2016.1
43. Hofbauer, L.C., Brueck, C.C., Singh, S.K., Dobnig, H.: Osteoporosis in patients with diabetes mellitus. J. Bone Miner. Res. **22**, 1317–1328 (2007)
44. Karim, L., Bouxsein, M.L.: Effect of type 2 diabetes-related non-enzymatic glycation on bone biomechanical properties. Bone **82**, 21–27 (2016). https://doi.org/10.1016/j.bone.2015.07.028
45. Farr, J.N., Khosla, S.: Determinants of bone strength and quality in diabetes mellitus in humans. Bone **82**, 28–34 (2016). https://doi.org/10.1016/j.bone.2015.07.027

46. Karim, L., Moulton, J., Van Vliet, M., Velie, K., Robbins, A., Malekipour, F., et al.: Bone microarchitecture, biomechanical properties, and advanced glycation end-products in the proximal femur of adults with type 2 diabetes. Bone 2018:#pagerange#. https://doi.org/10.1016/j.bone.2018.05.030
47. Tavakol, M., Vaughan, T.J.: The structural role of osteocalcin in bone biomechanics and its alteration in Type–2 diabetes. Sci. Rep. 1–10 (2020). https://doi.org/10.1038/s41598-020-73141-w
48. Creecy, A., Brown, K.L., Rose, K.L., Voziyan, P., Nyman, S.: Post-translational modifications in collagen type I of bone in a mouse model of aging. Bone 115763 (2020). https://doi.org/10.1016/j.bone.2020.115763
49. Rabotyagova, O.S., Cebe, P., Kaplan, D.L.: Collagen structural hierarchy and susceptibility to degradation by ultraviolet radiation. Mater. Sci. Eng. C **28**, 1420–1429 (2008). https://doi.org/10.1016/j.msec.2008.03.012
50. Wallace, J.M., Orr, B.G., Marini, J.C., Banaszak, M.M.: Nanoscale morphology of Type I collagen is altered in the Brtl mouse model of osteogenesis imperfecta. J. Struct. Biol. **173**, 146–152 (2011). https://doi.org/10.1016/j.jsb.2010.08.003
51. Fielder, M., Nair, A.K.: A computational study of mechanical properties of collagen based bio composites. Int. Biomech. **7**, 76–87 (2020). https://doi.org/10.1080/23335432.2020.1812428
52. Zimmermann, E.A., Busse, B., Ritchie, R.O.: The fracture mechanics of human bone: influence of disease and treatment. Bonekey. Rep. **4**, 1–13 (2015). https://doi.org/10.1038/bonekey.2015.112
53. Verma, A., Parashar, A., Packirisamy, M.: Atomistic modeling of graphene/hexagonal boron nitride polymer nanocomposites: a review. Wiley Interdisc. Rev. Comput. Mol. Sci. **8**(3), e1346 (2018)
54. Verma, A., Singh, V.K., Verma, S.K., Sharma, A.: Human hair: a biodegradable composite fiber–a review. Int. J. Waste Resour. **6**(206), 2 (2016)
55. Verma, A., Singh, V.K.: Mechanical, microstructural and thermal characterization of epoxy-based human hair–reinforced composites. J. Test. Eval. **47**(2), 1193–1215 (2019)
56. Verma, A., Parashar, A.: Structural and chemical insights into thermal transport for strained functionalised graphene: a molecular dynamics study. Mater. Res. Express **5**(11), 115605 (2018)
57. Verma, A., Negi, P., Singh, V.K.: Experimental analysis on carbon residuum transformed epoxy resin: chicken feather fiber hybrid composite. Polym. Compos. **40**(7), 2690–2699 (2019)
58. Verma, A., Gaur, A., Singh, V.K.: Mechanical properties and microstructure of starch and sisal fiber biocomposite modified with epoxy resin. Mater. Perform. Character. **6**(1), 500–520 (2017)
59. Verma, A., Parashar, A., Packirisamy, M.: Effect of grain boundaries on the interfacial behaviour of graphene-polyethylene nanocomposite. Appl. Surf. Sci. **470**, 1085–1092 (2019)
60. Verma, A., Budiyal, L., Sanjay, M.R., Siengchin, S.: Processing and characterization analysis of pyrolyzed oil rubber (from waste tires)-epoxy polymer blend composite for lightweight structures and coatings applications. Polym. Eng. Sci. **59**(10), 2041–2051 (2019)
61. Verma, A., Negi, P., Singh, V.K.: Physical and thermal characterization of chicken feather fiber and crumb rubber reformed epoxy resin hybrid composite. Adv. Civ. Eng. Mater. **7**(1), 538–557 (2018)
62. Verma, A., Negi, P., Singh, V.K.: Experimental investigation of chicken feather fiber and crumb rubber reformed epoxy resin hybrid composite: mechanical and microstructural characterization. J. Mech. Beh. Mater. **27**(3–4) (2018)
63. Chaurasia, A., Verma, A., Parashar, A., Mulik, R.S.: Experimental and computational studies to analyze the effect of h-BN nanosheets on mechanical behavior of h-BN/polyethylene nanocomposites. J. Phys. Chem. C **123**(32), 20059–20070 (2019)
64. Jain, N., Verma, A., Singh, V.K.: Dynamic mechanical analysis and creep-recovery behaviour of polyvinyl alcohol based cross-linked biocomposite reinforced with basalt fiber. Mater. Res. Express **6**(10), 105373 (2019)
65. Verma, A., Joshi, K., Gaur, A., Singh, V.K.: Starch-jute fiber hybrid biocomposite modified with an epoxy resin coating: fabrication and experimental characterization. J. Mech. Behav. Mater. **27**(5–6) (2018)

66. Verma, A., Kumar, R., Parashar, A.: Enhanced thermal transport across a bi-crystalline graphene–polymer interface: an atomistic approach. Phys. Chem. Chem. Phys. **21**(11), 6229–6237 (2019)
67. Verma, A., Singh, V.K.: Experimental investigations on thermal properties of coconut shell particles in DAP solution for use in green composite applications. J. Mater. Sci. Eng. **5**(3), 1000242 (2016)
68. Verma, A., Singh, V.K., Arif, M.: Study of flame retardant and mechanical properties of coconut shell particles filled composite. Res. Rev. J. Mater. Sci. **4**(3), 1–5 (2016)
69. Verma, A., Parashar, A., Jain, N., Singh, V.K., Rangappa, S.M., Siengchin, S.: Surface modification techniques for the preparation of different novel biofibers for composites. In: Biofibers and Biopolymers for Biocomposites, pp. 1–34. Springer, Cham (2020)
70. Rastogi, S., Verma, A., Singh, V.K.: Experimental response of nonwoven waste cellulose fabric–reinforced epoxy composites for high toughness and coating applications. Mater. Perform. Character. **9**(1), 151–172 (2020)
71. Bharath, K.N., Madhu, P., Gowda, T.G., Verma, A., Sanjay, M.R., Siengchin, S.: A novel approach for development of printed circuit board from biofiber based composites. Polym. Compos. **41**(11), 4550–4558 (2020)
72. Verma, A., Jain, N., Parashar, A., Gaur, A., Sanjay, M.R., Siengchin, S.: Lifecycle assessment of thermoplastic and thermosetting bamboo composites. In: Bamboo Fiber Composites, pp. 235–246. Springer, Singapore (2021)
73. Singh, K., Jain, N., Verma, A., Singh, V.K., Chauhan, S.: Functionalized graphite-reinforced cross-linked poly (vinyl alcohol) nanocomposites for vibration isolator application: morphology, mechanical, and thermal assessment. Mater. Perform. Charact. **9**(1), 215–230 (2020)
74. Verma, A., Parashar, A.: Molecular dynamics based simulations to study the fracture strength of monolayer graphene oxide. Nanotechnology **29**(11), 115706 (2018)
75. Verma, A., Parashar, A.: The effect of STW defects on the mechanical properties and fracture toughness of pristine and hydrogenated graphene. Phys. Chem. Chem. Phys. **19**(24), 16023–16037 (2017)
76. Verma, A., Parashar, A., Packirisamy, M.: Tailoring the failure morphology of 2D bicrystalline graphene oxide. J. Appl. Phys. **124**(1), 015102 (2018)
77. Singla, V., Verma, A., Parashar, A.: A molecular dynamics based study to estimate the point defects formation energies in graphene containing STW defects. Mater. Res. Express **6**(1), 015606 (2018)
78. Verma, A., Parashar, A.: Molecular dynamics based simulations to study failure morphology of hydroxyl and epoxide functionalised graphene. Comput. Mater. Sci. **143**, 15–26 (2018)
79. Verma, A., Zhang, W., van Duin, A.C.: ReaxFF reactive molecular dynamics simulations to study the interfacial dynamics between defective h-BN nanosheet and water nanodroplets. Phys. Chem. Chem. Phys. **23**, 10822–10834 (2021)
80. Verma, A., Parashar, A., Packirisamy, M.: Role of chemical adatoms in fracture mechanics of graphene nanolayer. Mater. Today Proceed. **11**, 920–924 (2019)
81. Verma, A., Parashar, A.: Characterization of 2D nanomaterials for energy storage. In: Recent Advances in Theoretical, Applied, Computational and Experimental Mechanics, pp. 221–226. Springer, Singapore (2020)
82. Verma, A., Jain, N., Avinash Parashar, V.K., Singh, S.M.R., Siengchin, S.: Design and Modeling of Lightweight Polymer Composite Structures in the book titled Lightweight Polymer Composite Structures: Design and Manufacturing Techniques published by the Taylor & Francis Group (CRC Press). Boca Raton, Chapter **7**, 193–224 (2020)
83. Verma, A., Jain, N., Avinash Parashar, V.K., Singh, S.M.R., Siengchin, S.: Lightweight Graphene Composite materials in the book titled Lightweight Polymer Composite Structures: Design and Manufacturing Techniques published by the Taylor & Francis Group (CRC Press). Boca Raton, Chapter **1**, 1–20 (2020)
84. Verma, A., Parashar, A., Singh, S.K., Jain, N., Sanjay, M.R., Siengchin, S.: Modeling and Simulation in Polymer Coatings in the book titled Polymer Coatings: Technologies and Applications published by the Taylor & Francis Group (CRC Press). Boca Raton, Chapter 16, 309–324 (2020)

85. Verma, A., Jain, N., Rastogi, S., Dogra, V., Sanjay, M.R., Siengchin, S., Mansour, R.: Mechanism, Anti-Corrosion Protection and Components of Anti-Corrosion Polymer Coatings in the book titled Polymer Coatings: Technologies and Applications published by the Taylor & Francis Group (CRC Press). Boca Raton, Chapter 4, 53–66 (2020)
86. Verma, A., Jain, N., Kalpana, S.M.R., Siengchin, S., Jawaid, M.: Natural Fibers Based Bio-Phenolic Composites in the book titled Phenolic Polymers Based Composite Materials published by the Springer Nature. Singapore, Chapter 10, 153–168 (2020)
87. Bharath, K.N., Madhu, P., Gowda, T.Y., Verma, A., Sanjay, M.R., Siengchin, S.: Mechanical and chemical properties evaluation of sheep wool fiber-reinforced vinylester and polyester composites. Mater. Perform. Charact. **10**(1), 99–109 (2021)
88. Marichelvam, M.K., Manimaran, P., Verma, A., Sanjay, M.R., Siengchin, S., Kandakodeeswaran, K., Geetha, M.: A novel palm sheath and sugarcane bagasse fiber based hybrid composites for automotive applications: an experimental approach. Polym. Compos. **42**(1), 512–521 (2021)
89. Chaudhary, A., Sharma, S., Verma, A.: Optimization of WEDM process parameters for machining of heat treated ASSAB'88 tool steel using Response surface methodology (RSM). Mater. Today: Proceed. (2021)
90. Chaudhary, A., Sharma, S., Verma, A.: WEDM machining of heat treated ASSAB '88 tool steel: a comprehensive experimental analysis. Mater. Today: Proceed (2021)
91. Verma, A., Singh, V.K.: Experimental characterization of modified epoxy resin assorted with almond shell particles. ESSENCE Int. J. Environ. Rehabil. Conserv. **36**
92. Verma, A., Samant, S.S.: Inspection of hydrodynamic lubrication in infinitely long journal bearing with oscillating journal velocity. J. Appl. Mech. Eng. **5**(3), 1–7 (2016)
93. Verma, A., Parashar, A., Jain, N., Singh, V.K., Rangappa, S.M., Siengchin, S.: Surface modification techniques for the preparation of different novel biofibers for composites. Biofibers Biopolym. Biocompos. 1–34 (2020)
94. Bisht, N., Verma, A., Chauhan, S., Singh, V.K.: Effect of functionalized silicon carbide nanoparticles as additive in cross-linked PVA based composites for vibration damping application. J. Vinyl Add. Tech. **27**(4), 920–932 (2021)
95. Kataria, A., Verma, A., Sanjay, M.R., Siengchin, S.: Molecular modeling of 2D graphene grain boundaries: mechanical and fracture aspects. Mater. Today: Proceed. (2021)
96. Arpitha, G.R., Verma, A., Sanjay, M.R., Siengchin, S.: Preparation and experimental investigation on mechanical and tribological performance of hemp-glass fiber reinforced laminated composites for lightweight applications. Adv. Civ. Eng. Mater. **10**(1), 427–439 (2021)
97. Deji, R., Verma, A., Kaur, N., Choudhary, B.C., Sharma, R.K.: Density functional theory study of carbon monoxide adsorption on transition metal doped armchair graphene nanoribbon. Mater. Today Proceed (2021)
98. Deji, R., Verma, A., Choudhary, B.C., Sharma, R.K.: New insights into NO adsorption on alkali metal and transition metal doped graphene nanoribbon surface: a DFT approach. J. Mol. Graph. Model. (2021). https://doi.org/10.1016/j.jmgm.2021.108109
99. Deji, R., Jyoti, R., Verma, A., Choudhary, B.C., Sharma, R.K.: A theoretical study of HCN adsorption and width effect on co-doped armchair Graphene Nanoribbon. Comput. Theor. Chem. 113592 (2022)

A Review on the Deformation Mechanism of Soft Tissue Collagen Molecules: An Atomistic Scale Experimental and Simulation Approaches



Ravinder Jhorar and Chhatar Singh Lamba

Abstract Understanding the structural and mechanical response of biological tissues is important for their diagnosis and treatment. The macroscopic mechanical behavior of tissues is correlated with their different hierarchy. The collagen main is main responsible for providing the complex mechanical behavior to soft tissue. Therefore, it becomes important to investigate the mechanical properties of collagen molecules. A significant amount of work has been done to characterize or predict the structural and mechanical behavior of collagen molecules in response to tissue level loading. Also, several studies are done to investigate the alteration of collagen molecule in diseased tissues. The available research works are either experimental based or simulation based. It is important to review the existing literature to summarize all experimental and simulation techniques, which will be helpful to future research work. Therefore, this article summarized the existing experimental and simulation techniques related to collagen molecules. Also, we have presented the literature where the effect of disease on the collagen molecules has been investigated. We have summarized the work related to experimental techniques such as Atomic force microscopy (AFM), Raman spectroscopy, attenuated total reflection Fourier transform infrared spectroscopy (ATR-FTIR), and fluorescence collagen hybrid peptide. Another side, this review work also presented the simulation method related to molecular dynamics. Numerous research works have been done to simulate the mechanical behavior (e.g., elastic, viscoelastic, and dynamic mechanical behavior) of collagen molecules using molecular dynamics. This review article will provide help to the research to explore their experimental and simulation expertise toward the biomechanics of soft tissues.

Keywords Biological tissue · Collagen molecules · AFM · Raman · ATR-FTIR · F-CHP · MD

R. Jhorar · C. S. Lamba
Department of Mechanical Engineering, Manipal University Jaipur, Jaipur, India

R. Jhorar (✉)
Mechanical Engineering, Delhi Skin and Entrepreneurship University, Delhi, India
e-mail: ravinder.jhorar@dseu.ac.in

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1 Introduction

The understanding of the biomechanical function of living connective soft tissues is important for tissues' treatment, pathological diagnostic, and device development. As connective tissues are composed of complex microstructure, the extensive accurate approaches are required for predicting the mechanical behavior of the tissue under and over physiological loading conditions [1–7].

1.1 Common Structure of Soft Tissue

Most soft tissues, such as skin, tendon, ligament, and aorta, consist of a dense network of collagen fibre along with elastin which are embedded with ground substance (e.g., proteoglycans and water) [4, 8–18]. Collagen fibres share about 70% dry weight of the tissue [8, 13, 14, 19–21] and are main responsible for tissue's mechanical strength. Collagen fibre is the bundle of fibrils whereas the staggered arrangement of triple-helical tropocollagen forms the fibril which is characterized by D-periodicity. The typical value of the D-periodicity in all soft is found 67 nm [10, 13, 14, 18, 22, 23]. In all type of connective tissues, collagen is the main to provide the mechanical strength, and also the arrangement of collagen fibre makes tissue nonlinear elastic and anisotropic. The different phases of collagen fibre deformations such as rotation, bending, straightening, and stretching are associated with the different phases of stress–strain diagram of the tissue (Fig. 1) [8–10, 13, 18, 24]. The structural arrangement of collagen fibres depends on the tissue functionality, for example, the collagen fibre in ligament and tendon is straight and aligned in their axial direction as the tendon/ligament is functionally active to support axial tensile load between the bones

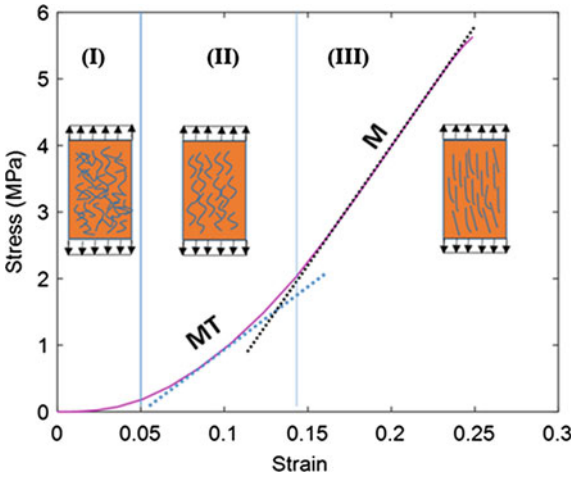


Fig. 1 Deformation phases of collagen fibres (I) rotation and bending (II) bending and straightening (III) Stretching. These different phases lead to a nonlinear response of tissue

or bone and mussels [25–34]: Aorta is composed of three layers, namely, the tunica intima, inner most layer of the arteries and composed of a single layer endothelial cells lining which is resting on a thin basal membrane. During homeostatic reaction, this layer stores baseline level of stress [4], the tunica media, middle layer of the arteries, and is composed of a complex three-dimensional network of smooth muscle cells, elastin, and buddle of collagen fibres. The collagen fibre in the media is aligned circumferentially and provides strength to media at lower and moderate value of tissue deformation [4, 19, 35]; the tunica adventitia, outer layer of the arteries, mainly consists of thick bundle of collagen fibre. Collagen fibre is arranged in helical manner in the adventitia layer. The adventitia layer is very compliant at a low value of stretch but acts like a stiff jacket tube at large stretch which protects arteries from stretching and rupture [4, 35]. Another important connective tissue of the human body is skin. Surface area wise, skin is the largest organ of the human body as it covers about 1.5–2.0 m^2 area of the body. Another side, it carries about 15% weight of the body. Skin is a prime protector against the external aggressions such as environmental, mechanical, and thermal [9, 11, 36–39]. Skin has a complex structure with three layers, namely, the epidermis (outer layer), the dermis (middle layer), and the hypodermis (inner layer). Out of these three layers, the dermis is the main to provide mechanical strength to the skin and is composed of elastin, collagen fibre, and ground substance [39, 40]. This layer also supports the epidermis and provides nourishment to it [40, 41]. Elastin fibres share is about 1–4% of the dry dermis weight, provides resilience to the dermis, and keeps the skin smooth [8, 42, 43]. However, collagen fibres contribute about 75% of the dry dermis weight and provide strength to the dermis at medium and large deformations. Further, the aim of this review is to summarize the nano and molecular level mechanical behavior of tissue and the required approach to investigate or simulate the nano/molecular response of the tissue. The lower hierarchy of the soft tissue is tropocollagen and is the main contributor toward the mechanical strength of tissues.

1.2 Structure of Tropocollagen

The tropocollagen or collagen molecules appear like rod and are made up of three polypeptide strands, each of which is a left-handed alpha helix (Fig. 2) [8, 44–47]. Initially, these three left-handed alpha helices are twisted together into a right-handed triple helix or procollagen with C-terminal and N-terminal ends. During, procollagen peptidase, C-terminal, and N-terminal ends are cleavage, and procollagen is converted into tropocollagen [48]. The tropocollagen is stabilized by the hydrogen bonds and there are also some covalent cross linking within the triple helices whereas their amount with the tropocollagen varies to form a different type of collagen in different soft tissues. In tropocollagen helix, the amino acids in every three chains are arranged regularly and generally follow the sequence of Gly-X-pro of Gly-X-Hyp [48, 49] (Fig. 2), where X is any of the various amino acid residues. The triple helix

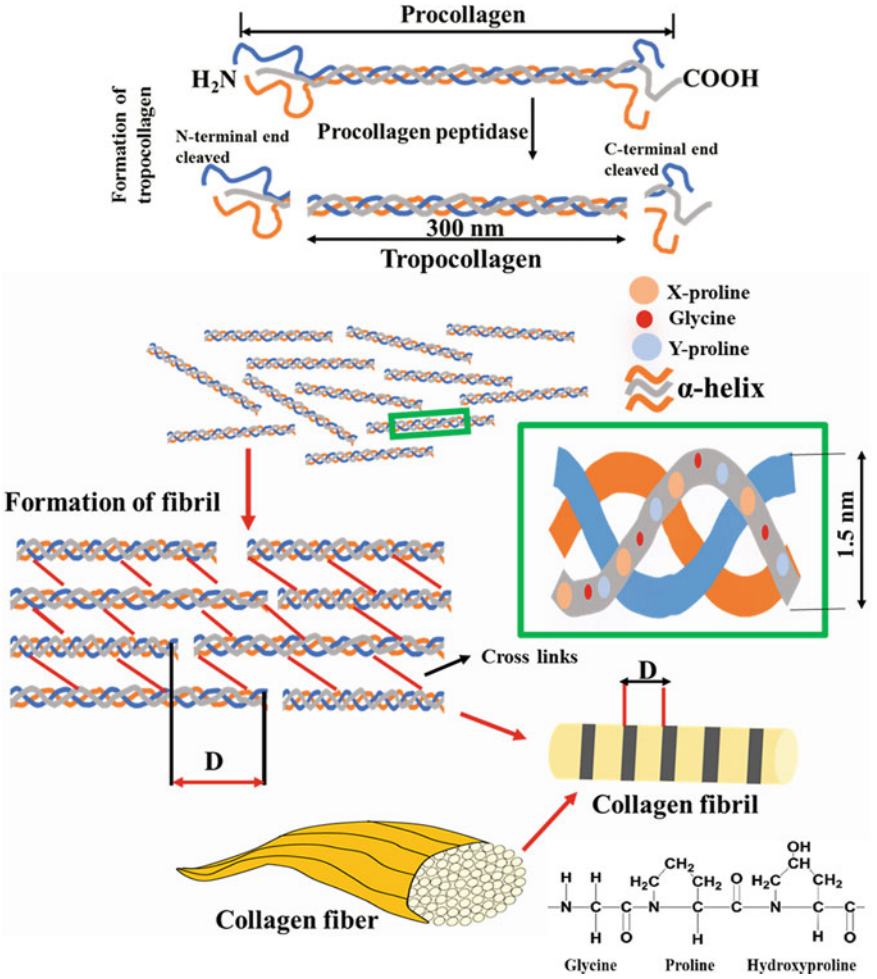


Fig. 2 Hierarchical schematic of collagen fibres from tropocollagen to collagen fibre

tightens under the starching, resisting tensile load, and making collagen inextensible. Glycine is the smallest residue of amino acid and plays a unique role in fibrous structural proteins.

This complex structure tropocollagen is interconnected through glycosaminoglycan cross link and arranged in a staggered manner to form the collagen fibrils. The typical length and diameter of tropocollagen are 300 and 1.5 nm. At molecular level of the connective tissues, tropocollagen is considered main to govern the biomechanical properties of the tissues. Also with different pathological conditions such as disease and age their structure may alter which affects the macroscale level and the mechanical behavior of the tissue. Also, it is believed that accumulation of damage in

the tissue due to physiological or over physiological loading starts from the tropocollagen lever. Numerous experimental and numerical studies are performed to capture the alteration in collagen molecules with age, disease, and mechanical loading. Also, several studies are focused to investigate the collagen molecules alteration and its correlation with biomechanics of the soft tissues. Further, along with tropocollagen the cross linking also contributes significantly to the mechanical properties of tissue and generally they are associated with the elastic response of the tissue.

1.3 Collagen Cross Link and Their Role in Mechanical Response of Tissue.

Generally, the collagen cross links are categorized into the enzymatic and non-enzymatic, depending on the origin of their formation. The enzymatic cross linking form during the collagen fibrillar assembly and these cross linkings are directed by cell [50]. The enzymes, such as lysyl oxidase and hydroxylase, act on particular molecular segments deamination the lysine and hydroxylysine residues located at the telopeptides of the collagen molecule [48]. The enzymatic cross link is responsible for stabilizing the fibrillar structure and is the cause of stiffening of tissues [51].

Another side, the non-enzymatic cross links (NE-xLR) are the results of instantaneous chemical reactions between collagen and glucose [50, 52, 53]. The formation of NE-xLR is initiated with a Maillard reaction [54], which is a reaction between amino acids and sugar [54]. Further, oxidation leads to the formation of advance glycation end products (AGEs). The cross links can form anywhere between the helical domains of adjacent collagen molecules [55]. These types of cross linking increase the stiffness very drastically [51, 55–58]. In addition to this, AGEs also play a significant role to develop the collagen related pathologies such as diabetes mellitus (type II) and kidney pathology [51, 55–60].

2 Different Types of Mechanical Response of the Tissue Associated with Nanoscale and Molecular Level

2.1 Elastic Response

Most of the soft tissues show nonlinear elastic behavior which is mainly governed by the deformation behavior of collagen fibrils, fibril elastic modulus, fibril structure, tropocollagen stability, and inter and intra cross linking of tropocollagen [8]. Any alteration at nano/molecular level may affect the elasticity of soft tissue. Further, the quality of these units of tissues affects their remodeling as the alteration at nano and molecular level is directly associated with the mechanotransduction pathway [6, 47,

50, 61–63]. Therefore, the understanding about the elastic and structural properties of lower hierarchy of soft tissue is important for disease diagnosis and its treatment.

2.2 Viscoelastic Properties

Viscoelastic properties of the soft tissues are defined by their strain rate dependent behavior, hysteresis loses under cyclic loading (energy dissipation), dissipation in stress with time (stress relaxation) value at constant amount hold strain, and increase in strain (creep) with time against the constant hold stress. Generally, the viscoelastic response of the soft tissues is model by the arrangement of spring dashpot where spring represents elastic part and dashpot presents viscous part. Viscoelastic properties provide damping to the tissue. The viscoelastic behavior of tissues comes from the interaction of different constituents such as water, collagen fibre, proteoglycans, fibre sliding, and fluid flow. But the viscoelastic behavior such as creep and stress relation is associated with the recruitment of collagen fibrils, tropocollagen folding-unfolding, and rearmament of collagen molecules as tissue extends under constant load or stress [64–67]. The viscoelastic properties of tissue alter with age and disease as aging and disease involves to change the collagen cross link conformation as well as collagen molecule structure [50, 63, 68–71]. Numerous, studies have been done to expose the underlying mechanism of viscoelasticity and its association with collagen molecules [64–67].

2.3 Poroelastic Property

Almost all soft tissues, water shares about 70% volume of the tissue and it is entrapped between the collagen molecules, fibrils, and fibre. During the stretching of the tissue, the collagen fibers come under the stress which creates a pressure gradient on the matrix. This pressure gradient pushes the fluid outside the tissue which results to decrease in volume. This fluid flow properties of tissue under the loading is known as Poroelastic behavior of the tissue [61, 72]. Generally, the fluid flow response of tissue is loading rate or loading frequency dependent. This behavior of the tissue is associated with the osmotic pressure of the tissue and mathematically Poroelastic behavior of tissue is modeled using Darcy's law [72]. It is also well documented that collagen confirmation alters with dehydration of the tissue.

3 Collagen Structural Mutation Related Changes in Mechanical Properties of Tissue

The collagen mutation alters the mechanical properties of the soft tissues such as collagen in Ehlers-Danlos syndrome and Alport syndrome lose their elasticity, flexibility, and structural integrity which results in connective tissue becoming more prone to damage. Further, in osteogenesis imperfecta, the small amino molecules Gly is replaced by another heavier amino acid in α-chain which impairs the folding of tropocollagen and results in brittle bone. In diabetic tissue collagen molecule lose their structural integrity such as loss in packing of collagen fibril, increase in fragmentation of fibrils, change in its D-periodicity, and degradation in flexibility and elasticity which ultimately affect the mechanical properties of the connective tissue [50, 73].

4 Deformation Mechanism of the Lower Hierarchy of the Soft Tissue

Generally, the deformation of collagen is associated with the stretching of tissue. At lower value of load/deformation the collagen fibre does not deform and deformation of elastin and ground substance is associated with the macroscopic deformation of tissue. Further, with increasing value of applied force or deformation collagen fibre start to rotate, bend, straight, and stretching. For more details please see the work of Wang [10]. During the stretching, the applied load or deformation transfer among the collagen fibrils through shearing between collagen fibrils and extracellular matrix and starching of Glycosaminoglycans cross links [26]. Once the fibrils are in sufficient stretch, the allied strain is transferred to the tropocollagen through cross linking and deformation of collagen molecules starts [9–11, 15, 26]. During the deformation of collagen molecules, the weaker bond of hydrogen break and once the load is removed these bonds again forms [10]. If the load or deformation is repetitive or its value is large, then the breakage of hydrogen bonds does not recover and collagen molecules become denatured which is known as molecular damage of collagen or soft tissue [11, 47].

Therefore, for capturing this complex deformation and damage response of lower hierarchy of the soft tissue, several experimental and molecular dynamic-based computation approach has been developed. These techniques and their outcomes are summarized in the upcoming sections.

5 Experimental Approach to Investigate the Nano and Molecular Structure of Tissues

5.1 Nano Mechanical Analysis for Various Soft and Hard Tissue (Atomic Force Microscopic)

Dwivedi et al. [11] utilized AFM to quantify the damage in collagen fibril of skin. Authors have measured the collagen fibril plastic strain by measuring the change in collagen fibril D-periodicity (plastic strain of collagen fibrils = $(D_{\text{after-loading}} - D_{\text{before-loading}})/D_{\text{before-loading}}$). The plastic strain accumulation of collagen fibrils was quantified corresponding to different values of cyclic frequencies. The accumulation of plastic strain in fibrils was found more corresponding to the lower frequency also the value of plastic strain was small than the value of tissue's permanent strain, which indicates that during the tissue deformation some part of applied strain is accumulated in the interfibrillar sliding (Fig. 3a, b). This study explores the capability of AFM to capture the nano level damage of tissue. Further, Jones et al. (2019) [74] coupled AFM with a reflected light module to capture the effect of aneurysm on the collagen fibril. They have observed clear abnormality in collagen fibrils due to disease where a reduction in D-periodicity and increase in collagen fibril. Further, a study performed on the expanded dog scalp [75] used AFM to quantify the effect of anisotropic expansion on the collagen fibril mechanical and structural properties. In this study AFM was used to measure the structural and mechanical properties of collagen fibrils. The obtained results of this study demonstrated that tissue expansion performed during skin grafting does not alter the morphology and mechanical properties of collagen fibrils as above-mentioned collagen fibril structural and mechanical parameters were not found significant different between expanded and control tissue. These results suggest that tissue expansion procedure does not alters the mechanotransduction of the tissue. For measuring the young modulus, force spectroscopy was used and Hertzian model was fitted with the contact region of force curve.

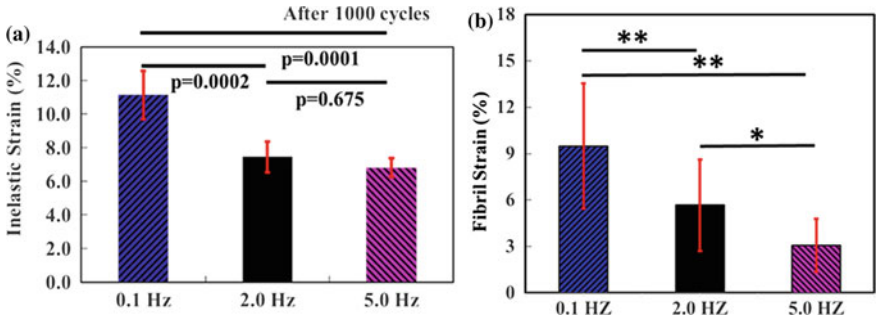


Fig. 3 **a** Inelastic strain at the tissue level, **b** plastic strain on the collagen fibrils. Adopted from Dwivedi et al. (2020) [11]

Argyropoulos et al. (2016) investigated the effect of diabetes on the collagen fibrils of the skin where skin was collected from the diabetic and healthy human groups. For comparing the mechanical properties of collagen between diabetic and control group, they have used AFM. Initially, they have determined the morphology of collagen fibrils and found that the collagen fibrils lose their native intactness, became loose, randomly oriented, and rough in diabetic skin. This morphological change of collagen fibrils affects the mechanotransduction pathway and then alters the remodeling of soft tissue during the infection or inflammation [50]. The increase roughness in collagen fibrils indicates that elevated matrix metalloproteinases (MMPs) may combust the collagen fibrils which results in their fragmentation and rough. This study showed that over formation of MMPs causes fragmentation of collagen fibrils and the impaired mechanical behavior of tissue. Further, they have observed that the key mechanical properties such as traction force and tensile strength of fibrils increase by 182% and 197%, respectively, while collagen fibril deformation decreased by 58% in the diabetic group than the control group.

Andriotis et al. [73] have investigated the relationship between structure and mechanical properties of collagen fibrils of osteogenesis imperfecta mouse model. In this study they have involved the 5-month-old mice. The tails of these mice were dissected and stored in gauze soaked in PBS solution Tendon from the tail were pulled using scalpel and tweezers. Each tendon was sectioned to expose the collagen fibrils and then scanned using the atomic force microscope. The AFM-based nano indentation was also performed on the hydrated, air dried, and ethanol dehydrated samples. The imaging and indentation of collagen fibrils was done in contact mode. The indentation modulus of collagen fibrils is measured using the Eq. 1–4.

$$E_r = \frac{1\sqrt{\pi}}{\alpha 2} \frac{S_c}{\sqrt{A_c(h)}} \tag{1}$$

where α is a dimensionless parameter which is close to unity [reference]. E_r is reduced modulus and defined as [reference],

$$\frac{1}{E_r} = \frac{(1 - \nu^2_{\text{sample}})}{E_{\text{sample}}} + \frac{(1 - \nu^2_{\text{indenter}})}{E_{\text{indenter}}} \tag{2}$$

where ν_{sample} and ν_{indenter} are Poisson's ratio of the sample and indenter, respectively, and E_{sample} and E_{indenter} the elastic modulus of sample and indenter, respectively. For AFM tip, the elastic modulus is very much large in comparison to sample modulus so the Eq. (2) is reduced to [reference].

$$\frac{1}{E_r} = \frac{(1 - \nu^2_{\text{indenter}})}{E_{\text{indenter}}} \tag{3}$$

After substituting Eq. (3) in Eq. (1), one can get the final form of elastic modulus.

$$E_{\text{sample}} = \frac{1\sqrt{\pi}}{\alpha 2}(1 - \nu^2_{\text{indenter}})\frac{S_c}{\sqrt{A_c(h)}} \tag{4}$$

Further, they also measured the swelling capacity of collagen fibrils. They have found that the mechanical response of collagen fibrils in OIM and WT are different between air dried and hydrated state. As in air dry state, the modulus of OIM fibrils is smaller than the modulus of WT fibrils whereas this trend was found to reverse in hydrated state. Further, the swelling capacity was found higher in WT specimen than OIM specimens. Another side the OIM collagen fibrils shrink more than the WT when treated chemically in ethanol.

It is well-known fact that the hydration influences the biomechanics of soft tissue strongly. Consequently, factors influencing the hydration in tissues, namely, osmolality and osmotic pressure, generated within tissue through proteoglycans and glycoproteins [26, 61]. A lot of work has been done on the macroscopic mechanical properties and their dependence on the hydration. Where the change in tissues' apparent mechanical properties such as stiffness, failure stress, and failure strain has been observed with change in tissue hydration [27, 76, 77]. The change in tissue level mechanical properties is associated with the change in tissue structural and mechanical properties. As discussed earlier that collagen fibrils are building block of most of the soft and hard tissue. It is also well established that collagen fibrils are chemically altered (e.g., enzymatic and non-enzymatic cross links), resulting in an increase in stiffness of soft tissues [78, 79]. Hence, it can be believed that amount of hydration can affect the structural and mechanical response of the collagen fibrils and then macro-mechanical properties of soft tissues. Using AFM-based indentation, Grant et al. showed one-fold increase in collagen fibril stiffness when exposing collagen fibrils from buffer to 1 M KCI at pH 7 as well as to increase ionic strength. Another side, another study did not observe a significant increase in the AFM-based tensile elastic modulus when collagen fibrils were exposed from buffer to 1 M KCL at pH7 [77]. The difference in the magnitude could be expected due to different loading conditions as Gant et al. [77] performed transverse indentation whereas Svensson et al. [80] applied longitudinal loading. These results indicate the anisotropic nature of collagen fibril or transversely isotropic. Therefore, it can be believed that tuning the mechanics of individual collagen fibrils can efficiently be achieved by adjusting fibril hydration through exerting osmotic pressure [81]. Hence, osmotic pressure can be achieved through colocalization of highly negative charged molecules such as glycosaminoglycans (GAGs), proteoglycans (PGs), and lipids [81]. A significant amount of work has been done where the mechanics of single collagen fibrils was explored using atomic force microscopy. In order to expose the effect of alteration in osmotic pressure at the level of individual collagen fibrils, Andriotis et al. [76] presented an AFM-based study where they have investigated the effect of hydration on the tensile properties of individual collagen fibrils within the physiological strain regime. In this study, others conducted atomic force microscopy experiments on individual collagen fibrils at varying osmotic pressures in a physiological tensile loading case. For more experimental details please refer the work of Andriotis et al. [76]. Initially they have measured the collagen shrinking and stiffness using the

AFM where the control collagen fibrils were fully submerged in an open fluid cell, allowing experiments in aqueous solution and then exposed to varying concentrations of a short-chain poly-ethylene glycol, PEG, 10 mM after that partially dehydrated upon increasing PEG concentration in a stepwise fashion. They have found that fibrils behave like a highly hygroscopic material as diameters of collagen fibrils increases significantly as going from air dried to a fully hydrated state in PBS at pH 7.5. This swelling nature of collagen fibrils is due to the many donor and acceptor sites of H-bond formation [82]. As, at the dried state structural water is tightly bound to the collagen molecules. However, collagen fibrils submerged in an aqueous solution are incorporated into the fibrillar structure which increase the collagen intermolecular distance [83] by reducing the free energy between the molecules due to its polar and highly dielectric nature. They also observed that swelling collagen fibrils contain about three times more water than ground substance. These volumetric changes of collagen fibrils can alter the osmotic pressure inside the collagen molecules, resulting in change in stiffness of the fibrils [83], as about 24-fold increase in elastic modulus was observed with an increase in PEG concentration. Also, these changes in fibril properties were found reversible once they were put back in their initial states. The mechanical properties of collagen fibrils were also found dependent on the concentration of collagen fibrils as changing pH from 5 to 7, the indentation modulus was found to change from 3.1 to 2 MPa [77]. Grant et al. [77] also found that the mechanical properties such as elastic modulus is strongly influenced by changing the ionic concentration. Further, the AFM -ased tensile testing shows that the elasticity of collagen fibrils increases about six-fold as the fibrils move from PBS to PEG [76] which could be due to partial dehydration of collagen fibrils due to the inclusion of PEG.

Therefore, dependency of indentation and tensile modulus on osmotic pressure/hydration could reflect the structure-material relationship. The collagen molecules deform due to their entropic and progressively energetic elasticity. When collagen fibril is stretched their deformation occurs due to collagen molecules rotation, straightening, molecular kink formation, and molecular uncoiling [84–87], these observations indicate that the deformation of collagen molecules will be less than the fibrils deformation, resulting in the young modulus of collagen molecules is greater than the tensile elastic modulus of collagen fibrils. Also, the strain required to unfold the collagen molecule decreases from PBS solution to PEG solution [76]. In fully hydrated state collagen molecules are apart due to a reduction in free binding energy and become more flexible whereas with an increase dehydration, the gap between the molecules decreases and free energy of binding increases, which indicates that the strain required for rotation, straightening, molecular kink, and uncoiling depends on the level of hydration/osmotic pressure [76]. Further, the collagen fibrils strain rate depends on the properties and is also influenced by the osmotic pressure where the modulus and peak force were found increasing with increase in strain rate and osmotic pressure. Further, Masic et al. [27] showed that upon removing the water from the collagen structure the tensile force increases up to 100 MPa, which is the result of decreasing in intermolecular spacing of collagen molecules. Over all this section reviewed the biomechanical properties of the tissue with respect to their

molecular hierarchy. Also, we summarized the capability of AFM to capture the alteration in the mechanical response of collagen fibrils under different environmental and pathological conditions. After reviewing the existing literature on collagen fibrils biomechanics, one can observed that the collagen mechanical properties are govern by its basic structural component such tropocollagen. There are other technique such as transmission electron microscopy (TEM) and small angle X-ray scattering techniques (SAXS) to characterize the collagen fibril [reference]. These technique are limited to the structural characterization of collagen fibrils. However, by coupling these techniques with micro tensile test setup, one can measure the deformation response of collagen fibrils under different environmental and physiological conditions [10]. Wang et al. [10] coupled the tensile test setup with SAXS to observe the deformation response of collagen fibrils against the macroscopic applied strain. They observed lesser strain in the collagen fibril than the applied value of strain, which indicates that a part of applied strain is accumulated in the intra- and interfibrillar sliding. They also proposed that this deformation mechanism at collagen fibril level provides blunting at the crack tip of the tissue which makes it stronger against the tear. A similar result was also observed by Dwivedi et al. [11] where the collagen fibrils' plastic strain was found less than the macroscopic permanent strain accumulated at the tissue level, also the plastic strain of collagen fibrils was found strain rate dependent (Fig. 3). Bell et al. used wide and small angle X-ray scattering along with a stretching device [88]. They proposed that such type of deformation response can play a crucial functional role in tissue which are rich in collagen fibrils. Dwivedi et al. [11] demonstrated that the deformation response of collagen fibrils are governed by its basic structural component tropocollagen. In next section, we have summarized the literature related to collagen molecules and the required technique to investigate molecular response of the collagen.

5.2 Raman Spectroscopy

Near-infrared Fourier transform Raman (NIR-FT-Raman) spectroscopy is most widely used to characterized the molecular structure of biological material. This technique has been adaptive due to its simplicity, less sample preparation, non-distractive nature, and non-chemical requirements. Raman spectroscopy allows the examination of molecular structure of tissue through light scattering analysis. Once the light incident on the sample, a part of incident light is scattered with change in frequency equal to the vibration frequency of the chemical bonds and this sift in frequency provides information about the vibrational mode in molecules. This energy of vibrational mode is related to the molecular structure, environmental conditions, atomic mass, bond type, molecular substituents, molecular geometry, and intermolecular interactions. All these affect the vibrational force constants and determine the value of vibrational energy. Numerous studies have used Raman spectroscopy to investigate the molecular structure of healthy and diseased tissues [1, 89–93], other studies used this technique to characterize the single cell [94] and to understand the effect of chemical

on the molecular structure of tissues [95]. Raman spectroscopy has also been explored in in vivo characterization of healthy and diseased tissues. There are other studies where Raman spectroscopy has been coupled with the stretching device to capture the strain-induced alteration in collagen molecules [96, 97]. Frushour and Koenig (1975) [98] investigated the Raman spectrum from unstrained native collagen fibre-derived animal sources. In this analysis, they have assigned several peaks according to specific groups, associated with amino acid backbone of collagen. Further, Greve et al. [99] summarize the several peaks of Raman spectrum, performed on skin tissue.

Raman spectra of biological tissues contain vibration spectra of amino acids, proteins, lipid, and water content (Table 1). The broad band between 3200 and 3600 cm^{-1} is associated with starching vibrational bond of water present in tissue and bending vibrational of proteins. An asymmetric peak at 2938 cm^{-1} is related to the stretching vibration band of the CH_2 and CH_3 side groups of amino acids and chain of fatty acids. The intense band with a maximum located around 1450 cm^{-1} is related to the deformation vibrational bend of CH_2 and CH_3. The available literature suggests that the intensity peaks within band of ~1600–1700 cm^{-1} are related to Amide I and peaks within band 1200–1300 cm^{-1} are associated with Amide III. These two peaks are restated to the organic components of the tissue. Further, peaks at ~1245 and ~1670 cm^{-1} are associated with random/unordered structures however, intensity peaks at ~1270 and ~1640 cm^{-1} are related to ordered structure [100–102]. In addition to this, peak at 1445 cm^{-1} is related to the total organic content, and majority of which is collagen [103]. Further, Unal et al. [102] demonstrated that the

Table 1 Position of different Raman spectrum peaks associated with different organic components

Assignments	Wavenumber (cm^{-1})	Compound name
ν(SS)	545	cysteine
ν(CC)	817	Protein backbone
ν(CC) or δ(CCH)	856	Proline
ν(CC) or δ(CCH)	877	Hydroxyproline
ν(CC)	919	Proline
ν(CC)	935	Proteins
ν(CC)	1004	phenylalanine
ν(CC)	1050–1150	lipids
ν(CN), δ(NH)	1255	Amide III
δ(CH)	1336	Desmosine, proteins
ν(CH_2, CH_3)	1450	Protein, lipids
ν(C=O),ν(CN), δ(NH)	1658	Amide I
ν(CH_2,CH_3)	2938	Proteins, lipids
ν(NH), ν(OH)	3200–3600	Amide A,B, Water

Note ν- stretching mode, δ-bending mode
Adopted from literature [96]

Raman spectral intensity ratios 1670/1640, 1245/1270, 1245/1454 are the representative of collagen integrity where these ratios changes in response to both thermal denaturation and monotonic mechanical damage [102]. They have shown that the increase in the 1670/1640 ratio is the results of increase in unordered-to-ordered collagen amount due to damage under mechanical load. The 1670/1640 ratio was found of great interest because it does not show sensitivity toward collagen orientation [102]. Further, similar group also explored this ratio to capture the molecular damage in collagen under fatigue and thermal loading [100].

Further, Raman spectrum peaks between 2700 and 3800 cm^{-1} are also the fuscous of research. Literature show the relation of several peaks in this region with water compartments of tissue. Specifically, the peaks at 3230 and 3325 cm^{-1} are found associated with organic matrix-bound water and organic matrix-bound water plus a contribution from the collagen backbone [100]. Unal et al. [102] reported that drying at 40 °C for 48 h the peak ratio 3220/2949 changes from its native value. Further, Flanagan et al. utilized the 1670/1640 and 3220/2949 ratio to capture the collagen molecular alteration caused by fatigue loading and irradiation. They have found that these two ratios are good biomarkers to capture the molecular damage/alteration caused by mechanical, thermal, or other environmental aggression. Raman spectroscopy has also been found capable to capture the alteration in molecular arrangement in between the stretching of tissue for both in vivo and ex vivo test. In an in vivo study, Glogowska et al. [96] coupled Raman spectroscopy with tensile test setup to capture the deformation response of collagen molecules in human skin where they have applied strain on the tissue and strain dependent alteration in collagen molecules of tissue has been captured. In this study they have focused in peaks of 1658 cm^{-1} (Amide I), 1271 and 1255 cm^{-1} (amide III), and 935 and 817 cm^{-1} (c–c stretching modes of the protein backbone). They observed that some of the bands exhibit a change in the position of the maximum with increase in strain which could be due to the change in interatomic distance under the stressing of chemical bonds. Under the large value of strain, they also observed denaturation of collagen molecules as the peak 1658 cm^{-1} associated with the tropocollagen sift toward the lower value. Another ex vivo study is performed on the aortic tissue under the Raman spectroscopy coupled with stretching devices [reference]. The results of this study show that unordered structure of collagen molecules increases with increase in applied stress as the peak at 1668 cm^{-1} sift from its native position. Based on these results, a correlation between Raman band sifting and the level of mechanical stress has established in this study. The obtained results through Raman spectroscopy-tensile test setup showed that Elastin alone participate in the transmission of low stresses in the circumferential direction, however both elastin and collagen take part in the transmission of physiological and higher stresses. The above reviewed literature indicate that collagen molecules are responsive to the mechanical load and which can be well captured using Raman spectroscopy alone or coupled with stretching device. Based on Fourier transformation, another devise is also commonly used to measure the alteration in molecular structure of tissue. This device is commonly named as attenuated total reflection Fourier transformation infrared spectroscopy (ATR-FTIR).

5.3 ATR-Ftir

The available studies shows that ATR-FTIR is widely used to identify the alteration in collagen molecules caused by different pathological conditions (e.g., age, diseases, etc.) and mechanical load. The assignment of peaks and bands to the chemical compounds is almost similar to the Raman spectrum. Some important peaks obtained using ATR-FTIR are summarized in Table 2. This data was borrowed from the work of Rabotyagova et al. [44].

Rabotyagova et al. [44] presented a novel study and explored the ATR-FTIR technique to quantify the collagen alteration caused by the ultraviolet radiation. They observed that the structural and functional changes in the collagen are related to the initial conformation and the energy of irradiation. Using the ATR-FTIR they have observed that the denaturation of native triple helix is most sensitive to UV 288 radiation. They showed that loss in triple helix of collagen is due to the breakage of hydrogen bonds through free radical mechanisms. They have suggested that to capture the collagen denature, the amide I region is a useful area of focus as a change in the relative intensity is conformation dependent. Further, they presented the deconvolution of Amide I to understand the qualitative and quantitate role of

Table 2 Position of different ATR-FTIR spectrum peaks associated with different organic components

Assignment	Position (cm^{-1})
Tyrosine side chain	1515
Amide II in a random coil	1530
Amide II in triple helix	1549
Carboxyl groups, N–H bending vibration, and C–N stretching vibration	1565
Left-handed 3–10 helix in denaturated state, partially beta-sheet region	1630
Random coil conformation: imide residues (amide I in random coil)	1646–1650
Random chain contribution of the non-proline and non-hydroxyproline	1660
Collagen in triple helix, with contribution from α-helix and beta-turns (protein amide stretching vibration of C_O)	1675
Intermolecular association. Helices of aggregated collagen-like peptides β-turns	1690
CH_3—stretching of methyl (assimetric)	2876
CH_2—stretching of methylene	2946
N–H stretch (amide 1)	3080
Asymmetric $–CH_2–N=$	3217–3180
N–H stretch	3330–3325
NH stretch coupled with H-bond from H_2O	3450
Intermolecular hydrogen bounds	3500–3470

Adopted from literature [44]

different structural components. Using the similar approach, Dwivedi et al. [reference] captured the collagen molecular sensitivity toward the loading rate. In this study, the cyclic experiments were performed on the pig skin at different loading frequencies, and molecular damage of collagen was assessed using ATR-FTIR. The obtained Amide I spectrum was deconvoluted, and peaks at 1632 and 1658 cm^{-1} were assigned to denatured α-helix and triple helix, respectively. After cyclic loading, the 1632/1658 ratio was found to increase from native to mechanical tested specimens and the increase in this ratio was significantly more for slow rate of loading. This study explored the capability of ATIR-FTIR to capture the mechanical damage of collagen molecules which will help to predict the remodeling of tissues. Further, our research group Sihota et al. [51, 55] utilized ATR-FTIR technique to investigate the effect of type two diabetes on the collagen structure of bone tissue [51]. Furthermore, they deconvoluted the Amide 1 peak to find the sub-peaks in this region. Later these sub-peaks were correlated with the collagen molecular constituents. They obtained total six sub-peaks at the position of 1610, 1630, 1645, 1660, 1678, and 1692 cm^{-1}. These peaks were deconvoluted based on the second derivatives of obtained spectrum. In this study different peak ratio were used to measure the alteration in molecular composition of tissue [51]. Uniyal et al. [104] explored ATR-FTIR technique to capture the effect of heat treatment on the molecular structure of collagen. Where they have observed increase in intensity of peak 1630 cm^{-1}, related to the denatured state of collagen molecules same time the value of peak intensity of 1660 cm^{-1}, related to triple helix was found to decrease with increase in temperature of heating (see Fig. 4).

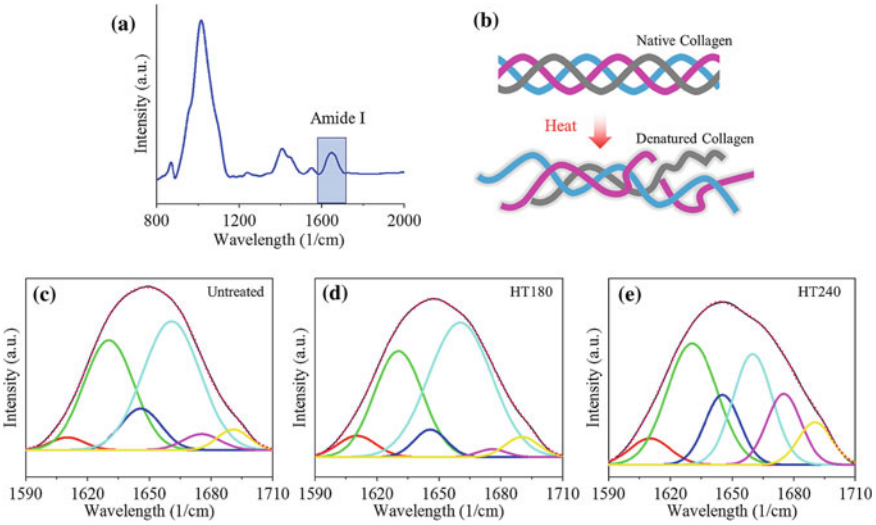


Fig. 4 **a** Amide I band (1590–1710 cm^{-1}). Adopted from Uniyal et al. [104]

Further dwivedi et al. [11] measured the strain rate sensitivity of collagen molecules denaturation in pig skin. They have used an intensity ratio of peaks 1632/1657. The obtained results are provided in Fig. 5.

Overall this brief literature survey directs about the use of ATR-FTIR to capture the molecular structure of the tissue. Other methods to capture the alteration of collagen molecules are based on the imaging techniques.

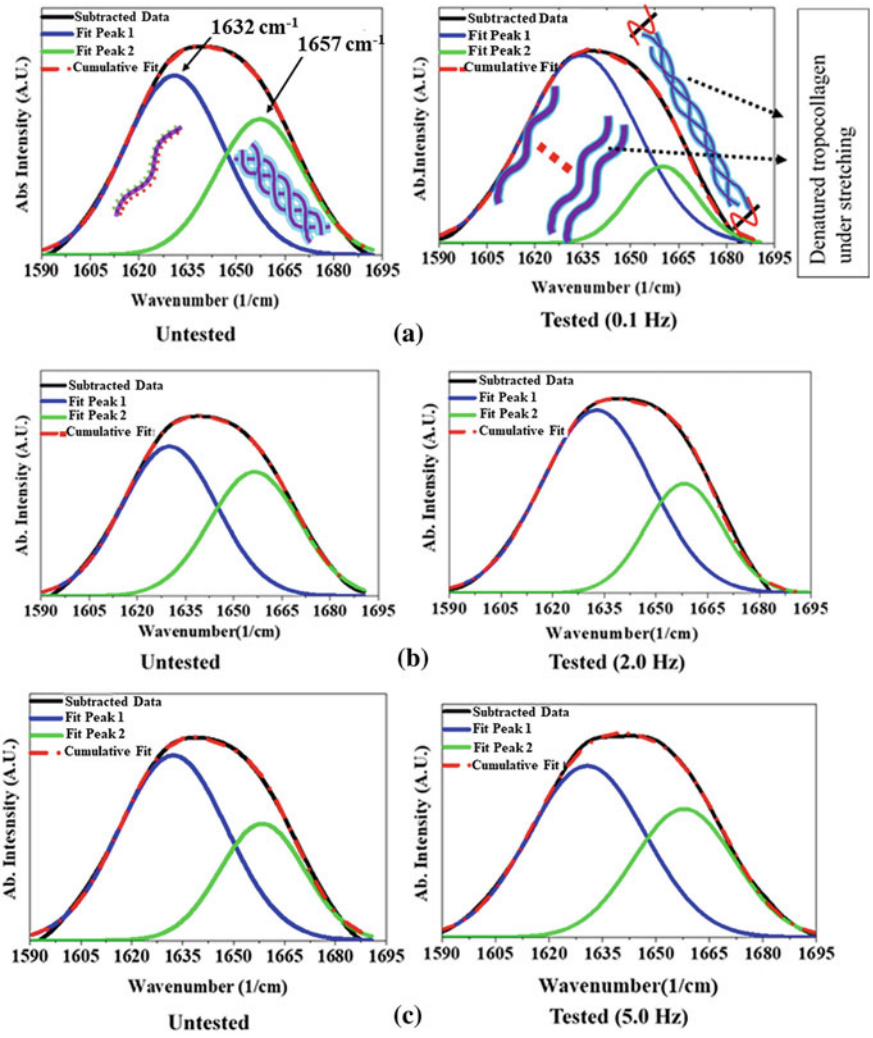
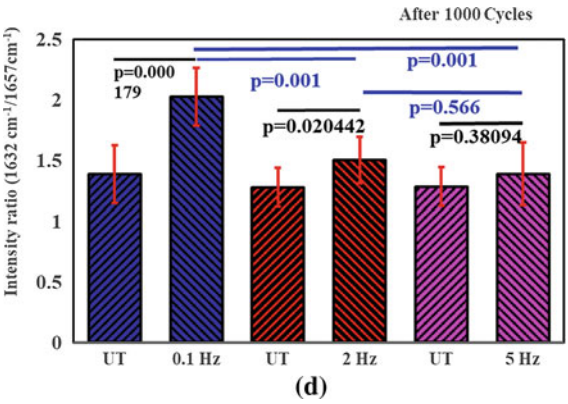


Fig. 5 Amide I spectra deconvoluted in two sub-peaks for **a** untested specimen, **b**, **c** tested specimens corresponding to 0.1, 2.0, and 5.0 Hz. **d** Comparison of intensity ratio (1632/1657 cm^{-1}) between tested and untested specimens. Adopted from Dwivedi et al. [11]

Fig. 5 (continued)



5.4 *Fluorescently Labeled Collagen Hybridizing Peptide (F-CHP)*

In order to demonstrate the unfolding or denaturation of collagen tropocollagen when subjected to a single quasistatic tensile overload event, Zitnay et al. [47] labeled the tendon tissue using F-CHP staining. CHP is a small synthetic peptide with glycine-proline-hydroxyproline residues that has a strong propensity to fold into the triple-helical supper secondary protein structure of native collagen molecules. This staining binds with denatured unfolded collagen α-chain [47]. The binding side of unfolded collagen molecules can be quantified using imaging techniques such as fluorescence or multiphoton microscope. Further, Zitnay et al. [47] presented to capture the cyclically natured collagen molecules using this F-CHP dye. They observed an increase in molecular damage with increase in number of fatigue cycles. This cyclically induced damage can propagate to the micro- and tissue-level injuries. Further, Hwang et al. [105] used F-CHP staining to investigate the remodeling procedure of the collagen, based on these observations they concluded the level of collagen remodeling and amount of denatured collagen is correlated with each other. They also concluded that F-CHP probe is not only useful in pathology and developmental biology research but also can be used histologically based disease diagnostic, staging, and therapeutic screening.

Based on the above-mentioned experimental results various computational techniques have been developed to predict the mechanical response of collagen molecules. A brief review on the computational methods is summarized next paragraphs.

6 Molecular Dynamics-Based Simulation to Predict the Molecular Level Mechanical Response of the Tissue

6.1 Quasistatic and Dynamic Simulation

Method to generate the collagen molecules.

Chang et al. [106] generated the full atomistic model to simulate the force effect on the cleavage site of the type 1. They have used real sequence of type I α-I and type I α-2 chains of mus musculus (type mouse) are used to generate the collagen molecules. In this model they used two α-1 chains and one α-2 chain where the entire α-1 and α-2 chains consist of 1014 residues with repeated G-X–Y triplets, and similar to tropocollagen this model excludes the C-terminal and N-terminal sequence. Segments of real sequences with 63 residues long centered at MMP-1 cleavage site of α-1 and α-2 chains are selected to construct the heterotrimer and homotrimer collagen molecules. The chosen sequence areas.

alpha-1:

GPPGPAGEKGSPGADGPAGSPGTPGPQGIAGQRGVVGLPGQRGER GFPGLPGPSGEPGKQGPS and

alpha-2:

GPPGFVGEKGPSGEPGTAGAPGTAGPQGLLGAPGILGLPGSRGER GLPGIAGALGEPGPLGIS.

Note that "–" indicates the scissile bond (the bond cut by MMP) at the cleavage site (G-IA for alpha-1 chain and G-LL for alpha-2 chain) of MMP-1. For more detail, please see the original research of Chang et al. [106].

The collagen molecules have been crated in THeBuScr (an interactive triple-helical collagen binding script) [107], this allows to build triple-helical molecules based on any specific amino acid sequence. The length of collagen molecule was set 17 nm and model is solved under a periodic water box composed TIP3 water molecules. Visual molecular dynamics (VMD) [108] is used to solve the model. The CHARMM [109] force field is used for this simulation. This force field is widely used for a variety of biochemical models of protein including collagen [86]. The energy is minimized using a conjugate gradient scheme before molecular dynamic simulation. In order to examine the effects of applied force on structural changes of heterotrimer and homotrimer collagen, a constant 100 pN force is applied to each chain. For more detail please refer to the original work of Chang et al. [106].

Further, Buehler (2005) [110] presented a study where the mechanical properties of tropocollagen under different loading conditions such as tension, compression, shear, and bending were simulated using the molecular dynamic simulation. In this modeling, he also predicted the yield of fracture strength of single tropocollagen and polypeptide. In this simulation, he utilized CHARMM force field function which was implemented in molecular dynamics program NAMD. He had also added an extension to the standard CHARMM force field to include a description

of hydroxyproline residue, apart from the 20 natural amino acids. Based on the obtained results, he suggested that for describing the large strain or hyperelastic properties, a correct description of bond behavior and its breaking process at large bond stretch is required. Further, similar research group [111] described a bottom-up atomistic modeling approach to study the collagen peptide behavior under two distinct impulsive displacement loads which include longitudinal and transverse inputs. For this simulation they used the model as described elsewhere [111]. In brief, based on X-ray diffraction observation, they prepared a $(GPO)_{20}$ peptide topology with triple-helical collagen building script (The BuScr) with length of about 18 nm is constructed. In order to consider the role of water in energy transfer, they have created a solvent box with dimensions of $40 \times 6 \times 6$ nm which is randomly ionized (0.5 mol/L) the peptide via Visual molecular dynamics, reaching a hydrated structure (HS). The equilibrium of dry and hydrated structure is performed via LAMMPS using the CHHARM force field that includes the hydroxyproline residue [111]. They used particle by particle-mesh solver with 10^{-4} kcal/mol-Angstrom accuracy. Using molecular dynamics approach, they fixed three backbone nitrogen atom (BB-N) at the extremity of the three chains of the peptide and loaded on the structure applied through other BB-N atoms at the opposite extreme. Two different types of loading, longitudinal, and transverse are applied during the simulation. Also peptide is loaded with 1 nm impulsive displacement. They have measured Young's modulus by the wave theory concepts where they assume the velocity of wave proportional to square root of Young's modulus and the constant of proportionality is the inverse of material density. Obtained results showed the directional decency of wave transmission and energy dissipation. They also observed that hydrated collagen peptides are more prone to dissipate energy.

Zhmurov et al. [112] used molecular dynamic simulation approach to understand the transition phenomenon of α-helical coiled coil to β-sheet. For this analysis, they performed pulling and ramped-force simulations. Pulling simulation has been performed on GPU using structural model of the whole human Fg molecules using implicit simulation. Another side, for Ramped force simulation, one end of molecule is constrained and the other end of molecule is stretched with time dependent force $f(t) = f(t)$n, where n is the unit vector along the longitudinal axis of the molecules. In this simulation force is increase linearly with the loading rate of $r_{f=ksp}v_f$ where $k_{sp} = 100$pN/nm is spring constant and v_f is loading speed. For modeling the α-to-β transition, they also performed the force camp simulation (please see the method section of the original work [112]. With the permission of authors, the results of this work are presented in Figs. 6 and 7). Based on this simulation their conclusion are: a dynamic transition is observed from the elastic regime to the plastic regime during the elongation of Fg coiled coils followed by a remarkable α-to-β phase transition; the observed plateau regions in the force spectra of proteins with high α-helical content, such as Fg, could serve as dynamic signature of α-to-β phase transition. The similar group Minin et al. simulated the dynamic transition of α-helical coiled coil to β-sheets in polypeptide coiled coil motifs [113]. The results of the study are

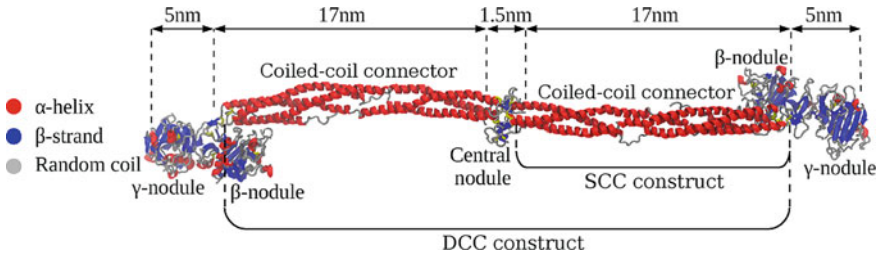


Fig. 6 The molecular dynamic structure presents two coiled coil connectors, a central nodule, γ-nodules, and β-nodules with all required dimensions. Adopted from Zhmurov et al. [112]

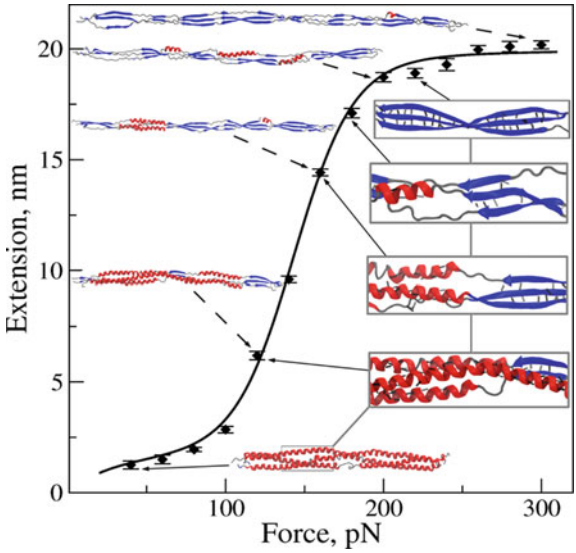


Fig. 7 The transition from α-to-β occurs in S-shape profile. Adopted from Zhmurov et al. [112]

presented in Figs. 8 and 9. In this study they have used Abeyaratne-Knowle formulation of phase transitions to develop a methodology to model mechanical and kinetic properties of protein coiled coils under the mechanical non-equilibrium conditions.

Zitnay et al. [47] used molecular dynamic simulation to predict the damage accumulation due to cyclic loading. For this simulation they model a representative homotrimeric collagen peptide composed of 20 residues of (Gly-Pro-Hyp) triplets has been generated by the triple-helical collagen building script. The simulation is performed in LAMMPS and using CHARMM force field in the simulation boxes of $25 \times 5 \times 5$ nm. The cyclic loading based on the maximum and minimum force has been designed in SMD. The minimum and maximum load values are 350 pN and 2085 pN. The obtained results show that the failure of tropocollagen is dominated by shear loading, which results unfolding of triple helices.

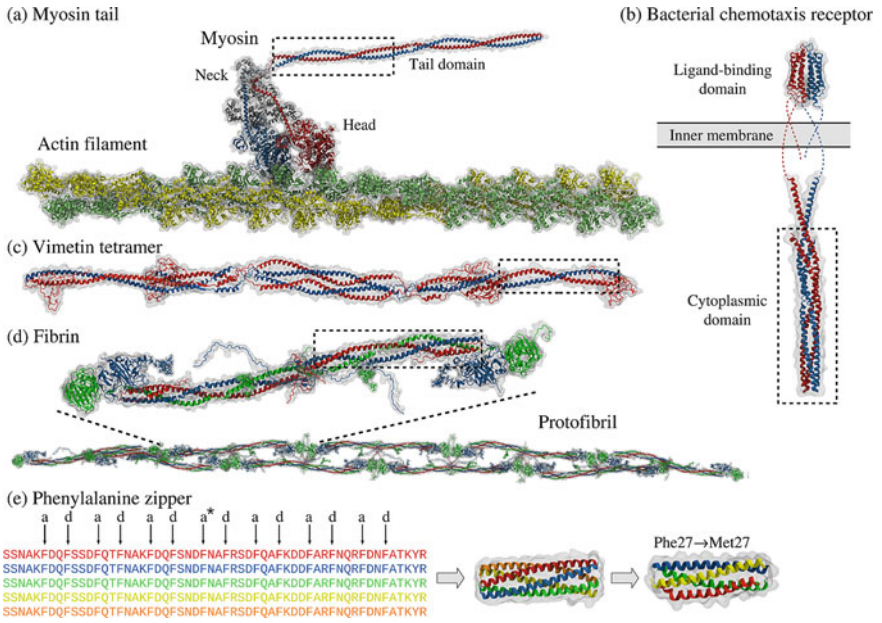


Fig. 8 Represents the model of collagen molecules used in molecular dynamic simulation [113]. For more details please refer original article of Minin et al. [113]

6.2 Viscoelastic Simulation Using Molecular Dynamics

Apart from the quasistatic and dynamic simulations, the viscoelastic behavior of the collagen molecules has also been simulated. Gautieri et al. [114] simulated the strain time response of collagen molecules. For this simulation they used collagen triplet model as described in above paragraphs. The creep simulation has been performed by fixing the N-terminal C_α atoms and other C-terminal and C_α are loaded instantaneous up to peak load and then hold for 10 ns. The load is applied in the range of 300–3000 pN. The obtained results showed the nonlinear behavior of single collagen molecules where linear modulus increases from 6 to 16 GPa when strain is increased up to 20%. The viscosity of single collagen molecules is significantly lesser than the tissue level viscosity, which indicates that the viscoelastic behavior of tissue is associated with other mechanisms. The similar analysis has also been performed by the other researchers [115].

7 Conclusion

Overall this article summarizes the different experimental techniques with their advancement in the field of biomechanics. As the biomechanics is a growing field for

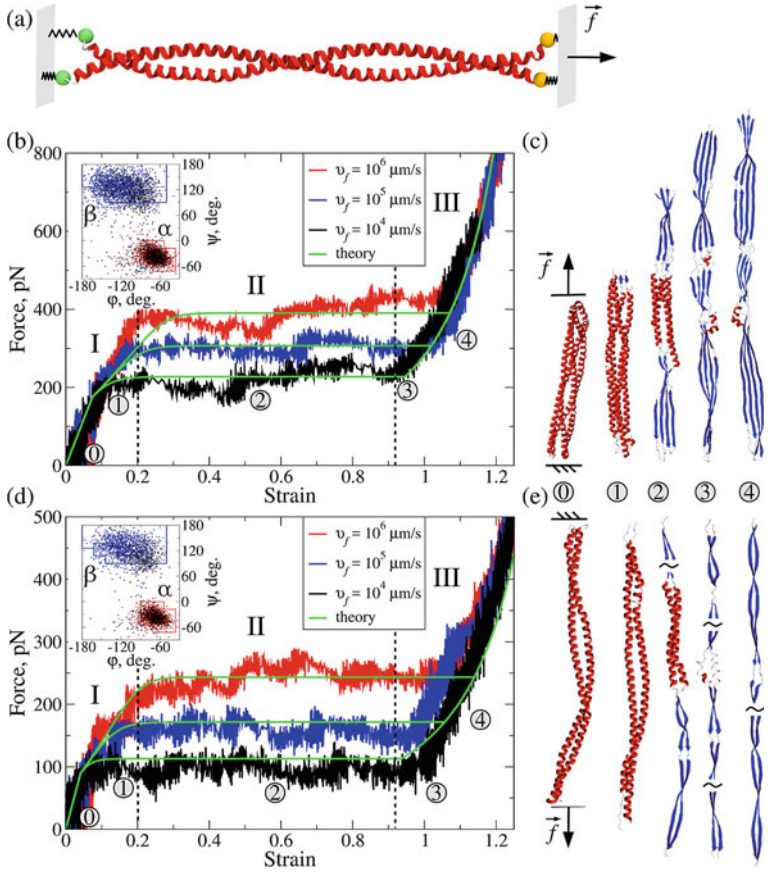


Fig. 9 α-to-β Transition in polypeptide coiled coils. Adopted from Minin et al. [113]

the researcher. The accurate understating of the mechanical and structural behavior of tissue particularly from the molecular level is important for diagnostic, treatment, and artificial tissue development. In this chapter focused on the molecular structure of the tissue, a building block of many connective tissues. There are several destructive and nondestructive techniques to characterize the molecular structure of the tissue. In this article we also present an explanation of the experimental techniques used to capture structural and mechanical responses of collagen fibrils. Apart from the experimental techniques, we also reviewed the computation techniques which are widely used to capture the mechanical and structural response of collagen molecules under different loading and physiological environment. With the advancement of computational facilities, the molecular dynamic becomes a favorite among the researcher to simulate the mechanical and structural behavior of collagen molecules.

References

1. Gasior-Głogowska, M., Komorowska, M., Hanuza, J., Maczka, M., Kobielarz, M.: Structural alteration of collagen fibres—spectroscopic and mechanical studies. Acta Bioeng. Biomech. **12**, 53–60 (2010)
2. Lee, D.H., Oh, J., Chung, J.H.: Glycosaminoglycan and proteoglycan in skin aging. J. Dermatol. Sci. (2016). https://doi.org/10.1016/j.jdermsci.2016.05.016
3. Analyzer, V.S., Vexler, A., Polyansky, I., Gorodetsky, R.: Evaluation of skin viscoelasticity and anisotropy by measurement of speed of shear wave propagation with. J. Invest. Dermatol. **113**, 732–739 (1999). https://doi.org/10.1046/j.1523-1747.1999.00751.x
4. Holzapfel, G.A., Sommer, G., Gasser, C.T., Regitnig, P.: Determination of layer-specific mechanical properties of human coronary arteries with nonatherosclerotic intimal thickening and related constitutive modeling. Am. J. Physiol. Circ. Physiol. **289**, H2048–H2058 (2005)
5. Gefen, A.: Plantar soft tissue loading under the medial metatarsals in the standing diabetic foot. Med. Eng. Phys. **25**, 491–499 (2003). https://doi.org/10.1016/S1350-4533(03)00029-8
6. Ledoux, W.R., Pai, S., Shofer, J.B., Wang, Y.N.: The association between mechanical and biochemical/histological characteristics in diabetic and non-diabetic plantar soft tissue. J. Biomech. **49**, 3328–3333 (2016). https://doi.org/10.1016/j.jbiomech.2016.08.021
7. Pai, S., Ledoux, W.R.: The shear mechanical properties of diabetic and non-diabetic plantar soft tissue. J. Biomech. **45**, 364–370 (2012). https://doi.org/10.1016/j.jbiomech.2011.10.021
8. Fung, Y.: Biomechanics: Mechanical Properties of Living Tissues. Springer Science & Business Media (2013)
9. Dwivedi, K.K., Lakhani, P., Kumar, S., Kumar, N.: The effect of strain rate on the stress relaxation of the pig dermis: a hyper-viscoelastic approach. J. Biomech. Eng. **142** (2020)
10. Yang, W., Sherman, V.R., Gludovatz, B., Schaible, E., Stewart, P., Ritchie, R.O., Meyers, M.A.: On the tear resistance of skin. Nat. Commun. (2015). https://doi.org/10.1038/ncomms7649
11. Dwivedi, K.K., Lakhani, P., Kumar, S., Kumar, N.: Frequency dependent inelastic response of collagen architecture of pig dermis under cyclic tensile loading: an experimental study. J. Mech. Behav. Biomed. Mater. 104030 (2020)
12. Lakhani, P., Dwivedi, K.K., Parashar, A., Kumar, N.: Non-Invasive in vivo quantification of directional dependent variation in mechanical properties for human skin. Front. Bioeng. Biotechnol. 979 (2021)
13. Lakhani, P., Dwivedi, K.K., Kumar, N.: Directional dependent variation in mechanical properties of planar anisotropic porcine skin tissue. J. Mech. Behav. Biomed. Mater. **104**, 103693 (2020)
14. Pissarenko, A., Yang, W., Quan, H., Brown, K.A., Williams, A., Proud, W.G., Meyers, M.A.: Tensile behavior and structural characterization of pig dermis. Acta Biomater. **86**, 77–95 (2019)
15. Pissarenko, A., Ruestes, C.J., Meyers, M.A.: Constitutive description of skin dermis: through analytical continuum and coarse-grained approaches for multi-scale understanding. Acta Biomater. (2020)
16. Gautieri, A., Pate, M.I., Vesentini, S., Redaelli, A., Buehler, M.J.: Hydration and distance dependence of intermolecular shearing between collagen molecules in a model microfibril. J. Biomech. **45**, 2079–2083 (2012). https://doi.org/10.1016/j.jbiomech.2012.05.047
17. Gah, A.: Determination of material models for arterial walls from uniaxial extension tests and histological structure. J. Theor. Biol. **238**, 290–302 (2006). https://doi.org/10.1016/j.jtbi.2005.05.006
18. Babaei, B., Velasquez-Mao, A.J., Thomopoulos, S., Elson, E.L., Abramowitch, S.D., Genin, G.M.: Discrete quasi-linear viscoelastic damping analysis of connective tissues, and the biomechanics of stretching. J. Mech. Behav. Biomed. Mater. **69**, 193–202 (2017). https://doi.org/10.1016/j.jmbbm.2016.12.013

19. Holzapfel, G.A., Niestrawska, J.A., Ogden, R.W., Reinisch, A.J., Schriefl, A.J.: Modelling non-symmetric collagen fibre dispersion in arterial walls. J. R. Soc. interface **12**, 20150188 (2015)
20. Sherman, V.R., Tang, Y., Zhao, S., Yang, W., Meyers, M.A.: Acta biomaterialia structural characterization and viscoelastic constitutive modeling of skin. Acta Biomaterialia **53**, 460–469 (2017)
21. Lanir, Y., Fung, Y.C.: Two-dimensional mechanical properties of rabbit skin-II. Experimental results. J. Biomech. (1974). https://doi.org/10.1016/0021-9290(74)90058-X
22. Lynch, B., Bancelin, S., Bonod-Bidaud, C., Gueusquin, J.B., Ruggiero, F., Schanne-Klein, M.C., Allain, J.M.: A novel microstructural interpretation for the biomechanics of mouse skin derived from multiscale characterization. Acta Biomater. **50**, 302–311 (2017). https://doi.org/10.1016/j.actbio.2016.12.051
23. Amabili, M., Asgari, M., Breslavsky, I.D., Franchini, G., Giovanniello, F., Holzapfel, G.A.: Microstructural and mechanical characterization of the layers of human descending thoracic aortas. Acta Biomater. **134**, 401–421 (2021)
24. Lynch, H.A.: Effect of fiber orientation and strain rate on the nonlinear uniaxial tensile material properties of tendon. J. Biomech. Eng. **125**, 726 (2003). https://doi.org/10.1115/1.1614819
25. Freedman, B.R., Sarver, J.J., Buckley, M.R., Voleti, P.B., Soslowsky, L.J.: Biomechanical and structural response of healing Achilles tendon to fatigue loading following acute injury. J. Biomech. **47**, 2028–2034 (2014)
26. Ahmadzadeh, H., Freedman, B.R., Connizzo, B.K., Soslowsky, L.J., Shenoy, V.B.: Micromechanical poroelastic finite element and shear-lag models of tendon predict large strain dependent Poisson's ratios and fluid expulsion under tensile loading. Acta Biomater. **22**, 83–91 (2015)
27. Masic, A., Bertinetti, L., Schuetz, R., Chang, S.-W., Metzger, T.H., Buehler, M.J., Fratzl, P.: Osmotic pressure induced tensile forces in tendon collagen. Nat. Commun. **6**, 1–8 (2015)
28. Böl, M., Ehret, A.E., Leichsenring, K., Ernst, M.: Tissue-scale anisotropy and compressibility of tendon in semi-confined compression tests. J. Biomech. **48**, 1092–1098 (2015)
29. Rigozzi, S., Müller, R., Stemmer, A., Snedeker, J.G.: Tendon glycosaminoglycan proteoglycan sidechains promote collagen fibril sliding—AFM observations at the nanoscale. J. Biomech. **46**, 813–818 (2013)
30. Nimeskern, L., Utomo, L., Lehtoviita, I., Fessel, G., Snedeker, J.G., Van, O.G.J.V.M., Müller, R., Stok, K.S.: Tissue composition regulates distinct viscoelastic responses in auricular and articular cartilage. J. Biomech. **49**, 344–352 (2016). https://doi.org/10.1016/j.jbiomech.2015.12.032
31. Snedeker, J.G., Foolen, J.: Tendon injury and repair–a perspective on the basic mechanisms of tendon disease and future clinical therapy. Acta Biomater. **63**, 18–36 (2017)
32. Abramowitch, S.D., Woo, S.L.: An improved method to analyze the stress relaxation of ligaments following a finite ramp time based on the quasi-linear viscoelastic theory. J. Biomech. Eng. **126** (2018). https://doi.org/10.1115/1.1645528
33. Abramowitch, S.D., Woo, S.L.-Y.: An improved method to analyze the stress relaxation of ligaments following a finite ramp time based on the quasi-linear viscoelastic theory. J. Biomech. Eng. **126**, 92–97 (2004)
34. Abramowitch, S.D., Zhang, X., Curran, M., Kilger, R.: A comparison of the quasi-static mechanical and non-linear viscoelastic properties of the human semitendinosus and gracilis tendons. Clin. Biomech. **25**, 325–331 (2010)
35. Holzapfel, G.A., Gasser, T.C., Ogden, R.W.: Comparison of a multi-layer structural model for arterial walls with a fung-type model. J. Biomech. Eng. **126** (2016). https://doi.org/10.1115/1.1695572
36. Groves, R.B., Coulman, S.A., Birchall, J.C., Evans, S.L.: An anisotropic, hyperelastic model for skin: experimental measurements, finite element modelling and identification of parameters for human and murine skin. J. Mech. Behav. Biomed. Mater. **18**, 167–180 (2013)
37. Liu, X., Cleary, J., German, G.K.: The global mechanical properties and multi-scale failure mechanics of heterogeneous human stratum corneum. Acta Biomater. **43**, 78–87 (2016)

38. Ottenio, M., Tran, D., Annaidh, A.N., Gilchrist, M.D., Bruyère, K.: Strain rate and anisotropy effects on the tensile failure characteristics of human skin. J. Mech. Behav. Biomed. Mater. **41**, 241–250 (2015)
39. Yang, W., Sherman, V.R., Gludovatz, B., Schaible, E., Stewart, P., Ritchie, R.O., Meyers, M.A.: On the tear resistance of skin. Nat. Commun. **6**, 6649 (2015)
40. Montagna, W.: The structure and function of skin. Elsevier (2012)
41. Griffiths, C., Barker, J., Bleiker, T., Chalmers, R., Creamer, D.: Rook's textbook of dermatology. Wiley (2016)
42. Joodaki, H., Panzer, M.B.: Skin mechanical properties and modeling: a review. Proc. Inst. Mech. Eng. Part H J. Eng. Med. **232**, 323–343 (2018)
43. Oxlund, H., Manschot, J., Viidik, A.: The role of elastin in the mechanical properties of skin. J. Biomech. **21**, 213–218 (1988)
44. Rabotyagova, O.S., Cebe, P., Kaplan, D.L.: Collagen structural hierarchy and susceptibility to degradation by ultraviolet radiation. Mater. Sci. Eng. C **28**, 1420–1429 (2008). https://doi.org/10.1016/j.msec.2008.03.012
45. Sherman, V.R., Yang, W., Meyers, M.A.: The materials science of collagen. J. Mech. Behav. Biomed. Mater. **52**, 22–50 (2015). https://doi.org/10.1016/j.jmbbm.2015.05.023
46. Pissarenko, A., Ruestes, C.J., Meyers, M.A.: Acta Biomaterialia constitutive description of skin dermis: through analytical continuum and coarse-grained approaches for multi-scale understanding. Acta Biomater. **106**, 208–224 (2020). https://doi.org/10.1016/j.actbio.2020.01.026
47. Zitnay, J.L., Jung, G.S., Lin, A.H., Qin, Z., Li, Y., Yu, S.M., Buehler, M.J., Weiss, J.A.: Accumulation of collagen molecular unfolding is the mechanism of cyclic fatigue damage and failure in collagenous tissues. Sci. Adv. **6**, eaba2795 (2020)
48. Fratzl, P.: Collagen: Structure and Mechanics, An introduction. In: Collagen, pp. 1–13. Springer (2008)
49. Shoulders, M.D., Raines, R.T.: Collagen structure and stability. Annu. Rev. Biochem. **78**, 929–958 (2009)
50. Argyropoulos, A.J., Robichaud, P., Balimunkwe, R.M., Fisher, G.J., Hammerberg, C., Yan, Y., Quan, T.: Alterations of dermal connective tissue collagen in diabetes: Molecular basis of aged-appearing skin. PLoS ONE **11**, 1–17 (2016). https://doi.org/10.1371/journal.pone.0153806
51. Sihota, P., Yadav, R.N., Dhiman, V., Bhadada, S.K., Mehandia, V., Kumar, N.: Investigation of diabetic patient's fingernail quality to monitor type 2 diabetes induced tissue damage. Sci. Rep. **9**, 3193 (2019)
52. Fisher, G.J., Quan, T., Purohit, T., Shao, Y., Cho, M.K., He, T., Varani, J., Kang, S., Voorhees, J.J.: Collagen fragmentation promotes oxidative stress and elevates matrix metalloproteinase-1 in fibroblasts in aged human skin. Am. J. Pathol. **174**, 101–114 (2009)
53. Spanheimer, R.G., Umpierrez, G.E., Stumpf, V.: Decreased collagen production in diabetic rats. Diabetes **37**, 371–376 (1988)
54. Dyer, D.G., Mccance, D.R., Baynes, J.W.: Accumulation of Maillard reaction products in skin collagen in diabetes and aging. J. Clin. Invest. **91**, 2463–2469 (1993)
55. Sihota, P. et al.: Investigation of mechanical, material and compositional determinants of human trabecular bone quality in type 2 diabetes. J. Clin. Endocrinol. Metab. (2021)
56. Brings, S., Zhang, S., Choong, Y.S., Hogl, S., Middleditch, M., Kamalov, M., Brimble, M.A., Gong, D., Cooper, G.J.S.: Diabetes-induced alterations in tissue collagen and carboxymethyllysine in rat kidneys: association with increased collagen-degrading proteinases and amelioration by Cu (II)-selective chelation. Biochim. Biophys. Acta (BBA)-Mol. Basis Dis. **1852**, 1610–1618 (2015)
57. Yadav, R.N., et al.: Prediction of mechanical properties of trabecular bone in patients with type 2 diabetes using damage based finite element method. J. Biomech. **123**, 110495 (2021)
58. Sihota, P., Yadav, R.N., Poleboina, S., Mehandia, V., Bhadada, S.K., Tikoo, K., Kumar, N.: Development of HFD-fed/low-dose STZ-treated female Sprague-Dawley rat model to investigate diabetic bone fragility at different organization levels. JBMR Plus **4**, e10379 (2020)

59. Hunt, H.B., Pearl, J.C., Diaz, D.R., King, K.B., Donnelly, E.: Bone tissue collagen maturity and mineral content increase with sustained hyperglycemia in the KK-Ay murine model of type 2 diabetes. J. Bone Miner. Res. **33**, 921–929 (2018)
60. Molsted, S., Tribler, J., Snorgaard, O.: Musculoskeletal pain in patients with type 2 diabetes. Diabetes Res. Clin. Pract. **96**, 135–140 (2012)
61. Ehret, A.E., Bircher, K., Stracuzzi, A., Marina, V., Zündel, M., Mazza, E.: Inverse poroelasticity as a fundamental mechanism in biomechanics and mechanobiology. Nat. Commun. **8**, 1–10 (2017)
62. Wang, Y.N., et al.: The effects of stretching on wool fibres as monitored by FT-Raman spectroscopy. Sci. Rep. **7**, 1–10 (2017). https://doi.org/10.1016/S0022-2860(97)00227-5
63. Mak, M.: Impact of crosslink heterogeneity on extracellular matrix mechanics and remodeling. Comput. Struct. Biotechnol. J. **18**, 3969–3976 (2020)
64. Ghodsi, H., Darvish, K.: Characterization of the viscoelastic behavior of a simplified collagen micro-fibril based on molecular dynamics simulations. J. Mech. Behav. Biomed. Mater. **63**, 26–34 (2016)
65. Anssari-Benam, A., Screen, H.R.C., Bucchi, A.: Insights into the micromechanics of stress-relaxation and creep behaviours in the aortic valve. J. Mech. Behav. Biomed. Mater. **93**, 230–245 (2019)
66. Screen, H.R.C.: Investigating load relaxation mechanics in tendon. J. Mech. Behav. Biomed. Mater. **1**, 51–58 (2008)
67. Gupta, H.S., Seto, J., Krauss, S., Boesecke, P., Screen, H.R.C.: In situ multi-level analysis of viscoelastic deformation mechanisms in tendon collagen. J. Struct. Biol. **169**, 183–191 (2010)
68. Sherifova, S., Sommer, G., Viertler, C., Regitnig, P., Caranasos, T., Smith, M.A., Griffith, B.E., Ogden, R.W., Holzapfel, G.A.: Failure properties and microstructure of healthy and aneurysmatic human thoracic aortas subjected to uniaxial extension with a focus on the media. Acta Biomater. **99**, 443–456 (2019)
69. LaCroix, A.S., Duenwald-Kuehl, S.E., Brickson, S., Akins, T.L., Diffee, G., Aiken, J., Vanderby, R., Lakes, R.S.: Effect of age and exercise on the viscoelastic properties of rat tail tendon. Ann. Biomed. Eng. **41**, 1120–1128 (2013)
70. Soni, A., Kumar, S., Kumar, N.: Effect of parametric uncertainties on fracture behavior of cortical bone using XIGA. Eng. Fract. Mech. **233**, 107079 (2020)
71. Soni, A., Negi, A., Kumar, S., Kumar, N.: An IGA based nonlocal gradient-enhanced damage model for failure analysis of cortical bone. Eng. Fract. Mech. **255**, 107976 (2021)
72. Wahlsten, A., Pensalfini, M., Stracuzzi, A., Restivo, G., Hopf, R., Mazza, E.: On the compressibility and poroelasticity of human and murine skin. Biomech. Model. Mechanobiol. **18**, 1079–1093 (2019)
73. Andriotis, O.G., Chang, S.W., Vanleene, M., Howarth, P.H., Davies, D.E., Shefelbine, S.J., Buehler, M.J., Thurner, P.J.: Structure–mechanics relationships of collagen fibrils in the osteogenesis imperfecta mouse model. J. R. Soc. Interface **12**, 20150701 (2015)
74. Jones, B., et al.: Collagen fibril abnormalities in human and mice abdominal aortic aneurysm. Acta Biomater. **110**, 129–140 (2020)
75. Aziz, J., Ahmad, M.F., Rahman, M.T., Yahya, N.A., Czernuszka, J., Radzi, Z.: AFM analysis of collagen fibrils in expanded scalp tissue after anisotropic tissue expansion. Int. J. Biol. Macromol. **107**, 1030–1038 (2018)
76. Andriotis, O.G., Desissaire, S., Thurner, P.J.: Collagen fibrils: nature's highly tunable nonlinear springs. ACS Nano **12**, 3671–3680 (2018)
77. Grant, C.A., Brockwell, D.J., Radford, S.E., Thomson, N.H.: Tuning the elastic modulus of hydrated collagen fibrils. Biophys. J. **97**, 2985–2992 (2009)
78. Eyre, D.R., Paz, M.A., Gallop, P.M.: Cross-linking in collagen and elastin. Annu. Rev. Biochem. **53**, 717–748 (1984)
79. Avery, N.C., Bailey, A.J.: Enzymic and non-enzymic cross-linking mechanisms in relation to turnover of collagen: relevance to aging and exercise. Scand. J. Med. Sci. Sport. **15**, 231–240 (2005)

80. Svensson, R.B., Hassenkam, T., Grant, C.A., Magnusson, S.P.: Tensile properties of human collagen fibrils and fascicles are insensitive to environmental salts. Biophys. J. **99**, 4020–4027 (2010)
81. Wells, J.D.: Salt activity and osmotic pressure in connective tissue. I. A study of solutions of dextran sulphate as a model system. Proc. R. Soc. London. Ser. B. Biol. Sci. **183**, 399–419 (1973)
82. Berendsen, H.J.C., Migchelsen, C.: Hydration structure of fibrous macromolecules. Ann. N. Y. Acad. Sci. **125**, 365–379 (1965)
83. Fullerton, G.D., Rahal, A.: Collagen structure: the molecular source of the tendon magic angle effect. J. Magn. Reson. Imaging An Off. J. Int. Soc. Magn. Reson. Med. **25**, 345–361 (2007)
84. Chang, S.-W., Shefelbine, S.J., Buehler, M.J.: Structural and mechanical differences between collagen homo-and heterotrimers: relevance for the molecular origin of brittle bone disease. Biophys. J. **102**, 640–648 (2012)
85. Orgel, J.P.R.O., Irving, T.C., Miller, A., Wess, T.J.: Microfibrillar structure of type I collagen in situ. Proc. Natl. Acad. Sci. **103**, 9001–9005 (2006)
86. Gautieri, A., Buehler, M.J., Redaelli, A.: Deformation rate controls elasticity and unfolding pathway of single tropocollagen molecules. J. Mech. Behav. Biomed. Mater. **2**, 130–137 (2009)
87. Buehler, M.J., Wong, S.Y.: Entropic elasticity controls nanomechanics of single tropocollagen molecules. Biophys. J. **93**, 37–43 (2007)
88. Bell, J.S., et al.: The hierarchical response of human corneal collagen to load. Acta Biomater. **65**, 216–225 (2018)
89. Frank, C.J., McCreary, R.L., Redd, D.C.B.: Raman spectroscopy of normal and diseased human breast tissues. Anal. Chem. **67**, 777–783 (1995). https://doi.org/10.1021/ac00101a001
90. Hanuza, J., Maczka, M., Gasior-Glogowska, M., Komorowska, M., Kobielarz, M., Będziński, R., Szotek, S., Maksymowicz, K., Hermanowicz, K.: FT-Raman spectroscopic study of thoracic aortic wall subjected to uniaxial stress. J. Raman Spectrosc. **41**, 1163–1169 (2010). https://doi.org/10.1002/jrs.2554
91. Liu, C.-H., et al.: Raman, fluorescence, and time-resolved light scattering as optical diagnostic techniques to separate diseased and normal biomedical media. J. Photochem. Photobiol. B Biol. **16**, 187–209 (1992)
92. Manoharan, R., Baraga, J.J., Feld, M.S., Rava, R.P.: Quantitative histochemical analysis of human artery using Raman spectroscopy. J. Photochem. Photobiol. B Biol. **16**, 211–233 (1992)
93. Van Gulick, L., Saby, C., Morjani, H., Beljebbar, A.: Age-related changes in molecular organization of type I collagen in tendon as probed by polarized SHG and Raman microspectroscopy. Sci. Rep. **9**, 1–12 (2019)
94. Wang, D., He, P., Wang, Z., Li, G., Majed, N., Gu, A.Z.: Advances in single cell Raman spectroscopy technologies for biological and environmental applications. Curr. Opin. Biotechnol. **64**, 218–229 (2020)
95. Anigbogu, A.N.C., Williams, A.C., Barry, B.W., Edwards, H.G.M.: Fourier transform Raman spectroscopy of interactions between the penetration enhancer dimethyl sulfoxide and human stratum corneum. Int. J. Pharm. **125**, 265–282 (1995)
96. Gasior-Głogowska, M. et al.: FT-Raman spectroscopic study of human skin subjected to uniaxial stress. J. Mech. Behav. Biomed. Mater. **18**, 240–252 (2013)
97. Masic, A., Bertinetti, L., Schuetz, R., Galvis, L., Timofeeva, N., Dunlop, J.W.C., Seto, J., Hartmann, M.A., Fratzl, P.: Observations of multiscale, stress-induced changes of collagen orientation in tendon by polarized Raman spectroscopy. Biomacromol **12**, 3989–3996 (2011). https://doi.org/10.1021/bm201008b
98. Frushour, B.G., Koenig, J.L.: Raman scattering of collagen, gelatin, and elastin. Biopolym. Orig. Res. Biomol. **14**, 379–391 (1975)
99. Greve, T.M., Andersen, K.B., Nielsen, O.F.: ATR-FTIR, FT-NIR and near-FT-Raman spectroscopic studies of molecular composition in human skin in vivo and pig ear skin in vitro. J. Spectrosc. **22**, 437–457 (2008)

100. Flanagan, C.D., Unal, M., Akkus, O., Rimnac, C.M.: Raman spectral markers of collagen denaturation and hydration in human cortical bone tissue are affected by radiation sterilization and high cycle fatigue damage. J. Mech. Behav. Biomed. Mater. **75**, 314–321 (2017). https://doi.org/10.1016/j.jmbbm.2017.07.016
101. Unal, M., Akkus, O.: Raman spectral classification of mineral-and collagen-bound water's associations to elastic and post-yield mechanical properties of cortical bone. Bone **81**, 315–326 (2015)
102. Unal, M., Jung, H., Akkus, O.: Novel Raman spectroscopic biomarkers indicate that postyield damage denatures bone's collagen. J. Bone Miner. Res. **31**, 1015–1025 (2016). https://doi.org/10.1002/jbmr.2768
103. Haugh, M.G., Jaasma, M.J., O'Brien, F.J.: The effect of dehydrothermal treatment on the mechanical and structural properties of collagen-GAG scaffolds. J. Biomed. Mater. Res. Part A An Off. J. Soc. Biomater. Jpn. Soc. Biomater. Aust. Soc. Biomater. Korean Soc. Biomater. **89**, 363–369 (2009)
104. Uniyal, P., Sihota, P., Kumar, N.: Effect of organic matrix alteration on strain rate dependent mechanical behaviour of cortical bone. J. Mech. Behav. Biomed. Mater. 104910 (2021)
105. Hwang, J., Huang, Y., Burwell, T.J., Peterson, N.C., Connor, J., Weiss, S.J., Yu, S.M., Li, Y.: In situ imaging of tissue remodeling with collagen hybridizing peptides. ACS Nano **11**, 9825–9835 (2017)
106. Chang, S.-W., Flynn, B.P., Ruberti, J.W., Buehler, M.J.: Molecular mechanism of force induced stabilization of collagen against enzymatic breakdown. Biomaterials **33**, 3852–3859 (2012)
107. Rainey, J.K., Goh, M.C.: An interactive triple-helical collagen builder. Bioinformatics **20**, 2458–2459 (2004)
108. Humphrey, W., Dalke, A., Schulten, K.: VMD: visual molecular dynamics. J. Mol. Graph. **14**, 33–38 (1996)
109. MacKerell, A.D., Jr., et al.: All-atom empirical potential for molecular modeling and dynamics studies of proteins. J. Phys. Chem. B **102**, 3586–3616 (1998)
110. Buehler, M.J.: Atomistic and continuum modeling of mechanical properties of collagen: elasticity, fracture, and self-assembly. J. Mater. Res. **21**, 1947–1961 (2006)
111. Milazzo, M., Jung, G.S., Danti, S., Buehler, M.J.: Wave propagation and energy dissipation in collagen molecules. ACS Biomater. Sci. Eng. **6**, 1367–1374 (2020)
112. Zhmurov, A., Kononova, O., Litvinov, R.I., Dima, R.I., Barsegov, V., Weisel, J.W.: Mechanical transition from α-helical coiled coils to β-sheets in fibrin (ogen). J. Am. Chem. Soc. **134**, 20396–20402 (2012)
113. Minin, K.A., Zhmurov, A., Marx, K.A., Purohit, P.K., Barsegov, V.: Dynamic transition from α-helices to β-sheets in polypeptide coiled-coil motifs. J. Am. Chem. Soc. **139**, 16168–16177 (2017)
114. Gautieri, A., Vesentini, S., Redaelli, A., Buehler, M.J.: Viscoelastic properties of model segments of collagen molecules. Matrix Biol. **31**, 141–149 (2012)
115. Gautieri, A., Vesentini, S., Redaelli, A., Ballarini, R.: Modeling and measuring visco-elastic properties: from collagen molecules to collagen fibrils. Int. J. Non. Linear. Mech. **56**, 25–33 (2013)

Introduction to Materials Studio Software for the Atomistic-Scale Simulations



Uday Shankar, Rupam Gogoi, Sushanta K. Sethi, and Akarsh Verma

Abstract We survey the atomistic simulations using Materials Studio (MS) software, which integrates many modules for molecular dynamics (MD) simulation. These modules are first used to construct the initial random coordinates of polymer models, while other modules can then perform desired MD simulations. Subsequently, such models are used by different modules like Blends, Conformers, and Forcite to estimate properties of interest such as interaction energy of contacting particles/molecules, surface energy, density, and solubility parameters. Different approaches are utilized to sample a fair physical distribution by employing the Metropolis or configurational bias technique. It is accomplished using the Sorption module of MS (i.e., helpful to estimate the thermodynamic characteristics of small molecules present in the matrix) and Amorphous Cell (i.e., used to construct simulation cells with periodic boundary conditions). Next, several other modules are used to perform simulations like annealing and associated techniques to optimize a function. Different modules such as Polymorph Predictor and Powder Solve are used to estimate the structure of crystal from molecular structure and X-ray diffraction, respectively; in addition, Adsorption Locator is implemented to obtain preferential

U. Shankar · R. Gogoi
Department of Polymer and Process Engineering, IIT Roorkee, Roorkee, India
e-mail: ushankar@pe.iitr.ac.in

R. Gogoi
e-mail: rgogoi@pe.iitr.ac.in

S. K. Sethi (✉)
Department of Metallurgical Engineering and Materials Science, IIT Bombay, Mumbai, India
e-mail: ssethi@pe.iitr.ac.in

R. Gogoi
Department of Chemistry, Imperial College London, London, UK

A. Verma
Department of Mechanical Engineering, University of Petroleum and Energy Studies, Dehradun 248007, India

Department of Mechanical Science and Bioengineering, Osaka University, Osaka 560-8531, Japan

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sites for adsorption. In this article, we have focused on the reports that primarily performed the atomistic simulations.

1 Introduction

It is difficult to exaggerate the importance of molecular dynamics (MD) simulations in theoretical modeling and the prediction of bulk and fundamental properties. An exhaustive study on these MD simulations would probably take many volumes. Many commercial and open-source tools are recently available which employ such computational methods. The primary goal is to identify the features and abilities of the software, i.e., Materials Studio (MS), in the area of MD simulations. Here, we narrow down the possibility of approaches to recently realized MS applications for property estimations.

BIOVIA MS [1] is a software application for modeling and simulations. It was initially designed and provided to users by BIOVIA (previously Accelrys), a software agency that focuses on computational chemistry, bioinformatics, chem-informatics, MD simulation, and quantum mechanics research. In their advanced research, researchers use this tool for several polymers/materials, like ZnO, carbon nanotubes, graphene, catalysts, metals, and ceramics, by top organizations such as academics, research centers, and industries. BIOVIA MS is a client–server model software package compatible with Windows and Linux-based PC clients and servers or clusters. It provides a broad group of options to model at the atomic level and further running their simulations. Several modules are available in MS not only for modeling but also for their visualization and post-processing. The graphic user interface of MS is evident and intuitive that offers a high-quality and user-friendly environment wherein one can easily plug any BIOVIA MS product. In the subsequent segments, several modules of MS have been briefly discussed along with relevant articles employing the same module in their investigations as an example wherever required.

MS software provides users with an integrated molecular modeling environment by providing a wide range of tools to model atomistic and coarse-grained models. It also provides a series of modules to simulate them at different lengths and time scales. It also enables the user to perform quantum mechanical approaches at a short length and time scale for estimating different electronic-based properties. Similarly, at higher time and length scales, a series of modules are available to simulate and exploit the statistics related to the molecular configuration. The report says that more than 200 research articles cite the MS software every year, as shown in Fig. 1.

The approaches surveyed can be divided into the following categories:

General configuration sampling as used in the Blends and Conformers submissions.

The Sorption and Amorphous Cell applications are based on conventional Metropolis and biassed sampling.

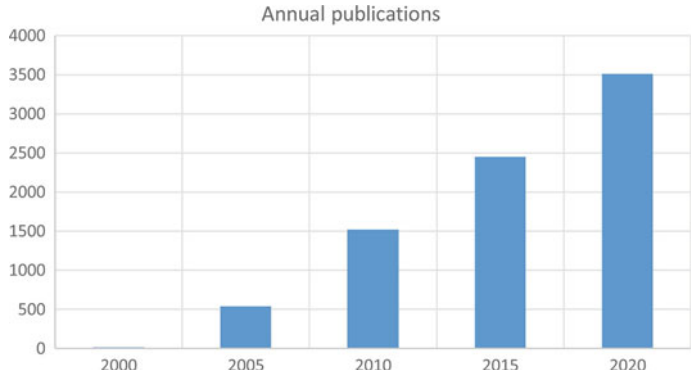


Fig. 1 Annual publications citing Materials Studio. Refurbished from [1]

MC as an optimization device, exploit in the Polymorph Interpreter, Powder Solve, and Adsorption Locator uses.

Further, a quick review of the concept underlying all of these strategies is given, backed by a more comprehensive argument of the submissions used within MS.

2 Modules and Their Applications

In 2000, MS software was first released. It included constructing systems like crystals and polymers, a module to analyze the diffraction pattern, and a 'Discover' module that permitted users to run a classical simulation engine to accomplish several jobs. Discover was selected as one of the original modules since the MD modeler continues to be one of the most critical groups in MS design. Further several other modules (or 'solvers') were added in subsequent releases, around one per year, including the quantum mechanical tools CASTEP and DMol3. In association with academia, similar modules with different abilities were designed which were assessed by users from different community groups and industries. The first version of Forcite was developed in the year 2002 which was likely to be a key event because the Forcite module was based on some of the old Cerius2 features, with new advanced optimizer tools, which opened the way for other familiar features, like Polymorph. In December 2003, QSAR and other statistical tools were added to MS that helped to create and manipulate data and construct numerous mathematical models based on those data. Next, in November 2006, the developers added the scripting feature to the software because the users have found more flexibility in combining features, controlling their interaction, and designing their sophisticated workflows. Such addition was a unique feature for the user engagements. More than twenty modules included a broad range of features by October 2009, when MS 5.0 was released. Next, a Mesocite parallel to Forcite was added to the software that helped model and simulated the systems at more considerable length and time scales. QMERA and VAMP

were also included, including the quantum mechanical calculations. The addition of new modules/features parallel with up-gradation of the existing modules with new techniques basis sets functionalities to help this software tool grow widely. Over the years, new development and enhancement of the software continued with the exciting addition of the DFTB + module (MS 6.0, December 2011) and the Cantera module (MS 2017 R2, December 2016).

The various modules of BIOVIA MS, present in the software, have been discussed in the subsequent sections. Going through these steps will definitely guide the readers to implement modeling using this software effectively.

2.1 Materials Visualizer

The materials visualizer comes up with a fast and interactive tool that helps build different graphical models (i.e., initial structure) of molecules, amorphous cells, crystalline materials, layers, interfaces, surfaces, and polymers. It can be used independently to construct, visualize, and edit the generated structures. One can also use this tool to view, analyze, and manipulate the developed models. This tool supports the software infrastructure and analysis tools in terms of providing data in various forms. It also helps to share and report the results or data easily.

2.2 Amorphous Cell

Once the initial structure has been generated, the next target is to construct a 3D model out of it. For this, the Amorphous Cell module available in MS software can be utilized to build the 3D structures with periodic boundary conditions in each x, y, and z-directions. Generally, a polymer chain comprises several segments wherein the torsion twisters are assigned to rotatable bonds. Later, amorphous cell construction is primarily automated using the initial constructed single polymeric chain into the desired number of N chains. The rotational degrees of freedom (DOF) are defined for backbone and side chains. Small segments with a smaller number of atoms are treated as rigid.

Generally, each chain segment is positioned using a biasing technique by sketching a specific number of head segment locations and orientations, estimating their (Boltzmann) weight, and weighted distribution. The algorithm considers all the atom and works by placing the consecutive component after the first chain segment has been placed. Boltzmann distribution helps to obtain a specific number of torsion angles, and also, the weights are calculated. The choice is made for the torsion angle in a chain segment, which specifies the location of the second segment. This step is continued till all the components are placed. Once this construction process is completed, the next target is to employ geometry optimization to remove any bad contacts or coordinate overlaps from the structure generated during the construction procedure. Finally,

the structure is now ready to equilibrate using MD, which comes under the Forcite module of MS.

Apart from the construction feature, the amorphous cell module also provides packing and limited layering assignments. The packing job is generally used to incorporate different chain segments inside a previously constructed amorphous cell. For instance, the addition of fillers to the base matrix and solvate of a protein or drug molecule. The limited layer assignment is helpful to construct structures with no crossing of molecules through the boundary in any direction. It is helpful to fold polymers inside a box. This task is typically utilized to construct layered composite systems like the layer builder option of MS. Constrained shear MD simulations are commonly carried out using such layered systems. The top and bottom layers are usually moved in different directions to apply shear to the intermediate layer. This limited layer building algorithm uses an external potential to restrict the substrate.

2.2.1 Applications of Amorphous Cell

In MD modeling employing the Amorphous Cell is considered to be the first step to build the given structures, and then simulated through different protocols or algorithms.

Water nanodroplet spreads on amorphous cellulose and polypropylene surfaces were examined by Liu et al. [2]. They constructed original polymer structures employing the Confined Layer feature in Amorphous Cell, then compacted and optimized them utilizing a layered structure with xenon crystals. The above method produced a level of molecular surface roughness, which closely matched experimental results. Furthermore, the estimated properties using simulations were consistent with that of the experimental findings. Water nanodroplet was observed to be spreading on the cellulose surfaces. The droplet dimension did not significantly vary and the observed contact angles were 106° and 33°, respectively, for polypropylene and amorphous cellulose surfaces. Several authors have utilized the MS software to construct spherical water/oil droplets first and then to place it above a surface for the assessment of phobic/philic nature [3–5].

Amorphous Cell was adopted to mimic membranes by Peng et al. [6] in which poly(vinyl alcohol) (PVA) was used and multi-walled carbon nanotubes (MW-CNT) were added which were wrapped by chitosan polymer. The pervaporation ability of the model system was investigated in relation to the free volume parameters. Addition of 1 wt.% MW-CNT improved the flux by three times the value with no MW-CNT. They observed that adding 1% nanotube material increased the permeation flux threefold. They aligned positron annihilation lifetime spectroscopy to their findings.

Moisture transport in epoxy resins was studied by Lin et al. [7]. They used amorphous cells to build epoxy-water boxes and estimated the water displacement with varying temperatures. The energy of desorption activation was compared to experimental values of moisture uptake.

The effect of the diffusion of cyclohexane and benzene due to the presence of graphite particles in a hybrid membrane of PVA-graphite was studied by Pan et al.

[8]. Polymer and graphite particles with varying quantities of hydroxyl and carboxyl groups were produced using the Amorphous Cell building method. The presence of graphite particles enhanced chain mobility, which was deduced by the glass transition temperature and mean square displacement values.

Hölck et al. [9] illustrated how the Packing job was applied. Using a cross-linking method, they established 3D networked epoxy molding materials.

2.3 Powder Solve

Powder Solve can use a crystal's powder diffraction pattern to determine its structure. In addition, the unit cell parameters are also required for the determination. (Another tool in Materials Studio can estimate these from the diffraction pattern.) The atoms (or stiff groupings of atoms) in the unit cell define the crystal structure. A potential structure's diffraction pattern is calculated and compared to experimental data, and a range of feasible structures is searched for matching.

2.3.1 Applications of Powder Solve

Ferrocene-based macrocycle molecules were prepared by Dinnebier et al. and incorporated Powder Solve to evaluate their properties and also employed Density-functional theory (DFT) which yielded optimized structural properties.

Previously unidentified structure for phase III solid methane was successfully determined by Neumann et al. [10]. Powder Solve used simulated annealing to investigate this challenging material, which required neutron diffraction. They proved that the crystal does not have a tetragonal space group, which was assumed earlier (and obtained the correct space group is C m c a).

Corma et al. [11] investigated ITQ-40, a zeolite with a highly open structure. They determined the structure using various methods, including electron diffraction, X-ray powder, and X-ray single-crystal data. The Powder Solve helped in obtaining the partial structure based on the powder data.

Luneau et al. [12] introduced a series of manganese-based materials, layered structures, and almost all ferromagnetic. Powder Solve was applied to solve the structure once more.

Dova et al. [13] focused on an iron-based compound with complicated spin-crossover characteristics. Synchrotron radiation was used to acquire powder diffraction data, then analyzed via Powder Solve's parallel tempering and a competing approach (a genetic algorithm).

2.4 Polymorph Predictor

Materials Studio's Polymorph Predictor module tries to forecast the most desirable shapes for a molecular crystal by beginning with the molecule itself. The essential premise yet the most probable structures are those with the least amount of energy–entropy is ignored. Although the crystal's energy may be easily calculated through forcefield approaches, the entropy is complex to calculate and ignore.

The Packing phase is the essential element of the method, wherein the space of potential crystal structures is explored for low-energy options. The crystal lattice characteristics (cell lengths and angles) and the location and orientation of each molecule within it define a crystal structure. A simulated annealing technique is employed because it produces a rough energy landscape with multiple local minima. The user enters the structure of the desired molecule and a forcefield. Furthermore, there should be room for interest groups.

2.4.1 Applications of Polymorph Predictor

Neumann et al. looked into the crystal structures of various tiny compounds such as ethane, ethylene, and acetylene. Polymorph Predictors were used to generate candidate crystal packings for these compounds, which were subsequently optimized via energies calculated using a combination of DFT and an empirical adjustment for the van der Waals (vdW) interaction.

Day et al. [14] contrasted anticipated with established experimental crystal structures for a sample of 50 tiny organic compounds. They observed that nearly half of the defined structures had energy, close to the calculated minimum.

Cross et al. [15] investigated diflunisal, a fluorinated aromatic carboxylic acid, using computational and experimental methods. They employed Polymorph Predictor to predict structure (on five space groups) and implemented the findings to assist their crystallization solvent selection.

A crystal comprising two distinct molecules, a cyclic phosphoric acid, and an ephedrine enantiomer was predicted by Leusen. Both molecular flexibility and chirality complicate this system.

Sarma et al. [16] used experimental results to supplement the Polymorph Predictor outcomes. The algorithm yielded a set of low-energy structures, which were subsequently rearranged considering existing structures of comparable molecules. It has the effect of augmenting Polymorph Predictor's enthalpic calculations with kinetic input inherent in the experimental findings.

3 Widely Used Forcefields Under Materials Studio (MS) Software

3.1 COMPASS

COMPASS is a strong forcefield widely used in molecular/atomistic simulations of condensed-phase materials. This forcefield has been designed for a wide range of materials and extensively utilized by numerous researchers [3, 17–22]. It is the first ab initio forcefield to be characterized and confirmed with condensed-phase parameters and ab initio and empirical data for molecules in isolation. As a result, this forcefield allows for the precise and instantaneous forecast of structural, conformational, vibrational, and thermophysical features for a wide variety of molecules in isolated and condensed phases and under a broad range of temperature and pressure circumstances. This forcefield is regularly modified and is under constant study. On the user interface discourse, the most current edition is labeled COMPASS.

3.2 COMPASS II

The COMPASS forcefield has been massively boosted with COMPASS II. COMPASS II expands on COMPASS's present scope to provide a far more significant number of drugs and molecules of concern to scientists working with ionic liquids. Prof. Huai Sun of Shanghai's Jiao Tong University contributed to the development of COMPASS II. The present COMPASS forcefield was integrated with current quantum mechanical computations with instruments created by Prof. Sun to get additional features and fix a few anomalies in the current COMPASS forcefield. By comparing equilibrium structures and vibrational frequencies, novel valence variables were revealed. It is how the training set that was used to estimate these parameters was expanded:

(i) COMPASS was used to evaluate a polymer database containing 430 homo- and copolymers, and 454 missing terms were discovered, denoted by 105 parts. The training set was expanded to include these species.
(ii) A similar analysis was performed on the NIST ionic liquid molecule database, and 40 species were included in the training set.
(iii) COMPASS was used to scan the Maybridge database, which had 59,465 molecules, and 1257 parts were included in the training set.

Forcite

Forcite is a molecular mechanics component that uses classical mechanics to calculate potential energy and geometry optimization of a random molecule and periodic systems. Forcite backing the COMPASS, UFF, and DREIDING forcefields. Because of its wide variety of forcefields, Forcite can tackle practically any material. In

addition to the advanced algorithm that applies these approaches successively, the geometry optimization algorithm includes the steepest descent, conjugate gradient, and quasi-Newton techniques. It enables extremely precise energy minimizations. The Forcite component will allow us to carry out a variety of molecular mechanic estimates on both molecular and periodic systems employing traditional forcefield-based simulation approaches. Single point energy estimation, geometry optimization, molecular dynamics, quench dynamics, anneal dynamics, shear, confined shear, cohesive energy density quantification, mechanical property analysis, and solvation free energy measurement are just a few of the jobs Forcite can undertake.

The Forcite energy job enables us to determine the total energy and provide inputs from various forcefield components, including bond and vdW terms, to the overall energy. The energy job aids in determining if a forcefield is appropriate for a chosen system. It can also show whether the structure has an acceptable geometry and, as a result, whether or not a geometry optimization is required.

The Forcite geometry optimization assignment offers us to adjust a structure's geometry until it meets specific requirements. It is accomplished by an iterative process wherein atomic coordinates and cell variables are modified until the structure's overall energy is minimized. As a result, the optimized structure correlates to a potential energy surface minimum. The number of computed forces and (if necessary) stresses are reduced by Forcite geometry optimization until they are fewer than specified convergence margins. It is feasible to provide an external model to characterize the system's behavior under tension or compression. Because the forces on an atom are computed from the potential energy expression, they depend on the chosen forcefield. Forcite geometry optimization acknowledges any symmetry elements specified for the system in crystal formations. As a result, the optimization is restricted with a limited number of degrees of freedom.

The Forcite dynamics job helps users simulate how atoms in a structure will move when forces are computed. Before executing a Forcite dynamics computation, a thermodynamic ensemble and associated variables like the simulation time step and temperature should be chosen. Experiments for fixed NVE settings are generally challenging to accomplish, drawing parallels between simulation and experiment challenge. Alternative ensembles can be formed using classical Legendre transformations, such as the canonical (or NVT) ensemble, allowing a system to exchange heat with the environment and maintain a constant temperature (T). You may mimic an ensemble using a thermostat application by changing Newton's equations.

Five thermostats are available from Forcite: velocity scale, Nose, Nose-Hoover-Langevin, Andersen, and Berendsen. Through an integration algorithm, the time step is a critical factor. A significant time step should be employed to create the ultimate exploitation of calculation time. Nevertheless, if the time step is too substantial, the integration method may become unstable and inaccurate. It usually shows up as a systematic drift in the constant of motion. Still, it can also result in the work failing suddenly because of the considerable energy difference between steps. Using the Forcite mechanical characteristics assignment, we can calculate mechanical characteristics for a single structure or a trajectory of structures. A Forcite physical properties evaluation can be done on a single structure or a series of structures produced by

dynamics and saved in a trajectory file (.arc, his, trj, and xtd). The physical properties are determined, averaged over all possible configurations, and published throughout the output text document.

3.3 Universal

The UFF has been categorized underclass I forcefield and parameterized on the element, hybridization, and connectivity that primarily covers all atoms present in the periodic table. Computational researchers have validated such forcefield through simulation studies for primary group compounds, metal complexes, and organic molecules. The most significant limitation of UFF is its accuracy compared with a forcefield that has been specifically developed for a group of molecules/compounds.

The cross-terms of UFF have been parameterized based on the hybridization of the considered set of atoms or molecules. It is composed of several factors like hybridization defined from atomic radii, bond, i.e., expressed by harmonic $\frac{1}{2}k(r-r_0)^2$ term, wherein the r_0 refers to an equilibrated bond length, and k represents the force constant; bond angle contribution includes triple term Fourier cosine series of the form $E_\theta = K_{ijk}\sum_{i=0}^{3} C_n \cos(n\theta)$; the dihedral angle term is of the form $E_\varnothing = K_{ijkl}\sum_{i=0}^{n} C_n \cos(n\varnothing_{ijkl})$; the inversion term is defined as $E_\omega = \sum_{i=0}^{3} C_n \cos(n\omega_{ijkl})$. Further, the non-bonded interactions consist of vdW term $E_{vdW} = \varepsilon_{ij}\left(-2\left(\frac{\sigma_{ij}}{r}\right)^6 + \left(\frac{\sigma_{ij}}{r}\right)^{12}\right)$ and electrostatic coupling $E_{el} = 332.0637\left(\frac{q_i q_j}{\varepsilon r_{ij}}\right)$.

3.4 CVFF

The CVFF is a generalized valence forcefield, and its parameters are described for water, amino acids, and several functional groups. Like other forcefields, this one has also gone under extensive validation tests. The CVFF shown in Eq. (1) is based on harmonic and in-harmonic terms consisting of the covalent interactions and the long-range vdW, ionic (Coulomb), and hydrogen bonds interactions.

$$E_{CVFF} = \sum_{bond} K_1(l-l_0)^2 + \sum_{angle} K_\theta(\theta-\theta_0)^2 + \sum_{dihedral} K_n[1+\cos(n\varphi-\varphi_0)] + \sum_{improper} K_\omega(1-cos2\omega)\cdot \sum_{non-bondvdW} +4\varepsilon\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] + \sum_{non-bondvdW} \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}} \quad (1)$$

Where

K_l = stiffness constants
K_θ = angle
K_n = dihedral
K_ω = improper terms.

The l, θ, φ, and ω represent the bond length, bond angle, dihedral angle, and improper torsion angle. Further ε and σ_{ij} are the LJ parameters that describe the vdW interaction, q_i refers to the partial charge of the corresponding atom describing the electrostatic interactions.

3.5 PCFF

The PCFF is a member of the consistent family of forcefields (COMPASS, PCFF, CFF, and CFF91) that comes under the 2nd generation forcefield. It is parameterized through a broad range of experimental observations for organic compounds, halogen atoms, ions, alkali metal cations, and numerous biochemically relevant divalent metal cations. It is derived from CFF91 and was intended for a broad range of polymers and organic materials. This forcefield is very useful for certain polymers, including polycarbonates, polysaccharides, melamine resins, and nucleic acid. It helps to predict properties like cohesive energy, mechanical property, heat capacity, elastic constant, compressibility. It handles electron delocalization in aromatic rings through a charging library rather than bond increments.

3.6 Comparative Study of Forcefields

Radosinski et al. [23] have extensively compared second and third-generation forcefields using combined Monte Carlo and MD simulations. The authors demonstrated that an INTERFACE forcefield provides a possibility of composing a reliable atomistic model to reproduce the change in density of poly(vinylalcohol) (PVA) matrix in a very narrow range of temperature (298–348 K), which further utilized to estimate the thermal coefficient of expansion with decent accuracy. The authors found the density to be 0.887, 1.248, 1.245, 1.235, and 1.246 g/cc with a relative error of 31.4, 3.49, 3.73, 4.47, and 3.62% for Universal, COMPASS, COMPASS II, PCFF, and INTERFACE FF, respectively, which is close to the experimentally observed value of 1.293 g/cc. The relative error % is acceptable for all forcefield predictions except Universal. Further, when they investigated the contributions of vdW and electrostatic energies, they found that significant contributions are coming from electrostatic interactions (depicted in Fig. 1). The Universal forcefield overestimated these electrostatic interactions, which caused to provide a higher relative error % in estimating the density (Fig. 2).

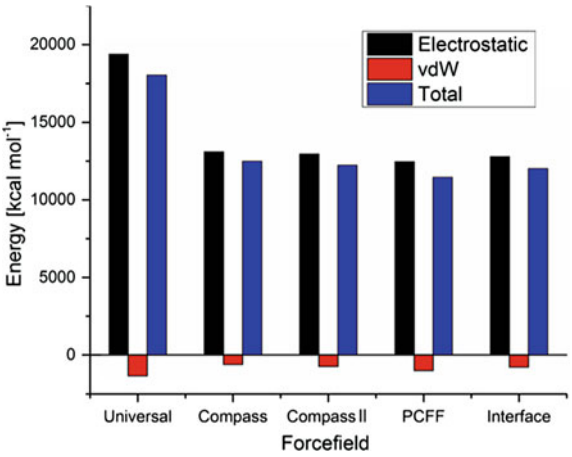


Fig. 2 Computed non-bonded energic contributions for PVA matrix by different forcefields. Refurbished from [23]

Hall et al. [24] have employed MD simulations on two models for two polymers, based on the monomers (1) 6,6′-bis(3,4-dihydro-3-methyl-2H-1,3-benzoxazinyl)isopropane and (2) 6,6′-bis(3,4-dihydro-3-methyl-2H-1,3-benzoxazinyl)sulphone. Their study aimed to validate the outcomes obtained by MS software with previously reported results obtained by Cerius and experimental investigations. For Cerius the Dreiding and MS, the UFF forcefield was employed in the study. The authors have observed good agreement of the obtained results; for instance, the recorded T_g in a range of 188–196 °C for polymer (1) and 133 °C for polymer (2), respectively, was within 11 K of the empirical value of 177 °C and 16 K of the empirical value of 117 °C. Additionally, molecular mechanics simulations suggest Young's modulus of 5.9 GPa for (1), which is higher than the empirically determined value of 4.3 GPa. The authors also have a wide experience in the field of molecular dynamics [25–42].

4 Conclusion

MS software consists of a comprehensive set of modules for the construction of desired models, their MD simulations, and post-processings. This tool is exploited by a broad range of users in a variety of ways ranging from traditional configuration to structural determinations. A wide range of forcefields are also available in all modules of MS to be used based on the materials selected. This tool also provides a feature to support complex workflows by combining different studies by using Perl-based Materialscript API. Nowadays this tool is widely being used effectively and extensively for more than a decade for several applications such as physical and chemical sciences and almost every industrial sector.

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References

1. Meunier, M., Robertson, S.: Materials Studio 20th anniversary. Mol. Simul. **47**, 537–539 (2021). https://doi.org/10.1080/08927022.2021.1892093
2. Liu, H., Li, Y., Krause, W.E., Rojas, O.J., Pasquinelli, M.A.: The soft-confined method for creating molecular models of amorphous polymer surfaces. J. Phys. Chem. B. **116**, 1570–1578 (2012). https://doi.org/10.1021/JP209024R
3. Sethi, S.K., Manik, G.: A combined theoretical and experimental investigation on the wettability of MWCNT filled PVAc-g-PDMS easy-clean coating. Prog. Org. Coatings. **151**, 106092 (2021). https://doi.org/10.1016/j.porgcoat.2020.106092
4. Sethi, S.K., Soni, L., Shankar, U., Chauhan, R.P., Manik, G.: A molecular dynamics simulation study to investigate poly(vinyl acetate)-poly(dimethyl siloxane) based easy-clean coating: an insight into the surface behavior and substrate interaction. J. Mol. Struct. **1202**, 127342 (2020). https://doi.org/10.1016/j.molstruc.2019.127342
5. Sethi, S.K., Singh, M., Manik, G.: A multi-scale modeling and simulation study to investigate the effect of roughness of a surface on its self-cleaning performance. Mol. Syst. Des. Eng. **5**, 1277–1289 (2020). https://doi.org/10.1039/d0me00068j
6. Peng, F., Pan, F., Sun, H., Lu, L., Jiang, Z.: Novel nanocomposite pervaporation membranes composed of poly(vinyl alcohol) and chitosan-wrapped carbon nanotube. J. Memb. Sci. **300**, 13–19 (2007). https://doi.org/10.1016/J.MEMSCI.2007.06.008
7. Kim, D.H., Kim, H.S.: Investigation of hygroscopic and mechanical properties of nanoclay/epoxy system: molecular dynamics simulations and experiments. Compos. Sci. Technol. **101**, 110–120 (2014). https://doi.org/10.1016/j.compscitech.2014.06.026
8. Pan, F., Peng, F., Jiang, Z.: Diffusion behavior of benzene/cyclohexane molecules in poly(vinyl alcohol)-graphite hybrid membranes by molecular dynamics simulation. Chem. Eng. Sci. **62**, 703–710 (2007). https://doi.org/10.1016/J.CES.2006.07.046
9. Chen, C.M., Chang, H.L., Lee, C.Y.: Improvement prediction on the dynamic performance of epoxy composite used in packaging by using nano-particle reinforcements in addition to 2-Hydroxyethyl Methacrylate Toughener, Materials **14**, 4193 (2021). https://doi.org/10.3390/MA14154193
10. Neumann, M.A., Press, W., Nöldeke, C., Asmussen, B., Prager, M., Ibberson, R.M.: The crystal structure of methane phase III. J. Chem. Phys. **119**, 1586 (2003). https://doi.org/10.1063/1.1580809
11. Corma, A., Díaz-Cabañas, M.J., Jiang, J., Afeworki, M., Dorset, D.L., Soled, S.L., Strohmaier, K.G.: Extra-large pore zeolite (ITQ-40) with the lowest framework density containing double four- And double three-rings. Proc. Natl. Acad. Sci. USA **107**, 13997–14002 (2010). https://doi.org/10.1073/PNAS.1003009107/-/DCSUPPLEMENTAL
12. Luneau, D., Borta, A., Chumakov, Y., Jacquot, J.F., Jeanneau, E., Lescop, C., Rey, P.: Molecular magnets based on two-dimensional Mn(II)–nitronyl nitroxide frameworks in layered structures. Inorg. Chim. Acta. **361**, 3669–3676 (2008). https://doi.org/10.1016/J.ICA.2008.03.109
13. Dova, E., Peschar, R., Sakata, M., Kato, K., Schenk, H.: High-spin- and low-spin-state structures of [Fe(chloroethyltetrazole)6](ClO4)2 from synchrotron powder diffraction data. Chem. A Eur. J. **12**, 5043–5052 (2006). https://doi.org/10.1002/chem.200501594
14. Day, G.M., Chisholm, J., Shan, N., Motherwell, W.D.S., Jones, W.: An assessment of lattice energy minimization for the prediction of molecular organic crystal structures. Cryst. Growth

Des. **4**, 1327–1340 (2004). https://doi.org/10.1021/CG0498148/SUPPL_FILE/CG0498148SI20040818_091635.PDF
15. Cross, W.I., Blagden, N., Davey, R.J., Pritchard, R.G., Neumann, M.A., Roberts, R.J., Rowe, R.C.: A whole output strategy for polymorph screening: combining crystal structure prediction, graph set analysis, and targeted crystallization experiments in the case of diflunisal. Cryst. Growth Des. **3**, 151–158 (2003). https://doi.org/10.1021/CG025589N/SUPPL_FILE/CG025589NSI20021121_021736.PDF
16. Sarma, J.A.R.P., Desiraju, G.R.: The supramolecular synthon approach to crystal structure prediction. Cryst. Growth Des. **2**, 93–100 (2002). https://doi.org/10.1021/CG015576U/SUPPL_FILE/CG015576U_S1.PDF
17. Sethi, S.K., Kadian, S., Anubhav, G., Chauhan, R.P., Manik, G.: Fabrication and analysis of ZnO quantum dots based easy clean coating: a combined theoretical and experimental investigation. Chem. Select **5**, 8942–8950 (2020). https://doi.org/10.1002/slct.202001092
18. Kumar, N., Manik, G.: Molecular dynamics simulations of polyvinyl acetate-perfluorooctane based anti-stain coatings. Polymer (Guildf). **100**, 194–205 (2016). https://doi.org/10.1016/j.polymer.2016.08.019
19. Gogoi, R., Sethi, S.K., Manik, G.: Surface functionalization and CNT coating induced improved interfacial interactions of carbon fiber with polypropylene matrix: a molecular dynamics study. Appl. Surf. Sci. **539** (2021). https://doi.org/10.1016/j.apsusc.2020.148162
20. Shankar, U., Sethi, S.K., Singh, B.P., Kumar, A., Manik, G., Bandyopadhyay, A.: Optically transparent and lightweight nanocomposite substrate of poly(methyl methacrylate-co-acrylonitrile)/MWCNT for optoelectronic applications: an experimental and theoretical insight. J. Mater. Sci. **56**(30), 17040–17061 (2021). https://doi.org/10.1007/S10853-021-06390-3
21. Agrawal, G., Samal, S.K., Sethi, S.K., Manik, G., Agrawal, R.: Microgel/silica hybrid colloids: bioinspired synthesis and controlled release application. Polymer (Guildf). **178**, 121599 (2019). https://www.sciencedirect.com/science/article/pii/S003238611930583X. Accessed 24 June 2019
22. Sethi, S.K., Soni, L., Manik, G.: Component compatibility study of poly(dimethyl siloxane) with poly(vinyl acetate) of varying hydrolysis content: an atomistic and mesoscale simulation approach. J. Mol. Liq. **272**, 73–83 (2018). https://doi.org/10.1016/J.MOLLIQ.2018.09.048
23. Radosinski, L., Labus, K.: Molecular modeling studies of structural properties of polyvinyl alcohol: a comparative study using INTERFACE force field. J. Mol. Model. **23** (2017). https://doi.org/10.1007/s00894-017-3472-z
24. Hall, S.A., Hamerton, I., Howlin, B.J., Mitchell, A.L.: Validating software and force fields for predicting the mechanical and physical properties of poly(bisbenzoxazine)s. **34**, 1259–1266 (2008). https://doi.org/10.1080/08927020802534835
25. Verma, A., Parashar, A., Packirisamy, M.: Atomistic modeling of graphene/hexagonal boron nitride polymer nanocomposites: a review. Wiley Interdiscip. Rev. Comput. Mol. Sci. **8**(3), e1346 (2018)
26. Verma, A., Parashar, A.: Structural and chemical insights into thermal transport for strained functionalised graphene: a molecular dynamics study. Mater. Res. Express **5**(11), 115605 (2018)
27. Verma, A., Parashar, A., Packirisamy, M.: Effect of grain boundaries on the interfacial behaviour of graphene-polyethylene nanocomposite. Appl. Surf. Sci. **470**, 1085–1092 (2019)
28. Chaurasia, A., Verma, A., Parashar, A., Mulik, R.S.: Experimental and computational studies to analyze the effect of h-BN nanosheets on mechanical behavior of h-BN/polyethylene nanocomposites. J. Phys. Chem. C **123**(32), 20059–20070 (2019)
29. Verma, A., Kumar, R., Parashar, A.: Enhanced thermal transport across a bi-crystalline graphene–polymer interface: an atomistic approach. Phys. Chem. Chem. Phys. **21**(11), 6229–6237 (2019)
30. Verma, A., Parashar, A.: Molecular dynamics based simulations to study the fracture strength of monolayer graphene oxide. Nanotechnology **29**(11), 115706 (2018)
31. Verma, A., Parashar, A.: The effect of STW defects on the mechanical properties and fracture toughness of pristine and hydrogenated graphene. Phys. Chem. Chem. Phys. **19**(24), 16023–16037 (2017)

32. Verma, A., Parashar, A., Packirisamy, M.: Tailoring the failure morphology of 2D bicrystalline graphene oxide. J. Appl. Phys. **124**(1), 015102 (2018)
33. Singla, V., Verma, A., Parashar, A.: A molecular dynamics based study to estimate the point defects formation energies in graphene containing STW defects. Mater. Res. Express **6**(1), 015606 (2018)
34. Verma, A., Parashar, A.: Molecular dynamics based simulations to study failure morphology of hydroxyl and epoxide functionalised graphene. Comput. Mater. Sci. **143**, 15–26 (2018)
35. Verma, A., Zhang, W., van Duin, A.C.: ReaxFF reactive molecular dynamics simulations to study the interfacial dynamics between defective h-BN nanosheet and water nanodroplets. Phys. Chem. Chem. Phys. **23**, 10822–10834 (2021)
36. Verma, A., Parashar, A., Packirisamy, M.: Role of chemical adatoms in fracture mechanics of graphene nanolayer. Mater. Today Proc. **11**, 920–924 (2019)
37. Verma, A., Parashar, A.: Characterization of 2D nanomaterials for energy storage. In: Recent Advances in Theoretical, Applied, Computational and Experimental Mechanics, pp. 221–226. Springer, Singapore (2020)
38. Verma, A., Parashar, A., Singh, S.K., Jain, N., Sanjay, M.R., Siengchin, S.: Modeling and simulation in polymer coatings. In: Polymer Coatings: Technologies and Applications, Chapter 16, pp. 309–324. Taylor & Francis Group (CRC Press), Boca Raton (2020)
39. Kataria, A., Verma, A., Sanjay, M.R., Siengchin, S.: Molecular modeling of 2D graphene grain boundaries: mechanical and fracture aspects. Mater. Today Proc. (2021)
40. Deji, R., Verma, A., Kaur, N., Choudhary, B.C., Sharma, R.K.: Density functional theory study of carbon monoxide adsorption on transition metal doped armchair graphene nanoribbon. Mater. Today Proc. (2021)
41. Deji, R., Verma, A., Choudhary, B.C., Sharma, R.K.: New insights into NO adsorption on alkali metal and transition metal doped graphene nanoribbon surface: a DFT approach. J. Mol. Graph. Model. (2021). https://doi.org/10.1016/j.jmgm.2021.108109
42. Deji, R., Jyoti, R., Verma, A., Choudhary, B.C., Sharma, R.K.: A theoretical study of HCN adsorption and width effect on co-doped armchair Graphene Nanoribbon. Comput. Theor. Chem. 113592 (2022)

Data-Driven Phase Selection, Property Prediction and Force-Field Development in Multi-Principal Element Alloys



Dishant Beniwal, Jhalak, and Pratik K. Ray

Abstract Multi-Principal Element Alloys (MPEAs) have brought a paradigm shift in the alloy design process and pose a significant challenge due to the astronomical and compositional space available for exploration. Since experimental and ab-initio methods are more suitable for targeted alloy design over a narrow composition range, data-driven methodologies have shown promise in the search for alloys with unique or improved properties. This chapter introduces the field of materials informatics by laying out the fundamentals of machine learning, i.e., the types of problems, dataset formulation, feature selection, and machine learning algorithms, as applied to materials science. It follows with a holistic review of the existing data-driven models targeted towards the prediction of phase selection, mechanical properties and ordering behaviour in MPEAs. It also discusses the methodology for the development of machine learning force fields that enable atomistic modelling of various phenomena, such as diffusion, phase transformations and mechanical deformation, in MPEAs.

Keywords Multi-Principal Element Alloys · Materials informatics · Machine learning · Phase selection · Mechanical properties · Force fields

D. Beniwal (✉) · Jhalak · P. K. Ray
Metallurgical & Materials Engineering, Indian Institute of Technology Ropar, Rupnagar, Punjab 140001, India
e-mail: dishant.beniwal@gmail.com

P. K. Ray
e-mail: pratik@iitrpr.ac.in

D. Beniwal
Mechanical Engineering, University of Petroleum and Energy Studies, Dehradun, Uttarakhand 248007, India

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1 Introduction

Multi-Principal Element Alloys (MPEAs) are compositionally complex alloys containing a large number of principal elements, often five or more, in high concentrations. While various terminologies are associated with these alloys, such as High-Entropy Alloys (HEAs), multicomponent alloys, compositionally complex alloys, and complex concentrated alloys, the central idea remains the same i.e., to explore the alloys in the center of multicomponent phase spaces. Contrary to the traditional viewpoint that mixing of multiple elements in high concentrations would lead to the formation of complex and brittle microstructures, in the 1980s, Prof. Brian Cantor started experimenting with mixing up to 20 elements and observed the surprising formation of a single-phase structure in five-component equiatomic CoCrFeMnNi. In 2004, research groups led by Prof. Brian Cantor and Prof. Jien-Wei Yeh reported the formation of simple- solid-solution (SS) phases in various equiatomic alloys wherein the stability of SS phases was attributed largely to the high configurational entropy of mixing [1, 2]. This new alloy design paradigm led to an explosion in MPEAs research resulting in the discovery of various alloy systems with unique and promising properties. While the initial interest in MPEAs was focused on finding systems that form simple SS phases, over the years it has branched into various domains involving the study of structure–property correlations, deformation mechanisms, alloy processing routes, and short-range ordering behaviour [3–11].

The major challenges associated with the exploration of MPEAs are—(a) an astronomical compositional space [8] (spanning more than 10^{100} unique compositions) that cannot be explored efficiently using experimental route or ab-initio calculations, and (b) a multitude of inter-atomic interactions (due to the presence of multiple elements in high concentrations) that need to be estimated accurately for computational modelling of MPEAs. Machine Learning (ML) based predictive models and force fields, driven by experimental results and ab-initio calculations, have evolved as a promising avenue for addressing these challenges. Predictive ML models, based on a variety of algorithms such as neural networks, random forests, support vector machines, and XGBoost, have been used for predicting phase selection, strength, hardness, Young's modulus, corrosion behaviour, and ordering behaviour in MPEAs. ML frameworks have also been used to design interatomic potentials for MPEAs to enable molecular-dynamics simulations for probing alloy properties.

This chapter introduces the fundamental concepts of data-driven methodologies that are essential for the training and validation of ML models, followed by a holistic review of the data-driven models that have been developed for exploring various aspects of MPEAs. The detailed methodology for ML based force-field development for MPEAs has also been included.

2 Machine Learning for Materials Science

Accelerating the development of new engineering materials has been a long-standing challenge. A number of initiatives in the recent years have attempted to address this in an attempt to go beyond the pioneering works of Michael J. Ashby [12, 13]. Primarily, one can divide the approaches into two different classes—(a) multi-scale modelling with a focus on Integrated Computational Materials Engineering (ICME) approaches [14] and (b) Data-driven Materials Informatics whose origins can be traced to the Materials Genome initiative [15–18]. Traditional computational approaches have been reasonably successful in alloy design [19]; however, recent advances in data science render Materials Informatics as a viable paradigm for alloy exploration and design. Here, our focus is on the second approach, i.e., Materials Informatics.

2.1 Materials Informatics

The case for Materials Informatics is quite compelling. Over the decades, metallurgists have favoured experimental approaches resulting in a fairly large volume of data. Recent advances in computational materials science have also allowed us to predict properties of materials with reasonable accuracy, albeit they tend to be time intensive. However, these computational methods allow us to augment the experimental data as and when needed. Consequently, it would appear that we are well-poised to usher in a new era in materials with the aid of data science.

2.1.1 An Overview of Materials Informatics

The standard approach towards finding data-driven solutions in materials research initiates with the curation of databases. The challenge here is to capture and combine data from large disparate datasets for well-established materials like steels and superalloys or to curate a database from relatively sparse data (as with HEAs, although with the recent emphasis on these materials, the available data keeps increasing exponentially). Modern trends in database curation rely on text-mining approaches involving Natural Language Processing (NLP) [20, 21]. An especially significant challenge arises in establishing the pedigree for data. Crystallographers have long faced this problem as they compiled databases of crystal structures. Today, many of the standard crystallographic files have a star-based rating system which indicates the data pedigree [22]. However, the crystallographers tend to use the traditional ICDD data or JCPDS cards to match patterns or use the relevant crystallographic information files for Rietveld refinement where one can "refine" the model structure according to experimental measurements. A quantitative aspect wherein the data quality factors directly into predicted structures or properties has not been explored, to the best of our knowledge. Data quality issues can arise in many forms—appropriate calibration

of equipment (e.g., how reliable is the heat treatment temperature reported for a grain growth study would depend on how well the furnace was calibrated and how long ago was it calibrated), the sensitivity of the measurement instruments, state of the art of the research field at the time of experimental investigations (e.g., phase diagrams of refractory metal silicides by Nowotny et al. had to be reworked since the earlier method of assessment did not incorporate the possible role of carbon in stabilizing the structures [23]), etc. The subject-matter experts (SME) are critical during the database curation phase. If sufficient data is available, this is a stage where one must look deeper into the data quality and discard the datasets that are of relatively lower quality.

Once a database is available, we typically train the ML models using these datasets. The training is done once the features that influence the target predictions are identified. Features are essentially physical descriptors or physics-based properties that can be used for fitting the experimental data. One typically divides a database into two parts—training dataset and test dataset. The model is trained using pre-defined learning algorithms on the training dataset and then used in conjunction with the test dataset for validating its accuracy. This is typically an iterative process involving multiple algorithms and sometimes multiple sets of training and test datasets. The model limitations are largely incurred by the training dataset. The quality of an ML model is only as good as the data it has been trained with.

2.1.2 Types of Problems

Problems in materials are diverse and often wrapped in complex layers and classes. While designing materials, we generally ask two questions—(i) what is the best material chemistry that is suited for us and (ii) how do we process the material in order to get the desired property? The first addresses the intrinsic properties of the material while the second addresses the extrinsic properties of the material. Suitable databases would then have to be curated depending on the type of properties. A different challenge presented by materials is that of non-linearity. On one hand, we have solid-solution alloys that follow Vegard's law; on the other hand, one has complex phase diagrams with multiple intermetallics. Vegard's law predicts that the volume of the alloy changes linearly as a function of the solute content. In other words, this is a case where linear variations in chemical composition are observed. The presence of multiple intermetallics, however, can result in non-linear (and even non-monotonic) variations in material properties [24]. Therefore, it is always a challenge to understand where one should expect a linear variation and where one expects a non-linear variation. Finally, there is the problem of feature selection. Many of the features can essentially be derived from a combination of other features—in some cases, this can lead to unfair biasing of the model while in other cases it leads to redundancy.

The challenge involving intrinsic vs extrinsic property predictions can be addressed primarily through the curation of appropriate database and feature engineering (discussed later in this chapter). Here, we focus on addressing the challenge

of non-linear variation of properties in materials at a philosophical level. From a purely statistical viewpoint, one can consider two classes of problems—(i) those based on "classification", especially where non-linearity is built in and (ii) those based on "regression" fitting over nearly linear regions. Let us look at specific examples where each class of problems becomes prominent as well as on overlap of these two classes.

Classification problems will typically involve features that "classifies" the material into different categories, usually using specific features. To exemplify this, let us consider the problem of crystal structure prediction. A material could be FCC, BCC, etc. Traditionally, this is partially addressed using the Hume-Rothery rules where crystal structures are assigned on the basis of valence electron concentrations (VEC). For instance, a VEC higher than 8 destabilizes the BCC structure in HEAs; it matters that the VEC is above 8, not *how much* above 8 [25], as shown in Fig. 1. Thus, a VEC of 8 acts as a classifier. Classification problems in ML can be of different types. The simplest of all would be the binary classification—e.g., either a material is FCC or it is not. Essentially, the outcome is like a binary, either a 0 or a 1. Algorithms such as decision trees, Naïve-Bayes or Support Vector Machine (SVM) are well-suited for such classification problems. If one were to expand the crystal structure prediction problem to include FCC, BCC and HCP, then one would have more than two classes possible. Such a problem would then constitute a multi-class situation, where there can be more than two classes, but the outcome will still only contain a single class. Continuing with the crystal structure prediction examples, one can consider multi-label classifications—an alloy can contain both FCC and BCC crystal structures if it is a two-phase alloy. Beniwal and Ray have recently illustrated how such classifications can be handled and quantified [25]. Usually, gradient boost algorithms or multi-label decision trees/random forests are used for addressing such a problem. Classification problems are quite significant in materials science. A phase diagram often contains multiple phases (and multi-phase field regions). Whenever a new phase emerges, there could be a significant change in the properties which results in a discontinuous jump in the target value of the properties as a function of the material composition, as we cross over a phase field. Essentially, classification approaches help us understand the non-linear variation of material properties.

Regression problems, on the other hand, are well-suited for linear variation. Solute-solution hardening in dilute alloys is a reasonable example. As long as phase

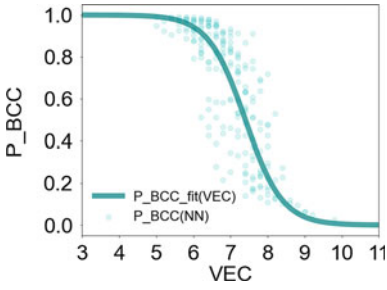


Fig. 1 Probability of occurrence of BCC phase (P_BCC) in equiatomic MPEAs as predicted by ML model [25] as a function of VEC

transformations are absent and strong ordering processes are not operative, the hardness and strength vary roughly linearly with the material chemistry. Even when the variation is monotonic and continuous, if not linear, regression approaches can be quite useful. A survey of the database can throw some trends in the property variation based on which the appropriate regression algorithm (e.g., linear regression, logistic regression) can be selected. In real materials, one typically has homogeneous phase fields over which regression algorithms work interspersed with phase boundaries across which non-monotonic and discontinuous jumps may occur. Hence, a combination of regression and classification algorithms is often used for modelling the material properties [24].

2.2 Databases

Despite the plethora of information, very few well-structured databases exist for materials. A large body of effort has gone into devising some very specific databases for individual research problems; however, they cater to only very specific problems and alloy systems. Hierarchical customizable platforms, like Granta (upon which the ASM phase diagram database is built), do exist; but their full potential is yet to be realized.

Since the initial interest in the Materials Genome efforts, a number of databases have been compiled based primarily on density-functional theory calculations. (e.g., AFLOW, Materials Project, NIMS-MatNavi database, etc.). Additionally, traditional materials databases include the phase diagram handbooks, crystallographic databases (ICDD, Pearson Crystal Structure Database, etc.) and thermodynamic databases (often proprietary, developed by companies such as ThermoCalc or CompuTherm—usually, these databases are intended to be used with the respective proprietary program). In addition to these well-established structured databases, a number of researchers and groups have published a large number of smaller sized datasets (e.g., the Fe-9Cr steel database [26]).

Given the nascent stage (compared to, say, biological sciences) of structured data availability for materials, the creation of well-structured databases is an important focus area in Materials Informatics. Primarily, one could construct databases in four ways—(i) manual data curation from the literature, (ii) high through-put assessment of the research literature using tools such as text-mining, (iii) creation of databases through computational tools (density-functional theory (DFT) based calculations, Molecular Dynamics simulations, Calphad calculations, etc.) and (iv) additional experiments for augmenting databases. Recent advances in experimental sciences have allowed us to rapidly generate data through high-fidelity combinatorial methods. For instance, instead of using diffusion couples, one could easily use diffusion multiples to extract interdiffusion coefficients for multiple combinations [27]. Similarly, techniques such as combinatorial nanocalorimetry allow us to rapidly obtain thermodynamic data from nanometric samples [28]. The utility of such combinatorial approaches has been demonstrated quite effectively in the design

of a number of novel materials ranging from high-entropy alloys [29] to magnets [30].

2.3 Feature Creation and Selection

Once the database is created, one must decide on a set of features with which the machine learning model will be trained. A feature is a measurable property of the material that we are trying to analyze, which can be expressed as a function of the input data. In a dataset, the features can appear as columns with each feature calculated for every compositional entry (for a study designed to study the composition dependence of a given property), as shown in Fig. 2 [25].

2.3.1 Feature Selection and Their Significance

Selecting the appropriate features is of paramount interest. The essence of solving materials engineering problems using machine learning is the ability to take a holistic look at what factors matter (should be included as a feature in the model) and what does not (can be safely excluded without compromising on the model accuracy). To illustrate this, let us take the example of metallic glasses. Early models of glass formation in metallic systems considered only one factor (or feature) at a time, such as—formation enthalpy [31], configurational and mismatch entropy [32], lattice strain and strain energy [33], valence electron concentration [34] or even dense-random packing of atoms in solute-centered clusters [35, 36]. Very few synergistic analytical models were proposed that could combine all of these factors [37, 38]. However, with the advent of machine learning enabled high throughput methods,

alloy_ name	phases	phase_ code	R_ delta	S_ config	VEC	Tm_ avg	E_ avg	E_ delta	EN_ Allen_ avg	H_ ch	H_ el
CoFeNiSi0.5	FCC+Im	4	0.03	0.01	8.29	1757.29	197.43	0.18	1.85	-10.83	1.39
Al0.5HfNbTaTiZr	BCC	2	0.05	0.01	4.27	2378.67	114.93	0.35	1.35	-7.52	3.43
CuMnNiSn0.75	FCC+Im	4	0.10	0.01	8.27	1328.88	156.39	0.39	1.83	-2.39	23.19
HfMo0.5NbSi0.3TiV0.5	BCC+Im	5	0.08	0.01	4.58	2382.35	140.95	0.50	1.40	-11.58	5.91
MoNbTaVW	BCC	2	0.03	0.01	5.40	2961.00	230.60	0.51	1.44	-4.69	2.43
HfMo0.5NbSi0.7TiV0.5	BCC+Im	5	0.10	0.01	4.53	2323.17	138.49	0.49	1.45	-21.91	6.01
CoFeMoNi1.2V	FCC+Im	4	0.04	0.01	7.69	2063.58	218.46	0.29	1.71	-8.95	5.46
CuMnNiSn	Im	7	0.11	0.01	8.00	1277.39	150.15	0.42	1.83	-2.72	27.12
AlMoNbTiV	BCC	2	0.04	0.01	4.60	2140.86	146.64	0.64	1.48	-12.73	1.61
Al0.158CuMnNi	FCC	1	0.04	0.01	9.02	1504.78	175.77	0.26	1.82	-3.47	1.19
AlCoCrFeMo0.4Ni	BCC+Im	5	0.06	0.01	7.11	1773.76	181.89	0.38	1.73	-11.75	5.14
MoNbTaTi0.5W	BCC	2	0.03	0.01	5.33	3020.78	239.33	0.51	1.42	-5.82	1.33
Al0.375CuMnNiZn0.375	FCC+Im	4	0.05	0.01	8.97	1390.48	161.36	0.33	1.78	-7.52	2.70
CuMnNiZn2	FCC+BCC	3	0.05	0.01	10.40	1197.94	145.88	0.36	1.73	-6.32	3.02
Al0.25NbTaTiV	BCC	2	0.04	0.01	4.65	2446.90	126.01	0.28	1.43	-4.68	2.21
CoCrCuFeMnNiTiV	FCC+BCC+Im	6	0.05	0.02	7.50	1811.20	166.63	0.26	1.71	-7.87	4.27
Hf0.5Mo0.5NbSi0.1TiZr	BCC+Im	5	0.07	0.01	4.49	2363.22	126.63	0.62	1.37	-4.69	5.29

Fig. 2 Structure of a database used for predicting phase selection in MPEAs [25]. It contains target information about each alloy (phases and corresponding phase_code) along with eight alloy descriptors (R_delta, S_config, and so on) that are used as independent variables to predict the target phase

newer models which combined all of these features with the appropriate weightage were formulated for bulk (non-equiatomic) metallic glasses [39] as well as high-entropy metallic glasses [40]. An example of relevant and (relatively) irrelevant features also surfaced in the work on solid-solution hardening by Roy et al. [41, 42], which conclusively demonstrated the lack of entropic contributions towards mechanical properties in HEAs while at the same time revealing that solute atom-dislocation energies and cohesive energies play a dominant role in the hardness and Young's modulus of HEAs. Feature engineering, therefore, forms a very critical aspect of Materials Informatics.

In order to identify the appropriate features, one needs to first apprehend the nature of the problem. In the world of materials, this offers two distinct possibilities, as discussed above—prediction of intrinsic properties or extrinsic properties. Intrinsic properties are a function of material chemistry. Therefore, the features used should reflect this. One needs to capture the properties and interactions between atoms in a perfect crystal. Thus, features such as formation enthalpies, cohesive energies, valence electrons, size asymmetries, etc. play an important role. As an example, one can consider the aforementioned example of predicting Young's modulus where cohesive energies play a major role. Alternatively, one could look into the examples of phase selection where VEC acts as a major classification feature. Capturing extrinsic properties is more challenging. This requires the ability to model defects in materials which are often a consequence of the processing conditions. Broadly, three approaches could be plausible depending on the available data and the target properties to be predicted. Namely, (i) one could adopt an approach where the features would be defect densities and energy barriers to the thermodynamic equilibrium state, coupled with input variables such as different processing parameters; (ii) one could plausibly quantify the final microstructure and correlate this to the properties and (iii) rely on intrinsic parameters, especially on those that play a key role in processing response (e.g., diffusion coefficients).

The final microstructure and crystallographic defects (and defect densities) are often a function of processing conditions. The processing conditions control the extent to which the material moves away from equilibrium. This can come through variables such as cooling rates (materials synthesis from liquid or vapor), heat treatment parameters such as temperature, time and heat treatment schedule (alloys being homogenized and brought closer to thermodynamic equilibrium) or strain and strain rates (solid-state processing often involving severe plastic deformation). An example of this approach in the case of solid-state processing was illustrated by optimization of critical velocities (which affects the strain rate, in turn) for cold-sprayed coatings [43], with the results being in reasonable agreement with the available (though simplified) analytical [44] and computational models [45].

If a database of microstructures is available, one could potentially tune the mechanical properties as a function of the microstructural features. Quantification of metallurgical microstructures remains a challenge though, despite the recent advances and success in correlating the mechanical behaviour with microstructure [46, 47]. The final approach involves using composition dependent parameters even for the prediction of extrinsic properties. This is exemplified in the recent efforts of

Beniwal et al., whereby the hardness has been captured effectively through a machine learning model including only composition dependent features [24]. It should be noted though that while the predicted hardness is in good agreement with cast alloys (closer to equilibrium) they significantly under-predict the hardness of mechanically alloyed and spark plasma sintered samples (which have a finer grain structure, having been processed through a non-equilibrium route). Nonetheless, trends are still captured by the model.

2.3.2 The Correlation Matrix and Multi-Collinearity

Before building any ML model, it is always advantageous to study the correlation between the various features and target variables present in the database. A correlation matrix is a table that expresses the correlation coefficients between variables and thus helps in visualizing the dependencies present in the data (especially if these dependencies are linear). Pearson's [48] and Spearman's [49] correlation coefficients are the most commonly used correlation metrics. Both these coefficients can vary between -1 to $+1$ wherein a value of $+1$ or -1 indicates a perfect linear relationship, positive and negative values represent direct and inverse relations respectively, and zero implies no correlation. Pearson's correlation is a measure of the linear relationship between features and assumes that the features are normally distributed, whereas Spearman's correlation is a measure of monotonicity of the relationship between two features and does not assume a normal distribution of features. Both Pearson's and Spearman's correlation are used only for continuous variables. If we have a binary variable, a 'point biserial' correlation coefficient [50] is used to measure the relationship between a binary variable and a continuous variable.

Multicollinearity between features is another important aspect of the database that must be probed before building ML models as it can have a significant impact on the perceived importance of a feature and thereby may lead to an incorrect interpretation of the learning. Multicollinearity occurs due to the presence of two or more variables in the database that are highly correlated to each other. This happens often due to the introduction of dependent variables that are a function of already existing independent variables in the dataset. There are generally two ways to root out multicollinear features: (a) through a detailed study of the correlation matrix and feature maps, and (b) through hierarchical clustering of features and creation of a dendrogram that groups collinear features based on a distance matrix [51].

2.4 Machine Learning Algorithms

While all ML methodologies are aimed at identifying relationships hidden in the data in one way or another, these are broadly classified into three categories: (a) supervised learning—a target (output) variable is predicted from a set of descriptors

(inputs) through the generation of a function that maps inputs to target, (b) unsupervised learning—there is no specific target variable, but instead inherent dependencies between variables are exposed through different techniques such as clustering of the population into groups, and (c) reinforcement learning—the model is trained to make specific decisions in such a way that it trains itself continuously through exposure to an environment wherein it is rewarded and punished from desired and undesired predictions respectively.

Some of the commonly used ML algorithms include linear regression, logistic regression, decision tree, support vector machine (SVM), naïve-bayes, k-nearest neighbours (kNN), K-Means clustering, random forest, gradient boosting and neural networks [52]. Amongst all these algorithms, neural networks have gained significant attention in the recent years due to their capability to: (a) learn both linear and non-linear relationships in data, (b) adapt to various kinds of input data structures like structured tables, images with spatial relationships, signals with temporal relationships and videos with both spatial and temporal relationships, and (c) perform both regression and classification tasks with extremely high accuracy. But despite these advantages, neural networks suffer from their lack of interpretability since it is difficult to make out the exact decision-making process learnt by the model. To address this, there are some methodologies that have been adopted to impart interpretability or explainability to these models [24, 53]. The choice of algorithm is often dictated by three factors: (a) the type of decision boundaries that are to be learnt, (b) the type of task i.e., regression or classification and the nature of the target variable, and (c) the amount of interpretability desired from the trained model.

3 Data-Driven Models for MPEAs

All supervised ML frameworks operate with an underlying hypothesis that the target property is fundamentally associated with the alloy features being used as an input for the model; and thus, the training process estimates these dependencies based on the relationships that are captured in the dataset. This approach has been used extensively in building ML models that predict various properties and behaviour of MPEAs.

3.1 Phase Selection

The mechanical properties of an alloy are strongly dependent on the phases present, and thus, a targeted alloy design approach requires prior knowledge of the phases that will form in an alloy. This becomes a challenge for MPEAs since the mixing of different elements can result in a variety of single-phase or multi-phase microstructures containing SS phases and Intermetallics (IM). Thus, the phase selection in MPEAs has been by far the most explored problem and numerous data-driven

methodologies have been implemented over the years to address it. These approaches have been classified into two categories here: (a) parameter-based approaches wherein a single or combination of parameters have been used to draw phase classification boundaries, and (b) ML based approaches wherein a group of features have been used to train ML algorithms for phase prediction.

3.1.1 Parameter-Based Phase Selection

Since the initial interest in MPEAs was focused on finding alloys with simple SS phases, a major portion of the earlier work in data-driven phase selection was centered around two key classifications: (hether the alloy will form only simple SS phases, or will it form IM phases and (b) whether the structure of SS phase will be FCC or BCC.

FCC/BCC structure prediction: In 2011, Guo et al. [54] proposed Valence Electron Count (VEC) as a criterion for delineating the stability of FCC and BCC SS wherein VEC $\geq$ 8, (6.7 $\leq$ VEC < 8) and VEC < 6.87 results in the formation of FCC, (FCC + BCC) and BCC SS phases respectively. This simple rule has stood the test of time and has been reinforced in numerous later works wherein VEC appears as the single most important factor in determining BCC versus FCC stability. But, at the same time, it is important to point out two limitations of this rule: (a) it cannot explain FCC/BCC stability variations in systems wherein VEC remains constant for e.g., Fe_x-$(AlCoCr_{0.5}Ni_{2.5})_{1-x}$ [55], and (b) there are exceptions to the bounds of 6.7 and 8 put on VEC since these have been estimated statistically and do not represent strict absolute boundaries for FCC/BCC stability. Recently, Beniwal and Ray [25] identified electronegativity (χ_{Allen}) and cohesive energy (E_{coh}) as additional parameters to guide the identification of FCC versus BCC stability domains based on binary-feature maps. They reported an improved empirical classification criterion: BCC (VEC < 6.8 and E_{coh} > 3.55 eV), FCC (VEC > 8 and E_{coh} < 5.25 eV) and BCC + FCC (6.8 < VEC < 8 and 3.55 eV < E_{coh} < 5.25 eV).

Predicting formation of simple-solid-solution versus intermetallic phases: The empirical and semi-empirical rules for predicting the formation of SS and IM phases in MPEAs are generally based on two types of parameters: (a) thermodynamic parameters such as configurational entropy of mixing (ΔS_{mix}) and mixing enthalpy (ΔH_{mix}), and (b) lattice distortion parameters such as size asymmetry, strain energy and geometric packing state. The configurational entropy of mixing is calculated as: $\Delta S_{mix} = -R\sum_{i=1}^{n} c_i \ln c_i$, where R is the universal gas constant and c_i is the concentration of ith element. The enthalpy of mixing is calculated using Takeuchi's extended regular solution model [56] as: $\Delta H_{mix} = \sum_{i=1, i\neq j}^{n} 4\Delta H_{ij}^{mix} c_i c_j$, where ΔH_{ij}^{mix} is the mixing enthalpy of equiatomic binary i–j calculated using Miedema's model [57, 58]. While various improvements have been proposed for ΔH_{mix} calculations in ternary systems [59–61], they are yet to be extended to multi-component alloys. Table 1 lists the existing parameter-based models for predicting the formation of simple SS phases in MPEAs along with the performance of each model as reported

Table 1 List of parameter-based models developed for phase classification in MPEAs, along with the description of parameters used and the performance of each model as measured by Tsai et al. [62]

Model Ref	Criterion	Description of parameters	Performance [62]	
			DoU (%)	Accuracy (%)
Yang and Zhang, 2012 [63]	$\Omega \geq 1.1$ and $\delta_r \leq 0.066$	$\Omega = T_m \Delta S_{\text{mix}} / \Delta H_{\text{mix}}$ $\delta_r = \sqrt{\sum_{i=1}^{n} c_i \left(1 - \frac{r_i}{\bar{r}}\right)^2}$ where, T_m and $\bar{r}$ are average melting temperature and average atomic radius respectively obtained using rule-of-mixture; c_i is the concentration of ith element; and n is the number of elements in alloy $\Delta S_{\text{mix}} = -R \sum_{i=1}^{n} c_i \ln c_i$ $\Delta H_{\text{mix}} = \sum_{i=1, i \neq j}^{n} 4 \Delta H_{ij}^{\text{mix}} c_i c_j$ where, $\Delta H_{ij}^{\text{mix}}$ is the mixing enthalpy of equiatomic binary *i-j* calculated using Miedema's model [57]	83	48
Guo et al. 2013 [64]	$-11.6 < \Delta H_{\text{mix}} < 3.2$ kJ mol^{-1} and $\delta_r < 0.066$	$\Delta H_{\text{mix}} = \sum_{i=1, i \neq j}^{n} 4 \Delta H_{ij}^{\text{mix}} c_i c_j$ where, $\Delta H_{ij}^{\text{mix}}$ is the mixing enthalpy of equiatomic binary *i–j* calculated using Miedema's model [57] $\delta_r = \sqrt{\sum_{i=1}^{n} c_i \left(1 - \frac{r_i}{\bar{r}}\right)^2}$ where, $\bar{r}$ is average atomic radius obtained using rule-of-mixture; c_i and r_i is the concentration and radius of ith element; and n is the number of elements in alloy	53	61

(continued)

Table 1 (continued)

Model Ref	Criterion	Description of parameters	Performance [62]	
			DoU (%)	Accuracy (%)
Wang et al. 2015 [65]	$-11.6 < \Delta H_{\text{mix}} < 3.2$ kJ mol^{-1} and $\gamma < 1.175$	$\gamma = \omega_S/\omega_L$ $\omega_S = (1 - \sqrt{((r_S + \overline{r})^2 - \overline{r}^2)/(r_S + \overline{r})^2})$ $\omega_L = (1 - \sqrt{((r_L + \overline{r})^2 - \overline{r}^2)/(r_L + \overline{r})^2})$ where, r_S and r_L are the radii of smallest and largest atoms in alloy	10	54
Singh et al. 2014 [66]	$\Lambda > 0.96$	$\Lambda = \Delta S_{\text{mix}}/{\delta_r}^2$ $\Delta S_{\text{mix}} = -R\sum_{i=1}^{n} c_i \ln c_i$ $\delta_r = \sqrt{\sum_{i=1}^{n} c_i \left(1 - \frac{r_i}{\overline{r}}\right)^2}$ where, $\overline{r}$ is average atomic radius obtained using rule-of-mixture; c_i and r_i is the concentration and radius of ith element; and n is the number of elements in alloy	−38	63

(continued)

Table 1 (continued)

Model Ref	Criterion	Description of parameters	Performance [62]	
			DoU (%)	Accuracy (%)
Ye et al. 2015 [67]	$\phi \geq 20$	$\phi = (\Delta S_{\text{mix}} - S_H)/\lvert S_E \rvert$ $S_H = \lvert \Delta H_{\text{mix}} \rvert / T_m$ where, S_E is the excess configurational entropy of mixing [68]	−23	61
Troparevsky et al. 2015 [69]	$-T_{\text{ann}} \Delta S_{\text{mix}} < \Delta H_f < 3.57$ meV/atom	ΔH_f is the formation enthalpy (ΔH_{ij}^{IM}) of most stable binary IM compound (i-j) possible in alloy ΔH_{ij}^{IM} values for various i-j pairs were calculated by authors using DFT calculations	−45	72
Senkov and Miracle 2016 [70]	$\kappa_1^{cr} > \Delta H_{IM}/\Delta H_{\text{mix}}$	$\kappa_1^{cr} = T_m \Delta S_{\text{mix}}(1 - \kappa_2)/\lvert \Delta H_{\text{mix}} \rvert$ $\kappa_2 = 0.6$ (assessed using a sub-lattice model)	−12	72
King et al. 2016 [71]	$\Phi \geq 1$	$\Phi = -\Delta G_{SS}/\lvert \Delta G_{\max} \rvert$ $\Delta G_{SS} = \Delta H_{\text{mix}} - T_m \Delta S_{\text{mix}}$ where, ΔH_{mix} includes chemical, elastic and structural contributions [57]; $\Delta G_{\max}$ is the highest (segregation) or lowest (intermetallic) Gibbs free energy amongst all binary systems that are possible from elements constituting the alloy	−84	63

by Tsai et al. [62] in a recent study. These rules lay out the conditions under which only simple SS phases will be observed i.e., outside these regions IM phases are predicted to appear in the alloy. While a variety of parameters are used in these rules, the key underlying principles comprise: (a) high ΔS_{mix} stabilizes the SS phases, (b) large negative ΔH_{mix} drives compound formation and a large positive ΔH_{mix} leads to clustering, (c) large asymmetry in atomic sizes destabilizes the SS phases due to the introduction of lattice strains, and (d) presence of atomic-pairs with high intermetallic formation tendency (i.e., high $\Delta H_{IM}/\Delta H_{\text{mix}}$ ratio) promotes the appearance of IM phases. It is important to note that while these rules give some general insights into SS versus IM stability in MPEAs, the performance of these parameter-based models is not very good as they exhibit low accuracy and a high degree of unbalance, as seen in Table 1. This is primarily due to: (a) the simplistic nature of these rules as they utilize only a few parameters that are insufficient to directly capture all of the physics associated with the alloying process, and (b) the small size of datasets used to formulate these rules. To overcome this, ML based phase prediction models have gained significant attention in recent years and have been discussed in the next section.

3.1.2 Machine Learning Driven Phase Selection

To overcome the limitations posed by parameter-based models, a large number of ML based models have been reported in recent years to predict phase selection in MPEAs [25, 42, 72–78], as listed in Table 2. A majority of these models are directed towards predicting whether the alloy will contain only simple SS phases or if it will contain IM phases; and only a few models focus on the identification of the crystal structure of SS phases [25, 42, 74]. The implementation of ML requires a database to start out with; and for phase selection, the database must contain phase information of each alloy obtained via. experimental characterization or first-principles calculations. The phase information is further encoded into integer labels, for e.g., (SS $\rightarrow$ 0, SS + IM $\rightarrow$ 1, and IM $\rightarrow$ 2) or (FCC $\rightarrow$ 0, BCC $\rightarrow$ 1, and FCC + BCC $\rightarrow$ 2), wherein these labels comprise the target space for model training. For a more exhaustive multi-label classification, the phase information can also be encoded as a 3-element binary vector as shown by Beniwal and Ray [25], wherein the vector-elements correspond to presence of FCC, BCC and IM phases, i.e., (FCC $\rightarrow$ [1,0,0], FCC + BCC $\rightarrow$ [1,1,0], FCC + BCC + IM $\rightarrow$ [1], and so on). Here, the model is trained to predict this 3-element binary vector for each alloy i.e., it predicts the probability of occurrence of each phase—FCC, BCC and IM. This approach allows a seven-label classification wherein both SS versus IM and FCC versus BCC classifications can be achieved without compromising the model accuracy [25].

For ML implementation, each alloy is represented through a set of descriptors that capture the atomic, chemical, physical or thermodynamic information about the alloy. An extensive list of such descriptors for MPEAs has been provided by Roy and Balasubramanian [79]. The features for the ML model are selected based on their statistical correlation and physical relevance to alloy phase stability, and

Table 2 List of ML based models for phase selection in MPEAs along with the details of classification labels, dataset, features and algorithms used in the model training

Model Ref	Classification labels	Training dataset	Features	ML algorithms	Classification accuracy
Islam et al. 2018 [72]	AM, SS, IM	Exp., 118 alloys	ΔH_{mix}, ΔS_{mix}, δ_r, VEC,$\Delta\chi$	Neural Network	83%
Huang et al. 2019 [73]	SS, SS + IM, IM	Exp., 401 alloys	ΔH_{mix}, ΔS_{mix}, δ_r, VEC,$\Delta\chi$	K-nearest neighbours, support vector machine, neural network	SS vs. IM: 86.7% SS + IM versus IM: 94.3% SS versus SS + IM: 78.9%
Li and Guo 2019 [74]	FCC, BCC, NSP (Not forming Single Phase SS)	Exp., 322 alloys	ΔH_{mix}, ΔS_{mix}, δ_r, VEC, $\Delta\chi$,T_m	Support vector machine	BCC:60% FCC:75% NSP:98%
Zhou et al. 2019 [75]	AM, SS, IM	Exp., 601 alloys (binary, ternary and higher-order alloys)	ΔH_{mix}, ΔS_{mix}, δ_r, VEC, $\Delta\chi$, T_m, $\overline{r}$, $\overline{\chi}$, δ_{T_m}, ΔK, ΔVEC, $(\Delta H_{\text{mix}})_{LM}$, $\overline{K}$	Neural network, convolutional neural network, support vector machine	AM:95.6% SS:97.8% IM:92.2%
Roy et al. 2020 [42]	FCC, BCC, FCC + BCC	Exp., 329 alloys	$\Delta\chi_{\text{Pauling}}$, ΔH_{mix}, ΔS_{mix}, δ_r, Λ, T_m, a_m, Ω, Δa,ΔT_m	Gradient boost classification	70%
Zhang et al. 2020 [76]	FCC, BCC, FCC + BCC, SS, NSS (not forming SS)	Exp., 550 alloys	A total of 70 material descriptors used	Linear discriminant analysis, decision tree, Naïve Bayes classification, neural network, random forest, support vector machine	FCC, BCC, FCC + BCC: 88.7% SS, NSS: 91.3%

(continued)

the parameter-based models discussed earlier provide a useful guide as to which descriptors are important for phase predictions. VEC, atomic size asymmetry, cohesive energy, enthalpy of mixing and entropy of mixing are amongst the most common features that appear repeatedly in ML models for phase prediction in MPEAs.

Table 2 (continued)

Model Ref	Classification labels	Training dataset	Features	ML algorithms	Classification accuracy
Jaiswal et al. 2021 [77]	FCC, BCC, FCC + BCC	Exp., 664 alloys	ΔH_{mix}, ΔS_{mix}, δ_r, VEC, $\Delta\chi$, T_m, $\overline{\chi}$,Ω	Logistic regression, random forest, support vector machine, decision tree, gradient boosting decision tree, neural network	86%
Lee et al. 2021 [78]	SS, IM, SS + IM, AM	Exp., 989 alloys	ΔH_{mix}, ΔS_{mix}, δ_r, VEC, $\Delta\chi$, T_m, $\overline{r}$, $\overline{\chi}$, δ_{T_m}, ΔK, ΔVEC, $(\Delta H_{\text{mix}})_{LM}$,$\overline{K}$	Deep Neural Network, Bayesian optimization, Generative Adversarial Network	SS: 89% SS + IM: 90% IM: 94% AM: 100%
Beniwal and Ray 2021 [25]	FCC, BCC, FCC + BCC, FCC + IM, BCC + IM, FCC + BCC + IM, IM	Exp., 323 alloys	δ_r^{cov}, δ_r^{met}, ΔH_{mix}, ΔH_{el}, ΔS_{mix}, VEC, ρ, δ_E, δ_G, δ_K, $\overline{\chi}_{\text{Allen}}$, E_{coh}, $\delta_{E_{coh}}$,δ_{V_m}	Deep Neural Network Ensemble	FCC: 90%, BCC: 93%, FCC + BCC: 70%, FCC + IM: 82%, BCC + IM: 77%, FCC + BCC + IM: 71%, IM:71%

The accuracy of each model is also mentioned. The description of each feature notation has been provided in the Appendix 1

Table 2 lists the various ML models, along with the information about dataset, features, algorithms, and performance, that have been developed over the last three years for predicting phase selection in MPEAs. These models exhibit a considerably higher classification accuracy as compared to the parameter-based models listed in Table 1. This is primarily due to two reasons: (a) the multi-dimensional nature of classification boundaries learnt by ML models due to the use of a large number of features, and (b) the use of larger datasets which amounts to more training examples for the model to learn from. Through continuous improvements in the ML models, a large pool of material descriptors is now available for use as model features and the feature selection process is often carried out based on statistical correlation, intuitive physical significance and/or feature-importance study. A systematic study to identify the multi-collinearity and physical-overlapping of these features is still lacking.

Also, the usual label classification approach only predicts the presence/absence of a phase and does not give any idea about the quantitative phase fractions. In a recent work, Beniwal and Ray [25] proposed an approach wherein the predicted

probabilities of occurrence of phases are treated as a measure of phase stabilities and are thereby transformed into phase fraction estimates. This was achieved through a combined use of 3-element binary vectors, that capture phase information, along with a softmax function in the output layer of the neural network that restricts the model outputs such that the summation of outputs (phase fractions) cannot exceed a value of 1. The ability of the model to capture both linear and non-linear variations in the phase fractions of FCC, BCC and IM phases, as a function of alloy composition, was demonstrated over a variety of MPEAs through comparison with experimental and CALPHAD results [25].

3.2 Mechanical Properties

Since MPEAs span an astronomical compositional space, the ability to predict mechanical properties is imperative for targeted alloy design. Various ML models, as listed in Table 3, have been reported in recent years for the prediction of mechanical properties such as hardness, Young's Modulus and elastic constants [29, 42, 80–83]. Table 3 also presents details concerning the dataset, features, algorithm and performance for each model. In one of the earlier works, Chang et al. [80] developed a simple neural network model to predict the Vickers hardness of Al-Co-Cr-Cu-Fe-Mn-Mo-Ni containing alloy systems. They used elemental concentrations as the model inputs, which makes the implementation easy but renders the model unusable for probing alloy systems that are away from this 8-element space used to train the model. Thus, all future works have used a more robust and general modelling approach wherein the compositional inputs are replaced with composition dependent physical and thermodynamic features that are relevant to alloy hardness. Since the hardness of an alloy is closely associated with the type of phases present, a significant overlap can be seen between the features that have been used for phase selection (Table 2) and hardness (Table 3) in MPEAs. A variety of ML algorithms, as listed in Table 3, have been used to model the Vickers hardness in MPEAs using physical descriptors (such as radius asymmetry, melting temperature, density, elastic modulus asymmetry and VEC) and thermodynamic descriptors (enthalpy and entropy of mixing). In a forthcoming work [24], Beniwal et al. have shown that neural network ensembles can accurately capture even the non-linear hardness variations in MPEAs that are driven by phase transformations and ordering behaviour. The predicted hardness variations closely mimic the phase stabilities and ordering parameters calculated through DFT calculations. Further, it has been shown that the physical origins of hardness in any given alloy system can be distilled using an explainability analysis that is driven by local-partial dependencies and stimulus–response study.

ML models, especially neural networks, often face criticism that even though they may perform exceptionally well in terms of statistical accuracy, the nature of fit may not capture the underlying physics associated with the manifestation of the target property. To address this concern, it is imperative to impart some sort of explainability or interpretability to the trained ML models. An important advance in this regard is

Table 3 List of ML based models for prediction of mechanical properties in MPEAs along with the details of target property, dataset, features, and algorithms used in the model training

Model Ref	Target property	Training dataset	Features	ML algorithms	Model performance
Chang et al., 2019 [80]	Vickers Hardness	Exp., 91 alloys	$c_{el}(el = Al, Co, Cr, Cu, Fe, Mn, Mo, Ni)$	Neural network	$R^2 = 0.94$; MAE = 36 HV
Rickman et al. 2019 [29]	Vickers Hardness	Exp., 82 alloys	ΔH_{mix}, ΔS_{mix}, δ_r, T_m, Ω, δ_E,VEC	Canonical-correlation analysis, Genetic algorithm	-
Wen et al. 2019 [81]	Vickers Hardness	Exp., 155 alloys	ΔH_{mix}, ΔS_{mix}, δ_r, Ω, Λ, VEC, $\Delta\chi$, γ, $(\Delta\chi)_{LM}$, e/a, E_{coh}, η, $(\delta)_{LM}$, A, F, w, $\overline{G}$, δ_G, $(\Delta G)_{LM}$,μ	Linear regression, polynomial regression, support vector regression, neural network, K-nearest neighbour	Best Model: Support Vector Regression RMSE = 68
Roy et al. 2020 [42]	Young's Modulus	Exp., 87 alloys	$\Delta\chi_{\text{Pauling}}$, ΔH_{mix}, ΔS_{mix}, δ_r, Λ, T_m, a_m, Ω, Δa,ΔT_m	Gradient Boost Regression	MAE = 36 GPa Avg. % Error = 18.2%
Khakurel et al., 2021 [83]	Young's Modulus	Exp., Two datasets (154 & 96 alloys)	$\Delta\chi_{\text{Pauling}}$, ΔH_{mix}, ΔS_{mix}, δ_r, Λ, T_m, a_m, Ω, Δa, ΔT_m,VEC	Gradient boosting, XGBoost, random forest, ada boost, support vector regression, regression (Lasso, Ridge & Gaussian)	Best Model: Gradient Boosting Only Refractory dataset: MAE = 6.2 GPa, R^2 = 0.9 All dataset: MAE = 10.4 GPa, $R^2 = 0.71$
Revi et al. 2021 [82]	Elastic Constants	Materials Project, 1229 binary alloys	$\overline{r}$, $\overline{\chi}$, $\overline{IP}$, $\overline{\rho}$, VEC, T_m, $\overline{K}$, $\overline{C}$, $\overline{k}$,$\overline{\nu}$	Linear regression, random forest, neural network	Best Model: Random Forest MAE = 23.5 (DFT)/ 19.3 (Exp.)
Yang et al. 2022 [84]	Vickers Hardness	Exp., 370 alloys	VEC, T_m, Δm, ΔCol,ΔV	Support vector regression	RMSE = 75 $R^2 = 0.88$

(continued)

Table 3 (continued)

Model Ref	Target property	Training dataset	Features	ML algorithms	Model performance
Beniwal et al. 2022 [24]	Vickers Hardness	Exp., 218 alloys	ΔH_{mix}, ΔS_{mix}, δ_r^{cov}, δ_r^{met}, δ_{V_m}, δ_E, δ_G, δ_K, VEC, $\overline{\rho}$	Neural Network Ensemble	RMSE = 70 $R^2 = 0.85$ MAE = 48.6 HV

The accuracy of each model is also mentioned, wherever available. The description of each feature notation has been provided in Appendix 1

an explainability analysis by Beniwal et al. [24] that integrates local-partial feature dependencies with a stimulus–response study to expose the exact decision-making process of the ML model as a function of composition variations. It enables the extraction of material specific insights into the physical origins of hardness that align well with the DFT calculations and conclusively shows that the nature of fit achieved is not just statistical in nature, but is driven by the underlying physics.

3.3 Ordering Behaviour

The ordering behaviour of alloys represents the phenomenon where the symmetry of the atomic arrangement is broken while the overall crystal structure remains the same. The ordering degree of an alloy can be described through Warren-Cowley short-range order (SRO) parameter [85] α_l^{ij} that is defined as: $\alpha_l^{ij} = 1 - \frac{P_l^{i|j}}{c_i}$, where c_i is the concentration of atom i, and $P_l^{i|j}$ is the probability of finding atom i at the l-th neighbour for a given atom j. The α_l^{ij} value is calculated for multiple atomic configurations followed by an ensemble averaging. This approach has some shortcomings: (i) it uses a matrix form, instead of a scalar, to represent all order parameters associated with atomic-pairs, and (ii) it poses a challenge in determining the neighbourhood distance scale that should be considered for obtaining a complete understanding of the ordering. To address these limitations, Yin et al. [86] developed a Variational Autoencoder (VAE)-based order parameter and applied it to assist the alloy design for various MPEAs.

VAEs are based on 3D convolutional neural networks (CNNs) and allow the encoding of the information into lower-dimensional latent variables in such a way that the data can be reconstructed through the decoding process [87, 88]. The model input comprises a cubic lattice where the atoms sit at BCC-lattice sites and each atom type was placed in a different channel. Each feature extraction (convolutional) layer, except for the final output layer, is followed by an average pooling layer. A dataset of 10^5 BCC-lattice based atomic configurations, comprising both ordered and disordered configurations, were generated using DFT-based Monte Carlo simulations at

different temperatures; and an 80% to 20% split was used for training and testing respectively. Manhattan Distance metric [89], calculated in the VAE latent space, was used as an order parameter for order–disorder phase transitions which satisfies the properties that: (i) it should have non-zero values in the ordered phase and approach zero in the disordered phase, and (ii) its second-order moment should peak at phase transitions. Yin et al. [86] applied this scalar VAE-based order parameter to study phase transitions in MoNbTaW, MoNbTaVW and $Al_xCoCrFeNi$ MPEAs; and observed that even though it cannot capture all the physical information related to phase transitions, it has the capability of quantifying the degree of ordering and phase-transition temperatures.

4 Atomistic Potential Development

Materials modelling approaches can be categorized as approaches based on quantum mechanical methods such as density-functional theory (DFT) and approaches based on semi-empirical analytical interatomic potentials. DFT calculations are based on electronic structure methods that provide force and energy calculations with high accuracy [90, 91]. These can be employed for a wide range of materials including single and multicomponent systems. Instead of their high accuracy, these cannot be used for large systems with more than a few hundred atoms as they demand high computational power [90].

Predictive materials modelling using atomistic simulations is revolutionizing diverse fields of materials including tribology, nano-particles self-assembly, catalysis and sensing. Atomistic simulations use interatomic potentials such as pair potentials, embedded-atom model (EAM) potentials, Tersoff potentials and reactive force potentials. These potentials are the sum of short-range or long-range interaction energies (potential energy) between the atoms which are a function of atomic positions, orientation and atomic charges [90]. Short-range interactions are bonded interactions due to stretching of bonds, angle bending and dihedral torsion whereas long-range interactions are between non-bonded atoms which can be due to dispersive or electrostatic forces [90].

Since last few decades complex solid-solutions including high-entropy alloys have gained attention due to their remarkable mechanical, corrosion resistant and tribological properties in harsh environments [2, 92, 93]. It is a challenge to design HEAs as numerous compositions can be produced with varying compositions of constituent elements. These are difficult to explore solely by experiments or by computational methods. Molecular dynamics can be used to effectively study such materials if suitable interatomic potentials are used. For most of the multicomponent systems, the interatomic potentials are unavailable. Researchers have been using machine learning techniques to develop potential for such materials [94].

4.1 Classical Methods

The interatomic potentials are mathematical formulations that deal with two-body, three-body and many body interaction energy terms or use bond-order concept. Stillinger–Weber [95] (SW) potential for covalent bonded systems and Vashishta [96] potentials use two-body or three-body interaction terms. Finnis–Sinclair [97] (FW), embedded-atom method [98] (EAM), and modified embedded-atom method [99] (MEAM) potentials for metallic systems are based on many body interactions treated as a function of atomic density. Interatomic potentials such as Abell [100], Tersoff [101, 102] and reactive empirical bond-order potential [103] (REBO) are based on the concept of bond-order. Although these potentials are widely used, they have some limitations. They do not allow charge transfer and emphasize on equilibrium configurations too much that may lead to errors in predicting energies for non-equilibrium systems [90]. Recently developed ReaxFF potentials and charged optimized many body potentials (COMB) have gained popularity. These potentials integrate the concepts of bond-order with charge equalization schemes.

To these interatomic potentials, a potential function is associated that depends on some global fitting parameters. This fitting parameter are identified by training the experimentally obtained datasets containing the information about material specific properties such as lattice constant, elastic constants, cohesive energy, defect formation energies and surface energies. Traditionally, the potentials are fitted directly commonly by using the least-squares fitting method to these properties for optimization after which all the parameters are fixed and can be used to predict energy and forces for different atomic configurations that are encountered during simulations. Other fitting techniques that are used include the minimal maximum error technique and interpolative moving least square technique [104]. This method of potential development utilizes physical models which are highly approximate with only a few adjustable parameters resulting in a lack in accuracy and predictive power [90, 105, 106]. It is difficult to get accurate predictions for materials with complex bonding such as Si with covalent as well as metallic bonding and carbon for which multiple metastable structures exist [106]. Although these potentials have limited accuracy, they are widely used as they demonstrate excellent results even for those atomic configurations which lie outside the training dataset if the chemical bonding within the system remains the same as it was assumed while developing the potentials [106]. Recently, the development of interatomic potentials using machine learning (ML) algorithms have gained popularity as it contains a large number of adjustable parameters resulting in more accuracy in energy predictions [106]. The method of potential development using ML is discussed in the next section.

4.2 Machine Learning Enabled Potential Development

Machine learning bridges between electronic and atomistic scales to obtain interatomic potentials. ML uses the dataset that is generated by quantum mechanical (commonly DFT) calculations instead of the experimental dataset that is used in the development of traditional potentials. Unlike traditional potentials that target only a specific set of properties, ML uses a function of atomic configurations that are represented by 3N-dimensional space known as potential energy surface [106] (PES). If the potential energy E_i is known, forces can be calculated for individual atoms as $F_i = \partial E_i / \partial r_i$, where r is the atomic position vector of atom *i*. E_i is the partitioned energy of an individual atom and the total energy of the system is the summation of atomic energies. ML potentials rely on local atomic environments which are defined by a set of position vectors. These position vectors that lie within a cut off radius are mapped on local energy using a potential function. The summation of local energy maps enables to reconstruct PES. The development of interatomic potentials which are functional of atomic positions and configurations depends on a set of parameters and starts with the selection of a model.

4.2.1 Model Selection

A wide range of models or functional forms such as Lennard–Jones (L-J) pairwise potential (that deals with the interaction between 2 bodies), EAM potentials, MEAM and many more are available and it is important to carefully select the functional form depending on the material properties and the environment for which potential is required [90, 107]. For example, L-J potentials do not take into account the directionality of bonding in covalent materials. This may ignore the energy contributions from many (3 or more) body interactions. Another example includes the successful use of EAM potential for bulk gold to describe its structural, thermodynamic and elastic properties and the same cannot be applied for gold nanoclusters as it exhibits a variety of structural configurations [108].

4.2.2 Training Dataset and Parameter Optimization

Once the model is selected, it is trained using a dataset so that it can capture material structure and properties at all free energies. ML utilizes the training dataset developed using electronic structure calculations to optimize the local and global parameters. A dataset of relevant properties such as bulk and surface structures, cohesive energies, elastic constants, surface and defect energies is used. The incorporation of surface configurations in the dataset enhances the ability of force fields to focus on the defect properties of the system as it captures the energy difference between various coordination systems [90]. A random sampling of the training dataset favours high energy structures; therefore, it is important to opt for a sampling method that results in

uniform energy distribution for a variety of configurations. This captures a large range of structural configurations including those which are far away from equilibrium. For example, Chan et. al [90] coupled genetic algorithms [109] with DFT calculations to develop hybrid bond-order potential. During the training of ML potentials, the loss function is minimized using the mean squared deviation from the reference values of the properties available in the dataset [106]. The training process is aimed to optimize the fitting parameters that result in a smooth PES. The use of ML algorithms simplifies optimization for large sized datasets. ML potentials are fitted to PES instead of directly fitting them to the material properties. Mapping of the local energy is done using a regression technique that uses a large number of adjustable parameters.

Local structural parameters or descriptors that consider the local environment of individual atoms are used for training the model. These parameters are vectors and are defined as a function of neighbouring atom's positions in single component systems. In multicomponent systems, the chemical identities of atoms are included. The accuracy of potentials depends on these functions so their proper selection is extremely important. To efficiently capture the different environments, a different descriptor and a complete set of descriptors to capture all structural features must be used [110]. However, a set with too many properties can lead to the generation of new descriptors for the same features leading to discontinuous energy predictions [111–113].

Algorithms such as the backpropagation method, the Levenberg–Marquardt, the Davidon-Fletcher-Powell (DFP), and the Broyden–Fletcher–Goldfarb–Shanno (BFGS) unconstrained optimization algorithms are commonly used for the optimization of fitting parameters [114, 115]. ML algorithms such as genetic algorithms optimize parameters after an unbiased global search that maintains diversity. This results in potentials with high accuracy and predictive powers [90]. ML potentials are usually based on purely numerical interpolation of reference energies where no physics of atomic interactions are considered except for the energy variation under translations, rotations and other atomic arrangements. This makes them less accurate in making energy predictions as compared to DFT calculations that capture the physics of the system. Recently, physically-informed machine learning potentials have started to gain attention. This is based on the fact that ML regression models are used to optimize parameters for physics-based interatomic potentials. Many researchers investigated the scope of combining ML based models with physics and developed ReaxFF, BOP and other potentials using this concept [90, 111, 116]. Singh et al. [94] developed interatomic potential for three element system MoTaW HEA using voxel-based (3D) Convolutional Neural Networks for predicting PES. This method enables the representation of huge datasets in the form of simple cellular structures bounded by a rectangular box. This captures the local atomic environment as well as other properties like bond length resulting in more accurate energy predictions.

4.2.3 Model Validation

It is important to check for any underfitting or overfitting of the dataset. Overfitting of the data produces oscillation between the reference data points resulting in inaccurate predictions. To avoid this, a validation set apart from the training dataset is used to make predictions and monitor any errors. Validation techniques such as k-fold cross validation are many times used to eliminate the chances of errors in the developed model [106].

The steps involved in potential generation using machine learning are summarized below:

(i) A model or functional form is selected based on the type of interaction between atoms such as ionic, metallic or covalent.
(ii) DFT calculations are performed to obtain a reference database.
(iii) Fitting is done for the optimization of fit-parameters.
(iv) Optimized force fields are obtained to perform MD simulations.

5 Conclusions and Future Perspectives

In summary, data-driven methodologies, targeted towards the exploration of MPEAs, have gained significant attention in the recent years. Various machine learning models have been developed to probe the phase selection, mechanical properties and ordering behaviour in MPEAs. Efforts have also been made to impart interpretability and explainability to these models to gain fundamental material specific insights into the underlying physics of the alloying process. Machine learning based force-field development has also surfaced as a robust way of modelling atomic interactions in MPEAs to further enable the atomistic simulation of key phenomena such as diffusion, phase transformations and deformation.

Going forward, there is a huge scope for both the improvement of current methodologies and for the development of novel techniques to extract insights from experimental and ab-initio data. These opportunities include the development of: (a) robust interpretation frameworks that can probe the cause-effect relationships to pinpoint the physical phenomenon responsible for linear/non-linear variations in model output, (b) search algorithms that can effectively comb through the astronomical compositional space using trained models to identify compositions of interest, (c) exhaustive and structured databases to consolidate the experimental and ab-initio findings that have been reported in the literature and make it available in a way that is readily machine-accessible, and (d) force fields for different MPEAs to enable atomistic modelling of these alloys so as to accelerate the exploration of these alloys.

Appendix 1

Description of features mentioned in Tables.

Feature description	Feature description
ΔH_{mix}: Described in Table 1	δ_K: Asymmetry in Bulk modulus: $\sqrt{\sum_{i=1}^{n} c_i \left(1 - \frac{K_i}{\overline{K}}\right)^2}$
ΔS_{mix}: Described in Table 1	$\overline{E_{\text{coh}}}$: Cohesive energy: $\sum_{i=1}^{n} c_i \ln E_{\text{coh},i}$
ΔH_{el}:Elastic enthalpy of mixing: $\sum_{i=1,i\neq j}^{n} 4\Delta H_{ij}^{el} c_i c_j$ where, ΔH_{ij}^{el} is the elastic enthalpy of mixing of equiatomic binary *i-j* calculated using Miedema's and Eshelby's model	H: Shear Modulus mismatch in strengthening model: $\sum_{i=1}^{n} \frac{c_i * \frac{2(G_i - \overline{G})}{(G_i + \overline{G})}}{1 + 0.5 * \lvert c_i * \frac{2(G_i - \overline{G})}{(G_i + \overline{G})} \rvert}$
δ_r: Described in Table 1	δ_{V_m}: Asymmetry in V_m: $\sqrt{\sum_{i=1}^{n} c_i \left(1 - \frac{V_{m,i}}{\overline{V_m}}\right)^2}$
$\overline{VEC}$: Valence Electron Count: $\sum_{i=1}^{n} c_i \ln VEC_i$	c_{el}: Concentration of element el
$\Delta\chi$: Electronegativity mismatch: $\sqrt{\sum_{i=1}^{n} c_i (\chi_i - \overline{\chi})^2}$	γ: Described in Table 1
$\overline{T_m}$: Melting temperature: $\sum_{i=1}^{n} c_i \ln T_{m,i}$	$(\Delta\chi)_{LM}$: Local mismatch in χ: $\sum_{i=1}^{n}\sum_{j=1,i\neq j}^{n} c_i c_j \lvert\chi_i - \chi_j\rvert$
$\overline{r}$: Average atomic radius: $\sum_{i=1}^{n} c_i \ln r_i$	$\overline{e/a}$: Itinerant electrons: $\sum_{i=1}^{n} c_i \ln(e/a)_i$
$\overline{\chi}$: Average electronegativity: $\sum_{i=1}^{n} c_i \ln \chi_i$	$\delta_{E_{\text{coh}}}$: Asymmetry in E_{coh}: $\sqrt{\sum_{i=1}^{n} c_i \left(1 - \frac{E_{\text{coh},i}}{\overline{E_{\text{coh}}}}\right)^2}$
δ_{T_m}: Asymmetry in T_m: $\sqrt{\sum_{i=1}^{n} c_i \left(1 - \frac{T_{m,i}}{\overline{T_m}}\right)^2}$	$(r)_{LM}$: Local mismatch in r: $\sum_{i=1}^{n}\sum_{j=1,i\neq j}^{n} c_i c_j \lvert r_i - r_j\rvert$
ΔK: Bulk Modulus mismatch: $\sqrt{\sum_{i=1}^{n} c_i \left(K_i - \overline{K}\right)^2}$	A: Energy term in strengthening model: $G * \delta_r * (1+\mu)/(1-\mu)$
ΔVEC: VEC mismatch: $\sqrt{\sum_{i=1}^{n} c_i \left(VEC_i - \overline{VEC}\right)^2}$	F: Peierls-Nabarro factor: $2G/(1-\mu)$
$(\Delta H_{\text{mix}})_{LM}$: Local mismatch in ΔH_{mix}: $\sum_{i=1}^{n}\sum_{j=1,i\neq j}^{n} c_i c_j \lvert\Delta H_{\text{mix}\,i} - \Delta H_{\text{mix}\,j}\rvert$	w: Sixth square of average work-function: $\overline{w}^6$
$\overline{K}$: Average Bulk Modulus: $\sum_{i=1}^{n} c_i \ln K_i$	$\overline{G}$: Average Shear Modulus: $\sum_{i=1}^{n} c_i \ln G_i$
Λ: Described in Table 1	$(\Delta G)_{LM}$: Local mismatch in G: $\sum_{i=1}^{n}\sum_{j=1,i\neq j}^{n} c_i c_j \lvert G_i - G_j\rvert$
$\overline{a}$: Average lattice parameter: $\sum_{i=1}^{n} c_i a_i$	μ: Lattice distortion energy: $0.5 * E * \delta_r$

(continued)

(continued)

Feature description	Feature description
Ω: Described in Table 1	$\overline{IP}$: Average First Ionization Potential: $\sum_{i=1}^{n} c_i \ln IP_i$
Δa: Lattice parameter mismatch: $\sqrt{\sum_{i=1}^{n} c_i (a_i - \overline{a})^2}$	$\overline{\rho}$: Average density: $\sum_{i=1}^{n} c_i \ln \rho_i$
ΔT_m: Melting temp. mismatch: $\sqrt{\sum_{i=1}^{n} c_i \left(T_{m,i} - \overline{T_m}\right)^2}$	$\overline{C}$: Average specific hear capacity: $\sum_{i=1}^{n} c_i \ln C_i$
δ_E: Asymmetry in Young's modulus: $\sqrt{\sum_{i=1}^{n} c_i \left(1 - \frac{E_i}{\overline{E}}\right)^2}$	$\overline{k}$: Average thermal conductivity: $\sum_{i=1}^{n} c_i \ln k_i$
δ_G: Asymmetry in Shear modulus: $\sqrt{\sum_{i=1}^{n} c_i \left(1 - \frac{G_i}{\overline{G}}\right)^2}$	$\overline{\nu}$: Average Poisson's ratio: $\sum_{i=1}^{n} c_i \ln \nu_i$
ΔCol: Column (periodic table) mismatch: $\sqrt{\sum_{i=1}^{n} c_i \left(Col_i - \overline{Col}\right)^2}$	Δm: Atomic weight mismatch: $\sqrt{\sum_{i=1}^{n} c_i (m_i - \overline{m})^2}$ ΔV: Specific volume mismatch: $\sqrt{\sum_{i=1}^{n} c_i \left(V_i - \overline{V}\right)^2}$

References

1. Cantor, B., Chang, I.T.H., Knight, P., Vincent, A.J.B.: Microstructural development in equiatomic multicomponent alloys. Mater. Sci. Eng., A **375–377**, 213–218 (2004). https://doi.org/10.1016/j.msea.2003.10.257
2. Yeh, J.-W., Chen, S.-K., Lin, S.-J., Gan, J.-Y., Chin, T.-S., Shun, T.-T., Tsau, C.-H., Chang, S.-Y.: Nanostructured high-entropy alloys with multiple principal elements: novel alloy design concepts and outcomes. Adv. Eng. Mater. **6**, 299–303 (2004). https://doi.org/10.1002/adem.200300567
3. Singh, P., Smirnov, A.V., Alam, A., Johnson, D.D.: First-principles prediction of incipient order in arbitrary high-entropy alloys: exemplified in $Ti_{0.25}$CrFeNiAlx. Acta Materialia. **189**, 248–254 (2020). https://doi.org/10.1016/j.actamat.2020.02.063
4. Zhang, Y., Zuo, T.T., Tang, Z., Gao, M.C., Dahmen, K.A., Liaw, P.K., Lu, Z.P.: Microstructures and properties of high-entropy alloys. Prog. Mater Sci. **61**, 1–93 (2014). https://doi.org/10.1016/j.pmatsci.2013.10.001
5. Singh, P., Gupta, S., Thimmaiah, S., Thoeny, B., Ray, P.K., Smirnov, A.V., Johnson, D.D., Kramer, M.J.: Vacancy-mediated complex phase selection in high entropy alloys. Acta Mater. **194**, 540–546 (2020). https://doi.org/10.1016/j.actamat.2020.04.063
6. Li, W., Xie, D., Li, D., Zhang, Y., Gao, Y., Liaw, P.K.: Mechanical behavior of high-entropy alloys. Progress Mater. Sci. **118**, 100777 (2021). https://doi.org/10.1016/j.pmatsci.2021.100777
7. Singh, P., Marshal, A., Smirnov, A.V., Sharma, A., Balasubramanian, G., Pradeep, K.G., Johnson, D.D.: Tuning phase-stability and short-range order through Al-doping in

(CoCrFeMn)100-xAlx high entropy alloys. Phys. Rev. Materials. **3**, 075002 (2019). https://doi.org/10.1103/PhysRevMaterials.3.075002
8. Cantor, B.: Multicomponent high-entropy Cantor alloys. Progress Mater. Sci. 100754 (2020). https://doi.org/10.1016/j.pmatsci.2020.100754
9. Singh, P., Smirnov, A.V., Johnson, D.D.: Ta-Nb-Mo-W refractory high-entropy alloys: Anomalous ordering behavior and its intriguing electronic origin, Phys. Rev. Mater. **2**, 055004 (2018). https://doi.org/10.1103/PhysRevMaterials.2.055004
10. Singh, R., Sharma, A., Singh, P., Balasubramanian, G., Johnson, D.D.: Accelerating computational modeling and design of high-entropy alloys. Nat. Comput. Sci. **1**, 54–61 (2021). https://doi.org/10.1038/s43588-020-00006-7
11. George, E.P., Raabe, D., Ritchie, R.O.: High-entropy alloys. Nat. Rev. Mater. **4**, 515–534 (2019). https://doi.org/10.1038/s41578-019-0121-4
12. Ashby, M.F., Cebon, D.: Materials selection in mechanical design. Le Journal de Physique IV. **3**, C7-1–C7-9
13. Ashby, M.F.: Multi-objective optimization in material design and selection. Acta Mater. **48**, 359–369 (2000)
14. Allison, J., Backman, D., Christodoulou, L.: Integrated computational materials engineering: a new paradigm for the global materials profession. Jom. **58**, 25–27 (2006)
15. Jain, A., Ong, S.P., Hautier, G., Chen, W., Richards, W.D., Dacek, S., Cholia, S., Gunter, D., Skinner, D., Ceder, G.: Commentary: the materials project: a materials genome approach to accelerating materials innovation. APL Mater. **1**, 011002 (2013)
16. Rajan, K.: Materials informatics. Mater. Today **8**, 38–45 (2005)
17. Rajan, K.: Materials informatics: the materials "gene" and big data. Annu. Rev. Mater. Res. **45**, 153–169 (2015)
18. de Pablo, J.J., Jackson, N.E., Webb, M.A., Chen, L.-Q., Moore, J.E., Morgan, D., Jacobs, R., Pollock, T., Schlom, D.G., Toberer, E.S., Analytis, J., Dabo, I., DeLongchamp, D.M., Fiete, G.A., Grason, G.M., Hautier, G., Mo, Y., Rajan, K., Reed, E.J., Rodriguez, E., Stevanovic, V., Suntivich, J., Thornton, K., Zhao, J.-C.: New frontiers for the materials genome initiative. NPJ Comput. Mater. **5**, 1–23 (2019). https://doi.org/10.1038/s41524-019-0173-4
19. Ray, P.K., Brammer, T., Ye, Y.Y., Akinc, M., Kramer, M.J.: A multi-stage hierarchical approach to alloy design. JOM. **62**, 25–29 (2010)
20. Tshitoyan, V., Dagdelen, J., Weston, L., Dunn, A., Rong, Z., Kononova, O., Persson, K.A., Ceder, G., Jain, A.: Unsupervised word embeddings capture latent knowledge from materials science literature. Nature **571**, 95–98 (2019)
21. McDonald, D.D.: Natural language generation. Handbook Nat. Lang. Process. **2**, 121–144 (2010)
22. Bergmann, J., Le Bail, A., Shirley, R., Zlokazov, V.: Renewed interest in powder diffraction data indexing. Zeitschrift Für Kristallographie-Crystalline Mater. **219**, 783–790 (2004)
23. Nunes, C.A., Sakidja, R., Dong, Z., Perepezko, J.H.: Liquidus projection for the Mo-rich portion of the Mo–Si–B ternary system. Intermetallics **8**, 327–337 (2000)
24. Beniwal, D., Singh, P., Gupta, S., Kramer, M.J., Johnson, D.D., Ray, P.K.: Distilling physical origins of hardness in Multi-Principal Element Alloys directly from ensemble neural network models. Manuscript Submitted for Publication. (n.d.)
25. Beniwal, D., Ray, P.K.: Learning phase selection and assemblages in high-entropy alloys through a stochastic ensemble-averaging model. Comput. Mater. Sci. **197**, 110647 (2021). https://doi.org/10.1016/j.commatsci.2021.110647
26. Verma, A.K., Hawk, J.A., Bruckman, L.S., French, R.H., Romanov, V., Carter, J.L.: Mapping multivariate influence of alloying elements on creep behavior for design of new martensitic steels. Metall. Mater. Trans. A. **50**, 3106–3120 (2019)
27. Zhao, J.-C., Zheng, X., Cahill, D.G.: High-throughput diffusion multiples. Mater. Today **8**, 28–37 (2005)
28. McCluskey, P.J., Vlassak, J.J.: Combinatorial nanocalorimetry. J. Mater. Res. **25**, 2086–2100 (2010)

29. Rickman, J.M., Chan, H.M., Harmer, M.P., Smeltzer, J.A., Marvel, C.J., Roy, A., Balasubramanian, G.: Materials informatics for the screening of multi-principal elements and high-entropy alloys. Nat Commun. **10**, 1–10 (2019). https://doi.org/10.1038/s41467-019-10533-1
30. Geng, J., Nlebedim, I.C., Besser, M.F., Simsek, E., Ott, R.T.: Bulk combinatorial synthesis and high throughput characterization for rapid assessment of magnetic materials: application of laser engineered net shaping (lens^{TM}). Jom. **68**, 1972–1977 (2016)
31. Murty, B.S., Ranganathan, S., Rao, M.M.: Solid state amorphization in binary Ti-Ni, Ti-Cu and ternary Ti-Ni-Cu system by mechanical alloying. Mater. Sci. Eng., A **149**, 231–240 (1992)
32. Bhatt, J., Dey, G.K., Murty, B.S.: Thermodynamic and topological modeling and synthesis of Cu-Zr-Ti-Ni-based bulk metallic glasses by mechanical alloying. Metall. Mater. Trans. A. **39**, 1543–1551 (2008)
33. Miracle, D.B., Senkov, O.N.: Topological criterion for metallic glass formation. Mater. Sci. Eng., A **347**, 50–58 (2003)
34. Han, G., Qiang, J., Li, F., Yuan, L., Quan, S., Wang, Q., Wang, Y., Dong, C., Haeussler, P.: The e/a values of ideal metallic glasses in relation to cluster formulae. Acta Mater. **59**, 5917–5923 (2011)
35. Miracle, D.B., Sanders, W.S., Senkov, O.N.: The influence of efficient atomic packing on the constitution of metallic glasses. Phil. Mag. **83**, 2409–2428 (2003)
36. Miracle, D.B.: A structural model for metallic glasses. Nat. Mater. **3**, 697–702 (2004)
37. Bhatt, J., Ray, P.K., Murty, B.S.: Understanding bulk metallic glass formation in Zr-Cu-Al system by thermodynamic approach. Trans. Indian Inst. Met. **60**, 323–330 (2007)
38. Ray, P.K., Chattopadhyay, K., Murty, B.S.: Influence of thermodynamics and local geometry on glass formation in Zr based alloys. Appl. Phys. Lett. **93**, 061903 (2008)
39. Tripathi, M.K., Chattopadhyay, P.P., Ganguly, S.: A predictable glass forming ability expression by statistical learning and evolutionary intelligence. Intermetallics **90**, 9–15 (2017)
40. Bajpai, A., Bhatt, J., Biswas, K., Gurao, N.P.: A new perspective to thermodynamical designing of high entropy bulk metallic glasses (HE-BMGs). Physica B **595**, 412350 (2020)
41. Roy, A., Sreeramagiri, P., Babuska, T., Krick, B., Ray, P.K., Balasubramanian, G.: Lattice distortion as an estimator of solid solution strengthening in high-entropy alloys. Mater. Char. **172**, 110877 (2021). https://doi.org/10.1016/j.matchar.2021.110877
42. Roy, A., Babuska, T., Krick, B., Balasubramanian, G.: Machine learned feature identification for predicting phase and Young's modulus of low-, medium- and high-entropy alloys. Scripta Mater. **185**, 152–158 (2020). https://doi.org/10.1016/j.scriptamat.2020.04.016
43. Wang, Z., Cai, S., Chen, W., Abd Ali, R., Jin, K.: Analysis of critical velocity of cold spray based on machine learning method with feature selection. J. Thermal Spray Technol. 1–13 (2021)
44. Palodhi, L., Singh, H.: On the dependence of critical velocity on the material properties during cold spray process. J. Therm. Spray Technol. **29**, 1863–1875 (2020)
45. Palodhi, L., Das, B., Singh, H.: Effect of particle size and morphology on critical velocity and deformation behavior in cold spraying. J. Mater. Eng. Perform. **30**, 8276–8288 (2021)
46. Swetlana, S., Khatavkar, N., Singh, A.K.: Development of Vickers hardness prediction models via microstructural analysis and machine learning. J. Mater. Sci. **55**, 15845–15856 (2020)
47. Cantwell, P.R., Tang, M., Dillon, S.J., Luo, J., Rohrer, G.S., Harmer, M.P.: Grain boundary complexions. Acta Mater. **62**, 1–48 (2014)
48. scipy.stats.pearsonr—SciPy v0.14.0 Reference Guide, (n.d.). https://docs.scipy.org/doc/scipy-0.14.0/reference/generated/scipy.stats.pearsonr.html#scipy.stats.pearsonr. Accessed 1 Dec 2021
49. scipy.stats.spearmanr—SciPy v0.14.0 Reference Guide, (n.d.). https://docs.scipy.org/doc/scipy-0.14.0/reference/generated/scipy.stats.spearmanr.html#scipy.stats.spearmanr. Accessed 1 Dec 2021
50. scipy.stats.pointbiserialr—SciPy v0.14.0 Reference Guide, (n.d.). https://docs.scipy.org/doc/scipy-0.14.0/reference/generated/scipy.stats.pointbiserialr.html#scipy.stats.pointbiserialr. Accessed 1 Dec 2021

51. Permutation Importance with Multicollinear or Correlated Features, Scikit-Learn. (n.d.). https://scikit-learn/stable/auto_examples/inspection/plot_permutation_importance_multicollinear.html. Accessed 2 Dec 2021
52. Géron, A.: Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, 2nd edn. O'Reilly Media, Beijing China, Sebastopol, CA (2019)
53. Lee, K., Ayyasamy, M., Delsa, P., Hartnett, T.Q., Balachandran, P.V.: Phase Classification of Multi-Principal Element Alloys via Interpretable Machine Learning. ArXiv:2108.05933 [Cond-Mat]. (2021). http://arxiv.org/abs/2108.05933. Accessed 2 Dec 2021
54. Guo, S., Ng, C., Lu, J., Liu, C.T.: Effect of valence electron concentration on stability of fcc or bcc phase in high entropy alloys. J. Appl. Phys. **109**, 103505 (2011). https://doi.org/10.1063/1.3587228
55. Liu, M., Xu, W., Zhang, S., Wang, Z., Wang, Z., Wang, B., Wang, D., Li, F.: Microstructures and hardnesses of $AlCoCr_{0.5}FexNi_{2.5}$ high entropy alloys with equal valence electron concentration. J. Alloys Compounds. **824**, 153881 (2020). https://doi.org/10.1016/j.jallcom.2020.153881
56. Takeuchi, A., Inoue, A.: Calculations of mixing enthalpy and mismatch entropy for ternary amorphous alloys. Mater. Trans. **41**, 1372–1378 (2000)
57. Miedema, A.R., de Châtel, P.F., de Boer, F.R.: Cohesion in alloys—fundamentals of a semi-empirical model. Physica B+C. **100**, 1–28 (1980). https://doi.org/10.1016/0378-4363(80)90054-6
58. Zhang, R.F., Zhang, S.H., He, Z.J., Jing, J., Sheng, S.H.: Miedema calculator: a thermodynamic platform for predicting formation enthalpies of alloys within framework of miedema's theory. Comput. Phys. Commun. **209**, 58–69 (2016). https://doi.org/10.1016/j.cpc.2016.08.013
59. Ray, P.K., Akinc, M., Kramer, M.J.: Applications of an extended Miedema's model for ternary alloys. J. Alloy. Compd. **489**, 357–361 (2010). https://doi.org/10.1016/j.jallcom.2009.07.062
60. Ray, P.K., Akinc, M., Kramer, M.J.: Estimation of formation enthalpies using an extended Miedema approach. In: Proceedings of the 22nd Annual Conference on 'Fossil Energy Materials', pp. 474–482. Pittsburgh, PA, USA (2008)
61. Mousavi, M.S., Abbasi, R., Kashani-Bozorg, S.F.: A thermodynamic approach to predict formation enthalpies of ternary systems based on Miedema's model. Metall Mat. Trans A. **47**, 3761–3770 (2016). https://doi.org/10.1007/s11661-016-3533-4
62. Li, J.-H., Tsai, M.-H.: Theories for predicting simple solid solution high-entropy alloys: classification, accuracy, and important factors impacting accuracy. Scripta Mater. **188**, 80–87 (2020). https://doi.org/10.1016/j.scriptamat.2020.06.064
63. Yang, X., Zhang, Y.: Prediction of high-entropy stabilized solid-solution in multi-component alloys. Mater. Chem. Phys. **132**, 233–238 (2012). https://doi.org/10.1016/j.matchemphys.2011.11.021
64. Guo, S., Hu, Q., Ng, C., Liu, C.T.: More than entropy in high-entropy alloys: forming solid solutions or amorphous phase. Intermetallics **41**, 96–103 (2013). https://doi.org/10.1016/j.intermet.2013.05.002
65. Wang, Z., Huang, Y., Yang, Y., Wang, J., Liu, C.T.: Atomic-size effect and solid solubility of multicomponent alloys. Scripta Mater. **94**, 28–31 (2015). https://doi.org/10.1016/j.scriptamat.2014.09.010
66. Singh, A.K., Kumar, N., Dwivedi, A., Subramaniam, A.: A geometrical parameter for the formation of disordered solid solutions in multi-component alloys. Intermetallics **53**, 112–119 (2014). https://doi.org/10.1016/j.intermet.2014.04.019
67. Ye, Y.F., Wang, Q., Lu, J., Liu, C.T., Yang, Y.: Design of high entropy alloys: a single-parameter thermodynamic rule. Scripta Mater. **104**, 53–55 (2015). https://doi.org/10.1016/j.scriptamat.2015.03.023
68. Ye, Y.F., Wang, Q., Lu, J., Liu, C.T., Yang, Y.: The generalized thermodynamic rule for phase selection in multicomponent alloys. Intermetallics **59**, 75–80 (2015). https://doi.org/10.1016/j.intermet.2014.12.011

69. Troparevsky, M.C., Morris, J.R., Kent, P.R.C., Lupini, A.R., Stocks, G.M.: Criteria for predicting the formation of single-phase high-entropy alloys. Phys. Rev. X. **5**, 011041 (2015). https://doi.org/10.1103/PhysRevX.5.011041
70. Senkov, O.N., Miracle, D.B.: A new thermodynamic parameter to predict formation of solid solution or intermetallic phases in high entropy alloys. J. Alloy. Compd. **658**, 603–607 (2016). https://doi.org/10.1016/j.jallcom.2015.10.279
71. King, D.J.M., Middleburgh, S.C., McGregor, A.G., Cortie, M.B.: Predicting the formation and stability of single phase high-entropy alloys. Acta Mater. **104**, 172–179 (2016). https://doi.org/10.1016/j.actamat.2015.11.040
72. Islam, N., Huang, W., Zhuang, H.L.: Machine learning for phase selection in multi-principal element alloys. Comput. Mater. Sci. **150**, 230–235 (2018). https://doi.org/10.1016/j.commatsci.2018.04.003
73. Huang, W., Martin, P., Zhuang, H.L.: Machine-learning phase prediction of high-entropy alloys. Acta Mater. **169**, 225–236 (2019). https://doi.org/10.1016/j.actamat.2019.03.012
74. Li, Y., Guo, W.: Machine-learning model for predicting phase formations of high-entropy alloys. Phys. Rev. Mater. **3**, 095005 (2019). https://doi.org/10.1103/PhysRevMaterials.3.095005
75. Zhou, Z., Zhou, Y., He, Q., Ding, Z., Li, F., Yang, Y.: Machine learning guided appraisal and exploration of phase design for high entropy alloys. Npj Comput Mater. **5**, 1–9 (2019). https://doi.org/10.1038/s41524-019-0265-1
76. Zhang, Y., Wen, C., Wang, C., Antonov, S., Xue, D., Bai, Y., Su, Y.: Phase prediction in high entropy alloys with a rational selection of materials descriptors and machine learning models. Acta Mater. **185**, 528–539 (2020). https://doi.org/10.1016/j.actamat.2019.11.067
77. Jaiswal, U.K., Vamsi Krishna, Y., Rahul, M.R., Phanikumar, G.: Machine learning-enabled identification of new medium to high entropy alloys with solid solution phases. Comput. Mater. Sci. **197**, 110623 (2021). https://doi.org/10.1016/j.commatsci.2021.110623
78. Lee, S.Y., Byeon, S., Kim, H.S., Jin, H., Lee, S.: Deep learning-based phase prediction of high-entropy alloys: optimization, generation, and explanation. Mater. Des. **197**, 109260 (2021). https://doi.org/10.1016/j.matdes.2020.109260
79. Roy, A., Balasubramanian, G.: Predictive descriptors in machine learning and data-enabled explorations of high-entropy alloys. Comput. Mater. Sci. **193**, 110381 (2021). https://doi.org/10.1016/j.commatsci.2021.110381
80. Chang, Y.-J., Jui, C.-Y., Lee, W.-J., Yeh, A.-C.: Prediction of the composition and hardness of high-entropy alloys by machine learning. JOM. **71**, 3433–3442 (2019). https://doi.org/10.1007/s11837-019-03704-4
81. Wen, C., Zhang, Y., Wang, C., Xue, D., Bai, Y., Antonov, S., Dai, L., Lookman, T., Su, Y.: Machine learning assisted design of high entropy alloys with desired property. Acta Mater. **170**, 109–117 (2019). https://doi.org/10.1016/j.actamat.2019.03.010
82. Revi, V., Kasodariya, S., Talapatra, A., Pilania, G., Alankar, A.: Machine learning elastic constants of multi-component alloys. Comput. Mater. Sci. **198**, 110671 (2021). https://doi.org/10.1016/j.commatsci.2021.110671
83. Khakurel, H., Taufique, M.F.N., Roy, A., Balasubramanian, G., Ouyang, G., Cui, J., Johnson, D.D., Devanathan, R.: Machine learning assisted prediction of the Young's modulus of compositionally complex alloys. Sci Rep. **11**, 17149 (2021). https://doi.org/10.1038/s41598-021-96507-0
84. Yang, C., Ren, C., Jia, Y., Wang, G., Li, M., Lu, W.: A machine learning-based alloy design system to facilitate the rational design of high entropy alloys with enhanced hardness. Acta Materialia. **222,** 117431 (2022). https://doi.org/10.1016/j.actamat.2021.117431
85. Owen, L.R., Playford, H.Y., Stone, H.J., Tucker, M.G.: A new approach to the analysis of short-range order in alloys using total scattering. Acta Mater. **115**, 155–166 (2016). https://doi.org/10.1016/j.actamat.2016.05.031
86. Yin, J., Pei, Z., Gao, M.C.: Neural network-based order parameter for phase transitions and its applications in high-entropy alloys. Nat Comput Sci. **1**, 686–693 (2021). https://doi.org/10.1038/s43588-021-00139-3

87. Doersch, C.: Tutorial on Variational Autoencoders (2021). http://arxiv.org/abs/1606.05908 [Cs, Stat]. Accessed 26 Nov 2021
88. Kingma, D.P., Welling, M.: Auto-Encoding Variational Bayes. (2014). http://arxiv.org/abs/1312.6114 [Cs, Stat]. Accessed 26 Nov 2021
89. Aggarwal, C.C., Hinneburg, A., Keim, D.A.: On the surprising behavior of distance metrics in high dimensional space. In: J. Van den Bussche, V. Vianu (eds.) Database Theory—ICDT 2001, pp. 420–434. Springer, Berlin, Heidelberg (2001). https://doi.org/10.1007/3-540-44503-X_27
90. Chan, H., Narayanan, B., Cherukara, M.J., Sen, F.G., Sasikumar, K., Gray, S.K., Chan, M.K.Y., Sankaranarayanan, S.K.R.S.: Machine learning classical interatomic potentials for molecular dynamics from first-principles training data. J. Phys. Chem. C. **123**, 6941–6957 (2019). https://doi.org/10.1021/acs.jpcc.8b09917
91. van de Walle, A., Asta, M.: High-throughput calculations in the context of alloy design. MRS Bull. **44**, 252–256 (2019). https://doi.org/10.1557/mrs.2019.71
92. Singh, P., Sharma, A., Smirnov, A.V., Diallo, M.S., Ray, P.K., Balasubramanian, G., Johnson, D.D.: Design of high-strength refractory complex solid-solution alloys. Npj. Comput. Mater. **4**, 16 (2018). https://doi.org/10.1038/s41524-018-0072-0
93. Gao, M.C.: Design of high-entropy alloys. In: M.C. Gao, J.-W. Yeh, P.K. Liaw, Y. Zhang (eds.) High-Entropy Alloys: Fundamentals and Applications, pp. 369–398. Springer International Publishing, Cham (2016). https://doi.org/10.1007/978-3-319-27013-5_11
94. Singh, R., Sharma, A., Bingol, O.R., Balu, A., Balasubramanian, G., Johnson, D.D., Sarkar, S.: 3D deep learning with voxelized atomic configurations for modeling atomistic potentials in complex solid-solution alloys (2018). http://arxiv.org/abs/1811.09724 [Cond-Mat, Physics:Physics, Stat]. Accessed 8 July 2021
95. Stillinger, F.H., Weber, T.A.: Computer simulation of local order in condensed phases of silicon. Phys. Rev. B. **31**, 5262–5271 (1985). https://doi.org/10.1103/PhysRevB.31.5262
96. Vashishta, P., Kalia, R.K., Rino, J.P., Ebbsjö, I.: Interaction potential for SiO_2: a molecular-dynamics study of structural correlations. Phys. Rev. B. **41**, 12197–12209 (1990). https://doi.org/10.1103/PhysRevB.41.12197
97. Rafii-Tabar, H., Sulton, A.P.: Long-range Finnis-Sinclair potentials for f.c.c. metallic alloys. Philos. Mag. Lett. **63**, 217–224 (1991). https://doi.org/10.1080/09500839108205994
98. Daw, M.S., Baskes, M.I.: Embedded-atom method: Derivation and application to impurities, surfaces, and other defects in metals. Phys. Rev. B. **29**, 6443–6453 (1984). https://doi.org/10.1103/PhysRevB.29.6443
99. Baskes, M.I.: Modified embedded-atom potentials for cubic materials and impurities. Phys. Rev. B. **46**, 2727–2742 (1992). https://doi.org/10.1103/PhysRevB.46.2727
100. Abell, G.C.: Empirical chemical pseudopotential theory of molecular and metallic bonding. Phys. Rev. B. **31**, 6184–6196 (1985). https://doi.org/10.1103/PhysRevB.31.6184
101. Tersoff, J.: New empirical approach for the structure and energy of covalent systems. Phys. Rev. B. **37**, 6991–7000 (1988). https://doi.org/10.1103/PhysRevB.37.6991
102. Tersoff, J.: Empirical interatomic potential for carbon, with applications to amorphous carbon. Phys. Rev. Lett. **61**, 2879–2882 (1988). https://doi.org/10.1103/PhysRevLett.61.2879
103. Brenner, D.W., Shenderova, O.A., Harrison, J.A., Stuart, S.J., Ni, B., Sinnott, S.B.: A second-generation reactive empirical bond order (REBO) potential energy expression for hydrocarbons. J. Phys.: Condens. Matter **14**, 783–802 (2002). https://doi.org/10.1088/0953-8984/14/4/312
104. Guo, Y., Kawano, A., Thompson, D.L., Wagner, A.F., Minkoff, M.: Interpolating moving least-squares methods for fitting potential energy surfaces: applications to classical dynamics calculations. J Chem Phys. **121**, 5091–5097 (2004). https://doi.org/10.1063/1.1777572
105. Lysogorskiy, Y., Hammerschmidt, T., Janssen, J., Neugebauer, J., Drautz, R.: Transferability of interatomic potentials for molybdenum and silicon. Modell. Simul. Mater. Sci. Eng. **27**, 025007 (2019). https://doi.org/10.1088/1361-651X/aafd13
106. Mishin, Y.: Machine-learning interatomic potentials for materials science. Acta Materialia. **214**, 116980 (2021). https://doi.org/10.1016/j.actamat.2021.116980

107. Erkoç, S.: Empirical potential energy functions used in the simulations of materials properties. In: Annual Reviews of Computational Physics IX, pp. 1–103. World Scientific (2001). https://doi.org/10.1142/9789812811578_0001
108. Lee, B.-J., Shim, J.-H., Baskes, M.I.: Semiempirical atomic potentials for the fcc metals Cu, Ag, Au, Ni, Pd, Pt, Al, and Pb based on first and second nearest-neighbor modified embedded atom method. Phys. Rev. B. (2003). https://doi.org/10.1103/PhysRevB.68.144112
109. McCall, J.: Genetic algorithms for modelling and optimisation. J. Comput. Appl. Math. **184**, 205–222 (2005)
110. Onat, B., Ortner, C., Kermode, J.R.: Sensitivity and dimensionality of atomic environment representations used for machine learning interatomic potentials. J. Chem. Phys. **153**, 144106 (2020). https://doi.org/10.1063/5.0016005
111. Drautz, R.: Atomic cluster expansion for accurate and transferable interatomic potentials. Phys. Rev. B. (2019). https://doi.org/10.1103/PhysRevB.99.014104
112. Kocer, E., Mason, J.K., Erturk, H.: Continuous and optimally complete description of chemical environments using Spherical Bessel descriptors. AIP Adv. **10**, 015021 (2020). https://doi.org/10.1063/1.5111045
113. Pozdnyakov, S.N., Willatt, M.J., Bartók, A.P., Ortner, C., Csányi, G., Ceriotti, M.: Incompleteness of atomic structure representations. Phys. Rev. Lett. (2020). https://doi.org/10.1103/PhysRevLett.125.166001
114. Fletcher, R.: Practical Methods of Optimization, 2nd edn. Wiley, n.d. https://www.wiley.com/en-in/Practical+Methods+of+Optimization%2C+2nd+Edition-p-9780471494638. Accessed 18 Nov 2021
115. Winkler, J.R.: Numerical Recipes in C: The Art of Scientific Computing, 2nd edn. Endeavour. 17, 201 (1993). https://doi.org/10.1016/0160-9327(93)90069-F
116. Cherukara, M.J., Narayanan, B., Kinaci, A., Sasikumar, K., Gray, S.K., Chan, M.K.Y., Sankaranarayanan, S.K.R.S.: Ab initio-based bond order potential to investigate low thermal conductivity of stanene nanostructures. J. Phys. Chem. Lett. **7**, 3752–3759 (2016). https://doi.org/10.1021/acs.jpclett.6b01562

Effect of Geometrical Parameters on Branched Cracks: A Finite Element Method-Based Computational Approach



Neeraj Bisht, Harshit Kumar, Virendra Singh, and Sakshi Chauhan

Abstract Branching of an existing crack in brittle materials is a common phenomenon. Mostly, the branching affects the service life of a component severely. It becomes important that an investigation be made into the parameters that can influence this phenomenon. In the current study, Finite element method has been used to investigate the effect of geometrical parameters on branched cracks. A rectangular plate with a symmetrically branched crack for different crack lengths, different biaxiality factors, and different inclination angles under static loading was studied. Mode I and mode II stress intensity factors' variations for all parameters were studied. Two types of effects on the stress intensity factor were observed; the intensification and shielding effect.

Keywords Branching · T-stress · Stress intensity factor · Finite element method

1 Introduction

Fracture mechanics difficulties commonly include branched cracks. The required requirements for crack branching are a crucial constant velocity and a critical stress intensity factor. Propagating cracks branching has been seen during fast unstable fracture of different brittle materials and stress corrosion cracking of high-strength steels. When a single fracture is exposed to both mode-I and mode-II stress, branched cracks have been seen. Furthermore, microscopic cracking frequently produces cracks with complex geometry. There are interactions between the many branching fractures. As a result, determining the specific fracture tip from which fast extension of the

N. Bisht (✉) · S. Chauhan
Mechanical Engineering Department, GBPUA&T, Pantnagar, India
e-mail: neerajbisht30@gmail.com; neerajbisht.me@gbpuat-tech.ac.in

H. Kumar
Design and Production Engineering, GBPUA&T, Pantnagar, India

V. Singh
Physics Department, GBPUA&T, Pantnagar, India

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branching crack occurs under specified applied stresses is critical. A variety of research has been conducted to determine the stress intensity variables at the terminals of branching cracks. The strain energy density criteria were utilized to determine the more susceptible branch of the crack from which brittle plate fracture occurs, as well as the critical load levels. In rapid fracture and stress corrosion cracking, symmetrical crack branching is common. The cracks may be symmetrically branched either simply or doubly branched. Stress intensity factor at crack tip of branched crack can be computed analytically or theoretically. It is necessary to understand the various parameters which influence the propagation and ultimate failure of branched cracks keeping in mind the above points, the following objectives are chosen:

(a) Effect of biaxiality factor on the SIF.
(b) Effect of crack ratios on the SIF.
(c) Effect of variation of included angle on the SIF.

2 Review of Literature

2.1 Evaluation of Stress Intensity Factor

A simplified method for determining fracture mechanics parameters with a quick mesh refinement method that allows a cracked body reaction to be accurately modeled on the given load was proposed [1]. Simplex triangular isoparametric components are implemented here in their creation, through simplicity and universality.

Conformal mapping method in 2D fracture mechanics calculations was proposed for the automatic generation of finite element meshes [2]. The conformal mapping results in a comparatively tiny, almost quadratic fracture at the base of the crack and a constant change to bigger component with greater range from the base of the crack.

The impact of proportional and non-proportional loading on mixed-mode fracture behavior was studied [3]. The anisotropic continuum damage mechanism is proposed, and the computed results from a non-proportional finite element program show that different modes of proportional and non-proportional loading have a significant impact on fracture parameters**.**

An element formulation for simultaneously obtaining variable-order singularities at two corner nodes of a four-noded quadrilateral was proposed [4]. The element may be used to represent a kinked crack with a short kink length.

Mode-I and non-self-similar crack development (mode-I and mode-II) scenarios using a series of tests and finite element analyses on compact stress specimens of various fracture lengths were conducted [5].

A finite element software for mode I case under biaxial loading that combines the analytical crack tip solution with a standard finite element analysis was created [6]. It has been established that keeping the second term of the series expansion for local stress is crucial. They demonstrated that the presence load applied parallel to

the fracture is shown only by continuous second terms, which disappears only under specified loading conditions.

A model for an approximate closed-form solution for fractures in an elastic material that are perpendicular to each other under mixed-mode loading was developed [7]. Under general mode I and II stresses, perpendicular fractures are commonly caused by torsional fatigue and the fatigue of the spline joint and shafts. For a precise assessment of fracture mechanics, knowledge of the variations in stress intensity components for this type of geometry and decreased geometry is essential. This type of loading and geometry has yet to be solved analytically. The superpositioning technique used by Kachanov to analyze various cracks was expanded and used to tackle the issue.

The body force technique (BFM), the impact of the contact between two cracks on the crack development behavior was assessed [8]. Taking into consideration the interference between two cracks, a simulation model was introduced to control the crack development cycle. This model created it feasible for distinct instances of the comparative place and duration of crack to evaluate various crack development behavior. He used the BEM to calculate the stress intensity factor. He concluded that the contact strength between these cracks is almost equal to that of a single coalesced fracture, even though they are independent cracks. He discovered a relationship between the number of cycles and SIF (K).

Multiple site damage in aging aircraft structures, which is defined as the emergence of tiny fatigue cracks at several locations was studied [9]. This work focused on this common structure and presented an analytical approach for determining the stress intensity factor of an infinite plate with many hole edge fractures. The stress function is evaluated using the features of complex variable functions. On numerous holes, the approximation superposition approach was used to tackle stress intensity factor difficulties. To change the procedure, the analogous crack was introduced.

The digital image correlation (DIC) approach to assess the in-plane stress intensity factors (SIFs), K_I and K_{II}, for contact between a half-plane with an edge fracture and an asymmetric tilted wedge was used [10]. To quantify the related displacements and strain fields, DIC requires two digital photographs of the same sample taken at separate phases of loading. For two distinct fracture angles (60° and 90°) and three different force magnitudes, the influence of wedge tip angles on SIF values was investigated. When the findings of the DIC approach were compared to those of the photoelasticity technique, it was discovered that the relative difference between the SIF values produced by the two methods was less than 3% for all of the examples analyzed. The half-plane with the 60° edge crack was solely under second mode fracture, according to a comparison of the two half-planes. The half-plane with a 90° edge crack, on the other hand, underwent mixed-mode fracture, with the first mode SIF, K_I, having a higher value than the second, K_{II}.

2.2 Evaluation of Crack Branching

The effect of branching fractures at tiny angles by evaluating the stress intensity parameters at the ends of a branched crack may have been the first study on branched cracks [11]. They came to the conclusion that the strategies used to solve these difficulties may be divided into two categories:

(a) Methods that make use of conformal mapping. A branching crack is mapped onto the basic geometry using this approach.
(b) Dislocation methods that produce singular integral equations directly.

Because the issue of stress intensities at the points of a branching fracture is undeniably essential, an analytical solution produces more reliable results, particularly for branched cracks with small angles.

Branch crack problems with random plane elasticity configuration through a new integral equation were studied [12]. The unknown functions were the usual dislocation functions in the new integral equation, and the resulting forces represent the right-hand terms of the integral equations. The problem of kinked crack can be regarded as a particular sort of branch crack issue, namely a crack problem with two branches. The branch crack issues mentioned above are restricted only to instances where the branch is a straight line.

The interaction of a symmetrical branching crack represented as three straight cracks intersecting at a same point, with each fracture being regarded as a continuous contribution of edge dislocations was studied [13]. The method of Gauss–Chebyshev quadrature is used to convert the resulting system of singular integral equations to a conventional matrix equation. The impacts of inclusion on stress intensity variables at branch tips and branch paths are examined. The branching path's variation of inclusion is also shown, and it demonstrates that with tougher inclusion, the branch angle grows greater.

The propagation of an edge fracture in a plate under plane stress and mode I loading was studied [14]. The start of the crack was investigated using a scanning electron microscope, which revealed the critical role of crack branching mode. The branching of fractures in an infinite rectangular plate occurs under plane stress and mode I loading, according to the findings. A system of cracks of any shape may be written as a system of singular integral equations by reducing the boundary-value issue of a fractured plate to a complex-valued singular integral equation. The stress intensity variables K_I and K_{II} were shown to be connected to the tips of the branching fracture using this kind of analysis.

The Griffith theorem can be used not only for Model-I crack development but also for separate crack-branching cases and theoretically validated the times for triple crack-branching, side-branching, symmetrical-branching, and the relatively well-known crack-kink of a Mode-I crack [15]. As a result, theoretical modeling was used to characterize the branching process for quasi-static Mode-I cracks. The branching hardness of multi-crack branching was determined, as well as the universal K-based criteria. The current approach proposes crack branching, which is in good accord with experimental evidence from the literature.

The limited branching crack data available can't be utilized to predict future crack growth or explain the delays observed in practice [16]. To solve this problem, the researchers employed specialized finite element (FE) and life assessment tools for fatigue. The path of propagation and related stress intensity factors (SIF) of kinked and bifurcated fractures, which can cause fracture retardation or even detention, were calculated in this work using a particular FE software.

A weighted function to derive an assessment model from the stress field sequence of the first crack tip to the second stage (WF) [17]. The third-order coefficients of the stress field series of the original crack tip were incorporated into this expression to improve the estimation accuracy for the kinked crack with limited kink length, and these parameters may be acquired via a fitting technique.

A method for dealing with the complex challenges of branching cracks was studied [18]. In addition, instead of using the continuous crack propagation rate, the XFEM now uses a unique crack propagation method in which the crack propagation time is a variable determined by two trial computations. Numerical tests show that the proposed approach is both feasible and precise. The numerical findings reveal that the improved XFEM is more precise than the normal XFEM at handling complicated branching crack situations. Meanwhile, a smooth crack path may be obtained using the innovative fracture propagation strategy.

A significant portion of the mysteries surrounding crack extension could be solved if fracture experiments could include a reasonable estimate of the stress conditions near the leading edge of a crack, particularly at points of rapid fracture onset and fracture arrest [19]. The impact of the test setup, loads, and crack length on the stresses at the crack end may be represented in terms of two parameters for fairly brittle tensile fractures in cases where a generalized plane-stress or a plane-strain analysis is suitable. One is an adjustable uniform stress parallel to the direction of fracture extension, while the other, the stress-intensity factor, is proportional to the square root of the force that tends to produce crack extension. Both parameters have a clear meaning and application in brittle-fracture mechanic research.

The fracture toughness (KIC) of silicon nitride and fused silica using a Vickers indentation finite element model with a virtual crack closure techniques (VCCT) [20]**.** The elastic modulus and yield stress of the materials required for finite element simulation were calculated using dimensionless expressions developed specifically for ceramic materials, revealing approximate numerical relationships between indentation responses and elastoplastic properties of ceramic materials. Calculating the stress intensity factor (KI) at the fracture tip yielded the computed KIC. The equi-KI crack front was obtained by consecutive simulation and modification since the computed stress intensity factor (KI) varied markedly along an ideal semi-circle crack front**.** In comparison, the estimated KIC values of quiv-KI crack fronts were in good agreement with the two materials' reference KIC, but the semi-circle crack fronts had considerable inaccuracies. The results showed that using the finite element method (FEM) and VCCT, indentation fracture toughness could be accurately calculated and that obtaining an equi-KI crack front was critical to the accuracy of computed fracture toughness.

The vertical displacement, critical Euler buckling stress, and vibration behavior of a damaged beam were estimated [21]. There is a border on the first eigenvalue of free vibration results where the first eigenvalue of free vibration does not change if the center of the crack is located on that border, and after that border, the first eigenvalue of free vibration increases, which is a counterexample relationship of critical Euler buckling load and first eigenvalue of free vibration.

2.3 Evaluation of T-Stress

The evaluation of *T*-stress from FEA for mode I and mixed-mode I/II loading was done [22]. T-stress was calculated using finite element analysis in this work. It suggested methods for determining T-stress without needing to compute stress intensity parameters for every mixed-mode I/II loading. The case study demonstrates that the T-stress may be estimated in mixed-mode issues utilizing the stresses or displacements along with the two faces of the fracture without calculating stress intensity components. This is possible if the contribution of mode II direct stresses is first reduced by superposition of the stresses along the crack's two sides. It was demonstrated that for pure shear loading of a single edge notched specimen subjected to varied loading circumstances and different combinations of mode I and mode II loadings, a considerable T-stress may be achieved unless when the loading is totally antisymmetric. The magnitude of the biaxiality ratio B in mode II for real loading circumstances is substantially larger than that in mode I loading, according to a finite element study of the mixed-mode specimen. In this scenario, the biaxiality ratio in mode II reduces as fracture length rises, whereas B increases as crack length increases in mode I.

Mode I cracks subjected to large *T*-stresses in a rectangular plate were studied [23]. The modified maximum tangential stress criteria, which was first developed for mixed-mode loading, was used to investigate theoretically brittle fracture for mode I fractures in this research. From the case study, it was observed that, for large positive T–stress, it shows that the initiation of brittle fracture in mode I cracks can occur at angles other than directly ahead of crack. It was also observed that the fracture angle increases for larger values of the biaxiality ratio B, larger magnitudes of the normalized radius of process zone α, or both. The apparent fracture toughness decreases once B increases beyond a critical value. Along the crack line, the T-stress component of tangential stress disappears. When the T-stress surpasses a threshold value, however, the maximum tangential stress no longer lies along the path of the original fracture. As a result, specimens with a significant T-stress might expect a deviation in the angle of start of fracture. For increasing amounts of T-stress, the deviation angle rises. Theoretical studies reveal that as the angle of the fracture deviates, the apparent fracture toughness drops dramatically. When there is a change in angle, fracture toughness is reduced dramatically.

A large range of T-stress values for different specimen configurations was studied [24]. For a CT specimen made up of steel, positive and negative values are obtained. T-stress varied from −0.8 to –0.1 stress value along the y-direction. It was also observed that along the ligament of the specimen the value of T-stress was not constant. To solve this problem, the effective T-stress, which is the average value of the T-stress distribution at the effective distance supplied by the volumetric method, has been developed. From the above research done in the calculation of T-stress, the following conclusion will be figured out:

(a) The T-stress value is affected by the specimen shape, specimen, loading mechanism, and fracture size. When the specimen loading mode and geometry change from tension to bending, the T-stress increases from a high negative value to a low negative or positive value [25]
(b) The stress intensity factor over the T ratio increases nonlinearly with non-dimensional crack length [26].
(c) The sign and amount of the T-stress significantly alter the size and form of the plane strain crack tip plastic zone [27, 28]. T-stress, whether positive or negative, expands the plastic zone in comparison to a no-T-stress environment. In plane strain, the plastic zone was oriented along with the crack extension when

    $T > 0$ and in the opposite direction when $T < 0$.

    a. It was discovered that under the Paris law regime, the fatigue crack development rate lowers as T increases [29].
    b. Analytical and experimental studies indicate that the T-stress might be used as a constraint measure prior to the crack tip. As (–T) increases, fracture toughness increases [30–32].

The T-stress calculation methods in fracture mechanics computation were investigated [33]. The goal of the study was to provide a numerical study defining the T-stress in 2D utilizing the FEM and the software ANSYS 15.0. According to the example study, the T-stress readings for a CT specimen in the range $a/W = 0.2$–0.6 are positive (tension stress). It was also discovered that the T-stress does not remain constant throughout the ligament ($\theta = 0°$) and crack mouth ($\theta = 180°$) forward of the crack tip in CT specimens. It was also discovered that the non-singular terms for a crack are not negligible as the distance from the fracture tip grows. This result demonstrates that the T-stress value was not constant for a crack depth ratio fluctuation, and a theory cannot be formed with a distribution. The Maleski approach yields satisfactory results for $\theta = 0°$ and $\theta = 180°$.

Using a common numerical example, the convergence, accuracy, and domain independence of the integral connection numerically were studied, and then evaluate the impact of crack settings and boundary conditions on T-stress through two more cases [34]. The current study may be further developed to investigate T-stress for cracked non-homogeneous materials under both thermomechanical and mechanical stresses. Three numerical cases are assessed on non-compliant uniform mathematical covers with all domain boundaries to validate the suggested model and the calculated

results are consistent with the present alternatives. Besides the computational techniques, experimental techniques like photoelasticity have also been used for solving fracture problems [35, 36]

In view of the literature survey, it can be said that:

1. No studies have focused on the concept of crack branching under biaxial loading.
2. There are few studies focusing on asymmetric crack branching.

3 Materials and Methods

The material and technique utilized to explore the influence of branching crack configurations such as crack inclination angle, biaxial factor, and crack length ratio on fracture parameters such as stress intensity factor and T-stress are described in this chapter.

3.1 Material

To evaluate the fracture parameters of a component, it is very important to know its mechanical properties. Properties of the specimen used are listed in Table 1.

Steel is an iron–carbon alloy with less than 2% carbon and 1% manganese, as well as some other elements. Its application is in auto and construction equipment, some household appliances, cargo ships, and medical scalpels, among other things. It may be recycled several times without causing damage to the environment.

Rectangular thin plate with branched crack is used in FEM in the current study. Figure 1 shows the various instances of branched crack configurations. The plate thickness, width, and length are 1 mm, 10 mm, and 20 mm, respectively. Branching of cracks is done under biaxial loading for the center of the plate. OA is the primary crack and OB & OC are taken as secondary crack. Specimen geometry investigates the impacts on fracture parameters due to the existence of secondary branched cracks. The symmetrical secondary crack in this geometry is inclined at an angle to the primary crack. Several effects can be observed with the use of geometry such as effect of crack of inclination angle ($\alpha = 0°$–$75°$), biaxial factor ($\beta = 0$–2), and crack length ratio ($a_1/a_2 = 0.5$–2.5).

where

L 20 mm, length of the specimen

Table 1 Material properties [22]

Material	Young's modulus (GPa)	Poisson's ratio	Density (kg/m^3)
Steel	210	0.3	8050

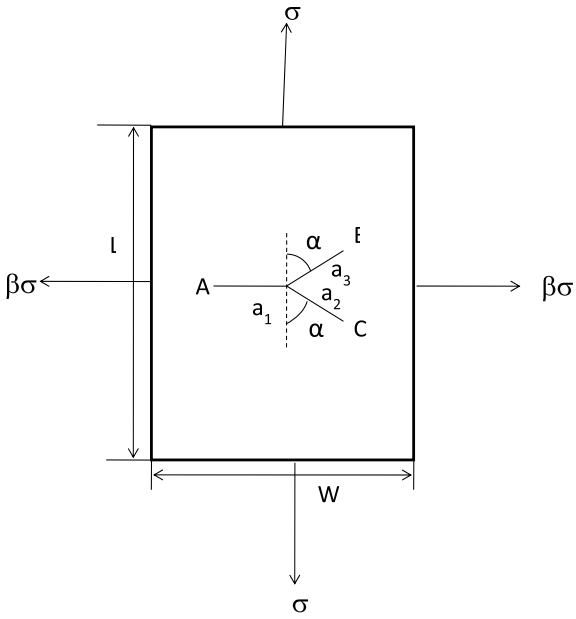


Fig. 1 Rectangular plate with branched cracks

Table 2 Geometry specifications and loading parameters

SI. No	a_1 (mm)	a_2 (mm)	a_3 (mm)	W (mm)	L (mm)	β	α (degree)
1	2	2	2	10	20	0.5	0–75
2	2	2	2	10	20	0–2	45
3	2	0.5–2.5	0.5–2.5	10	20	0.5	30

w 10 mm, width of the specimen
α angle between symmetrically inclined crack with loading axis
σ load applied in y-direction
βσ load applied in x-direction
A, B, C Crack tips

Here, β is the biaxial load factor (Table 2).

3.2 Stress Intensity Factor

3.2.1 Mode I Crack

A polynomial formula for a finite plate with a center fracture under tensile stress for solutions of a center cracked plate in tension [37].

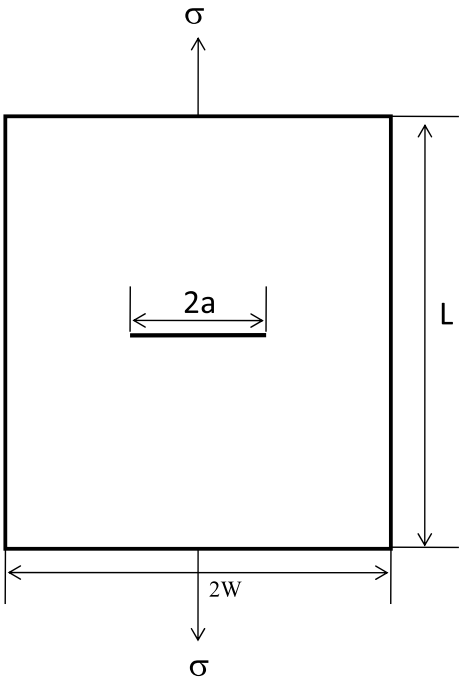


Fig. 2 Rectangular plate with single-center crack under uniaxial loading

$$K_{I(O)} = \sigma\sqrt{(\pi a)}[\sec\left(\frac{\pi a}{2w}\right)]^{\frac{1}{2}}[1 - 0.025(\frac{a}{w})^2 + 0.06\left(\frac{a}{w}\right)^4] \quad (1)$$

where 2w is the width of plate, 2a is the length of crack, and σ is the tensile stress as shown in Fig. 2.

3.2.2 Inclined Crack

The analytical expression of the stress intensity factors for infinite plate when a crack is oriented at an angle (α) from the applied normal stress are given as.

For uniaxial loading:

$$K_{IA} = K_{I(O)}cos^2\alpha \quad (2)$$

$$K_{IIA} = K_{I(O)}sin(\alpha)cos(\alpha) \quad (3)$$

where $K_{I(A)}$, $K_{II(A)}$ are mode I and mode II stress intensity factors of inclined crack, respectively (Figs. 3 and 4).

For biaxial loading

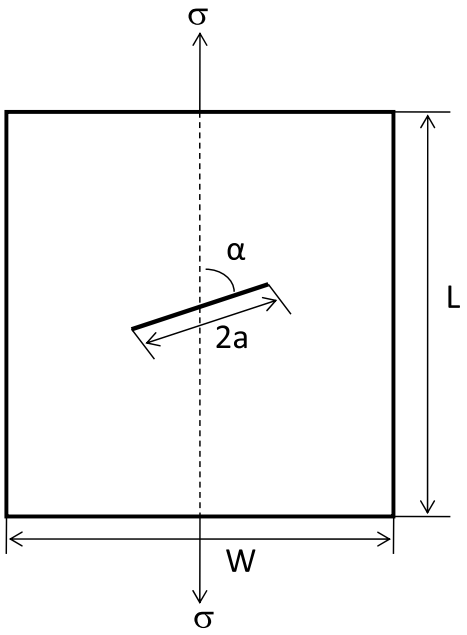


Fig. 3 Rectangular plate with central inclined crack under uniaxial loading

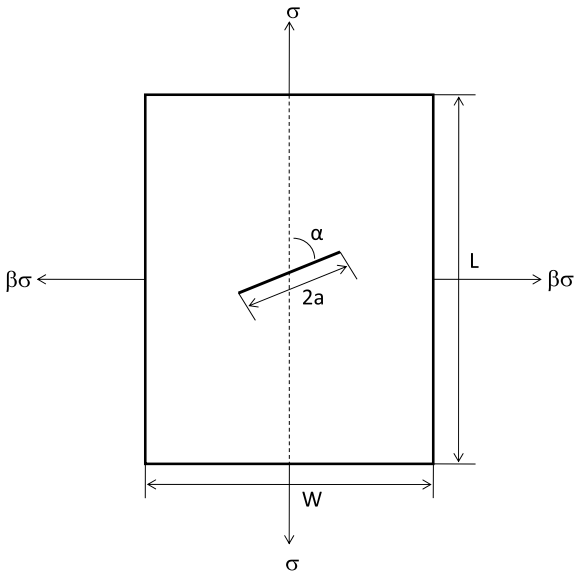


Fig. 4 Rectangular plate with central inclined crack under biaxial loading

$$K_{IA} = K_{I(O)}(cos^2(\alpha) + \beta sin^2(\alpha)) \tag{4}$$

$$K_{IIA} = K_{I(O)}(sin(\alpha)cos(\alpha))(1 - \beta) \tag{5}$$

where $\beta = \sigma_2/\sigma_1$, tension stresses in the direction Oy and Ox are $\sigma_1 = \sigma$ and $\sigma_2 = \beta\sigma$.

3.3 Specimen Geometry

Rectangular thin plate with branched crack is used in FEM in the current study. Figure 5 shows the various instances of branched crack configurations. The plate thickness, width, and length are 1 mm, 10 mm, and 20 mm, respectively. Branching of cracks is done under biaxial loading for the center of the plate. OA is the primary crack and OB & OC are taken as secondary crack. Specimen geometry investigates the impacts on fracture parameters due to the existence of secondary branched cracks. The symmetrical secondary crack in this geometry is inclined at an angle to the primary crack. Several effects can be observed with the use of geometry such as

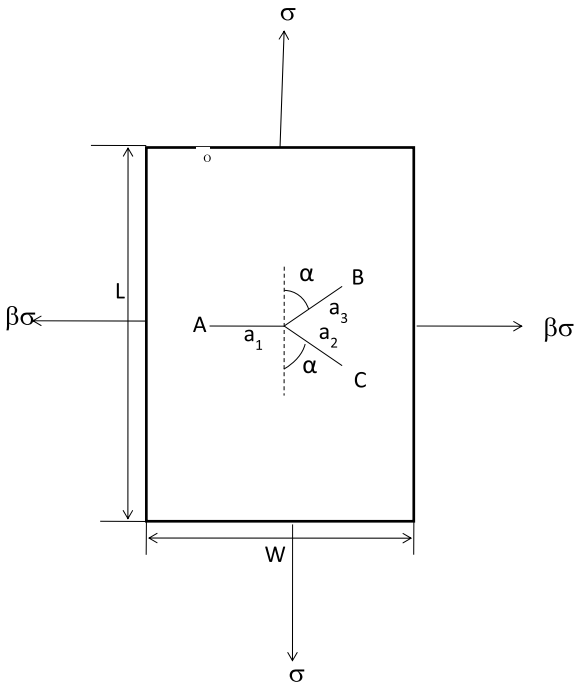


Fig. 5 Rectangular plate with branched cracks

effect of crack of inclination angle ($\alpha = 0°–75°$), biaxial factor ($\beta = 0–2$) and crack length ratio ($a_1/a_2 = 0.5–2.5$).

where

L	20 mm, length of the specimen
w	10 mm, width of the specimen
α	angle between symmetrically inclined crack with loading axis
σ	load applied in y-direction
$\beta\sigma$	load applied in x-direction
A, B, C	Crack tips

Here, β is biaxial load factor.

3.4 Method

For a thin steel plate under plane strain condition containing branched cracks, stress intensity factors were calculated using the ANSYS software and its variation for different conditions and orientations of cracks were analyzed. For verifying the validity, we use theoretical method in which stress intensity factor is compared with the result obtained analytically.

3.4.1 Finite Element Analysis Software (ANSYS)

Finite element analysis is a numerical method by which a complex system is divided into very small parts called components. ANSYS is a software package for the particular purpose of evaluation of finite elements. The software uses governing equations and solving the behavior of these components; producing a detailed description of how the structure functions as a whole. It is then possible to present these outcomes in tabulated or graphical forms. Typically, this sort of assessment is used to develop and optimize a scheme, that is too complicated for hand analysis. Because of their geometry, scale, or governing equations, systems that can fit into this category are too complicated. The product solves a wide range of engineering issues. Finite element analysis involves three stages: pre-processing, processing, and post-processing. All the three stages are interlinked with each other. Stages of finite element analysis are shown in figures with the help of a given flowchart. These are the steps of graphical user interface in ANSYS. In ANSYS APDL at the beginning of the analysis, we have to specify the preference like ANSYS Fluid thermal or structural. These options are given in the preference section in ANSYS APDL.

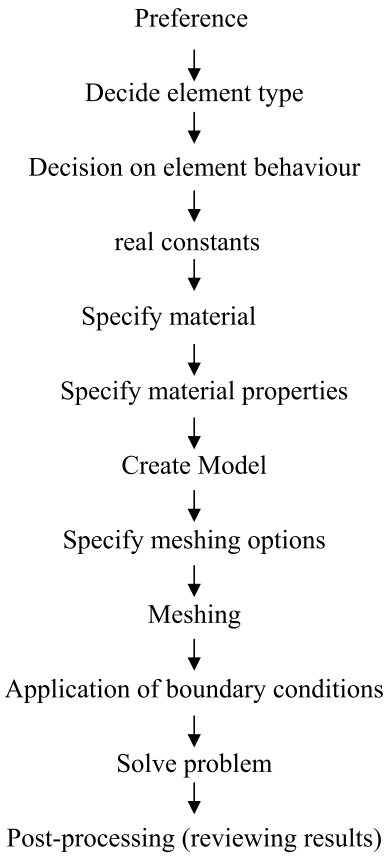


Fig. 6 Steps of FEM

3.4.2 Steps Involved

See Fig. 6.

3.4.3 Assumptions

The dimensions of steel plate are shown in Fig. 2. Plate is made up of steel with Young's modulus 210GPa and Poison's ratio 0.33. In the analysis, plane strain conditions are assumed. A 2-D analysis using PLANE 82 is performed. For generating the 2-D model, mesh solid modeling is used. The SIFs are obtained through post-processing command and used in the analysis of branched crack.

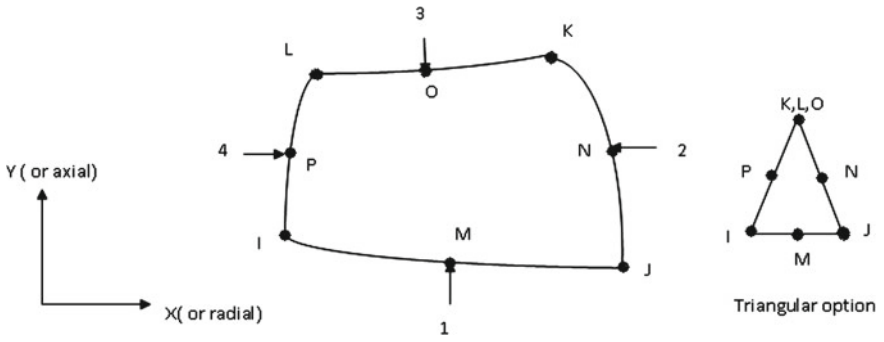


Fig. 7 Plane 82 element with 8-nodes

3.4.4 Types of Elements Used

The PLANE 82 elements utilized in the study are a higher-order variant of the 2-D, four-node element (Plane 42) depicted in the picture. It produces more accurate results for mixed (quadrilateral–triangular) automated meshes and may be utilized for unusual forms without losing much precision. The eight-node components are ideally suited to modeling curved boundaries since they have appropriate displacement forms.

It is also defined by eight nodes with two degrees of freedom at each node translation in x and y nodal directions. It can be utilized as a planar or axisymmetric element. The element has plasticity, creeping, swelling and stiffness of stress, large deflection, and large strain capacity (Fig. 7).

Data may be retrieved in tabular or contour plot form for all nodes and elements. A route is manually established to retrieve the SIFs by choosing the five nodes at the fracture face. The SIFs are retrieved once the route has been defined. The contour chart may be used to derive the von Mises distribution around the fracture tip, which can then be utilized for further investigation.

3.4.5 Boundary Condition

To answer a mechanics issue, specific criteria surrounding the border of the substance under study must be specified. The setting of a known value for a displacement or a related load is a boundary condition for the model. You can set either the load or the displacement for a certain node, but not both.

3.4.6 Discretization

This is the first step in FEM and hence it is an important process. In this step, body or continuum is divided or discredited into elements. The elements are non-overlapping

Table 3 Discretization of specimen geometry for inclination angle

Inclination angle (α), degree	0°	15°	30°	45°	60°	75°
Element	333	357	372	393	413	435
Nodes	955	1092	1098	1113	1115	1156

Table 4 Discretization of specimen geometry for biaxial factor

Biaxial factor (β)	0	0.5	1	1.5	2
Element	414	414	414	414	414
Nodes	1156	1156	1156	1156	1156

Table 5 Discretization of specimen geometry for crack length

Crack length ratio (a_1/a_2)	0.5	1	1.5	2	2.5
Element	422	429	436	443	446
Nodes	1180	1195	1220	1239	1260

and can be of different shapes and sizes. The elements are connected to each other through node. The discrediting of continuum in element and node is called mesh generation. Dividing the body into more number of elements is always preferable because more perfect solutions are obtained.

Number of Elements and Nodes Created During the Meshing of the Different Geometries

See Tables 3, 4 and 5.

3.4.7 Solution

The solver subroutine in the finite element software solves the simultaneous set of equations generated by the finite element technique during the solution phase of the analysis. Solving the nodal degree of freedom values, such as displacements and rotations, usually yields the principal solution. Then derived results, such as stresses and strains are calculated at the integral points. In ANSYS, primary results are called nodal solutions and derived results are called element solutions. Several methods of solving the system of simultaneous equations are available. Some methods are better for larger model others are faster for nonlinear analysis, and some allow to distribute the solution by parallel computation.

Commercial finite element programs solve these equations in batch mode. The frontal direct solution method is commonly used because it is more efficient for finite element analysis. When the analysis is nonlinear the equation s must be solved

repeatedly thus increasing the computational time significantly. The preconditioned conjugate gradient solver operates similarly to the Jacobi conjugate gradient solver with four exceptions:

- For structural solid components, the preconditioned conjugate gradient solver is typically 4–10 times quicker than the Jacobi conjugate gradient solver and roughly ten times faster for shell elements. This prevents the issue from growing in magnitude.
- The preconditioned conjugate gradient solver uses the EMAT file.
- As a preconditioner, the Jacobi conjugate gradient solver employs the global matrix's diagonal.
- Because it keeps two matrices in memory, the preconditioned conjugate gradient solver often takes twice as much memory as the Jacobi conjugate gradient solver.
- The stiffness matrix's preconditioner, which has about the same size.
- The symmetric non-zero part of the stiffness matrix.

3.5 Numerical Approach of T-stress Analysis

For evaluation of T-stress, displacement method is used which is given below:

$$\text{For mode I } T = (E * [Ux(x) - Ux(0)])/(x) \tag{6}$$

$$\text{For mode II } T = (E * [Ux(x, -\pi) - Ux(x, \pi)])/(2x) \tag{7}$$

Here, E is the Young's modulus and x is the distance of the first row of the element. U_x is thedisplacement in x-direction, and x is the distance of the first row of the element from the crack tip.

3.6 Crack Analysis

For a linear elastic fracture mechanics study, the stress intensity factor at a crack may be calculated (using the KCALC command). A fit of the nodal displacements in the region of the fracture is used in the study. The actual displacements at and near a crack for linear elastic materials are:

$$u = \frac{K_I}{4G}\left(\sqrt{\frac{r}{2\pi}}\right)\left[(2k-1)cos_2^{\theta} - cos_2^{3\theta}\right) - \frac{K_{II}}{4G}\left(\frac{\sqrt{r}}{2\pi}\right)\left[(2k+3)sin_2^{\theta} + sin_2^{3\theta}\right] + O(r) \tag{8}$$

$$v = \frac{K_I}{4G}\left(\sqrt{\frac{r}{2\pi}}\right)\left[(2k-1)sin_2^{\theta} - sin_2^{3\theta}\right) - \frac{K_{II}}{4G}\left(\frac{\sqrt{r}}{2\pi}\right)\left[(2k+3)cos_2^{\theta} + cos_2^{3\theta}\right] + O(r) \tag{9}$$

$$w = \frac{2K_{III}}{G}\left(\sqrt{\frac{r}{2\pi}}\right)\left[(sin_2^{\theta})\right] + O(r) \tag{10}$$

where

u, v, w	displacements in a local Cartesian coordinate system as shown in Fig. 5
r, θ	coordinates in a local cylindrical coordinate system also shown in Fig. 5
G	shear modulus.
K_I, K_{II}	Mode I and Mode II stress intensity factors.
v	Poisson's ratio.
0(r)	terms of order r or higher.

Evaluating equations at θ = ± 180° and neglecting the higher-order terms

$$u = \frac{K_{II}}{2G}\left(\sqrt{\frac{r}{2\pi}}\right)(1+k) \tag{11}$$

$$v = \frac{K_I}{2G}\left(\sqrt{\frac{r}{2\pi}}\right)(k) \tag{12}$$

$$w = \frac{2K_{III}}{G}\left(\sqrt{\frac{r}{2\pi}}\right) \tag{13}$$

Equations for full crack model are as follows:

$$K_I = \left(\sqrt{2\pi}\right)\left(\frac{G}{k}\right)\left(\frac{|\Delta v|}{\sqrt{r}}\right) \tag{14}$$

$$K_{II} = \left(\sqrt{2\pi}\right)\left(\frac{G}{1+k}\right)\left(\frac{|\Delta u|}{\sqrt{r}}\right) \tag{15}$$

$$K_{III} = \left(\sqrt{2\pi}\right)G\left(\frac{|\Delta w|}{\sqrt{r}}\right) \tag{16}$$

where Δ *v*, Δ*u*, and Δ*w* are the motions of one crack face with respect to the other.

On the nodal displacements and locations k = 3–4*v* if plane strain or axisymmetric; (3-*v*) if plane stress; where *v* is Poisson's ratio the final factor $\frac{\Delta v}{\sqrt{r}}$ is calculated (Fig. 8).

$$\left(\frac{|\Delta v|}{\sqrt{r}}\right) = A + Br \tag{17}$$

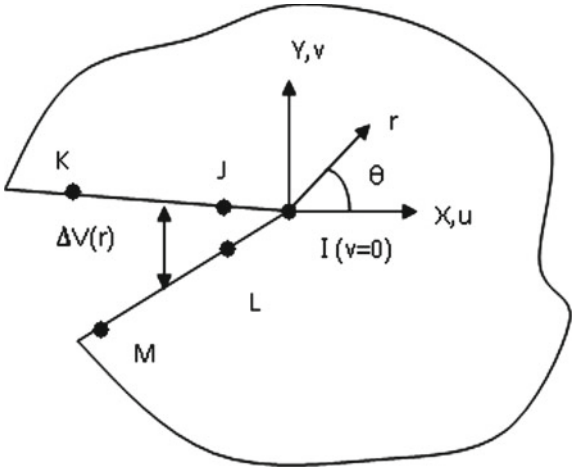


Fig. 8 Representation of displacements to be used in analysis

At points J and K. R approach 0,

$$lim_{r\to 0}\left(\frac{|\Delta v|}{\sqrt{r}}\right) = A \tag{18}$$

The equation becomes

$$K_I = \left(\sqrt{2\pi}\right)\left(\frac{2GA}{k}\right) \tag{19}$$

Similarly,

$$\left(\frac{|\Delta u|}{\sqrt{r}}\right) = C + Dr \tag{20}$$

$$lim_{r\to 0}\left(\frac{|\Delta u|}{\sqrt{r}}\right) = C \tag{21}$$

The equation becomes

$$K_{II} = \left(\sqrt{2\pi}\right)\left(\frac{GC}{1+k}\right) \tag{22}$$

and

$$\left(\frac{|w|}{\sqrt{r}}\right) = E + Fr \tag{23}$$

$$lim_{r\to 0}\left(\frac{|\Delta w|}{\sqrt{r}}\right) = E \tag{24}$$

The equation becomes

$$K_{III} = \left(\sqrt{2\pi}\right) G\left(\frac{|\Delta w|}{\sqrt{r}}\right) \tag{25}$$

3.7 General Postprocessor

All forms of solution data, such as von Mises stress, primary stress, deformation, maximum stress, translations, failure criteria, Stress intensity factor, and J-integral, may be acquired using the POST 1 command. Data may be retrieved in tabular or contour plot form for all nodes and elements. To acquire the SIFs, a route is manually established by selecting five nodes at the fracture face. After locating the SIF, we proceed to the contour plot, where we obtain the von Mises distribution of the crack tip, which may be exported as a JPEG file. The von Mises distribution of a branched crack in a rectangular plate is shown in Fig. 6 (Figs. 9 and 10).

4 Results and Discussion

4.1 Validation

Validation of the FEM with published results is critical for the acceptance of the finite element approach in computation. The theoretical results of a single central and inclined crack using biaxial loading are compared with FEM results in this study. The crack angle affects the stress intensity component in theoretical validation. The findings are compared to the theoretical result [35]. As shown in the figure, the fluctuation of mode I and mode II stress intensity factors with crack inclination angle derived by theoretical approach and FEM, respectively. The results of KI and KII produced using a finite element technique and the commercial programmed ANSYS are remarkably similar. It means that any complicated fracture arrangement may be modeled using ANSYS software to estimate the stress intensity factor (Figs. 11 and 12).

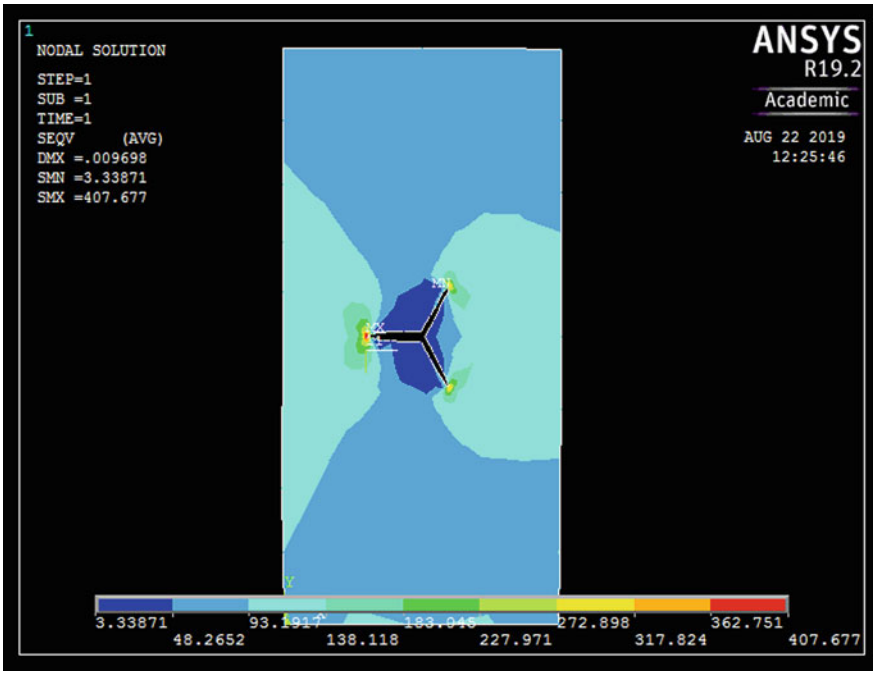


Fig. 9 Distribution of von Mises stress for a branched crack in a plate

4.1.1 Validation of Inclined Crack Theoretically with Ansys Under Biaxial Loading

See Table 6.

where

L 20 mm, length of the specimen
W 10 mm, width of the specimen
α angle between inclined crack with loading axis
σ load applied in y-direction (100MPa)
βσ load applied in x-direction

Here, β is Biaxial load factor taken as 0.5
2a= Crack length in m

4.1.2 Validation of Single Central Crack Theoretically with Ansys Under Biaxial Loading

where.

L 20 mm, length of the specimen

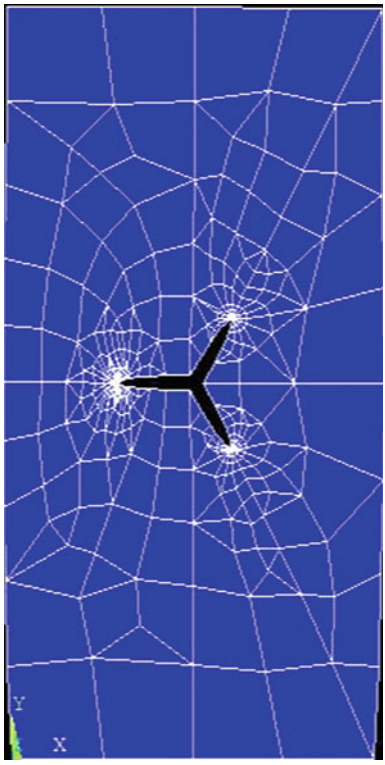


Fig. 10 Discretized image of branched crack

W 10 mm, width of the specimen
σ load applied in y-direction (100MPa)
βσ load applied in x-direction

Here, β is Biaxial load factor taken as 0.5
2a= Crack length in mm
SIF from analytical solution of the single central crack of mode I by using Eq. 4 is 250.66 (Figs. 13 and 14) (Table 7).

4.1.3 Validation for T-Stress

The specimen width ratio, i.e., a/W = 0.4 and W/L = 0.5 for a specified geometry as shown in Fig. 15, under plane strain condition having a crack length of 2 mm, where length and width of the specimen are 20 mm and 10 mm, respectively. A tensile stress of 70 MPa is applied to it as shown in Fig. 15. T-Stress was calculated using ANSYS software. T-Stress was obtained for the geometry and the results obtained were compared with the result obtained [22].

For T-stress calculation, **displacement method** was used. The T/σ by displacement method is **0.55** [22]**,** whereas from the displacement method normalized, T/σ

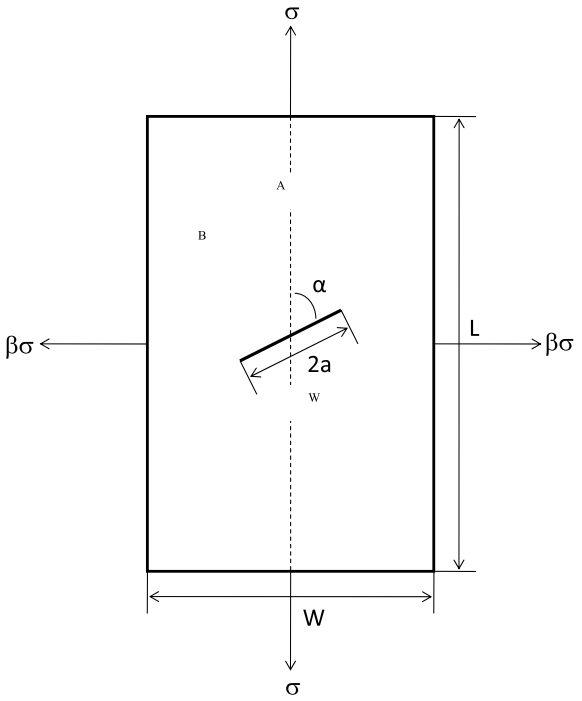


Fig. 11 Rectangular plate with single central inclined crack

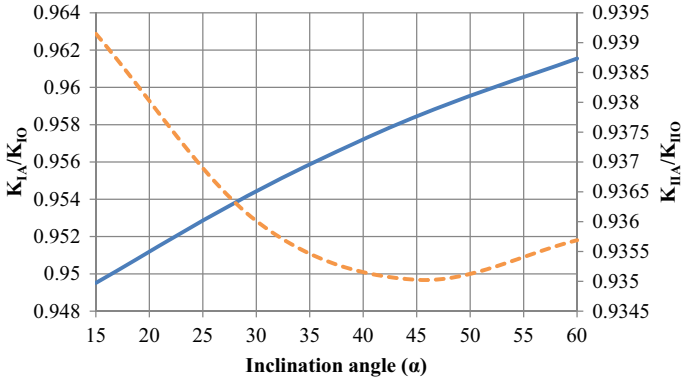


Fig. 12 Variation of stress intensity factor with inclination angle

obtained is **0.578**. The percentage error between the above two methods is **3.84%**. It is observed that results obtained from FEM using commercial software ANSYS are very close to the results [22] as shown in Fig. 16. It means finite element approach using ANSYS software can be used to determine T-stress for any complex geometry.

Table 6 Observation table for validation of SIF by theoretically and FEM

Angle	(Theoretical) K_{IA}	(Theoretical) K_{IIA}	(FEM) K_{IO}	(FEM) K_{IIO}	% error K_I	% error K_{II}
15	133.72	31.33	136.83	32.36	2.817	3.679
30	156.66	54.27	160.14	56.98	3.674	3.436
45	187.99	62.6	191.14	64.95	2.455	3.048
60	219.32	54.27	224.09	56.01	3.598	3.173

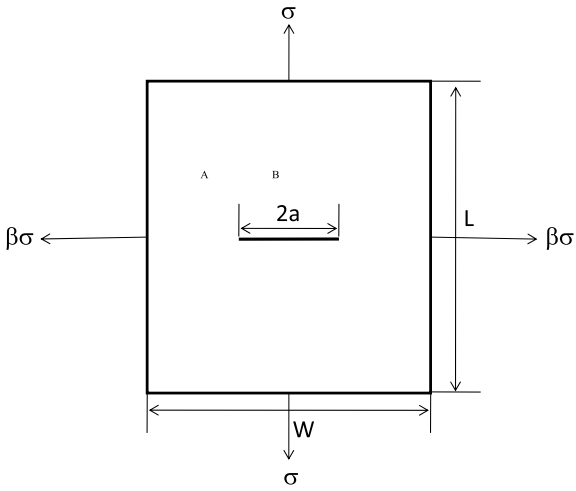


Fig. 13 Rectangular plate with single central crack

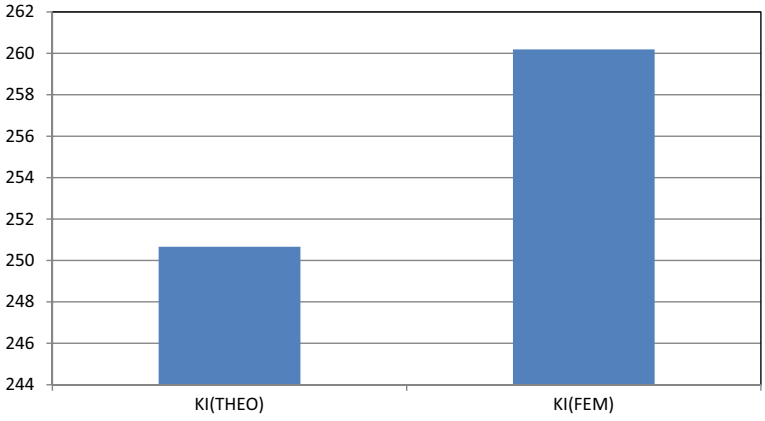


Fig. 14 Comparison of $K_{I(THEO)}$ and $K_{I(FEM)}$

Table 7 Observation table of K_I (Theoretical) and K_I (FEM)

K_I (THEORETICAL)	K_I (FEM)	% error OF K_I
250.66	262.19	3.8%

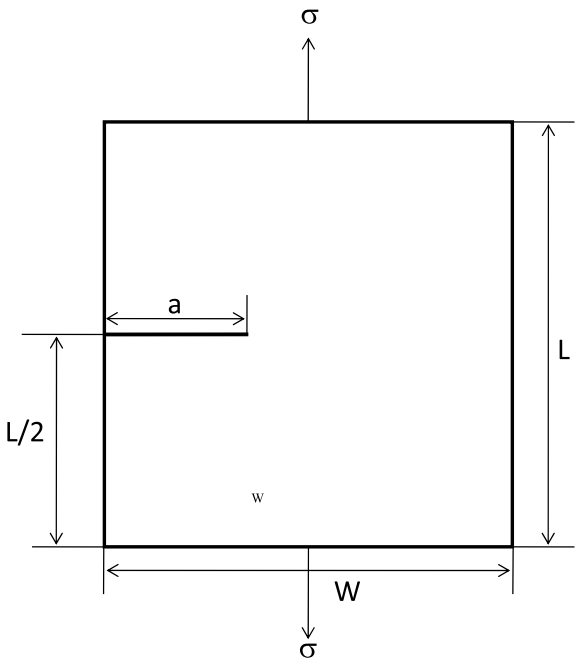


Fig. 15 Rectangular plate with side edge crack

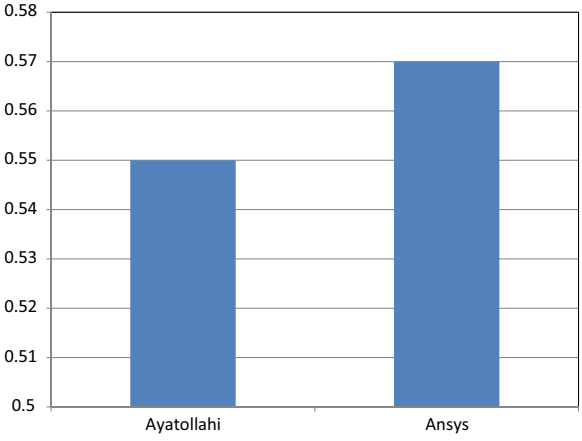


Fig. 16 Values of T/σ for the side edge crack by [22] and FEM Implementation (ANSYS)

4.1.4 Grid Independence Test

Based on the error analysis described in Fig. 17 the finite element modeling parameters are selected for inclined crack. For this, the radius of first row of crack tip element was varied. The density of finite element modeling mesh is changed to evaluate stress intensity factor by taking the number of elements of first row as 16 and the radius of first row as a/8 where *a* is the crack length measured as 2 mm. Next, the radius of

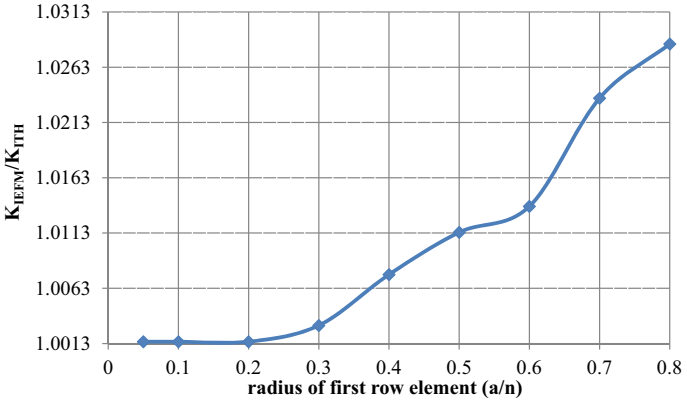


Fig. 17 Variation of K_{IEFM}/K_{ITH} with the radius of first-row element

the first row (a/n) around the crack tip was varied, taking n = 8, 10, 12, 16, and 20 with 32 elements. Finite element approach was carried out for calculating the stress intensity factor. Theoretically, K_I is computed from the relation given by **Kumar Prashant (2009)**. It is clearly seen from the graph as shown below in Fig. 17 that on average the calculation with the mesh size as 0.2 mm yields the least error, i.e. closest to unity, and after that the graph becomes linear.

4.2 Branched Crack

Geometry shown in Fig. 18 was analyzed and various parameters like T-stress and stress intensity factor were evaluated. The results obtained are discussed in the subsequent section.

4.3 Effect of Crack Inclination Angle (α) on SIF and T-Stress

4.3.1 Effect of Crack Inclination Angle (α) on SIF for Crack Tip A

The effect of crack inclination angle for the geometry having branched cracks of same crack length under same loading condition ($p_1 = p_2$) and $\beta = 0.5$ as shown in Fig. 19 are discussed in this section. As the angle increases, mode I SIF increases, and it becomes equal to K_O when the angle is 75°, the increment in mode I is 36.6%, whereas mode II SIF decreases as the crack inclination angle increases the decrement in mode II is 56.3%. It can be said that when the branched crack tip is far from the primary crack tip it shields the primary crack tip. It results in mixed-mode loading conditions which results in increasing the structural integrity of the structure.

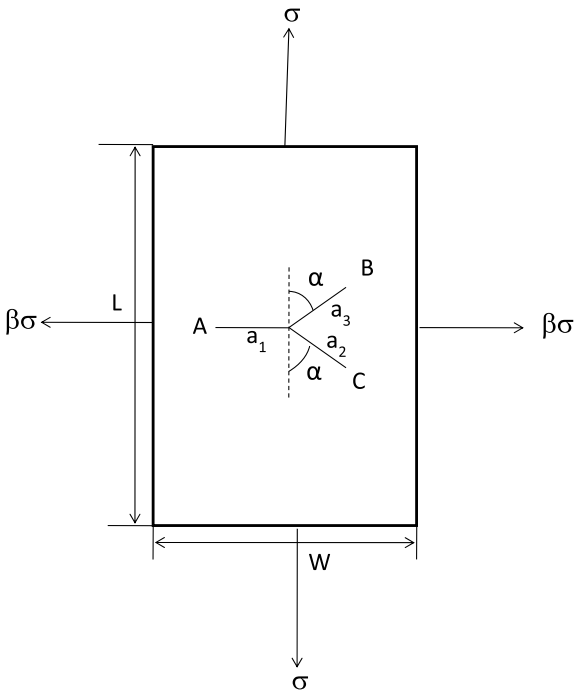


Fig. 18 Rectangular plate with branched cracks

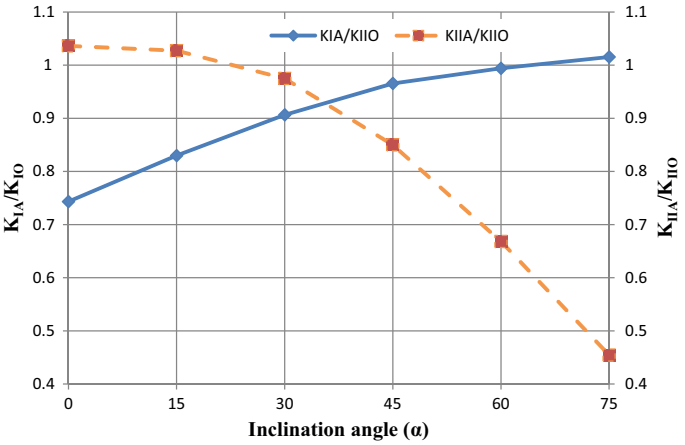


Fig. 19 Variation of K_I and K_{II} with crack inclination angle (α) at crack tip A

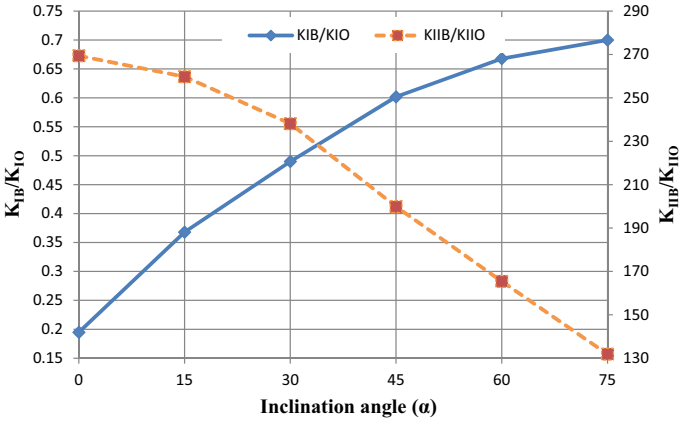


Fig. 20 Variation of K_I and K_{II} with crack inclination angle (α) for crack tip B

However, as the two tips start moving towards each other, mode I SIF increases. Mode II SIF is minimal and makes no significant contribution to failure.

4.3.2 Effect of Crack Inclination Angle (α) on SIF for Crack Tip B

Figure 20 shows the effect of crack inclination angle (α) on SIF behavior of crack tip B. It shows that the normalized mode I SIF (K_I/K_O) increases as the crack inclination angle is increased from 0° to 75°, whereas mode II SIF decreases as the crack inclination angle is increased. It shows that, as the crack is inclined more towards the higher loading axis, mode I stress intensity factor increases. The normalized mode II SIF decreases with an increase in inclination angle for crack tip B. From the results it is clearly shown that at the tip of branched crack there is a mixed-mode loading, however, counter to crack tip A, here mode II is highly pronounced and keeps on increasing as the crack tip B moves closer to crack tip A.

4.3.3 Effect of Crack Inclination Angle (α) on T-Stress

The effect of crack inclination angle (α) of crack tip B on normalized T/σ of primary crack tip A is observed. It shows that the change in inclination angle (α = 0–75) of crack tip B will affect the T-stress. T/σ value is increased with an increase in crack inclination angle (α) as shown in Fig. 21. For α = 0°, the value of T/σ is 0.58 and, for α =75°, the value of T/σ is 0.79. The change in T/σ value is 36.21%, whereas for crack tip, A T/σ value is decreased with an increase in crack inclination angle (α). It can be said that for crack tip B the T/σ is quite considerable for a higher inclination angle and cannot be neglected to predict the failure of the specimen. This will produce the enhancement effect in the material as the stress value is increased.

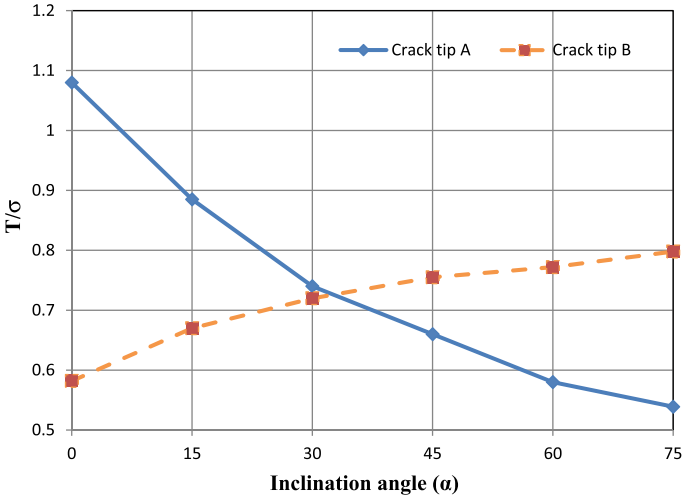


Fig. 21 Variation of T-Stress with crack Inclination angle (α)

4.3.4 Correlation Between SIF and T-Stress for Crack Tip A

The correlation between stress intensity factor and T-stress. Figure 22 shows that they are directly correlated to each other. The above results show that, for a crack tip A of branched crack, if stress intensity factor increase, then T-stress also increases. It can be seen that the non-singular term T becomes significant when the specimen is under mixed-mode load. However, as the mode mixing changes to pure mode I, the term becomes insignificant. It can be said that the T-stress can be an important parameter

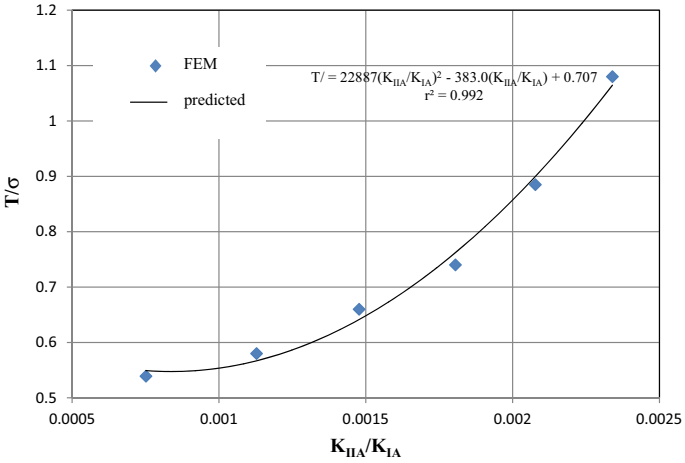


Fig. 22 Correlation between SIF and T-stress for crack tip A

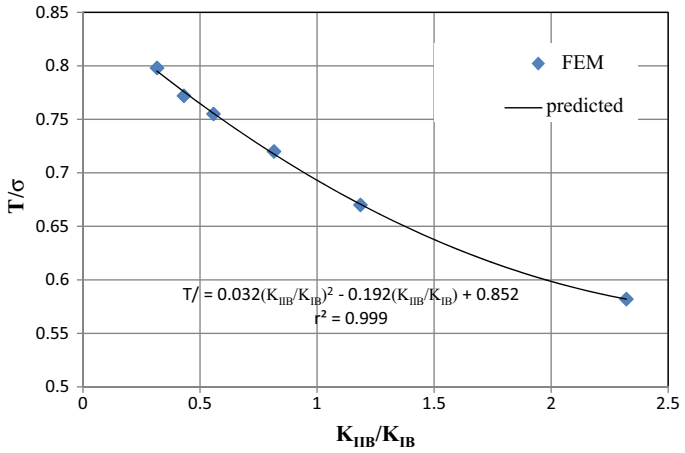


Fig. 23 Correlation between SIF and T-stress for crack tip B

for predicting failures and using these types of correlations it can be further used to replace the SIF.

4.3.5 Correlation Between SIF and T-Stress for Crack Tip B

The correlation between stress intensity factor and T-stress. In Fig. 23, it shows that they are directly correlated to each other. It shows that, for a branched crack at different inclination angles, the stress intensity factor increases when T-stress decreases. It can be said that T-stress is highly influenced by the configurations of crack.

4.4 Effect of Biaxial Load Factor (β) on SIF and T-Stress

4.4.1 Effect of Biaxial Load Factor (β) on SIF for Crack Tip A

The effect of biaxial load factor β on stress intensity factor K_I and K_{II} is shown in Fig. 24 for same crack length a = 2 mm and at same inclination angle α = 45° and for a/w = 0.1 of crack tip A of a branched crack. It shows that the effect of biaxial load factor on normalized mode I and mode II SIF is observed. The figure shows that the value of normalized mode I SIF increases linearly as the biaxial load factor increases from 0 to 2.5. The change in the value is 5.05%, whereas the normalized mode II SIF gradually increases as the horizontal tensile load increases from 0 to 2.5. Mode II SIF increases by 0.70%.

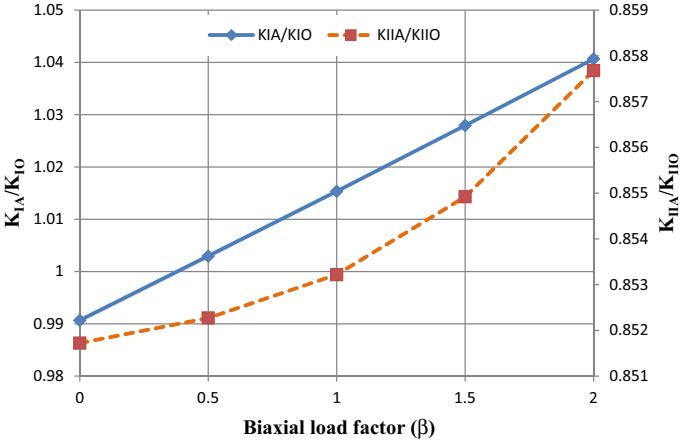


Fig. 24 Variation of K_I and K_{II} with biaxial load factor (β) for crack tip A

4.4.2 Effect of Biaxial Factor (β) on SIF for Crack Tip B

The effect of biaxial load factor β on stress intensity factor K_I and K_{II} as shown in Fig. 25 for same crack length a = 2 mm and at inclination angle α = 45° for a/w = 0.1 for crack tip B of a branched crack. It shows that the effect of biaxial load factor on mode I and mode II SIF is observed. The figure shows that the value of normalized mode I SIF decreases as the biaxial load factor increases from β = 0 to 2. From β = 0 to 2, the change in value is 18.60%. Due to the effect of branched crack, whereas the normalized mode II SIF increases from β = 0 to 0.5 linearly with

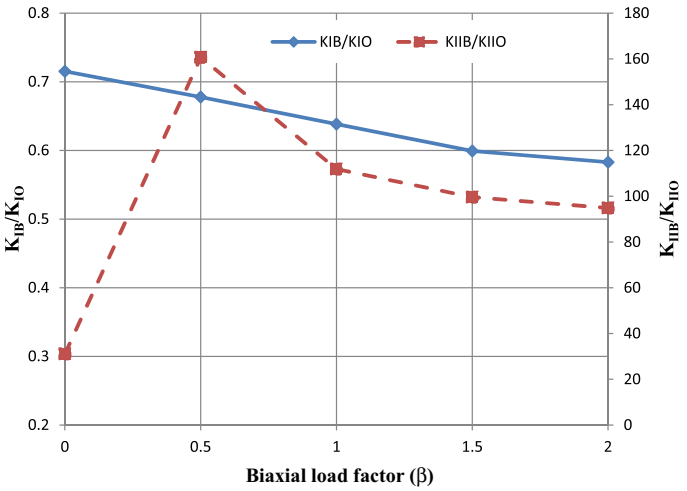


Fig. 25 Variation of K_I and K_{II} with biaxial load factor (β) for crack tip B

416.7%, because at $\beta = 0$ the horizontal tensile load is less so it acts as uniaxial loading on geometry, whereas from 0.5 to 2.5, the normalized mode II SIF decreases as the horizontal tensile load increases with 17.9% decrement. It can be said that there is a shielding effect on the primary crack due to branching when biaxial load varies from $\beta = 0.5$ to 2. However, mode II SIF is minimal and makes no significant contribution to failure.

It means that when the horizontal load exceeds the vertical load, the shear component of the stress at the crack tip becomes insignificant.

4.4.3 Effect of Biaxial Factor (β) on T-Stress

The effect of biaxial load factor (β) on T-stress of crack tip B on normalized T/σ value of primary crack is observed. Figure 26 shows that the change in the biaxial factor of symmetrically inclined cracks in which $a_{2=}a_3 = 2$ mm will affect T-stress. T/σ value increases as the biaxial load factor increases, whereas at crack tip A the value of normalized T-stress decreases. For $\beta = 0$, the value of T/σ is 0.47 and, for $\beta = 2$, the value of T/σ is 0.76. The change in T/σ value is 61.72%. Therefore, on increasing the biaxial load factor, the T-stress for crack tip B increases, and this will produce a shielding effect on primary crack tip.

For $\beta = 0$, the value of T/σ is 0.47 and, for $\beta = 2$, the value of T/σ is 0.76. The change in T/σ value is 38%. Therefore, on increasing the biaxial load factor, the T-stress increases of symmetrically inclined crack, and this produces an enhancement effect in the material.

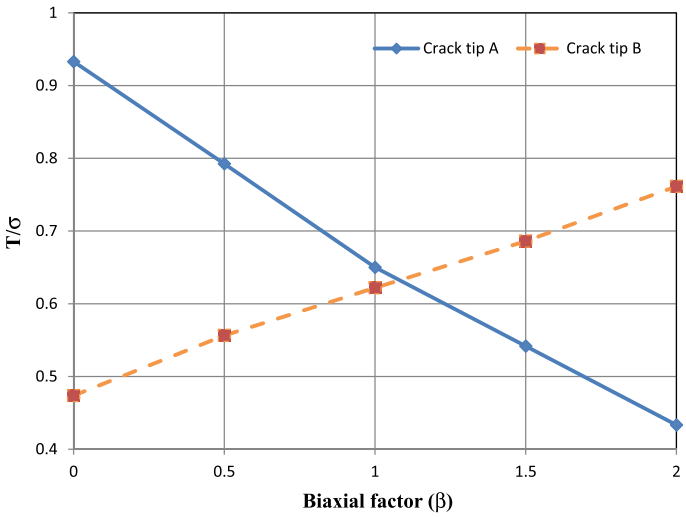


Fig. 26 Variation of T-Stress with biaxial factor (β)

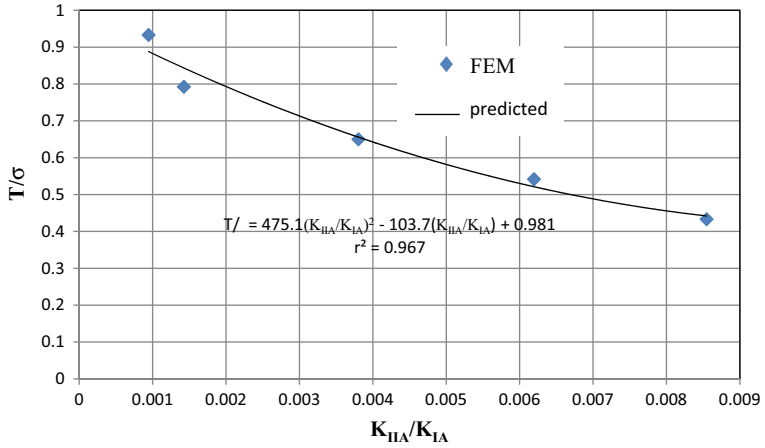


Fig. 27 Correlation between SIF and T-stress for crack tip A

4.4.4 Correlation Between SIF and T-Stress for Crack Tip A

The correlation between stress intensity factor and T-stress. Figure 27 shows that they are directly correlated to each other. It showed that, for a branched crack at different biaxial factors, the stress intensity factor increases when T-stress decreases. T/σ value decreases as the biaxial load factor increases from $\beta = 0$ to 2, whereas at crack tip A, the value of T-stress decreases.

4.4.5 Correlation Between SIF and T-Stress for Crack Tip B at Different Biaxial Factors

The correlation between stress intensity factor and T-stress. Figure 28 shows the polynomial relations. Results show that for a branched crack at a different biaxial factor β, the stress intensity factor increases linearly with an increase in T-stress The result was obtained for crack tip B, whereas with decreasing mode mixity, T-stress is increasing.

4.5 Effect of Crack Length Ratio (a_1/a_2) on SIF and T-Stress

4.5.1 Effect of Crack Length Ratio (a_1/a_2) on SIF for Crack Tip A

The effect of crack length ratio (a_1/a_2) on normalized mode I and mode II SIF is shown in Fig. 29 at $\alpha = 30$ and $\beta = 0.5$ for crack tip A. It shows that normalized mode I SIF increases as the crack length ratio increases from 0.5 to 2.5. Mode I

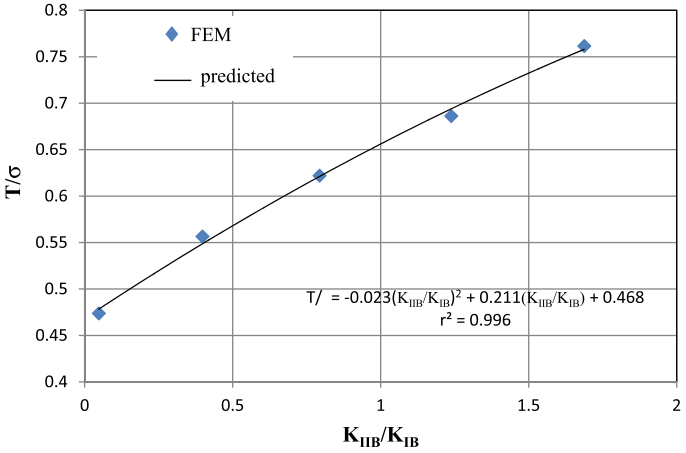


Fig. 28 Correlation between SIF and T-stress for crack tip B

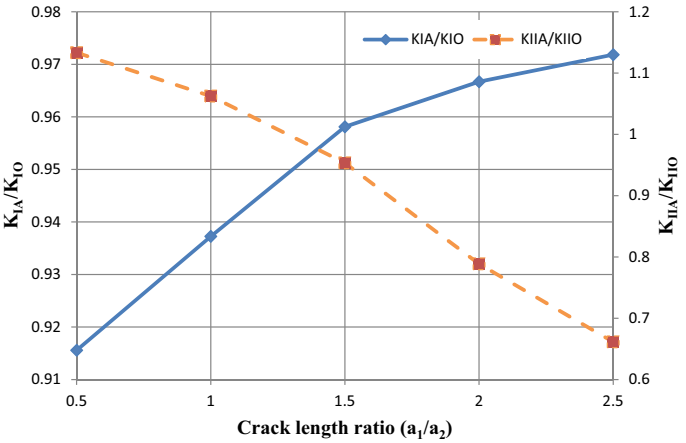


Fig. 29 Variation of K_I and K_{II} with crack length ratio (a_1/a_2) for crack tip A

increases by 6.53%, whereas the normalized mode II SIF decreases from 0.5 to 2.5. Mode II decreases with 50.3%. Results show that when normalized mode I attains the maximum value of SIF, mode II attains the maximum value of stress intensity factor at the points. It can be said that stress intensity factor depends on geometry factors such as crack length or size factors. Due to the crack tip B, the enhancement effect will produce on the primary crack.

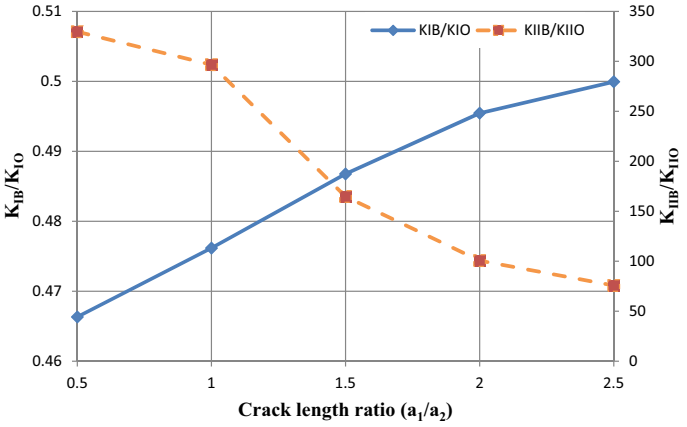


Fig. 30 Variation of K_I and K_{II} with crack length ratio (a_1/a_2) for crack tip B

4.5.2 Effect of Crack Ratio (a_1/a_2) on SIF for Crack Tip B

The effect of crack length ratio (a_1/a_2) on normalized mode I and mode II SIFs is shown in Fig. 30 at $\alpha = 30$ and $\beta = 1$ for symmetrical branched cracks. It shows that for mode I SIF increases as the crack length ratio increases from (a_1/a_2) = 0.5 to 2.5, whereas the normalized mode II SIF decreases from (a_1/a_2) = 0.5 to 2.5 at equal tension load. From Fig. 26, it can be said that stress intensity factor depends on geometry factors such as crack length or size factors. Results show that when normalized mode I attains the maximum value of SIF, mode II attains the maximum value of stress intensity factor at that point. For mode I (a_1/a_2) = 0.5, the value is 0.46 and at (a_1/a_2) = 0.25 is 0.49, so the normalized stress intensity factor increases with 6.52% and for mode II the decrement with 77.02%.

The change in K_I/K_0 from $a_1/a_2 = 0.5$ to $a_1/a_2 = 2.5$ is 6%. The change in K_{II}/K_{IIO} from $a_1/a_2 = 0.5$ to $a_1/a_2 = 2.5$ is 77%. However, because the value of mode II SIF is much lower than that of mode I SIF, mode II SIF will have little influence on the total shielding effect created by the secondary crack. It may be deduced that the greater the secondary crack size, the greater the shielding effect. Mode II developed however is substantially low and can be neglected.

4.5.3 Effect of Crack Length Ratio (a_1/a_2) on T-Stress

The effect of crack ratio a_1/a_2 of branched crack on normalized T/σ is observed. For crack tip B, T/σ value is increased as the crack ratio a_1/a_2 is increased, whereas at crack tip A, the value of T-stress decreases as shown in Fig. 31. But for crack tip B at $a_1/a_2 = 0.5$, the value of T/σ is 0.17 and, for $a_1/a_{2=}2.5$, the value of T/σ is 1.01. It shows that T/σ will have a considerable amount of stress value for all ratios (a_1/a_2

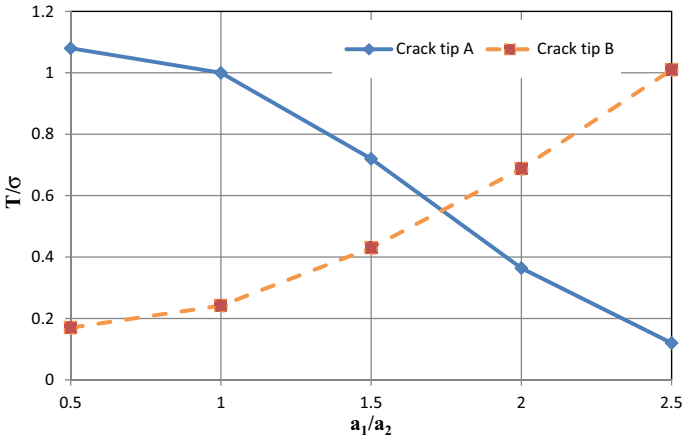


Fig. 31 Variation of T-Stress with crack length ratio (a_1/a_2)

= 0.5–2.5), it increases as the crack ratio increases. Due to an increase in the crack ratio, T-stress increases, and due to this, an enhancement effect is produced.

4.5.4 Correlation Between SIF and T-Stress for Crack Tip A

The correlation between stress intensity factor and T-stress. Figure 32 shows that, from (a_1/a_2 = 0.5–2.5), the stress intensity factor increases when the T-stress decreases gradually.

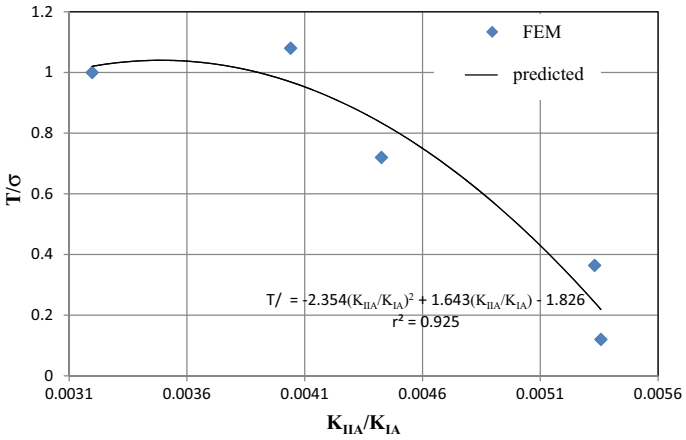


Fig. 32 Correlation between SIF and T-stress for crack tip A

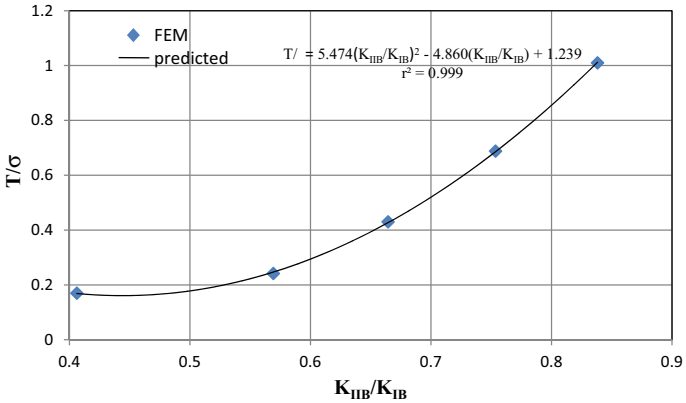


Fig. 33 Correlation between SIF and T-stress for crack tip B

4.5.5 Correlation Between SIF and T-Stress for Crack Tip B

The correlation between stress intensity factor and T-stress. Figure 33 shows the gradual relationship. Results show that, for a branched crack at a different crack length ratio ($a_1/a_2 = 0.5$–2.5), the stress intensity factor increases with an increase in T-stress. It can be said that T-stress is highly influenced by the configuration of cracks. The result was obtained for crack tip B where with decreasing mode mixity, T-stress is increasing.

4.6 Von Mises Distribution

The distribution of von Mises stresses around the crack tip for various parameters such as biaxial factor k, crack length ratio (a_2/a_1), and inclination angle (α) at uniform load level 100 MPa. Von Mises stresses are calculated by taking midpoint or element stress. Identical boundary conditions crack tip elements size and full mesh model has been applied for all crack configurations discussed in this work. Figure 34a–f shows the effect of inclination angle (α) on the distribution of von Mises stress around the crack tip. Figure 35a–e shows the effect of biaxial factor (β) on the distribution of von Mises stress around the tip. Similarly, Fig. 36a–e shows the effect of crack length ratio (a_1/a_2) on von Mises distribution.

4.6.1 Effect of Crack Inclination Angle (α) on Distribution of Von Mises Stress

From result, it is clear that the equivalent distribution of von Mises stress is shown in Fig. 34 for various angles. It shows that the distribution of von Mises stress increases

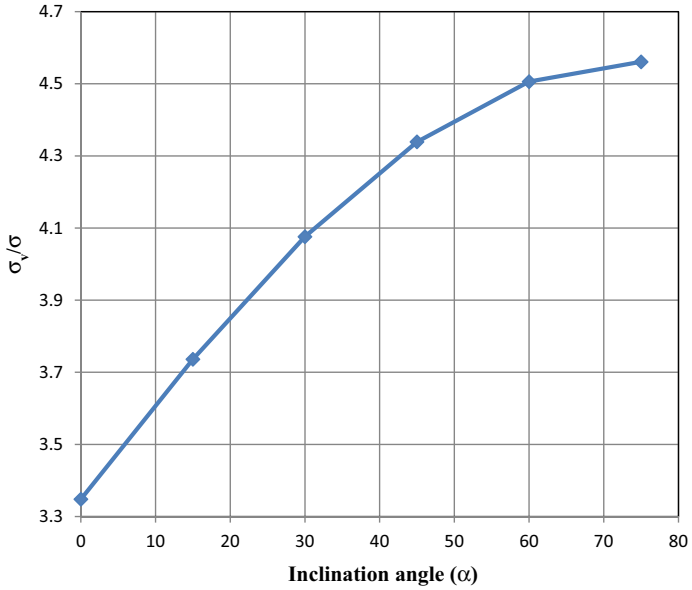


Fig. 34 Variation of von Mises with inclination angle (α). **a** Distribution of von Mises stress for a/W = 0.2, β = 0.5, S = 0.5, α = 0°, **b** distribution of von Mises stress for a/W = 0.2, β = 0.5, S = 0.5, α = 15°, **c** distribution of von Mises stress for a/W = 0.2, β = 0.5, S = 0.5, α = 30°, **d** distribution of von Mises stress for a/W = 0.2, β = 0.5, S = 0.5, α = 45°, **e** distribution of von Mises stress for a/W = 0.2, β = 0.5, S = 0.5, α = 60°, **f** distribution of von Mises stress for a/W = 0.2, β = 0.5, S = 0.5, α = 75°

with the increase in crack angle from 0° to 75°. For α = 15°, the von Mises value is 334.8 MPa and at 75° is 465.11 MPa. The change in the value is 28%. The figure shows the von Mises distribution from α = 0° to 75°.

4.6.2 Effect of Biaxial Factor (β) on Distribution of Von Mises Stress

The effect of biaxial factor (β) on Distribution of von Mises stress are shown in Fig. 35. From the results it is clear that the von Mises stresses increases as the biaxial factor increases. From figure it shows that β = 0 to 2 increases. This indicates that due to increase of horizontal tension load, the yielding at the crack tip increases which helps in increase in stress intensity factor.

4.6.3 Effect of Crack Length (a_1/a_2) on Distribution of Von Mises Stress

The von Mises stresses distribution for branched crack having crack ratio from a_1/a_2 = 0.5 to 2.5 is shown in Fig. 36. From results it is clear that the equivalent von Mises

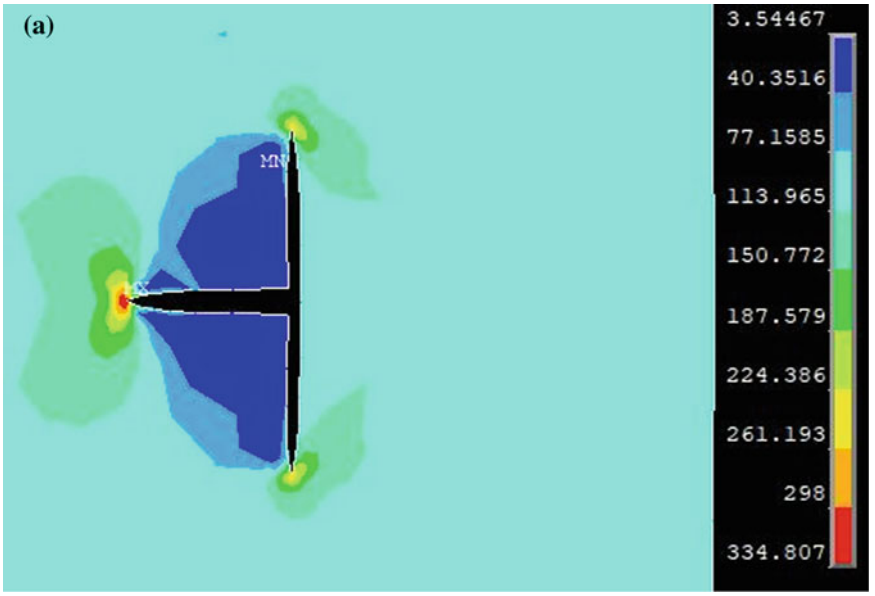


Fig. 34 (continued)

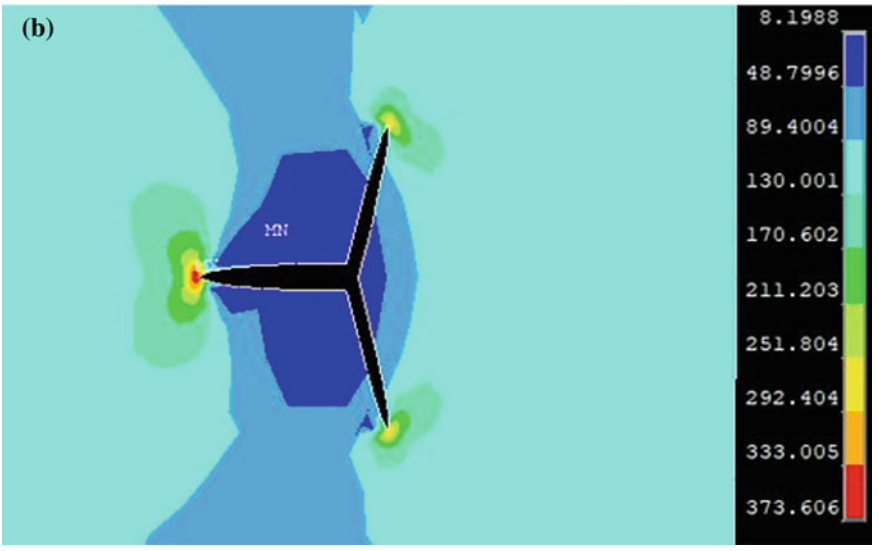


Fig. 34 (continued)

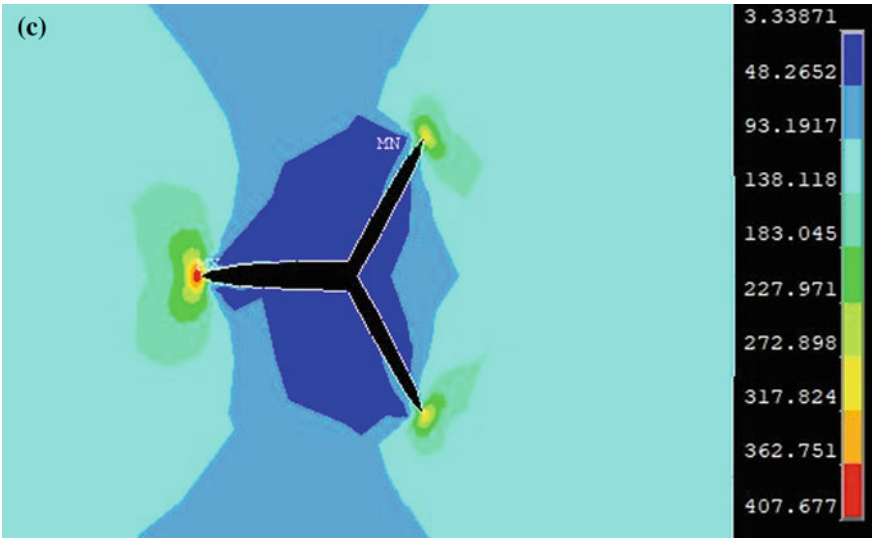


Fig. 34 (continued)

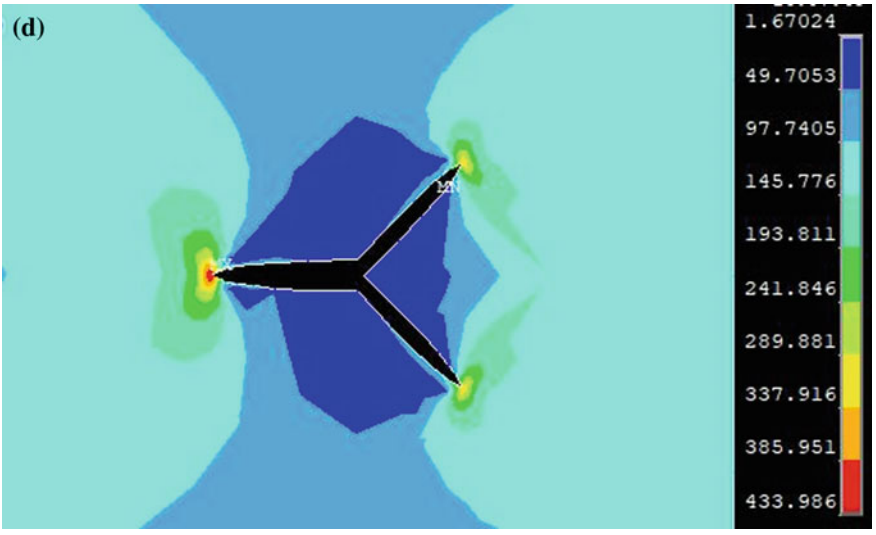


Fig. 34 (continued)

stresses increases as the crack ratio increases. It shows that yielding at the crack tip decreases with increase in crack length.

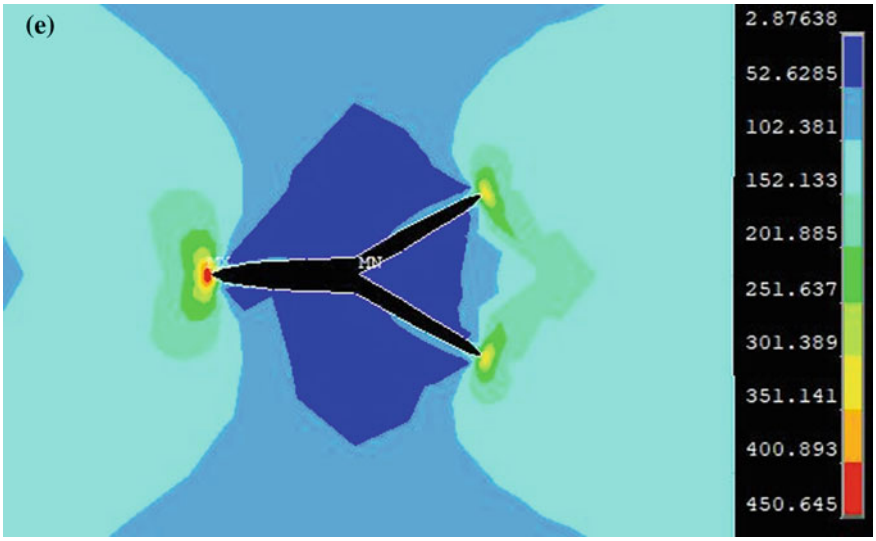


Fig. 34 (continued)

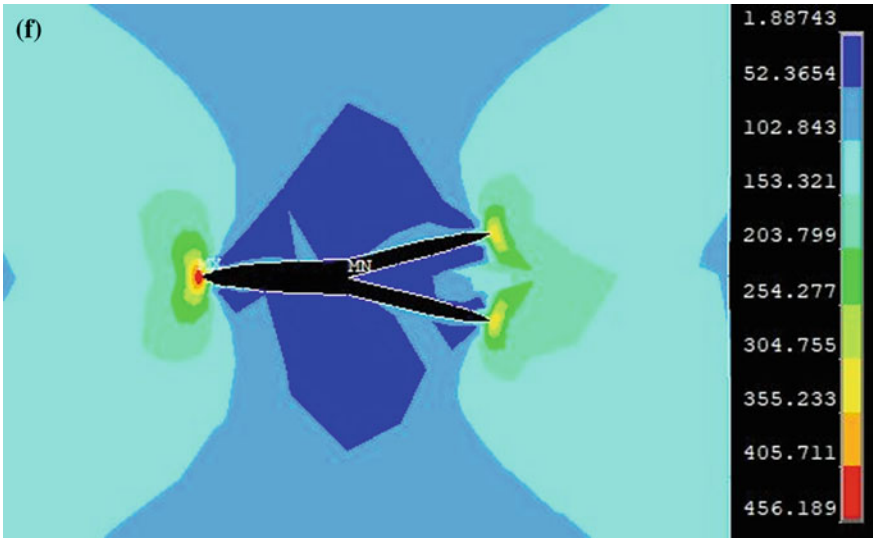


Fig. 34 (continued)

5 Conclusions

Crack inclination angle (α): The effect of crack inclination angle for the geometry having branched cracks of same crack length under same loading condition ($p_1 = p_2$)

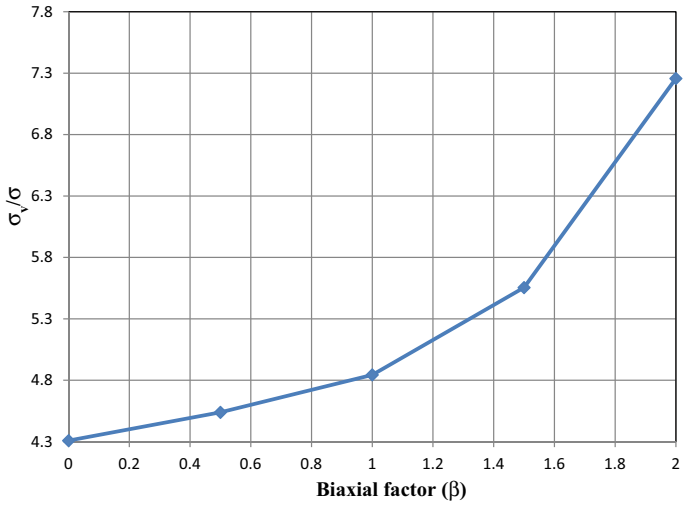


Fig. 35 Variation of von Mises on biaxial factor (β). **a** Distribution of von Mises stress for a/W = 0.2, β = 0, S = 0.5, α = 45°, **b** distribution of von Mises stress for a/W = 0.2, β = 0.5, S = 0.5, α = 45°, **c** distribution of von Mises stress for a/W = 0.2, β = 1, S = 0.5, α = 45°, **d** distribution of von Mises stress for a/W = 0.2, β = 1.5, S = 0.5, α = 45°. **e** distribution of von Mises stress for a/W = 0.2, β = 2, S = 0.5, α = 45°

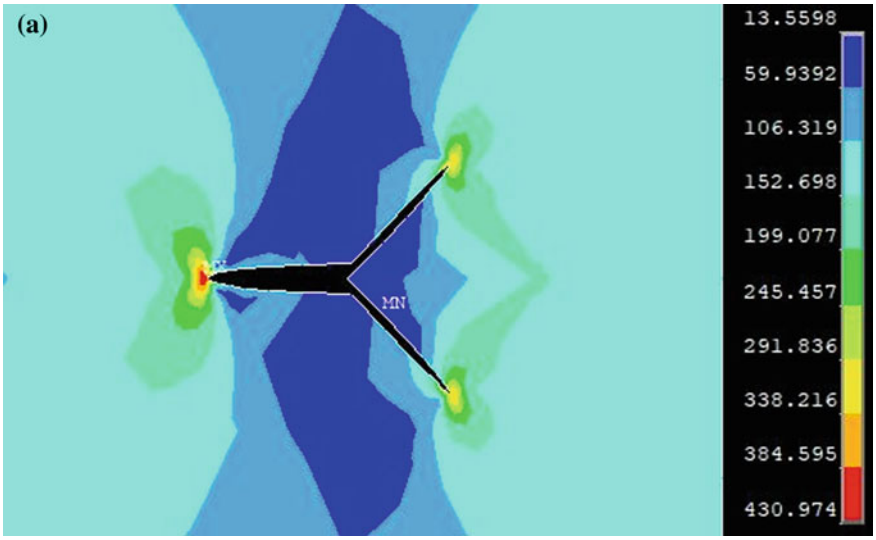


Fig. 35 (continued)

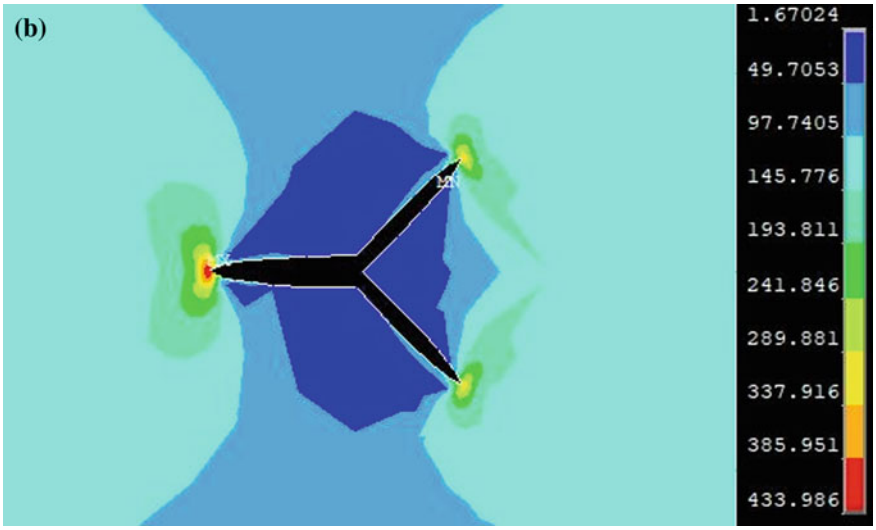


Fig. 35 (continued)

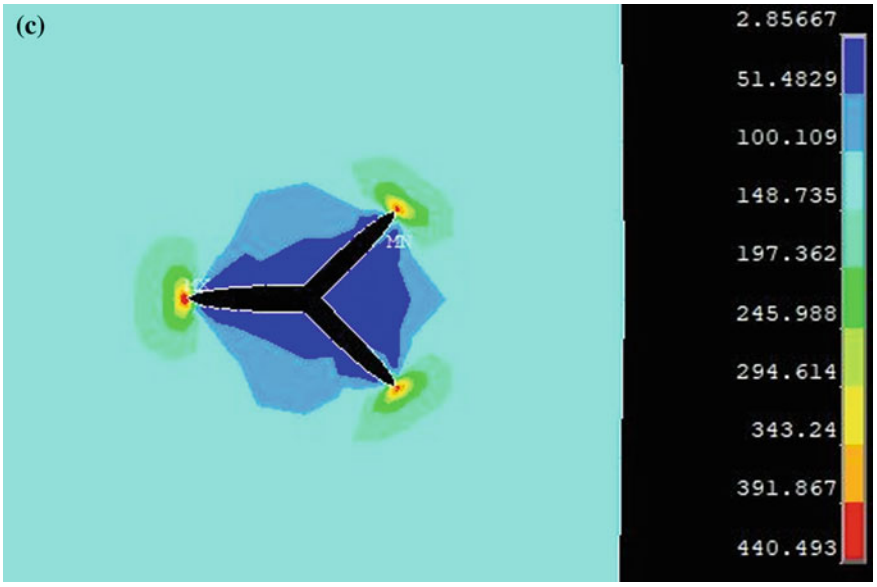


Fig. 35 (continued)

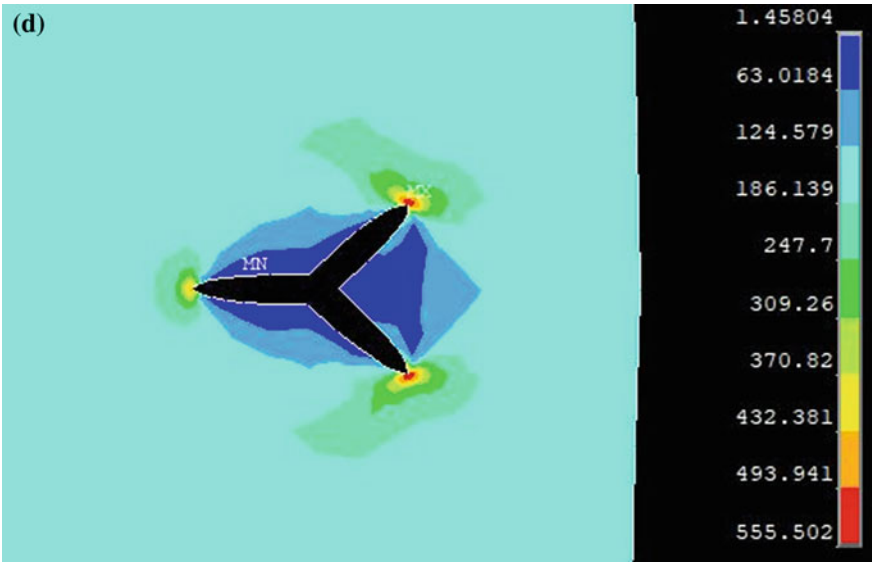


Fig. 35 (continued)

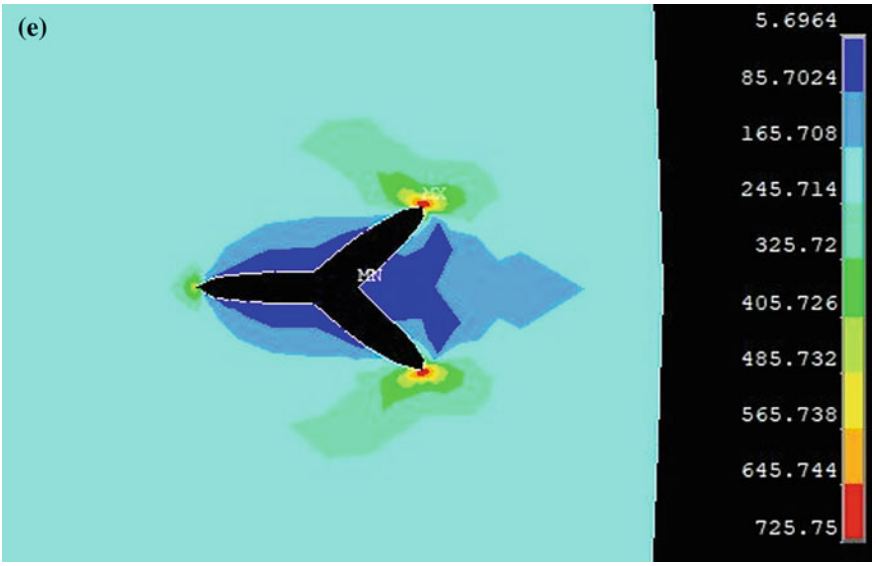


Fig. 35 (continued)

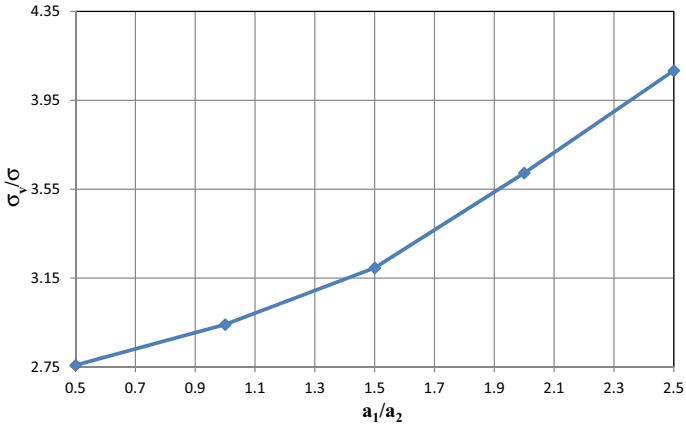


Fig. 36 Variation of von Mises with crack length ratio (a_1/a_2). **a** Distribution of von Mises stress for a/W = 0.2, β = 0.5, S = 0.5, α = 30°, a_1/a_2 = 0.5, **b** distribution of von Mises stress for a/W = 0.2, β = 0.5, S = 0.5, α = 30°, a_1/a_2 = 1, **c** distribution of von Mises stress for a/W = 0.2, β = 0.5, S = 0.5, α = 30°, a_1/a_2 = 1.5, **d** distribution of von Mises stress for a/W = 0.2, β = 0.5, S = 0.5, α = 30°, a_1/a_2 = 2, **e** distribution of von Mises stress for a/W = 0.2, β = 0.5, S = 0.5, α = 30°, a_1/a_2 = 2.5

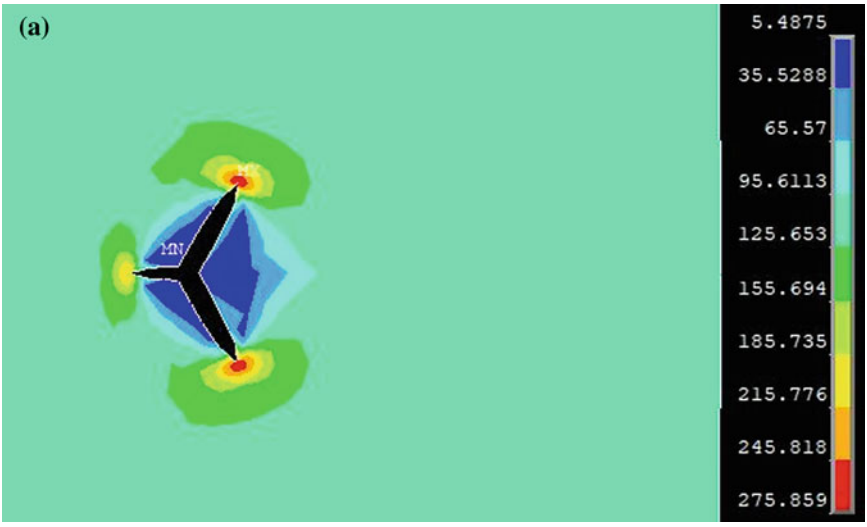


Fig. 36 (continued)

and β = 0.5 was observed. For crack tip A, it is found that the value of normalized mode I Stress intensity factor increases as the crack inclination angle increases from 0° to 75°, whereas normalized mode II SIF decreases as the crack inclination angle increases. For mode I, the increase in value was found to be 36.48% and, for mode

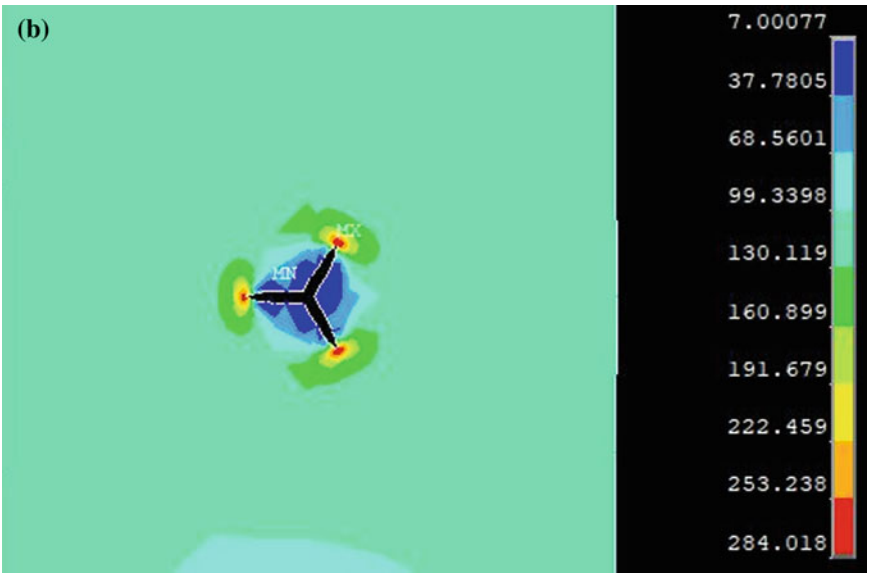


Fig. 36 (continued)

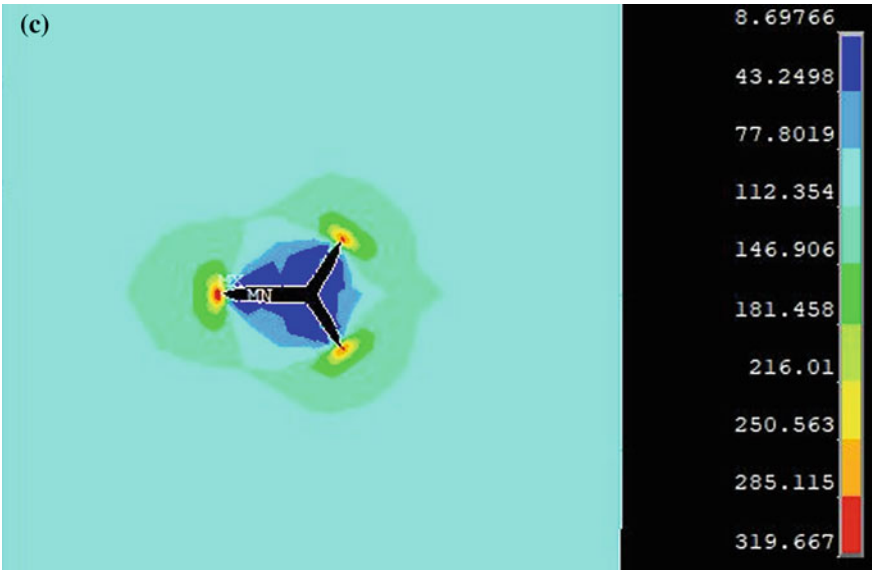


Fig. 36 (continued)

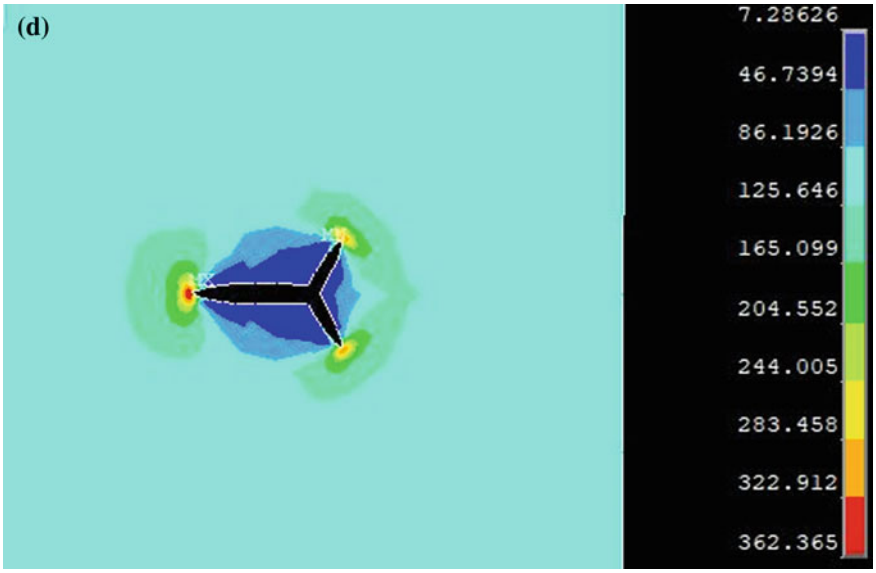


Fig. 36 (continued)

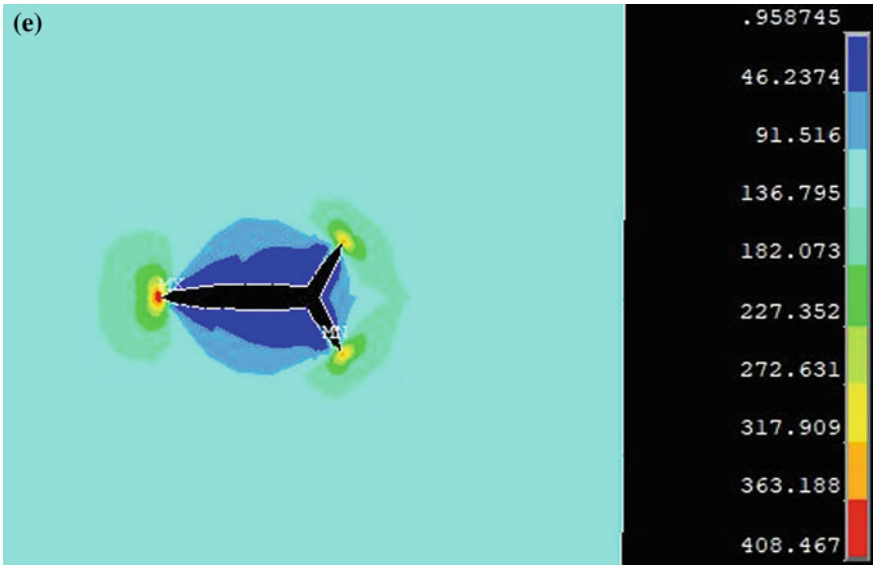


Fig. 36 (continued)

II, the decrease was found to be 56.31%. For crack tip B the value of normalized mode I SIF increases as the crack inclination angle is increased from 0° to 75° which comes out to be 260.30%, whereas the normalized mode II SIF decreases as the crack inclination angle increases, i.e., 51.05%. It shows that, as the crack is inclined more towards the higher loading axis, mode I stress intensity factor increases. The normalized mode II SIF decreases with an increase in the inclination angle of the secondary crack.

Biaxial factor (β): The effect of biaxial load factor β on stress intensity factors K_I and K_{II} for crack length a = 2 mm and at inclination angle $\alpha = 45°$ and, for a/w = 0.1, crack tip A showed that the value of normalized mode I SIF increases linearly as the biaxial load factor increased from 0 to 2.5, whereas the normalized mode II SIF gradually increases as the horizontal tensile load increases from 0 to 2.5. For mode I, the increase in value was found to be 5.05% and, for mode II, the increase was found to be 0.69%. For crack tip B, the effect of biaxial load factor on normalized mode I and mode II SIFs showed that the value of normalized mode I SIF decreases as the biaxial load factor increased from 0 to 2.5 which is nearly 18.60%. Whereas, the normalized mode II SIF increases by 416.69% from ($\beta = 0$ to 0.5) linearly because at $\beta = 0$ the horizontal tensile load is less so it acts as uniaxial loading on geometry whereas from ($\beta = 0.5$ to 2.5), the normalized mode II SIF decreases with 40.95%, as the horizontal tensile load increases.

Crack length ratio (a_1/a_2): The effect of crack length ratio (a_1/a_2) on normalized mode I and mode II SIF at $\alpha = 30$ and $\beta = 1$ for crack tip A, showed that normalized mode I SIF increases as the crack length ratio increases from ($a_1/a_2 = 0.5$ to 2.5), whereas the normalized mode II SIF decreases. For mode I, the increase in value was found to be 6.06% and, for mode II, the decrease in value was found to be 41.74%. For crack tip B, it is shown that normalized mode I SIF increases as the crack length ratio increases from ($a_1/a_2 = 0.5$ to 2.5) which was found to be 7.08% increase, whereas the normalized mode II SIF decreases from 0.5 to 2.5 at equal tension load that comes out to be 77.01%.

6 Future Aspects

- More fracture parameters can be calculated for biaxial loading.
- T-stress can be evaluated using other experimental techniques like DIC, photoelasticity, strain gauges, etc.
- Various laws of fracture mechanics can be applied for T-stress like the Paris Law.

References

1. Sedmak, A.: Finite element evaluation of fracture mechanics parametres using rapid mesh refinement. Adv. Fract. Res. **2**, 1095–1106 (1984)
2. Haber, O., Grebner, H., Hofler, A.: Mesh generation by conformal mapping in two-dimensional fracture problems. In: Proceedings of the 7th International Conference on Fracture, vol. 3, pp. 2091–2098 (1989)
3. Wang, J., Chow, C.L.: Mixed mode ductile fracture studies with nonproportional loading based on continuum damage mechanics. J. Eng. Mater. Technol. **111**(2), 204–209 (1989)
4. Dutta, B.K., Kakodkar, A., Maiti, S.: Use of two singular point finite elements in the analysis of kinked cracks. In: Proceedings of the 7th International Conference on Fracture, vol. 3, pp. 2315–2322 (1989)
5. Mohanty, D.K., Maiti, S.K.M.A.: Experimental and finite element studies on mode I and mixed mode (I and II) stable crack growth-I, experimental. Eng. Fract. Mech. **37**, 1237–1250 (1990)
6. Haefele, P.M., Lee, J.D.: Combination of finite element analysis and analytical crack tip solution for mixed mode fracture. Eng. Fract. Mech. **50**(5–6), 849–868 (1995)
7. Hasan, M.S., Jordan, E.: An approximate analysis of perpendicular cracks under general in-plane loading. Eng. Fract. Mech. **51**(2), 167–191 (1995)
8. Kamaya, M.: A crack growth evaluation method for interacting multiple cracks. Int. J. Ser. A Solid Mech. Mater. Eng. **46**(1), 15–23 (2003)
9. Zhao, J., Xie, L., Liu, J., Zhao, Q.: Method for stress intensity factor calculation of infinite plate containing multiple hole-edge cracks. Int. J. Fatigue **35**, 2–9 (2012)
10. Dehnavia, M.Y., Khaleghianb, S., Emamic, A., Tehranid, M., Soltania, N.: Utilizing digital image correlation to determine stress intensity factors. Polym. Test. **2**, 3–18 (2014)
11. Theocaris, P.S., Kardomateas, G.A., Andrianopoulos, N.P.: Experimental study of the T-criterion in ductile fractures. Eng. Fract. Mech. **17**(5), 439–447 (1983)
12. Cheung, Y.K., Chen, Y.Z.: Solutions of branch crack problems in plane elasticity by using a new integral equation approach. Eng. Fract. Mech. **28**(1), 31–41 (1987)
13. Lam, K.Y., Ong, P.P., Wude, N.: Interaction between a circular inclusion and a symmetrically branched crack. Theoret. Appl. Fract. Mech. **28**(3), 197–211 (1998)
14. Theocaris, P.S.: The elastic field around the crack tip measured by scanning electromicroscopy. Eng. Fract. Mech. **37**(4), 739–751 (1990)
15. Xie, Y.J., Li, J., Hu, X.Z., Wang, X.H., Cai, M., Wang, W.: Modelling of multiple crack-branching from Mode-I crack-tip in isotropic solids. Eng. Fract. Mech. **109**, 105–116 (2013)
16. Meggiolaro, M.A., Miranda, A.C.O., Castro, J.T.P., Martha, L.F.: Stress intensity factor equations for branched crack growth. Eng. Fract. Mech. **72**(17), 2647–2671 (2005)
17. Li, Y., Sun, T., Gao, Q., Tan, C.: A stress intensity factor estimation method for kinked crack. Eng. Fract. Mech. **188**, 202–216 (2018)
18. Chen, J.W., Zhou, X.P.: The enhanced extended finite element method for the propagation of complex branched cracks. Eng. Anal. Boundary Elem. **104**, 46–62 (2019)
19. Irwin, G.R.: Analysis of stress strains near the end of a crack traversing plate. Trans. ASME J. Appl. Mech. **24**, 361–364 (1957)
20. Sun, L., Ma, D., Wang, L., Shi, X., Wang, J., Chen, W.: Determining indentation fracture toughness of ceramics by finite element method using virtual crack closure technique. Eng. Fract. Mech. **197**, 151–159 (2018)
21. Inglis, C.E.: Stresses in plate due to the presence of cracks and sharp corners. Trans. Inst. Naval Arch.
22. Ayatollahi, M.R., Pavier, M.J., Smith, D.J.: Determination of T-stress from finite element analysis for mode I and mixed mode I/II loading. Int. J. Fract. **91**, 283–298 (1998)
23. Ayatollahi, M.R., Pavier, M.J., Smith, D.J.: Mode I cracks subjected to large T-stresses. Int. J. Fract. **117**, 159–177 (2002)
24. Meliani, M.H., Azari1, Z., Pluvinage, G., Matvienko, Y.G.: New approach for the T-stress estimation for specimens with a U-notch. Theor. Appl. Fract. Mech. **88**, 29–41 (2015)

25. Matvienko, Yu.G.: In: 17th European Conference on Fracture, 2–5 Sept, Brno, Czech Republic (2008)
26. Sherry, A.H., France, C.C., Goldthorpe, M.R.: Calcuations of fracture parameters in mixed mode loading. Fatigue Fract. Eng. Mater. Struct. **18**, 141–155 (1995)
27. Rice, J.R.: T-stress evaluations in multiple crack interaction. J. Mech. Phys. Solids **22**, 17–26 (1974)
28. Larsson, S.G., Carlsson, A.J.: T-stress evaluations multiple crack interaction. J. Mech. Phys. Solids **21**, 263–278 (1973)
29. Kabiri, M.R.: T-stress evaluations of mixed mode I/II fracture specimens and T-effects on mixed mode failure of aluminium. Theoret. Appl. Fract. Mech. **102**, 33–52 (2003)
30. Sumpter, J.D.S.: Analytical determination of T-stress. ASTM STP **1171**, 492–502 (1993)
31. Chao, Y.J., Lam, P.S., Zhang, L.: Interaction between branched cracks. Theoret. Appl. Fract. Mech. **30**, 75–86 (1998)
32. Hancock, J.W., Reuter, W.G., Parks, D.M.: Analytical determination of T-stress. In: ASTM STP, vol. 1171, pp. 21–40 (1993)
33. Bouledroua, O., Meliani, M.H., Pluvinage, G.: A review of T-stress calculation methods in fracture mechanics computation. A-Sci. Fondamentales et Eng. **18**, 20–30 (2016)
34. Zhang, H.H., Liu, S.M., Han, S.Y., Fan, L.F.: Computation of T-stresses for multiple-branched and intersecting cracks with the numerical manifold method. Eng. Anal. Boundary Elem. **107**, 149–158 (2019)
35. Gope, P.C., Bisht, N., Singh, V.K.: Influence of crack offset distance on interaction of multiple collinear and offset edge cracks in a rectangular plate. Theoret. Appl. Fract. Mech. **70**, 19–29 (2014)
36. Bisht, N., Gope, P.C., Panwar, K.: Influence of crack offset distance on the interaction of multiple cracks on the same side in a rectangular plate. Frattura ed Integrità Strutturale **9**(32), 1–12 (2015)
37. Nhu, N.H., Giang, N.T.: Calculation of fracture mechanics parameter via FEM for some cracked plate under different loads. Vietnam J. Mech. **8**, 22–29 (2006)